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(54) **CD4-SPECIFIC ANTIBODY CONSTRUCTS AND COMPOSITIONS AND USES THEREOF**

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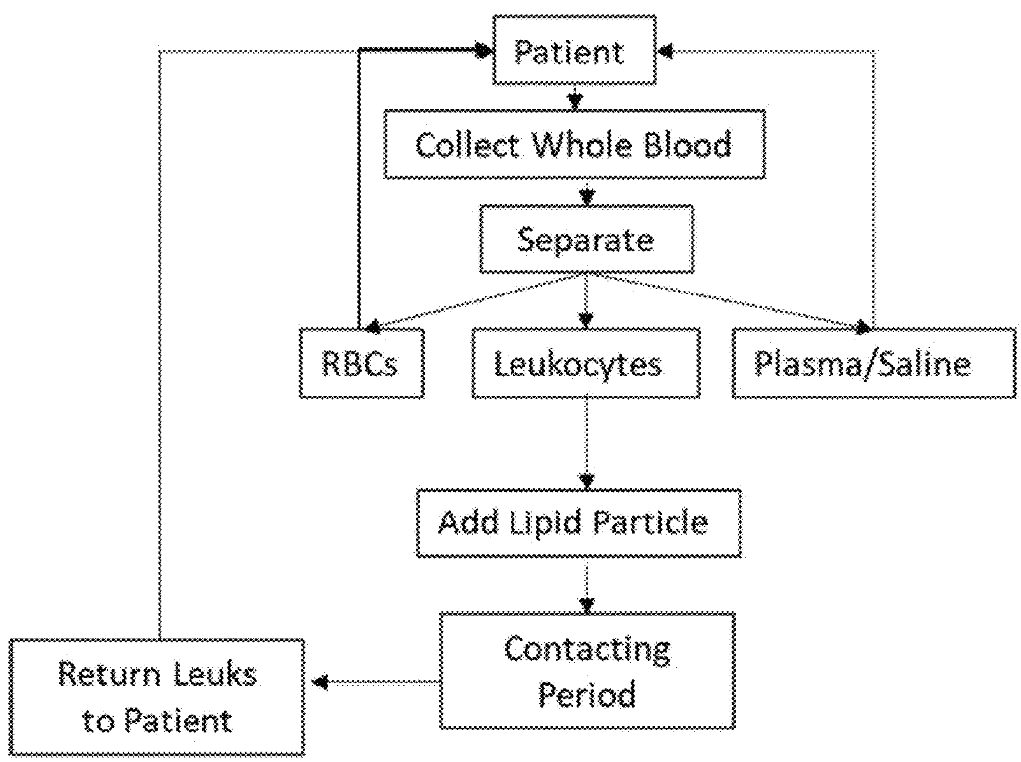
(57) **ABSTRACT**

Disclosed herein are antibodies and antigen binding fragments thereof that specifically bind human CD4. Also disclosed are fusion proteins comprising a glycoprotein G of the Paramyxoviridae family and CD4 antibodies for targeting and transducing cells expressing CD4. Viral vectors and other compositions containing the fusion proteins, as well as methods of using the fusion proteins, are also disclosed.

Specification includes a Sequence Listing.



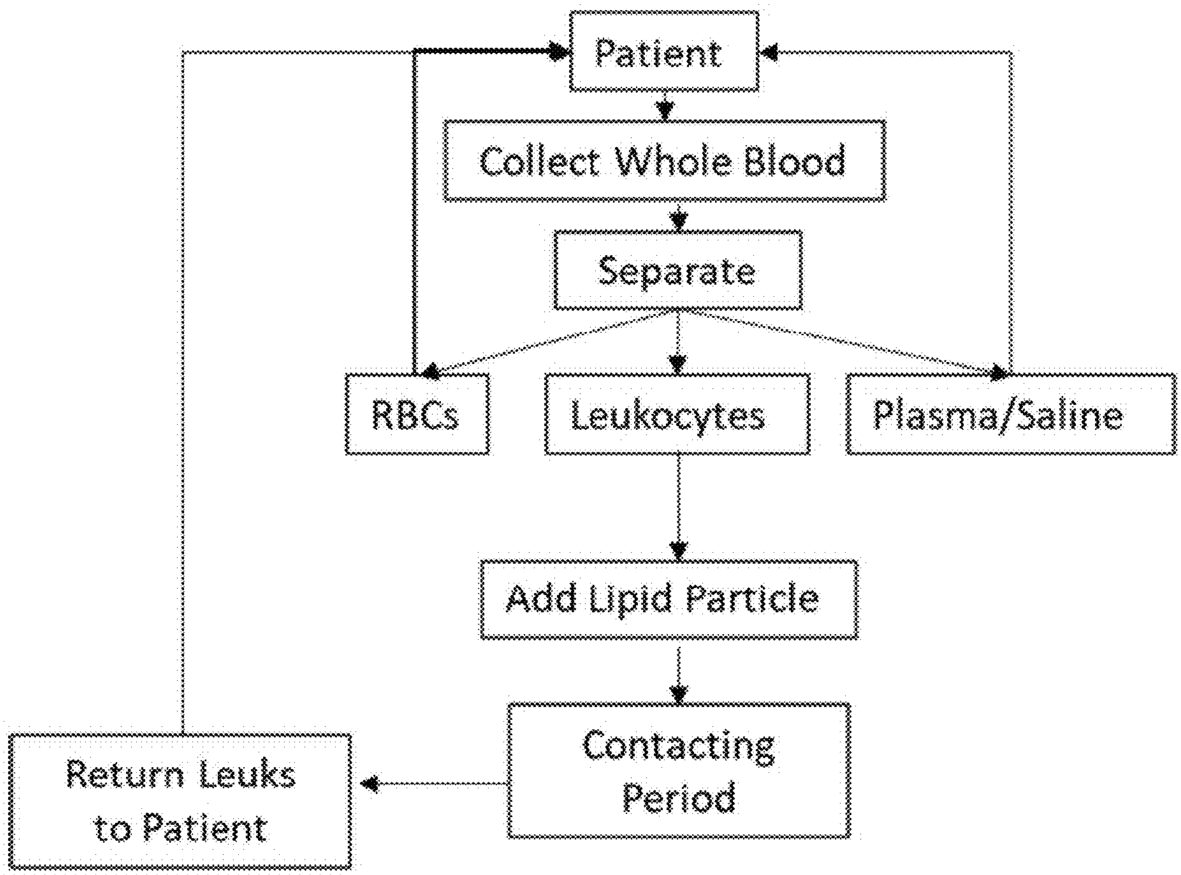

Patient
Collect Whole Blood
Separate
RBCs
Leukocytes
Plasma/Saline
Add Lipid Particle
Contacting Period
Return Leuks to Patient

FIG. 1

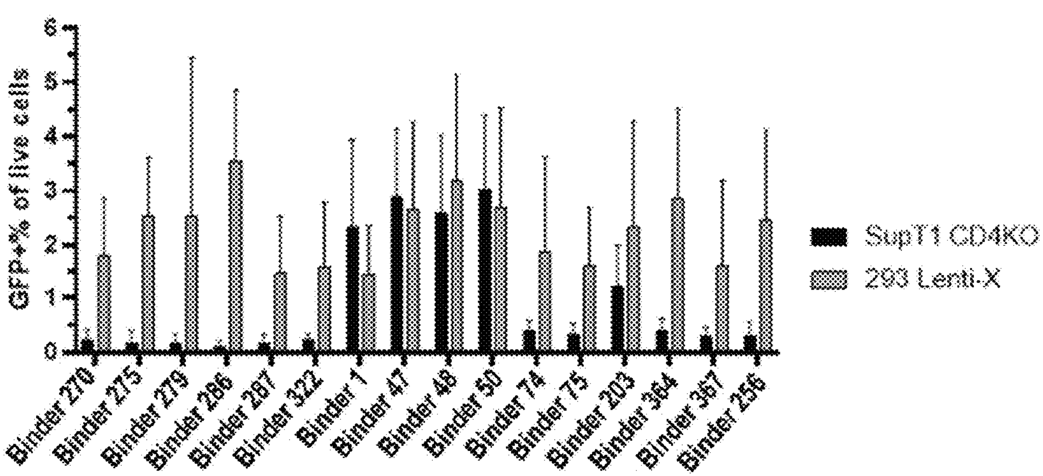
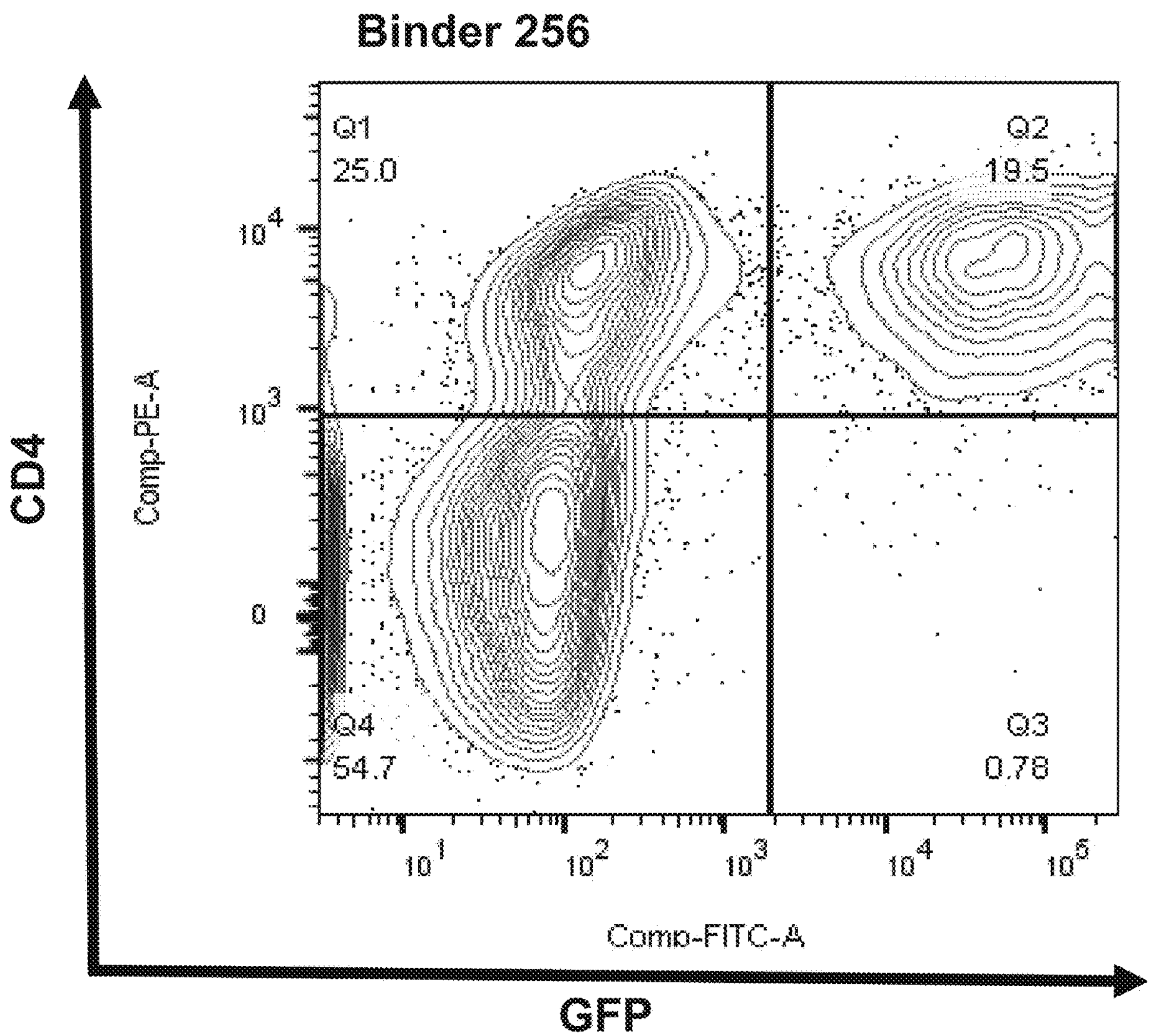

GFP+% of live cells
6
5
4
3
2
1
0
Binder 270
Binder 275
Binder 279
Binder 286
Binder 287
Binder 322
Binder 1
Binder 47
Binder 48
Binder 50
Binder 74
Binder 75
Binder 203
Binder 364
Binder 367
Binder 256
SupT1 CD4KO
293 Lenti-X

FIG. 2



Binder 256
Q1
25.0
Q2
19.5
Q4
54.7
Q3
0.78
CD4
Comp-PE-A
10^4
10^3
0
10^1
10^2
10^3
10^4
10^5
Comp-FITC-A
GFP

FIG. 3

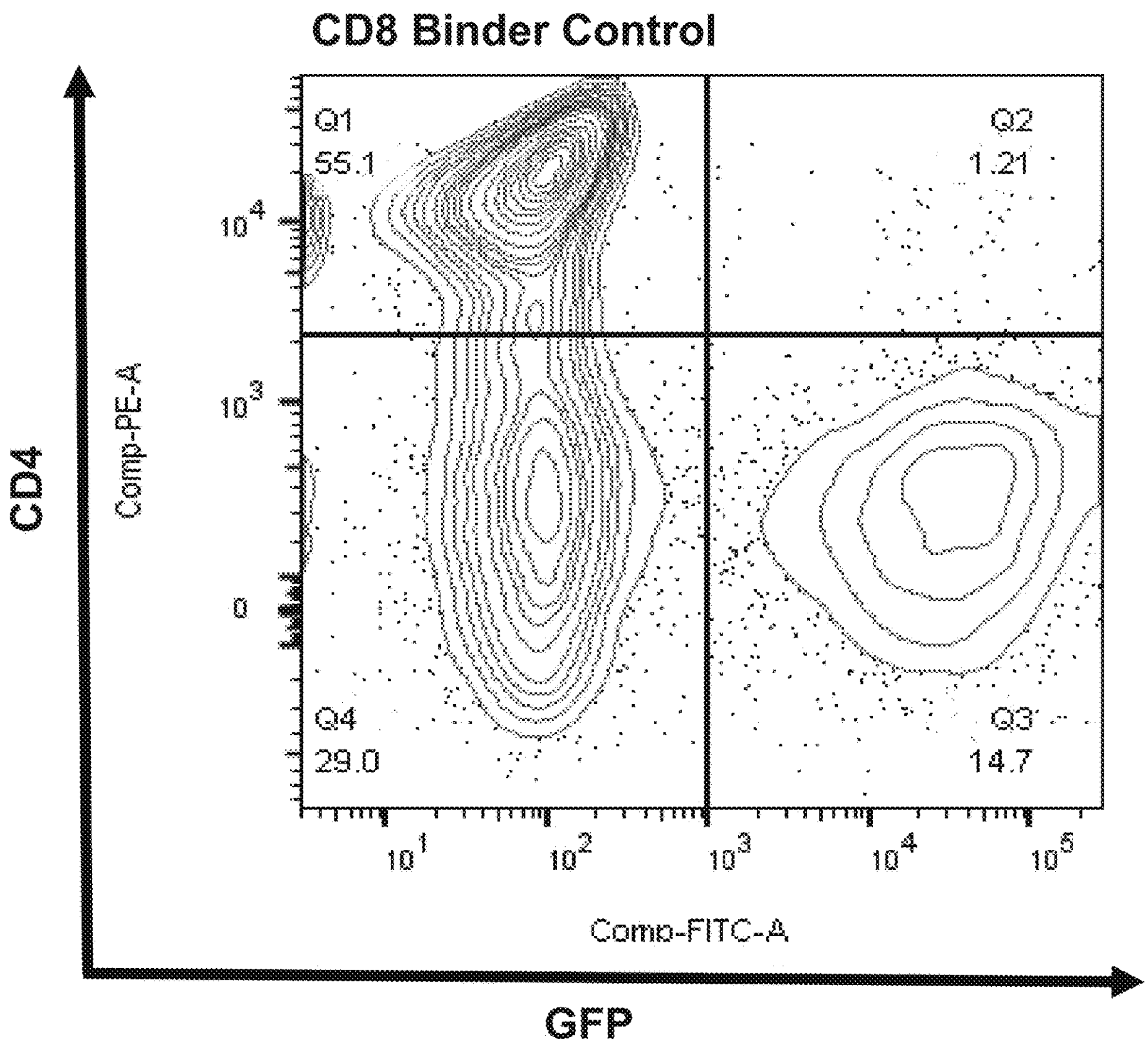

CD8 Binder Control
Q1
55.1
Q2
1.21
Q4
29.0
Q3
14.7
10^4
10^3
0
Comp-PE-A
CD4
10^1
10^2
10^3
10^4
10^5
Comp-FITC-A
GFP

FIG. 3 cont.

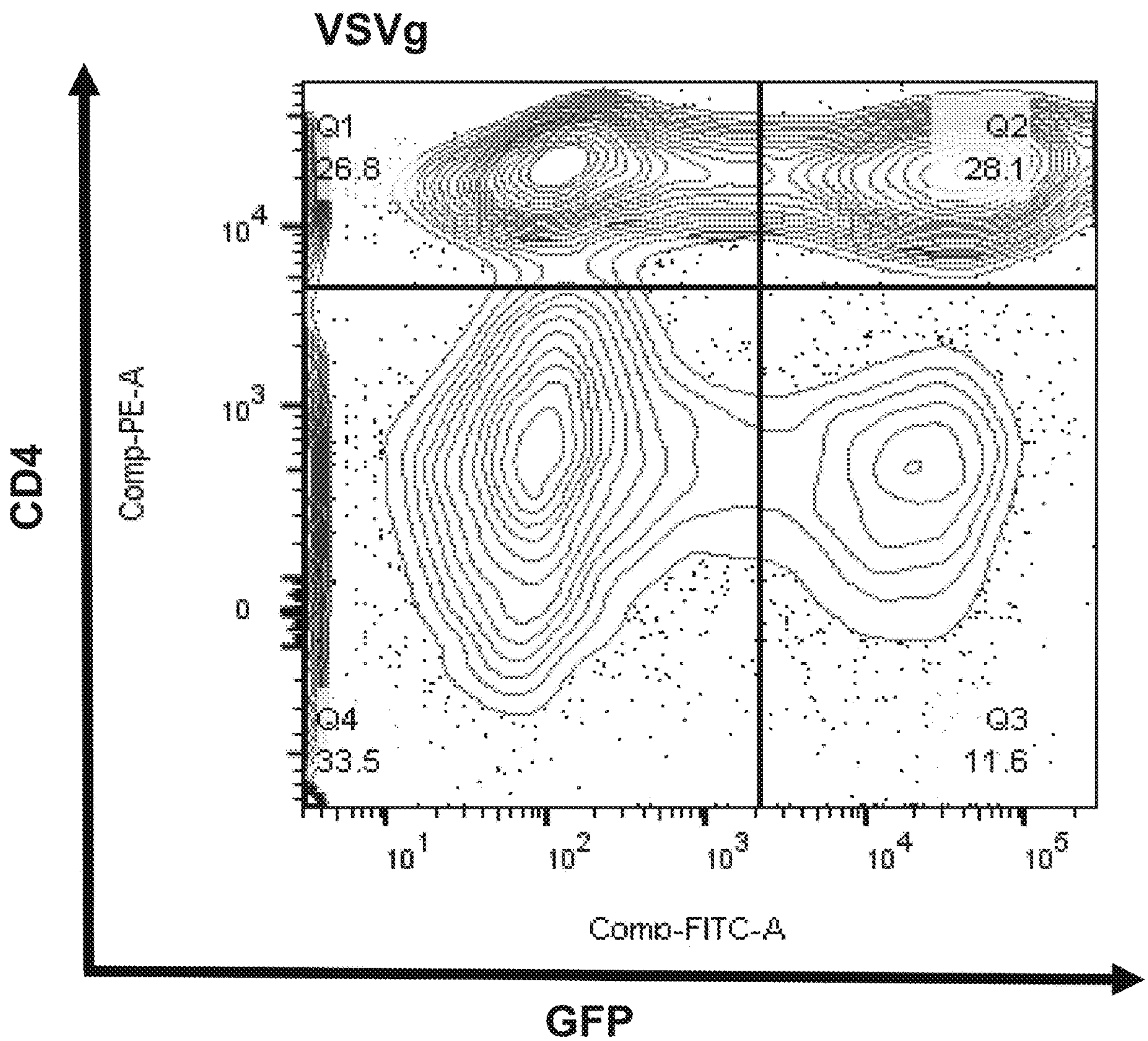

VSVg
Q1
26.8
Q2
28.1
Q4
33.5
Q3
11.6
CD4
Comp-PE-A
GFP
Comp-FITC-A
10^4
10^3
0
10^1
10^2
10^3
10^4
10^5

FIG. 3 cont.

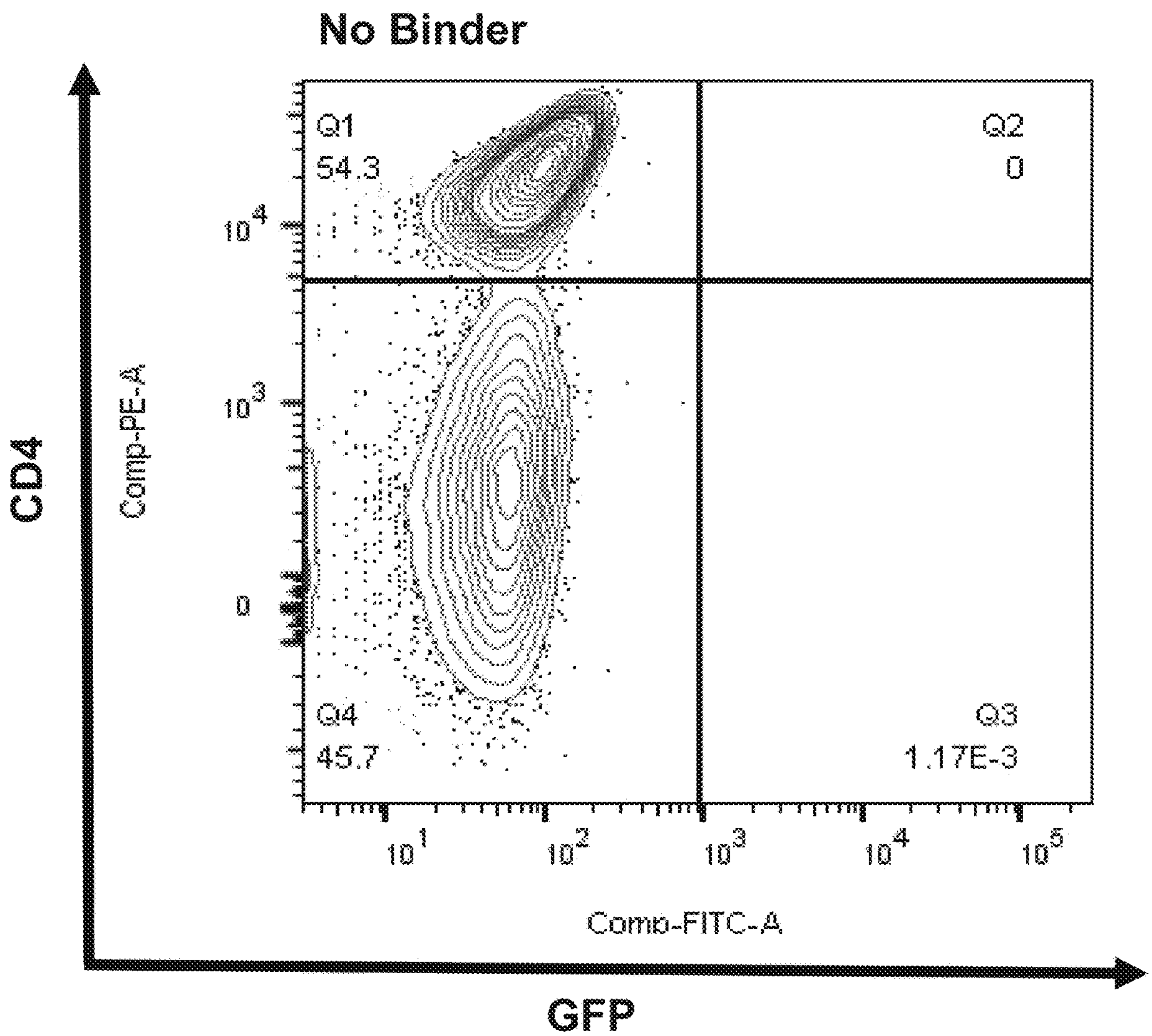

No Binder
Q1
54.3
Q2
0
Q4
45.7
Q3
1.17E-3
CD4
Comp-PE-A
GFP
Comp-FITC-A
0
10^3
10^4
10^1
10^2
10^5

FIG. 3 cont.

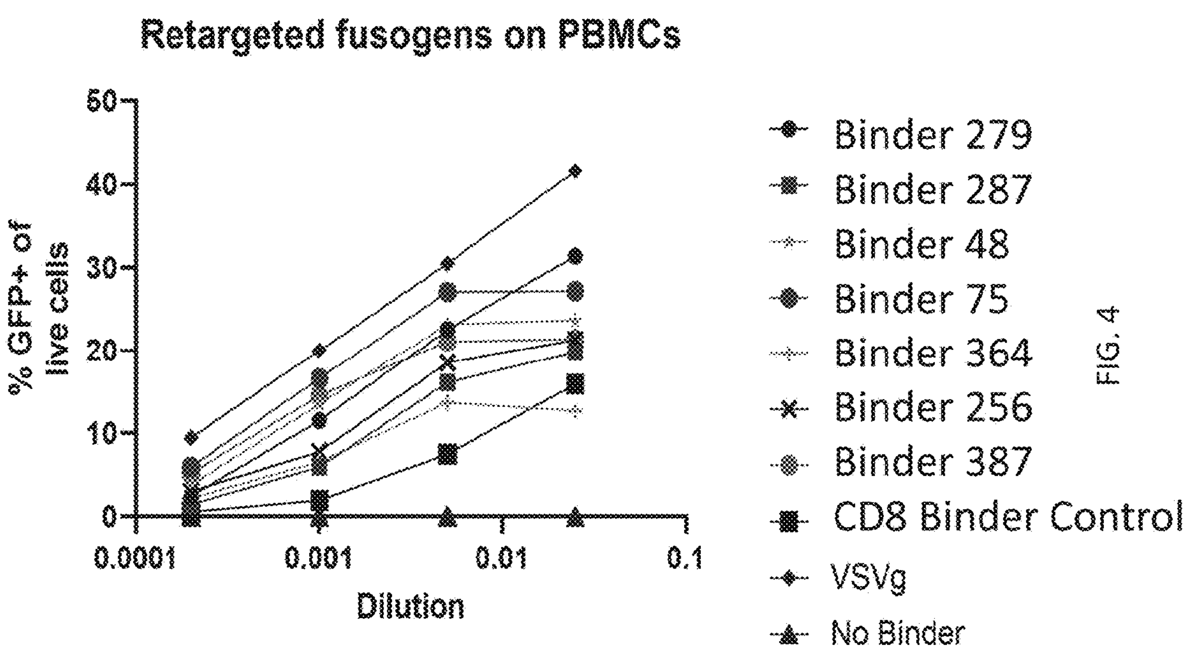
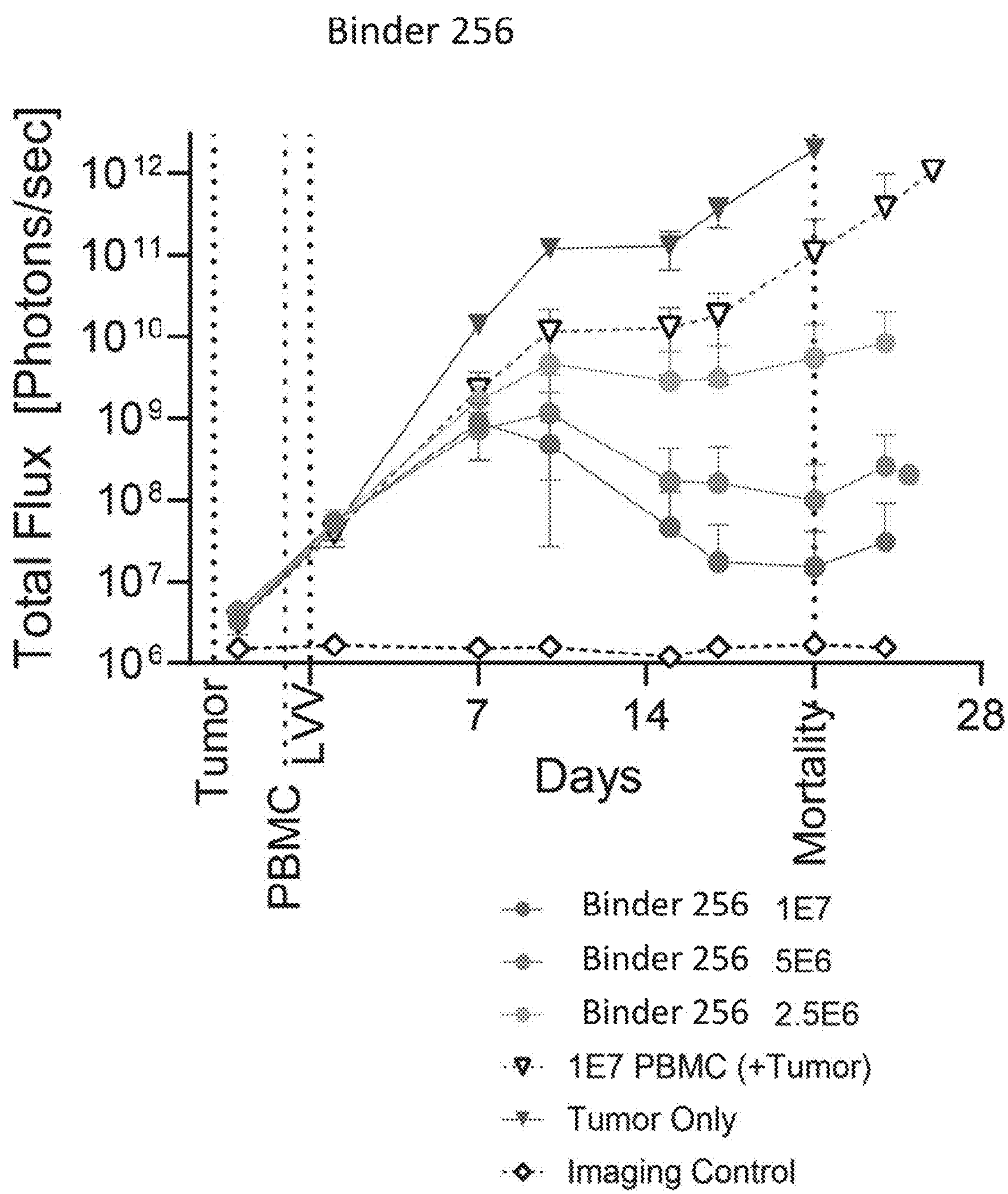

Retargeted fusogens on PBMCs
% GFP+ of live cells
50
40
30
20
10
0
0.0001
0.001
0.01
0.1
Dilution
Binder 279
Binder 287
Binder 48
Binder 75
Binder 364
Binder 256
Binder 387
CD8 Binder Control
VSVg
No Binder

FIG. 4



Binder 256
Total Flux [Photons/sec]
10^{12}
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{6}
Tumor
PBMC
LVV
7
14
Mortality
28
Days
Binder 256 1E7
Binder 256 5E6
Binder 256 2.5E6
1E7 PBMC (+Tumor)
Tumor Only
Imaging Control

FIG.5A

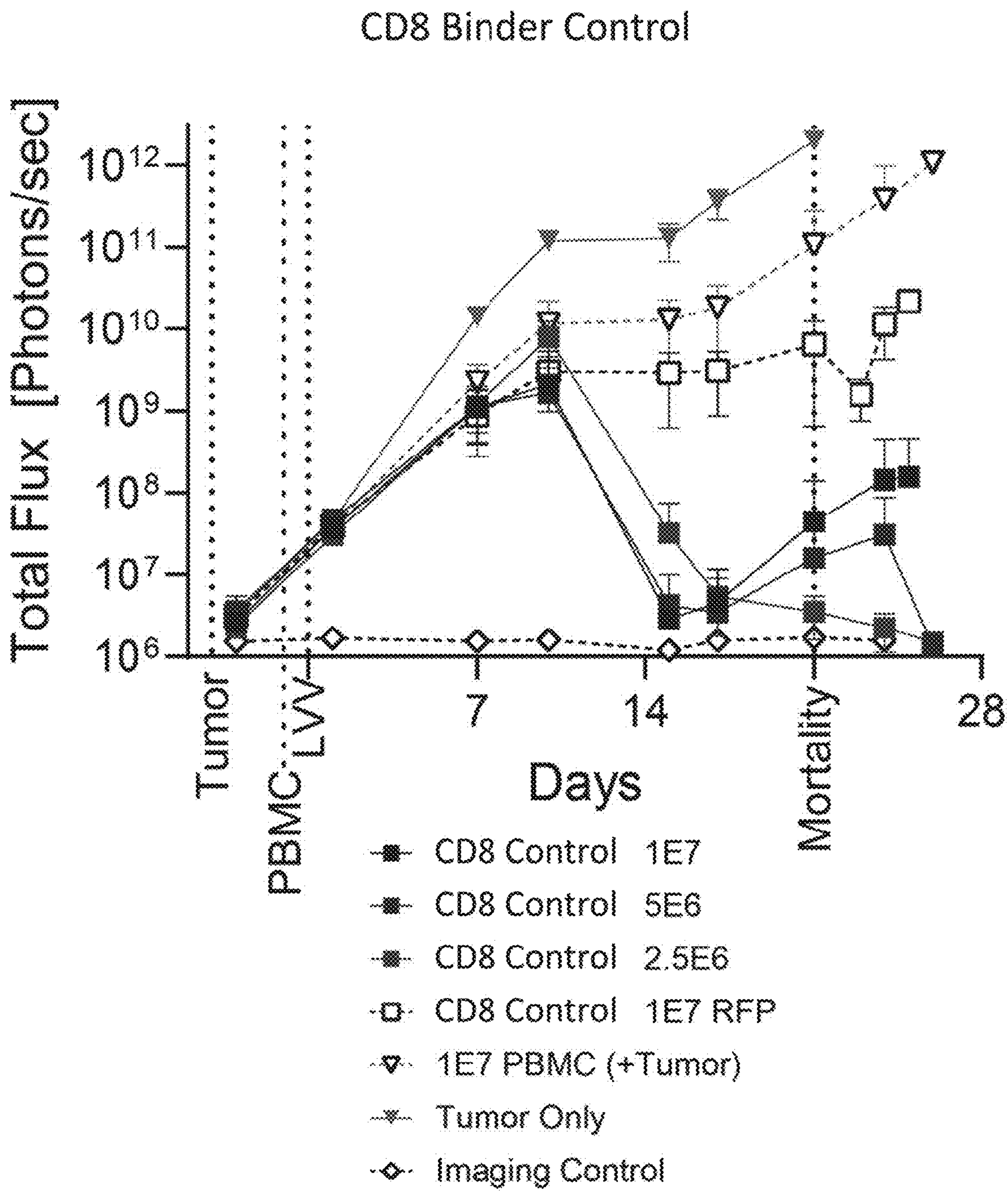

CD8 Binder Control
Total Flux [Photons/sec]
10^12
10^11
10^10
10^9
10^8
10^7
10^6
Tumor
PBMC
LVV
7
14
Mortality
28
Days
CD8 Control 1E7
CD8 Control 5E6
CD8 Control 2.5E6
CD8 Control 1E7 RFP
1E7 PBMC (+Tumor)
Tumor Only
Imaging Control

FIG.5B

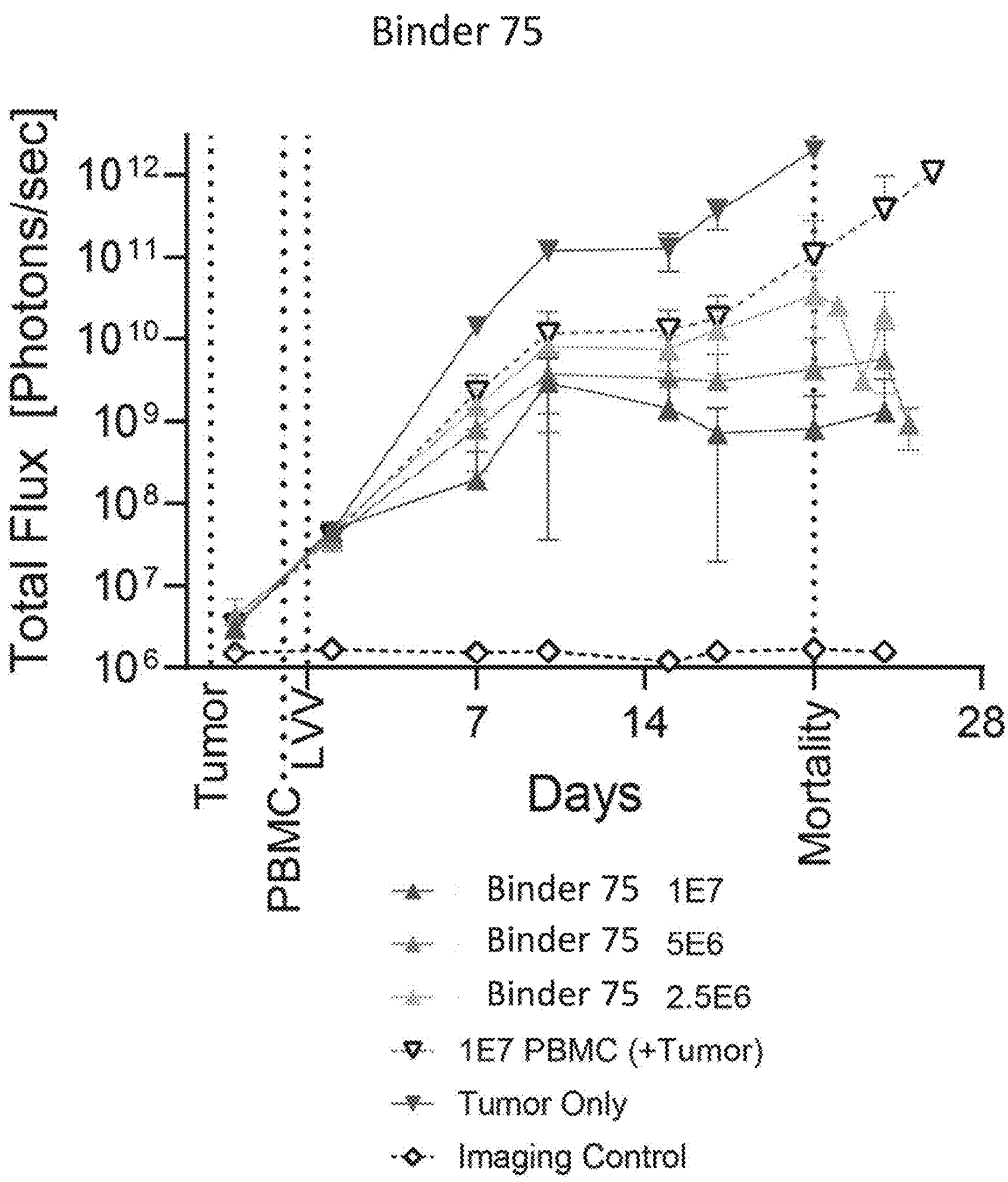

Binder 75
Total Flux [Photons/sec]
10^{12}
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{6}
Tumor
PBMC
LVV
7
14
Mortality
28
Days
Binder 75 1E7
Binder 75 5E6
Binder 75 2.5E6
1E7 PBMC (+Tumor)
Tumor Only
Imaging Control

FIG.5C

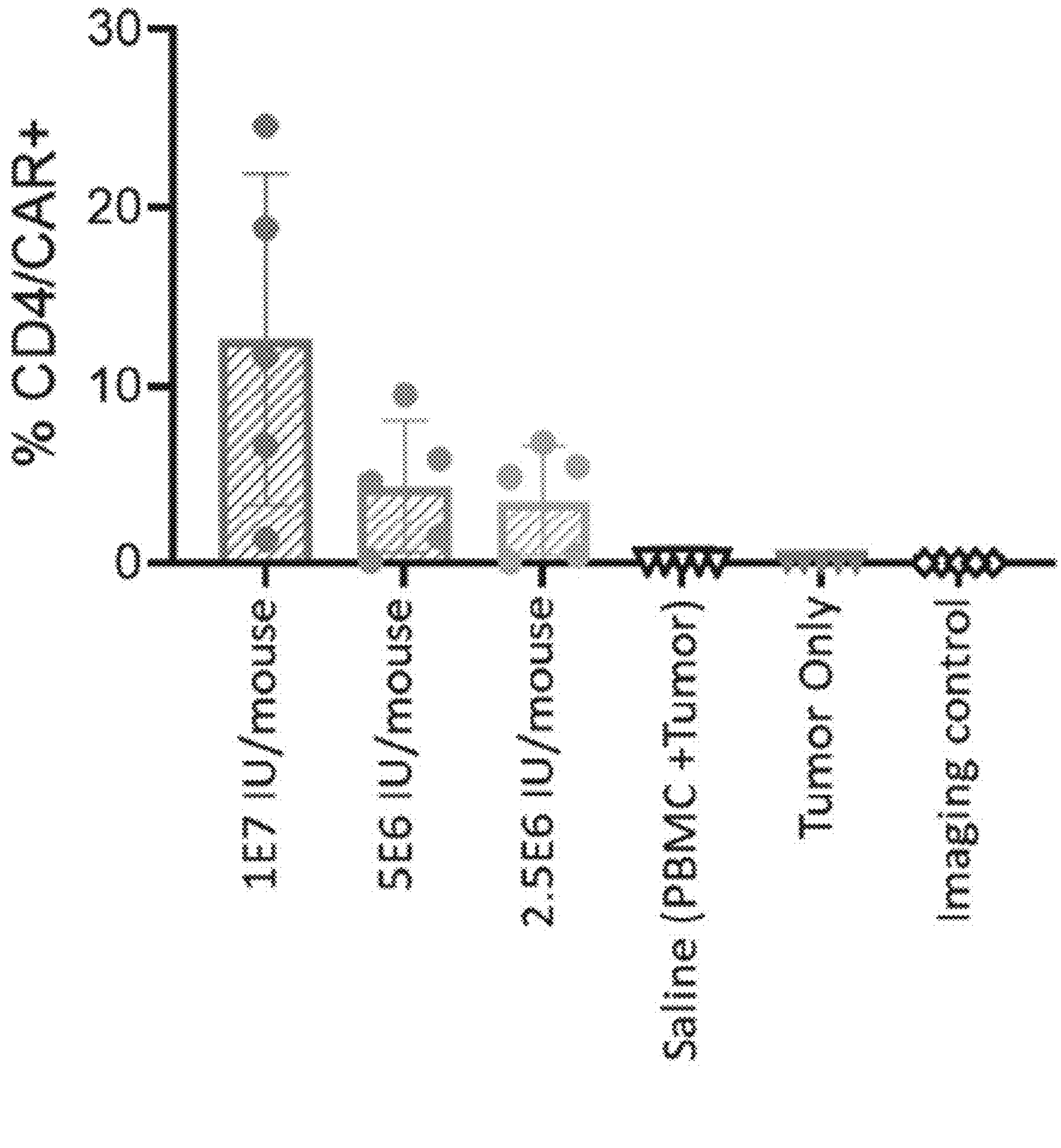

% CD4/CAR+
30
20
10
0
1E7 IU/mouse
5E6 IU/mouse
2.5E6 IU/mouse
Saline (PBMC +Tumor)
Tumor Only
Imaging control

FIG. 5D

CD4-SPECIFIC ANTIBODY CONSTRUCTS AND COMPOSITIONS AND USES THEREOF

FIELD

[0001] The present disclosure relates to antibodies or antigen binding fragments thereof that specifically bind human CD4. Also disclosed are fusion proteins comprising an envelope glycoprotein G, H, HN, and/or an F protein of the Paramyxoviridae family. Also disclosed are fusosomes comprising an envelope glycoprotein G, H, and/or an F protein of the Paramyxoviridae family. Fusosomes in one embodiment are gene therapy vectors pseudotyped with an envelope glycoprotein, including envelope glycoproteins G, H, HN and/or an F protein of the Paramyxoviridae family. Also disclosed are an envelope glycoprotein G, H, HN, and/or an F protein of the Paramyxoviridae family and a CD4 antibody, or an antigen binding fragment thereof, for targeting and transducing cells expressing CD4. Viral vectors and other compositions containing the fusion proteins, antibodies, or antigen binding fragments thereof, as well as methods of using the fusosomes, fusion proteins, antibodies, or antigen binding fragments thereof are also disclosed.

SUMMARY

[0002] CD4 (cluster of differentiation 4) is a transmembrane glycoprotein that serves as a co-receptor for the T cell receptor (TCR). CD4 serves multiple functions in immune responses against both external and internal challenges. In T cells, the CD4 co-receptor functions primarily to bind to a major histocompatibility complex (MHC) molecule to facilitate T cell signaling and aid with cytotoxic T cell antigen interactions. While CD4 is predominantly expressed on the surface of helper T cells, it can also be found on natural killer cells, cortical thymocytes, and dendritic cells. The CD4 molecule is also used as a marker for cytotoxic T cell populations.

[0003] T lymphocytes are common targets in gene therapy, even more so since chimeric antigen receptor (CAR) T cells have reached the clinic. Current approaches for T cell engineering mainly rely on ex vivo gene transfer methods. Following their isolation from either healthy donors or patients, lymphocytes are activated and subsequently transduced by lentiviral vectors. The modified lymphocytes are then expanded and either used in functional in vivo assays or used for in vivo applications.

[0004] Ex vivo modification of T lymphocytes, however, has its disadvantages. The complexity of the overall procedure, cost of the manufacturing process, and prolonged ex vivo culture negatively impact the quality of the final product. Methods that improve T lymphocyte engineering that use in vivo delivery platforms are needed.

[0005] In vivo delivery platforms using fusogenic glycoproteins of viral vectors have been shown to be beneficial for targeting, binding, and transducing cells of interest. Certain fusogenic glycoproteins, however, may not be sufficiently stable or expressed on the surface of the viral vector. Thus, improved fusogenic glycoproteins, fusosomes and viral vectors containing those glycoproteins are needed. The provided disclosure addresses this need.

[0006] The present disclosure provides an antibody or antigen binding fragment thereof that specifically binds CD4, comprising certain heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and/or light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3). Another embodiment is an antibody or antigen binding fragment thereof specifically binding CD4, comprising certain heavy (VH) and/or light (VL) chain variable regions. The disclosure likewise provides for isolated nucleotides, vectors, and host cells comprising the anti-CD4 antibody or antigen binding fragment thereof.

[0007] The present disclosure also provides a fusion protein comprising a glycoprotein G (G protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein), or a biologically active portion thereof of the Paramyxoviridae family and at least one disclosed CD4 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment is fused to the C-terminus of the G protein or the biologically active portion thereof.

[0008] The present disclosure also provides a fusosome comprising at least one antibody or antigen binding fragment thereof that specifically binds CD4, and at least one fusogen. The antibody or antigen binding fragment thereof that specifically binds CD4 may be any antibody or antigen binding fragment thereof that specifically binds CD4, including any antibody or antigen binding fragment thereof that specifically binds CD4 described herein. In some embodiments the fusogen and the antibody or antigen binding fragment thereof that specifically binds CD4 are linked within the fusosome, for example via a linker sequence. In some embodiments the fusogen and the antibody or antigen binding fragment thereof that specifically binds CD4 are not linked within the fusosome. In some embodiments the fusogen and the antibody or antigen binding fragment thereof that specifically binds CD4 are operably linked. The fusogen may be any fusogen, including any fusogen described herein. In some embodiments the fusogen is a G protein, including any G protein described herein. The present disclosure also provides a viral vector comprising a F protein molecule or biologically active portion thereof of the Paramyxoviridae family, an envelope glycoprotein G (G protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein), or a biologically active portion thereof of the Paramyxoviridae family, and at least one disclosed CD4 antibody or antigen binding fragment thereof, wherein the antibody or antigen binding fragment thereof is attached to the C-terminus of the G protein or the biologically active portion thereof.

[0009] The present disclosure likewise relates to methods of selectively modulating and transducing CD4+ T cells using the disclosed fusosomes or viral vectors. Also disclosed are methods of delivering an exogenous agent to a subject, comprising administering to the subject the disclosed fusosomes or viral vectors, in which the fusosomes or viral vector further comprises an exogenous agent. The present disclosure also relates to methods of treating cancer in a subject, comprising administering to the subject the disclosed viral vectors, and corresponding first and second medical uses.

[0010] The present disclosure also provides compositions comprising the fusosomes or fusion proteins or viral vectors of the invention, comprising an antibody or antigen binding fragment thereof that specifically binds CD4, for use as a medicament.

[0011] The present disclosure also provides compositions comprising the fusosomes or fusion proteins or viral vectors of the invention, comprising an antibody or antigen binding fragment thereof that specifically binds CD4, for use in a method of treating cancer.

BRIEF DESCRIPTION OF DRAWINGS

[0012] FIG. **1** shows an exemplary system for administration of a lentiviral vector comprising a CD4 binding agent to a subject.

[0013] FIG. **2** shows off-target transduction of certain CD4 binders using CD4 knockout SupT1 cells and HEK-293T cells, as assessed by measuring percentage of GFP-expressing cells with flow cytometry.

[0014] FIG. **3** shows the flow cytometry data for CD4 retargeted fusogens on PBMCs.

[0015] FIG. **4** shows the percent of GFP+ cells of retargeted fusogens in donor PBMCs.

[0016] FIG. **5**A shows tumor burden at Day 21 in CD19+ tumor bearing mice treated with 2.5E6, 5E6, or 1E7 integrating units (IU) of Binder 256, as assessed by bioluminescence imaging.

[0017] FIG. **5**B shows tumor burden at Day 21 in CD19+ tumor bearing mice treated with 2.5E6, 5E6, or 1E7 IU of a CD8 Binder Control, as assessed by bioluminescence imaging.

[0018] FIG. **5**C shows tumor burden at Day 21 in CD19+ tumor bearing mice treated with 2.5E6, 5E6, or 1E7 integrating units IU of Binder 75, as assessed by bioluminescence imaging.

[0019] FIG. **5**D shows the percentage of CD4+ T cells that express CAR at Day 15 in CD19+ tumor bearing mice treated with 2.5E6, 5E6, or 1E7 integrating units (IU) of CD4-targeted CD19 CAR fusosomes, as assessed by flow cytometry.

DETAILED DESCRIPTION

[0020] Unless defined otherwise, all terms of art, notations, and other technical and scientific terms or terminology used herein are intended to have the same meaning as is commonly understood by one of ordinary skill in the art to which the claimed subject matter pertains. In some cases, terms with commonly understood meanings are defined herein for clarity and/or for ready reference, and the inclusion of such definitions herein should not necessarily be construed to represent a substantial difference over what is generally understood in the art, unless such differences are expressly noted.

[0021] Unless defined otherwise, all technical and scientific terms, acronyms, and abbreviations used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure pertains. Unless indicated otherwise, abbreviations and symbols for chemical and biochemical names is per IUPAC-IUB nomenclature. Unless indicated otherwise, all numerical ranges are inclusive of the values defining the range as well as all integer values in-between.

[0022] As used herein, the articles "a" and "an" refer to one or to more than one (i.e. to at least one) of the grammatical object of the article. By way of example, "an element" means one element or more than one element.

[0023] As used herein, the term "about" will be understood by persons of ordinary skill in the art and will vary to some extent on the context in which it is used. In some embodiments, the term "about" when referring to a measurable value such as an amount, a temporal duration, and the like, is meant to encompass art-accepted variations based on standard errors in making such measurements. In some embodiments, the term "about" when referring to such values, is meant to encompass variations of ±20% or ±10%, more preferably ±5%, even more preferably ±1%, and still more preferably ±0.1% from the specified value, as such variations are appropriate to perform the disclosed methods.

[0024] As used herein, "CD4" or "cluster of differentiation 4" refers to a transmembrane glycoprotein which is a specific marker for a subclass of T cells (which includes helper T cells). The CD4 protein acts as a co-receptor together with the T cell receptor (TCR) to recognize antigen presentation by MHC class II cells. CD4 plays a role in the development of T cells and activation of mature T cells.

[0025] As used herein, "affinity" refers to the strength of the sum total of noncovalent interactions between a single binding site of a molecule (e.g., an antibody) and its binding partner (e.g., an antigen). The affinity of a molecule for its partner can generally be represented by the equilibrium dissociation constant (K_D) (or its inverse equilibrium association constant, KA). Affinity can be measured by common methods known in the art, including those described herein. See, for example, surface plasmon resonance methods described in Pope M. E., Soste M. V., Eyford B. A., Anderson N. L., Pearson T. W., (2009) *J. Immunol. Methods.* 341(1-2):86-96, and methods described therein.

[0026] As used herein, "antibody" is meant in a broad sense and includes immunoglobulin molecules including monoclonal antibodies including murine, human, humanized, and chimeric antibodies, antibody fragments, bispecific or multispecific antibodies formed from at least two intact antibodies or antibody fragments, dimeric, tetrameric or multimeric antibodies, single chain antibodies, and any other modified configuration of the immunoglobulin molecule that comprises an antigen recognition site of the required specificity.

[0027] Immunoglobulins can be assigned to five major classes, namely IgA, IgD, IgE, IgG, and IgM, depending on the heavy chain constant domain amino acid sequence. IgA and IgG are further sub-classified to IgA1, IgA2, IgG1, IgG2, IgG3, and IgG4. Antibody light chains of any vertebrate species can be assigned to one of two types, namely kappa (κ) and lambda (Λ), based on the amino acid sequences of their constant domains.

[0028] As used herein, "antigen binding fragment" or "antibody fragment" refers to a portion of an immunoglobulin molecule that retains the heavy chain and/or the light chain antigen binding site, such as the heavy chain complementarity determining regions (HCDR) 1 (HCDR1), 2 (HCDR2), and 3 (HCDR3), the light chain complementarity determining regions (LCDR) 1 (LCDR1), 2 (LCDR2), and 3 (LCDR3), the heavy chain variable region (VH), or the light chain variable region (VL). Antibody fragments include a Fab fragment (a monovalent fragment comprising the VL or the VH); a F (ab) 2 fragment (a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region); a Fd fragment comprising the VH and CH1 domains; a Fv fragment comprising the VL and VH domains of a single arm of an antibody; a dAb fragment, which comprises a VH domain; and a variable domain (e.g., VNAR, VHH, etc.) from, e.g., human, shark, or camelid origin. VH and VL domains can be engineered and linked together via one or more synthetic linkers to form various types of single chain antibody designs in which the VH/VL domains pair intramolecularly, or intermolecularly in those cases in which the VH and VL domains are expressed by separate single chain antibody constructs, to form a monovalent antigen binding site, such as a single-chain Fv (scFv) or diabody. Such antibody fragments may be obtained using well known techniques and the fragments may be characterized in the same manner as are intact antibodies.

[0029] An antibody variable region comprises a "framework" region interrupted by three "antigen binding sites." The antigen binding sites are defined using various terms, including, for example (i) "Complementarity Determining Regions" (CDRs), three in the VH (HCDR1, HCDR2, HCDR3) and three in the VL (LCDR1, LCDR2, LCDR3) (Wu and Kabat, *J Exp Med* 132:211-50, 1970; Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md., 1991), and (ii) "Hypervariable regions," "HVR," or "HV," three in the VH (H1, H2, H3) and three in the VL (L1, L2, L3) (Chothia and Lesk *Mol Biol* 196:901-17, 1987). Other terms include "IMGT-CDRs" (Lefranc et al., *Dev Comparat Immunol* 27:55-77, 2003) and "Specificity Determining Residue Usage" (SDRU) (Almagro Mol Recognit, 17:132-43, 2004). The International ImMunoGeneTics (IMGT) database (http://www_imgt org) provides a standardized numbering and definition of antigen-binding sites. The correspondence between CDRs, HVs, and IMGT delineations is described in Lefranc et al., *Dev Comparat Immunol* 27:55-77, 2003.

[0030] The term "framework," or "FR" or "framework sequence" refers to the remaining sequences of a variable region other than those sequences defined to be antigen binding sites. Because the antigen binding site can be defined by various terms as described above, the exact amino acid sequence of a framework depends on how the antigen-binding site was defined.

[0031] The term "CDR" denotes a complementarity determining region as defined by at least one manner of identification to one of skill in the art. The precise amino acid sequence boundaries of a given CDR or FR can be readily determined using any of a number of well-known schemes, including those described by Kabat et al. (1991), "Sequences of Proteins of Immunological Interest," 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD ("Kabat" numbering scheme); Al-Lazikani et al., (1997) JMB 273,927-948 ("Chothia" numbering scheme); MacCallum et al., J. Mol. Biol. 262:732-745 (1996), "Antibody-antigen interactions: Contact analysis and binding site topography," J. Mol. Biol. 262, 732-745." ("Contact" numbering scheme); Lefranc M P et al., "IMGT unique numbering for immunoglobulin and T cell receptor variable domains and Ig superfamily V-like domains," *Dev Comp Immunol,* 2003 January; 27(1):55-77 ("IMGT" numbering scheme); Honegger A and Plückthun A, "Yet another numbering scheme for immunoglobulin variable domains: an automatic modeling and analysis tool," J Mol Biol, 2001 Jun. 8; 309(3):657-70, ("Aho" numbering scheme); and Martin et al., "Modeling antibody hypervariable loops: a combined algorithm," PNAS, 1989, 86(23):9268-9272, ("AbM" numbering scheme).

[0032] The boundaries of a given CDR or FR may vary depending on the scheme used for identification. For example, the Kabat scheme is based on structural alignments, while the Chothia scheme is based on structural information. Numbering for both the Kabat and Chothia schemes is based upon the most common antibody region sequence lengths, with insertions accommodated by insertion letters, for example, "30a," and deletions appearing in some antibodies. The two schemes place certain insertions and deletions ("indels") at different positions, resulting in differential numbering. The Contact scheme is based on analysis of complex crystal structures and is similar in many respects to the Chothia numbering scheme. The AbM scheme is a compromise between the Kabat and Chothia definitions based on that used by Oxford Molecular's AbM antibody modeling software.

[0033] In some embodiments, CDRs are defined in accordance with any of the Chothia numbering schemes, the Kabat numbering scheme, the IMGT numbering scheme, a combination of Kabat, IMGT, and Chothia, the AbM definition, and/or the contact definition. A sdAb variable domain comprises three CDRs, designated CDR1, CDR2, and CDR3. Table 1, below, lists exemplary position boundaries of CDR-H1, CDR-H2, CDR-H3 as identified by Kabat, Chothia, AbM, and Contact schemes, respectively. For CDR-H1, residue numbering is listed using both the Kabat and Chothia numbering schemes. FRs are located between CDRs, for example, with FR-H1 located before CDR-H1, FR-H2 located between CDR-H1 and CDR-H2, FR-H3 located between CDR-H2 and CDR-H3 and so forth. It is noted that because the shown Kabat numbering scheme places insertions at H35A and H35B, the end of the Chothia CDR-H1 loop when numbered using the shown Kabat numbering convention varies between H32 and H34, depending on the length of the loop.

TABLE 1

Boundaries of CDRs according to various numbering schemes.

CDR	Kabat	Chothia	AbM	Contact
CDR-H1 (Kabat Numbering[1])	H31--H35B	H26--H32 . . . 34	H26--H35B	H30--H35B
CDR-H1 (Chothia Numbering[2])	H31--H35	H26--H32	H26--H35	H30--H35
CDR-H2	H50--H65	H52--H56	H50--H58	H47--H58
CDR-H3	H95--H102	H95--H102	H95--H102	H93--H101

[1]Kabat et al. (1991), "Sequences of Proteins of Immunological Interest," 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD
[2]Al-Lazikani et al., (1997) JMB 273,927-948.

[0034] Thus, unless otherwise specified, a "CDR" or "complementary determining region," or individual specified CDRs (e.g., CDR-H1, CDR-H2, CDR-H3), of a given antibody or region thereof, such as a variable region thereof, should be understood to encompass a (or the specific) complementary determining region as defined by any of the aforementioned schemes. For example, where it is stated that a particular CDR (e.g., a CDR-H3) contains the amino acid sequence of a corresponding CDR in a given sdAb amino acid sequence, it is understood that such a CDR has a sequence of the corresponding CDR (e.g., CDR-H3) within the sdAb, as defined by any of the aforementioned schemes. It is understood that any antibody, such as a sdAb, includes CDRs and such can be identified according to any of the other aforementioned numbering schemes or other numbering schemes known to a skilled artisan.

[0035] As used herein, "Fv" refers to the minimum antibody fragment which contains a complete antigen-recognition and antigen-binding site. This region comprises a dimer of one heavy chain and one light chain variable domain in tight, non-covalent association. It is in this configuration that the three hypervariable regions of each variable domain interact to define an antigen-binding site on the surface of the VH-VL dimer. Collectively, the six hypervariable regions confer antigen-binding specificity to the antibody. However, even a single variable domain (or half of an Fv comprising only three hypervariable regions specific for an antigen) may have the ability to recognize and bind an antigen, although at a lower affinity than the entire binding site.

[0036] As used herein, "single-chain Fv" or "scFv" antibody fragments comprise the VH and VL domains of an antibody, wherein these domains are present in a single polypeptide chain. Preferably, the Fv polypeptide further comprises a linker (e.g., a polypeptide linker) between the VH and VL domains which enables the scFv to form the desired structure for antigen binding. For a review of scFv see Pluckthun in The Pharmacology of Monoclonal Antibodies, vol. 113, Rosenburg and Moore eds., Springer-Verlag, New York, pp. 269-315 (1994).

[0037] As used herein, "VHH" or "VHH antibodies" refer to single domain antibodies that comprise the variable (antigen binding) domain of the heavy chain antibody (HCAb or hcIgG) molecules produced by Camelidae family mammals (e.g., llamas, camels, and alpacas).

[0038] As used herein, "VNAR" or "VNAR antibodies" refer to single domain antibodies that comprise the variable (antibody binding) domain of the shark immunoglobulin new antigen receptors (IgNARs).

[0039] As used herein, the term "specifically binds" to a target molecule, such as an antigen, means that a binding molecule, such as a single domain antibody, reacts or associates more frequently, more rapidly, with greater duration, and/or with greater affinity with a particular target molecule than it does with alternative molecules. A binding molecule, such as a sdAb or scFv, "specifically binds" to a target molecule if it binds with greater affinity, avidity, more readily, and/or with greater duration than it binds to other molecules. It is understood that a binding molecule, such as a sdAb or scFv, that specifically binds to a first target may or may not specifically bind to a second target. As such, "specific binding" does not necessarily require (although it can include) exclusive binding.

[0040] As used herein, "percent (%) sequence identity" with respect to an amino acid or nucleic acid sequence is defined as the percentage of amino acid or nucleic acid residues in a candidate sequence that are identical with the amino acid or nucleic acid residues in another amino acid or nucleic acid sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Percent identity between nucleic acid sequences may be determined using a suite of commonly used and freely available sequence comparison algorithms provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) (Altschul, S. F. et al. (1990) J. Mol. Biol. 215:403-410), which is available from several sources, including the NCBI, Bethesda, Md., and on the Internet at http://www.ncbi.nlm.nih.gov/BLAST/. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared.

[0041] An amino acid substitution may include but is not limited to the replacement of one amino acid in a polypeptide with another amino acid. Exemplary substitutions are shown in Table 2. Amino acid substitutions may be introduced into an antibody of interest and the products screened for a desired activity, for example, retained/improved binding.

TABLE 2

Original Residue	Exemplary Substitutions
Ala (A)	Val; Leu; Ile
Arg (R)	Lys; Gln; Asn
Asn (N)	Gln; His; Asp, Lys; Arg
Asp (D)	Glu; Asn
Cys (C)	Ser; Ala
Gln (Q)	Asn; Glu
Glu (E)	Asp; Gln
Gly (G)	Ala
His (H)	Asn; Gln; Lys; Arg
Ile (I)	Leu; Val; Met; Ala; Phe; Norleucine
Leu (L)	Norleucine; Ile; Val; Met; Ala; Phe
Lys (K)	Arg; Gln; Asn
Met (M)	Leu; Phe; Ile
Phe (F)	Trp; Leu; Val; Ile; Ala; Tyr
Pro (P)	Ala
Ser (S)	Thr
Thr (T)	Val; Ser
Trp (W)	Tyr; Phe
Tyr (Y)	Trp; Phe; Thr; Ser
Val (V)	Ile; Leu; Met; Phe; Ala; Norleucine

[0042] Amino acids may be grouped according to common side-chain properties:

[0043] (1) hydrophobic: Norleucine, Met, Ala, Val, Leu, Ile;

[0044] (2) neutral hydrophilic: Cys, Ser, Thr, Asn, Gln;

[0045] (3) acidic: Asp, Glu;

[0046] (4) basic: His, Lys, Arg;

[0047] (5) residues that influence chain orientation: Gly, Pro;

[0048] (6) aromatic: Trp, Tyr, Phe.

[0049] Non-conservative substitutions will entail exchanging a member of one of these classes for another class. The term, "corresponding to" with reference to nucleotide or amino acid positions of a sequence, such as set forth in the Sequence Listing, refers to nucleotide or amino acid positions identified upon alignment with a target sequence based on structural sequence alignment or using a standard alignment algorithm, such as the GAP algorithm. For example, corresponding residues of a similar sequence (e.g. fragment or species variant) can be determined by alignment to a reference sequence by structural alignment methods. By aligning the sequences, one skilled in the art can identify corresponding residues, for example, using conserved and identical amino acid residues as guides.

[0050] The term "isolated" as used herein refers to a molecule that has been separated from at least some of the components with which it is typically found in nature or produced. For example, a polypeptide is referred to as "isolated" when it is separated from at least some of the components of the cell in which it was produced. When a polypeptide is secreted by a cell after expression, physically separating the supernatant containing the polypeptide from the cell that produced it is considered to be "isolating" the polypeptide. Similarly, a polynucleotide is referred to as "isolated" when it is not part of the larger polynucleotide (such as, for example, genomic DNA or mitochondrial DNA, in the case of a DNA polynucleotide) in which it is typically found in nature, or is separated from at least some of the components of the cell in which it was produced. Thus, a DNA polynucleotide that is contained in a vector inside a host cell may be referred to as "isolated."

[0051] As used herein, "lipid particle" refers to any biological or synthetic particle that contains a bilayer of amphipathic lipids enclosing a lumen or cavity. Typically, a lipid particle does not contain a nucleus. Examples of lipid particles include nanoparticles, viral-derived particles, or cell-derived particles. Such lipid particles include, but are not limited to, viral particles (e.g. lentiviral particles), virus-like particles, viral vectors (e.g., lentiviral vectors), exosomes, enucleated cells, vesicles (e.g., microvesicles, membrane vesicles, extracellular membrane vesicles, plasma membrane vesicles, and giant plasma membrane vesicles), apoptotic bodies, mitoparticles, pyrenocytes, or lysosomes. In some embodiments, a lipid particle is a fusosome. In some embodiments, the lipid particle is not a platelet.

[0052] As used herein a "biologically active portion," such as with reference to a protein such as a G protein or an F protein, refers to a portion of the protein that exhibits or retains an activity or property of the full-length of the protein. For example, a biologically active portion of an F protein retains fusogenic activity in conjunction with the G protein when each are embedded in a lipid bilayer. A biologically active portion of the G protein retains fusogenic activity in conjunction with an F protein when each is embedded in a lipid bilayer. The retained activity can include 10%-150% or more of the activity of a full-length or wild-type F protein or G protein. Examples of biologically active portions of F and G proteins include truncations of the cytoplasmic domain, e.g. truncations of up to 1, 2, 3, 4, 5, 6, 7, 8 9, 10, 11, 12, 13, 14, 15, 20, 22, 25, 30, 33, 34, 35, or more contiguous amino acids, see e.g. Khetawat and Broder 2010 Virology Journal 7:312; Witting et al. 2013 Gene Therapy 20:997-1005; published international; patent application No. WO/2013/148327.

[0053] As used herein, "G protein" refers to an envelope attachment glycoprotein G or biologically active portion thereof of the Paramyxoviridae family. "F protein" refers to a fusion protein F or biologically active portion thereof of the Paramyxoviridae family. "H protein" refers to an envelope attachment protein with haemagglutination activity. Morbilliviruses attachment proteins are designated H proteins. "HN protein" refers to an envelope attachment protein with haemagglutination-neuraminidase activity. Respiroviruses, rubulaviruses and avulaviruses attachment proteins are designated HN proteins. H, HN, and G proteins are cell attachment proteins that span the viral envelope and project from the surface as spikes. These proteins bind to proteins on the surface of target cells to facilitate cell entry.

[0054] The F and G proteins may be from a henipavirus, a Hendra (HeV) virus, or a Nipah (NiV) virus, and may be a wild-type protein or may be a variant thereof that exhibits reduced binding for the native binding partner. The F (fusion) and G (attachment) glycoproteins mediate cellular entry of Nipah virus. The G protein initiates infection by binding to the cellular surface receptor ephrin-B2 (EphB2) or EphB3. The subsequent release of the viral genome into the cytoplasm is mediated by the action of the F protein, which induces the fusion of the viral envelope with cellular membranes. The efficiency of transduction of targeted lipid particles can be improved by engineering hyperfusogenic mutations in one or both of the F protein (such as NiV-F) and G protein (such as NiV-G).

[0055] As used herein, "fusosome" refers to a particle containing a bilayer of amphipathic lipids enclosing a lumen or cavity and a fusogen that interacts with the amphipathic lipid bilayer. In some embodiments, the fusosome comprises a nucleic acid. In some embodiments, the fusosome is a membrane enclosed preparation. In some embodiments, the fusosome is derived from a source cell. In some embodiments the fusosme is a vector. In some embodiments the fusosome is an integrating vector. In some embodiments the fusosome is a viral vector. In some embodiments the fusosome is a lipid particle, including a targeted lipid particle, including any lipid particle or targeted lipid particle described herein. As used herein, "fusosome composition" refers to a composition comprising one or more fusosomes.

[0056] As used herein, "fusogen" refers to an agent or molecule that creates an interaction between two membrane enclosed lumens. In embodiments, the fusogen facilitates fusion of the membranes. In other embodiments, the fusogen creates a connection, e.g., a pore, between two lumens (e.g., a lumen of a retroviral vector and a cytoplasm of a target cell). In some embodiments, the fusogen comprises a complex of two or more proteins, e.g., wherein neither protein has fusogenic activity alone. In some embodiments, the fusogen comprises a targeting domain.

[0057] As used herein, a "re-targeted fusogen" refers to a fusogen that comprises a targeting moiety having a sequence that is not part of the naturally-occurring form of the fusogen. In embodiments, the fusogen comprises a different targeting moiety relative to the targeting moiety in the naturally-occurring form of the fusogen. In embodiments, the naturally-occurring form of the fusogen lacks a targeting domain, and the re-targeted fusogen comprises a targeting moiety that is absent from the naturally-occurring form of the fusogen. In embodiments, the fusogen is modified to comprise a targeting moiety. In embodiments, the fusogen comprises one or more sequence alterations outside of the targeting moiety relative to the naturally-occurring form of the fusogen, e.g., in a transmembrane domain, fusogenically active domain, or cytoplasmic domain.

[0058] As used herein, a "targeted envelope protein" refers to a polypeptide that contains a G protein (G protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein), of the Paramyxoviridae family attached to a single domain antibody (sdAb) variable domain, such as a VL or VH sdAb, a scFv, a nanobody, a camelid VHH domain, a shark VNAR, or fragments thereof, that target a molecule on a desired cell type. In some such embodiments, the attachment may be direct or indirect via a linker, such as a polypeptide linker. The "targeted envelope protein" may also be referred to as a "fusion protein" comprising the G protein and antibodies or antigen binding fragments of the disclosure in which the antibody or antigen binding fragment is fused to the C-terminus of the G protein or a biologically active portion thereof.

[0059] As used herein, a "targeted lipid particle" refers to a lipid particle that contains a targeted envelope protein embedded in the lipid bilayer, e.g., targeting CD4. Such targeted lipid particles can be a viral particle, a virus-like particle, a nanoparticle, a vesicle, an exosome, a dendrimer, a lentivirus, a viral vector, an enucleated cell, a microvesicle, a membrane vesicle, an extracellular membrane vesicle, a plasma membrane vesicle, a giant plasma membrane vesicle, an apoptotic body, a mitoparticle, a pyrenocyte, a lysosome, another membrane enclosed vesicle, a lentiviral vector, a viral based particle, a virus like particle (VLP), or a cell derived particle.

[0060] As used herein, a "retroviral nucleic acid" refers to a nucleic acid containing at least the minimal sequence requirements for packaging into a retrovirus or retroviral vector, alone or in combination with a helper cell, helper virus, or helper plasmid. In some embodiments, the retroviral nucleic acid further comprises or encodes an exogenous agent, a positive target cell-specific regulatory element, a non-target cell-specific regulatory element (TCSRE), or a negative TCSRE. In some embodiments, the retroviral nucleic acid comprises one or more of (e.g., all of) a 5' LTR (e.g., to promote integration), U3 (e.g., to activate viral genomic RNA transcription), R (e.g., a Tat-binding region), U5, a 3' LTR (e.g., to promote integration), a packaging site (e.g., psi (')), and RRE (e.g., to bind to Rev and promote nuclear export). The retroviral nucleic acid can comprise RNA (e.g., when part of a virion) or DNA (e.g., when being introduced into a source cell or after reverse transcription in a recipient cell). In some embodiments, the retroviral nucleic acid is packaged using a helper cell, helper virus, or helper plasmid which comprises one or more of (e.g., all of) gag, pol, and env.

[0061] As used herein, a "target cell" refers to a cell of a type to which it is desired that a targeted lipid particle delivers an exogenous agent. In embodiments, a target cell is a cell of a specific tissue type or class, e.g., an immune effector cell, e.g., a T cell. In some embodiments, a target cell is a diseased cell, e.g., a cancer cell. In some embodiments, the fusogen, e.g., a re-targeted fusogen, leads to preferential delivery of the exogenous agent to a target cell compared to a non-target cell.

[0062] As used herein a "non-target cell" refers to a cell of a type to which it is not desired that a targeted lipid particle delivers an exogenous agent. In some embodiments, a non-target cell is a cell of a specific tissue type or class. In some embodiments, a non-target cell is a non-diseased cell, e.g., a non-cancerous cell. In some embodiments, the fusogen, e.g., a re-targeted fusogen, leads to lower delivery of the exogenous agent to a non-target cell compared to a target cell.

[0063] The term "effective amount" as used herein means an amount of a pharmaceutical composition which is sufficient to significantly and positively modify the symptoms and/or conditions to be treated (e.g., provide a positive clinical response). The effective amount of the targeted lipid particles of the disclosure for use in a pharmaceutical composition will vary with the particular condition being treated, the severity of the condition, the duration of treatment, the nature of concurrent therapy, the particular lipid particle being employed, the particular pharmaceutically-acceptable excipient(s) and/or carrier(s) utilized, and like factors within the knowledge and expertise of the attending physician.

[0064] An "exogenous agent" as used herein with reference to a targeted lipid particle, refers to an agent that is neither comprised by nor encoded in the corresponding wild-type virus or fusogen made from a corresponding wild-type source cell. In some embodiments, the exogenous agent does not naturally exist, such as a protein or nucleic acid that has a sequence that is altered (e.g., by insertion, deletion, or substitution) relative to a naturally occurring protein. In some embodiments, the exogenous agent does not naturally exist in the source cell. In some embodiments, the exogenous agent exists naturally in the source cell but is exogenous to the virus. In some embodiments, the exogenous agent does not naturally exist in the recipient cell. In some embodiments, the exogenous agent exists naturally in the recipient cell, but is not present at a desired level or at a desired time. In some embodiments, the exogenous agent comprises DNA, RNA, or protein.

[0065] As used herein, a "promoter" refers to a cis-regulatory DNA sequence that, when operably linked to a gene coding sequence, drives transcription of the gene. The promoter may comprise one or more transcription factor binding sites. In some embodiments, a promoter works in concert with one or more enhancers which are distal to the gene.

[0066] As used herein, "operably linked" refers to a polynucleotide sequence that is joined to a regulatory region sequence in a manner that allows expression of the polynucleotide sequence. A regulatory region sequence directs transcription of a polynucleotide sequence, and can include enhancer sequences, response elements, protein recognition sites, inducible elements, promoter control elements, 5' and 3' untranslated regions protein binding sequences, transcriptional start sites, termination sequences, polyadenylation sequences, and introns.

[0067] As used herein, a composition refers to any mixture of two or more products, substances, or compounds, including cells. It may be a solution, a suspension, a liquid, a powder, a paste, aqueous, non-aqueous, or any combination thereof.

[0068] As used herein, the term "pharmaceutically acceptable" refers to a material, such as a carrier or diluent, which does not abrogate the biological activity or properties of a therapeutic compound, and is relatively nontoxic, i.e., the material may be administered to an individual without causing undesirable biological effects or interacting in a deleterious manner with any of the components of the composition in which it is contained.

[0069] As used herein, the term "pharmaceutical composition" refers to a mixture of at least one targeted lipid particle of the disclosure with other chemical components, such as carriers, stabilizers, diluents, dispersing agents, suspending agents, thickening agents, and/or excipients. The pharmaceutical composition facilitates administration of the targeted lipid particle to an organism. Multiple techniques of administering targeted lipid particles of the disclosure exist in the art including, but not limited to, intravenous, oral, aerosol, parenteral, ophthalmic, pulmonary, and topical administration.

[0070] A "disease" or "disorder" as used herein refers to a condition for which treatment is needed and/or desired.

[0071] As used herein, the terms "treat," "treating," or "treatment" refer to ameliorating a disease or disorder, e.g., slowing or arresting or reducing the development of the disease or disorder or reducing at least one of the clinical symptoms thereof. For purposes of this disclosure, ameliorating a disease or disorder can include obtaining a beneficial or desired clinical result that includes, but is not limited to, any one or more of: alleviation of one or more symptoms, diminishment of extent of disease, preventing or delaying spread (for example, metastasis, for example metastasis to the lung or to the lymph node) of disease, preventing or delaying recurrence of disease, delay or slowing of disease progression, amelioration of the disease state, inhibiting the disease or progression of the disease, inhibiting or slowing the disease or its progression, arresting its development, and remission (whether partial or total).

[0072] The terms "individual" and "subject" are used interchangeably herein to refer to an animal; for example a mammal. The terms include human and veterinary animals. In some embodiments, methods of treating animals, including, but not limited to, humans, rodents, simians, felines, canines, equines, bovines, porcines, ovines, caprines, mammalian laboratory animals, mammalian farm animals, mammalian sport animals, and mammalian pets, are provided. The animal can be male or female and can be any suitable age, including infant, juvenile, adolescent, adult, and geriatric. In some examples, an "individual" or "subject" refers to an animal in need of treatment for a disease or disorder. In some embodiments, the animal to receive the treatment is a "patient," designating the fact that the animal has been identified as having a disorder of relevance to the treatment, or being at adequate risk of contracting the disorder. In particular embodiments, the animal is a human, such as a human patient.

CD4-Specific Antibodies

[0073] Described herein are novel antibodies and antigen binding fragments thereof that specifically target and bind CD4. In some embodiments, the antibodies or antigen binding fragments thereof cross-react with cynomolgus (or "cyno") or *M. nemestrina* CD4. In some embodiments, the antibodies or antigen binding fragments thereof are single-chain variable fragments (scFvs) composed of the antigen-binding domains derived from the heavy (VH) and the light (VL) chains of an IgG molecule and connected via a linker domain. In some embodiments, the antibodies or antigen binding fragments thereof are VHHs or VNARs that correspond to the antigen binding domains of the camelid and shark IgG molecules, respectively. The present disclosure also provides polynucleotides encoding the antibodies and fragments thereof, vectors, and host cells, and methods of using the antibodies or antigen binding fragments thereof. In some embodiments, e.g., the antibodies or antigen binding fragments thereof fuse to a glycoprotein (G Protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein) of the Paramyxoviridae family for targeted binding and transduction to cells.

[0074] Sequences for exemplary antibodies and antigen binding fragments of the disclosure using the Kabat numbering scheme are shown in Tables 19-22 below. Sequences for exemplary HCDRs and LCDRs of the disclosure are shown in Table 22.

[0075] The sequences for the disclosed VH and VL domains are provided in Tables 20-21. Tables 23-24 provided herein show the CDR sequences of the disclosed antibodies and antigen binding fragments thereof using both Chothia and IMGT numbering schemes, respectively. The full CD4 binder sequences of the variant CD4 scFvs and VHHs of the disclosure are shown in Table 19.

[0076] In some embodiments, an antibody or antigen binding fragment thereof capable of binding CD4 is disclosed, comprising a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3), and the light chain variable region comprises three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3). In some embodiments, the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 comprise amino acid sequences of any one of the SEQ ID NOs recited in Table 22. In some embodiments, the heavy chain variable region (VH) comprises an amino acid sequence of any one of SEQ ID NOs: 256-511, 9447-9576, or 14000-14002 (Table 20) and the light chain variable region (VL) comprises an amino acid sequence of any one of SEQ ID NOs: 512-766 or 9577-9706 (Table 21).

[0077] In another embodiment, the antibody or antigen binding fragment thereof comprises a VH having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to a sequence selected from SEQ ID NOs: 256-511, 9447-9576, or 14000-14002.

[0078] In another embodiment, the antibody or antigen binding fragment thereof comprises a VL having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to a sequence selected from SEQ ID NOs: 512-766 or 9577-9706.

[0079] In another embodiment, the antibody or antigen binding fragment comprises a VH having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to a sequence selected from SEQ ID NOs: 256-511, 9447-9576, or 14000-14002 and a VL having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to a sequence selected from SEQ ID NOs: 512-766 or 9577-9706.

[0080] In another embodiment, the antibody or antigen binding fragment thereof comprises a VH having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to SEQ ID NO: 256.

[0081] In another embodiment, the antibody or antigen binding fragment thereof comprises the VH of SEQ ID NO: 304 and the VL of SEQ ID NO: 559.

[0082] In another embodiment, the antibody or antigen binding fragment thereof comprises the VH of SEQ ID NO: 331 and the VL of SEQ ID NO: 586.

[0083] In another embodiment, the antibody or antigen binding fragment thereof comprises the VH of SEQ ID NO: 9554 and the VL of SEQ ID NO: 9684.

[0084] In another embodiment, the antibody or antigen binding fragment thereof comprises the VH of SEQ ID NO: 256.

[0085] In another embodiment, the antibody or antigen binding fragment thereof comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of SEQ ID NOs: 1308, 1822, 2336, 5672, 6182, 6692, respectively.

[0086] In another embodiment, the antibody or antigen binding fragment thereof comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of SEQ ID NOs: 1376, 1890, 2404, 5740, 6250, 6760, respectively.

[0087] In another embodiment, the antibody or antigen binding fragment thereof comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of SEQ ID NOs: 10074, 10336, 10598, 12300, 12560, 12820, respectively.

[0088] In another embodiment, the antibody or antigen binding fragment thereof comprises the HCDR1, HCDR2, and HCDR3 of SEQ ID NOs: 1535, 2049, 2563, respectively.

[0089] In some embodiments, the single domain antibody is human or humanized. In some embodiments, the single domain antibody or portion thereof is naturally occurring. In some embodiments, the single domain antibody or portion thereof is synthetic.

[0090] In some embodiments, the single domain antibodies are antibodies whose complementary determining regions are part of a single domain polypeptide. In some embodiments, the single domain antibody is a heavy chain only antibody variable domain. In some embodiments, the single domain antibody does not include light chains.

[0091] In various embodiments, any of the antibodies or antigen binding fragments described herein comprise a heavy chain constant region and a light chain constant region. The heavy chain constant region may be an IgG, IgM, IgA, IgD, or IgE isotype, or a derivative or fragment thereof that retains at least one effector function of the intact heavy chain. The heavy chain constant region may be a human IgG isotype. The heavy chain constant region may be a human IgG1 or human IgG4 isotypes. The heavy chain constant region may be a human IgG1 isotype. The light chain constant region may be a human kappa light chain or lambda light chain or a derivative or fragment thereof that retains at least one effector function of the intact light chain. The light chain constant region may be a human kappa light chain.

[0092] In various embodiments, any of the disclosed antibodies or antigen binding fragments may be a rodent antibody or antigen binding fragment thereof, a chimeric antibody or an antigen binding fragment thereof, a CDR-grafted antibody or an antigen binding fragment thereof, or a humanized antibody or an antigen binding fragment thereof. In another embodiment, any of the disclosed antibodies or antigen binding fragments comprises human or human-derived heavy and light chain variable regions, including human frameworks or human frameworks with one or more backmutations. In various embodiments, any of the disclosed antibodies or antigen binding fragments may be a Fab, Fab', F(ab')2, Fd, scFv, (scFv)2, scFv-Fc, VHH, or Fv fragment.

[0093] Antibodies whose heavy chain CDR, light chain CDR, VH, or VL amino acid sequences differ insubstantially from those shown in Tables 19-24 are encompassed within the scope of the disclosure. Typically, this involves one or more conservative amino acid substitutions with an amino acid having similar charge, hydrophobic, or stereo chemical characteristics in the antigen-binding site or in the framework without adversely altering the properties of the antibody. Conservative substitutions may also be made to improve antibody properties, for example stability or affinity. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 amino acid substitutions can be made to the VH or VL sequence. For example, a "conservative amino acid substitution" may involve a substitution of a native amino acid residue with a nonnative residue such that there is little or no effect on the polarity or charge of the amino acid residue at that position. Desired amino acid substitutions can be determined by those skilled in the art at the time such substitutions are desired. For example, amino acid substitutions can be used to identify important residues of the molecule sequence, or to increase or decrease the affinity of the molecules described herein. The following eight groups contain amino acids that are conservative amino acid substitutions for one another: 1) Alanine (A), Glycine (G); 2) Aspartic acid (D), Glutamic acid (E); 3) Asparagine (N), Glutamine (Q); 4) Arginine (R), Lysine (K); 5) Isoleucine (I), Leucine (L), Methionine (M), Valine (V); 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W); 7) Serine(S), Threonine (T); and 8) Cysteine (C), Methionine (M).

[0094] In some embodiments, the antibody or antigen binding fragment binding CD4 is a single-chain variable fragment. In embodiments involving a single polypeptide containing both a heavy chain variable region and a light chain variable region, both orientations of these variable regions are contemplated. In some cases, the heavy chain variable region is on the N-terminal side of the light chain variable region, which means the heavy chain variable region is closer to the N-terminus of the polypeptide. In other cases, the light chain variable region is on the N-terminal side of the heavy chain variable region, which means the light chain variable region is closer to the N-terminus of the polypeptide than the heavy chain variable region.

[0095] In some embodiments, the scFv binding proteins comprise a linker. In some embodiments, the linker is between the heavy chain variable region (VH) and the light chain variable region (VL) (or vice versa). In some embodiments, the linker comprises the amino acid sequence of GS, GGS, GGGS (SEQ ID NO: 14125), GGGGS (SEQ ID NO: 9294), GGGGGS (SEQ ID NO: 9292), any one of SEQ ID NOs: 9312-9315, or combinations thereof. Substitutions to introduce new disulfide bonds are also within the scope of the disclosure, e.g., by making substitutions G44C in the VH FR 2 and G100C in the VL FR4.

[0096] In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to human CD4 with an affinity constant (K_D) of about 1 nM to about 900 nM. In some embodiments, the K_D to human CD4 is about 5 nM to about 500 nM, about 6 nM to about 10 nM, about 11 nM to about 20 nM, about 25 nM to about 40 nM, about 40 nM to about 60 nM, about 70 nM to about 90 nM, about 100 nM to about 120 nM, about 125 nM to about 140 nM, about 145 nM to about 160 nM, about 170 nM and to about 200 nM, about 210 nM to about 250 nM, about 260 nM to about 300 nM, about 310 nM to about 350 nM, about 360 nM to about 400 nM, about 410 nM to about 450 nM, and about 460 nM to about 500 nM. In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to human CD4 with an affinity constant (K_D) of 500 nM, 400 nM, 300 nM, 200 nM, 100 nM, 50 nM, 20 nM, or 10 nM or lower. In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to human CD4 and CD4 of a non-human primate including cynomolgus, *M. mulatta* (rhesus monkey), or *M. nemestrina* CD4 with comparable binding affinity (K_D).

[0097] In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to a non-human primate, cynomolgus, *M. mulatta* (rhesus monkey), or *N. nemestrina* CD4. In some embodiments, the anti-CD4 antibody or antigen binding binds to mouse, dog, pig, etc., CD4. In some embodiments, the K_D to a non-human primate, cynomolgus or *M. nemestrina* CD4 is about 5 nM to about 500 nM, about 6 nM to about 10 nM, about 11 nM to about 20 nM, about 25 nM to about 40 nM, about 40 nM to about 60 nM, about 70 nM to about 90 nM, about 100 nM to about 120 nM, about 125 nM to about 140 nM, about 145 nM to about 160 nM, about 170 nM and to about 200 nM, about 210 nM to about 250 nM, about 260 nM to about 300 nM, about 310 nM to about 350 nM, about 360 nM to about 400 nM, about 410 nM to about 450 nM, and about 460 nM to about 500 nM. In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to cynomolgus or *M. nemestrina* CD4 with an affinity constant (K_D) of 500 nM, 400 nM, 300 nM, 200 nM, 100 nM, 50 nM, 20 nM, or 10 nM or lower.

[0098] An antibody or antigen binding fragment thereof that specifically binds CD4 refers to an antibody or binding fragment that preferentially binds to CD4, respectively, over other antigen targets. As used herein, references to an antibody that "specifically binds CD4" are interchangeable with an "anti-CD4" antibody or an "antibody that binds CD4." In some embodiments, the antibody or binding fragment capable of binding to CD4 does so with higher affinity for that antigen than others. In some embodiments, the antibody or binding fragment capable of binding CD4 binds to that antigen with a K_D of at least about 10-1, 10-2, 10-3, 10-4, 10-5, 10-6, 10-7, 10-8, 10-9, 10-10, 10-11, 10-12 or greater (or any value in between), e.g., as measured by surface plasmon resonance or other methods known to those skilled in the art.

[0099] Another embodiment of the disclosure is an isolated polynucleotide encoding any of the antibody heavy chain variable regions or the antibody light chain variable regions of the disclosure. Certain exemplary polynucleotides are disclosed herein, however, other polynucleotides which, given the degeneracy of the genetic code or codon preferences in a given expression system, encode the antibodies or antigen binding fragments thereof of the disclosure are also within the scope of the disclosure. The polynucleotide sequences encoding a VH or a VL or a fragment thereof of the antibody or antigen binding fragments thereof of the disclosure can be operably linked to one or more regulatory elements, such as a promoter and enhancer, that allow expression of the nucleotide sequence in the intended host cell. The polynucleotide may be a cDNA.

[0100] Another embodiment of the disclosure is a vector comprising the polynucleotide of the disclosure. Such vectors may be plasmid vectors, viral vectors, vectors for baculovirus expression, transposon-based vectors, or any other vector suitable for introduction of the polynucleotide of the disclosure into a given organism or genetic background by any means. For example, polynucleotides encoding light and heavy chain variable regions of the antibodies of the disclosure, optionally linked to constant regions, may be inserted into expression vectors. The light and heavy chains can be cloned in the same or different expression vectors. The DNA segments encoding immunoglobulin chains may be operably linked to control sequences in the expression vector(s) that ensure the expression of immunoglobulin polypeptides. Such control sequences include signal sequences, promoters (e.g., naturally associated or heterologous promoters), enhancer elements, and transcription termination sequences, and are chosen to be compatible with the host cell chosen to express the antibody. Once the vector has been incorporated into the appropriate host, the host is maintained under conditions suitable for high level expression of the proteins encoded by the incorporated polynucleotides.

[0101] Suitable expression vectors are typically replicable in the host organisms either as episomes or as an integral part of the host chromosomal DNA. Commonly, expression vectors contain selection markers such as ampicillin-resistance, hygromycin-resistance, tetracycline resistance, kanamycin resistance, or neomycin resistance to permit detection of those cells transformed with the desired DNA sequences. Suitable vectors, promoter, and enhancer elements are known in the art; many are commercially available for generating subject recombinant constructs.

[0102] Another embodiment of the disclosure is a host cell comprising the vector of the disclosure. The term "host cell" refers to a cell into which a vector has been introduced. It is understood that the term host cell is intended to refer not only to the particular subject cell but to the progeny of such a cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not be identical to the parent cell, but are still included within the scope of the term "host cell" as used herein. Such host cells may be eukaryotic cells, prokaryotic cells, plant cells, or archaeal cells. *Escherichia coli*, bacilli, such as *Bacillus subtilis*, and other Enterobacteriaceae, such as *Salmonella, Serratia*, and various *Pseudomonas* species are examples of prokaryotic host cells. Other microbes, such as yeast, are also useful for expression. *Saccharomyces* (e.g., *S. cerevisiae*) and *Pichia* are examples of suitable yeast host cells. Exemplary eukaryotic cells may be of mammalian, insect, avian, or other animal origins.

Fusion Proteins Targeting CD4

[0103] Also provided herein are fusion proteins targeting CD4 that may be exposed on the surface on a lipid particle or viral vector. In some embodiments the fusion protein comprises an envelope glycoprotein G, H, and/or an F protein of the Paramyxoviridae family. In some embodiments, the fusion protein contains a henipavirus envelope attachment glycoprotein G (G protein) or a biologically active portion thereof and a single domain antibody (sdAb) variable domain or a single chain variable fragment (scFv). The sdAb variable domain or scFv can be linked directly or indirectly to the G protein. In particular embodiments, the sdAb variable domain or scFv is linked to the C-terminus (C-terminal amino acid) of the G protein or the biologically active portion thereof. The linkage can be via a peptide linker, such as a flexible peptide linker. Table 25 provides a list of non-limiting examples of G proteins. Exemplary full length fusion protein sequences of the disclosure are disclosed in Table 19.

[0104] In some embodiments, the G protein is of the Paramyxoviridae family. In some embodiments the G protein is a Henipavirus G protein or a biologically active portion thereof. In some embodiments, the Henipavirus G protein is a Hendra (HeV) virus G protein, a Nipah (NiV) virus G-protein (NIV-G), a Cedar (CedPV) virus G-protein, a Mojiang virus G-protein, a bat Paramyxovirus G-protein, or a biologically active portion thereof. In some embodiments, the fusion protein is glycoprotein GP64 of baculovirus, or glycoprotein GP64 variant E45K/T259A. Non-limiting examples of G proteins include those disclosed in Table 25.

[0105] In some embodiments, the attachment G proteins are type II transmembrane glycoproteins containing an N-terminal cytoplasmic tail (e.g., corresponding to amino acids 1-49 of SEQ ID NO: 9266), a transmembrane domain (e.g., corresponding to amino acids 50-70 of SEQ ID NO:

9266), and an extracellular domain containing an extracellular stalk (e.g., corresponding to amino acids 71-187 of SEQ ID NO: 9266), and a globular head (corresponding to amino acids 188-602 of SEQ ID NO: 9266). In such embodiments, the N-terminal cytoplasmic domain is within the inner lumen of the lipid bilayer and the C-terminal portion is the extracellular domain that is exposed on the outside of the lipid bilayer. Regions of the stalk in the C-terminal region (e.g. corresponding to amino acids 159-167 of NiV-G) have been shown to be involved in interactions with F protein and triggering of F protein fusion (Liu et al. 2015 J of Virology 89:1838). In wild-type G protein, the globular head mediates receptor binding to henipavirus entry receptors eprhin B2 and ephrin B3, but is dispensable for membrane fusion (Brandel-Tretheway et al. Journal of Virology. 2019. 93(13)e00577-19). In particular embodiments herein, tropism of the G protein is altered by linkage of the G protein or biologically active fragment thereof (e.g. cytoplasmic truncation) to a sdAb variable domain. Binding of the G protein to a binding partner can trigger fusion mediated by a compatible F protein or a biologically active portion thereof. G protein sequences disclosed herein are predominantly disclosed as expressed sequences including an N-terminal methionine required for start of translation. As such N-terminal methionines are commonly cleaved co- or post-translationally, the mature protein sequences for all G protein sequences disclosed herein are also contemplated as lacking the N-terminal methionine.

[0106] G glycoproteins are highly conserved among henipavirus species. For example, the G proteins of NiV and HeV viruses share 79% amino acid identity. Studies have shown a high degree of compatibility among G proteins with F proteins of different species as demonstrated by heterotypic fusion activation (Brandel-Tretheway et al. *Journal of Virology.* 2019). As described further below, a targeted lipid particle can contain heterologous G and F proteins from different species.

[0107] In some embodiments, the G protein has a sequence set forth in any of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037, or is a functionally active variant or biologically active portion thereof that has a sequence that is at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% identical to any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037. In particular embodiments, the G protein or functionally active variant or biologically active portion is a protein that retains fusogenic activity in conjunction with a Henipavirus F protein, such as an F protein (e.g. NiV-F or HeV—F). Fusogenic activity includes the activity of the G protein in conjunction with a Henipavirus F protein to promote or facilitate fusion of two membrane lumens, such as the lumen of the targeted lipid particle having embedded in its lipid bilayer a henipavirus F and G protein, and a cytoplasm of a target cell, e.g. a cell that contains a surface receptor or molecule that is recognized or bound by the targeted envelope protein. In some embodiments, the F protein and G protein are from the same Henipavirus species (e.g. NiV-G and NiV-F). In some embodiments, the F protein and G protein are from different Henipavirus species (e.g. NiV-G and HeV-F).

[0108] In particular embodiments, the G protein has the sequence of amino acids set forth in SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037, or is a functionally active variant thereof or a biologically active portion thereof that retains fusogenic activity. In some embodiments, the functionally active variant comprises an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037 and retains fusogenic activity in conjunction with a Henipavirus F protein (e.g., NiV-F or HeV-F). In some embodiments, the biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037 and retains fusogenic activity in conjunction with a Henipavirus F protein (e.g., NiV-F or HeV-F).

[0109] Reference to retaining fusogenic activity includes activity (in conjunction with a Henipavirus F protein) that is at or about 10% to at or about 150% or more of the level or degree of binding of the corresponding wild-type G protein, such as set forth in any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037, such as at least or at least about 10% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 15% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 20% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 25% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 30% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 35% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 40% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 45% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 50% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 55% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 60% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 65% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 70% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 75% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 80% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 85% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 90% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 95% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 100% of the level or degree of fusogenic activity of the corresponding wild-type G protein, or such as at least or at least about 120% of the level or degree of fusogenic activity of the corresponding wild-type G protein.

[0110] In some embodiments, the G protein is a mutant G protein that is a functionally active variant or biologically active portion containing one or more amino acid mutations, such as one or more amino acid insertions, deletions, substitutions, or truncations. In some embodiments, the mutations described herein relate to amino acid insertions, deletions, substitutions, or truncations of amino acids compared to a reference G protein sequence. In some embodiments, the reference G protein sequence is the wild-type sequence of a G protein or a biologically active portion thereof. In some embodiments, the functionally active variant or the biologically active portion thereof is a mutant of a wild-type Hendra (HeV) virus G protein, a wild-type Nipah (NiV) virus G-protein (NiV-G), a wild-type Cedar (CedPV) virus G-protein, a wild-type Mojiang virus G-protein, a wild-type bat Paramyxovirus G-protein, or biologically active portions thereof. In some embodiments, the wild-type G protein has the sequence set forth in any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037.

[0111] In some embodiments, the G protein is a mutant G protein that is a biologically active portion that is an N-terminally and/or C-terminally truncated fragment of a wild-type Hendra (HeV) virus G protein, a wild-type Nipah (NiV) virus G-protein (NiV-G), a wild-type Cedar (CedPV) virus G-protein, a wild-type Mojiang virus G-protein, or a wild-type bat Paramyxovirus G-protein. In particular embodiments, the truncation is an N-terminal truncation of all or a portion of the cytoplasmic domain. In some embodiments, the mutant G protein is a biologically active portion that is truncated and lacks up to 49 contiguous amino acid residues at or near the N-terminus of the wild-type G protein, such as a wild-type G protein set forth in any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037. In some embodiments, the mutant G protein is truncated and lacks up to 49 contiguous amino acids, such as up to 49, 48, 47, 46, 45, 44, 43, 42, 41, 40, 30, 38, 37, 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2 or 1 contiguous amino acid(s) at the N-terminus of the wild-type G protein.

[0112] In some embodiments, the G protein is a wild-type Nipah virus G (NiV-G) protein or a wild-type Hendra virus G protein, or is a functionally active variant or biologically active portion thereof. In some embodiments, the G protein is a NiV-G protein that has the sequence set forth in SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295, or is a functional variant or a biologically active portion thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295.

[0113] In some embodiments, the G protein is a mutant NiV-G protein that is a biologically active portion of a wild-type NiV-G. In some embodiments, the biologically active portion is an N-terminally truncated fragment. In some embodiments, the mutant NiV-G protein is truncated and lacks up to 5 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 6 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 7 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 8 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 9 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 10 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 11 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 12 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 13 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 14 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 15 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 16 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 17 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 18 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 19 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 20 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 21 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 22 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 23 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 24 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID

NO:9295), up to 25 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 26 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 27 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 28 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 29 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 30 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 31 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 32 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 33 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 34 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 35 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 36 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 37 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 38 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 39 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 40 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 41 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 42 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 43 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 44 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 618, or SEQ ID NO: 628), or up to 45 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295).

[0114] In some embodiments, the NiV-G protein is a biologically active portion that does not contain a cytoplasmic domain. In some embodiments, the NiV-G protein without the cytoplasmic domain is encoded by SEQ ID NO:9289.

[0115] In some embodiments, the mutant NiV-G protein comprises a sequence set forth in any of SEQ ID NOs: 601-606, 629-634, 612, 622, or 637, or is a functional variant thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, or at least at or about 87%, at least at or about 88%, or at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NOs: 9267-9269, 9296-9301, 9277, 9289, 9304.

[0116] In some embodiments, the mutant NiV-G protein has a 5 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9267 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9267, or as set forth in SEQ ID NO:9296 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9296 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9296.

[0117] In some embodiments, the mutant NiV-G protein has a 10 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9268 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9268, or such as set forth in SEQ ID NO:9297 or a functional variant thereof having at least at or about 80%, at least at or about

81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9297.

[0118] In some embodiments, the mutant NiV-G protein has a 15 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9269 or a functional variant thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9269, or such as set forth in SEQ ID NO:9298 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9298.

[0119] In some embodiments, the mutant NiV-G protein has a 20 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295) such as set forth in SEQ ID NO:9270, or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9270, or such as set forth in SEQ ID NO:9299 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9299.

[0120] In some embodiments, the mutant NiV-G protein has a 25 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9271 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9271, or such as set forth in SEQ ID NO:9300 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9300.

[0121] In some embodiments, the mutant NiV-G protein has a 30 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9273 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9273, or such as set forth in SEQ ID NO:9301 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9301.

[0122] In some embodiments, the mutant NiV-G protein has a 33 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295) or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9277, or such as set forth in SEQ ID NO:9302 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9302.

[0123] In some embodiments, the mutant NiV-G protein has a 34 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9277 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9277, or such as set forth in SEQ ID NO:9302 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9302.

[0124] In a preferred embodiment, the NiV-G protein has a 34 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295) and one or more amino acid substitutions corresponding to amino acid substitutions selected from E501A, W504A, Q530A, and E533A with reference to the numbering set forth in SEQ ID NO:9285.

[0125] In some embodiments, the mutant NiV-G protein lacks the N-terminal cytoplasmic domain of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO: 9295), such as set forth in SEQ ID NO:9289 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9289.

[0126] In some embodiments, the mutant G protein is a mutant HeV-G protein that has the sequence set forth in SEQ ID NO:9275 or 9303, or is a functional variant or biologically active portion thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9275 or 9303.

[0127] In some embodiments, the G protein is a mutant HeV-G protein that is a biologically active portion of a wild-type HeV-G. In some embodiments, the biologically active portion is an N-terminally truncated fragment. In some embodiments, the mutant HeV-G protein is truncated and lacks up to 5 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 6 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 7 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 8 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 9 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 10 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 11 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 12 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 13 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 14 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 15 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 16 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 17 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 18 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 19 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 20 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 21 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 22 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 23 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 24 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 25 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 26 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 27 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 28 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 29 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 30 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein

(SEQ ID NO:9275 or 9303), up to 31 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 32 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 33 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 34 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 35 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 36 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 37 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 38 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 39 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 40 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 41 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 42 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 43 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 44 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), or up to 45 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303).

[0128] In some embodiments, the HeV-G protein is a biologically active portion that does not contain a cytoplasmic domain. In some embodiments, the mutant HeV-G protein lacks the N-terminal cytoplasmic domain of the wild-type HeV-G protein (SEQ ID NO: 9275 or 9303), such as set forth in SEQ ID NO:9303 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9303.

[0129] In some embodiments, the G protein or the functionally active variant or biologically active portion thereof binds to Ephrin B2 or Ephrin B3. In some aspects, the G protein has the sequence of amino acids set forth in any one of SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO: 9287, or SEQ ID NO:9288, or is a functionally active variant thereof or a biologically active portion thereof that is able to bind to Ephrin B2 or Ephrin B3. In some embodiments, the functionally active variant or biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO: 9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, and retains binding to Ephrhin B2 or B3.

[0130] Reference to retaining binding to Ephrin B2 or B3 includes binding that is at least or at least about 5% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 10% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO: 9295, SEQ ID NO: 9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 15% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 20% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 25% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO: 9295, SEQ ID NO: 9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion, 30% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO: 9288, or a functionally active variant or biologically active portion thereof, 35% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 40% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 45% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 50% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO: 9295, SEQ ID NO: 9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 55% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ

ID NO:9288, or a functionally active variant or biologically active portion thereof, 60% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 65% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO: 9295, SEQ ID NO: 9287, or SEQ ID NO: 9288, or a functionally active variant or biologically active portion thereof, 70% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, such as at least or at least about 75% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, such as at least or at least about 80% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, such as at least or at least about 85% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, such as at least or at least about 90% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, or such as at least or at least about 95% of the level or degree of binding of the corresponding wild-type protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof. In some embodiments, the G protein is NiV-G or a functionally active variant or biologically active portion thereof and binds to Ephrin B2 or Ephrin B3.

[0131] In some aspects, the NiV-G has the sequence of amino acids set forth in SEQ ID NO: 9266, SEQ ID NO:9285, or SEQ ID NO:9295, or is a functionally active variant thereof or a biologically active portion thereof that is able to bind to Ephrin B2 or Ephrin B3. In some embodiments, the functionally active variant or biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO: 9295 and retains binding to Eprhin B2 or B3. Exemplary biologically active portions include N-terminally truncated variants lacking all or a portion of the cytoplasmic domain, e.g. 1 or more, such as 1 to 49 contiguous N-terminal amino acid residues, e.g. set forth in any one of SEQ ID NOS: 9267-9272, 9289, and 9296-9301.

[0132] Reference to retaining binding to Ephrin B2 or B3 includes binding that is at least or at least about 5% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, 10% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, 15% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, 20% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9285, or SEQ ID NO:92954, 25% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, 30% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in S SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 35% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO: 9295, 40% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO: 9295, 45% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 50% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 55% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 60% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 65% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 70% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, such as at least or at least about 75% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9285, or SEQ ID NO:9295, such as at least or at least about 80% of the level or degree of binding of the corresponding wild-type NIV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, such as at least or at least about 85% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO: 9295, such as at least or at least about 90% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, or such as at least or at least about 95% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9285, or SEQ ID NO:9295.

[0133] In some embodiments, the G protein is HeV-G or a functionally active variant or biologically active portion thereof and binds to Ephrin B2 or Ephrin B3. In some aspects, the HeV-G has the sequence of amino acids set forth in SEQ ID NO:9275 or 9303, or is a functionally active variant thereof or a biologically active portion thereof that is able to bind to Ephrin B2 or Ephrin B3. In some embodiments, the functionally active variant or biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9275 or 9303 and retains binding to Eprhin B2 or B3. Exemplary biologically active portions include N-terminally truncated variants lacking all or a portion of the cytoplasmic domain, e.g. 1 or more, such as 1 to 49 contiguous N-terminal amino acid residues, e.g. set forth in any one of SEQ ID NO:9290.

[0134] Reference to retaining binding to Ephrin B2 or B3 includes binding that is at least or at least about 5% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 10% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 15% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 20% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 25% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 30% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 35% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 40% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 45% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 50% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 55% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 60% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 65% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 70% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, such as at least or at least about 75% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, such as at least or at least about 80% of the level or degree of binding of the corresponding wild-type NIV-G, such as set forth in SEQ ID NO:9275 or 9303, such as at least or at least about 85% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, such as at least or at least about 90% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, or such as at least or at least about 95% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303.

[0135] In some embodiments, the G protein or the biologically thereof is a mutant G protein that exhibits reduced binding for the native binding partner of a wild-type G protein. In some embodiments, the mutant G protein or the biologically active portion thereof is a mutant of wild-type Niv-G and exhibits reduced binding to one or both of the native binding partners Ephrin B2 or Ephrin B3. In some embodiments, the mutant G-protein or the biologically active portion, such as a mutant NiV-G protein, exhibits reduced binding to the native binding partner. In some embodiments, the reduced binding to Ephrin B2 or Ephrin B3 is reduced by greater than at or about 5%, at or about 10%, at or about 15%, at or about 20%, at or about 25%, at or about 30%, at or about 40%, at or about 50%, at or about 60%, at or about 70%, at or about 80%, at or about 90%, or at or about 100%.

[0136] In some embodiments, the mutations described herein improve transduction efficiency. In some embodiments, the mutations described herein allow for specific targeting of other desired cell types that are not Ephrin B2 or Ephrin B3. In some embodiments, the mutations described herein result in at least the partial inability to bind at least one natural receptor, such as to reduce the binding to at least one of Ephrin B2 or Ephrin B3. In some embodiments, the mutations described herein interfere with natural receptor recognition.

[0137] In some embodiments, the mutant NiV-G protein or the biologically active portion thereof is truncated and lacks up to 5 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 6 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 7 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 8 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 9 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 10 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 11 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 12 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 13 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 14 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 15 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 16 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 17 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 18 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 19 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 20 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 21 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 22 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 23 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID

NO:9285), 24 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 25 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 26 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 27 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 28 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 29 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 30 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 31 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 32 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 33 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 34 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 35 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 36 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 37 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 38 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 39 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), or 40 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285).

[0138] In some embodiments, the G protein contains one or more amino acid substitutions in a residue that is involved in the interaction with one or both of Ephrin B2 and Ephrin B3. In some embodiments, the amino acid substitutions correspond to mutations E501A, W504A, Q530A, and E533A with reference to numbering set forth in SEQ ID NO:9285.

[0139] In some embodiments, the G protein is a mutant G protein containing one or more amino acid substitutions selected from the group consisting of E501A, W504A, Q530A, and E533A with reference to numbering set forth in SEQ ID NO:9285. In some embodiments, the G protein is a mutant G protein that contains one or more amino acid substitutions selected from the group consisting of E501A, W504A, Q530A, and E533A with reference to SEQ ID NO:9285 or a biologically active portion thereof containing an N-terminal truncation. In some embodiments, the G protein is a mutant G protein that contains one or more amino acid substitutions selected from the group consisting of E501A, W504A, Q530A, and E533A in combination with any one of the N-terminal truncations disclosed above with reference to SEQ ID NO:9285 or a biologically active portion thereof. In some embodiments, any of the mutant G proteins described above contains one, two, three, or all four amino acid selected from the group consisting of E501A, W504A, Q530A, and E533A with reference to numbering set forth in SEQ ID NO:9285, in all pairwise and triple combinations thereof.

[0140] In some embodiments, the mutant NiV-G protein has the amino acid sequence set forth in SEQ ID NO:9273 or 9302 or an amino acid sequence having at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9273 or 9302. In particular embodiments, the G protein has the sequence of amino acids set forth in SEQ ID NO:9273 or 9302.

[0141] In some embodiments, the targeted envelope protein contains a G protein or a functionally active variant or biologically active portion thereof and an sdAb variable domain, in which the targeted envelope protein exhibits increased binding for another molecule that is different from the native binding partner of a wild-type G protein. In some embodiments, the other molecule is a protein expressed on the surface of desired target cell. In some embodiments, the increased binding to the other molecule is increased by greater than at or about 25%, at or about 30%, at or about 40%, at or about 50%, at or about 60%, at or about 70%, at or about 80%, at or about 90%, or at or about 100%. In particular embodiments, the binding confers re-targeted binding compared to the binding of a wild-type G protein in which a new or different binding activity is conferred.

[0142] In some embodiments, the C-terminus of the single domain antibody is attached to the C-terminus of the G protein or biologically active portion thereof. In some embodiments, the N-terminus of the single domain antibody is exposed on the exterior surface of the lipid bilayer. In some embodiments, the N-terminus of the single domain antibody binds to a cell surface molecule of a target cell. In some embodiments, the single domain antibody specifically binds to a cell surface molecule present on a target cell. In some embodiments, the cell surface molecule is a protein, glycan, lipid, or low molecular weight molecule.

[0143] In some embodiments, the cell surface molecule of a target cell is an antigen or portion thereof. In some embodiments, the single domain antibody or portion thereof is an antibody having a single monomeric domain antigen binding/recognition domain that is able to bind selectively to a specific antigen. In some embodiments, the single domain antibody binds an antigen present on a target cell.

[0144] Exemplary cells include immune effector cells, peripheral blood mononuclear cells (PBMCs) such as lymphocytes (T cells, B cells, natural killer cells) and monocytes, granulocytes (neutrophils, basophils, eosinophils), macrophages, dendritic cells, cytotoxic T lymphocytes, polymorphonuclear cells (also known as PMNs, PMLs, or PMNLs), stem cells, embryonic stem cells (ESs or ECSs), neural stem cells, mesenchymal stem cells (MSCs), hematopoietic stem cells (HSCs), human myogenic stem cells, muscle-derived stem cells (MuStem), limbal epithelial stem cells, cardio-myogenic stem cells, cardiomyocytes, progenitor cells, allogenic cells, resident cardiac cells, induced pluripotent stem cells (iPSs or iPSCs), adipose-derived or phenotypic modified stem or progenitor cells, CD133+ cells, aldehyde dehydrogenase-positive cells (ALDH+), umbilical cord blood (UCB) cells, peripheral blood stem cells (PBSCs), neurons, neural progenitor cells, pancreatic beta cells, glial cells, or hepatocytes.

[0145] In some embodiments, the target cell is a cell of a target tissue. The target tissue can include liver, lungs, heart, spleen, pancreas, gastrointestinal tract, kidney, testes, ovaries, brain, reproductive organs, central nervous system, peripheral nervous system, skeletal muscle, endothelium, inner ear, or eye.

[0146] In some embodiments, the target cell is a muscle cell (e.g., skeletal muscle cell), kidney cell, liver cell (e.g. hepatocyte), or a cardiac cell (e.g. cardiomyocyte). In some embodiments, the target cell is a cardiac cell, e.g., a cardiomyocyte (e.g., a quiescent cardiomyocyte), a hepatoblast (e.g., a bile duct hepatoblast), an epithelial cell, a T cell (e.g. a naive T cell), a macrophage (e.g., a tumor infiltrating macrophage), or a fibroblast (e.g., a cardiac fibroblast).

[0147] In some embodiments, the target cell is a tumor-infiltrating lymphocyte, a T cell, a neoplastic or tumor cell, a virus-infected cell, a stem cell, a central nervous system (CNS) cell, a hematopoietic stem cell (HSC), a liver cell or a fully differentiated cell. In some embodiments, the target cell is a CD3+ T cell, a CD4+ T cell, a CD8+ T cell, a hepatocyte, a hematopoietic stem cell, a CD34+ hematopoietic stem cell, a CD105+ hematopoietic stem cell, a CD117+ hematopoietic stem cell, a CD105+ endothelial cell, a B cell, a CD20+ B cell, a CD19+ B cell, a cancer cell, a CD133+ cancer cell, an EpCAM+ cancer cell, a CD19+ cancer cell, a Her2/Neu+ cancer cell, a GluA2+ neuron, a GluA4+ neuron, a NKG2D+ natural killer cell, a SLC1A3+ astrocyte, a SLC7A10+ adipocyte, or a CD30+ lung epithelial cell.

[0148] In some embodiments, the target cell is an antigen presenting cell, an MHC class II+ cell, a professional antigen presenting cell, an atypical antigen presenting cell, a macrophage, a dendritic cell, a myeloid dendritic cell, a plasmacytoid dendritic cell, a CD11c+ cell, a CD11b+ cell, a splenocyte, a B cell, a hepatocyte, a endothelial cell, or a non-cancerous cell. In some embodiments, the cell surface molecule is any one of CD4.

[0149] In some embodiments, the G protein or functionally active variant or biologically active portion thereof is linked directly to the sdAb variable domain (e.g., a VHH) or scFv. In some embodiments, the targeted envelope protein is a fusion protein that has the following structure: (N'-single domain antibody-C')-(C'-G protein-N'). In some embodiments, the targeted envelope protein is a fusion protein that has the following structure: (N'-scFv-C')-(C'-G protein-N').

[0150] In some embodiments, the G protein or functionally active variant or biologically active portion thereof is linked indirectly via a linker to the sdAb variable domain or scFv. In some embodiments, the linker is a peptide linker, such as a polypeptide linker. In some embodiments, the linker is a chemical linker.

[0151] In some embodiments, the linker is a peptide linker and the targeted envelope protein is a fusion protein containing the G protein or functionally active variant or biologically active portion thereof linked via a peptide linker to the sdAb variable domain or svFv. In some embodiments, the targeted envelope protein is a fusion protein that has the following structure: (N'-single domain antibody-C')-Linker-(C'-G protein-N'). In some embodiments, the targeted envelope protein is a fusion protein that has the following structure: (N'-scFv-C')-Linker-(C'-G protein-N'). In some embodiments, the peptide linker is a polypeptide linker up to 65 amino acids in length. In some embodiments, the peptide linker comprises from or from about 2 to 65 amino acids, 2 to 60 amino acids, 2 to 56 amino acids, 2 to 52 amino acids, 2 to 48 amino acids, 2 to 44 amino acids, 2 to 40 amino acids, 2 to 36 amino acids, 2 to 32 amino acids, 2 to 28 amino acids, 2 to 24 amino acids, 2 to 20 amino acids, 2 to 18 amino acids, 2 to 14 amino acids, 2 to 12 amino acids, 2 to 10 amino acids, 2 to 8 amino acids, 2 to 6 amino acids, 6 to 65 amino acids, 6 to 60 amino acids, 6 to 56 amino acids, 6 to 52 amino acids, 6 to 48 amino acids, 6 to 44 amino acids, 6 to 40 amino acids, 6 to 36 amino acids, 6 to 32 amino acids, 6 to 28 amino acids, 6 to 24 amino acids, 6 to 20 amino acids, 6 to 18 amino acids, 6 to 14 amino acids, 6 to 12 amino acids, 6 to 10 amino acids, 6 to 8 amino acids, 8 to 65 amino acids, 8 to 60 amino acids, 8 to 56 amino acids, 8 to 52 amino acids, 8 to 48 amino acids, 8 to 44 amino acids, 8 to 40 amino acids, 8 to 36 amino acids, 8 to 32 amino acids, 8 to 28 amino acids, 8 to 24 amino acids, 8 to 20 amino acids, 8 to 18 amino acids, 8 to 14 amino acids, 8 to 12 amino acids, 8 to 10 amino acids, 10 to 65 amino acids, 10 to 60 amino acids, 10 to 56 amino acids, 10 to 52 amino acids, 10 to 48 amino acids, 10 to 44 amino acids, 10 to 40 amino acids, 10 to 36 amino acids, 10 to 32 amino acids, 10 to 28 amino acids, 10 to 24 amino acids, 10 to 20 amino acids, 10 to 18 amino acids, 10 to 14 amino acids, 10 to 12 amino acids, 12 to 65 amino acids, 12 to 60 amino acids, 12 to 56 amino acids, 12 to 52 amino acids, 12 to 48 amino acids, 12 to 44 amino acids, 12 to 40 amino acids, 12 to 36 amino acids, 12 to 32 amino acids, 12 to 28 amino acids, 12 to 24 amino acids, 12 to 20 amino acids, 12 to 18 amino acids, 12 to 14 amino acids, 14 to 65 amino acids, 14 to 60 amino acids, 14 to 56 amino acids, 14 to 52 amino acids, 14 to 48 amino acids, 14 to 44 amino acids, 14 to 40 amino acids, 14 to 36 amino acids, 14 to 32 amino acids, 14 to 28 amino acids, 14 to 24 amino acids, 14 to 20 amino acids, 14 to 18 amino acids, 18 to 65 amino acids, 18 to 60 amino acids, 18 to 56 amino acids, 18 to 52 amino acids, 18 to 48 amino acids, 18 to 44 amino acids, 18 to 40 amino acids, 18 to 36 amino acids, 18 to 32 amino acids, 18 to 28 amino acids, 18 to 24 amino acids, 18 to 20 amino acids, 20 to 65 amino acids, 20 to 60 amino acids, 20 to 56 amino acids, 20 to 52 amino acids, 20 to 48 amino acids, 20 to 44 amino acids, 20 to 40 amino acids, 20 to 36 amino acids, 20 to 32 amino acids, 20 to 28 amino acids, 20 to 26 amino acids, 20 to 24 amino acids, 24 to 65 amino acids, 24 to 60 amino acids, 24 to 56 amino acids, 24 to 52 amino acids, 24 to 48 amino acids, 24 to 44 amino acids, 24 to 40 amino acids, 24 to 36 amino acids, 24 to 32 amino acids, 24 to 30 amino acids, 24 to 28 amino acids, 28 to 65 amino acids, 28 to 60 amino acids, 28 to 56 amino acids, 28 to 52 amino acids, 28 to 48 amino acids, 28 to 44 amino acids, 28 to 40 amino acids, 28 to 36 amino acids, 28 to 34 amino acids, 28 to 32 amino acids, 32 to 65 amino acids, 32 to 60 amino acids, 32 to 56 amino acids, 32 to 52 amino acids, 32 to 48 amino acids, 32 to 44 amino acids, 32 to 40 amino acids, 32 to 38 amino acids, 32 to 36 amino acids, 36 to 65 amino acids, 36 to 60 amino acids, 36 to 56 amino acids, 36 to 52 amino acids, 36 to 48 amino acids, 36 to 44 amino acids, 36 to 40 amino acids, 40 to 65 amino acids, 40 to 60 amino acids, 40 to 56 amino acids, 40 to 52 amino acids, 40 to 48 amino acids, 40 to 44 amino acids, 44 to 65 amino acids, 44 to 60 amino acids, 44 to 56 amino acids, 44 to 52 amino acids, 44 to 48 amino acids, 48 to 65 amino acids, 48 to 60 amino acids, 48 to 56 amino acids, 48 to 52 amino acids, 50 to 65 amino acids, 50 to 60 amino acids, 50 to 56 amino acids, 50 to 52 amino acids, 54 to 65 amino acids, 54 to 60 amino acids, 54 to 56 amino acids, 58 to 65 amino acids, 58 to 60 amino acids, or 60 to 65 amino acids. In some embodiments, the peptide linker is a peptide that is 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, or 65 amino acids in length.

[0152] In particular embodiments, the linker is a flexible peptide linker. In some such embodiments, the linker is 1-20 amino acids, such as 1-20 amino acids predominantly composed of glycine. In some embodiments, the linker is 1-20 amino acids, such as 1-20 amino acids predominantly composed of glycine and serine. In some embodiments, the linker is a flexible peptide linker containing amino acids Glycine and Serine, referred to as GS-linkers. In some embodiments, the peptide linker includes the sequences GS, GGS, GGGGS (SEQ ID NO:9294), GGGGGS (SEQ ID NO:9292) or combinations thereof. In some embodiments, the peptide linker is a polypeptide linker that has the sequence (GGS) n (SEQ ID NO: 14126), wherein n is 1 to 10. In some embodiments, the peptide linker is a polypeptide linker that has the sequence (GGGGS) n, (SEQ ID NO:9293) wherein n is 1 to 10. In some embodiments, the peptide linker is a polypeptide linker that has the sequence (GGGGGS) n (SEQ ID NO:9284), wherein n is 1 to 6.

[0153] Also provided herein are polynucleotides comprising a nucleic acid sequence encoding a targeted envelope protein. In some embodiments, the polynucleotides comprise a nucleic acid sequence encoding a G protein or biologically active portion thereof. In some embodiments, the polynucleotides further comprise a nucleic acid sequence encoding a single domain antibody (sdAb) variable domain or scFv or biologically active portion thereof. The polynucleotides may include a sequence of nucleotides encoding any of the targeted envelope proteins described above. The polynucleotide can be a synthetic nucleic acid. Also provided are expression vectors containing any of the provided polynucleotides.

[0154] In some of any embodiments, expression of natural or synthetic nucleic acids is typically achieved by operably linking a nucleic acid encoding the gene of interest to a promoter and incorporating the construct into an expression vector. In some embodiments, vectors are suitable for replication and integration in eukaryotes. In some embodiments, cloning vectors contain transcription and translation terminators, initiation sequences, and promoters useful for expression of the desired nucleic acid sequence. In some of any embodiments described herein, a plasmid comprises a promoter suitable for expression in a cell.

[0155] In some embodiments, the polynucleotides contain at least one promoter that is operatively linked to control expression of the targeted envelope protein containing the G protein and the single domain antibody (sdAb) variable domain or scFv. For expression of the targeted envelope protein, at least one element in each promoter functions to position the start site for RNA synthesis. The best-known example of this is the TATA box, but in some promoters lacking a TATA box, such as the promoter for the mammalian terminal deoxynucleotidyl transferase gene and the promoter for the SV40 genes, a discrete element overlying the start site itself helps to fix the place of initiation.

[0156] In some embodiments, additional promoter elements, e.g., enhancers, regulate the frequency of transcriptional initiation. In some embodiments, additional promoter elements are located in the region 30-110 bp upstream of the start site, although a number of promoters have been shown to contain functional elements downstream of the start site as well. In some embodiments, spacing between promoter elements frequently is flexible, so that promoter function is preserved when elements are inverted or moved relative to one another. In some embodiments, such as with the thymidine kinase (tk) promoter, the spacing between promoter elements is increased to 50 bp apart before activity begins to decline. In some embodiments, depending on the promoter, individual elements function either cooperatively or independently to activate transcription.

[0157] A promoter may be one naturally associated with a gene or polynucleotide sequence, as may be obtained by isolating the 5' non-coding sequences located upstream of the coding segment and/or exon. Such a promoter can be referred to as "endogenous." Similarly, an enhancer may be one naturally associated with a polynucleotide sequence, located either downstream or upstream of that sequence. Alternatively, certain advantages will be gained by positioning the coding polynucleotide segment under the control of a recombinant or heterologous promoter, which refers to a promoter that is not normally associated with a polynucleotide sequence in its natural environment. A recombinant or heterologous enhancer refers also to an enhancer not normally associated with a polynucleotide sequence in its natural environment. Such promoters or enhancers may include promoters or enhancers of other genes, and promoters or enhancers isolated from any other prokaryotic, viral, or eukaryotic cell, and promoters or enhancers not "naturally occurring," i.e., containing different elements of different transcriptional regulatory regions, and/or mutations that alter expression. In addition to producing nucleic acid sequences of promoters and enhancers synthetically, sequences may be produced using recombinant cloning and/or nucleic acid amplification technology, including PCR, in connection with the compositions disclosed herein.

[0158] In some embodiments, a suitable promoter is the immediate early cytomegalovirus (CMV) promoter sequence. In some embodiments, the promoter sequence is a strong constitutive promoter sequence capable of driving high levels of expression of any polynucleotide sequence operatively linked thereto. In some embodiments, a suitable promoter is Elongation Growth Factor-la (EF-I a). In some embodiments, other constitutive promoter sequences are also used, including, but not limited to the simian virus 40 (SV40) early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, MoMuLV promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the hemoglobin promoter, and the creatine kinase promoter.

[0159] In some embodiments, the promoter is an inducible promoter. In some embodiments, the inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence to which it is operatively linked when such expression is desired, or turning off the expression when expression is not desired. In some embodiments, inducible promoters comprise a metallothionine promoter, a glucocorticoid promoter, a progesterone promoter, and a tetracycline promoter.

[0160] In some embodiments, exogenously controlled inducible promoters are used to regulate expression of the G protein and single domain antibody (sdAb) variable domain or scFv. For example, radiation-inducible promoters, heatinducible promoters, and/or drug-inducible promoters can be used to selectively drive transgene expression in, for example, targeted regions. In such embodiments, the location, duration, and level of transgene expression is regulated by the administration of the exogenous source of induction.

[0161] In some embodiments, expression of the targeted envelope protein containing a G protein and single domain antibody (sdAb) variable domain or scFv is regulated using a drug-inducible promoter. For example, in some cases, the promoter, enhancer, or transactivator comprises a Lac operator sequence, a tetracycline operator sequence, a galactose operator sequence, a doxycycline operator sequence, a rapamycin operator sequence, a tamoxifen operator sequence, or a hormone-responsive operator sequence, or an analog thereof. In some instances, the inducible promoter comprises a tetracycline response element (TRE). In some embodiments, the inducible promoter comprises an estrogen response element (ERE), which can activate gene expression in the presence of tamoxifen. In some instances, a drug-inducible element, such as a TRE, is combined with a selected promoter to enhance transcription in the presence of drug, such as doxycycline. In some embodiments, the drug-inducible promoter is a small molecule-inducible promoter.

[0162] Any of the provided polynucleotides can be modified to remove CpG motifs and/or to optimize codons for translation in a particular species, such as human, canine, feline, equine, ovine, bovine, etc. species. In some embodiments, the polynucleotides are optimized for human codon usage (i.e., human codon-optimized). In some embodiments, the polynucleotides are modified to remove CpG motifs. In other embodiments, the provided polynucleotides are modified to remove CpG motifs and are codon-optimized, such as human codon-optimized. Methods of codon optimization and CpG motif detection and modification are well-known. Typically, polynucleotide optimization enhances transgene expression, increases transgene stability and preserves the amino acid sequence of the encoded polypeptide.

[0163] In order to assess the expression of the targeted envelope protein, the expression vector to be introduced into a cell can also contain either a selectable marker gene or a reporter gene or both to facilitate identification and selection of expressing particles, e.g. viral particles. In other embodiments, the selectable marker is carried on a separate piece of DNA and used in a co-transfection procedure. Both selectable markers and reporter genes may be flanked with appropriate regulatory sequences to enable expression in the host cells. Useful selectable markers are known in the art and include, for example, antibiotic-resistance genes, such as neo and the like.

[0164] Reporter genes are used for identifying potentially transfected cells and for evaluating the functionality of regulatory sequences. Reporter genes that encode for easily assayable proteins are well known in the art. In general, a reporter gene is a gene that is not present in or expressed by the recipient organism or tissue and that encodes a protein whose expression is manifested by some easily detectable property, e.g., enzymatic activity. Expression of the reporter gene is assayed at a suitable time after the DNA has been introduced into the recipient cells.

[0165] Suitable reporter genes may include genes encoding luciferase, beta-galactosidase, chloramphenicol acetyl transferase, secreted alkaline phosphatase, or the green fluorescent protein gene (see, e.g., Ui-Tei et al., 2000, FEBS Lett. 479:79-82). Suitable expression systems are well known and may be prepared using well known techniques or obtained commercially. Internal deletion constructs may be generated using unique internal restriction sites or by partial digestion of non-unique restriction sites. Constructs may then be transfected into cells that display high levels of the desired polynucleotide and/or polypeptide expression. In general, the construct with the minimal 5' flanking region showing the highest level of expression of reporter gene is identified as the promoter. Such promoter regions may be linked to a reporter gene and used to evaluate agents for the ability to modulate promoter-driven transcription.

Lipid Particles Targeting CD4

[0166] Also provided herein are targeted lipid particles (e.g. targeting CD4), such as targeted viral vectors, that comprise a F protein molecule or biologically active portion thereof of the Paramyxoviridae family, and a fusion protein comprising (i) an envelope attachment glycoprotein G (G protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein), or a biologically active portion thereof of the Paramyxoviridae family, and (ii) a single domain antibody (sdAb) variable domain or scFv, wherein the single domain antibody variable domain or scFv is attached to the C-terminus of the G protein or the biologically active portion, wherein each is exposed on the outer surface of the targeted lipid particle. In particular embodiments, the provided targeted lipid particles exhibit fusogenic activity, which is mediated by the targeted envelope protein that facilitates binding to a target cell and contains the G protein or biologically active portion thereof, and the F protein or biologically active portion thereof that is involved in facilitating the merger or fusion of the two lumens of the lipid particle and the target cell membranes. Table 25 provides non-limiting examples of G and F proteins for use in the targeted lipid particles of the disclosure.

[0167] In some embodiments, the targeted lipid particle provided herein (e.g. targeted lentiviral vector) has increased or greater expression of the targeted envelope protein compared to a reference lipid particle (e.g. reference lentiviral vector) that incorporates a similar envelope protein but that is fused to an alternative targeting moiety other than a sdAb variable domain, such as a single chain variable fragment (scFv). In some embodiments, the targeted lipid particles are produced by pseudotyping of viral vectors (e.g lentiviral particles) following co-transfection of the packaging cells with the transfer, envelope, and gag-pol plasmids.

[0168] In some embodiments, the expression is increased by at or greater than 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 125%, 150%, 200%, 300%, 400%, 500% or more, compared to a reference lipid particle (e.g. reference lentiviral vector), e.g. a reference lipid particle containing a similar envelope protein but that is fused to an scFv. In some examples, the expression is increased by at or greater than 1.5-fold, 2-fold, 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, 15-fold, 20-fold, 30-fold or more, compared to a reference lipid particle (e.g. reference lentiviral vector), e.g. a reference lipid particle containing a similar envelope protein but that is fused to an scFv. In some embodiments, expression is assayed in vitro using flow cytometry, e.g. FACs. In some embodiments, expression can be depicted as the number or density of targeted envelope protein on the surface of a targeted lipid particle (e.g. targeted lentiviral vector). In some embodiments, expression is depicted as the mean fluorescent intensity (MFI) of surface expression of the targeted envelope protein on the surface of a targeted lipid particle (e.g. targeted lentiviral vector). In some embodiments, expression is depicted as the percent of lipid particle (e.g. lentiviral vectors) in a population that are surface positive for the targeted envelope protein.

[0169] In some embodiments, in a population of targeted lipid particles (e.g. targeted lentiviral vectors) greater than at or about 50% of the lipid particles are surface positive for the targeted envelope protein. For example, in a population of provided targeted lipid particle (e.g. targeted lentiviral vectors) greater than at or about 55%, greater than at or about 60%, greater than at or about 65%, greater than at or about 70%, or greater than at or about 75% of the viral vectors in the population are surface positive for the targeted envelope protein.

[0170] In some embodiments, titer of the targeted lipid particles following introduction into target cells, such as by transduction (e.g. transduced cells), is increased compared to titer into the same target cells of reference lipid particles (e.g. reference lentiviral vector) that incorporate a similar envelope protein but fused to an alternative targeting moiety other than a sdAb variable domain, such as a single chain variable fragment (scFv). Typically, the alternative targeting moiety recognizes or binds the same target molecule as the sdAb variable domain of the targeted envelope protein of the targeted lipid particles. In some embodiments, the titer is increased by at or greater than 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 125%, 150%, 200%, 300%, 400%, 500% or more, compared to titer of a reference lipid particle (e.g. reference lentiviral vector), e.g. a reference lipid particle containing a similar envelope protein but that is fused to an scFv. In some examples, the titer is increased by at or greater than 1.5-fold, 2-fold, 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, 15-fold, 20-fold, 30-fold or more, compared to the titer of a reference lipid particle (e.g. reference lentiviral vector), e.g. a reference viral vector containing a similar envelope protein but that is fused to an scFv. In some embodiments, the titer of the targeted lipid particles in target cells (e.g. transduced cells) is greater than at or about 1×10^6 transduction units (TU)/mL. For example, the titer of the targeted lipid particles in target cells (e.g. transduced cells) is greater than at or about 2×10^6 TU/mL, greater than at or about 3×10^6 TU/mL, greater than at or about 4×10^6 TU/mL, greater than at or about 5×10^6 TU/mL, greater than at or about 6×10^6 TU/mL, greater than at or about 7×10^6 TU/mL, greater than at or about 8×10^6 TU/mL, greater than at or about 9×10^6 TU/mL, or greater than at or about 1×10^7 TU/mL.

A. F Proteins

[0171] In some embodiments, the targeted lipid particle comprises one or more fusogens, e.g. F proteins of the Paramyxoviridae family. In some embodiments, the targeted lipid particle contains an exogenous or overexpressed fusogen. In some embodiments, the fusogen is disposed in the lipid bilayer. In some embodiments, the fusogen facilitates the fusion of the targeted particle's lipid bilayer to a membrane. In some embodiments, the membrane is a plasma cell membrane.

[0172] In some embodiments, fusogens comprise protein based, lipid based, and chemical based fusogens. In some embodiments, the targeted lipid particle comprises a first fusogen comprising a protein fusogen and a second fusogen comprising a lipid fusogen or chemical fusogen. In some embodiments, the fusogen binds a fusogen binding partner on a target cell surface.

[0173] In some embodiments, the fusogen comprises a protein with a hydrophobic fusion polypeptide domain. In some embodiments the fusogen comprises an F protein of the Paramyxoviridae family. In some embodiments the fusogen contains a Nipah virus protein F, a measles virus F protein, a tupaia paramyxovirus F protein, a paramyxovirus F protein, a Hendra virus F protein, a Henipavirus F protein, a Morbilivirus F protein, a respirovirus F protein, a Sendai virus F protein, a rubulavirus F protein, or an avulavirus F protein, or a biologically active portion thereof.

[0174] In some embodiments, the fusion protein is a hemagglutinin-neuraminidase (HN) of the Paramyxoviridae family and/or F protein of the Paramyxoviridae family. In some embodiments, the respiratory paramyxovirus is a Sendai virus. The HN and F glycoproteins of Sendai viruses function to attach to sialic acids via the HN protein, and to mediate cell fusion for entry to cells via the F protein. In some embodiments, the fusion protein is a F and/or HN protein from the murine parainfluenza virus type 1 (See e.g., U.S. Pat. No. 10,704,061).

[0175] In some embodiments, the N-terminal hydrophobic fusion polypeptide domain of the F protein molecule or biologically active portion thereof is exposed on the outside of a lipid bilayer.

[0176] F proteins of henipaviruses are encoded as F_0 precursors containing a signal polypeptide (e.g. corresponding to amino acid residues 1-26 of SEQ ID NO: 592). Following cleavage of the signal polypeptide, the mature F_0 (e.g. SEQ ID NO: 593) is transported to the cell surface, then endocytosed and cleaved by cathepsin L (e.g. between amino acids 109-110 of SEQ ID NO: 592) into the mature fusogenic subunits F1 (e.g. corresponding to amino acids 110-546 of SEQ ID NO:9258; set forth in SEQ ID NO:9261) and F2 (e.g. corresponding to amino acid residues 27-109 of SEQ ID NO:1; set forth in SEQ ID NO:9260). The F1 and F2 subunits are associated by a disulfide bond and recycled back to the cell surface. The F1 subunit contains the fusion polypeptide domain located at the N terminus of the F1 subunit (e.g., corresponding to amino acids 110-129 of SEQ ID NO:9258) where it is able to insert into a cell membrane to drive fusion. In some cases, fusion activity is blocked by association of the F protein with G protein, until G engages with a target molecule resulting in its disassociation from F and exposure of the fusion polypeptide to mediate membrane fusion.

[0177] Among different henipavirus species, the sequence and activity of the F protein is highly conserved. For example, the F protein of NiV and HeV viruses share 89% amino acid sequence identity. Further, in some cases, the henipavirus F proteins exhibit compatibility with G proteins from other species to trigger fusion (Brandel-Tretheway et al. Journal of Virology. 2019. 93(13):e00577-19). In some aspects of the provided targeted lipid particle, the F protein is heterologous to the G protein, i.e., the F and G protein or biologically active portions thereof are from different henipavirus species. For example, the F protein is from Hendra virus and the G protein is from Nipah virus. In other aspects, the F protein can be a chimeric F protein containing regions of F proteins from different species of Henipavirus. In some embodiments, switching a region of amino acid residues of the F protein from one species of Henipavirus to another can result in fusion to the G protein of the species comprising the amino acid insertion. (Brandel-Tretheway et al. 2019). In some cases, the chimeric F protein contains an extracellular domain from one henipavirus species and a transmembrane and/or cytoplasmic domain from a different henipavirus species. For example, the F protein may contain an extracellular domain of Hendra virus and a transmembrane/cytoplasmic domain of Nipah virus. F protein sequences disclosed herein are predominantly disclosed as expressed sequences including an N-terminal signal sequence. Such N-terminal signal sequences are commonly cleaved co- or post-translationally, thus the mature protein sequences for all F protein sequences disclosed herein are also contemplated as lacking the N-terminal signal sequence.

[0178] In some embodiments, the F protein is encoded by a polynucleotide sequence that encodes the sequence set forth by any one of SEQ ID NOs: 592, 593, 608, 614-616, or 641-644, or is a functionally active variant or a biologically active portion thereof that has a sequence that is at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% identical to any one of SEQ ID NOS: 592, 593, 608, 614-616, or 641-644. In particular embodiments, the F protein or the functionally active variant or biologically active portion thereof retains fusogenic activity in conjunction with a Henipavirus G protein, such as a G protein set forth herein. Fusogenic activity includes the activity of the F protein in conjunction with a Henipavirus G protein to promote or facilitate fusion of two membrane lumens, such as the lumen of the targeted lipid particle having embedded in its lipid bilayer a henipavirus F and G protein, and a cytoplasm of a target cell, e.g., a cell that contains a surface receptor or molecule that is recognized or bound by the targeted envelope protein. In some embodiments, the F protein and G protein are from the same Henipavirus species (e.g. NiV-G and NiV-F). In some embodiments, the F protein and G protein are from different Henipavirus species (e.g., NiV-G and HeV-F). In particular embodiments, the F protein of the functionally active variant or biologically active portion retains the cleavage site cleaved by cathepsin L (e.g., corresponding to the cleavage site between amino acids 109-110 of SEQ ID NO:9258).

[0179] In particular embodiments, the F protein has the sequence of amino acids set forth in SEQ ID NO:9258, SEQ ID NO:9259, SEQ ID NO:9274, SEQ ID NO:9281, SEQ ID NO: 9282, SEQ ID NO:9283, SEQ ID NO:9308, SEQ ID NO:9309, SEQ ID NO:9310, or SEQ ID NO:9311 or is a functionally active variant thereof or a biologically active portion thereof that retains fusogenic activity. In some embodiments, the functionally active variant comprises an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9258, SEQ ID NO:9259, SEQ ID NO:9274, SEQ ID NO:9281, SEQ ID NO:9282, SEQ ID NO:9283, SEQ ID NO: 9308, SEQ ID NO: 9309, SEQ ID NO:9310, or SEQ ID NO:9311 and retains fusogenic activity in conjunction with a Henipavirus G protein (e.g., NiV-G or HeV-G). In some embodiments, the biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9258, SEQ ID NO:9259, SEQ ID NO:9274, SEQ ID NO:9281, SEQ ID NO:9282, SEQ ID NO:9283, SEQ ID NO:9308, SEQ ID NO:9309, SEQ ID NO:9310, or SEQ ID NO: 9311 and retains fusogenic activity in conjunction with a Henipavirus G protein (e.g., NiV-G or HeV-G).

[0180] Reference to retaining fusogenic activity includes activity (in conjunction with a Henipavirus G protein) that is at or about 10% to at or about 150% or more of the level or degree of binding of the corresponding wild-type F protein, such as set forth in SEQ ID NO:9258, SEQ ID NO:9259, SEQ ID NO:9274, SEQ ID NO:9281, SEQ ID NO: 9282, SEQ ID NO: 9283, SEQ ID NO:9308, SEQ ID NO:9309, SEQ ID NO:9310, or SEQ ID NO:9311, such as at least or at least about 10% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 15% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 20% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 25% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 30% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 35% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 40% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 45% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 50% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 55% of the level or degree of fusogenic activity of the corresponding wild-type f protein, such as at least or at least about 60% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 65% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 70% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 75% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 80% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 85% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 90% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 95% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 100% of the level or degree of fusogenic activity of the corresponding wild-type F protein, or such as at least or at least about 120% of the level or degree of fusogenic activity of the corresponding wild-type F protein.

[0181] In some embodiments, the F protein is a mutant F protein that is a functionally active fragment or a biologically active portion containing one or more amino acid mutations, such as one or more amino acid insertions, deletions, substitutions, or truncations. In some embodiments, the mutations described herein relate to amino acid insertions, deletions, substitutions, or truncations of amino acids compared to a reference F protein sequence. In some embodiments, the reference F protein sequence is the wild-type sequence of an F protein or a biologically active portion thereof. In some embodiments, the mutant F protein or the biologically active portion thereof is a mutant of a wild-type Hendra (Hev) virus F protein, a Nipah (NiV) virus F-protein, a Cedar (CedPV) virus F protein, a Mojiang virus F protein, or a bat Paramyxovirus F protein. In some embodiments, the wild-type F protein is encoded by a sequence of nucleotides that encodes any one of SEQ ID NO: 592, 593, 608, 614-616, or 641-644.

[0182] In some embodiments, the mutant F protein is a biologically active portion of a wild-type F protein that is an N-terminally and/or C-terminally truncated fragment. In some embodiments, the mutant F protein or the biologically active portion of a wild-type F protein thereof comprises one or more amino acid substitutions. In some embodiments, the mutations described herein improve transduction efficiency. In some embodiments, the mutations described herein increase fusogenic capacity. Exemplary mutations include any as described, see e.g. Khetawat and Broder 2010 Virology Journal 7:312; Witting et al. 2013 Gene Therapy 20:997-1005; published international; patent application No. WO/2013/148327.

[0183] In some embodiments, the mutant F protein is a biologically active portion that is truncated and lacks up to 20 contiguous amino acid residues at or near the C-terminus of the wild-type F protein, such as a wild-type F protein encoded by a sequence of nucleotides encoding the F protein set forth in any one of SEQ ID NOS: 592, 593, 608, or 614-616. In some embodiments, the mutant F protein is truncated and lacks up to 19 contiguous amino acids, such as up to 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 contiguous amino acid(s) at the C-terminus of the wild-type F protein.

[0184] In some embodiments, the F protein or the functionally active variant or biologically active portion thereof comprises an F1 subunit or a fusogenic portion thereof. In some embodiments, the F1 subunit is a proteolytically cleaved portion of the F0 precursor. In some embodiments, the F0 precursor is inactive. In some embodiments, the cleavage of the F0 precursor forms a disulfide-linked F1+F2 heterodimer. In some embodiments, the cleavage exposes the fusion polypeptide and produces a mature F protein. In some embodiments, the cleavage occurs at or around a single basic residue. In some embodiments, the cleavage occurs at Arginine 109 of NiV-F protein. In some embodiments, cleavage occurs at Lysine 109 of the Hendra virus F protein.

[0185] In some embodiments, the F protein is a wild-type Nipah virus F (NiV-F) protein or is a functionally active variant or biologically active portion thereof. In some embodiments, the F_0 precursor is encoded by a sequence of nucleotides encoding the sequence set forth in SEQ ID NO:9258. The encoding nucleic acid can encode a signal polypeptide sequence that has the sequence MVVILDKRCY CNLLILILMI SECSVG (SEQ ID NO:9291) or another signal polypeptide sequence. In some embodiments, the F protein has the sequence set forth in SEQ ID NO:9259. In some examples, the F protein is cleaved into an F1 subunit comprising the sequence set forth in SEQ ID NO:9261 and an F2 subunit comprising the sequence set forth in SEQ ID NO:9260.

[0186] In some embodiments, the F protein is a NiV-F protein that is encoded by a sequence of nucleotides encoding the sequence set forth in SEQ ID NO:9258, or is a functionally active variant or biologically active portion thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9258. In some embodiments, the NiV-F-protein has the sequence of set forth in SEQ ID NO:9259, or is a functionally active variant or a biologically active portion thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9259. In particular embodiments, the F protein or the functionally active variant or biologically active portion thereof retains the cleavage site cleaved by cathepsin L (e.g., corresponding to the cleavage site between amino acids 109-110 of SEQ ID NO:9258).

[0187] In some embodiments, the F protein or the functionally active variant or the biologically active portion thereof includes an F1 subunit that has the sequence set forth in SEQ ID NO:9261, or an amino acid sequence having, at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89% at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9261.

[0188] In some embodiments, the F protein or the functionally active variant or biologically active portion thereof includes an F2 subunit that has the sequence set forth in SEQ ID NO: 9260, or an amino acid sequence having, at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89% at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9260.

[0189] In some embodiments, the F protein is a mutant NiV-F protein that is a biologically active portion thereof that is truncated and lacks up to 20 contiguous amino acid residues at or near the C-terminus of the wild-type NiV-F protein (e.g., set forth SEQ ID NO: 9259). In some embodiments, the mutant NiV-F protein comprises an amino acid sequence set forth in SEQ ID NO:9262. In some embodiments, the mutant NiV-F protein has a sequence that has at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9262. In some embodiments, the mutant F protein contains an F1 protein that has the sequence set forth in SEQ ID NO:9263. In some embodiments, the mutant F protein has a sequence that has at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9263.

[0190] In some embodiments, the F protein is a mutant NiV-F protein that is a biologically active portion thereof that comprises a 20 amino acid truncation at or near the C-terminus of the wild-type NiV-F protein (SEQ ID NO:9259); and a point mutation on an N-linked glycosylation site. In some embodiments, the mutant NiV-F protein comprises an amino acid sequence set forth in SEQ ID NO:9264. In some embodiments, the mutant NiV-F protein has a sequence that has at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9264.

[0191] In some embodiments, the F protein is a mutant NiV-F protein that is a biologically active portion thereof that comprises a 22 amino acid truncation at or near the C-terminus of the wild-type NiV-F protein (SEQ ID NO:9259). In some embodiments, the NiV-F protein comprises an amino acid sequence set forth in SEQ ID NO:9265. In some embodiments, the NiV-F protein has a sequence with at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9265. In particular embodiments, the variant F protein is a mutant Niv-F protein that has the sequence of amino acids set forth in SEQ ID NO:9280. In some embodiments, the NiV-F protein has a sequence with at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9280.

[0192] It has been reported that the henipavirus F proteins from various species exhibit compatibility with G proteins from other species to trigger fusion (Brandel-Tretheway et al. Journal of Virology. 2019. 93(13):e00577-19). In some aspects of the provided lentiviral vector, the F protein is heterologous to the G protein, i.e. the F and G protein or biologically active portions are from different henipavirus species. For example, the G protein is from Hendra virus and the F protein is a NiV-F as described. In other aspects, the F and/or G protein can be a chimeric F and/or G protein containing regions of F and/or G proteins from different species of Henipavirus. In some embodiments, replacing a portion of the F protein with amino acids from a heterologous sequence of Henipavirus results in fusion to the G protein with the heteroglous sequence. (Brandel-Tretheway et al. 2019). In some cases, the chimeric F and/or G protein contains an extracellular domain from one henipavirus species and a transmembrane and/or cytoplasmic domain from a different henipavirus species. For example, the F protein contains an extracellular domain of Hendra virus and a transmembrane/cytoplasmic domain of Nipah virus.

B. Lipid Bilayer

[0193] In some embodiments, the targeted lipid particle includes a naturally derived bilayer of amphipathic lipids that encloses a lumen or cavity. In some embodiments, the targeted lipid particle comprises a lipid bilayer as the outermost surface. In some embodiments, the lipid bilayer encloses a lumen. In some embodiments, the lumen is aqueous. In some embodiments, the lumen is in contact with the hydrophilic head groups on the interior of the lipid bilayer. In some embodiments, the lumen is a cytosol. In some embodiments, the cytosol contains cellular components present in a source cell. In some embodiments, the cytosol does not contain cellular components present in a source cell. In some embodiments, the lumen is a cavity. In some embodiments, the cavity contains an aqueous environment. In some embodiments, the cavity does not contain an aqueous environment.

[0194] In some aspects, the lipid bilayer is derived from a source cell during a process to produce a lipid-containing particle. In some embodiments, the lipid bilayer includes membrane components of the cell from which the lipid bilayer is produced, e.g., phospholipids, membrane proteins, etc. In some embodiments, the lipid bilayer includes a cytosol that includes components found in the cell from which the lipid bilayer is produced, e.g., solutes, proteins, nucleic acids, etc., but not all of the components of a cell, e.g., it lacks a nucleus. In some embodiments, the lipid bilayer is considered to be exosome-like. The lipid particle may vary in size, and in some instances have a diameter ranging from 30 and 300 nm, such as from 30 and 150 nm, and including from 40 to 100 nm.

[0195] In some embodiments, the lipid bilayer is a viral envelope. In some embodiments, the viral envelope is obtained from a source cell. In some embodiments, the viral envelope is obtained from the source cell plasma membrane. In some embodiments, the lipid bilayer is obtained from a membrane other than the plasma membrane of a host cell. In some embodiments, the viral envelope lipid bilayer is embedded with viral proteins, including viral glycoproteins.

[0196] In other aspects, the lipid bilayer includes synthetic lipid complex. In some embodiments, the lipid bilayer is a liposome that includes a synthetic lipid complex. In some embodiments, the lipid particle is a vesicular structure characterized by a phospholipid bilayer membrane and an inner aqueous medium. In some embodiments, the lipid bilayer has multiple lipid layers separated by aqueous medium. In some embodiments, the lipid bilayer forms spontaneously when phospholipids are suspended in an excess of aqueous solution. In some examples, the lipid components undergo self-rearrangement before the formation of closed structures and entrap water and dissolved solutes between the lipid bilayers. In some embodiments the lipid bilayer is a fusosome.

[0197] In some embodiments, a targeted envelope protein and fusogen, such as any described above including any that are exogenous or overexpressed relative to the source cell, is disposed in the lipid bilayer.

[0198] In some embodiments, the targeted lipid particle comprises several different types of lipids. In some embodiments, the lipids are amphipathic lipids. In some embodiments, the amphipathic lipids are phospholipids. In some embodiments, the phospholipids comprise phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol, and phosphatidylserine. In some embodiments, the lipids comprise DMPC, DOPC, and DSPC.

[0199] In some embodiments, the bilayer is comprised of one or more lipids of the same or different type. In some embodiments, the source cell comprises a cell selected from CHO cells, BHK cells, MDCK cells, C3H 10T1/2 cells, FLY cells, Psi-2 cells, BOSC 23 cells, PA317 cells, WEHI cells, COS cells, BSC 1 cells, BSC 40 cells, BMT 10 cells, VERO cells, W138 cells, MRC5 cells, A549 cells, HT1080 cells, 293 cells, 293T cells, B-50 cells, 3T3 cells, NIH3T3 cells, HepG2 cells, Saos-2 cells, Huh7 cells, Hela cells, W163 cells, 211 cells, and 211A cells.

C. Exogenous Agent

[0200] In some embodiments, the targeted lipid particle further comprises an agent that is exogenous relative to the source cell (also referred to herein as a "cargo" or "payload"). In some embodiments, the exogenous agent is a small molecule, a protein, or a nucleic acid (e.g., a DNA, a chromosome (e.g. a human artificial chromosome), an RNA, e.g., an mRNA or miRNA). In some embodiments, the exogenous agent or cargo encodes a cytosolic protein. In some embodiments the exogenous agent or cargo comprises or encodes a membrane protein. In some embodiments, the exogenous agent or cargo comprises a therapeutic agent. In some embodiments, the therapeutic agent is chosen from one or more of a protein, e.g., an enzyme, a transmembrane protein, a receptor, an antibody; a nucleic acid, e.g., DNA, a chromosome (e.g. a human artificial chromosome), RNA, mRNA, siRNA, miRNA; or a small molecule.

[0201] In some embodiments, the exogenous agent is present in at least, or no more than, 10, 20, 50, 100, 200, 500, 1,000, 2,000, 5,000, 10,000, 20,000, 50,000, 100,000, 200,000, 500,000, 1,000,000, 5,000,000, 10,000,000, 50,000,000, 100,000,000, 500,000,000, or 1,000,000,000 copies. In some embodiments, the targeted lipid particle has an altered, e.g., increased or decreased level of one or more endogenous molecules, e.g., protein or nucleic acid (e.g., in some embodiments, endogenous relative to the source cell, and in some embodiments, endogenous relative to the target cell), e.g., due to treatment of the source cell, e.g., mammalian source cell with a siRNA or gene editing enzyme. In some embodiments, the endogenous molecule is present in at least, or no more than, 10, 20, 50, 100, 200, 500, 1,000, 2,000, 5,000, 10,000, 20,000, 50,000, 100,000, 200,000, 500,000, 1,000,000, 5,000,000, 10,000,000, 50,000,000, 100,000,000, 500,000,000, or 1,000,000,000 copies. In some embodiments, the endogenous molecule (e.g., an RNA or protein) is present at a concentration of at least 1, 2, 3, 4, 5, 10, 20, 50, 100, 500, 10^3, 5.0×10^3, 10^4, 5.0×10^4, 10^5, 5.0×10^5, 10^6, 5.0×10^6, 1.0×10^7, 5.0×10^7, or 1.0×10^8, greater than its concentration in the source cell. In some embodiments, the endogenous molecule (e.g., an RNA or protein) is present at a concentration of at least 1, 2, 3, 4, 5, 10, 20, 50, 100, 500, 10^3, 5.0×10^3, 10^4, 5.0×10^4, 10^5, 5.0×10^5, 10^6, 5.0×10^6, 1.0×10^7, 5.0×10^7, or 1.0×10^8 less than its concentration in the source cell.

[0202] In some embodiments, the targeted lipid particle (e.g., targeted viral vector) delivers to a target cell at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, or 99% of the cargo (e.g., a therapeutic agent, e.g., an exogenous therapeutic agent) comprised by the targeted lipid particle. In some embodiments, the targeted lipid particle that fuses with the target cell(s) delivers to the target cell an average of at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, or 99% of the cargo (e.g., a therapeutic agent, e.g., an exogenous therapeutic agent) comprised by the targeted lipid particle that fuses with the target cell(s). In some embodiments, the targeted lipid particle composition delivers to a target tissue at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, or 99% of the cargo (e.g., a therapeutic agent, e.g., an exogenous therapeutic agent) comprised by the targeted lipid particle composition.

[0203] In some embodiments, the exogenous agent or cargo is not expressed naturally in the cell from which the targeted lipid particle is derived. In some embodiments, the exogenous agent or cargo is expressed naturally in the cell from which the viral vector is derived. In some embodiments, the exogenous agent or cargo is loaded into the targeted lipid particle via expression in the cell from which the viral vector is derived (e.g. expression from DNA or mRNA introduced via transfection, transduction, or electroporation). In some embodiments, the exogenous agent or cargo is expressed from DNA integrated into the genome or maintained episomally. In some embodiments, expression of the exogenous agent or cargo is constitutive. In some embodiments, expression of the exogenous agent or cargo is induced. In some embodiments, expression of the exogenous agent or cargo is induced immediately prior to generating the targeted lipid particle. In some embodiments, expression of the exogenous agent or cargo is induced at the same time as expression of the fusogen.

[0204] In some embodiments, the exogenous agent or cargo is loaded into the viral vector via electroporation into the viral vector itself or into the cell from which the viral vector is derived. In some embodiments, the exogenous agent or cargo is loaded into the viral vector via transfection (e.g., of a DNA or mRNA encoding the cargo) into the viral vector itself or into the cell from which the viral vector is derived.

[0205] In some embodiments, the exogenous agent or cargo includes one or more nucleic acid sequences, one or more amino acid sequences, a combination of nucleic acid sequences and/or amino acid sequences, one or more organelles, and any combination thereof. In some embodiments, the exogenous agent or cargo includes one or more cellular components. In some embodiments, the exogenous agent or cargo includes one or more cytosolic and/or nuclear components.

[0206] In some embodiments, the exogenous agent or cargo includes a nucleic acid, e.g., DNA, nDNA (nuclear DNA), mtDNA (mitochondrial DNA), protein coding DNA, gene, operon, chromosome, genome, transposon, retrotransposon, viral genome, intron, exon, modified DNA, mRNA (messenger RNA), tRNA (transfer RNA), modified RNA, microRNA, siRNA (small interfering RNA), tmRNA (transfer messenger RNA), rRNA (ribosomal RNA), mtRNA (mitochondrial RNA), snRNA (small nuclear RNA), small nucleolar RNA (snoRNA), SmY RNA (mRNA trans-splicing RNA), gRNA (guide RNA), TERC (telomerase RNA component), aRNA (antisense RNA), cis-NAT (Cis-natural antisense transcript), CRISPR RNA (crRNA), lncRNA (long noncoding RNA), piRNA (piwi-interacting RNA), shRNA (short hairpin RNA), tasiRNA (trans-acting siRNA), eRNA (enhancer RNA), satellite RNA, pcRNA (protein coding RNA), dsRNA (double stranded RNA), RNAi (interfering RNA), circRNA (circular RNA), reprogramming RNAs, aptamers, and any combination thereof. In some embodiments, the nucleic acid is a wild-type nucleic acid. In some embodiments, the nucleic acid is a mutant nucleic acid. In some embodiments the nucleic acid is a fusion or chimera of multiple nucleic acid sequences.

[0207] In some embodiments, the exogenous agent or cargo includes a nucleic acid. For example, the exogenous agent or cargo may comprise RNA to enhance expression of an endogenous protein, or a siRNA or miRNA that inhibits protein expression of an endogenous protein. For example, the endogenous protein may modulate structure or function in the target cells. In some embodiments, the cargo includes a nucleic acid encoding an engineered protein that modulates structure or function in the target cells. In some embodiments, the exogenous agent or cargo is a nucleic acid that targets a transcriptional activator that modulate structure or function in the target cells.

[0208] In some embodiments, the exogenous agent or cargo includes a polypeptide, e.g., enzymes, structural proteins, signaling proteins, regulatory proteins, transport proteins, sensory proteins, motor proteins, defense proteins, storage proteins, transcription factors, antibodies, cytokines, hormones, catabolic proteins, anabolic proteins, proteolytic proteins, metabolic proteins, kinases, transferases, hydrolases, lyases, isomerases, ligases, enzyme modulator proteins, protein binding polypeptides, lipid binding proteins, membrane fusion proteins, cell differentiation proteins, epigenetic proteins, cell death proteins, nuclear transport proteins, nucleic acid binding proteins, reprogramming proteins, DNA editing proteins, DNA repair proteins, DNA recombination proteins, transposase proteins, DNA integration proteins, targeted endonucleases (e.g. Zinc-finger nucleases, transcription-activator-like nucleases (TALENs), cas9 and homologs thereof), recombinases, and any combination thereof. In some embodiments the protein targets a protein in the cell for degradation. In some embodiments the protein targets a protein in the cell for degradation by localizing the protein to the proteasome. In some embodiments, the protein is a wild-type protein. In some embodiments, the protein is a mutant protein. In some embodiments the protein is a fusion or chimeric protein.

[0209] In some embodiments, the exogenous agent or cargo includes a small molecule, e.g., ions (e.g. Ca^{2+}, C1−, Fe^{2+}), carbohydrates, lipids, reactive oxygen species, reactive nitrogen species, isoprenoids, signaling molecules, heme, peptide cofactors, electron accepting compounds, electron donating compounds, metabolites, ligands, and any combination thereof. In some embodiments the small molecule is a pharmaceutical that interacts with a target in the cell. In some embodiments the small molecule targets a protein in the cell for degradation. In some embodiments the small molecule targets a protein in the cell for degradation by localizing the protein to the proteasome. In some embodiments that small molecule is a proteolysis targeting chimera molecule (PROTAC).

[0210] In some embodiments, the exogenous agent or cargo includes a mixture of proteins, nucleic acids, or metabolites, e.g., multiple amino acids, multiple nucleic acids, multiple small molecules; combinations of nucleic acids, amino acids, and small molecules; ribonucleoprotein complexes (e.g. Cas9-gRNA complex); multiple transcription factors, multiple epigenetic factors, reprogramming factors (e.g. Oct4, Sox2, cMyc, and Klf4); multiple regulatory RNAs; and any combination thereof.

[0211] In some embodiments, the exogenous agent or cargo includes one or more organelles, e.g., chondriosomes, mitochondria, lysosomes, nucleus, cell membrane, cytoplasm, endoplasmic reticulum, ribosomes, vacuoles, endosomes, spliceosomes, polymerases, capsids, acrosome, autophagosome, centriole, glycosome, glyoxysome, hydrogenosome, melanosome, mitosome, myofibril, cnidocyte, peroxisome, proteasome, vesicle, stress granule, networks of organelles, and any combination thereof.

[0212] In some embodiments, the exogenous agent encodes a therapeutic agent or a diagnostic agent. In some embodiments, the therapeutic agent is a chimeric antigen receptor (CAR) or T-cell receptor (TCR). In some embodiments, the CAR targets a tumor antigen selected from CD19, CD20, CD22, or BCMA. In another embodiment, the CAR is engineered to comprise an intracellular signaling domain of the T cell antigen receptor complex zeta chain (e.g., CD3 zeta). In a preferred embodiment, the intracellular domain is selected from a CD137 (4-1BB) signaling domain, a CD28 signaling domain, and a CD3zeta signaling domain.

D. Methods of Generating Targeted Lipid Particles Derived from Virus

[0213] Provided herein are targeted lipid particles that are derived from virus, such as viral particles or virus-like particles, including those derived from retroviruses or lentiviruses. In some embodiments, the targeted lipid particle's bilayer of amphipathic lipids is or comprises the viral envelope. In some embodiments, the targeted lipid particle's bilayer of amphipathic lipids is or comprises lipids derived from a producer cell. In some embodiments, the viral envelope comprises a fusogen, e.g., a fusogen that is endogenous to the virus or a pseudotyped fusogen. In some embodiments, the targeted lipid particle's lumen or cavity comprises a viral nucleic acid, e.g., a retroviral nucleic acid, e.g., a lentiviral nucleic acid. In some embodiments, the viral nucleic acid is a viral genome. In some embodiments, the targeted lipid particle further comprises one or more viral non-structural proteins, e.g., in its cavity or lumen. In some embodiments, the targeted lipid particle is or comprises a virus-like particle (VLP). In some embodiments, the VLP does not comprise an envelope. In some embodiments, the VLP comprises an envelope.

[0214] In some embodiments, the viral particle or virus-like particle, such as a retrovirus or retrovirus-like particle, comprises one or more of a Gag polyprotein, polymerase (e.g., Pol), integrase (IN, e.g., a functional or non-functional variant), protease (PR), and a fusogen. In some embodiments, the targeted lipid particle further comprises Rev. In some embodiments, one or more of the aforesaid proteins are encoded in the retroviral genome, and in some embodiments, one or more of the aforesaid proteins are provided in trans, e.g., by a helper cell, helper virus, or helper plasmid.

In some embodiments, the targeted lipid particle nucleic acid (e.g., retroviral nucleic acid) comprises one or more of the following nucleic acid sequences: 5' LTR (e.g., comprising U5 and lacking a functional U3 domain), Psi packaging element (Psi), Central polypurine tract (cPPT) Promoter operatively linked to the payload gene, payload gene (optionally comprising an intron before the open reading frame), Poly A tail sequence, WPRE, and 3' LTR (e.g., comprising U5 and lacking a functional U3). In some embodiments the targeted lipid particle nucleic acid further comprises one or more insulator elements. In some embodiments, the recognition sites are situated between the poly A tail sequence and the WPRE.

[0215] In some embodiments, the targeted lipid particle comprises supramolecular complexes formed by viral proteins that self-assemble into capsids. In some embodiments, the targeted lipid particle is a viral particle or virus-like particle derived from viral capsids. In some embodiments, the targeted lipid particle is a viral particle or virus-like particle derived from viral nucleocapsids. In some embodiments, the targeted lipid particle comprises nucleocapsid-derived proteins that retain the property of packaging nucleic acids. In some embodiments, the viral particles or virus-like particles comprise only viral structural glycoproteins. In some embodiments, the targeted lipid particle does not contain a viral genome.

[0216] In some embodiments, the targeted lipid particle packages nucleic acids from host cells during the expression process. In some embodiments, the nucleic acids do not encode any genes involved in virus replication. In particular embodiments, the targeted lipid particle is a virus-like particle, e.g. retrovirus-like particle such as a lentivirus-like particle, that is replication defective.

[0217] In some cases, the targeted lipid particle is a viral particle that is morphologically indistinguishable from the wild type infectious virus. In some embodiments, the viral particle presents the entire viral proteome as an antigen. In some embodiments, the viral particle presents only a portion of the proteome as an antigen.

[0218] In some embodiments, the viral particle or virus-like particle is produced utilizing proteins (e.g., envelope proteins) from a virus within the Paramyxoviridae family. In some embodiments, the Paramyxoviridae family comprises members within the Henipavirus genus. In some embodiments, the Henipavirus is or comprises a Hendra (HeV) or a Nipah (NiV) virus. In particular embodiments, the viral particles or virus-like particles incorporate a targeted envelope protein and fusogen.

[0219] In some embodiments, viral particles or virus-like particles is produced in multiple cell culture systems including bacteria, mammalian cell lines, insect cell lines, yeast, and plant cells.

[0220] Suitable cell lines which can be used include, for example, CHO cells, BHK cells, MDCK cells, C3H 10T1/2 cells, FLY cells, Psi-2 cells, BOSC 23 cells, PA317 cells, WEHI cells, COS cells, BSC 1 cells, BSC 40 cells, BMT 10 cells, VERO cells, W138 cells, MRC5 cells, A549 cells, HT1080 cells, 293 cells, 293T cells, B-50 cells, 3T3 cells, NIH3T3 cells, HepG2 cells, Saos-2 cells, Huh7 cells, Hela cells, W163 cells, 211 cells, 211A cells, and cyno and *Macaca nemestrina* cell lines. In embodiments, the packaging cells are 293 cells, 293T cells, or A549 cells.

[0221] In some embodiments, a source cell line includes a cell line which is capable of producing recombinant retroviral particles, comprising a producer cell line and a transfer vector construct comprising a packaging signal. Methods of preparing viral stock solutions are illustrated by, e.g., Y. Soneoka et al. (1995) *Nucl. Acids Res.* 23:628-633, and N. R. Landau et al. (1992) *J. Virol.* 66:5110-5113, which are incorporated herein by reference.

[0222] In some embodiments, the assembly of a viral particle or virus-like particle is initiated by binding of the core protein to a unique encapsidation sequence within the viral genome (e.g. UTR with stem-loop structure). In some embodiments, the interaction of the core with the encapsidation sequence facilitates oligomerization.

[0223] In some embodiments, the targeted lipid particle is a virus-like particle which comprises a sequence that is devoid of or lacking viral RNA. In some embodiments, such particles are the result of removing or eliminating the viral RNA from the sequence. In some embodiments, this is achieved by using an endogenous packaging signal binding site on Gag. In some embodiments, the endogenous packaging signal binding site is on Pol. In some embodiments, the RNA which is to be delivered will contain a cognate packaging signal. In some embodiments, a heterologous binding domain (which is heterologous to Gag) located on the RNA to be delivered, and a cognate binding site located on Gag or Pol, is used to ensure packaging of the RNA to be delivered. In some embodiments, the heterologous sequence is non-viral or it could be viral, in which case it is derived from the same virus or a different virus. In some embodiments, the vector particles could be used to deliver therapeutic RNA, in which case functional integrase and/or reverse transcriptase is not required. In some embodiments, the vector particles could also be used to deliver a therapeutic gene of interest, in which case Pol is typically included. In some embodiments, the retroviral nucleic acid comprises one or more of (e.g., all of): a 5' promoter (e.g., to control expression of the entire packaged RNA), a 5' LTR (e.g., that includes R (polyadenylation tail signal) and/or U5 which includes a primer activation signal), a primer binding site, a Psi packaging signal, a RRE element for nuclear export, a promoter directly upstream of the transgene to control transgene expression, a transgene (or other exogenous agent element), a polypurine tract, and a 3' LTR (e.g., that includes a mutated U3, a R, and U5). In some embodiments, the retroviral nucleic acid further comprises one or more of a cPPT, a WPRE, and/or an insulator element.

[0224] A retrovirus typically replicates by reverse transcription of its genomic RNA into a linear double-stranded DNA copy and subsequently covalently integrates its genomic DNA into a host genome. Illustrative retroviruses suitable for use in particular embodiments, include, but are not limited to: Moloney murine leukemia virus (M-MuLV), Moloney murine sarcoma virus (MOMSV), Harvey murine sarcoma virus (HaMuSV), murine mammary tumor virus (MuMTV), gibbon ape leukemia virus (GaLV), feline leukemia virus (FLV), spumavirus, Friend murine leukemia virus, Murine Stem Cell Virus (MSCV), Rous Sarcoma Virus (RSV), and other lentiviruses.

[0225] In some embodiments the retrovirus is a Gammaretrovirus. In some embodiments the retrovirus is an Epsilonretrovirus. In some embodiments the retrovirus is an Alpharetrovirus. In some embodiments the retrovirus is a Betaretrovirus. In some embodiments the retrovirus is a Deltaretrovirus. In some embodiments the retrovirus is a

Lentivirus. In some embodiments the retrovirus is a Spumaretrovirus. In some embodiments the retrovirus is an endogenous retrovirus.

[0226] Illustrative lentiviruses include, but are not limited to: HIV (human immunodeficiency virus; including HIV type 1, and HIV type 2); visna-maedi virus (VMV) virus; the caprine arthritis-encephalitis virus (CAEV); equine infectious anemia virus (EIAV); feline immunodeficiency virus (FIV); bovine immune deficiency virus (BIV); and simian immunodeficiency virus (SIV). In some embodiments, HIV based vector backbones (i.e., HIV cis-acting sequence elements) are used.

[0227] In some embodiments, a vector herein is a nucleic acid molecule capable transferring or transporting another nucleic acid molecule. The transferred nucleic acid is generally linked to, e.g., inserted into, the vector nucleic acid molecule. A vector may include sequences that direct autonomous replication in a cell, or may include sequences sufficient to allow integration into host cell DNA. Useful vectors include, for example, plasmids (e.g., DNA plasmids or RNA plasmids), transposons, cosmids, bacterial artificial chromosomes, and viral vectors. Useful viral vectors include, e.g., replication defective retroviruses and lentiviruses.

[0228] In some embodiments, a viral vector comprises a nucleic acid molecule (e.g., a transfer plasmid) that includes virus-derived nucleic acid elements that typically facilitate transfer of the nucleic acid molecule or integration into the genome of a cell or to a viral particle that mediates nucleic acid transfer. Viral particles will typically include various viral components and sometimes also host cell components in addition to nucleic acid(s). In some embodiments, a viral vector comprises e.g., a virus or viral particle capable of transferring a nucleic acid into a cell, or the transferred nucleic acid (e.g., as naked DNA). In some embodiments, a viral vectors and transfer plasmids comprise structural and/or functional genetic elements that are primarily derived from a virus. A retroviral vector can comprise a viral vector or plasmid containing structural and functional genetic elements, or portions thereof, that are primarily derived from a retrovirus. A lentiviral vector can comprise a viral vector or plasmid containing structural and functional genetic elements, or portions thereof, including LTRs that are primarily derived from a lentivirus.

[0229] In embodiments, a lentiviral vector (e.g., lentiviral expression vector) comprises a lentiviral transfer plasmid (e.g., as naked DNA) or an infectious lentiviral particle. With respect to elements such as cloning sites, promoters, regulatory elements, heterologous nucleic acids, etc., it is to be understood that the sequences of these elements can be present in RNA form in lentiviral particles and can be present in DNA form in DNA plasmids.

[0230] In some embodiments, the viral vector further comprises a vector-surface targeting moiety which specifically binds to a target ligand. In some embodiments, the vector-surface targeting moiety is a polypeptide. In some embodiments, a nucleic acid encoding the Paramyxovirus envelope protein (e.g. G protein) is modified with a targeting moiety to specifically bind to a target molecule on a target cells. In some embodiments, the targeting moiety is any targeting protein, including but not necessarily limited to antibodies and antigen binding fragments thereof.

[0231] In some embodiments, in the vectors described herein at least part of one or more protein coding regions that contribute to or are essential for replication are absent compared to the corresponding wild-type virus. In some embodiments, the viral vector is replication-defective. In some embodiments, the vector is capable of transducing a target non-dividing host cell and/or integrating its genome into a host genome.

[0232] In some embodiments, different cells differ in their usage of particular codons. In some embodiments, this codon bias corresponds to a bias in the relative abundance of particular tRNAs in the cell type. In some embodiments, by altering the codons in the sequence so that they are tailored to match with the relative abundance of corresponding tRNAs, it is possible to increase expression. In some embodiments, it is possible to decrease expression by deliberately choosing codons for which the corresponding tRNAs are known to be rare in the particular cell type. In some embodiments, an additional degree of translational control is available. An additional description of codon optimization is found, e.g., in WO 99/41397, which is herein incorporated by reference in its entirety.

[0233] Conventional techniques for generating retrovirus vectors (and, in particular, lentivirus vectors) with or without the use of packaging/helper vectors are known to those skilled in the art and may be used to generate targeted lipid particles according to the present disclosure. (See, e.g., Derse and Newbold 1993 Virology 194:530-6; Maury et al. 1994 Virology 200:632-42; Wanisch et al. 2009. Mol Ther. 1798:1316-1332; Martarano et al. 1994 J. Virol. 68:3102-11; Naldini et al., (1996a, 1996b, and 1998); Zufferey et al., 1999, J. Virol., 73:2886; Huang et al., Mol. Cell. Biol., 5:3864; Liu et al., 1995, Genes Dev., 9:1766; Cullen et al., 1991. J. Virol. 65:1053; and Cullen et al., 1991. Cell 58:423; Dull et al., 1998, U.S. Pat. Nos. 6,013,516; and 5,994,136; PCT patent applications WO 99/15683, WO 98/17815, WO 99/32646, and WO 01/79518). Conventional techniques relating to packaging vectors and producer cells known in the art may also be used according to the present disclosure. (See, e.g., Yao et al, 1998; Jones et al, 2005.)

[0234] Provided herein are targeted lipid particles that comprise a naturally derived membrane. In some embodiments, the naturally derived membrane comprises membrane vesicles prepared from cells or tissues. In some embodiments, the targeted lipid particle comprises a vesicle that is obtainable from a cell. In some embodiments, the targeted lipid particle comprises a microvesicle, an exosome, a membrane enclosed body, an apoptotic body (from apoptotic cells), a particle (which may be derived from e.g. platelets), an ectosome (derivable from, e.g., neutrophiles and monocytes in serum), a prostatosome (obtainable from prostate cancer cells), or a cardiosome (derivable from cardiac cells).

[0235] In some embodiments, the source cell is an endothelial cell, a fibroblast, a blood cell (e.g., a macrophage, a neutrophil, a granulocyte, a leukocyte), a stem cell (e.g., a mesenchymal stem cell, an umbilical cord stem cell, bone marrow stem cell, a hematopoietic stem cell, an induced pluripotent stem cell e.g., an induced pluripotent stem cell derived from a subject's cells), an embryonic stem cell (e.g., a stem cell from embryonic yolk sac, placenta, umbilical cord, fetal skin, adolescent skin, blood, bone marrow, adipose tissue, erythropoietic tissue, hematopoietic tissue), a myoblast, a parenchymal cell (e.g., hepatocyte), an alveolar cell, a neuron (e.g., a retinal neuronal cell), a precursor cell (e.g., a retinal precursor cell, a myeloblast, myeloid precursor cells, a thymocyte, a meiocyte, a megakaryoblast, a promegakaryoblast, a melanoblast, a lymphoblast, a bone marrow precursor cell, a normoblast, or an angioblast), a progenitor cell (e.g., a cardiac progenitor cell, a satellite cell, a radial glial cell, a bone marrow stromal cell, a pancreatic progenitor cell, an endothelial progenitor cell, a blast cell), or an immortalized cell (e.g., HeEa, HEK293, HFF-I, MRC-5, WI-38, IMR 90, IMR 91, PER.C6, HT-1080, or BJ cell). In some embodiments, the source cell is other than a 293 cell, HEK cell, human endothelial cell, or a human epithelial cell, monocyte, macrophage, dendritic cell, or stem cell.

[0236] In some embodiments, the targeted lipid particle has a density of <1, 1-1.1, 1.05-1.15, 1.1-1.2, 1.15-1.25, 1.2-1.3, 1.25-1.35, or >1.35 g/ml. In some embodiments, the targeted lipid particle composition comprises less than 0.01%, 0.05%, 0.1%, 0.5%, 1%, 1.5%, 2%, 2.5%, 3%, 4%, 5%, or 10% source cells by protein mass, or less than 0.01%, 0.05%, 0.1%, 0.5%, 1%, 1.5%, 2%, 2.5%, 3%, 4%, 5%, or 10% of cells having a functional nucleus.

[0237] In embodiments, the targeted lipid particle has a size, or the population of targeted lipid particles have an average size, that is less than about 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, of that of the source cell.

[0238] In some embodiments the targeted lipid particle comprises an extracellular vesicle, e.g., a cell-derived vesicle comprising a membrane that encloses an internal space and has a smaller diameter than the cell from which it is derived. In embodiments the extracellular vesicle has a diameter from 20 nm to 1000 nm. In embodiments the targeted lipid particle comprises an apoptotic body, a fragment of a cell, a vesicle derived from a cell by direct or indirect manipulation, a vesiculated organelle, and a vesicle produced by a living cell (e.g., by direct plasma membrane budding or fusion of the late endosome with the plasma membrane). In embodiments the extracellular vesicle is derived from a living or dead organism, explanted tissues or organs, or cultured cells.

[0239] In embodiments, the targeted lipid particle comprises a nanovesicle, e.g., a cell-derived small (e.g., from 20 to 250 nm in diameter, or from 30 to 150 nm in diameter) vesicle comprising a membrane that encloses an internal space, and which is generated from said cell by direct or indirect manipulation. The production of nanovesicles can, in some instances, result in the destruction of the source cell. The nanovesicle may comprise a lipid or fatty acid and a polypeptide.

[0240] In embodiments, the targeted lipid particle comprises an exosome. In embodiments, the exosome is a cell-derived small (e.g., from 20 to 300 nm in diameter, or from 40 to 200 nm in diameter) vesicle comprising a membrane that encloses an internal space, and which is generated from said cell by direct plasma membrane budding or by fusion of the late endosome with the plasma membrane. In embodiments, production of exosomes does not result in the destruction of the source cell. In embodiments, the exosome comprises a lipid or fatty acid and a polypeptide.

[0241] In some embodiments, the targeted lipid particle is derived from a source cell with a genetic modification which results in increased expression of an immunomodulatory agent. In some embodiments, the immunosuppressive agent is on an exterior surface of the cell. In some embodiments, the immunosuppressive agent is incorporated into the exterior surface of the targeted lipid particle. In some embodiments, the targeted lipid particle comprises an immunomodulatory agent attached to the surface of the solid particle by a covalent or non-covalent bond.

A. Generation of Cell-Derived Particles

[0242] In some embodiments, targeted lipid particles are generated by inducing budding of an exosome, microvesicle, membrane vesicle, extracellular membrane vesicle, plasma membrane vesicle, giant plasma membrane vesicle, apoptotic body, mitoparticle, pyrenocyte, lysosome, or other membrane enclosed vesicle.

[0243] In some embodiments, targeted lipid particles are generated by inducing cell enucleation. Enucleation may be performed using assays such as genetic, chemical (e.g., using Actinomycin D, see Bayona-Bafaluy et al., "A chemical enucleation method for the transfer of mitochondrial DNA to p° cells" Nucleic Acids Res. 2003 Aug. 15; 31(16): e98), or mechanical methods (e.g., squeezing or aspiration, see Lee et al., "A comparative study on the efficiency of two enucleation methods in pig somatic cell nuclear transfer: effects of the squeezing and the aspiration methods." Anim Biotechnol. 2008; 19(2):71-9), or combinations thereof.

[0244] In some embodiments, the targeted lipid particles are generated by inducing cell fragmentation. In some embodiments, cell fragmentation is performed using the following methods, including, but not limited to: chemical methods, mechanical methods (e.g., centrifugation (e.g., ultracentrifugation, or density centrifugation), freeze-thaw, or sonication), or combinations thereof.

[0245] In some embodiments, the targeted lipid particle is a microvesicle. In some embodiments the microvesicle has a diameter of about 100 nm to about 2000 nm. In some embodiments, a targeted lipid particle comprises a cell ghost. In some embodiments, a vesicle is a plasma membrane vesicle, e.g., a giant plasma membrane vesicle.

[0246] In some embodiments, a characteristic of a targeted lipid particle is described by comparison to a reference cell. In embodiments, the reference cell is the source cell. In embodiments, the reference cell is a HeLa, HEK293, HFF-1, MRC-5, WI-38, IMR 90, IMR 91, PER.C6, HT-1080, or BJ cell. In some embodiments, for example when the source cell used to make the targeted lipid particle is not available for testing after the targeted lipid particle is made, a characteristic of a population of targeted lipid particle is described by comparison to a population of reference cells, e.g., a population of source cells, or a population of HeLa, HEK293, HFF-1, MRC-5, WI-38, IMR 90, IMR 91, PER.C6, HT-1080, or BJ cells.

Pharmaceutical Compositions

[0247] The present disclosure also provides, in some aspects, a pharmaceutical composition comprising the targeted lipid particle (e.g., targeted viral vectors) composition described herein and a pharmaceutically acceptable carrier. The pharmaceutical compositions can include any of the described targeted lipid particles.

[0248] In some embodiments, the targeted lipid particle meets a pharmaceutical or good manufacturing practices (GMP) standard. In some embodiments, the targeted lipid particle is made according to good manufacturing practices (GMP). In some embodiments, the targeted lipid particle has a pathogen level below a predetermined reference value, e.g., is substantially free of pathogens. In some embodiments, the targeted lipid particle has a contaminant level below a predetermined reference value, e.g., is substantially free of contaminants. In some embodiments, the targeted lipid particle has low immunogenicity.

[0249] In some embodiments, provided herein are the use of pharmaceutical compositions to practice the methods of the disclosure. Such a pharmaceutical composition may comprise at least one targeted lipid particle of the disclosure in a form suitable for administration to a subject, or the pharmaceutical composition may comprise at least one targeted lipid particle of the disclosure and one or more pharmaceutically acceptable carriers, one or more additional ingredients, or some combination of these.

[0250] In some embodiments, the relative amounts of the targeted lipid particle, the pharmaceutically acceptable carrier, and any additional ingredients in a pharmaceutical composition of the disclosure will vary, depending upon the identity, size, and condition of the subject treated and further depending upon the route by which the composition is to be administered. In some embodiments, the composition comprises from 0.1% to 100% (w/w) of the targeted lipid particles of the disclosure.

[0251] In some embodiments, pharmaceutical compositions that are useful in the methods of the disclosure are suitably developed for intravenous, intratumoral, oral, rectal, vaginal, parenteral, topical, pulmonary, intranasal, buccal, ophthalmic, or another route of administration. In some embodiments, a composition useful within the methods of the disclosure are directly administered to the skin, vagina or any other tissue of a mammal. In some embodiments, formulations include liposomal preparations, resealed erythrocytes containing the targeted lipid particles of the disclosure, and immunologically based formulations. In some embodiments, the route(s) of administration will be readily apparent to the skilled artisan and will depend upon any number of factors including the type and severity of the disease being treated, the type and age of the veterinary or human subject being treated, and the like.

[0252] In some embodiments, formulations of the pharmaceutical compositions described herein are prepared by any method known or hereafter developed in the art of pharmacology. In some embodiments, preparatory methods include the step of bringing the targeted lipid particles of the disclosure into association with a carrier or one or more other accessory ingredients, and then, if necessary or desirable, shaping or packaging the product into a desired single- or multi-dose unit.

[0253] In some embodiments, a "unit dose" is a discrete amount of the pharmaceutical composition comprising a predetermined amount of the targeted lipid particles of the disclosure. In some embodiments, the amount is generally equal to the dosage that would be administered to a subject or a convenient fraction of such a dosage such as, for example, one-half or one-third of such a dosage. In some embodiments, the unit dosage form is for a single daily dose or one of multiple daily doses (e.g., about 1 to 4 or more times per day). In some embodiments, when multiple daily doses are used, the unit dosage form is the same or different for each dose.

[0254] In some embodiments, although the descriptions of pharmaceutical compositions provided herein are principally directed to pharmaceutical compositions that are suitable for ethical administration to humans, it will be understood by the skilled artisan that such compositions are generally suitable for administration to animals of all sorts. In some embodiments, modification of pharmaceutical compositions suitable for administration to humans in order to render the compositions suitable for administration to various animals is well understood, and the ordinarily skilled veterinary pharmacologist may design and perform such modification with merely ordinary, if any, experimentation. In some embodiments, subjects to which administration of the pharmaceutical compositions of the disclosure is contemplated include humans and other primates, mammals including commercially relevant mammals such as cattle, pigs, horses, sheep, cats, and dogs.

[0255] In some of any embodiments, the compositions of the disclosure are formulated using one or more pharmaceutically acceptable excipients or carriers. In some embodiments, the pharmaceutical compositions of the disclosure comprise a therapeutically effective amount of a targeted lipid particle of the disclosure and a pharmaceutically acceptable carrier. In some embodiments, pharmaceutically acceptable carriers that are useful, include, but are not limited to, glycerol, water, saline, ethanol, and other pharmaceutically acceptable salt solutions such as phosphates and salts of organic acids. Examples of these and other pharmaceutically acceptable carriers are described in Remington's Pharmaceutical Sciences (1991, Mack Publication Co., New Jersey).

[0256] In some embodiments, the carrier is a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and vegetable oils. In some embodiments, the proper fluidity is maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. In some embodiments, prevention of the action of microorganisms is achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal, and the like. In some embodiments, it is preferable to include isotonic agents, for example, sugars, sodium chloride, or polyalcohols such as mannitol and sorbitol, in the composition. In some embodiments, prolonged absorption of the injectable compositions is brought about by including in the composition an agent that delays absorption, for example, aluminum monostearate or gelatin. In some embodiments, the pharmaceutically acceptable carrier is not DMSO alone.

[0257] In some embodiments, formulations are employed in admixtures with conventional excipients, i.e., pharmaceutically acceptable organic or inorganic carrier substances suitable for oral, vaginal, parenteral, nasal, intravenous, subcutaneous, enteral, or any other suitable mode of administration, known to the art. In some embodiments, the pharmaceutical preparations are sterilized and, if desired, mixed with auxiliary agents, e.g., lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure buffers, coloring, flavoring, and/or aromatic substances and the like. In some embodiments, pharmaceutical preparations are also combined with other active agents, e.g., other analgesic agents.

[0258] In some embodiments, "additional ingredients" include, but are not limited to, one or more of the following: excipients; surface active agents; dispersing agents; inert diluents; granulating and disintegrating agents; binding agents; lubricating agents; sweetening agents; flavoring agents; coloring agents; preservatives; physiologically degradable compositions such as gelatin; aqueous vehicles and solvents; oily vehicles and solvents; suspending agents; dispersing or wetting agents; emulsifying agents, demulcents; buffers; salts; thickening agents; fillers; emulsifying agents; antioxidants; antibiotics; antifungal agents; stabilizing agents; and pharmaceutically acceptable polymeric or hydrophobic materials. In some embodiments, "additional ingredients" that are included in the pharmaceutical compositions of the disclosure are known in the art and described, for example in Genaro, ed. (1985, Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pa.), which is incorporated herein by reference.

[0259] In some embodiments, the composition of the disclosure comprises a preservative from about 0.005% to 2.0% by total weight of the composition. In some embodiments, the preservative is used to prevent spoilage in the case of exposure to contaminants in the environment. In some embodiments, examples of preservatives useful in accordance with the disclosure included but are not limited to those selected from the group consisting of benzyl alcohol, sorbic acid, parabens, imidurea and combinations thereof. In some embodiments, a particularly preferred preservative is a combination of about 0.5% to 2.0% benzyl alcohol and 0.05% to 0.5% sorbic acid.

[0260] In some embodiments, liquid suspensions are prepared using conventional methods to achieve suspension of the targeted lipid particles of the disclosure in an aqueous or oily vehicle. In some embodiments, aqueous vehicles include, for example, water, and isotonic saline. In some embodiments, oily vehicles include, for example, almond oil, oily esters, ethyl alcohol, vegetable oils such as *arachis*, olive, sesame, or coconut oil, fractionated vegetable oils, and mineral oils such as liquid paraffin. In some embodiments, liquid suspensions further comprise one or more additional ingredients including, but not limited to, suspending agents, dispersing or wetting agents, emulsifying agents, demulcents, preservatives, buffers, salts, flavorings, coloring agents, and sweetening agents. In some embodiments, oily suspensions further comprise a thickening agent. In some embodiments, suspending agents include, but are not limited to, sorbitol syrup, hydrogenated edible fats, sodium alginate, polyvinylpyrrolidone, gum tragacanth, gum acacia, and cellulose derivatives such as sodium carboxymethylcellulose, methylcellulose, hydroxypropylmethylcellulose. In some embodiments, dispersing or wetting agents include, but are not limited to, naturally-occurring phosphatides such as lecithin, condensation products of an alkylene oxide with a fatty acid, with a long chain aliphatic alcohol, with a partial ester derived from a fatty acid and a hexitol, or with a partial ester derived from a fatty acid and a hexitol anhydride (e.g., polyoxyethylene stearate, heptadecaethyleneoxycetanol, polyoxyethylene sorbitol monooleate, and polyoxyethylene sorbitan monooleate, respectively). Known emulsifying agents include, but are not limited to, lecithin, and acacia. Known preservatives include, but are not limited to, methyl, ethyl, or n-propyl-para-hydroxybenzoates, ascorbic acid, and sorbic acid. Known sweetening agents include, for example, glycerol, propylene glycol, sorbitol, sucrose, and saccharin. Known thickening agents for oily suspensions include, for example, beeswax, hard paraffin, and cetyl alcohol.

[0261] In some embodiments, liquid solutions of the targeted lipid particles of the disclosure in aqueous or oily solvents are prepared in substantially the same manner as liquid suspensions, the primary difference being that the targeted lipid particles of the disclosure is dissolved, rather than suspended in the solvent. As used herein, an "oily" liquid is one which comprises a carbon-containing liquid molecule and which exhibits a less polar character than water. In some embodiments, liquid solutions of the pharmaceutical composition of the disclosure comprise each of the components described with regard to liquid suspensions, it being understood that suspending agents will not necessarily aid dissolution of the targeted lipid particles of the disclosure in the solvent. In some embodiments, aqueous solvents include, for example, water, and isotonic saline. In some embodiments, oily solvents include, for example, almond oil, oily esters, ethyl alcohol, vegetable oils such as *arachis*, olive, sesame, or coconut oil, fractionated vegetable oils, and mineral oils such as liquid paraffin.

[0262] In some embodiments, powdered and granular formulations of a pharmaceutical preparation of the disclosure are prepared using known methods. In some embodiments, formulations are administered directly to a subject, used, for example, to form tablets, to fill capsules, or to prepare an aqueous or oily suspension or solution by addition of an aqueous or oily vehicle thereto. In some of any embodiments, formulations further comprise one or more of dispersing or wetting agent, a suspending agent, and a preservative. Additional excipients, such as fillers and sweetening, flavoring, or coloring agents, are also included in these formulations.

[0263] In some embodiments, a pharmaceutical composition of the disclosure is also prepared, packaged, or sold in the form of oil-in-water emulsion or a water-in-oil emulsion. In some embodiments, the oily phase is a vegetable oil such as olive or arachis oil, a mineral oil such as liquid paraffin, or a combination of these. In some embodiments, compositions further comprise one or more emulsifying agents such as naturally occurring gums such as gum acacia or gum tragacanth, naturally-occurring phosphatides such as soybean or lecithin phosphatide, esters or partial esters derived from combinations of fatty acids and hexitol anhydrides such as sorbitan monooleate, and condensation products of such partial esters with ethylene oxide such as polyoxyethylene sorbitan monooleate. In some embodiments, emulsions also contain additional ingredients including, for example, sweetening or flavoring agents.

Methods of Treatment

[0264] In some embodiments, the targeted lipid particles (e.g. targeted viral vectors) provided herein, or pharmaceutical compositions thereof as described herein are administered to a subject, e.g. a mammal, e.g. a human. In such embodiments, the subject is at risk of, has a symptom of, or is diagnosed with or identified as having, a particular disease or condition. In some embodiments, the subject has cancer. In some embodiments, the subject has an infectious disease. In some embodiments, the targeted lipid particle contains nucleic acid sequences encoding an exogenous agent for treating the disease or condition in the subject. For example, the exogenous agent is one that targets or is specific for a protein of a neoplastic cells and the targeted lipid particle is administered to a subject for treating a tumor or cancer in the subject. In another example, the exogenous agent is an inflammatory mediator or immune molecule, such as a cytokine, and targeted lipid particle is administered to a subject for treating any condition in which it is desired to modulate (e.g., increase) the immune response, such as a cancer or infectious disease. In some embodiments, the targeted lipid particle is administered in an effective amount or dose to effect treatment of the disease, condition, or disorder. Provided herein are uses of any of the provided targeted lipid particles in such methods and treatments, and in the preparation of a medicament in order to carry out such therapeutic methods. In some embodiments, the methods are carried out by administering the targeted lipid particle or compositions comprising the same, to the subject having, having had, or suspected of having the disease or condition or disorder. In some embodiments, the methods thereby treat the disease or condition or disorder in the subject. Also provided herein are uses of any of the compositions, such as pharmaceutical compositions provided herein, for the treatment of a disease, condition or disorder associated with a particular gene or protein targeted by or provided by the exogenous agent.

[0265] In some embodiments, the provided methods or uses involve administration of a pharmaceutical composition comprising oral, inhaled, transdermal or parenteral (including intravenous, intratumoral, intraperitoneal, intramuscular, intracavity, and subcutaneous) administration. In some embodiments, the targeted lipid particle is administered alone or formulated as a pharmaceutical composition. In some embodiments, the targeted lipid particle or compositions described herein are administered to a subject, e.g., a mammal, e.g., a human. In some of any embodiments, the subject is at risk of, has a symptom of, or is diagnosed with or identified as having, a particular disease or condition (e.g., a disease or condition described herein). In some embodiments, the disease is a disease or disorder. In some embodiments, the disease is a B cell malignancy.

[0266] In some embodiments, the targeted lipid particles are administered in the form of a unit-dose composition, such as a unit dose oral, parenteral, transdermal, or inhaled composition. In some embodiments, the compositions are prepared by admixture and are adapted for oral, inhaled, transdermal, or parenteral administration, and as such may be in the form of tablets, capsules, oral liquid preparations, powders, granules, lozenges, reconstitutable powders, injectable, and infusible solutions or suspensions, or suppositories or aerosols.

[0267] In some embodiments, the regimen of administration affects what constitutes an effective amount. In some embodiments, the therapeutic formulations are administered to the subject either prior to or after a diagnosis of disease. In some embodiments, several divided dosages, as well as staggered dosages are administered daily or sequentially, or the dose is continuously infused, or is a bolus injection. In some embodiments, the dosages of the therapeutic formulations are proportionally increased or decreased as indicated by the exigencies of the therapeutic or prophylactic situation.

[0268] In some embodiments, the administration of the compositions of the present disclosure to a subject, preferably a mammal, more preferably a human, are carried out using known procedures, at dosages and for periods of time effective to prevent or treat disease. In some embodiments, an effective amount of the targeted lipid particle of the disclosure necessary to achieve a therapeutic effect varies according to factors such as the activity of the particular lipid particle employed; the time of administration; the rate of excretion; the duration of the treatment; other drugs, compounds or materials used in combination with the targeted lipid particle of the disclosure; the state of the disease or disorder, age, sex, weight, condition, general health and prior medical history of the subject being treated, and like factors well-known in the medical arts. In some embodiments, the dosage regimens are adjusted to provide the optimum therapeutic response. In some embodiments, several divided doses are administered daily or the dose is proportionally reduced as indicated by the exigencies of the therapeutic situation. One of ordinary skill in the art would be able to study the relevant factors and make the determination regarding the effective amount of the therapeutic targeted lipid particle of the disclosure without undue experimentation.

[0269] In some embodiments, dosage levels of the targeted lipid particles in the pharmaceutical compositions of this disclosure are varied so as to obtain an amount that is effective to achieve the desired therapeutic response for a particular subject, composition, and mode of administration, without being toxic to the subject.

[0270] A medical doctor, e.g., physician or veterinarian, having ordinary skill in the art may readily determine and prescribe the effective amount of the pharmaceutical composition required. In some embodiments, the physician or veterinarian could start doses of the targeted lipid particles of the disclosure employed in the pharmaceutical composition at levels lower than that required in order to achieve the desired therapeutic effect and gradually increase the dosage until the desired effect is achieved.

[0271] In some embodiments, the term "container" includes any receptacle for holding the pharmaceutical composition. In some embodiments, the container is the packaging that contains the pharmaceutical composition. In other embodiments, the container is not the packaging that contains the pharmaceutical composition, i.e., the container is a receptacle, such as a box or vial that contains the packaged pharmaceutical composition or unpackaged pharmaceutical composition and the instructions for use of the pharmaceutical composition. It should be understood that the instructions for use of the pharmaceutical composition may be contained on the packaging containing the pharmaceutical composition, and as such the instructions form an increased functional relationship to the packaged product. In some embodiments, instructions contain information pertaining to the pharmaceutical composition's ability to perform its intended function, e.g., treating or preventing a disease in a subject, or delivering an imaging or diagnostic agent to a subject.

[0272] In some embodiments, routes of administration of any of the compositions disclosed herein include oral, nasal, rectal, parenteral, sublingual, transdermal, transmucosal (e.g., sublingual, lingual, (trans) buccal, (trans) urethral, vaginal (e.g., trans- and perivaginally), (intra) nasal, and (trans) rectal), intravesical, intrapulmonary, intraduodenal, intragastrical, intrathecal, subcutaneous, intramuscular, intradermal, intra-arterial, intravenous, intrabronchial, inhalation, and topical administration.

[0273] In some of any embodiments, suitable compositions and dosage forms include, for example, tablets, capsules, caplets, pills, gel caps, troches, dispersions, suspensions, solutions, syrups, granules, beads, transdermal patches, gels, powders, pellets, magmas, lozenges, creams, pastes, plasters, lotions, discs, suppositories, liquid sprays for nasal or oral administration, dry powder or aerosolized formulations for inhalation, compositions and formulations for intravesical administration, and the like.

[0274] In some embodiments, the targeted lipid particle composition comprising an exogenous agent or cargo, are used to deliver such exogenous agent or cargo to a cell tissue or subject. In some embodiments, delivery of a cargo by administration of a targeted lipid particle composition described herein modify cellular protein expression levels. In certain embodiments, the administered composition directs upregulation (via expression in the cell, delivery in the cell, or induction within the cell) of one or more cargo (e.g., a polypeptide or mRNA) that provide a functional activity which is substantially absent or reduced in the cell in which the polypeptide is delivered. In some embodiments, the missing functional activity is enzymatic, structural, or regulatory in nature. In some embodiments, the administered composition directs up-regulation of one or more proteins that increases (e.g., synergistically) a functional activity which is present but substantially deficient in the cell in which the protein is upregulated. In some of any embodiments disclosed herein, the administered composition directs downregulation of (via expression in the cell, delivery in the cell, or induction within the cell) one or more cargo (e.g., a protein, siRNA, or miRNA) that repress a functional activity which is present or upregulated in the cell in which the protein, siRNA, or miRNA is delivered. In some embodiments, the upregulated functional activity is enzymatic, structural, or regulatory in nature. In some embodiments, the administered composition directs downregulation of one or more proteins that decreases (e.g., synergistically) a functional activity which is present or upregulated in the cell in which the protein is downregulated. In some embodiments, the administered composition directs upregulation of certain functional activities and downregulation of other functional activities.

[0275] In some of any embodiments, the targeted lipid particle composition (e.g., one comprising mitochondria or DNA) mediates an effect on a target cell, and the effect lasts for at least 1, 2, 3, 4, 5, 6, or 7 days, 2, 3, or 4 weeks, or 1, 2, 3, 6, or 12 months. In some embodiments (e.g., wherein the targeted viral vector composition comprises an exogenous protein), the effect lasts for less than 1, 2, 3, 4, 5, 6, or 7 days, 2, 3, or 4 weeks, or 1, 2, 3, 6, or 12 months.

[0276] In some of any embodiments, the targeted lipid particle composition described herein is delivered ex-vivo to a cell or tissue, e.g., a human cell or tissue. In embodiments, the composition improves function of a cell or tissue ex-vivo, e.g., improves cell viability, respiration, or other function (e.g., another function described herein).

[0277] In some embodiments, the composition is delivered to an ex vivo tissue that is in an injured state (e.g., from trauma, disease, hypoxia, ischemia or other damage).

[0278] In some embodiments, the composition is delivered to an ex-vivo transplant (e.g., a tissue explant or tissue for transplantation, e.g., a human vein, a musculoskeletal graft such as bone or tendon, cornea, skin, heart valves, nerves; or an isolated or cultured organ, e.g., an organ to be transplanted into a human, e.g., a human heart, liver, lung, kidney, pancreas, intestine, thymus, eye). In some embodiments, the composition is delivered to the tissue or organ before, during and/or after transplantation.

[0279] In some embodiments, the composition is delivered, administered, or contacted with a cell, e.g., a cell preparation. In some embodiments, the cell preparation is a cell therapy preparation (a cell preparation intended for administration to a human subject). In embodiments, the cell preparation comprises cells expressing a T-cell receptor (TCR) or chimeric antigen receptor (CAR), e.g., expressing a recombinant CAR. The cells expressing the CAR may be, e.g., T cells, Natural Killer (NK) cells, cytotoxic T lymphocytes (CTL), regulatory T cells. In embodiments, the cell preparation is a neural stem cell preparation. In embodiments, the cell preparation is a mesenchymal stem cell (MSC) preparation. In embodiments, the cell preparation is a hematopoietic stem cell (HSC) preparation. In embodiments, the cell preparation is an islet cell preparation.

[0280] In some embodiments, the viral vector comprising an anti-CD4 sdAb or scFv composition described herein is used to deliver a CAR or TCR. In some embodiments, the viral vector transduces a cell expressing CD4 (e.g., a CD4+ T cell) and expresses and amplifies the CAR or TCR. The amplified CAR or TCR T cells then mediate targeted cell killing. Thus, the disclosure includes the use of viral vector comprising an anti-CD4 scFv fusogen construct to elicit an immune response specific to the antigen binding moiety of the CAR or TCR. In some embodiments, the CAR is used to target a tumor antigen selected from CD19, CD20, CD22, or BCMA. In another embodiment, the CAR is engineered to comprise an intracellular signaling domain of the T cell antigen receptor complex zeta chain (e.g., CD3 zeta). In a preferred embodiment, the intracellular domain is selected from a CD137 (4-1BB) signaling domain, a CD28 signaling domain, and a CD3zeta signaling domain.

Engineered Receptor Payloads

[0281] In some embodiments, the targeted lipid particles (e.g. targeted viral vectors) disclosed herein encode an engineered receptor. In some embodiments, the cells for use in or administered in connection with the provided methods contain or are engineered to contain an engineered receptor, e.g., an engineered antigen receptor, such as a chimeric antigen receptor (CAR). Also provided are populations of such cells, compositions containing such cells and/or enriched for such cells, such as in which cells of a certain type such as T cells or CD4+ cells are enriched or selected. Among the compositions are pharmaceutical compositions and formulations for administration, such as for adoptive cell therapy. Also provided are therapeutic methods for administering the cells and compositions to subjects, e.g., patients, in accord with the provided methods, and/or with the provided articles of manufacture or compositions.

[0282] In some embodiments, gene transfer is accomplished without first stimulating the cells, such as by combining it with a stimulus that induces a response such as proliferation, survival, and/or activation, e.g., as measured by expression of a cytokine or activation marker, followed by introduction of the nucleic acids, e.g., by transduction, into the stimulated cells, and optionally incubation or expansion in culture to numbers sufficient for clinical applications.

[0283] The viral vectors may express recombinant receptors, such as antigen receptors including chimeric antigen receptors (CARs), and other antigen-binding receptors such as transgenic T cell receptors (TCRs). Also among the receptors are other chimeric receptors.

a. Chimeric Antigen Receptors (CARs)

[0284] In some embodiments of the provided methods and uses, chimeric receptors, such as a CARs, contain one or more domains that combine an antigen- or ligand-binding domain (e.g. antibody or antibody fragment) that provides specificity for a desired antigen (e.g., tumor antigen) with intracellular signaling domains. In some embodiments, the intracellular signaling domain is a stimulating or an activating intracellular domain portion, such as a T cell stimulating or activating domain, providing a primary activation signal or a primary signal. In some embodiments, the intracellular signaling domain contains or additionally contains a costimulatory signaling domain to facilitate effector functions. In some embodiments, chimeric receptors when genetically engineered into immune cells modulate T cell activity, and, in some cases, modulate T cell differentiation or homeostasis, thereby resulting in genetically engineered cells with improved longevity, survival and/or persistence in vivo, such as for use in adoptive cell therapy methods.

[0285] Exemplary antigen receptors, including CARs, and methods for engineering and introducing such receptors into cells, include those described, for example, in W0200014257, WO2013126726, WO2012/129514, WO2014031687, WO2013/166321, WO2013/071154, WO2013/123061, U.S. patent app. Pub. Nos. US2002131960, US2013287748, US20130149337, U.S. Pat. Nos. 6,451,995, 7,446,190, 8,252,592, 8,339,645, 8,398,282, 7,446,179, 6,410,319, 7,070,995, 7,265,209, 7,354,762, 7,446,191, 8,324,353, and 8,479,118, and European patent app. No. EP2537416, and/or those described by Sadelain et al., Cancer Discov. 2013 April; 3(4): 388-398; Davila et al. (2013) PLOS ONE 8(4):e61338; Turtle et al., Curr. Opin. Immunol., 2012 October; 24(5): 633-39; Wu et al., Cancer, 2012 Mar. 18(2): 160-75. In some aspects, the antigen receptors include a CAR as described in U.S. Pat. No. 7,446,190, and those described in WO/2014055668. Examples of the CARs include CARs as disclosed in any of the aforementioned publications, such as WO2014031687, U.S. Pat. Nos. 8,339,645, 7,446,179, US 2013/0149337, U.S. Pat. Nos. 7,446,190, 8,389,282, Kochenderfer et al., (2013) Nature Reviews Clinical Oncology, 10, 267-276; Wang et al. (2012) J. Immunother. 35(9): 689-701; and Brentjens et al., Sci Transl Med. 2013 5(177). See also WO2014031687, U.S. Pat. Nos. 8,339,645, 7,446,179, US 2013/0149337, U.S. Pat. Nos. 7,446,190, and 8,389,282. The recombinant receptors, such as CARs, generally include an extracellular antigen binding domain, such as a portion of an antibody molecule, generally a variable heavy (VH) chain region and/or variable light (VL) chain region of the antibody, e.g., an scFv antibody fragment. In some embodiments, the antigen binding domain of the CAR molecule comprises an antibody, an antibody fragment, an scFv, a Fv, a Fab, a (Fab')2, a single domain antibody (SdAb), a VH or VL domain, or a camelid VHH domain.

[0286] In some embodiments, the antigen targeted by the receptor is a polypeptide. In some embodiments, it is a carbohydrate or other molecule. In some embodiments, the antigen is selectively expressed or overexpressed on cells of the disease or condition, e.g., the tumor or pathogenic cells, as compared to normal or non-targeted cells or tissues. In other embodiments, the antigen is expressed on normal cells and/or is expressed on the engineered cells.

[0287] In some embodiments, the antigen targeted by the receptor includes antigens associated with a B cell malignancy, such as any of a number of known B cell marker. In some embodiments, the antigen targeted by the receptor is CD20, CD19, CD22, ROR1, CD45, CD47, CD21, CD5, CD33, Igkappa, Iglambda, CD79a, CD79b or CD30.

[0288] In some embodiments, the chimeric antigen receptor includes an extracellular portion containing an antibody or antibody fragment. In some aspects, the chimeric antigen receptor includes an extracellular portion containing the antibody or fragment and an intracellular signaling domain. In some embodiments, the antibody or fragment includes an scFv.

[0289] In some embodiments, the antigen targeted by the antigen-binding domain is CD19. In some aspects, the antigen-binding domain of the recombinant receptor, e.g., CAR, and the antigen-binding domain binds, such as specifically binds or specifically recognizes, a CD19, such as a human CD19. In some embodiments, the scFv contains a VH and a VL derived from an antibody or an antibody fragment specific to CD19. In some embodiments, the antibody or antibody fragment that binds CD19 is a mouse derived antibody such as FMC63 and SJ25C1. In some embodiments, the antibody or antibody fragment is a human antibody, e.g., as described in U.S. Patent Publication No. US 2016/0152723.

[0290] In some embodiments, the antigen is CD19. In some embodiments, the scFv contains a VH and a VL derived from an antibody or an antibody fragment specific to CD19. In some embodiments, the antibody or antibody fragment that binds CD19 is a mouse derived antibody such as FMC63 and SJ25C1. In some embodiments, the antibody or antibody fragment is a human antibody, e.g., as described in U.S. Patent Publication No. US 2016/0152723.

[0291] In some embodiments, the scFv is derived from FMC63. FMC63 generally refers to a mouse monoclonal IgGI antibody raised against Nalm-1 and -16 cells expressing CD19 of human origin (Fing, N. R., et al. (1987). Leucocyte typing III. 302).

[0292] In some embodiments, the antigen targeted by the antigen-binding domain is BCMA. In some aspects, the antigen-binding domain of the recombinant receptor, e.g., CAR, and the antigen-binding domain binds, such as specifically binds or specifically recognizes, a BCMA, such as a human BCMA. In some embodiments, the antigen-binding domain is a fully human VH sdAb disclosed in US2020/0138865 (disclosed herein by reference in its entirety), e.g., FHVH74, FHVH32, FHVH33, or FHVH93.

Antigen Binding Domain (ABD) Targets an Antigen Characteristic of a Neoplastic or Cancer Cell

[0293] In some embodiments, the antigen binding domain (ABD) targets an antigen characteristic of a neoplastic cell. In other words, the antigen binding domain targets an antigen expressed by a neoplastic or cancer cell. In some embodiments, the ABD binds a tumor associated antigen. In some embodiments, the antigen characteristic of a neoplastic cell (e.g., antigen associated with a neoplastic or cancer cell) or a tumor associated antigen is selected from a cell surface receptor, an ion channel-linked receptor, an enzyme-linked receptor, a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, histidine kinase associated receptor, epidermal growth factor receptors (EGFR) (including ErbB1/EGFR, ErbB2/HER2, ErbB3/HER3, and ErbB4/

HER4), fibroblast growth factor receptors (FGFR) (including FGF1, FGF2, FGF3, FGF4, FGF5, FGF6, FGF7, FGF18, and FGF21), vascular endothelial growth factor receptors (VEGFR) (including VEGF-A, VEGF-B, VEGF-C, VEGF-D, and PIGF), RET Receptor and the Eph Receptor Family (including EphA1, EphA2, EphA3, EphA4, EphA5, EphA6, EphA7, EphA8, EphA9, EphA10, EphB1, EphB2. EphB3, EphB4, and EphB6), CXCR1, CXCR2, CXCR3, CXCR4, CXCR6, CCR1, CCR2, CCR3, CCR4, CCR5, CCR6, CCR8, CFTR, CIC-1, CIC-2, CIC-4, CIC-5, CIC-7, CIC-Ka, CIC-Kb, Bestrophins, TMEM16A, GABA receptor, glycin receptor, ABC transporters, NAV1.1, NAV1.2, NAV1.3, NAV1.4, NAV1.5, NAV1.6, NAV1.7, NAV1.8, NAV1.9, sphingosin-1-phosphate receptor (S1P1R), NMDA channel, transmembrane protein, multispan transmembrane protein, T-cell receptor motifs, T-cell alpha chains, T-cell β chains, T-cell γ chains, T-cell δ chains, CCR7, CD3, CD4, CD5, CD7, CD8, CD11b, CD11c, CD16, CD19, CD20, CD21, CD22, CD25, CD28, CD34, CD35, CD40, CD45RA, CD45RO, CD52, CD56, CD62L, CD68, CD80, CD95, CD117, CD127, CD133, CD137 (4-1BB), CD163, F4/80, IL-4Ra, Sca-1, CTLA-4, GITR, GARP, LAP, granzyme B, LFA-1, transferrin receptor, NKp46, perforin, CD4+, Th1, Th2, Th17, Th40, Th22, Th9, Tfh, canonical Treg. FoxP3+, Tr1, Th3, Treg17, TREG; CDCP, NT5E, EpCAM, CEA, gpA33, mucins, TAG-72, carbonic anhydrase IX, PSMA, folate binding protein, gangliosides (e.g., CD2, CD3, GM2), Lewis-γ^2, VEGF, VEGFR 1/2/3, αVβ3, α5β1, ErbB1/EGFR, ErbB1/HER2, ErB3, c-MET, IGF1R, EphA3, TRAIL-R1, TRAIL-R2, RANKL, FAP, Tenascin, PDL-1, BAFF, HDAC, ABL, FLT3, KIT, MET, RET, IL-1β, ALK, RANKL, mTOR, CTLA-4, IL-6, IL-6R, JAK3, BRAF, PTCH, Smoothened, PIGF, ANPEP, TIMP1, PLAUR, PTPRJ, LTBR, ANTXR1, folate receptor alpha (FRa), ERBB2 (Her2/neu), EphA2, IL-13Ra2, epidermal growth factor receptor (EGFR), mesothelin, TSHR, CD19, CD123, CD22, CD30, CD171, CS-1, CLL-1, CD33, EGFRvIII, GD2, GD3, BCMA, MUC16 (CA125), L1CAM, LeY, MSLN, IL13Rα1, L1-CAM, Tn Ag, prostate specific membrane antigen (PSMA), ROR1, FLT3, FAP, TAG72, CD38, CD44v6, CEA, EPCAM, B7H3, KIT, interleukin-11 receptor a (IL-11Ra), PSCA, PRSS21, VEGFR2, LewisY, CD24, platelet-derived growth factor receptor-beta (PDGFR-beta), SSEA-4, CD20, MUC1, NCAM, Prostase, PAP, ELF2M, Ephrin B2, IGF-1 receptor, CAIX, LMP2, gpl00, bcr-abl, tyrosinase, Fucosyl GM1, sLe, GM3, TGS5, HMWMAA, o-acetyl-GD2, folate receptor beta, TEM1/CD248, TEM7R, CLDN6, GPRC5D, CXORF61, CD97, CD179a, ALK, Polysialic acid, PLACI, GloboH, NY-BR-1, UPK2, HAVCR1, ADRB3, PANX3, GPR20, LY6K, OR51E2, TARP, WT1, NY-ESO-1, LAGE-la, MAGE-A1, legumain, HPV E6, E7, ETV6-AML, sperm protein 17, XAGE1, Tie 2, MAD-CT-1, MAD-CT-2, major histocompatibility complex class I-related gene protein (MR1), urokinase-type plasminogen activator receptor (uPAR), Fos-related antigen 1, p53, p53 mutant, prostein, survivin, telomerase, PCTA-1/Galectin 8, MelanA/MART1, Ras mutant, hTERT, sarcoma translocation breakpoints, ML-IAP, ERG (TMPRSS2 ETS fusion gene), NA17, PAX3, androgen receptor, cyclin B1, MYCN, RhoC, TRP-2, CYPIB I, BORIS, SART3, PAX5, OY-TES1, LCK, AKAP-4, SSX2, RAGE-1, human telomerase reverse transcriptase, RU1, RU2, intestinal carboxyl esterase, mut hsp70-2, CD79a, CD79b, CD72, LAIR1, FCAR, LILRA2, CD300LF, CLEC12A, BST2, EMR2, LY75, GPC3, FCRL5, IGLL1, a neoantigen, CD133, CD15, CD184, CD24, CD56, CD26, CD29, CD44, HLA-A, HLA-B, HLA-C, (HLA-A, B,C) CD49f, CD151 CD340, CD200, tkrA, trkB, or trkC, or an antigenic fragment or antigenic portion thereof.

ABD Targets an Antigen Characteristic of a T Cell

[0294] In some embodiments, the antigen binding domain targets an antigen characteristic of a T cell. In some embodiments, the ABD binds an antigen associated with a T cell. In some instances, such an antigen is expressed by a T cell or is located on the surface of a T cell. In some embodiments, the antigen characteristic of a T cell or the T cell associated antigen is selected from a cell surface receptor, a membrane transport protein (e.g., an active or passive transport protein such as, for example, an ion channel protein, a pore-forming protein, etc.), a transmembrane receptor, a membrane enzyme, and/or a cell adhesion protein characteristic of a T cell. In some embodiments, an antigen characteristic of a T cell is a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, histidine kinase associated receptor, AKT1; AKT2; AKT3; ATF2; BCL10; CALM1; CD3D (CD30); CD3E (CD38); CD3G (CD3γ); CD4; CD8; CD28; CD45; CD80 (B7-1); CD86 (B7-2); CD247 (CD32); CTLA-4 (CD152); ELK1; ERK1 (MAPK3); ERK2; FOS; FYN; GRAP2 (GADS); GRB2; HLA-DRA; HLA-DRB1; HLA-DRB3; HLA-DRB4; HLA-DRB5; HRAS; IKBKA (CHUK); IKBKB; IKBKE; IKBKG (NEMO); IL2; ITPR1; ITK; JUN; KRAS2; LAT; LCK; MAP2K1 (MEK1); MAP2K2 (MEK2); MAP2K3 (MKK3); MAP2K4 (MKK4); MAP2K6 (MKK6); MAP2K7 (MKK7); MAP3K1 (MEKK1); MAP3K3; MAP3K4; MAP3K5; MAP3K8; MAP3K14 (NIK); MAPK8 (JNK1); MAPK9 (JNK2); MAPK10 (JNK3); MAPK11 (p383); MAPK12 (p38γ); MAPK13 (p380); MAPK14 (p38a); NCK; NFAT1; NFAT2; NFKB1; NFKB2; NFKBIA; NRAS; PAK1; PAK2; PAK3; PAK4; PIK3C2B; PIK3C3 (VPS34); PIK3CA; PIK3CB; PIK3CD; PIK3R1; PKCA; PKCB; PKCM; PKCQ; PLCY1; PRF1 (Perforin); PTEN; RAC1; RAF1; RELA; SDF1; SHP2; SLP76; SOS; SRC; TBK1; TCRA; TEC; TRAF6; VAV1; VAV2; or ZAP70.

ABD Targets an Antigen Characteristic of an Autoimmune or Inflammatory Disorder

[0295] In some embodiments, the antigen binding domain targets an antigen characteristic of an autoimmune or inflammatory disorder. In some embodiments, the ABD binds an antigen associated with an autoimmune or inflammatory disorder. In some instances, the antigen is expressed by a cell associated with an autoimmune or inflammatory disorder. In some embodiments, the autoimmune or inflammatory disorder is selected from chronic graft-vs-host disease (GVHD), lupus, arthritis, immune complex glomerulonephritis, goodpasture syndrome, uveitis, hepatitis, systemic sclerosis or scleroderma, type I diabetes, multiple sclerosis, cold agglutinin disease, Pemphigus vulgaris, Grave's disease, autoimmune hemolytic anemia, Hemophilia A, Primary Sjogren's Syndrome, thrombotic thrombocytopenia purrpura, neuromyelits optica, Evan's syndrome, IgM mediated neuropathy, cryoglobulinemia, dermatomyositis, idiopathic thrombocytopenia, ankylosing spondylitis, bullous pemphigoid, acquired angioedema, chronic urticarial, antiphospholipid demyelinating polyneuropathy, and autoimmune thrombocytopenia or neutropenia or pure red cell aplasias, while exemplary non-limiting examples of alloimmune diseases include allosensitization (see, for example, Blazar et al., 2015, Am. J. Transplant, 15(4):931-41) or xenosensitization from hematopoietic or solid organ transplantation, blood transfusions, pregnancy with fetal allosensitization, neonatal alloimmune thrombocytopenia, hemolytic disease of the newborn, sensitization to foreign antigens such as can occur with replacement of inherited or acquired deficiency disorders treated with enzyme or protein replacement therapy, blood products, and gene therapy. In some embodiments, the antigen characteristic of an autoimmune or inflammatory disorder is selected from a cell surface receptor, an ion channel-linked receptor, an enzyme-linked receptor, a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, or histidine kinase associated receptor.

[0296] In some embodiments, an antigen binding domain of a CAR binds to a ligand expressed on B cells, plasma cells, or plasmablasts. In some embodiments, an antigen binding domain of a CAR binds to CD10, CD19, CD20, CD22, CD24, CD27, CD38, CD45R, CD138, CD319, BCMA, CD28, TNF, interferon receptors, GM-CSF, ZAP-70, LFA-1, CD3 gamma, CD5 or CD2. See, e.g., US 2003/0077249; WO 2017/058753; WO 2017/058850, the contents of which are herein incorporated by reference.

ABD Targets an Antigen Characteristic of Senescent Cells

[0297] In some embodiments, the antigen binding domain targets an antigen characteristic of senescent cells, e.g., urokinase-type plasminogen activator receptor (uPAR). In some embodiments, the ABD binds an antigen associated with a senescent cell. In some instances, the antigen is expressed by a senescent cell. In some embodiments, the CAR is used for treatment or prophylaxis of disorders characterized by the aberrant accumulation of senescent cells, e.g., liver and lung fibrosis, atherosclerosis, diabetes and osteoarthritis.

ABD Targets an Antigen Characteristic of an Infectious Disease

[0298] In some embodiments, the antigen binding domain targets an antigen characteristic of an infectious disease. In some embodiments, the ABD binds an antigen associated with an infectious disease. In some instances, the antigen is expressed by a cell affected by an infectious disease. In some embodiments, wherein the infectious disease is selected from HIV, hepatitis B virus, hepatitis C virus, Human herpes virus, Human herpes virus 8 (HHV-8, Kaposi sarcoma-associated herpes virus (KSHV)), Human T-lymphotrophic virus-1 (HTLV-1), Merkel cell polyomavirus (MCV), Simian virus 40 (SV40), Epstein-Barr virus, CMV, human papillomavirus. In some embodiments, the antigen characteristic of an infectious disease is selected from a cell surface receptor, an ion channel-linked receptor, an enzyme-linked receptor, a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, histidine kinase associated receptor, HIV Env, gpl20, or CD4-induced epitope on HIV-1 Env.

ABD Binds to a Cell Surface Antigen of a Cell

[0299] In some embodiments, an antigen binding domain binds to a cell surface antigen of a cell. In some embodiments, a cell surface antigen is characteristic of (e.g., expressed by) a particular or specific cell type. In some embodiments, a cell surface antigen is characteristic of more than one type of cell.

[0300] In some embodiments, a CAR antigen binding domain binds a cell surface antigen characteristic of a T cell, such as a cell surface antigen on a T cell. In some embodiments, an antigen characteristic of a T cell is a cell surface receptor, a membrane transport protein (e.g., an active or passive transport protein such as, for example, an ion channel protein, a pore-forming protein, etc.), a transmembrane receptor, a membrane enzyme, and/or a cell adhesion protein characteristic of a T cell. In some embodiments, an antigen characteristic of a T cell is a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, or histidine kinase associated receptor.

[0301] In some embodiments, an antigen binding domain of a CAR binds a T cell receptor. In some embodiments, a T cell receptor is AKT1; AKT2; AKT3; ATF2; BCL10; CALM1; CD3D (CD30); CD3E (CD38); CD3G (CD3γ); CD4; CD8; CD28; CD45; CD80 (B7-1); CD86 (B7-2); CD247 (CD3ζ); CTLA-4 (CD152); ELK1; ERK1 (MAPK3); ERK2; FOS; FYN; GRAP2 (GADS); GRB2; HLA-DRA; HLA-DRB1; HLA-DRB3; HLA-DRB4; HLA-DRB5; HRAS; IKBKA (CHUK); IKBKB; IKBKE; IKBKG (NEMO); IL2; ITPR1; ITK; JUN; KRAS2; LAT; LCK; MAP2K1 (MEK1); MAP2K2 (MEK2); MAP2K3 (MKK3); MAP2K4 (MKK4); MAP2K6 (MKK6); MAP2K7 (MKK7); MAP3K1 (MEKK1); MAP3K3; MAP3K4; MAP3K5; MAP3K8; MAP3K14 (NIK); MAPK8 (JNK1); MAPK9 (JNK2); MAPK10 (JNK3); MAPK11 (p38B); MAPK12 (p38γ); MAPK13 (p380); MAPK14 (p38a); NCK; NFAT1; NFAT2; NFKB1; NFKB2; NFKBIA; NRAS; PAK1; PAK2; PAK3; PAK4; PIK3C2B; PIK3C3 (VPS34); PIK3CA; PIK3CB; PIK3CD; PIK3R1; PKCA; PKCB; PKCM; PKCQ; PLCY1; PRF1 (Perforin); PTEN; RAC1; RAF1; RELA; SDF1; SHP2; SLP76; SOS; SRC; TBK1; TCRA; TEC; TRAF6; VAV1; VAV2; or ZAP70.

Transmembrane Domain

[0302] In some embodiments, the CAR transmembrane domain comprises at least a transmembrane region of the alpha, beta or zeta chain of a T cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, or functional variant thereof. In some embodiments, the transmembrane domain comprises at least a transmembrane region(s) of CD4, 4-1BB/CD137, CD28, CD34, CD4, FcεRIγ, CD16, OX40/CD134, CD3ζ, CD3ε, CD3γ, CD3δ, TCRα, TCRβ, TCRζ, CD32, CD64, CD64, CD45, CD5, CD9, CD22, CD37, CD80, CD86, CD40, CD40L/CD154, VEGFR2, FAS, and FGFR2B, or functional variant thereof. antigen binding domain binds

Signaling Domain or Plurality of Signaling Domains

[0303] In some embodiments, a CAR described herein comprises one or at least one signaling domain selected from one or more of B7-1/CD80; B7-2/CD86; B7-H1/PD-L1;

B7-H2; B7-H3; B7-H4; B7-H6; B7-H7; BTLA/CD272; CD28; CTLA-4; Gi24/VISTA/B7-H5; ICOS/CD278; PD-1; PD-L2/B7-DC; PDCD6); 4-1BB/TNFSF9/CD137; 4-1BB Ligand/TNFSF9; BAFF/BLyS/TNFSF13B; BAFF R/TNFRSF13C; CD27/TNFRSF7; CD27 Ligand/TNFSF7; CD30/TNFRSF8; CD30 Ligand/TNFSF8; CD40/TNFRSF5; CD40/TNFSF5; CD40 Ligand/TNFSF5; DR3/TNFRSF25; GITR/TNFRSF18; GITR Ligand/TNFSF18; HVEM/TNFRSF14; LIGHT/TNFSF14; Lymphotoxin-alpha/TNF-beta; OX40/TNFRSF4; OX40 Ligand/TNFSF4; RELT/TNFRSF19L; TACI/TNFRSF13B; TL1A/TNFSF15; TNF-alpha; TNF RII/TNFRSF1B); 2B4/CD244/SLAMF4; BLAME/SLAMF8; CD2; CD2F-10/SLAMF9; CD48/SLAMF2; CD58/LFA-3; CD84/SLAMF5; CD229/SLAMF3; CRACC/SLAMF7; NTB-A/SLAMF6; SLAM/CD150); CD2; CD7; CD53; CD82/Kai-1; CD90/Thyl; CD96; CD160; CD200; CD300a/LMIR1; HLA Class I; HLA-DR; Ikaros; Integrin alpha 4/CD49d; Integrin alpha 4 beta 1; Integrin alpha 4 beta 7/LPAM-1; LAG-3; TCL1A; TCL1B; CRTAM; DAP12; Dectin-1/CLEC7A; DPPIV/CD26; EphB6; TIM-1/KIM-1/HAVCR; TIM-4; TSLP; TSLP R; lymphocyte function associated antigen-1 (LFA-1); NKG2C, a CD3 zeta domain, an immunoreceptor tyrosine-based activation motif (ITAM), CD27, CD28, 4-1BB, CD134/OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, a ligand that specifically binds with CD83, or functional fragment thereof.

[0304] In some embodiments, the at least one signaling domain comprises a CD3 zeta domain or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof. In other embodiments, the at least one signaling domain comprises (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; and (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof. In yet other embodiments, the at least one signaling domain comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; and (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof. In some embodiments, the at least one signaling domain comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof; and (iv) a cytokine or costimulatory ligand transgene.

[0305] In some embodiments, the at least two signaling domains comprise a CD3 zeta domain or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof. In other embodiments, the at least two signaling domains comprise (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; and (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof. In yet other embodiments, the at least one signaling domain comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; and (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof. In some embodiments, the at least two signaling domains comprise a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof; and (iv) a cytokine or costimulatory ligand transgene.

[0306] In some embodiments, the at least three signaling domains comprise a CD3 zeta domain or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof. In other embodiments, the at least three signaling domains comprise (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; and (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof. In yet other embodiments, the least three signaling domains comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; and (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof. In some embodiments, the at least three signaling domains comprise a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof; and (iv) a cytokine or costimulatory ligand transgene.

[0307] In some embodiments, the CAR comprises a CD3 zeta domain or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof. In some embodiments, the CAR comprises (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; and (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof.

[0308] In some embodiments, the CAR comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; and (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof.

[0309] In some embodiments, the CAR comprises (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof, and/or (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof.

[0310] In some embodiments, the CAR comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof; and (iv) a cytokine or costimulatory ligand transgene.

Domain which Upon Successful Signaling of the CAR Induces Expression of a Cytokine Gene

[0311] In some embodiments, a first, second, third, or fourth generation CAR further comprises a domain which upon successful signaling of the CAR induces expression of a cytokine gene. In some embodiments, a cytokine gene is endogenous or exogenous to a target cell comprising a CAR which comprises a domain which upon successful signaling of the CAR induces expression of a cytokine gene. In some embodiments, a cytokine gene encodes a pro-inflammatory cytokine. In some embodiments, a cytokine gene encodes IL-1, IL-2, IL-9, IL-12, IL-18, TNF, or IFN-gamma, or functional fragment thereof. In some embodiments, a domain which upon successful signaling of the CAR induces expression of a cytokine gene is or comprises a transcription factor or functional domain or fragment thereof. In some embodiments, a domain which upon successful signaling of the CAR induces expression of a cytokine gene is or comprises a transcription factor or functional domain or fragment thereof. In some embodiments, a transcription factor or functional domain or fragment thereof is or comprises a nuclear factor of activated T cells (NFAT), an NF-kB, or functional domain or fragment thereof. See, e.g., Zhang. C. et al., Engineering CAR-T cells. Biomarker Research. 5:22 (2017); WO 2016126608; Sha, H. et al. Chimaeric antigen receptor T-cell therapy for tumour immunotherapy. Bioscience Reports Jan. 27, 2017, 37 (1).

[0312] In some embodiments, the CAR further comprises one or more spacers, e.g., wherein the spacer is a first spacer between the antigen binding domain and the transmembrane domain. In some embodiments, the first spacer includes at least a portion of an immunoglobulin constant region or variant or modified version thereof. In some embodiments, the spacer is a second spacer between the transmembrane domain and a signaling domain. In some embodiments, the second spacer is an oligopeptide, e.g., wherein the oligopeptide comprises glycine and serine residues such as but not limited to glycine-serine doublets. In some embodiments, the CAR comprises two or more spacers, e.g., a spacer between the antigen binding domain and the transmembrane domain and a spacer between the transmembrane domain and a signaling domain.

[0313] In some embodiments, any one of the cells described herein comprises a nucleic acid encoding a CAR or a first generation CAR. In some embodiments, a first generation CAR comprises an antigen binding domain, a transmembrane domain, and signaling domain. In some embodiments, a signaling domain mediates downstream signaling during T cell activation.

[0314] In some embodiments, the methods and compositions disclosed herein comprise a nucleic acid encoding a CAR or a second generation CAR. In some embodiments, a second generation CAR comprises an antigen binding domain, a transmembrane domain, and two signaling domains. In some embodiments, a signaling domain mediates downstream signaling during T cell activation. In some embodiments, a signaling domain is a costimulatory domain. In some embodiments, a costimulatory domain enhances cytokine production, CAR-T cell proliferation, and/or CAR-T cell persistence during T cell activation.

[0315] In some embodiments, any one of the compositions and methods described herein comprises a nucleic acid encoding a CAR or a third generation CAR. In some embodiments, a third generation CAR comprises an antigen binding domain, a transmembrane domain, and at least three signaling domains. In some embodiments, a signaling domain mediates downstream signaling during T cell activation. In some embodiments, a signaling domain is a costimulatory domain. In some embodiments, a costimulatory domain enhances cytokine production, CAR-T cell proliferation, and or CAR-T cell persistence during T cell activation. In some embodiments, a third generation CAR comprises at least two costimulatory domains. In some embodiments, the at least two costimulatory domains are not the same.

[0316] In some embodiments, any one of the compositions and methods described herein comprises a nucleic acid encoding a CAR or a fourth generation CAR. In some embodiments, a fourth generation CAR comprises an antigen binding domain, a transmembrane domain, and at least two, three, or four signaling domains. In some embodiments, a signaling domain mediates downstream signaling during T cell activation. In some embodiments, a signaling domain is a costimulatory domain. In some embodiments, a costimulatory domain enhances cytokine production, CAR-T cell proliferation, and or CAR-T cell persistence during T cell activation.

Abd Comprising an Antibody or Antigen-Binding Portion Thereof

[0317] In some embodiments, a CAR antigen binding domain is or comprises an antibody or antigen-binding portion thereof. In some embodiments, a CAR antigen binding domain is or comprises an scFv or Fab. In some embodiments, a CAR antigen binding domain comprises an scFv or Fab fragment of a CD19 antibody; CD22 antibody; T-cell alpha chain antibody; T-cell β chain antibody; T-cell γ chain antibody; T-cell δ chain antibody; CCR7 antibody; CD3 antibody; CD4 antibody; CD5 antibody; CD7 antibody; CD8 antibody; CD11b antibody; CD11c antibody; CD16 antibody; CD20 antibody; CD21 antibody; CD25 antibody; CD28 antibody; CD34 antibody; CD35 antibody; CD40 antibody; CD45RA antibody; CD45RO antibody; CD52 antibody; CD56 antibody; CD62L antibody; CD68 antibody; CD80 antibody; CD95 antibody; CD117 antibody; CD127 antibody; CD133 antibody; CD137 (4-1 BB) antibody; CD163 antibody; F4/80 antibody; IL-4Ra antibody; Sca-1 antibody; CTLA-4 antibody; GITR antibody GARP antibody; LAP antibody; granzyme B antibody; LFA-1 antibody; MR1 antibody; uPAR antibody; or transferrin receptor antibody.

[0318] In some embodiments, a CAR comprises a signaling domain which is a costimulatory domain. In some embodiments, a CAR comprises a second costimulatory domain. In some embodiments, a CAR comprises at least two costimulatory domains. In some embodiments, a CAR comprises at least three costimulatory domains. In some embodiments, a CAR comprises a costimulatory domain selected from one or more of CD27, CD28, 4-1BB, CD134/OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, a ligand that specifically binds with CD83. In some embodiments, if a CAR comprises two or more costimulatory domains, two costimulatory domains are different. In some embodiments, if a CAR comprises two or more costimulatory domains, two costimulatory domains are the same.

[0319] In addition to the CARs described herein, various chimeric antigen receptors and nucleotide sequences encoding the same are known in the art and would be suitable for fusosomal delivery and reprogramming of target cells in vivo and in vitro as described herein. See, e.g., WO2013040557; WO2012079000; WO2016030414; Smith T, et al., Nature Nanotechnology. 2017. DOI: 10.1038/NNANO.2017.57, the disclosures of which are herein incorporated by reference.

Additional Descriptions of CARS

[0320] In certain embodiments, the compositions and methods comprise a polynucleotide encoding a CAR. CARs (also known as chimeric immunoreceptors, chimeric T cell receptors, or artificial T cell receptors) are receptor proteins that have been engineered to give host cells (e.g., T cells) the new ability to target a specific protein. The receptors are chimeric because they combine both antigen-binding and T cell activating functions into a single receptor. The polycistronic vector of the present disclosure may be used to express one or more CARs in a host cell (e.g., a T cell) for use in therapies against various target antigens. The CARs expressed by the one or more expression cassettes may be the same or different. In these embodiments, the CAR comprises an extracellular binding domain (also referred to as a "binder") that specifically binds a target antigen, a transmembrane domain, and an intracellular signaling domain. In certain embodiments, the CAR further comprises one or more additional elements, including one or more signal peptides, one or more extracellular hinge domains, and/or one or more intracellular costimulatory domains. Domains may be directly adjacent to one another, or there may be one or more amino acids linking the domains. The nucleotide sequence encoding a CAR may be derived from a mammalian sequence, for example, a mouse sequence, a primate sequence, a human sequence, or combinations thereof. In the cases where the nucleotide sequence encoding a CAR is non-human, the sequence of the CAR may be humanized. The nucleotide sequence encoding a CAR may also be codon-optimized for expression in a mammalian cell, for example, a human cell. In any of these embodiments, the nucleotide sequence encoding a CAR may be at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to any of the nucleotide sequences disclosed herein. The sequence variations may be due to codon-optimalization, humanization, restriction enzyme-based cloning scars, and/or additional amino acid residues linking the functional domains, etc.

[0321] In certain embodiments, the CAR comprises a signal peptide at the N-terminus. Non-limiting examples of signal peptides include CD4 signal peptide, IgK signal peptide, and granulocyte-macrophage colony-stimulating factor receptor subunit alpha (GMCSFR-α, also known as colony stimulating factor 2 receptor subunit alpha (CSF2RA)) signal peptide, and variants thereof, the amino acid sequences of which are provided in Table 3 below.

TABLE 3

Exemplary sequences of signal peptides

SEQ ID NO:	Sequence	Description
14003	MALPVTALLLPLALLLHA ARP	CD8α signal peptide
14004	METDTLLLWVLLLWVPGS TG	IgK signal peptide
14005	MLLLVTSLLLCELPHPAF LLIP	GMCSFR-α (CSF2RA) signal peptide

[0322] In certain embodiments, the extracellular binding domain of the CAR comprises one or more antibodies specific to one target antigen or multiple target antigens. The antibody may be an antibody fragment, for example, an scFv, or a single-domain antibody fragment, for example, a VHH. In certain embodiments, the scFv may comprise a heavy chain variable region (V_H) and a light chain variable region (V_L) of an antibody connected by a linker. The V_H and the V_L may be connected in either order, i.e., V_H-linker-V_L or V_L-linker-V_H. Non-limiting examples of linkers include Whitlow linker, (G4S) n (n can be a positive integer, e.g., 1, 2, 3, 4, 5, 6, etc. (SEQ ID NO: 14127)) linker, and variants thereof. In certain embodiments, the antigen is an antigen that is exclusively or preferentially expressed on tumor cells, or an antigen that is characteristic of an autoimmune or inflammatory disease. Exemplary target antigens include, but are not limited to, CD5, CD19, CD20, CD22, CD23, CD30, CD70, Kappa, Lambda, and B cell maturation agent (BCMA), G-protein coupled receptor family C group 5 member D (GPRC5D) (associated with leukemias); CS1/SLAMF7, CD38, CD138, GPRC5D, TACI, and BCMA (associated with myelomas); GD2, HER2, EGFR, EGFRvIII, B7H3, PSMA, PSCA, CAIX, CD171, CEA, CSPG4, EPHA2, FAP, FRa, IL-13Ra, Mesothelin, MUC1, MUC16, and ROR1 (associated with solid tumors). In any of these embodiments, the extracellular binding domain of the CAR is codon-optimized for expression in a host cell or have variant sequences to increase functions of the extracellular binding domain.

[0323] In certain embodiments, the CAR comprises a hinge domain, also referred to as a spacer. The terms "hinge" and "spacer" may be used interchangeably in the present disclosure. Non-limiting examples of hinge domains include CD4 hinge domain, CD28 hinge domain, IgG4 hinge domain, IgG4 hinge-CH2-CH3 domain, and variants thereof, the amino acid sequences of which are provided in Table 4 below.

TABLE 4

Exemplary sequences of hinge domains

SEQ ID NO:	Sequence	Description
14006	TTTPAPRPPTPAPTI-ASQPLSL RPEACRPAAGGAVHTRGLDFACD	CD8α hinge domain
14007	IEVMYPPPYLDNEKSNGTIIHVK GKHLCP-SPLFPGPSKP	CD28 hinge domain
14013	AAAIEVMYPPPYLDNEKSNGTII HVKGKHLCP-SPLFPGPSKP	CD28 hinge domain
14008	ESKYGPPCPPCP	IgG4 hinge domain
14009	ESKYGPPCPSCP	IgG4 hinge domain
14010	ESKYGPPCPPCPAPEFLGGPSVF LFPPKPKD-TLMISRTPEVTCVV VDVSQEDPEVQFNWY-VDGVEVH NAKTKPREEQFNSTYRVVSVLTV LHQDWLNGKEYKCKVSNKGLPSS IEK-TISKAKGQPREPQVYTLPP -SQEEMTKNQVSLTCLVKGFYPS DIAVEWESNGQPENNYKTTPPVL DSDGSFFLYSRL-TVDKSRWQEG NVFSCSVM-HEALHNHYTQKSLS LSLGK	IgG4 hinge-CH2-CH3 domain

[0324] In certain embodiments, the transmembrane domain of the CAR comprises a transmembrane region of the alpha, beta, or zeta chain of a T cell receptor, CD28, CD38, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, or a functional variant thereof, including the human versions of each of these sequences. In other embodiments, the transmembrane domain comprises a transmembrane region of CD4, 4-1BB/CD137, CD28, CD34, CD8α, CD8β, FcεRIγ, CD16, OX40/CD134, CD3ζ, CD3ε, CD3γ, CD3δ, TCRα, TCRβ, TCRζ, CD32, CD64, CD64, CD45, CD5, CD9, CD22, CD37, CD80, CD86, CD40, CD40L/CD154, VEGFR2, FAS, and FGFR2B, or a functional variant thereof, including the human versions of each of these sequences. Table 5 provides the amino acid sequences of a few exemplary transmembrane domains.

TABLE 5

Exemplary sequences of transmembrane domains

SEQ ID NO:	Sequence	Description
14011	IYIWAPLAGTCGVL LLSLVITLYC	CD8α transmembrane domain
14012	FWVLVVVGGVLACY SLLVTVAFIIFWV	CD28 transmembrane domain
14014	MFWVLVVVGGVLAC YSLLVTVAFIIFWV	CD28 transmembrane domain

[0325] In certain embodiments, the intracellular signaling domain and/or intracellular costimulatory domain of the CAR comprises one or more signaling domains selected from B7-1/CD80, B7-2/CD86, B7-H1/PD-L1, B7-H2, B7-H3, B7-H4, B7-H6, B7-H7, BTLA/CD272, CD28, CTLA-4, Gi24/VISTA/B7-H5, ICOS/CD278, PD-1, PD-L2/B7-DC, PDCD6, 4-1BB/TNFSF9/CD137, 4-1BB Ligand/TNFSF9, BAFF/BLyS/TNFSF13B, BAFF R/TNFRSF13C, CD27/TNFRSF7, CD27 Ligand/TNFSF7, CD30/TNFRSF8, CD30 Ligand/TNFSF8, CD40/TNFRSF5, CD40/TNFSF5, CD40 Ligand/TNFSF5, DR3/TNFRSF25, GITR/TNFRSF18, GITR Ligand/TNFSF18, HVEM/TNFRSF14, LIGHT/TNFSF14, Lymphotoxin-alpha/TNFβ, OX40/TNFRSF4, OX40 Ligand/TNFSF4, RELT/TNFRSF19L, TACI/TNFRSF13B, TL1A/TNFSF15, TNFα, TNF RII/TNFRSF1B, 2B4/CD244/SLAMF4, BLAME/SLAMF8, CD2, CD2F-10/SLAMF9, CD48/SLAMF2, CD58/LFA-3, CD84/SLAMF5, CD229/SLAMF3, CRACC/SLAMF7, NTB-A/SLAMF6, SLAM/CD150, CD2, CD7, CD53, CD82/Kai-1, CD90/Thyl, CD96, CD160, CD200, CD300a/LMIR1, HLA Class I, HLA-DR, Ikaros, Integrin alpha 4/CD49d, Integrin alpha 4 beta 1, Integrin alpha 4 beta 7/LPAM-1, LAG-3, TCL1A, TCL1B, CRTAM, DAP12, Dectin-1/CLEC7A, DPPIV/CD26, EphB6, TIM-1/KIM-1/HAVCR, TIM-4, TSLP, TSLP R, lymphocyte function associated antigen-1 (LFA-1), NKG2C, CD3ζ, an immunoreceptor tyrosine-based activation motif (ITAM), CD27, CD28, 4-1BB, CD134/OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, a ligand that specifically binds with CD83, and a functional variant thereof including the human versions of each of these sequences. In some embodiments, the intracellular signaling domain and/or intracellular costimulatory domain comprises one or more signaling domains selected from a CD37 domain, an ITAM, a CD28 domain, 4-1BB domain, or a functional variant thereof. Table 6 provides the amino acid sequences of a few exemplary intracellular costimulatory and/or signaling domains. In certain embodiments, as in the case of tisagenlecleucel as described below, the CD34 signaling domain of SEQ ID NO:14017 has a mutation, e.g., a glutamine (Q) to lysine (K) mutation, at amino acid position 14 (see SEQ ID NO:14018).

TABLE 6

Exemplary sequences of intracellular costimulatory and/or signaling domains

SEQ ID NO:	Sequence	Description
14015	KRGRKKLLY-IFKQPFMRPVQ TTQEEDGCSCRF-PEEEEGGC EL	4-1BB costimulatory domain
14016	RSKRS-RLLHSDYMNMTPRR PGPTRKHYQPYAPPRDFAAY RS	CD28 costimulatory domain
14017	RVKFSRSADAPA-YQQGQNQ LYNELNLGR-REEYDVLDKR RGRDPEMGGKPRR-KNPQEG LYNEL-QKDKMAEAYSEIGM KGERRRGKGHDGLYQGLSTA TKDTYDALHMQALPPR	CD3ζ signaling domain
14018	RVKFSRSADAPA-YKQGQNQ LYNELNLGR-REEYDVLDKR RGRDPEMGGKPRR-KNPQEG LYNEL-QKDKMAEAYSEIGM KGERRRGKGHDGLYQGLSTA TKDTYDALHMQALPPR	CD3ζ signaling domain (with Q to K mutation at position 14)

[0326] In certain embodiments where the polycistronic vector encodes two or more CARs, the two or more CARs comprise the same functional domains, or one or more different functional domains, as described. For example, the two or more CARs comprise different signal peptides, extracellular binding domains, hinge domains, transmembrane domains, costimulatory domains, and/or intracellular signaling domains, in order to minimize the risk of recombination due to sequence similarities. Or, alternatively, the two or more CARs comprise the same domains. In the cases where the same domain(s) and/or backbone are used, it is optional to introduce codon divergence at the nucleotide sequence level to minimize the risk of recombination.

CD19 CAR

[0327] In some embodiments, the CAR is a CD19 CAR ("CD19-CAR"), and in these embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR. In some embodiments, the CD19 CAR comprises a signal peptide, an extracellular binding domain that specifically binds CD19, a hinge domain, a transmembrane domain, an intracellular costimulatory domain, and/or an intracellular signaling domain in tandem.

[0328] In some embodiments, the signal peptide of the CD19 CAR comprises a CD4 signal peptide. In some embodiments, the CD4 signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO:14003 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID

NO:14003. In some embodiments, the signal peptide comprises an IgK signal peptide. In some embodiments, the IgK signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14004 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14004. In some embodiments, the signal peptide comprises a GMCSFR-α or CSF2RA signal peptide. In some embodiments, the GMCSFR-α or CSF2RA signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14005 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14005.

[0329] In some embodiments, the extracellular binding domain of the CD19 CAR is specific to CD19, for example, human CD19. The extracellular binding domain of the CD19 CAR can be codon-optimized for expression in a host cell or to have variant sequences to increase functions of the extracellular binding domain. In some embodiments, the extracellular binding domain comprises an immunogenically active portion of an immunoglobulin molecule, for example, an scFv.

[0330] In some embodiments, the extracellular binding domain of the CD19 CAR comprises an scFv derived from the FMC63 monoclonal antibody (FMC63), which comprises the heavy chain variable region (V_H) and the light chain variable region (V_L) of FMC63 connected by a linker. FMC63 and the derived scFv have been described in Nicholson et al., Mol. Immun. 34(16-17): 1157-1165 (1997) and PCT Application Publication No. WO2018/213337. In some embodiments, the amino acid sequences of the entire FMC63-derived scFv (also referred to as FMC63 scFv) and its different portions are provided in Table 7 below. In some embodiments, the CD19-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14019, 14020, or 14025, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO:14019, 14020, or 14025. In some embodiments, the CD19-specific scFv comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14021-14023 and 14026-14028. In some embodiments, the CD19-specific scFv comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14021-14023. In some embodiments, the CD19-specific scFv comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14026-14028. In any of these embodiments, the CD19-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the CD19 CAR comprises or consists of the one or more CDRs as described herein.

[0331] In some embodiments, the linker linking the V_H and the V_L portions of the scFv is a Whitlow linker having an amino acid sequence set forth in SEQ ID NO:14024. In some embodiments, the Whitlow linker is replaced by a different linker, for example, a 3×G4S linker (SEQ ID NO: 9313) having an amino acid sequence set forth in SEQ ID NO: 14030, which gives rise to a different FMC63-derived scFv having an amino acid sequence set forth in SEQ ID NO:14029. In certain of these embodiments, the CD19-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO:14029 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO:14029.

TABLE 7

Exemplary sequences of anti-CD19 scFv and components

SEQ ID NO:	Amino Acid Sequence	Description
14019	DIQMTQTTSSLSASLGDRVTIS-CRASQDISKY-LNWYQQKPDGTVKLLI-YHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGN-TLPYTFGGGTKLEIT-GSTSGSGKPGSGEGSTKGEV-KLQESGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQP-PRKGLELGVIWGSET-TYYNSALKSRLTIIKDNSKSQVFLK-MNSLQTDDTAIYYCAKHYYYGGSYAMDYWGQGTSVTVSS	Anti-CD19 FMC63 scFv entire sequence, with Whitlow linker
14020	DIQMTQTTSSLSASLGDRVTIS-CRASQDISKY-LNWYQQKPDGTVKLLI-YHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGN-TLPYTFGGGTKLEIT	Anti-CD19 FMC63 scFv light chain variable region
14021	QDISKY	Anti-CD19 FMC63 scFv light chain CDR1
	HTS	Anti-CD19 FMC63 scFv light chain CDR2

TABLE 7-continued

Exemplary sequences of anti-CD19 scFv and components

SEQ ID NO:	Amino Acid Sequence	Description
14023	QQGNTLPYT	Anti-CD19 FMC63 scFv light chain CDR3
14024	GSTSGSGKPGSGEGSTKG	Whitlow linker
14025	EVKLQESGPGLVAP-SQSLSV TCTVSGVSLPDY-GVSWIRQP-PRKGLEWLGVIWGSET-TYYN SALKSRLTIIKDNSKSQVFLK MNSLQTDD-TAIYYCAKHYYY GGSYAMDYWGQGTSVTVSS	Anti-CD19 FMC63 scFv heavy chain variable region
14026	GVSLPDYG	Anti-CD19 FMC63 scFv heavy chain CDR1
14027	IWGSETT	Anti-CD19 FMC63 scFv heavy chain CDR2
14028	AKHYYYGGSYAMDY	Anti-CD19 FMC63 scFv heavy chain CDR3
14029	DIQMTQTTSSLSASLGDRVTIS-CRASQDISKY-LNWYQQKPDGT VKLLI-YHTSRLHSGVPSRFSG SGSGTDYSLTISNLEQEDIATY FCQQGN-TLPYTFGGGTKLEIT-GGGGSGGGGSGGGGSEV-KLQE SGPGLVAPSQSLSVTCTVSGVS LPDYGVSWIRQP-PRKGLEWLG VIWGSET-TYYNSALKSRLTII KDNSKSQVFLK-MNSLQTDDTA IYYCAKHYYYGGSYAMDYWGQG TSVTVSS	Anti-CD19 FMC63 scFv entire sequence, with 3xG$_4$S linker (SEQ ID NO: 9313)
14030	GGGGSGGGGSGGGGS	3xG$_4$S linker

[0332] In some embodiments, the extracellular binding domain of the CD19 CAR is derived from an antibody specific to CD19, including, for example, SJ25C1 (Bejcek et al., Cancer Res. 55:2346-2351 (1995)), HD37 (Pezutto et al., J. Immunol. 138(9):2793-2799 (1987)), 4G7 (Meeker et al., Hybridoma 3:305-320 (1984)), B43 (Bejcek (1995)), BLY3 (Bejcek (1995)), B4 (Freedman et al., 70:418-427 (1987)), B4 HB12b (Kansas & Tedder, J. Immunol. 147:4094-4102 (1991); Yazawa et al., Proc. Natl. Acad. Sci. USA 102: 15178-15183 (2005); Herbst et al., J. Pharmacol. Exp. Ther. 335:213-222 (2010)), BU12 (Callard et al., J. Immunology, 148(10):2983-2987 (1992)), and CLB-CD19 (De Rie Cell. Immunol. 118:368-381 (1989)). In any of these 10 embodiments, the extracellular binding domain of the CD19 CAR comprises or consists of the V_H, the V_L, and/or one or more CDRs of any of the antibodies.

[0333] In some embodiments, the hinge domain of the CD19 CAR comprises a CD4 hinge domain, for example, a human CD4 hinge domain. In some embodiments, the CD4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14006 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14006. In some embodiments, the hinge domain comprises a CD28 hinge domain, for example, a human CD28 hinge domain. In some embodiments, the CD28 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO:14007 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO:14007. In some embodiments, the hinge domain comprises an IgG4 hinge domain, for example, a human IgG4 hinge domain. In some embodiments, the IgG4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO:14008 or SEQ ID NO:14009, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14008 or SEQ ID NO: 14009. In some embodiments, the hinge domain comprises a IgG4 hinge-Ch2-Ch3 domain, for example, a human IgG4 hinge-Ch2-Ch3 domain. In some embodiments, the IgG4 hinge-Ch2-Ch3 domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14010 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14010.

[0334] In some embodiments, the transmembrane domain of the CD19 CAR comprises a CD4 transmembrane domain, for example, a human CD4 transmembrane domain. In some embodiments, the CD4 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14011 or an amino acid sequence that is at least 80%

identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14011. In some embodiments, the transmembrane domain comprises a CD28 transmembrane domain, for example, a human CD28 transmembrane domain. In some embodiments, the CD28 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14012 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14012.

[0335] In some embodiments, the intracellular costimulatory domain of the CD19 CAR comprises a 4-1BB costimulatory domain. 4-1BB, also known as CD137, transmits a potent costimulatory signal to T cells, promoting differentiation and enhancing long-term survival of T lymphocytes. In some embodiments, the 4-1BB costimulatory domain is human. In some embodiments, the 4-1BB costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14015 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14015. In some embodiments, the intracellular costimulatory domain comprises a CD28 costimulatory domain. CD28 is another co-stimulatory molecule on T cells. In some embodiments, the CD28 costimulatory domain is human. In some embodiments, the CD28 costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14016 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14016. In some embodiments, the intracellular costimulatory domain of the CD19 CAR comprises a 4-1BB costimulatory domain and a CD28 costimulatory domain as described.

[0336] In some embodiments, the intracellular signaling domain of the CD19 CAR comprises a CD3 zeta (2) signaling domain. CD37 associates with T cell receptors (TCRs) to produce a signal and contains immunoreceptor tyrosine-based activation motifs (ITAMs). The CD37 signaling domain refers to amino acid residues from the cytoplasmic domain of the zeta chain that are sufficient to functionally transmit an initial signal necessary for T cell activation. In some embodiments, the CD3ζ signaling domain is human. In some embodiments, the CD37 signaling domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14017 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14017.

[0337] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR, including, for example, a CD19 CAR comprising the CD19-specific scFv having sequences set forth in SEQ ID NO: 14019 or SEQ ID NO: 14029, the CD4 hinge domain of SEQ ID NO:14006, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the CD19 CAR additionally comprises a signal peptide (e.g., a CD4 signal peptide) as described.

[0338] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR, including, for example, a CD19 CAR comprising the CD19-specific scFv having sequences set forth in SEQ ID NO: 14019 or SEQ ID NO: 14029, the IgG4 hinge domain of SEQ ID NO:14008 or SEQ ID NO: 14009, the CD28 transmembrane domain of SEQ ID NO:14012, the 4-1 BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the CD19 CAR additionally comprises a signal peptide (e.g., a CD4 signal peptide) as described.

[0339] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR, including, for example, a CD19 CAR comprising the CD19-specific scFv having sequences set forth in SEQ ID NO: 14019 or SEQ ID NO: 14029, the CD28 hinge domain of SEQ ID NO: 14007, the CD28 transmembrane domain of SEQ ID NO: 14012, the CD28 costimulatory domain of SEQ ID NO: 14016, the CD33 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the CD19 CAR additionally comprises a signal peptide (e.g., a CD4 signal peptide) as described.

[0340] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR as set forth in SEQ ID NO: 14031 or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the nucleotide sequence set forth in SEQ ID NO: 14031 (see Table 8). The encoded CD19 CAR has a corresponding amino acid sequence set forth in SEQ ID NO: 14032 or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14032, with the following components: CD4 signal peptide, FMC63 scFv (V_L-Whitlow linker-V_H), CD4 hinge domain, CD4 transmembrane domain, 4-1BB costimulatory domain, and CD3ζ signaling domain.

[0341] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a commercially available embodiment of CD19 CAR. Non-limiting examples of commercially available embodiments of CD19 CARs expressed and/or encoded by T cells include tisagenlecleucel, lisocabtagene maraleucel, axicabtagene ciloleucel, and brexucabtagene autoleucel.

[0342] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding tisagenlecleucel or portions thereof. Tisagenlecleucel comprises a CD19 CAR with the following components: CD4 signal peptide, FMC63 scFv (V_L-3×G4S linker-V_H), CD4 hinge domain, CD4 transmembrane domain, 4-1BB costimulatory domain, and CD37 signaling domain. The nucleotide and amino acid sequence of the CD19 CAR in tisagenlecleucel are provided in Table 8, with annotations of the sequences provided in Table 9.

[0343] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding lisocabtagene maraleucel or portions thereof. Lisocabtagene maraleucel comprises a CD19 CAR with the following components: GMCSFR-α or CSF2RA signal peptide, FMC63 scFv (V_L-Whitlow linker-V_H), IgG4 hinge domain, CD28 transmembrane domain, 4-1BB costimulatory domain, and CD37 signaling domain. The nucleotide and amino acid sequence of the CD19 CAR in lisocabtagene maraleucel are provided in Table 8, with annotations of the sequences provided in Table 10.

[0344] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding axicabtagene ciloleucel or portions thereof. Axicabtagene ciloleucel comprises a CD19 CAR with the following components: GMCSFR-α or CSF2RA signal peptide, FMC63 scFv (V_L-Whitlow linker-V_H), CD28 hinge domain, CD28 transmembrane domain, CD28 costimulatory domain, and CD33 signaling domain. The nucleotide and amino acid sequence of the CD19 CAR in axicabtagene ciloleucel are provided in Table 8, with annotations of the sequences provided in Table 11.

[0345] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding brexucabtagene autoleucel or portions thereof. Brexucabtagene autoleucel comprises a CD19 CAR with the following components: GMCSFR-α signal peptide, FMC63 scFv, CD28 hinge domain, CD28 transmembrane domain, CD28 costimulatory domain, and CD3ζ signaling domain.

[0346] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR as set forth in SEQ ID NO: 14033, 14035, or 14037, or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the nucleotide sequence set forth in SEQ ID NO: 14033, 14035, or 14037. The encoded CD19 CAR has a corresponding amino acid sequence set forth in SEQ ID NO: 14034, 14036, or 14038, respectively, or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14034, 14036, or 14038, respectively.

TABLE 8

Exemplary sequences of CD19 CARs

SEQ ID NO:	Sequence	Description
14031	atggccttaccagtgaccgccttgctcctgccgct ggcctt-gctgctccac-gccgccaggccggacat ccagatgacacagactacatcctccctgtctgcct ctctgggagacagagtcaccatcagtt-gcagggc aagtcaggacatt-agtaaatatttaaattggtat cagcagaaaccagatggaactgttaaactcctgat ctaccatacatcaagattacactcag-gagtccca tcaaggttcag-tggcagtgggtctggaacagatt attctctcaccattagcaacctggagcaagaagat attgccacttacttttt-gccaacagggtaatacgc ttccgtacac-gttcggaggggggaccaagctgga gatcacaggctccacctctggatccggcaagcccg -gatctggcgagggatccaccaagggcgaggtgaa actg-cag-gagtcaggacctggcctggtggcgcc ctcacagagcctgtccgtcacatgcactgtctcag gggtctcattacccgac-tatggtgtaagctggat -tcgccagcctccacgaaagggtctggagtggctg ggagtaatatggggtagtgaaaccacatacta-ta attcagctctcaaatccagactgac-catcatcaa ggacaactccaagagccaagttttcttaaaaatga acagtctg-caaactgatgacacagccatttacta ctgtgccaaacattattactacggtggtagctatg ctatggac-tactggggccaaggaacctcagtcac -cgtctcctcaaccacgacgccagcgccgcgacca ccaacaccggcgcccaccatcgcgtcg-cagcccc tgtccctgcgcccagaggcgtgccggccagcggcg gggggcgcagtgcacacgagggggctg-gacttcg cctgtga-tatctacatctgggcgcccttggccgg gacttgtggggtccttctcctgtcactggttatca ccctttactgcaaacgggg-cagaaagaaactcct gtata-tattcaaacaaccatttatgagaccagta caaactactcaagaggaagatggctgtagctgccg atttccagaagaa-gaagaaggaggatgtgaactg agag-tgaagttcagcaggagcgcagacgcccccg cgtaccagcagggccagaaccagctctataacgag ctcaatctag-gacgaagagaggagtacgatgttt t-ggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaaccctcaggaaggcc	Exemplary CD19 CAR nucleotide sequence

TABLE 8-continued

Exemplary sequences of CD19 CARs		
SEQ ID NO:	Sequence	Description
	tg-tacaatgaactgcagaaagataa-gatggcgg aggcctacagtgagattgggatgaaaggcgagcgc cggaggggcaaggggcacgatggcctttac-cagg gtctcagtacagccac-caaggacacctacgacgc ccttcacatgcaggccctgccccctcgc	
14032	MAL-PVTALLLPLALLLHAARPDIQMTQTTSSLSA SLGDRVTISCRASQDISKY-LNWYQQKPDGTVKLL I-YHTSRLHSGVPSRFSGSGSGTDYS-LTISNLEQ EDIATYFCQQGNTLPYTFGGGTKLEITGSTSGSGK PGSGEGSTKGEV-KLQESGPGLVAP-SQSLSVTCT VSGVSLPDY-GVSWIRQPPRKGLEWLGVIWGSETT YYNSALKSRLTIIKDNSKSQVFLKMNSLQTDD-TA IYYCAKHYYYGGSYAMDYWGQGTSVTVSSTTTPAP RPPTPAPTI-ASQPLSLRPEACRPAAGGAVHTRGL D-FACDIYIWAPLAG-TCGVLLLSLVITLYCKRGR KKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEE-G GCELRVKFSRSADAPA-YQQGQNQLYNELNLGR-R EEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNEL-Q KDKMAEAYSEIGMKGERRRGKGH-DGLYQGLSTAT KDTYDALHMQALPPR	Exemplary CD19 CAR amino acid sequence
14033	atggccttaccagtgaccgccttgctcctgccgct ggcctt-gctgctccac-gccgccaggccggacat ccagatgacacagactacatcctccctgtctgcct ctctgggagacagagtcaccatcagtt-gcagggc aagtcaggacatt-agtaaatatttaaattggtat cagcagaaaccagatggaactgttaaactcctgat ctaccatacatcaagattacactcag-gagtccca tcaaggttcag-tggcagtgggtctggaacagatt attctctcaccattagcaacctggagcaagaagat attgccacttacttt-gccaacagggtaatacgc ttccgtacac-gttcggaggggggaccaagctgga gatcacaggtggcggtggctcgggcggtggtgggt cgggtggcggcg-gatctgaggtgaaactgcagga gtcag-gacctggcctggtggcgccctcacagagc ctgtccgtcacatgcactgtctcaggggtctcatt acccgactatggtgtaa-gctggattcgccagcct ccac-gaaagggtctggagtggctgggagtaatat ggggtagtgaaaccacatactataattcagctctc aaatccagactgac-catcatcaaggacaactcca a-gagccaagttttcttaaaaatgaacagtctgca aactgatgacacagccatttactactgtgccaaac attattactac-ggtggtagctatgctatggac-t actggggccaaggaacctcagtcaccgtctcctca accacgacgccagcgccgcgaccaccaacac-cgg cgcccaccatcgcgtcg-cagcccctgtccctgcg cccagaggcgtgccggccagcggcggggggcgcag tgcacacgagggggctg-gacttcgcctgtga-ta tctacatctgggcgcccttggccgggacttgtggg gtccttctcctgtcactggttatcaccctttactg caaacgggg-cagaaagaaactcctgtata-tatt caaacaaccatttatgagaccagtacaaactactc aagaggaagatggctgtagctgccgatttccagaa gaa-gaagaaggaggatgtgaactgagag-tgaag ttcagcaggagcgcagacgcccccgcgtacaagca gggccagaaccagctctataacgagctcaatctag -gacgaagagaggagtacgatgtttt-ggacaaga gacgtggccgggaccctgagatggggggaaagccg agaaggaagaaccctcaggaaggcctg-tacaatg aactgcagaaagataa-gatggcggaggcctacag tgagattgggatgaaaggcgagcgccggaggggca aggggcacgatggcctttac-cagggtctcagtac agccac-caaggacacctacgacgcccttcacatg caggccctgccccctcgc	Tisagenlecleucel CD19 CAR nucleotide sequence
14034	MAL-PVTALLLPLALLLHAARPDIQMTQTTSSLSA SLGDRVTISCRASQDISKY-LNWYQQKPDGTVKLL I-YHTSRLHSGVPSRFSGSGSGTDYS-LTISNLEQ EDIATYFCQQGNTLPYTFGGGTKLEITGGGGSGGG GSGGGGSEV-KLQESGPGLVAP-SQSLSVTCTVSG VSLPDYGVSWIRQP-PRKGLEWLGVIWGSETTYYN SALKSRLTIIKDNSKSQVFLKMNSLQTDD-TAIYY	Tisagenlecleucel CD19 CAR amino acid sequence

TABLE 8-continued

Exemplary sequences of CD19 CARs		
SEQ ID NO:	Sequence	Description
	CAKHYYYGGSYAMDYWGQGTSVTVSSTTTPAPRPP TPAPTI-ASQPLSLRPEACRPAAGGAVHTRGLD-F ACDIYIWAPLAG-TCGVLLLSLVITLYCKRGRKKL LYIFKQPFMRPVQTTQEEDGCSCRFPEEEE-GGCE LRVKFSRSADAPA-YKQGQNQLYNELNLGR-REEY DVLDKRRGRDPEMGGKPRRKNPQEGLYNEL-QKDK MAEAYSEIGMKGERRRGKGH-DGLYQGLSTATKDT YDALHMQALPPR	
14035	atgctgctgctggtgac-cagcctgctgctgtgcg agctgccccaccccgcctttctgctgatccccgac atccagatgacccagaccac-ctccagcctgagcg ccagcctgggcgac-cgggtgaccatcagctgccg ggccagccaggacatcagcaagtacctgaactggt atcagcagaagcccgacgg-caccgtcaagctgct gatctac-cacaccagccggctgcacagcggcgtg cccagccggtttagcggcagcggctccggcaccga ctacagcctgac-catctccaacctggaacaggaa ga-tatcgccacctacttttgccagcagggcaaca cactgccctacacctttggcggcggaacaaagctg gaaatcaccgg-cagcacctccggcagcggcaa-g cctggcagcggcgagggcagcaccaagggcgaggt gaagctgcaggaaa-gcggccctggcctggtggcc cccagccagagcctgagcgtgacctgcaccgtgag cggcgtgagcctgcccgactac-ggcgtgagctgg atccgg-cagcccccccaggaagggcctggaatggc tgggcgtgatctggggcagcgagaccacctactac aacagcgccctgaa-gagccggctgac-catcatc aaggacaacagcaagagccaggtgttcctgaagat gaacagcctgcagaccgacgacac-cgccatctac tactgcgccaagcactactactac-ggcggcagct acgccatggactactggggccagggcaccagcgtg accgtgagcagcgaatctaagtacggac-cgccct gcccccctt-gccctatgttctgggtgctggtggt ggtcggaggcgtgctggcctgctacagcctgctgg tcaccgtggccttcatcatcttttt-gggtgaaacg gggcagaaa-gaaactcctgtatatattcaaacaa ccatttatgagaccagtacaaactactcaagagga agatggctgtagctgccgat-ttccagaagaagaa gaaggag-gatgtgaactgcgggtgaagttcagca gaagcgccgacgcccctgcctaccagcagggccag aatcagctgtacaac-gagctgaacctgggcagaa gggaa-gagtacgacgtcctggataagcggagagg ccgggaccctgagatgggcggcaagcctcggcgga agaaccccag-gaaggcctgtataacgaactg-c agaaagacaagatggccgaggcctacagcgagatc ggcatgaagggcgagcggaggcggggcaagggcca c-gacggcctgtatcagggcctgtccac-cgccac caaggatacctacgacgccctgcacatgcaggccc tgcccccaagg	Lisocabtagene maraleucel CD19 CAR nucleotide sequence
14036	MLLLVTSLLLCELPHPAFL-LIPDIQMTQTTSSLS ASLGDRVTIS-CRASQDISKY-LNWYQQKPDGTVK LLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQ EDIATY-FCQQGNTLPYTFGGGTKLEIT-GSTSGS GKPGSGEGSTKGEVKLQESGPGLVAPSQSLSVTCT VSGVSLPDY-GVSWIRQPPRKGLEWLGVIWGSET-TYYNSALKSRLTIIK-DNSKSQVFLKMNSLQTDDT AIYYCAKHYYYGGSYAMDYWGQGTSVTVSSESKYG PPCPPCPMFWVLVVVGGVLACYSLLVTVAFI-IFW VKRGRKKLLY-IFKQPFMRPVQTTQEEDGCSCRFP EEEEGGCELRVKFSRSADAPA-YQQGQNQLYNELN LGR-REEYDVLDKRRGRDPEMGGKPRR-KNPQEGL YNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGL STAT-KDTYDALHMQALPPR	Lisocabtagene maraleucel CD19 CAR amino acid sequence
14037	atgcttctcctggtgacaagccttctgctctgtga gttac-cacacccag-cattcctcctgatcccaga catccagatgacacagactacatcctccctgtctg cctctctgggagacagagtcac-catcagttgcag ggcaagtcag-gacattagtaaatatttaaattgg tatcagcagaaaccagatggaactgttaaactcct gatctaccatacatcaagat-tacactcaggag-t	Axicabtagene ciloleucel CD19 CAR nucleotide sequence

TABLE 8-continued

Exemplary sequences of CD19 CARs

SEQ ID NO:	Sequence	Description
	cccatcaaggttcagtggcagtgggtctggaacag attattctctcaccattagcaacctggagcaagaa gatattgccac-ttacttttgccaacagggtaa-t acgcttccgtacacgttcggaggggggactaagtt ggaaataacaggctccacctctggatccggcaagc ccg-gatctggcgagggatccac-caagggcgagg tgaaactgcaggagtcaggacctggcctggtggcg ccctcacagagcctgtccgtcacatgcac-tgtct caggggtctcattacccgac-tatggtgtaagctg gattcgccagcctccacgaaagggtctggagtggc tgggagtaatatggggtagtgaaac-cacatacta -taattcagctctcaaatccagactgaccatcatc aaggacaactccaagagccaagttttcttaaaaat gaacagtctg-caaactgatgacacagccatttac tactgtgccaaacattattactacggtggtagcta tgctatggac-tactggggtcaaggaacctcagtc ac-cgtctcctcagcggccgcaattgaagttatgt atcctcctccttacctagacaatgagaagagcaat ggaac-cattatccatgtgaaagggaaacaccttt -gtccaagtcccctatttcccggaccttctaagcc cttttgggtgctggtggtggttgggggagtcctgg cttgctatagctt-gctagtaacag-tggccttta ttattttctgggtgaggagtaagaggagcaggctc ctgcacagtgac-tacatgaacatgactccccgcc gccccgggcccacccg-caagcattac-cagccct atgccccaccacgcgacttcgcagcctatcgctcc agagtgaagttcagcaggagcgcagac-gcccccg cgtaccagcagggccagaaccagctc-tataacga gctcaatctaggacgaagagaggagtacgatgttt tggacaagagacgtggccgggaccctga-gatggg gggaaagccgagaaggaa-gaaccctcaggaaggc ctgtacaatgaactgcagaaagataagatggcgga ggcctacagtgagattgg-gatgaaaggcgagcgc cggaggggcaaggggg-cacgatggcctttaccagg gtctcagtacagccaccaaggacacctacgac-gc ccttcacatgcaggccctgccccctcgc	
14038	MLLLVTSLLLCELPHPAFL-LIPDIQMTQTTSSLS ASLGDRVTIS-CRASQDISKY-LNWYQQKPDGTVK LLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQ EDIATY-FCQQGNTLPYTFGGGTKLEIT-GSTSGS GKPGSGEGSTKGEVKLQESGPGLVAPSQSLSVTCT VSGVSLPDY-GVSWIRQPPRKGLEWLGVIWGSET- TYYNSALKSRLTIIK-DNSKSQVFLKMNSLQTDDT AIYYCAKHYYYGGSYAMDYWGQGTSVTVSSAAAIE VMYPPPYLDNEKSNGTIIHVKGKHLCPSPLF-PGP SKPFWVLVVVGGVLACYSLLVTVAFII-FWVRSKR SRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDFAAY RSRVKFSRSADAPA-YQQGQNQLYNELNLGR-REE YDVLDKRRGRD-PEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRGKGHDGLYQGLSTAT-KD TYDALHMQALPPR	Axicabtagene ci- loleucel CD19 CAR amino acid sequence

TABLE 9

Annotation of tisagenlecleucel CD19 CAR sequences

Feature	Nucleotide Sequence Position	Amino Acid Sequence Position
CD8a signal peptide	1-63	1-21
FMC63 scFv (V_L-3xG_4S linker-V_H)	64-789	22-263
CD8α hinge domain	790-924	264-308
CD8α transmembrane domain	925-996	309-332
4-1BB costimulatory domain	997-1122	333-374
CD3ζ signaling domain	1123-1458	375-486

TABLE 10

Annotation of lisocabtagene maraleucel CD19 CAR sequences

Feature	Nucleotide Sequence Position	Amino Acid Sequence Position
GMCSFR-a signal peptide	1-66	1-22
FMC63 scFv (V_L-Whitlow linker-V_H)	67-801	23-267
IgG4 hinge domain	802-837	268-279
CD28 transmembrane domain	838-921	280-307
4-1BB costimulatory domain	922-1047	308-349
CD3ζ signaling domain	1048-1383	350-461

TABLE 11

Annotation of axicabtagene ciloleucel CD19 CAR sequences		
Feature	Nucleotide Sequence Position	Amino Acid Sequence Position
CSF2RA signal peptide	1-66	1-22
FMC63 scFv (V_L-Whitlow linker-V_H)	67-801	23-267
CD28 hinge domain	802-927	268-309
CD28 transmembrane domain	928-1008	310-336
CD28 costimulatory domain	1009-1131	337-377
CD3ζ signaling domain	1132-1467	378-489

[0347] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding CD19 CAR as set forth in SEQ ID NO: 14033, 14035, or 14037, or at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the nucleotide sequence set forth in SEQ ID NO: 14033, 14035, or 14037. The encoded CD19 CAR has a corresponding amino acid sequence set forth in SEQ ID NO: 14034, 14036, or 14038, respectively, or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14034, 14036, or 14038, respectively.

CD20 CAR

[0348] In some embodiments, the CAR is a CD20 CAR ("CD20-CAR"), and in these embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR. CD20 is an antigen found on the surface of B cells as early at the pro-B phase and progressively at increasing levels until B cell maturity, as well as on the cells of most B-cell neoplasms. CD20 positive cells are also sometimes found in cases of Hodgkins disease, myeloma, and thymoma. In some embodiments, the CD20 CAR comprises a signal peptide, an extracellular binding domain that specifically binds CD20, a hinge domain, a transmembrane domain, an intracellular costimulatory domain, and/or an intracellular signaling domain in tandem.

[0349] In some embodiments, the signal peptide of the CD20 CAR comprises a CD4 signal peptide. In some embodiments, the CD4 signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO:14003 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO:14003. In some embodiments, the signal peptide comprises an IgK signal peptide. In some embodiments, the IgK signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14004 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14004. In some embodiments, the signal peptide comprises a GMCSFR-α or CSF2RA signal peptide. In some embodiments, the GMCSFR-α or CSF2RA signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14005 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14005.

[0350] In some embodiments, the extracellular binding domain of the CD20 CAR is specific to CD20, for example, human CD20. The extracellular binding domain of the CD20 CAR is codon-optimized for expression in a host cell or to have variant sequences to increase functions of the extracellular binding domain. In some embodiments, the extracellular binding domain comprises an immunogenically active portion of an immunoglobulin molecule, for example, an scFv.

[0351] In some embodiments, the extracellular binding domain of the CD20 CAR is derived from an antibody specific to CD20, including, for example, Leu16, IF5, 1.5.3, rituximab, obinutuzumab, ibritumomab, ofatumumab, tositumumab, odronextamab, veltuzumab, ublituximab, and ocrelizumab. In any of these embodiments, the extracellular binding domain of the CD20 CAR comprises or consists of the V_H, the V_L, and/or one or more CDRs of any of the antibodies.

[0352] In some embodiments, the extracellular binding domain of the CD20 CAR comprises an scFv derived from the Leu16 monoclonal antibody, which comprises the heavy chain variable region (V_H) and the light chain variable region (V_L) of Leu16 connected by a linker. See Wu et al., Protein Engineering. 14(12):1025-1033 (2001). In some embodiments, the linker is a 3×G4S linker (SEQ ID NO: 9313). In other embodiments, the linker is a Whitlow linker as described herein. In some embodiments, the amino acid sequences of different portions of the entire Leu16-derived scFv (also referred to as Leu16 scFv) and its different portions are provided in Table 12 below. In some embodiments, the CD20-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14039, 14040, or 14044, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14039, 14040, or 14044. In some embodiments, the CD20-specific scFv comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14041-14043, 14045 and 14046. In some embodiments, the CD20-specific scFv comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14041-14043. In some embodiments, the CD20-specific scFv comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14045-14046. In any of these embodiments, the CD20-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the CD20 CAR comprises or consists of the one or more CDRs as described herein.

TABLE 12

Exemplary sequences of anti-CD20 scFv and components

SEQ ID NO:	Amino Acid Sequence	Description
14039	DIVLTQSPAILSASPGEKVTMT-CRASSSVNYMDWYQKKPGSSPKP-WIYATSNLAS-GVPARFSGSGSGTSYSLTISRVEAEDAATYYCQQWS-FNPPTFGGGTKLEIKGSTSGSGKPGSGEGSTKGEVQLQQSGAELVKP-GASVKMSCKASGYTFTSYN-MHWVKQTPGQGLEWIGAI-YPGNGDTSYNQKFKGKATLTADKSSSTAYMQLSSLTSED-SADYYCARSNYYGSSYWFFDVW-GAGTTVTVSS	Anti-CD20 Leu16 scFv entire sequence, with Whitlow linker
14040	DIVLTQSPAILSASPGEKVTMT-CRASSSVNYMDWYQKKPGSSPKP-WIYATSNLAS-GVPARFSGSGSGTSYSLTISRVEAEDAATYYCQQWS-FNPPTFGGGTKLEIK	Anti-CD20 Leu16 scFv light chain variable region
14041	RASSSVNYMD	Anti-CD20 Leu16 scFv light chain CDR1
14042	ATSNLAS	Anti-CD20 Leu16 scFv light chain CDR2
14043	QQWSFNPPT	Anti-CD20 Leu16 scFv light chain CDR3
14044	EVQLQQSGAELVKPGASVKMSCK-ASGYTFTSYN-MHWVKQTPGQGLEWIGAIYPGNG-DTSYNQKFKGKATLTADKSSSTAYMQLSSLTSED-SADYYCARSNYYGSSYWFFDVW-GAGTTVTVSS	Anti-CD20 Leu16 scFv heavy chain
14045	SYNMH	Anti-CD20 Leu16 scFv heavy chain CDR1
14046	AIYPGNGDTSYNQKFKG	Anti-CD20 Leu16 scFv heavy chain CDR2

[0353] In some embodiments, the hinge domain of the CD20 CAR comprises a CD4 hinge domain, for example, a human CD4 hinge domain. In some embodiments, the CD4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14006 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14006. In some embodiments, the hinge domain comprises a CD28 hinge domain, for example, a human CD28 hinge domain. In some embodiments, the CD28 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14007 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14007. In some embodiments, the hinge domain comprises an IgG4 hinge domain, for example, a human IgG4 hinge domain. In some embodiments, the IgG4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14008 or SEQ ID NO: 14009, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14008 or SEQ ID NO: 14009. In some embodiments, the hinge domain comprises a IgG4 hinge-Ch2-Ch3 domain, for example, a human IgG4 hinge-Ch2-Ch3 domain. In some embodiments, the IgG4 hinge-Ch2-Ch3 domain comprises or consists of an amino acid sequence set forth in SEQ ID NO:14010 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14010.

[0354] In some embodiments, the transmembrane domain of the CD20 CAR comprises a CD4 transmembrane domain, for example, a human CD4 transmembrane domain. In some embodiments, the CD4 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14011 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least

99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14011. In some embodiments, the transmembrane domain comprises a CD28 transmembrane domain, for example, a human CD28 transmembrane domain. In some embodiments, the CD28 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14012 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14012.

[0355] In some embodiments, the intracellular costimulatory domain of the CD20 CAR comprises a 4-1BB costimulatory domain, for example, a human 4-1BB costimulatory domain. In some embodiments, the 4-1BB costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14015 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14015. In some embodiments, the intracellular costimulatory domain comprises a CD28 costimulatory domain, for example, a human CD28 costimulatory domain. In some embodiments, the CD28 costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14016 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14016.

[0356] In some embodiments, the intracellular signaling domain of the CD20 CAR comprises a CD3 zeta (2) signaling domain, for example, a human CD3ζ signaling domain. In some embodiments, the CD32 signaling domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14017 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14017.

[0357] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the CD4 hinge domain of SEQ ID NO: 14006, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD3ζ signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0358] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the CD28 hinge domain of SEQ ID NO: 14007, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0359] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the IgG4 hinge domain of SEQ ID NO: 14008 or SEQ ID NO: 14009, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0360] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the CD4 hinge domain of SEQ ID NO: 14006, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0361] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the CD28 hinge domain of SEQ ID NO: 14007, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0362] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the IgG4 hinge domain of SEQ ID NO: 14008 or SEQ ID NO: 14009, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

CD22 CAR

[0363] In some embodiments, the CAR is a CD22 CAR ("CD22-CAR"), and in these embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR. CD22, which is a transmembrane protein found mostly on the surface of mature B cells that functions as an inhibitory receptor for B cell receptor (BCR) signaling. CD22 is expressed in 60-70%

of B cell lymphomas and leukemias (e.g., B-chronic lymphocytic leukemia, hairy cell leukemia, acute lymphocytic leukemia (ALL), and Burkitt's lymphoma) and is not present on the cell surface in early stages of B cell development or on stem cells. In some embodiments, the CD22 CAR comprises a signal peptide, an extracellular binding domain that specifically binds CD22, a hinge domain, a transmembrane domain, an intracellular costimulatory domain, and/or an intracellular signaling domain in tandem.

[0364] In some embodiments, the signal peptide of the CD22 CAR comprises a CD4 signal peptide. In some embodiments, the CD4 signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14003 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14003. In some embodiments, the signal peptide comprises an IgK signal peptide. In some embodiments, the IgK signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14004 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14004. In some embodiments, the signal peptide comprises a GMCSFR-α or CSF2RA signal peptide. In some embodiments, the GMCSFR-α or CSF2RA signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14005 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14005.

[0365] In some embodiments, the extracellular binding domain of the CD22 CAR is specific to CD22, for example, human CD22. The extracellular binding domain of the CD22 CAR is codon-optimized for expression in a host cell or to have variant sequences to increase functions of the extracellular binding domain. In some embodiments, the extracellular binding domain comprises an immunogenically active portion of an immunoglobulin molecule, for example, an scFv.

[0366] In some embodiments, the extracellular binding domain of the CD22 CAR is derived from an antibody specific to CD22, including, for example, SM03, inotuzumab, epratuzumab, moxetumomab, and pinatuzumab. In any of these embodiments, the extracellular binding domain of the CD22 CAR comprises or consists of the V_H, the V_L, and/or one or more CDRs of any of the antibodies.

[0367] In some embodiments, the extracellular binding domain of the CD22 CAR comprises an scFv derived from the m971 monoclonal antibody (m971), which comprises the heavy chain variable region (V_H) and the light chain variable region (V_L) of m971 connected by a linker. In some embodiments, the linker is a 3×G4S linker (SEQ ID NO: 9313). In other embodiments, the Whitlow linker is used instead. In some embodiments, the amino acid sequences of the entire m971-derived scFv (also referred to as m971 scFv) and its different portions are provided in Table 13 below. In some embodiments, the CD22-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14047, 14048, or 14052, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14047, 14048, or 14052. In some embodiments, the CD22-specific scFv comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14049-14051 and 14053-14055. In some embodiments, the CD22-specific scFv comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14049-14051. In some embodiments, the CD22-specific scFv comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14053-14055. In any of these embodiments, the CD22-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the CD22 CAR comprises or consists of the one or more CDRs as described herein.

[0368] In some embodiments, the extracellular binding domain of the CD22 CAR comprises an scFv derived from m971-L7, which is an affinity matured variant of m971 with significantly improved CD22 binding affinity compared to the parental antibody m971 (improved from about 2 nM to less than 50 pM). In some embodiments, the scFv derived from m971-L7 comprises the V_H and the V_L of m971-L7 connected by a 3×G4S linker (SEQ ID NO: 9313). In other embodiments, the Whitlow linker is used instead. In some embodiments, the amino acid sequences of the entire m971-L7-derived scFv (also referred to as m971-L7 scFv) and its different portions are provided in Table K below. In some embodiments, the CD22-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14056, 14057, or 14061, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14056, 14057, or 14061. In some embodiments, the CD22-specific scFv comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14058-14060 and 14062-14064. In some embodiments, the CD22-specific scFv comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14058-14060. In some embodiments, the CD22-specific scFv comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14062-14064. In any of these embodiments, the CD22-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the CD22 CAR comprises or consists of the one or more CDRs as described herein.

TABLE 13

Exemplary sequences of anti-CD22 scFv and components		
SEQ ID NO:	Amino Acid Sequence	Description
14047	QVQLQQSGPGLVKP-SQTLSLTCAISGDSVSS-NSAAWNWIRQSPSR-GLEWLGRTYYRSKWYNDYAVSVK SRITINPDTSKNQFSLQLNSVTPED-TAVYYCAREVTGDLEDAFD-IWGQGTMVTVSSGGGGSGGGGSG GGGSDIQMTQSPSSLSASVG-DRVTITCRASQTI-WSYLNWYQQRPGKAPNLLIYAAS-SLQSGVPSRFSGRGSGTDFTLTISS LQAEDFA-TYYCQQSYSIPQTFGQGTKLEIK	Anti-CD22 m971 scFv entire sequence, with 3xG_4S linker (SEQ ID NO: 9313)
14048	QVQLQQSGPGLVKP-SQTLSLTCAISGDSVSS-NSAAWNWIRQSPSR-GLEWLGRTYYRSKWYNDYAVSVK SRITINPDTSKNQFSLQLNSVTPED-TAVYYCAREVTGDLEDAFD-IWGQGTMVTVSS	Anti-CD22 m971 scFv heavy chain variable region
14049	GDSVSSNSAA	Anti-CD22 m971 scFv heavy chain CDR1
14050	TYYRSKWYN	Anti-CD22 m971 scFv heavy chain CDR2
14051	AREVTGDLEDAFDI	Anti-CD22 m971 scFv heavy chain CDR3
14052	DIQMTQSPSSLSASVG-DRVTITCRASQTI-WSYLNWYQQRPGKAPNLLIYAAS-SLQSGVPSRFSGRGSGTDFTLTISS LQAEDFA-TYYCQQSYSIPQTFGQGTKLEIK	Anti-CD22 m971 scFv light chain
14053	QTIWSY	Anti-CD22 m971 scFv light chain CDR1
	AAS	Anti-CD22 m971 scFv light chain CDR2
14055	QQSYSIPQT	Anti-CD22 m971 scFv light chain CDR3
14056	QVQLQQSGPGMVKP-SQTLSLTCAISGDSVSS-NSVAWNWIRQSPSR-GLEWLGRTYYRST-WYNDYAVSMKSRITINPDTNKNQFS LQLNSVTPEDTAVYYCAREV-TGDLEDAFD-IWGQGTMVTVSSGGGGSGGGGSG GGGSDIQMIQSPSSLSASVG-DRVTITCRASQTI-WSYLNWYRQRPGEAPNLLIYAAS-SLQSGVPSRFSGRGSGTDFTLTISS LQAEDFA-TYYCQQSYSIPQTFGQGTKLEIK	Anti-CD22 m971-L7 scFv entire sequence, with 3xG_4S linker (SEQ ID NO: 9313)
14057	QVQLQQSGPGMVKP-SQTLSLTCAISGDSVSS-NSVAWNWIRQSPSR-GLEWLGRTYYRST-WYNDYAVSMKSRITINPDTNKNQFS LQLNSVTPEDTAVYYCAREV-TGDLEDAFDIWGQGTMVTVSS	Anti-CD22 m971-L7 scFv heavy chain vari-able region
14058	GDSVSSNSVA	Anti-CD22 m971-L7 scFv heavy chain CDR1

TABLE 13-continued

Exemplary sequences of anti-CD22 scFv and components		
SEQ ID NO:	Amino Acid Sequence	Description
14059	TYYRSTWYN	Anti-CD22 m971-L7 scFv heavy chain CDR2
14060	AREVTGDLEDAFDI	Anti-CD22 m971-L7 scFv heavy chain CDR3
14061	DIQMIQSPSSLSASVG-DRVTITCRASQTI-WSYLNWYRQRPGEAPNLLIYAAS-SLQSGVPSRFSGRGSGTDFTLTISS LQAEDFA-TYYCQQSYSIPQTFGQGTKLEIK	Anti-CD22 m971-L7 scFv light chain variable region
14062	QTIWSY	Anti-CD22 m971-L7 scFv light chain CDR1
	AAS	Anti-CD22 m971-L7 scFv light chain CDR2
14064	QQSYSIPQT	Anti-CD22 m971-L7 scFv light chain CDR3

[0369] In some embodiments, the extracellular binding domain of the CD22 CAR comprises immunotoxins HA22 or BL22. Immunotoxins BL22 and HA22 are therapeutic agents that comprise an scFv specific for CD22 fused to a bacterial toxin, and thus can bind to the surface of the cancer cells that express CD22 and kill the cancer cells. BL22 comprises a dsFv of an anti-CD22 antibody, RFB4, fused to a 38-kDa truncated form of Pseudomonas exotoxin A (Bang et al., Clin. Cancer Res., 11:1545-50 (2005)). HA22 (CAT8015, moxetumomab pasudotox) is a mutated, higher affinity version of BL22 (Ho et al., J. Biol. Chem., 280(1): 607-17 (2005)). Suitable sequences of antigen binding domains of HA22 and BL22 specific to CD22 are disclosed in, for example, U.S. Pat. Nos. 7,541,034; 7,355,012; and 7,982,011.

[0370] In some embodiments, the hinge domain of the CD22 CAR comprises a CD4 hinge domain, for example, a human CD4 hinge domain. In some embodiments, the CD4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14006 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14006. In some embodiments, the hinge domain comprises a CD28 hinge domain, for example, a human CD28 hinge domain. In some embodiments, the CD28 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14007 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14007. In some embodiments, the hinge domain comprises an IgG4 hinge domain, for example, a human IgG4 hinge domain. In some embodiments, the IgG4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14008 or SEQ ID NO: 14009, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14008 or SEQ ID NO: 14009. In some embodiments, the hinge domain comprises a IgG4 hinge-Ch2-Ch3 domain, for example, a human IgG4 hinge-Ch2-Ch3 domain. In some embodiments, the IgG4 hinge-Ch2-Ch3 domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14010 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14010.

[0371] In some embodiments, the transmembrane domain of the CD22 CAR comprises a CD4 transmembrane domain, for example, a human CD4 transmembrane domain. In some embodiments, the CD4 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14011 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14011. In some embodiments, the transmembrane domain comprises a CD28 transmembrane domain, for example, a human CD28 transmembrane domain. In some embodiments, the CD28 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO:14012 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14012.

[0372] In some embodiments, the intracellular costimulatory domain of the CD22 CAR comprises a 4-1BB costimulatory domain, for example, a human 4-1BB costimulatory domain. In some embodiments, the 4-1BB costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14015 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14015. In some embodiments, the intracellular costimulatory domain comprises a CD28 costimulatory domain, for example, a human CD28 costimulatory domain. In some embodiments, the CD28 costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14016 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14016.

[0373] In some embodiments, the intracellular signaling domain of the CD22 CAR comprises a CD3 zeta (2) signaling domain, for example, a human CD37 signaling domain. In some embodiments, the CD34 signaling domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14017 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14017.

[0374] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the CD4 hinge domain of SEQ ID NO:9, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0375] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the CD28 hinge domain of SEQ ID NO: 14007, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0376] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the IgG4 hinge domain of SEQ ID NO: 14008 or SEQ ID NO: 14009, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD33 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0377] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the CD4 hinge domain of SEQ ID NO:9, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0378] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the CD28 hinge domain of SEQ ID NO: 14007, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0379] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the IgG4 hinge domain of SEQ ID NO: 14008 or SEQ ID NO: 14009, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

BCMA CAR

[0380] In some embodiments, the CAR is a BCMA CAR ("BCMA-CAR"), and in these embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a BCMA CAR. BCMA is a tumor necrosis family receptor (TNFR) member expressed on cells of the B cell lineage, with the highest expression on terminally differentiated B cells or mature B lymphocytes. BCMA is involved in mediating the survival of plasma cells for maintaining long-term humoral immunity. The expression of BCMA has been recently linked to a number of cancers, such as multiple myeloma, Hodgkin's and non-Hodgkin's lymphoma, various leukemias, and glioblastoma. In some embodiments, the BCMA CAR comprises a signal peptide, an extracellular binding domain that specifically binds BCMA, a hinge domain, a transmembrane domain, an intracellular costimulatory domain, and/or an intracellular signaling domain in tandem.

[0381] In some embodiments, the signal peptide of the BCMA CAR comprises a CD4 signal peptide. In some embodiments, the CD4 signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14003 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14003. In some embodiments, the signal peptide comprises an IgK signal peptide. In some embodiments, the IgK signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14004 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14004. In some embodiments, the signal peptide comprises a GMCSFR-α or CSF2RA signal peptide. In some embodiments, the GMCSFR-α or CSF2RA signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14005 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14005.

[0382] In some embodiments, the extracellular binding domain of the BCMA CAR is specific to BCMA, for example, human BCMA. The extracellular binding domain of the BCMA CAR can be codon-optimized for expression in a host cell or to have variant sequences to increase functions of the extracellular binding domain.

[0383] In some embodiments, the extracellular binding domain comprises an immunogenically active portion of an immunoglobulin molecule, for example, an scFv. In some embodiments, the extracellular binding domain of the BCMA CAR is derived from an antibody specific to BCMA, including, for example, belantamab, erlanatamab, teclistamab, LCAR-B38M, and ciltacabtagene. In any of these embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the V_H, the V_L, and/or one or more CDRs of any of the antibodies.

[0384] In some embodiments, the extracellular binding domain of the BCMA CAR comprises an scFv derived from C11D5.3, a murine monoclonal antibody as described in Carpenter et al., Clin. Cancer Res. 19(8):2048-2060 (2013). See also PCT Application Publication No. WO2010/104949. The C11D5.3-derived scFv may comprise the heavy chain variable region (V_H) and the light chain variable region (V_L) of C11D5.3 connected by the Whitlow linker, the amino acid sequences of which is provided in Table 14 below. In some embodiments, the BCMA-specific extracellular binding domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14065, 14066, or 14067, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14065, 14066, or 14067. In some embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14067-14069 and 14071-14073. In some embodiments, the BCMA-specific extracellular binding domain comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14067-14069. In some embodiments, the BCMA-specific extracellular binding domain comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14071-14073. In any of these embodiments, the BCMA-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the one or more CDRs as described herein.

[0385] In some embodiments, the extracellular binding domain of the BCMA CAR comprises an scFv derived from another murine monoclonal antibody, C12A3.2, as described in Carpenter et al., Clin. Cancer Res. 19(8):2048-2060 (2013) and PCT Application Publication No. WO2010/104949, the amino acid sequence of which is also provided in Table 14 below. In some embodiments, the BCMA-specific extracellular binding domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14074, 14075, or 14079, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14074, 14075, or 14079. In some embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14076-14078 and 14080-14082. In some embodiments, the BCMA-specific extracellular binding domain comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14076-14078. In some embodiments, the BCMA-specific extracellular binding domain comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14080-14082. In any of these embodiments, the BCMA-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the one or more CDRs as described herein.

[0386] In some embodiments, the extracellular binding domain of the BCMA CAR comprises a murine monoclonal antibody with high specificity to human BCMA, referred to as BB2121 in Friedman et al., Hum. Gene Ther. 29(5):585-601 (2018)). See also, PCT Application Publication No. WO2012163805.

[0387] In some embodiments, the extracellular binding domain of the BCMA CAR comprises single variable fragments of two heavy chains (VHH) that bind to two epitopes of BCMA as described in Zhao et al., J. Hematol. Oncol. 11(1):141 (2018), also referred to as LCAR-B38M. See also, PCT Application Publication No. WO2018/028647.

[0388] In some embodiments, the extracellular binding domain of the BCMA CAR comprises a fully human heavy-chain variable domain (FHVH) as described in Lam et al., Nat. Commun. 11(1):283 (2020), also referred to as FHVH33. See also, PCT Application Publication No. WO2019/006072. The amino acid sequences of FHVH33 and its CDRs are provided in Table 14 below. In some embodiments, the BCMA-specific extracellular binding domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14083 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least

98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14083. In some embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14084-14086. In any of these embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the one or more CDRs as described herein.

[0389] In some embodiments, the extracellular binding domain of the BCMA CAR comprises an scFv derived from CT103A (or CAR0085) as described in U.S. Pat. No. 11,026,975 B2, the amino acid sequence of which is provided in Table 14 below. In some embodiments, the BCMA-specific extracellular binding domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14087, 14088, or 14092, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14087, 14088, or 14092. In some embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14089-14091 and 14093-14095. In some embodiments, the BCMA-specific extracellular binding domain comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14089-14091. In some embodiments, the BCMA-specific extracellular binding domain comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14093-14095. In any of these embodiments, the BCMA-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the one or more CDRs as described herein.

[0390] Additionally, CARs and binders directed to BCMA have been described in U.S. Application Publication Nos. 2020/0246381 A1 and 2020/0339699 A1.

TABLE 14

Exemplary sequences of anti-BCMA binder and components

SEQ ID NO:	Amino Acid Sequence	Description
14065	DIVLTQSPASLAMSLGKRATIS-CRASESVSVIGAHLIHWYQQKPGQPPKLLIYLASN-LETGVPARFSGSGSGTDFT-LTIDPVEEDDVAIYSCLQSRIFPRT-FGGGTKLEIKGSTSGSGKPGSGEGSTKGQIQLVQSGPELKKPGETVKISCKASGYTFTDYSINWVKRAPGKGLKWMGWIN-TETREPAYAYDFRGRFAFSLETSASTAYLQINNLKYEDTATYFCAL-DYSYAMDYWGQGTSVTVSS	Anti-BCMA C11D5.3 scFv entire sequence, with Whitlow linker
14066	DIVLTQSPASLAMSLGKRATIS-CRASESVSVIGAHLIHWYQQKPGQPPKLLIYLASN-LETGVPARFSGSGSGTDFT-LTIDPVEEDDVAIYSCLQSRIFPRT-FGGGTKLEIK	Anti-BCMA C11D5.3 scFv light chain variable region
14067	RASESVSVIGAHLIH	Anti-BCMA C11D5.3 scFv light chain CDR1
14068	LASNLET	Anti-BCMA C11D5.3 scFv light chain CDR2
14069	LQSRIFPRT	Anti-BCMA C11D5.3 scFv light chain CDR3
14070	QIQLVQSGPELKKPGETVKISCK-ASGYTFTDYSINWVKRAPGKGLK-WMGWIN-TETREPAYAYDFRGRFAFSLETSASTAYLQINNLKYEDTATYFCAL-DYSYAMDYWGQGTSVTVSS	Anti-BCMA C11D5.3 scFv heavy chain vari-able region
14071	DYSIN	Anti-BCMA C11D5.3 scFv heavy chain CDR1

TABLE 14-continued

Exemplary sequences of anti-BCMA binder and components		
SEQ ID NO:	Amino Acid Sequence	Description
14072	WINTETREPAYAYDFRG	Anti-BCMA C11D5.3 scFv heavy chain CDR2
14073	DYSYAMDY	Anti-BCMA C11D5.3 scFv heavy chain CDR3
14074	DIVLTQSPPSLAMSLGKRATIS-CRASESVTILGSHLI-YWYQQKPGQPPTLLIQ-LASNVQTGVPARFSGSGSRTDFTL TIDPVEEDDVAVYYCLQSRTIPRT-FGGGTKLEIKGSTSGSGKPGSGEG STKGQIQLVQSGPELKKPGETVKIS CKASGYTFRHYSMNWVKQAPGKG LKWMGRINTESGVPIYADD-FKGRFAFSVETSASTAYL-VINNLKDEDTASYFCSNDYLYSLD-FWGQGTALTVSS	Anti-BCMA C12A3.2 scFv entire sequence, with Whitlow linker
14075	DIVLTQSPPSLAMSLGKRATIS-CRASESVTILGSHLI-YWYQQKPGQPPTLLIQ-LASNVQTGVPARFSGSGSRTDFTL TIDPVEEDDVAVYYCLQSRTIPRT-FGGGTKLEIK	Anti-BCMA C12A3.2 scFv light chain variable region
14076	RASESVTILGSHLIY	Anti-BCMA C12A3.2 scFv light chain CDR1
14077	LASNVQT	Anti-BCMA C12A3.2 scFv light chain CDR2
14078	LQSRTIPRT	Anti-BCMA C12A3.2 scFv light chain CDR3
14079	QIQLVQSGPELKKPGETVKISCK-ASGYTFRHYSMNWVKQAPGKGLK-WMGRINTESGVPIYADDFKGRFAF SVETSASTAYL-VINNLKDEDTASYFCSNDYLYSLD-FWGQGTALTVSS	Anti-BCMA C12A3.2 scFv heavy chain variable region
14080	HYSMN	Anti-BCMA C12A3.2 scFv heavy chain CDR1
14081	RINTESGVPIYADDFKG	Anti-BCMA C12A3.2 scFv heavy chain CDR2
14082	DYLYSLDF	Anti-BCMA C12A3.2 scFv heavy chain CDR3
14083	EVQLLESGGGLVQPGGSLRLS-CAASGFTFSSYAMSWVR-QAPGKGLEWVSSISGSGDYIY-YADSVKGRFTISRDISKNTLYLQMN SLRAEDTAVYYCAKEGTGANSSLA DYRGQGTLVTVSS	Anti-BCMA FHVH33 entire sequence
14084	GFTFSSYA	Anti-BCMA FHVH33 CDR1
14085	ISGSGDYI	Anti-BCMA FHVH33 CDR2
14086	AKEGTGANSSLADY	Anti-BCMA FHVH33 CDR3

TABLE 14-continued

Exemplary sequences of anti-BCMA binder and components

SEQ ID NO:	Amino Acid Sequence	Description
14087	DIQMTQSPSSLSASVG-DRVTITCRASQSIS-SYLNWYQQKPGKAPKLLIYAAS-SLQSGVPSRFSGSGSGTDFTLTISS LQPEDFA-TYYCQQKYDLLTFGGGTKVEIKGST SGSGKPGSGEGSTKGQLQLQESG PGLVKPSETLSLTCTVSGGSIS-SSSYYWGWIRQPPGKGLEWIG-SISYSGSTYYNPSLKSRVTISVDTSK NQFSLKLSSVTAADTAVYYCARDR GDTILDVWGQGTMVTVSS	Anti-BCMA CT103A scFv entire sequence, with Whitlow linker
14088	DIQMTQSPSSLSASVG-DRVTITCRASQSIS-SYLNWYQQKPGKAPKLLIYAAS-SLQSGVPSRFSGSGSGTDFTLTISS LQPEDFA-TYYCQQKYDLLTFGGGTKVEIK	Anti-BCMA CT103A scFv light chain variable region
14089	QSISSY	Anti-BCMA CT103A scFv light chain CDR1
	AAS	Anti-BCMA CT103A scFv light chain CDR2
14091	QQKYDLLT	Anti-BCMA CT103A scFv light chain CDR3
14092	QLQLQESGPGLVKP-SETLSLTCTVSGGSIS-SSSYYWGWIRQPPGKGLEWIG-SISYSGSTYYNPSLKSRVTISVDTSK NQFSLKLSSVTAADTAVYYCARDR GDTILDVWGQGTMVTVSS	Anti-BCMA CT103A scFv heavy chain variable region
14093	GGSISSSSYY	Anti-BCMA CT103A scFv heavy chain CDR1
14094	ISYSGST	Anti-BCMA CT103A scFv heavy chain CDR2
14095	ARDRGDTILDV	Anti-BCMA CT103A scFv heavy chain CDR3

[0391] In some embodiments, the hinge domain of the BCMA CAR comprises a CD4 hinge domain, for example, a human CD4 hinge domain. In some embodiments, the CD4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14006 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14006. In some embodiments, the hinge domain comprises a CD28 hinge domain, for example, a human CD28 hinge domain. In some embodiments, the CD28 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14007 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14007. In some embodiments, the hinge domain comprises an IgG4 hinge domain, for example, a human IgG4 hinge domain. In some embodiments, the IgG4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14008 or SEQ ID NO: 14009, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14008 or SEQ ID NO: 14009. In some embodiments, the hinge domain comprises a IgG4 hinge-Ch2-Ch3 domain, for example, a human IgG4 hinge-Ch2-Ch3 domain. In some embodiments, the IgG4 hinge-Ch2-Ch3 domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14010 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14010.

[0392] In some embodiments, the transmembrane domain of the BCMA CAR comprises a CD4 transmembrane domain, for example, a human CD4 transmembrane domain. In some embodiments, the CD4 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14011 or an amino acid sequence that is at least

80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14011. In some embodiments, the transmembrane domain comprises a CD28 transmembrane domain, for example, a human CD28 transmembrane domain. In some embodiments, the CD28 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14012 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14012.

[0393] In some embodiments, the intracellular costimulatory domain of the BCMA CAR comprises a 4-1BB costimulatory domain, for example, a human 4-1BB costimulatory domain. In some embodiments, the 4-1BB costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14015 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14015. In some embodiments, the intracellular costimulatory domain comprises a CD28 costimulatory domain, for example, a human CD28 costimulatory domain. In some embodiments, the CD28 costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14016 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14016.

[0394] In some embodiments, the intracellular signaling domain of the BCMA CAR comprises a CD3 zeta (ζ) signaling domain, for example, a human CD3ζ signaling domain. In some embodiments, the CD37 signaling domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14017 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14017.

[0395] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a BCMA CAR, including, for example, a BCMA CAR comprising any of the BCMA-specific extracellular binding domains as described, the CD4 hinge domain of SEQ ID NO: 14006, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the BCMA CAR additionally comprises a signal peptide (e.g., a CD4 signal peptide) as described.

[0396] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a BCMA CAR, including, for example, a BCMA CAR comprising any of the BCMA-specific extracellular binding domains as described, the CD4 hinge domain of SEQ ID NO: 14006, the CD4 transmembrane domain of SEQ ID NO: 14011, the CD28 costimulatory domain of SEQ ID NO: 14016, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the BCMA CAR additionally comprises a signal peptide as described.

[0397] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a BCMA CAR as set forth in SEQ ID NO: 14096 or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the nucleotide sequence set forth in SEQ ID NO: 14096 (see Table 15). The encoded BCMA CAR has a corresponding amino acid sequence set forth in SEQ ID NO: 14097 or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14097, with the following components: CD4 signal peptide, CT103A scFv (V_L-Whitlow linker-V_H), CD4 hinge domain, CD4 transmembrane domain, 4-1BB costimulatory domain, and CD3ζ signaling domain.

[0398] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a commercially available embodiment of BCMA CAR, including, for example, idecabtagene vicleucel (ide-cel, also called bb2121). In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding idecabtagene vicleucel or portions thereof. Idecabtagene vicleucel comprises a BCMA CAR with the following components: the BB2121 binder, CD4 hinge domain, CD4 transmembrane domain, 4-1BB costimulatory domain, and CD37 signaling domain.

TABLE 15

Exemplary sequences of BCMA CARs

SEQ ID NO:	Sequence	Description
14096	atggccttaccagtgaccgccttgctcctgccgctggcctt-gctgctccac-gccgccaggccggacatccagatgacccagtctccatc ctccctgtctgcatctgtaggagacagagtcaccatcac-ttgccgggcaagtcagagcatt-agcagctatttaaattggtatcagcagaaaccagggaaa gcccctaagctcctgatctatgctgcatccagttt-gcaaagtggggtcccatcaaggttcag-	Exemplary BCMA CAR nucleotide sequence

TABLE 15-continued

Exemplary sequences of BCMA CARs		
SEQ ID NO:	Sequence	Description
	tggcagtggatctgggacagatttcactctcaccatcagc agtctgcaacctgaagattttgcaacttactactgtcag-caaaaatacgacctcctcac-ttttggcggagggaccaaggttgagatcaaaggcagca ccagcggctccggcaagcctggctctggcgagggcag-cacaaagggacagctgcagctgcag-gagtcgggcccaggactggtgaagccttcggagaccct gtccctcacctgcactgtctctggtggctccatcagcag-tagtagttactactggggctg-gatccgccagcccccagggaaggggctggagtggattg ggagtatctcctatagtgggagcacctacta-caacccgtccctcaagagtcgagtcac-catatccgtagacacgtccaagaaccagttctccctgaag ctgagttctgtgaccgccgcagacacggcggtg-tactactgcgccagagatcgtggagacac-catactagacgtatggggtcagggtacaatggtcaccgtc agctcattcgtgcccgtgttcctgcccgccaaacctac-caccacccctgcccctagac-ctcccaccccagccccaacaatcgccagccagcctctgt ctctgcggcccgaagcctgtagacctgctgccggcg-gagccgtgcacaccagaggcctg-gacttcgcctgcgacatctacatctgggcccctctggccg gcac-ctgtggcgtgctgctgctgagcctggtgatcaccctg-tactgcaaccaccg-gaacaaacggggcagaaagaaactcctgtatatattca aacaaccatttatgagaccagtacaaactactcaagag-gaagatggctgtagctgccgat-ttccagaagaagaagaaggaggatgtgaactgagagtg aagttcagcagatccgccgacgcccctgcctaccag-cagggacagaaccagctgtacaac-gagctgaacctgggcagacgggaagagtacgacgtgct ggacaagcggagaggccgggaccccgagatgggcg-gaaagcccagacggaagaacccccag-gaaggcctgtataacgaactgcagaaagacaagatgg ccgaggcctacagcgagatcgg-catgaagggcgagcggaggcgcggcaagggccac-gatggcctgtaccagggcctgagcaccgccaccaagga cacctacgacgccctg-cacatgcaggccctgccccccaga	
14097	MAL-PVTALLLPLALLLHAARPDIQMTQSPSSLS ASVGDRVTITCRASQSIS-SYLNWYQQKPGKAPKLLIYAAS-SLQSGVPSRFSGSGSGTDFT-LTISSLQPEDFATYYCQQKYDLLTFGGGT KVEIKGSTSGSGKPGSGEGSTKGQLQLQ ESGPGLVKPSETLSLTCTVSGGSIS-SSSYYWGWIRQPPGKGLEWIG-SISYSGSTYYNPSLKSRVTISVDTSKNQF SLKLSSVTAADTAVYYCARDRGDTIL-DVWGQGTMVTVSS-FVPVFLPAKPTTTPAPRPPTPAP-TIASQPLSLRPEACRPAAGGAVHTRGLDF ACDIYIWAPLAGTCGVLLLSLVITLYC-NHRNKRGRKKLLY-IFKQPFMRPVQTTQEEDGCSCRFPEEEE GGCELRVKFSRSADAPA-YQQGQNQLYNELNLGR-REEYDVLDKRRGRDPEMGGKPRR-KNPQEGLYNELQKDKMAEAYSEIGMKGE RRRGKGHDGLYQGLSTAT-KDTYDALHMQALPPR	Exemplary BCMA CAR amino acid sequence

[0399] In some embodiments, the antibody portion of the recombinant receptor, e.g., CAR, further includes a spacer between the transmembrane domain and extracellular antigen binding domain. In some embodiments, the spacer includes at least a portion of an immunoglobulin constant region, such as a hinge region, e.g., an IgG4 hinge region, and/or a CH1/CL and/or Fc region. In some embodiments, the constant region or portion is of a human IgG, such as IgG4 or IgGI. In some aspects, the portion of the constant region serves as a spacer region between the antigen-recognition component, e.g., scFv, and transmembrane domain. The spacer can be of a length that provides for increased responsiveness of the cell following antigen binding, as compared to in the absence of the spacer. Exemplary spacers include, but are not limited to, those described in Hudecek et al. (2013) *Clin. Cancer Res.,* 19:3153, WO2014031687, U.S. Pat. No. 8,822,647 or published app. No. US 2014/0271635. In some embodiments, the constant region or portion is of a human IgG, such as IgG4 or IgGI.

[0400] In some embodiments, the antigen receptor comprises an intracellular domain linked directly or indirectly to the extracellular domain. In some embodiments, the chimeric antigen receptor includes a transmembrane domain linking the extracellular domain and the intracellular signaling domain. In some embodiments, the intracellular signaling domain comprises an ITAM. For example, in some aspects, the antigen recognition domain (e.g. extracellular domain) generally is linked to one or more intracellular signaling components, such as signaling components that mimic activation through an antigen receptor complex, such as a TCR complex, in the case of a CAR, and/or signal via another cell surface receptor. In some embodiments, the chimeric receptor comprises a transmembrane domain linked or fused between the extracellular domain (e.g. scFv) and intracellular signaling domain. Thus, in some embodiments, the antigen-binding component (e.g., antibody) is linked to one or more transmembrane and intracellular signaling domains.

[0401] In some embodiments, a transmembrane domain that naturally is associated with one of the domains in the receptor, e.g., CAR, is used. In some instances, the transmembrane domain is selected or modified by amino acid substitution to avoid binding of such domains to the transmembrane domains of the same or different surface membrane proteins to minimize interactions with other members of the receptor complex.

[0402] The transmembrane domain in some embodiments is derived either from a natural or from a synthetic source. Where the source is natural, the domain in some aspects is derived from any membrane-bound or transmembrane protein. Transmembrane regions include those derived from (i.e. comprise at least the transmembrane region(s) of) the alpha, beta or zeta chain of the T-cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154. Alternatively, the transmembrane domain in some embodiments is synthetic. In some aspects, the synthetic transmembrane domain comprises predominantly hydrophobic residues such as leucine and valine. In some aspects, a triplet of phenylalanine, tryptophan and valine will be found at each end of a synthetic transmembrane domain. In some embodiments, the linkage is by linkers, spacers, and/or transmembrane domain(s). In some aspects, the transmembrane domain contains a transmembrane portion of CD28.

[0403] In some embodiments, the extracellular domain and transmembrane domain is linked directly or indirectly. In some embodiments, the extracellular domain and transmembrane are linked by a spacer, such as any described herein. In some embodiments, the receptor contains extracellular portion of the molecule from which the transmembrane domain is derived, such as a CD28 extracellular portion.

[0404] Among the intracellular signaling domains are those that mimic or approximate a signal through a natural antigen receptor, a signal through such a receptor in combination with a costimulatory receptor, and/or a signal through a costimulatory receptor alone. In some embodiments, a short oligo- or polypeptide linker, for example, a linker of 2 to 10 amino acids in length, such as one containing glycines and serines, e.g., glycine-serine doublet, is present and forms a linkage between the transmembrane domain and the cytoplasmic signaling domain of the CAR.

[0405] T cell activation is in some aspects described as being mediated by two classes of cytoplasmic signaling sequences: those that initiate antigen-dependent primary activation through the TCR (primary cytoplasmic signaling sequences), and those that act in an antigen-independent manner to provide a secondary or co-stimulatory signal (secondary cytoplasmic signaling sequences). In some aspects, the CAR includes one or both of such signaling components.

[0406] The receptor, e.g., the CAR, generally includes at least one intracellular signaling component or components. In some aspects, the CAR includes a primary cytoplasmic signaling sequence that regulates primary activation of the TCR complex. Primary cytoplasmic signaling sequences that act in a stimulatory manner may contain signaling motifs which are known as immunoreceptor tyrosine-based activation motifs or ITAMs. Examples of ITAM containing primary cytoplasmic signaling sequences include those derived from CD3 zeta chain, FcR gamma, CD3 gamma, CD3 delta and CD3 epsilon. In some embodiments, cytoplasmic signaling molecule(s) in the CAR contain(s) a cytoplasmic signaling domain, portion thereof, or sequence derived from CD3 zeta.

[0407] In some embodiments, the receptor includes an intracellular component of a TCR complex, such as a TCR CD3 chain that mediates T-cell activation and cytotoxicity, e.g., CD3 zeta chain. Thus, in some aspects, the antigen-binding portion is linked to one or more cell signaling modules. In some embodiments, cell signaling modules include a CD3 transmembrane domain, CD3 intracellular signaling domains, and/or other CD transmembrane domains. In some embodiments, the intracellular component is or includes a CD3-zeta intracellular signaling domain. In some embodiments, the intracellular component is or includes a signaling domain from a Fc receptor gamma chain. In some embodiments, the receptor, e.g., CAR, includes the intracellular signaling domain and further includes a portion, such as a transmembrane domain and/or hinge portion, of one or more additional molecules such as CD8, CD4, CD25, or CD16. For example, in some aspects, the CAR or other chimeric receptor is a chimeric molecule of CD3-zeta (CD3-z) or Fc receptor gamma and a portion of one of CD8, CD4, CD25 or CD16.

[0408] In some embodiments, upon ligation of the CAR or other chimeric receptor, the cytoplasmic domain or intracellular signaling domain of the receptor activates at least one of the normal effector functions or responses of the immune cell, e.g., T cell engineered to express the CAR. For example, in some contexts, the CAR induces a function of a T cell such as cytolytic activity or T-helper activity, such as secretion of cytokines or other factors. In some embodiments, a truncated portion of an intracellular signaling domain of an antigen receptor component or costimulatory molecule is used in place of an intact immunostimulatory chain, for example, if it transduces the effector function signal. In some embodiments, the intracellular signaling domain or domains include the cytoplasmic sequences of the T cell receptor (TCR), and in some aspects also those of co-receptors that in the natural context act in concert with such receptors to initiate signal transduction following antigen receptor engagement.

[0409] In the context of a natural TCR, full activation generally requires not only signaling through the TCR, but also a costimulatory signal. Thus, in some embodiments, to promote full activation, a component for generating secondary or co-stimulatory signal is also included in the CAR. In other embodiments, the CAR does not include a component for generating a costimulatory signal. In some aspects, an additional CAR is expressed in the same cell and provides the component for generating the secondary or costimulatory signal.

[0410] In some embodiments, the chimeric antigen receptor contains an intracellular domain of a T cell costimulatory molecule. In some embodiments, the CAR includes a signaling domain and/or transmembrane portion of a costimulatory receptor, such as CD28, 4-1BB, 0X40, DAP10, and ICOS. In some aspects, the same CAR includes both the activating and costimulatory components. In some embodiments, the chimeric antigen receptor contains an intracellular domain derived from a T cell costimulatory molecule or a functional variant thereof, such as between the transmembrane domain and intracellular signaling domain. In some aspects, the T cell costimulatory molecule is CD28 or 41BB.

[0411] In some embodiments, the activating domain is included within one CAR, whereas the costimulatory component is provided by another CAR recognizing another antigen. In some embodiments, the CARs include activating or stimulatory CARs, costimulatory CARs, both expressed on the same cell (see WO2014/055668). In some aspects, the cells include one or more stimulatory or activating CARs and/or a costimulatory CAR. In some embodiments, the cells further include inhibitory CARs (iCARs, see Fedorov et al., Sci. Transl. Medicine, 5(215) (December 2013), such as a CAR recognizing an antigen other than the one associated with and/or specific for the disease or condition whereby an activating signal delivered through the disease-targeting CAR is diminished or inhibited by binding of the inhibitory CAR to its ligand, e.g., to reduce off-target effects.

[0412] In certain embodiments, the intracellular signaling domain comprises a CD28 transmembrane and signaling domain linked to a CD3 (e.g., CD3-zeta) intracellular domain. In some embodiments, the intracellular signaling domain comprises a chimeric CD28 and CD137 (4-1BB, TNFRSF9) co-stimulatory domains, linked to a CD3 zeta intracellular domain.

[0413] In some embodiments, the CAR encompasses one or more, e.g., two or more, costimulatory domains and an activation domain, e.g., primary activation domain, in the cytoplasmic portion. Exemplary CARs include intracellular components of CD3-zeta, CD28, and 4-1BB.

[0414] In some embodiments the intracellular signaling domain includes intracellular components of a 4-1BB signaling domain and a CD3-zeta signaling domain. In some embodiments, the intracellular signaling domain includes intracellular components of a CD28 signaling domain and a CD3zeta signaling domain.

[0415] In some embodiments, the CAR comprises an extracellular antigen binding domain (e.g., antibody or antibody fragment, such as an scFv) that binds to an antigen (e.g. tumor antigen), a spacer (e.g. containing a hinge domain, such as any as described herein), a transmembrane domain (e.g. any as described herein), and an intracellular signaling domain (e.g. any intracellular signaling domain, such as a primary signaling domain or costimulatory signaling domain as described herein). In some embodiments, the intracellular signaling domain is or includes a primary cytoplasmic signaling domain. In some embodiments, the intracellular signaling domain additionally includes an intracellular signaling domain of a costimulatory molecule (e.g., a costimulatory domain). Examples of exemplary components of a CAR are described in Table 16. In provided aspects, the sequences of each component in a CAR can include any combination listed in Table 16.

TABLE 16

Component	Sequence	SEQ ID NO:
Extracellular binding domain		
Anti-CD19 scFv (FMC63)	DIQMTQTTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDG TVKLLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATY FCQQGNTLPYTFGGGTKLEITGSTSGSGKPGSGEGSTKGEV KLQESGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQPPRKG LEWLGVIWGSETTYYNSALKSRLTIIKDNSKSQVFLKMNSLQT DDTAIYYCAKHYYYGGSYAMDYWGQGTSVTVSS	14098
Anti-CD19 scFv (FMC63)	DIQMTQTTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDG TVKLLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATY FCQQGNTLPYTFGGGTKLEITGGGGSGGGGSGGGGSEVKL QESGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQPPRKGLE WLGVIWGSETTYYNSALKSRLTIIKDNSKSQVFLKMNSLQTDD TAIYYCAKHYYYGGSYAMDYWGQGTSVTVSS	14099
Anti-BCMA sdAb (FHVH74)	QVQLVESGGGLVQPGGSLRLSCAASGFTFTNHAMSWVRQA PGKGLELVSSISGNGRTTYYADSVKGRFTISRDISKNTLDLQM NSLRAEDTAVYYCAKDGGETLVDSRGQGTLVTVSS	14100
Anti-BCMA sdAb (FHVH32)	QVQLVESGGGLVQPGGSLRLSCAASGFTFSSHAMTWVRQAP GKGLEWVAAISGSGDFTHYADSVKGRFTISRDNSKNTVSLQM NNLRAEDTAVYYCAKDEDGGSLLGYRGQGTLVTVSS	14101

TABLE 16-continued

Component	Sequence	SEQ ID NO:
Anti-BCMA sdAb (FHVH33)	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAP GKGLEWVSSISGSGDYIYYADSVKGRFTISRDISKNTLYLQMN SLRAEDTAVYYCAKEGTGANSSLADYRGQGTLVTVSS	14102
Anti-BCMA sdAb (FHVH93)	EVQLLESGGGLIQPGGSLRLSCAASGFTFSSHAMTWVRQAP GKGLEWVSAISGSGDYTHYADSVKGRFTISRDNSKNTVYLQM NSLRAEDSAVYYCAKDEDGGSLLGHRGQGTLVTVSS	14103
Spacer (e.g. hinge)		
IgG4 Hinge	ESKYGPPCPPCP	14104
CD8 Hinge	TTTPAPRPPTPAPTIASQPLSLRPE	14105
CD28 Transmembrane	IEVMYPPPYLDNEKSNGTIIHVKGKHLCPSPLFPGPSKP	14106
CD8	ACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLY C	14107
CD28	FWVLVVVGGVLACYSLLVTVAFIIFWV	14108
CD28	FWVLVVVGGVLACYSLLVTVAFIIFWV	14109
Costimulatory domain		
CD28	RSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDFAAYRS	14110
4-1BB	KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCEL	14111
Primary Signaling Domain		
	RVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERR RGKGHDGLYQGLSTATKDTYDALHMQALPPR	14112
CD3zeta	RVKFSRSADAPAYKQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERR RGKGHDGLYQGLSTATKDTYDALHMQALPPR	14113

[0416] In some embodiments, the antigen receptor further includes a marker and/or cells expressing the CAR or other antigen receptor further include a surrogate marker, such as a cell surface marker, which is used to confirm transduction or engineering of the cell to express the receptor. In some aspects, the marker includes all or part (e.g., truncated form) of CD34, a NGFR, or epidermal growth factor receptor, such as truncated version of such a cell surface receptor (e.g., tEGFR). In some embodiments, the nucleic acid encoding the marker is operably linked to a polynucleotide encoding a linker sequence, such as a cleavable linker sequence, e.g., T2A. For example, a marker, and optionally a linker sequence, can be any as disclosed in published patent application No. WO2014031687. For example, the marker can be a truncated EGFR (tEGFR) that is, optionally, linked to a linker sequence, such as a T2A cleavable linker sequence.

[0417] In some embodiments, the marker is a molecule, e.g., cell surface protein, not naturally found on T cells or not naturally found on the surface of T cells, or a portion thereof. In some embodiments, the molecule is a non-self molecule, e.g., non-self protein, i.e., one that is not recognized as "self" by the immune system of the host into which the cells will be adoptively transferred.

[0418] In some embodiments, the marker serves no therapeutic function and/or produces no effect other than to be used as a marker for genetic engineering, e.g., for selecting cells successfully engineered. In other embodiments, the marker is a therapeutic molecule or molecule otherwise exerting some desired effect, such as a ligand for a cell to be encountered in vivo, such as a costimulatory or immune checkpoint molecule to enhance and/or dampen responses of the cells upon adoptive transfer and encounter with ligand.

[0419] In some cases, CARs are referred to as first, second, and/or third generation CARs. In some aspects, a first generation CAR is one that solely provides a CD3-chain induced signal upon antigen binding; in some aspects, a second-generation CAR is one that provides such a signal and costimulatory signal, such as one including an intracellular signaling domain from a costimulatory receptor such as CD28 or CD 137; in some aspects, a third generation CAR is one that includes multiple costimulatory domains of different costimulatory receptors.

[0420] For example, in some embodiments, the CAR contains an antibody, e.g., an antibody fragment, a transmembrane domain that is or contains a transmembrane portion of CD28 or a functional variant thereof, and an intracellular signaling domain containing a signaling portion of CD28 or functional variant thereof and a signaling portion of CD3 zeta or functional variant thereof. In some embodiments, the CAR contains an antibody, e.g., antibody fragment, a transmembrane domain that is or contains a transmembrane portion of CD28 or a functional variant thereof, and an intracellular signaling domain containing a signaling portion of a 4-IBB or functional variant thereof and a signaling portion of CD3 zeta or functional variant thereof. In some such embodiments, the receptor further includes a spacer containing a portion of an Ig molecule, such as a human Ig molecule, such as an Ig hinge, e.g. an IgG4 hinge, such as a hinge-only spacer.

[0421] In some aspects, the spacer contains only a hinge region of an IgG, such as only a hinge of IgG4 or IgG. In other embodiments, the spacer is or contains an Ig hinge, e.g., an IgG4-derived hinge, optionally linked to a CH2 and/or CH3 domains. In some embodiments, the spacer is an Ig hinge, e.g., an IgG4 hinge, linked to CH2 and CH3 domains. In some embodiments, the spacer is an Ig hinge, e.g., an IgG4 hinge, linked to a CH3 domain only. In some embodiments, the spacer is or comprises a glycine-serine rich sequence or other flexible linker such as known flexible linkers.

[0422] For example, in some embodiments, the CAR includes an antibody such as an antibody fragment, including scFvs, a spacer, such as a spacer containing a portion of an immunoglobulin molecule, such as a hinge region and/or one or more constant regions of a heavy chain molecule, such as an Ig-hinge containing spacer, a transmembrane domain containing all or a portion of a CD28-derived transmembrane domain, a CD28-derived intracellular signaling domain, and a CD3 zeta signaling domain. In some embodiments, the CAR includes an antibody or fragment, such as scFv, a spacer such as any of the Ig-hinge containing spacers, a CD28-derived transmembrane domain, a 4-IBB-derived intracellular signaling domain, and a CD3 zeta-derived signaling domain.

[0423] The recombinant receptors, such as CARs, expressed by the cells administered to the subject generally recognize or specifically bind to a molecule that is expressed in, associated with, and/or specific for the disease or condition or cells thereof being treated. Upon specific binding to the molecule, e.g., antigen, the receptor generally delivers an immunostimulatory signal, such as an ITAM-transduced signal, into the cell, thereby promoting an immune response targeted to the disease or condition. For example, in some embodiments, the cells express a CAR that specifically binds to an antigen expressed by a cell or tissue of the disease or condition or associated with the disease or condition.

b. T Cell Receptors Antigen Receptors (TCRs)

[0424] In some embodiments, engineered cells, such as T cells, used in connection with the provided methods, uses, articles of manufacture or compositions are cells that express a T cell receptor (TCR) or antigen-binding portion thereof that recognizes a protein epitope or T cell epitope of a target protein, such as an antigen of a tumor, viral or autoimmune protein.

[0425] In some embodiments, a "T cell receptor" or "TCR" is a molecule that contains variable α and β chains (also known as TCRalpha and TCRbeta, respectively) or a variable γ and δ chains (also known as TCRgamma and TCRdelta, respectively), or antigen-binding portions thereof, and which is capable of specifically binding to a polypeptide bound to an MHC molecule. In some embodiments, the TCR is in the $\alpha\beta$ form. Typically, TCRs that exist in $\alpha\beta$ and $\gamma\delta$ forms are generally structurally similar, but T cells expressing them may have distinct anatomical locations or functions. A TCR can be found on the surface of a cell or in soluble form. Generally, a TCR is found on the surface of T cells (or T lymphocytes) where it is generally responsible for recognizing antigens bound to major histocompatibility complex (MHC) molecules.

[0426] Unless otherwise stated, the term "TCR" should be understood to encompass full TCRs as well as antigen-binding portions or antigen-binding fragments thereof. In some embodiments, the TCR is an intact or full-length TCR, including TCRs in the ab form or gd form. In some embodiments, the TCR is an antigen-binding portion that is less than a full-length TCR but that binds to a specific peptide bound in an MHC molecule, such as binds to an MHC-peptide complex. In some cases, an antigen-binding portion or fragment of a TCR can contain only a portion of the structural domains of a full-length or intact TCR, but yet is able to bind the peptide epitope, such as MHC-peptide complex, to which the full TCR binds. In some cases, an antigen-binding portion contains the variable domains of a TCR, such as variable a chain and variable b chain of a TCR, sufficient to form a binding site for binding to a specific MHC-peptide complex. Generally, the variable chains of a TCR contain complementarity determining regions involved in recognition of the peptide, MHC and/or MHC-peptide complex.

c. Multi-Targeting

[0427] In some embodiments, the cells used in connection with the provided methods, uses, articles of manufacture and compositions include cells employing multi-targeting strategies, such as expression of two or more genetically engineered receptors on the cell, each recognizing the same of a different antigen and typically each including a different intracellular signaling component. Such multi-targeting strategies are described, for example, in WO 2014055668 (describing combinations of activating and costimulatory CARs, e.g., targeting two different antigens present individually on off-target, e.g., normal cells, but present together only on cells of the disease or condition to be treated) and Fedorov et al., Sci. Transl. Medicine, 5(215) (2013) (describing cells expressing an activating and an inhibitory CAR, such as those in which the activating CAR binds to one antigen expressed on both normal or non-diseased cells and cells of the disease or condition to be treated, and the inhibitory CAR binds to another antigen expressed only on the normal cells or cells which it is not desired to treat).

[0428] For example, in some embodiments, the cells include a receptor expressing a first genetically engineered antigen receptor (e.g., CAR) which is capable of inducing an activating or stimulatory signal to the cell, generally upon specific binding to the antigen recognized by the first receptor, e.g., the first antigen. In some embodiments, the cell further includes a second genetically engineered antigen receptor (e.g., CAR), e.g., a chimeric costimulatory receptor, which is capable of inducing a costimulatory signal to the immune cell, generally upon specific binding to a second antigen recognized by the second receptor. In some embodiments, the first antigen and second antigen are the same. In some embodiments, the first antigen and second antigen are different.

[0429] In some embodiments, the first and/or second genetically engineered antigen receptor (e.g. CAR) is capable of inducing an activating signal to the cell. In some embodiments, the receptor includes an intracellular signaling component containing ITAM or ITAM-like motifs. In some embodiments, the activation induced by the first receptor involves a signal transduction or change in protein expression in the cell resulting in initiation of an immune response, such as ITAM phosphorylation and/or initiation of ITAM-mediated signal transduction cascade, formation of an immunological synapse and/or clustering of molecules near the bound receptor (e.g. CD4 or CD8, etc.), activation of one or more transcription factors, such as NF-KB and/or AP-1, and/or induction of gene expression of factors such as cytokines, proliferation, and/or survival.

[0430] In some embodiments, the first and/or second receptor includes intracellular signaling domains or regions of costimulatory receptors such as CD28, CD137 (4-1BB), OX40, and/or ICOS. In some embodiments, the first and second receptor include an intracellular signaling domain of a costimulatory receptor that are different. In some embodiments, the first receptor contains a CD28 costimulatory signaling region and the second receptor contain a 4-IBB co-stimulatory signaling region or vice versa.

[0431] In some embodiments, the first and/or second receptor includes both an intracellular signaling domain containing ITAM or ITAM-like motifs and an intracellular signaling domain of a costimulatory receptor.

[0432] In some embodiments, the first receptor contains an intracellular signaling domain containing ITAM or ITAM-like motifs and the second receptor contains an intracellular signaling domain of a costimulatory receptor. The costimulatory signal in combination with the activating signal induced in the same cell is one that results in an immune response, such as a robust and sustained immune response, such as increased gene expression, secretion of cytokines and other factors, and T cell mediated effector functions such as cell killing.

[0433] In some embodiments, neither ligation of the first receptor alone nor ligation of the second receptor alone induces a robust immune response. In some aspects, if only one receptor is ligated, the cell becomes tolerized or unresponsive to antigen, or inhibited, and/or is not induced to proliferate or secrete factors or carry out effector functions. In some such embodiments, however, when the plurality of receptors are ligated, such as upon encounter of a cell expressing the first and second antigens, a desired response is achieved, such as full immune activation or stimulation, e.g., as indicated by secretion of one or more cytokine, proliferation, persistence, and/or carrying out an immune effector function such as cytotoxic killing of a target cell.

[0434] In some embodiments, the two receptors induce, respectively, an activating and an inhibitory signal to the cell, such that binding by one of the receptors to its antigen activates the cell or induces a response, but binding by the second inhibitory receptor to its antigen induces a signal that suppresses or dampens that response. Examples are combinations of activating CARs and inhibitory CARs or iCARs. Such a strategy may be used, for example, in which the activating CAR binds an antigen expressed in a disease or condition but which is also expressed on normal cells, and the inhibitory receptor binds to a separate antigen which is expressed on the normal cells but not cells of the disease or condition.

[0435] In some embodiments, the multi-targeting strategy is employed in a case where an antigen associated with a particular disease or condition is expressed on a non-diseased cell and/or is expressed on the engineered cell itself, either transiently (e.g., upon stimulation in association with genetic engineering) or permanently. In such cases, by requiring ligation of two separate and individually specific antigen receptors, specificity, selectivity, and/or efficacy may be improved.

[0436] In some embodiments, the plurality of antigens, e.g., the first and second antigens, are expressed on the cell, tissue, or disease or condition being targeted, such as on the cancer cell. In some aspects, the cell, tissue, disease or condition is multiple myeloma or a multiple myeloma cell. In some embodiments, one or more of the plurality of antigens generally also is expressed on a cell which it is not desired to target with the cell therapy, such as a normal or non-diseased cell or tissue, and/or the engineered cells themselves. In such embodiments, by requiring ligation of multiple receptors to achieve a response of the cell, specificity and/or efficacy is achieved.

d. Chimeric Auto-Antibody Receptor (CAAR)

[0437] In some embodiments, the recombinant receptor is a chimeric autoantibody receptor (CAAR). In some embodiments, the CAAR binds, e.g., specifically binds, or recognizes, an autoantibody. In some embodiments, a cell expressing the CAAR, such as a T cell engineered to express a CAAR, is used to bind to and kill autoantibody-expressing cells, but not normal antibody expressing cells. In some embodiments, CAAR-expressing cells are used to treat an autoimmune disease associated with expression of self-antigens, such as autoimmune diseases. In some embodiments, CAAR-expressing cells target B cells that ultimately produce the autoantibodies and display the autoantibodies on their cell surfaces, marking these B cells as disease-specific targets for therapeutic intervention. In some embodiments, CAAR-expressing cells are used to efficiently target and kill the pathogenic B cells in autoimmune diseases by targeting the disease-causing B cells using an antigen-specific chimeric autoantibody receptor. In some embodiments, the recombinant receptor is a CAAR, such as any described in U.S. Patent Application Pub. No. US 2017/0051035.

[0438] In some embodiments, the CAAR comprises an autoantibody binding domain, a transmembrane domain, and one or more intracellular signaling region or domain (also interchangeably called a cytoplasmic signaling domain or region). In some embodiments, the intracellular signaling region comprises an intracellular signaling domain. In some embodiments, the intracellular signaling domain is or comprises a primary signaling domain, a signaling domain that is capable of stimulating and/or inducing a primary activation signal in a T cell, a signaling domain of a T cell receptor (TCR) component (e.g. an intracellular signaling domain or region of a CD3-zeta) chain or a functional variant or signaling portion thereof), and/or a signaling domain comprising an immunoreceptor tyrosine-based activation motif (ITAM).

[0439] In some embodiments, the autoantibody binding domain comprises an autoantigen or a fragment thereof. The choice of autoantigen can depend upon the type of autoantibody being targeted. For example, the autoantigen may be chosen because it recognizes an autoantibody on a target cell, such as a B cell, associated with a particular disease state, e.g. an autoimmune disease, such as an autoantibody-mediated autoimmune disease. In some embodiments, the autoimmune disease includes pemphigus vulgaris (PV). Exemplary autoantigens include desmoglein 1 (Dsgl) and Dsg3.

[0440] In some embodiments, the encoded nucleic acid is operatively linked to a "positive target cell-specific regulatory element" (or positive TCSRE). In some embodiments, the positive TCSRE is a functional nucleic acid sequence. In some embodiments, the positive TCSRE comprises a promoter or enhancer. In some embodiments, the TCSRE is a nucleic acid sequence that increases the level of an exogenous agent in a target cell. In some embodiments, the positive target cell-specific regulatory element comprises a T cell-specific promoter, a T cell-specific enhancer, a T cell-specific splice site, a T cell-specific site extending half-life of an RNA or protein, a T cell-specific mRNA nuclear export promoting site, a T cell-specific translational enhancing site, or a T cell-specific post-translational modification site. In some embodiments, the T cell-specific promoter is a promoter described in Immgen consortium, herein incorporated by reference in its entirety, e.g., the T cell-specific promoter is an IL2RA (CD25), LRRC32, FOXP3, or IKZF2 promoter. In some embodiments, the T cell-specific promoter or enhancer is a promoter or enhancer described in Schmidl et al., Blood. 2014 Apr. 24; 123(17): e68-78., herein incorporated by reference in its entirety. In some embodiments, the T cell-specific promoter is a transcriptionally active fragment of any of the foregoing. In some embodiments, the T-cell specific promoter is a variant having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity to any of the foregoing.

[0441] In some embodiments, the encoded nucleic acid is operatively linked to a "negative target cell-specific regulatory element" (or negative TCSRE). In some embodiments, the negative TCSRE is a functional nucleic acid sequence. In some embodiments, the negative TCSRE is a miRNA recognition site that causes degradation of inhibition of the viral vector in a non-target cell. In some embodiments, the exogenous agent is operatively linked to a "non-target cell-specific regulatory element" (or NTCSRE). In some embodiments, the NTCSRE comprises a nucleic acid sequence that decreases the level of an exogenous agent in a non-target cell compared to in a target cell. In some embodiments, the NTCSRE comprises a non-target cell-specific miRNA recognition sequence, non-target cell-specific protease recognition site, non-target cell-specific ubiquitin ligase site, non-target cell-specific transcriptional repression site, or non-target cell-specific epigenetic repression site. In some embodiments, the NTCSRE comprises a tissue-specific miRNA recognition sequence, tissue-specific protease recognition site, tissue-specific ubiquitin ligase site, tissue-specific transcriptional repression site, or tissue-specific epigenetic repression site. In some embodiments, the NTCSRE comprises a non-target cell-specific miRNA recognition sequence, non-target cell-specific protease recognition site, non-target cell-specific ubiquitin ligase site, non-target cell-specific transcriptional repression site, or non-target cell-specific epigenetic repression site.

[0442] In some embodiments, the NTCSRE comprises a non-target cell-specific miRNA recognition sequence and the miRNA recognition sequence is able to be bound by one or more of miR3 1, miR363, or miR29c. In some embodiments, the NTCSRE is situated or encoded within a transcribed region encoding the exogenous agent, optionally wherein an RNA produced by the transcribed region comprises the miRNA recognition sequence within a UTR or coding region.

[0443] In some embodiments, the viral vector comprising an anti-CD4 scFv or sdAb composition described herein are administered to a subject, e.g., a mammal, e.g., a human. In such embodiments, the subject is at risk of, has a symptom of, or is diagnosed with or identified as having, a particular disease or condition (e.g., a disease or condition described herein).

[0444] In some aspects, resting or non-activated T cells are contacted with a viral vector of the disclosure (e.g., a retroviral vector or lentiviral vector) that includes a CD4 binding agent. The contacting may be performed in vitro (e.g., with T cells derived from a healthy donor or a donor in need of cellular therapy) or in vivo by administration of the viral vector to a subject. In some embodiments the process comprises a) obtaining whole blood from the subject; b) collecting the fraction of blood containing leukocyte components including CD4+ T cells; c) contacting the leukocyte components including CD4+ T cells with a composition comprising the lentiviral vector to create a transfection mixture; and d) reinfusing the contacted leukocyte components including CD4+ T cells and/or the transfection mixture to the subject, thereby administering the lipid particle and/or payload gene to the subject. In some embodiments, the T cells (e.g. CD4+ T cells) are not activated during the method. In some embodiments, step (c) of the method is carried out for no more than 24 hours, e.g., no more than 20, 16, 12, 8, 6, 5, 4, 3, 2, or 1 hour.

[0445] In some embodiments, the method according to the present disclosure is capable of delivering a lentiviral particle to an ex vivo system. The method includes the use of a combination of various apheresis machine hardware components, a software control module, and a sensor module to measure citrate or other solute levels in-line to ensure the maximum accuracy and safety of treatment prescriptions, and the use of replacement fluids designed to fully exploit the design of the system according to the present methods. It is understood that components described for one system according to the present invention can be implemented within other systems according to the present invention as well.

[0446] In some embodiments, the method for administration of the lentiviral vector to the subject comprises the use of a blood processing set for obtaining whole blood from the subject, a separation chamber for collecting the fraction of blood containing leukocyte components including CD4+ T cells, a contacting container for contacting the CD4+ T cells with the composition comprising the lentiviral vector, and a further fluid circuit for reinfusion of CD4+ T cells to the patient. In some embodiments, the method further comprises any of i) a washing component for concentrating T cells, and ii) a sensor and/or module for monitoring cell density and/or concentration. In some embodiments, the methods allow processing of blood directly from the patient, transduction with the lentiviral vector, and reinfusion directly to the patient without any steps of selection for T cells or for CD4+ T cells. Further the methods also can be carried out without cryopreserving or freezing any cells before or between any one or more of the steps, such that there is no step of formulating cells with a cryoprotectant, e.g. DMSO. In some embodiments, the provided methods do not include a lymphodepletion regimen. In some embodiments, the method including steps (a)-(d) are carried out for a time of no more than 24 hours, such as between 2 hours and 12 hours, for example 3 hours to 6 hours.

[0447] In some embodiments, the method for administration of the lentiviral vector to the subject comprises the use of a blood processing set for obtaining whole blood from the subject, a separation chamber for collecting the fraction of blood containing leukocyte components including CD4+ T cells, a contacting container for contacting the CD4+ T cells with the composition comprising the lentiviral vector, and a further fluid circuit for reinfusion of CD4+ T cells to the patient. In some embodiments, the method further comprises any of i) a washing component for concentrating T cells, and ii) a sensor and/or module for monitoring cell density and/or concentration. In some embodiments, the methods allow processing of blood directly from the patient, transduction with the lentiviral vector, and reinfusion directly to the patient without any steps of selection for T cells or for CD4+ T cells. Further the methods also can be carried out without cryopreserving or freezing any cells before or between any one or more of the steps, such that there is no step of formulating cells with a cryoprotectant, e.g. DMSO. In some embodiments, the provided methods do not include a lymphodepletion regimen. In some embodiments, the method including steps (a)-(d) are carried out for a time of no more than 24 hours, such as between 2 hours and 12 hours, for example 3 hours to 6 hours.

[0448] Also provided herein are systems for administration of a lentiviral vector comprising a CD4 binding agent to a subject. An exemplary system for administration is shown in FIG. **1**.

[0449] In some embodiments, the resting or non-activated T cells are not treated with one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines. In some embodiments, the resting or non-activated T cells are not treated with any of one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines.

[0450] In additional aspects, the application includes methods of administration to a subject of a viral vector that includes an anti-CD4 binding agent, wherein the subject is not administered or has not been administered a T cell activating treatment. In some embodiments, the T cell activating treatment includes one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines. In some embodiments, the subject is not administered or has not been administered any of one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines. In some embodiments, the T cell activating treatment is lymphodepletion. In certain embodiments, the subject is not administered or has not been administered the T cell activating treatment within 1 month before or after administration of the viral vector. In some embodiments, the subject is not administered or has not been administered the T cell activating treatment within 1 month before administration of the viral vector, such as within or at or about 4 weeks, 3 weeks, 2 weeks or 1 weeks, such as at or about 1 day, 2 days, 3 days, 4 days, 5 days, 6 days or 7 days before administration of the viral vector. In some embodiments, the subject is not administered the T cell activating treatment within 1 month after administration of the viral vector, such as within or at or about 4 weeks, 3 weeks, 2 weeks or 1 weeks, such as at or about 1 day, 2 days, 3 days, 4 days, 5 days, 6 days or 7 days after administration of the viral vector.

[0451] In some aspects, the viral vectors of the disclosure do not include one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines.

[0452] The use of anti-CD3 antibodies is well-known for activation of T cells. The anti-CD3 antibodies can be of any species, e.g., mouse, rabbit, human, humanized, or camelid. Exemplary antibodies include OKT3, CRIS-7, I2C the anti-CD3 antibody included in DYNABEADS Human T-Activator CD3/CD28 (Thermo Fisher), and the anti-CD3 domains of approved and clinically studied molecules such as blinatumomab, catumaxomab, fotetuzumab, teclistamab, ertumaxomab, epcoritamab, talquetamab, odronextamab, cibistamab, obrindatamab, tidutamab, duvortuxizumab, solitomab, eluvixtamab, pavurutamab, tepoditamab, vibecotamab, plamotamab, glofitamab, etevritamab, and tarlatamab.

[0453] In some embodiments, the one or more T cell costimulatory molecules include CD28 ligands (e.g., CD80 and CD86); antibodies that bind to CD28 such as CD28.2, the anti-CD28 antibody included in DYNABEADS Human T-Activator CD3/CD28 (Thermo Fisher) and anti-CD28 domains disclosed in US2020/0199234, US2020/0223925, US2020/0181260, US2020/0239576, US2020/0199233, US2019/0389951, US2020/0299388, US2020/0399369, and US2020/0140552; CD137 ligand (CD137L); anti-CD137 antibodies such as urelumab and utomilumab; ICOS ligand (ICOS-L); and anti-ICOS antibodies such as feladilimab, vopratelimab, and the anti-ICOS domain of izuralimab.

[0454] In some embodiments, the one or more T cell activating cytokines include IL-2, IL-7, IL-15, IL-21, interferons (e.g., interferon-gamma), and functional variants and modified versions thereof.

[0455] Lymhpodepletion may be induced by various treatments that destroy lymphocytes and T cells in the subject. For example, the lymphodepletion may include myeloablative chemotherapies, such as fludarabine, cyclophosphamide, bendamustine, and combinations thereof. Lymphodepletion may also be induced by irradiation (e.g., full-body irradiation) of the subject.

[0456] In some embodiments, the source of targeted lipid particles is the same subject that is administered a targeted lipid particle composition. In other embodiments, they are different. In some embodiments, the source of targeted lipid particles and recipient tissue is autologous (from the same subject) or heterologous (from different subjects). In some embodiments, the donor tissue for targeted lipid particle compositions described herein is a different tissue type than the recipient tissue. In some embodiments, the donor tissue is muscular tissue and the recipient tissue is connective tissue (e.g., adipose tissue). In other embodiments, the donor tissue and recipient tissue are of the same or different type, but from different organ systems.

[0457] In some embodiments, the targeted lipid particles (e.g, viral vector) composition described herein are administered to a subject having a cancer, an autoimmune disease, an infectious disease, a metabolic disease, a neurodegenerative disease, or a genetic disease (e.g., enzyme deficiency). In some embodiments, the subject is in need of regeneration.

[0458] In some embodiments, the cancer is a T cell-mediated cancer. In another embodiment, the antigen binding moiety portion of a CAR is designed to treat a particular cancer. In some embodiments, the targeted lipid particle is used to treat cancers and disorders including but not limited to non-Hodgkin lymphoma (NHL), acute lymphocytic leukemia (ALL), chronic lymphocytic leukemia (CLL), multiple myeloma, and the like. In some embodiments, the targeted lipid particle is used to treat B cell malignancies, e.g., refractory B cell malignancies.

[0459] In some embodiments, the targeted lipid particle is co-administered with an inhibitor of a protein that inhibits membrane fusion. For example, Suppressyn is a human protein that inhibits cell-cell fusion (Sugimoto et al., "A novel human endogenous retroviral protein inhibits cell-cell fusion" *Scientific Reports* 3:1462 DOI: 10.1038/srep01462). In some embodiments, the targeted lipid particle is co-administered with an inhibitor of sypressyn, e.g., a siRNA or inhibitory antibody.

EXAMPLES

[0460] The present disclosure may be further described by the following non-limiting examples, in which standard techniques known to the skilled artisan and techniques analogous to those described in these examples may be used where appropriate. It is understood that the skilled artisan will envision additional embodiments consistent with the disclosure provided herein.

Example 1 Characterization of CD4 Binders

[0461] Binders were selected that demonstrated detectable CD4 binding in solution. To assay the ability of the binders to direct cell-specific transduction, the binders were used to generate binder (e.g., as scFv)-Nipah G glycoprotein fusions as described in WO2017182585 for pseudotyping of lentiviral vectors. The Nipah G-linker-binder construct was codon optimized for expression in human cells and subcloned into an expression vector for lentivirus generation.

[0462] Crude lentiviral production was performed as follows: HEK-293LX cells were plated 24 hours in advance of transfection. On the day of transfection, HEK-293LX cells were transfected with a lentiviral packaging plasmid, the lentiviral transfer plasmid encoding GFP (pSFFV-GFP), and the plasmids encoding for Nipah G protein retargeted for CD4 receptor targeting (NIV-G (CD4)) and Nipah F fusion protein (NiV-Fd22). To harvest the lentivirus, supernatant was removed from the HEK293LX cells and spun at 1000×g for 5 minutes. The supernatant was removed and immediately added to CD4-positive target cells or T cells, or frozen at −80° C. for later use. Lentiviral vectors were produced in both adherent cells and in suspension. In certain experiments, the vectors were filtered using a 0.45 μm filter and concentrated by ultracentrifugation.

[0463] To estimate activity of the CD4-retargeted vectors, the supernatants were diluted 1:5 and used to transduce SupT1 cells. After 5 days, the transduced cells were assayed for GFP expression by flow cytometry. The percentage of live cells expressing GFP is shown in Table 17, column "Single point SupT1". Titer of certain CD4-retargeted vectors was determined by multi-point dilution of vector for transduction of SupT1 cells (adherent and suspension production) (Table 17, columns "Multiple point SupT1 Adh." and "Multiple point SupT1 Susp."). Titer was similarly determined using HEK-293T cells overexpressing *Macaca nemestrina* CD4 to estimate cross-reactivity of the vectors (Table 17, column "Multiple point 293oeNemestrinaCD4").

TABLE 17

Binder	Single point SupT1 1:5 dilution GFP (% GFP)	Multiple point SupT1 Adh. titer [Log(TU/mL)]	Multiple point SupT1 Susp. titer [Log(TU/mL)]	Multiple point 293oeNemestri-naCD4 Adh. titer [Log(TU/mL)]
265	24.1 ± 10.8	5.8 ± 0.25		
266	13.1 ± 1.6	5.51 ± 0.05		
267	10.1 ± 5.8	5.27 ± 0.25	4.7	
263	4.5 ± 7			
268	34.7 ± 7.8	6.1 ± 0.09	5.65 ± 0.57	
270	56.5 ± 12.1	6.53 ± 0.3	5.32 ± 0.23	
271	52.9 ± 11.8	6.51 ± 0.21	5.38 ± 0.28	
273	44 ± 12.2	6.29 ± 0.26	5.7 ± 0.51	5.31 ± 0.76
274	23 ± 18	5.8 ± 0.36	5.61 ± 0.13	
275	53.9 ± 3.2	6.46 ± 0.24	5.45 ± 0.14	
276	9.4 ± 0.1	5.35 ± 0.05		
277	4.2 ± 1.5	5.10		
278	36.6 ± 5	6.12 ± 0.04	5.3	
279	37.5 ± 34.8	6.13 ± 0.67	5.71 ± 0.19	
264	11.3 ± 0			
257	47.3 ± 1.1	6.1 ± 0.15		
280	25.9 ± 2.1	6 ± 0.04	5.34 ± 0.12	
282	12.8 ± 1.1	5.51 ± 0.05	5.19 ± 0.12	
283	41.1 ± 8.2	6.26 ± 0.11	5.24 ± 0.09	
284	17.9 ± 2.3	5.65 ± 0.07		
285	5.9 ± 1.8	5.24		
286	59.3 ± 25.7	6.23 ± 0.39	5.46 ± 0.03	6.81 ± 0.06
258	49.6 ± 0.8	6.1 ± 0.17		
287	38.8 ± 22.8	5.96 ± 0.49	5.28 ± 0.2	
288	5.3 ± 0.5	5.18		
289	9.8 ± 2.8	5.39 ± 0.12		

TABLE 17-continued

Binder	Single point SupT1 1:5 dilution GFP (% GFP)	Multiple point SupT1 Adh. titer [Log(TU/mL)]	Multiple point SupT1 Susp. titer [Log(TU/mL)]	Multiple point 293oeNemestri-naCD4 Adh. titer [Log(TU/mL)]
290	27.7 ± 14.5	5.91 ± 0.4	5.36 ± 0.19	
291	0.6 ± 0.5			
260	0.2 ± 0			
292	21.8 ± 8.3	5.9 ± 0.3	5.63 ± 0.29	6.59 ± 0.03
293	3.9 ± 0.3			
294	11.6 ± 2.3	5.45 ± 0.08		
295	11.3 ± 8.9	5.74 ± 0.28	5.2 ± 0.19	
296	5.8 ± 2.4	5.10		
261	3.1 ± 0			
297	31.8 ± 4.4	5.89 ± 0.26	5.34 ± 0.14	
298	19.9 ± 0.3	5.79 ± 0.12		
300	16.3 ± 4.1	5.59 ± 0.12		
301	28.2 ± 8.7	5.93 ± 0.21	5.34 ± 0.2	
302	0.2 ± 0.2			
304	20.7 ± 9.9	5.74 ± 0.28	5.15 ± 0.11	
305	30.7 ± 9.9	6.01 ± 0.15	5.26 ± 0.26	
306	0.3 ± 0			
307	4.7 ± 2	5.24		
314	25 ± 4.5	5.89 ± 0.12	5.43 ± 0.21	
315	21.3 ± 5.9	5.79 ± 0.14	5.61 ± 0.37	
316	0.2 ± 0.1			
317	12.5 ± 4.4	5.41 ± 0.15	4.85	
319	35.4 ± 17.2	6.16 ± 0.11	5.31 ± 0.18	
320	14 ± 7.4	5.44 ± 0.3	5.08 ± 0.21	
322	59.7 ± 10.3	6.44 ± 0.2	5.57 ± 0.26	
323	20.5 ± 4.5	5.77 ± 0.13		
324	14.2 ± 6.9	5.41 ± 0.18	5.02 ± 0.42	4.62
325	21.4 ± 5.1	5.82 ± 0.17	5.44 ± 0.14	
326	43.7 ± 7.7	6.25 ± 0.15	5.54 ± 0.25	
327	0.3 ± 0			
262	0 ± 0			
328	23.5 ± 9.1	5.76 ± 0.16	5.49 ± 0.28	
329	10.5 ± 2.4	5.34 ± 0.15	4.92 ± 0.11	
330	14.1 ± 0.8	5.56 ± 0.02	5.17 ± 0.1	
331	24.6± 1.2	5.92 ± 0.04	5.21 ± 0.06	
333	22.9 ± 13.6	5.89 ± 0.26	4.85	5.78 ± 1
334	5.5 ± 2.1	5.24		5.25 ± 0.1
335	17.6 ± 2.3	5.64 ± 0.05	5.21 ± 0.12	
336	0.1 ± 0.1			
337	27.4 ± 6.5	6.01 ± 0.11	5.25 ± 0.07	
339	23.1 ± 5.2	5.8 ± 0.13	5.09 ± 0.09	
340	13 ± 2.2	5.43 ± 0.15	5.15 ± 0.08	
109	17.7 ± 3.5	5.66 ± 0.12	5.17 ± 0.1	
110	0.4 ± 0.1			
111	57.3 ± 5.4	6.41 ± 0.1	5.62 ± 0.11	
112	7.6 ± 1.4	5.28 ± 0.07	5.18	5.37 ± 0.02
1	79.2 ± 6	6.25 ± 0.5	5.84 ± 0.13	
113	45 ± 32.1	6.54 ± 0.26	5.68 ± 0.23	4.68 ± 0.08
114	65.2 ± 9.8	6.6 ± 0.11	5.6 ± 0.08	
115	27.8 ± 5.9	5.91 ± 0.14	5.44 ± 0.06	
23	59.9 ± 7	6.51 ± 0.12	5.69 ± 0.26	
116	51.6 ± 9.4	6.4 ± 0.17	5.51 ± 0.05	
117	54.1 ± 14	6.45 ± 0.26	5.66 ± 0.25	5.41
2	32.1 ± 3	6.04 ± 0.03	5.61 ± 0.09	
118	61.2 ± 2.3	6.57 ± 0	5.6	
119	55.4 ± 13.1	6.38 ± 0.23	5.74 ± 0.3	4.73
120	42.3 ± 8.6	6.21 ± 0.06	5.46 ± 0.03	
121	0.6 ± 0.6			
122	67.1 ± 10.1	6.72 ± 0.07	5.54	
123	56.9 ± 2.6	6.57 ± 0	5.65 ± 0.13	
38	59.9 ± 14.6	6.51 ± 0.26	5.88 ± 0.31	4.86 ± 0.28
39	53.1 ± 23.1	6.36 ± 0.24	5.67 ± 0.41	5.96 ± 0.76
40	61.8 ± 11.3	6.43 ± 0.21	5.88 ± 0.3	5.39 ± 0.71
41	64.9 ± 7.6	6.52 ± 0.24	5.8 ± 0.4	5.42 ± 0.78
42	61.8 ± 10.1	6.49 ± 0.18	5.95 ± 0.4	5.39 ± 0.69
43	45.2 ± 10.2	6.21 ± 0.1	5.48 ± 0.2	4.68
44	58.3 ± 8.5	6.44 ± 0.2	5.65 ± 0.21	4.81 ± 0.11
45	8.2 ± 1.3	5.25 ± 0.08	5.19 ± 0.27	4.68
46	71.7 ± 11.7	6.56 ± 0.24	5.78 ± 0.18	5.86 ± 0.9
47	77.9 ± 11.1	6.6 ± 0.21	5.77 ± 0.21	5.44 ± 0.92
48	80.7 ± 9.1	6.57 ± 0.47	5.8 ± 0.22	5.53 ± 0.78
49	52.4 ± 13.3	6.36 ± 0.24	5.59 ± 0.27	4.87 ± 0.07

TABLE 17-continued

Binder	Single point SupT1 1:5 dilution GFP (% GFP)	Multiple point SupT1 Adh. titer [Log(TU/mL)]	Multiple point SupT1 Susp. titer [Log(TU/mL)]	Multiple point 293oeNemestri-naCD4 Adh. titer [Log(TU/mL)]
50	77.5 ± 8.8	6.68 ± 0.22	5.76 ± 0.23	5.26 ± 0.96
51	59.4 ± 18.2	6.51 ± 0.3	5.74 ± 0.25	4.6 ± 0.18
52	59 ± 10.7	6.39 ± 0.19	5.57 ± 0.22	5.56 ± 1.02
54	65.3 ± 9.5	6.53 ± 0.24	5.7 ± 0.27	5.2 ± 0.93
55	42.6 ± 2.8	5.94 ± 0.24		5.95 ± 0.2
56	71.8 ± 3.7	6.5 ± 0.15		
3	1.2 ± 0.1			
4	1 ± 0			
6	1.1 ± 0.1			
7	1.2 ± 0.4			
8	1 ± 0.2			
10	0 ± 0			
11	1.2 ± 0.2			
12	1.3 ± 0.2			
13	0.1 ± 0			
57	15.5 ± 8.2	5.49 ± 0.21	5.09 ± 0.13	5.17 ± 0.21
125	5 ± 2.3	5.24		
129	0.2 ± 0			
130	0.2 ± 0.1			
131	0.1 ± 0			
132	0.1 ± 0.1			
134	0.2 ± 0.1			
64	0.1 ± 0.1			
65	0 ± 0			
135	11.4 ± 4.8	5.35 ± 0.2	5.19 ± 0.07	4.56
136	14.3 ± 5.2	5.45 ± 0.2	5.31 ± 0.12	4.62
137	24.7 ± 4.2	5.94 ± 0.06	5.57 ± 0.24	4.56
138	17.4 ± 2.1	5.69 ± 0.1		5.44 ± 0.02
15	38.2 ± 8.6	6.1 ± 0.15	5.4 ± 0.2	5.56 ± 0.83
66	9.1 ± 2.3	5.28 ± 0.12	5.06 ± 0.23	
67	23.9 ± 4.8	5.91 ± 0.11	5.22 ± 0.32	4.68
68	25.5 ± 1.2	5.94 ± 0	5.36 ± 0.16	4.68
69	19.7 ± 4.2	5.75 ± 0.17	5.13 ± 0.16	4.52 ± 0.06
24	0.2 ± 0			
139	0.1 ± 0.1			
140	0 ± 0			
141	10.2 ± 5	5.29 ± 0.23	5 ± 0.1	4.56
142	4.6 ± 1.3	5.18		5.29 ± 0.05
70	9.9 ± 2.9	5.38 ± 0.13	5.3 ± 0.08	5.37 ± 0.02
25	0.1 ± 0			
144	0.1 ± 0.1			
145	0 ± 0			
146	56.1 ± 8.2	6.43 ± 0.13	5.76 ± 0.07	
147	8.7 ± 0.8	5.33 ± 0.04		
148	53.3 ± 14.3	6.35 ± 0.2	5.91 ± 0.23	4.83 ± 0.14
71	62.1 ± 9.5	6.39 ± 0.21	6.04 ± 0.4	5.22 ± 0.54
72	25.5 ± 0.3	5.84 ± 0.06	5.24	
73	55.4 ± 23.3	6.41 ± 0.24	6.13 ± 0.28	5.83 ± 0.55
74	58.5 ± 25.4	6.53 ± 0.23	5.99 ± 0.32	5.76 ± 0.55
75	56.3 ± 28.5	6.38 ± 0.38	6.18 ± 0.33	5.49 ± 0.6
149	0.5 ± 0.3			5.13
150	0.1 ± 0.1			
26	1.2 ± 0.1			5.24 ± 0.03
77	3.8 ± 2.4	4.96 ± 0.17	4.63 ± 0.04	5.34 ± 0.08
151	0.1 ± 0			
152	0.2 ± 0.1			
153	0.1 ± 0.1			
154	5 ± 0.5	5.10		
155	2.7 ± 0.4			
156	10.3 ± 1.8	5.41 ± 0.09		4.95 ± 0.1
80	6.1 ± 1.3	5.14 ± 0.09	5.24 ± 0.12	
82	15 ± 2	5.57 ± 0.04	5.34 ± 0.14	
17	0.1 ± 0.1			
157	0.1 ± 0			
27	0.1 ± 0			
158	0 ± 0			

TABLE 17-continued

Binder	Single point SupT1 1:5 dilution GFP (% GFP)	Multiple point SupT1 Adh. titer [Log(TU/mL)]	Multiple point SupT1 Susp. titer [Log(TU/mL)]	Multiple point 293oeNemestrinaCD4 Adh. titer [Log(TU/mL)]
159	10.9 ± 1.7	5.44 ± 0.06		
83	20.2 ± 7	5.75 ± 0.19	5.25 ± 0.1	4.62
160	0.1 ± 0			
28	1.3 ± 0.4			4.99 ± 0.05
29	0.4 ± 0.1			
161	0 ± 0			
162	0 ± 0			
18	11.7 ± 4.8	5.41 ± 0.17	5.13 ± 0.06	
85	23.1 ± 1.7	5.85 ± 0.05	5.05 ± 0.18	
164	1 ± 0.1			
165	1.3 ± 0.6			
87	0 ± 0			
32	0.8 ± 0.4			
166	17.1 ± 0.5	5.75 ± 0.14		
88	36.3 ± 11.5	6.15 ± 0.21	5.37 ± 0.16	5.16 ± 0.19
168	37.6 ± 13.2	5.99 ± 0.28	5.51 ± 0.11	5.42 ± 0.9
89	48.3 ± 16.1	6.19 ± 0.28	5.49 ± 0.22	5.52 ± 1.01
19	0.1 ± 0			
91	0 ± 0			
92	0 ± 0			
93	41.1 ± 12.7	6.12 ± 0.19	5.88	6.36 ± 0.28
176	0.4 ± 0.3			
177	0.4 ± 0.1			
178	0 ± 0			
33	0.1 ± 0.1			
179	0.1 ± 0.1			
34	0.1 ± 0.1			
180	14.6 ± 4	5.55 ± 0.11		
95	39.2 ± 20.5	6 ± 0.22	4.77 ± 0.1	5.79 ± 1.02
181	0 ± 0			
182	19.4 ± 3.6	5.76 ± 0.14	5.24 ± 0.08	5.02 ± 0.65
97	36.1 ± 14.4	6.06 ± 0.26	5.13 ± 0.09	5.99
99	5.1 ± 0.1	4.91 ± 0.15	4.78	
100	6.5 ± 0.8	5.03 ± 0.17	5.03 ± 0.17	
36	11.8 ± 3.4	5.47 ± 0.12	5.13 ± 0.1	4.94 ± 0.31
101	35.7 ± 10.9	6.07 ± 0.2	5.31 ± 0.16	5.35 ± 0.64
185	8 ± 2.5	5.22 ± 0.19	5.32 ± 0.13	
103	32 ± 8.1	6.04 ± 0.14	5.35 ± 0.17	
192	15.4 ± 7.7	5.49 ± 0.26	5.17 ± 0.1	4.88 ± 0.46
193	0.1 ± 0.1			
195	14.1 ± 7.2	5.42 ± 0.28	5.17 ± 0.08	
104	31.4 ± 10.6	6 ± 0.23	5.24 ± 0.09	5.56 ± 1.2
37	0 ± 0			
198	0 ± 0			
200	0 ± 0.1			
201	0 ± 0			
202	0 ± 0			
203	66.1 ± 13.8	6.67 ± 0.23	5.7 ± 0.26	5.6 ± 0.98
204	0 ± 0			
206	0.4 ± 0.1			
208	0 ± 0			
210	0 ± 0			
211	16.4 ± 8	5.6 ± 0.31		6.61 ± 0.06
213	9 ± 0.2	5.35 ± 0		
216	24.1 ± 14.8	5.7 ± 0.45	4.95 ± 0.21	5.66 ± 1.28
217	0 ± 0			
218	0.1 ± 0.1			
219	0±0			
220	0 ± 0			
22	0 ± 0			
341	0 ± 0			
342	0.4 ± 0			
343	22 ± 3.1	5.86 ± 0.15		
344	11 ± 2.8	5.43 ± 0.11		5.35
345	38.5 ± 4.3	6.06 ± 0.27		6.45 ± 0.04
346	1 ± 0.2			
347	0 ± 0			
348	21.6 ± 2.4	5.9 ± 0.18		
349	0 ± 0			
350	41.2 ± 13.6	6.13 ± 0.34	5.1	
351	33.3 ± 10	5.89 ± 0.49	5.3 ± 0.08	6.02 ± 0.09
352	3.1 ± 0			

TABLE 17-continued

Binder	Single point SupT1 1:5 dilution GFP (% GFP)	Multiple point SupT1 Adh. titer [Log(TU/mL)]	Multiple point SupT1 Susp. titer [Log(TU/mL)]	Multiple point 293oeNemestri-naCD4 Adh. titer [Log(TU/mL)]
353	20.7 ± 0.5	5.84 ± 0.06		
354	0.6 ± 0.1			
355	18.3 ± 3.1	5.74 ± 0.19		
356	2.4 ± 0.4			
357	0.9 ± 0.1			5.41
358	39.6 ± 15	6.19 ± 0.17	5.1	
359	0.9 ± 0.1			
360	0.2 ± 0.1			
361	12.7 ± 2	5.49 ± 0.07		
362	8.6 ± 1.3	5.35 ± 0.07		
363	13.3 ± 3	5.51 ± 0.1		
364	60.5 ± 12.3	6.44 ± 0.36	5.8	5.75 ± 0.69
259	54.9 ± 6.8	6.24 ± 0.13		5.97 ± 0.12
365	2.4 ± 0.3			
366	20.5 ± 2	5.83 ± 0.14		
367	51.8 ± 16.3	6.54 ± 0.19	5.49 ± 0.07	
368	7 ± 0.4	5.24 ± 0		
369	37.7 ± 18.8	6.13 ± 0.25	5.33 ± 0.04	6.59 ± 0.24
370	11 ± 1	5.44 ± 0.06		
371	5.9 ± 0.8	5.14 ± 0.06		5.4 ± 0.07
372	2.5 ± 0.7			
373	0.1 ± 0.1			
374	24 ± 4.2	6.02 ± 0.11		
375	24.6 ± 0.5	6.03 ± 0.04		
376	8.8 ± 1.1	5.35 ± 0.07		
377	0.7 ± 0.1			
378	21 ± 0.2	5.88 ± 0	5.18	
379	45.9 ± 12.1	6.23 ± 0.24	5.64 ± 0.02	
380	6.7 ± 0.7	5.21 ± 0.05		
381	1.4 ± 0.7			
382	49 ± 0.2	6.2 ± 0.16	5.1	6.67 ± 0.07
383	11.3 ± 0.9	5.46 ± 0.03		
384	51.2 ± 1.1	6.23 ± 0.18	5.24	
385	17.6 ± 8.9	5.62 ± 0.26		6.57
386	45.1 ± 13.3	6.25 ± 0.15	5.24 ± 0.09	6.24 ± 0.09
223	2.4 ± 0.4			
224				6.45 ± 0.04
106	0.9 ± 0.1			6.45 ± 0.04
225	1.5 ± 0.1			
226	0.1 ± 0			
228	15.5 ± 1.1	5.39 ± 0.16		6.21 ± 0.1
229	0.6 ± 0.4			
107	0.1 ± 0.1			
238	0±0			
239	0 ± 0			
240	0.1 ± 0			
241	41.1 ± 2.1	5.77 ± 0.24		5.99 ± 0.11
243	1.7 ± 0.6			
256	60.5 ± 11.3	6.31 ± 0.4	6.37 ± 1.18	5.91 ± 1.33

[0464] Titer was also determined on SupT1 cells following concentration of the viral vector compositions (Table 26).

TABLE 26

Binder	Modality	CD4 domain specificity	Nemestrina Cross-Reactivity	SupT1 titer (crude, adherent, PE) TU/mL	SupT1 titer (crude, suspension, PE) TU/mL	SupT1 titer (concentrated, suspension, PE) TU/mL [conc factor]*
387	DARPin	Domain 1	Cross-reactive	6.50E6 ± 1.47E6	2.33E6*	1.2E8 [200X]
256 (VHH4)*	VHH	Domain 2	Cross-reactive	4.73E6 ± 2.79E6	2.34E6*	5.22E7 [200X]
388*	VHH	Domain 3/4*	Cross-reactive	1.18E ± 1.26E6	Too low*	5.45E6 [200]

TABLE 26-continued

Binder	Modality	CD4 domain specificity	Nemestrina Cross-Reactivity	SupT1 titer (crude, adherent, PE) TU/mL	SupT1 titer (crude, suspension, PE) TU/mL	SupT1 titer (concentrated, suspension, PE) TU/mL [conc factor]*
279	scFv	Domain 3/4	No cross-reactivity	2.85E6 ± 3.35E6	5.50E5 ± 2.46E5	1.43E8 [200X]
286	scFv	Domain 3/4	Cross-reactive	2.27E6 ± 1.78E6	2.88E5 ± 1.77E4	5.47E7 [200X]
287	scFv	Domain 2	No cross-reactivity	1.73E6 ± 1.29E6	1.94E5 ± 8.40E4	3.83E7 [200X]
322	scFv	Domain 3/4	No cross-reactivity	3.03E6 ± 1.42E6	4.40E5 ± 3.25E5	4.66E7 [200X]
47	scFv	Domain 1/2	Cross-reactive	4.39E6 ± 1.98E6	6.47E5 ± 3.21E5	6.86E7 [200X]
48	scFv	Domain 1/2	Cross-reactive	5.91E6 ± 4.47E6	6.93E5 ± 3.22E5	8.75E7 [200X]
75	scFv	Domain 2	Cross-reactive	3.80E6 ± 1.75E6	1.44E6	1.36E8 [200X]
364	scFv	Domain 2	Cross-reactive	3.41E6 ± 1.70E6	7.42E5	5.67E7 [200X]

[0465] Off-target transduction directed by certain binders was determined using CD4 knockout SupT1 cells and HEK-293T cells, which were determined to be negative for CD4 expression. CD4-retargeted vectors expressing GFP were produced in either adherent or suspension culture, as described above, and used to transduce CD4 knockout SupT1 cells and HEK-293T cells at a single dilution. The percentage of GFP-expressing cells was determined by flow cytometry (FIG. **2**).

[0466] Transduction efficiency on human PBMCs and Pan T cells was also determined for CD4-targeted lentiviral vectors. Concentrated vector was produced as described above and used to transduce human PBMCs from 3 donors, and the transduced cells were assayed for GFP expression by flow cytometry five days after transduction. The flow cytometry results are presented in FIG. **3**. The percent of live cells that were GFP+ are shown in FIG. **4** The vectors showed a strong specificity for CD4+ cells vs. CD4− cells.

[0467] Binding kinetics of certain CD4 binders were assayed using biolayer interferometry (BLI). The CD4 binders were expressed as homodimers with mouse Fc. Human CD4-Fc was used as the capture reagent. Kinetic parameters are shown in Table 18 below.

TABLE 18

Binder	Conc [nM]	K_D [nM]	K_{on} [1/Ms]	K_{off} [1/s]
387	250	7	2.47E+04	1.75E−05
256	500	5.14	8.95E+04	4.60E−04
279	125	5	4.17E+04	2.10E−04
286	125	13.8	7.19E+02	2.70E−04
322	62.5	1.7	9.94E+04	1.75E−04
47	62.5	0.3	4.79E+05	5.70E−05
75	62.5	0.7	3.12E+05	2.05E−04
364	62.5	7.54	3.04E+04	2.30E−06

Example 2. Transduction of Resting T Helper Cells Using a CD4 Targeted Fusogen to Generate CAR T Cells

[0468] A CD19-specific CAR encoding 4-1BB and the CD3zeta endodomains (CD 19 CAR) was generated to examine CD4+ CAR T transduction efficiency and functionality. PBMCs were thawed and activated with anti-CD3/anti-CD28 beads and exposed to GFP-expressing CD4 fusosomes (Binder 256), and specificity of targeting CD4+ T cells was measured by flow cytometry.

[0469] CD19 CAR fusosomes targeting CD4 were used to test transduction efficiency against activated (CD3/CD28 or IL-7 treated) or resting CD4+ T cells, and to measure T cell function against CD19+ and CD19 CRISPR/Cas9-knockout lymphoma cells (Nalm-6) (e.g., tumor co-culture and rechallenge assays and cytokine production) in vitro. Vector copy number (VCN) was determined by a multiplex digital droplet polymerase chain reaction (ddPCR) assay and reported as copies per diploid genome (c/dg). CD4-targeted CD 19 CAR fusosomes could efficiently transduce both activated (34%+ 1.5% CD4+ CAR+; 0.54±0.18 c/dg), and resting CD4-selected T cells, albeit at a lower expression and integration level (20%±0.5% CD4+ CAR+; 0.28±0.14 c/dg). Resting CD4-transduced CART cells demonstrated specific cytotoxicity and cytokine production (GM-CSF, IFN-γ, TNF-α, IL-2, IL-6, and IL-10) against CD19+ Nalm-6 cells, but did not recognize CD19 knockout tumor cells. In long-term co-culture assays (9-day) with repetitive stimulation with fresh tumor cells (3×), CD4+CD 19 CAR T cells transduced without prior activation continued to show potent tumor cell killing.

[0470] CD4-specific fusosomes encoding LVV were observed to efficiently deliver an integrating CAR payload to resting and activated CD4+ T cells. Modified CD4+ CAR T cells demonstrated potent anti-tumor activity against CD19+ tumor cells. These data are consistent with a finding that targeting the CD4 co-receptor through in vivo delivery using a novel pseudotyped LVV can produce functional CAR T cells.

Example 3. CD4-Targeted Fusosomes Reduce CD19+ Tumor Burden In Vivo

[0471] CD4-targeted CD19 CAR fusosomes (lentiviral vector) were generated substantially as described above and assessed for their ability to reduce tumor burden in vivo. The fusosomes were pseudotyped with Nipah virus fusogen retargeted with CD4 Binder 256, CD4 Binder 75, and a CD8 Binder Control. NSG mice (n=5/group) were injected with

1E6 Nalm6-Luc leukemia B cells via intravenous (IV) tail injection, followed three days later by an IV tail injection of 1E7 human peripheral blood mononuclear cells (hPBMC). A day after hPBMC injection, 2.5E6, 5E6, or 1E7 integrating units (IU) of CD4-targeted CD19 CAR fusosomes were injected into separate groups of mice via the same route and volume. Beginning 1 day following fusosome injection, Nalm6 tumor progression was tracked via bioluminescent imaging (BLI) weekly throughout the duration of the study. Tumor growth data is represented as total flux (photons/sec). In addition, peripheral blood was collected from all animals on study day 15 to assess the presence of CAR+ T− cells in peripheral blood using flow cytometry. The CD19 CAR contained an anti-scFv directed against CD19 and an intracellular signaling domain containing intracellular components of 4-1BB and CD3-zeta.

[0472] As shown in FIGS. **5**A and **5**B, dose dependent tumor control was observed with Binder 256 and CD8 Control Binder at Day 21. Binder 75 showed a reduced ability to control Nalm6 growth, as shown in FIG. **5**C. As shown in FIG. **5**D, expression of the CAR in CD4+ T cells was dose-dependent. These data indicate that in vivo delivery of a CD19 CAR transgene payload with CD4-targeted fusosomes in CD19+ tumor bearing mice demonstrates robust production of CAR T cells and CD19+ tumor eradication.

TABLE 19

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEIFPGSGHTSFNEKFKGKATFTADTSSNTAYIQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	1	1
QVQLQQSGAELMKPGASVKISCKATGYTLSSYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	2	2
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGATSYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGEYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVFAATYLADGVPSRFSGSGSGTQYSLKINSLQSEDFGNYYCQHFWGTPWTFGGGTKLEIK	3	3
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	4	4
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTSYNQKFKGKATFTVDTSSSTAYMHFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQVLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGSPWTFGGGTKLEIK	5	5
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPRLLVYAATNLADGVPSRFSGSGSGTQYSLKITSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	6	6
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGANGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	7	7
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQVLVYAATNVADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	8	8
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASDNIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	9	9

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPRLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	10	10
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLAVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	11	11
LVKTGASVKISCKASGYSFTGFYMHWVKQSHGKGLEWIGYISSYNGATGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVFAATYLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	112	12
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGATGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGSPWTFGGGTKLEIK	13	13
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGTTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAYWGQGTLVTVSAGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWFQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTFSSMEAEDAATYYCQQWSSNPLYTFGGGTKLEIK	14	14
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGGGTKLELK	15	15
QVQLQQPGAELIKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARPGGYGFVYWGQGTLVTVSAGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTINNLESEDFADYYCLQFYEFPLTFGAGTKLELK	16	16
EVQLQQSGAELVKPGASVKLSCTPSGFNIKDTSLHWVKQGPEQGLEWIGRIDPANGNTKYDPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCARGPDDGYFYYYSMDYWGQGTSVTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	17	17
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMGWINTYTGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDMATYFCARKYYDYEFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIVMTQSPSSLAMSVGQKVTMSCKSSQSLLNSSNQKNYLAWYQQKPGQSPKLLVYFASTRESGVPDRFIGSGSGTDFTLTISSVQAEDLADYFCQQHYSTPLTFGAGTKLELK	18	18
QVQLQESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARHEEANWAWFAYWGQGTLVTVSAGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISRMEAEDAATYYCQQRSSFPYTFGGGTKLEIK	19	19
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQANDTAIYYCASYYGSSRSYWYLDVWGAGTTVTVSSGGGSGGGGSGGGGSSIVMTQTPKFLLVSAGDRVTITCKASQSVSNDVAWYQQKPGQSPKLLIYYASNRYTGVPDRFTGSGYGTDFTFTISTVQAEDLAVYFCQQDYTSLPTFGAGTKLEIK	20	20
QVKLVESGGDLVKPGGSLKLSCATSGFTFSSYGMSWVRQTPDKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEGTAMYYCARHNYSNWDWFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIQMTQSPSSLSASLGDTITITCHASQNINVWLSWYQQKPGNIPKLLIYKASNLHTGVPSRFSGSGSGTGFTLTISSLQPEDIATYYCQQGQSYPLTFGGGTNLEIK	21	21

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
EVQLQQSGAELVKPGASVKLSCTPSGFNIKDTSLHWVKQGPEQGLEWIGRIDPAN GNTKYDPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCARGPDDGYFYYYSMD YWGQGTSVTVSSGGGSGGGGSGGGGSETTVTQSPASLSMAIGEKVTIRCITSTDI DDDMNWYQQKPGEPPKLLISEGNSLRPGVPSRFSSSGYGTDFVFTIENMLSEDVA DYYCLQSDNLPLTFGAGTKLELK	22	22
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC LQYDEFPPTFGAGTKLELK	23	23
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPGQGLEWIGLIDP SDSETHYNQVFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCATYDVYYRFAYWG QGTLVTVSAGGGSGGGGSGGGGSDIQMTQSPSSLSASLGGKVTITCKASQDINKY IAWYQHKPGKGPSLLIHYTSTLQPGIPSRFSGSGSGRDYSFSISNLEPEDIATYYCLQY DNLMYTFGGGTKLEIK	24	24
QVQLQQPGAELVRPGASVKLSCKASGYTFTSYWINWVKQRPGQGLEWIGNIYPS DSYTNYNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCTRGNYIDYWGQGT TLTVSSGGGSGGGGSGGGGSENVLTQSPAIMSASLGEKVTMSCRASSSVNYMY WYQQKSDASPKLWIYYTSNLAPGVPGRFSGSGSGNSYSLTISSMEGEDAATYYCQ QFTSSHTFGGGTKLEIK	25	25
QVQLKESGPGLVAPSQSLSIPCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAG GTTNYNSALMSRLSISKDNSKSQVFLKMYSLQTDDTAMYYCARGDGYDDGYAM DYWGQGTSVTVSSGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRASQ SVSTSSYSYMHWYQQKPGQPPKLLIKYASNLESGVPARFSGSGSGTDFTLNIHPVE EEDTATYYCQHSWEIPLTFGAGTKLELK	26	26
QVQLQQSGAELVKPGASVKLSCKASGSTFTTYYIYWVKQRPGQGLEWIGEINPSN GGTNFNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCTSYYTHETYYYAMDY WGQGTSVTVSSGGGSGGGGSGGGGSETTVTQSPASLSVATGEKVSIRCMTSIDID DDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENTLSEDVADY YCLQSDNMPFTFGSGTKLEIK	27	27
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMQWVKQRPGQGLEWIGEIDP SDSYTNYNQKFKGKATLTVDTSSSTAYMQLSSLTSEDSAVYYCARAEYGYGNYPW FAYWGQGTLVTVSAGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSST GAVTTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQ TEDDAMYFCALWYSTHWVFGGGTKLTVL	28	28
QVQLQQPGAELVKPGASVKVSCKASGYTFTSYWMHWVKQRPGQGLEWIGRIHP SDSDTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAIPYYYGGWYFDV WGTGTTVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGETVTLTCRSSTGAV TTSNYANWVQEKPDHLFTGLIGGTNNRAPGVPARFSGSLIGDKAALTITGAQTED EAIYFCALWYSNHLFGSGTKVTVL	29	29
EVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARLYDAHWDYFDYW GQGTTLTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGETVTLTCRSSTGAVTT SNYANWVQEKPDHLFTGLIGGTNNRAPGVPARFSGSLIGDKAALTITGAQTEDEAI YFCALWYSNHWVFGGGTKLTVL	30	30
QVQLQQPGSVLVRPGASVKLSCKASGYTFTSSWMHWAKQRPGQGLEWIGEIHP NSGNTNYNEKFKGKATLTVDTSSSTAYVDLSSLTSEDSAVYYCAIYYDYDAYYFDYW GQGTTLTVSSGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVK NLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYC LQFYEFPYTFGGGTKLEIK	31	31
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCANPYYGYDVGY WGQGTTLTVSSGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSV SYMYWYQQKPGSSPRLLIYDTSNLASGVPVRFSGSGSGTSYSLTISRMEAEDAATY YCQQWSSYPLTFGAGTKLELK	32	32
EVQLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCTRHDDSSYDWFAYW GQGTLVTVSAGGGSGGGGSGGGGSETTVTQSPASLSVATGEKVTIRCITSTDIDDD	33	33

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
MNWYQQKPGEPPKLLISEGNTLRPGVPSRFFSGGYGTDFVFTIENTLSEDVADYYC LQSDNMPLMFGAGTKLELK		
QVQLQQPGAELVKPGASVKLSCKASGYTFTNYWMHWVKQRPGQGLEWIGMIH PNSGTTNYNEKFKSKATLTVDKSSSTTYMQLISLTSEDSAVYYCARFGDGYHFDYW GQGTTLTVSSGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTITCSASSSVSY MHWFQQRPGTSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISRMEAEDAATYY CQQRSTYPTFGGGTKLEIK	34	34
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMGWIN TNTGEPTYAEEFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARWYPYFDYWGQ GTTLTVSSGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKNL NWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCLQ FYEFPYTFGGGTKLEIK	35	35
QVQLQQSGAELAKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGYINP SSGYTKYNQKFKDKATLTADKSSSTAYMQLSSLTYEDSAVYYCARSDGSSGNWYF DVWGTGTTVTVSSGGGSGGGGSGGGGSDIQMTQSPASLSASVGETVTITCRASG NIHNYLAWYQQKQGKSPQLLVYNAKTLADGVPSRFSGSGSGTQYSLKINSLQPED FGSYYCQHFWSTPWTFGGGTKLEIK	36	36
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYGNYDYAM DYWGQGTSVTVSSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASE NIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDF GSYYCQHFWGTPYTFGGGTKLEIK	37	37
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTSYNEKFKDKATFTADTSSNTAFMQLSSLTSEDSAVYYCARRAYGYDEGFDYW GQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLQYDEFPLTFGGGTKLEIKR	38	38
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVRQAPGHGLEWIGEVLPG SGSTSYNEKFKGRATFTADTSSNTAYMQLSSLRSEDSAVYYCARRAYGYDEGFDY WGQGTTVTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQ DINGYLSWYQQKPGKSPQTLLYRANRLVDGVPSRFRGSGSGQDYTLTISSLEYED MGTYYCLQYDEFPPTFGGGTKLELKR	39	39
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVRQAPGHGLEWIGEILPGS GRTSYIEKFKGRATFTADTSSNTAYMQLSSLRSEDSAVYYCARRGYGYDEGFDYW GQGTTVTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWYQQKPGKSPKTLIYRAKRLVDGVPSRFSGSGSGQDYTLTISSLEYEDMGT YYCLQYDEFPPTFGGGTKLELKR	40	40
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEVLPG SGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDY WGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQ DINGYLSWFQQKPGKSPQTLLYRANRLVDGVPSRFRGSGSGQDYSLTISSLEYEDM GIYYCLQYDEFPPTFGAGTKLELKR	41	41
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GRTSYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWG QGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLSWFQQKPGKSPKTLIYRAKRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC LQYDEFPPTFGAGTKLELKR	42	42
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDGGFDYW GQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLHYDEFPPTFGAGTKLELKR	43	43
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS DSTSYNEKFKGKTTFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDYW GQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLQYDEFPPTFGAGTKLELKR	44	44
QVQLQQSGAELMKPGASVKISCKATAYTFSIYWIEWVKQRPGHGLEWIGEILPGS GSTNYNEKVKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDGGFDY WGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQ	45	45

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
DINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR		
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEIFPGSGHTSFNEKFKGKATFTADTSSNTAYIQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	46	46
QVQLQQSGAELMKPGASVKMSCKATGYTFSNYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGSTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	47	47
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	48	48
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVQQRPGHGLEWIGEILPGSGYTSYIEQFKGKATFTADTSSNTAYMQLGSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFHQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	49	49
QVQLQQSGAELMKPGASVKISCKATGYTLSSYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	50	50
QVQLQQSGAELMKPGASVKISCKGTGYTFSSYWIEWVKQRPGHGLEWIGEISPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	51	51
QVQLQQSGAELMKPGASVKISCKATGYTFGTYWIEWVKQRPGHGLEWIGEILPGSGTPNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDAGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGFYYCLQYDEFPPTFGAGTKLELKR	52	52
QVQLQQSGAELMKPGASVKISFKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTSCNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSIYASLGERVTITCKASQDINSYLNWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGVGTKLELKR	53	53
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGRTSYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	54	54
QVQLQQSGAELMKPGASVKMSCKATGYTFSNYWIEWVKQRPGHCLEWIGEILPGSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGSTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGCGTKLELKR	55	55
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHCLEWIGEILPGSGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGCGTKLELKR	56	56
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGG	57	57

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
GGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASDNIYSNLAWYQQKQG KSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPW TFGGGTKLEIKR		
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGATSYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGEYFDYWGQGTTLTVSSGG GGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQG KSPQLLVFAATYLADGVPSRFSGSGSGTQYSLKINSLQSEDFGNYYCQHFWGTPW TFGGGTKLEIKR	58	58
QVQLKQSGPGLVQPSQSLSITCTVSGFSLSSYGVHWVRQSPGKALEWLGVIWRG GSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLQYDEFPPTFGAGAKLELKR	59	59
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTSYGVHWVRQSPGKGLEWLGVIWRG GSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDSAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASEN VVTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLA DYHCGQSYSYPFTFGSGTKLEIKR	60	60
QVQLKQSGPGPVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWRG GSTDNNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASEN VVTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLA DYHCGQSYSYPFTFGSGTKLEIKR	61	61
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTRYGVHWVRQSPGKGLEWLGVIWRG GSTDHNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASEN VVTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLA DYHCGQSYSYPFTFGSGTKLEIKR	62	62
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTTYGVHWVRQSPGKGLEWLGVIWRG GSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASEN VVTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLA DYHCGQSYSYPFTFGSGTKLEIKR	63	63
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAY WGQGTLVTVSAGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASS SVSYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDA ATYYCQQWSSNPLYTFGGGTKLEIKR	64	64
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGTTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAY WGQGTLVTVSAGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASS SVSYMHWFQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTFSSMEAEDA ATYYCQQWSSNPLYTFGGGTKLEIKR	65	65
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPG SGSTYYNEKFKGKATLTADKSSNTVYMQLSSLTSEDSAVYFCARRGERGPWFAYW GQGTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVD YDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEE DAATYYCQQSNEDPLTFGAGTKLELKR	66	66
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPG SGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYW GQGTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVD YDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEE DAATYYCQQSNEDPLTFGAGTKLELKR	67	67
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPG SGSSYYNEKFRGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYW GQGTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVD YDGDSYMNWYQQKPGQPPQLLIYAASNLQSGIPARFSGSGSGTDFTLNIHPVEEE DAATYYCQQSNEDPLTFGAGTKLELKR	68	68

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQKTGQGLEWIGEIYPG SGSSYYNEKFKGKATLTADKSSNTAYIQLSSLTSEDSAVYFCARRGERGPWFAYWG QGTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDY DGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFGGSGSGTDFTLNIHPVEEED AATYYCQQSNEDPLTFGAGTKLELKR	69	69
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSPNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARPGGYGFAYW GQGTLVTVSAGGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDI VKNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADY YCLQFYEFPLTFGAGTKLELKR	70	70
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPG SGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQ GTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYD GDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDA ATYYCQQSNKDPLTFGAGTKLELKR	71	71
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPG SGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQ GTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYD GDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDA ATYYCQQSNKDPLTFGAGTKLELKR	72	72
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPG SGSNYYNEKFKGKAIMTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWG QGTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDY DGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEED AATYYCQQSNEDPLTFGAGTKLELKR	73	73
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPG SGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQ GTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYD GDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDA ATYYCQQSNKDPFTFGAGTKLELKR	74	74
/QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVRQAPGQGLEWIGEIYP GSGSSYYNEKFKGRATLTADKSSNTAYMQLSSLRSEDSAVYFCARPGDLGFAYWG QGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPSSLAVSLGQRATISCKASQSVDY DGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLTIHPVEEEDA ATYYCQQSNKDPFTFGGGTKLELKR	75	75
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPG SGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQ GTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYD GDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDA ATYYCQQSNKDPFTFGAGTKLELKR	76	76
QVQLKESGPGLVAPSQSLSITCTVSGFSLTNYGVHWVRQPPGKGLEWLGVVWAG GITNYNWALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCARGDGYDDGYAM DYWGQGTSVTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRAS QSVSTSSYSYMHWYQQKPGQAPKLLIKYASNLESGVPARFSGSGSGTDFTLNIHPV EEEDTATYYCQHSWEIPLTFGAGAKLELKR	77	77
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVLWAG GITNYNSALMSRLSIRKDNSKSQVFLKMYSLHTDDTAMYYCARGDGYDDGYAMD YWGQGTSVTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRASQ SVSTSSYSYMHWYQQKPGQPPKLLIKYASNLESGVPARFSGSGSGTDFTLNVHPV EEEDTATYYCQHSWEIPLTFGAGTKLELKR	78	78
QVQLKESGPGLVAPSQSLSIPCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAG GTTNYNSALMSRLSISKDNSKSQVFLKMYSLQTDDTAMYYCARGDGYDDGYAM DYWGQGTSVTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRAS QSVSTSSYSYMHWYQQKPGQPPKLLIKYASNLESGVPARFSGSGSGTDFTLNIHPV EEEDTATYYCQHSWEIPLTFGAGTKLELKR	79	79
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGWIDPE NGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNAPLLRYSSAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGA VTTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQTED DAMYFCALWYSTHYVFGGGTKVTVLR	80	80

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYIHWVKQRPEQGLEWIGWIDPEN GDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNAPLLRYSSSMDYW GQGTSVTVSSGGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGAV TTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQTEDD AMYFCALWYSTHYVFGGGTKVTVLR	81	81
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGWIDPE NGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNVALLRYSSAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGA VTTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQTED DAMYFCALWYSTHYVFGGGTKVTVLR	82	82
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPG SGSTYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYW GQGTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVD YDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEE DAATYYCQQSNEDPLTFGAGTKLELKR	83	83
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMQWVKQRPGQGLEWIGEIDP SDSYTNYNQKFKGKATLTVDTSSSTAYMQLSSLTSEDSAVYYCARAEYGYGNYPW FAYWGQGTLVTVSAGGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSS TGAVTTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGA QTEDDAMYFCALWYSTHWVFGGGTKLTVLR	84	84
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMGWIN TYTGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDMATYFCARKYYDYEFAYW GQGTLVTVSAGGGGSGGGGSGGGGSDIVMTQSPSSLAMSVGQKVTMSCKSSQS LLNSSNQKNYLAWYQQKPGQSPKLLVYFASTRESGVPDRFIGSGSGTDFTLTISSV QAEDLADYFCQQHYSTPLTFGAGTKLELKR	85	85
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPE TGGTAYNQKFKVKAILTADKSSSTAYMELRSLTSEDSAVYYCTRLGDYDVMDYWG QGTSVTVSSGGGGSGGGGSGGGGSDIQMTQTTSSLSASLGDRVTISCRASQDISN YLNWYQQKPDGTVKLLIYYTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFC QQDNTLPRTFGGGTKLEIKR	86	86
QVQLQQPGSVLVRPGASVKLSCKASGYTFTSSWMHWAKQRPGQGLEWIGEIHP NSGNTNYNEKFKGKATLTVDTSSSTAYVDLSSLTSEDSAVYYCAIYYDYDAYYFDYW GQGTTLTVSSGGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIV KNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYY CLQFYEFPYTFGGGTKLEIKR	87	87
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPG SGSNYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCAREEKIYFDYWGQG TTLTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDG DSYMNWYRQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAAT YYCQQSNEDPWTFGGGTKLEIKR	88	88
EVQLQQSGTVLARPGASVKMSCKTSGYTFTSYWMHWIKQRPGQGLEWIGAIYP GNSDTTYNQKFKGKAKLTAVTSASTAYMELSSLTNEDSAVYYCTSLITTAYYFDYW GQGTTLTVSSGGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIV KNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYY CLQFYEFPLTFGAGTKLELKR	89	89
QVQLQESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARHEEANWAWFAY WGQGTLVTVSAGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTITCSASSS VSYMHWFQQKPGTSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISRMEAEDAA TYYCQQRSSFPYTFGGGTKLEIKR	90	90
QVQLQQSGPQLVSPGASVKISCKASGYSFTNYWMHWVKQRPGQGLEWIGMID PSDSETRLNQQFKDKATLTVDESSSTAYMQLSSPTSEDSAVYYCAIPYYAMDYWG QGTSVTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDIN SYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYY CLQYDEFPLTFGAGTKLELKR	91	91
QVQLQQPGSVLVRPGASVKLSCKASGYTFTSSWMHWAKQRPGQGLEWIGEIHP NSGNTNYNEKNKGKATLTVDTSSSTAYVDLSSLTSEDSAVYYCATYYGNYVWYFD VWGAGTSVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASLGERVSLTCRASQ DIHGYLNLFQQKPGETIKHLIYETSNLDSGVPKRFSGSRSGSDYSLIIGSLESEDFADY YCLQYASSPLTFGAGTKLELKR	92	92

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCASYGSSYWYFDV WGTGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSHKFMSTSVGDRVSITCKASQ DVGTAVAWYQQKPGQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISNVQSED LADYFCQQYSSYPFTFGSGTKLEIKR	93	93
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQANDTAIYYCASYYGSSRSYWYLDV WGAGTTVTVSSGGGGSGGGGSGGGGSSIVMTQTPKFLLVSAGDRVTITCKASQS VSNDVAWYQQKPGQSPKLLIYYASNRYTGVPDRFTGSGYGTDFTFTISTVQAEDL AVYFCQQDYTSLPTFGAGTKLEIKR	94	94
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARQNDSSWAWFAY WGQGTLVTVSAGGGGSGGGGSGGGGSETTVTQSPASLSVATGEKVTIRCITSTDI DDDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENTLSEDVA DYYCLQSDNMPLTFGAGTKLELKR	95	95
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCALPYSNYGWYFD VWGTGTTVTVSSGGGGSGGGGSGGGGSDIQMTQTTSSLSASLGDRVTISCRASQ DISNYLNWYQQKPDGTVKLLIYYTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIAT YFCQQGNTLPFTFGSGTKLEIKR	96	96
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPS DSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSYWYFDV WGTGTTVTVSSGGGGSGGGGSGGGGSDIQMTQTTSSLSASLGDRVTISCSASQG ISNYLNWYQQKPDGTVKLLIYYTSSLHSGVPSRFSGSGSGTDYSLTISNLEPEDIATY YCQQYSKLPYTFGGGTKLEIKR	97	97
EVQLQQSGAELVKPGASVKLSCTASGFNIKDYYMHWVKQRTEQGLEWIGRIDPE DGETKYAPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCAAYGNSAWFAYWG QGTLVTVSAGGGGSGGGGSGGGGSDIVMTQSPATLSVTPGDRVSLSCRASQSIS DYLHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGSDFTLSINSVEPEDVGVYYC QNGHSFPWTFGGGTKLEIKR	98	98
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVRQAPGKGLEWMGWIN TNTGEPTYAEEFKGRFAFSLETSASTAYLQINNLRNEDTATYFCARWYPYFDYWGQ GTTVTVSSGGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKN LNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYTLTISNLESEDFATYYCLQ FYEFPYTFGGGTKLEIKR	99	99
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMGWIN TNTGEPTYAEEFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARWYPYFDYWGQ GTTLTVSSGGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKN LNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCL QFYEFPYTFGGGTKLEIKR	100	100
QVQLQQSGAELAKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGYINP SSGYTKYNQKFKDKATLTADKSSSTAYMQLSSLTYEDSAVYYCARSDGSSGNWYF DVWGTGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPASLSASVGETVTITCRAS GNIHNYLAWYQQKQGKSPQLLVYNAKTLADGVPSRFSGSGSGTQYSLKINSLQPE DFGSYYCQHFWSTPWTFGGGTKLEIKR	101	101
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAG GSTNYNSALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCAREGGYTGYFDVW GAGTTVTVSSGGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGAVT TSNYANWVQEKPDHLFTGLIGGTSYRAPGVPVRFSGSLIGDKAALTITGAQTEDDA MYFCALWYSTHYVFGGGTKVTVLR	102	102
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPS DSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAYSNYVPYYAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASLGERVSLTCRASQE ISGYLSWLQQKPDGTIKRLIYAASTLDSGVPKRFSGSRSGSDYSLTISSLESEDFADYY CLQYASYPWTFGGGTKLEIKR	103	103
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYGNYDYAM DYWGQGTSVTVSSGGGGSGGGGSGGGGSDIVMTQSPATLSVTPGDRVSLSCRA	104	104

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
SQSISDYLHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGSDFTLSINSVEPEDV GVYYCQNGHSFPYTFGGGTKLEIKR		
EVQLQQSGAELVRPGALVKLSCKASGFNIKDYFMHWVKQRPEQCLEWIGWIDPE TDNTIYDPKFQGKASITADTSSNTAYLQLSSLTSEDTAVYYCARSGNMGFTYWGQ GTLVTVSAGGGGSGGGGSGGGGSENVLTQSPAIMSASLGEKVTMSCRASSSVNY MYWYQQKSDASPKLWIYYTSNLAPGVPARFSGSGSGNSYSLTISSMEGEDAATYY CQQFTSSPSTFGCGTKLEIKR	105	105
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKCLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCASQGGSSWGAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSETTVTQSPASLSMAIGEKVTIRCITNTDI DDDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENMLSEDVA DYYCLQSDNLPLTFGCGTKLELKR	106	106
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKCLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARKGYGYDWYFDVW GTGTTVTVSSGGGGSGGGGSGGGGSQLVLTQSSSASFSLGASAKLTCTLSSQHSTY TIEWYQQQPLKPPKYVMELKKDGSHSTGDGIPDRFSGSSSGADRYLSISNIQPEDE AIYICGVGDTIKEQFVYVFGCGTKVTVLG	107	107
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQCLEWIGEIDP SDSYTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARSSYYYYAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSS VSYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAA TYYCQQWSSNPLTFGCGTKLELKR	108	108
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDGGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC LHYDEFPPTFGAGTKLELK	109	109
QVQLQQSGAELMKPGASVKISFKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTNYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLNWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC LQYDEFPPTFGAGTKLELK	110	110
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS DSTSYNEKFKGKTTFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC LQYDEFPPTFGAGTKLELK	111	111
QVQLQQSGAELMKPGASVKISCKATAYTFSIYWIEWVKQRPGHGLEWIGEILPGS GSTNYNEKVKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDGGFDY WGQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLQYDEFPPTFGAGTKLELK	112	112
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC PQYVESPPTFGAGTKLELK	113	113
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC LQYDEFPPTFGAGTKLELK	114	114
QVQLQQSGAELMKPGASVKMSCKATGYTFSNYWIEWVKQRPGHGLEWIGEILP GSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDY WGQGSTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLQYDEFPPTFGAGTKLELK	115	115
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVQQRPGHGLEWIGEILPG SGYTSYIEQFKGKATFTADTSSNTAYMQLGSLTSEDSAVYYCARRGYGYDEGFDY WGQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI	116	116

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
NSYLSWFHQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLQYDEFPPTFGAGTKLELK		
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTSYNEKFKDKATFTADTSSNTAFMQLSSLTSEDSAVYYCARRAYGYDEGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC LQYDEFPLTFGGGTKLEIK	117	117
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEVLPG SGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDY WGQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NGYLSWFQQKPGKSPQTLLYRANRLVDGVPSRFRGSGSGQDYSLTISSLEYEDMGI YYCLQYDEFPPTFGAGTKLELK	118	118
QVQLQQSGAELMKPGASVKISCKGTGYTFSSYWIEWVKQRPGHGLEWIGEISPGS GSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS YLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC LQYDEFPPTFGAGTKLELK	119	119
QVQLQQSGAELMKPGASVKISCKATGYTFGTYWIEWVKQRPGHGLEWIGEILPG SGTPNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDAGFDY WGQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGF YYCLQYDEFPPTFGAGTKLELK	120	120
QVQLQQSGAELMKPGASVKISFKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTSCNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYW GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSIYASLGERVTITCKASQDINSY LNWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCL QYDEFPPTFGVGTKLELK	121	12.
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GRTSYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWG QGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSY LSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCL QYDEFPPTFGAGTKLELK	122	122
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GRTSYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWG QGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSY LSWFQQKPGKSPKTLIYRAKRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCL QYDEFPPTFGAGTKLELK	123	123
QVQLKQSGPGLVQPSQSLSITCTVSGFSLSSYGVHWVRQSPGKALEWLGVIWRG GSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDIN SYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYY CLQYDEFPPTFGAGAKLELK	124	124
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTSYGVHWVRQSPGKGLEWLGVIWRG GSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDSAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASENV VTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLAD YHCGQSYSYPFTFGSGTKLEIK	125	125
QVQLKQSGPGPVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWRG GSTDNNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASENV VTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLAD YHCGQSYSYPFTFGSGTKLEIK	126	126
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTRYGVHWVRQSPGKGLEWLGVIWRG GSTDHNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASENV VTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLAD YHCGQSYSYPFTFGSGTKLEIK	127	127
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTTYGVHWVRQSPGKGLEWLGVIWRG GSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASENV	128	128

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
VTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLAD YHCGQSYSYPFTFGSGTKLEIK		
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTSYGVHWVRQSPGKGLEWLGVIWRG GSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASENV VTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLAD YHCGQSYSYLIHVRFGSGTKLEIK	129	129
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTRYGVHWVRQSPGKGLEWLGVIWRG GSTDHNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYW GQGTSVTVSSGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASENV VTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLAD YHCGQSYSYLIHVRFGSGTKLEIK	130	130
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAY WGQGTLVTVSAGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSV SYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAAT YYCQQWSSNPLYTFGGGTKLEIK	131	131
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAY WGQGTLVTVSAGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSV SYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAAT YYCQQWSSNPHVHVFGGGTKLEIK	132	132
QVQLQQPGAELVKPGASVKLSCKASGYTFTTYWMHWVKQRPGQGLEWIGMIH PNSDNTNYNEKFKSKATLTVDKSSSTAYIQLSSLTSEDSAVYYCARWGDGYSFAYW GQGTLVTVSAGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSVS YMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAATY YCQQWSSNPLYTFGGGTKLEIK	133	133
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGNTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAY WGQGTLVTVSAGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSV SYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAAT YYCQQWSSNPLYTFGGGTKLEIK	134	134
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPG SGSTYYNEKFKGKATLTADKSSNTVYMQLSSLTSEDSAVYFCARRGERGPWFAYW GQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDY DGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEED AATYYCQQSNEDPLTFGAGTKLELK	135	135
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPG SGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYW GQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDY DGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEED AATYYCQQSNEDPLTFGAGTKLELK	136	136
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPG SGSSYYNEKFRGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYW GQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDY DGDSYMNWYQQKPGQPPQLLIYAASNLQSGIPARFSGSGSGTDFTLNIHPVEEED AATYYCQQSNEDPLTFGAGTKLELK	137	137
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQKTGQGLEWIGEIYPG SGSSYYNEKFKGKATLTADKSSNTAYIQLSSLTSEDSAVYFCARRGERGPWFAYWG QGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYD GDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFGGSGSGTDFTLNIHPVEEEDA ATYYCQQSNEDPLTFGAGTKLELK	138	138
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPGQGLEWIGMID PSDSETHYNQMFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCATYDGYYRFAY WGQGTLVTVSAGGGSGGGGSGGGGSDIQMTQSPSSLSASLGGKVTITCKASQDI NKYIAWYQHKPGKGPRLLIHYTSTLQPGIPSRFSGSGSGRDYSFSISNLEPEDIATYY CLQYDILMYTFGGGTKLEIK	139	139
QVQLQQPGAELVRPGASVRLSCKASGYTFTNYWMNWVKQRPGQGLEWIGMID PSDSETHFNQMFKDKATLTVDKSSSTAYMQVSSLTSEDSAVYYCATYDIYYRFAYW	140	140

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
GQGTLVTVSAGGGSGGGGSGGGGSDIQMTQSPSSLSASLGGKVTITCKASQDINKYIAWYQHKPGKGPRLLIHYTSTLQPGIPSRFSGSGSGRDYSFSISNLEPEDIATYYCLQYDILMYTFGGGTKLEIK		
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSDSTNYNEKFKSKATLTVDKSSSTAYMHLSSLTSEDSAVYYCARPGGYGFADWGQGTLVTVSAGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELK	141	141
QVQLQQPGAELVKPGASVKLSCKASGYTFTTYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARPGGYGFTYWGQGTLVTVSAGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELK	142	142
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSPNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARPGGYGFAYWGQGTLVTVSAGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELK	143	143
QVQLQQPGAELVRPGASVKLSCKASGYTFTSYWINWVKQRPGQGLEWIGNIYPSDSYTNYNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCTRGNYIDYWGQGTTLTVSSGGGSGGGGSGGGGSENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWYQQKSDASPKLWIYYTSNLAPGVPARFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSHTFGGGTKLEIK	144	144
QVQLQQPGAELVRPGASVKLSCKASGYTFTDYWINWVKQRPGQGLEWIGNIYPSDSYTNYNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCTRGNYIDYWGQGTTLTVSSGGGSGGGGSGGGGSENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWYQQKSDASPKLWIYYTSNLAPGVPARFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSHTFGGGTKLEIK	145	145
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNKDPLTFGAGTKLELK	146	146
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSNYYNEKFKGKAIMTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELK	147	147
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNKDPFTFGAGTKLELK	148	148
QVQLKESGPGLVAPSQSLSITCTVSGFSLTNYGVHWVRQPPGKGLEWLGVVWAGGITNYNWALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCARGDGYDDGYAMDYWGQGTSVTVSSGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQAPKLLIKYASNLESGVPARFSGSGSGTDFTLNIHPVEEEDTATYYCQHSWEIPLTFGAGAKLELK	149	149
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVLWAGGITNYNSALMSRLSIRKDNSKSQVFLKMYSLHTDDTAMYYCARGDGYDDGYAMDYWGQGTSVTVSSGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQPPKLLIKYASNLESGVPARFSGSGSGTDFTLNVHPVEEEDTATYYCQHSWEIPLTFGAGTKLELK	150	150
QVQLQQSGPQLVSPGASVKISCKASGYSFTSYWMYWVKQRPGQGLEWIGMIDPSDSETRLNQKFKDRATLTVDKSSSTAYMQLSSPTSEDSAVYYCARTRNYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQTPSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYSTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPWTFGGGTKLEIK	151	151

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLQQSGPQLVSPGASVKISCKASGYSFTSYWMYWVKQRPGQGLEWIGMIDPSDSETRLNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCARTRNYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQTPSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYSTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNALPWTFGGGTKLEIK	152	152
QVQLQQSGPQLVSPGASVKISCKASGYSFTSYWMYWVKQRPGQGLEWIGMIDPSDSETRLNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCARTRNYWGQGTSLTVSSGGGSGGGGSGGGGSDIQMTQTPSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYSTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPWTFGGGTKLEIK	153	153
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNAPLLRYSSAMDYWGQGTSVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVL	154	154
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYIHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNAPLLRYSSSMDYWGQGTSVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVL	155	155
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNVALLRYSSAMDYWGQGTSVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVL	156	156
QVQLQQSGAELVKPGASVKLSCKASGYTFSNYYVYWVKQRPGQGLEWIGEINPSNGDTNFNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYFCTSYYTHEAYYYAMDCWGQGTSVTVSSGGGSGGGGSGGGGSETTVTQSPASLSVATGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENTLSEDVADYYCLQSDNMPFTFGSGTKLEIK	157	157
QVQLQQSGAELVRPGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGRIDPEDGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCTPYSIYDAMDYWGQGTSVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYFGGGTKVTVL	158	158
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSTYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELK	159	159
QVQLQQSGAELVRPGVSVKISCKGSGYSFTDYGMHWVKQSHAKSLEWIGVISTYYGDASYNQKFKGKATMTVDKSSSTAYMELARLTSEDSAIYYCARQMDYDYTYYYAMDYWGQGTSVTVSSGGGSGGGGSGGGGSDIVMSQSPSSLAVSVGEKVTMSCKSSQSLLYSTNQKNYLAWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGSGTDFTLTISSVKAEDLAVYYCQQYYSYPPWTFGGGTKLEIK	160	160
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWNGEIYPGSGSTYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARMDGPWFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPPTFGGGTKLEIK	161	161
QVTLKVSGPGILQPSQTLGLACTFSGISLSTSGMGLSWLRQPSGKALEWLASIWNNDNYYNPSLKSRLTISKETSNNQVFLKLTSVDTADSTTYYCAWRPYYRYDSFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIQMTQSPASLAASVGETVTITCRASENIYYSLAWYQQKQGKSPQLLIYNANSLEDGVPSRFSGSGSGTQYSMKINSMQPEDTATYFCKQAYDVPYTFGGGTKLEIK	162	162
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPETGGTAYNQKFKVKAILTADKSSSTAYMELRSLTSEDSAVYYCTRLGDYDVMDYWGQGTSVTVSSGGGSGGGGSGGGGSDIQMTQTTSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYYTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQDNTLPRTFGGGTKLEIK	163	163

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGEIDPSDSYTNYNQKFKGKSTLTVDKSSSTAYMQLSSLTSEDSAVYYCARAGRYGSSFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIQMTQTTSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYYTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPWTFGGGTKLEIK	164	164
QVTLKESGPGILQPSQTLSLTCSFSGFSLSTSGMGVSWIRQPSGKGLEWLAHIYWDDDKRYNPSLKSRLTISKDTSRNQGFLKITSVDTADTATYYCAGRPDDYDGAWFPYWGQGTLVTVSAGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	165	165
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSNYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCAREEKIYFDYWGQGTTLTVSSGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYRQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPWTFGGGTKLEIK	166	166
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARYDGYWFDYWGQGTTLTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTNNRAPGVPARFSGSLIGDKAALTITGAQTEDEAIYFCALWYSNHWVFGGGTKLTVL	167	167
EVQLQQSGTVLARPGASVKMSCKTSGYTFTSYWMHWIKQRPGQGLEWIGAIYPGNSDTTYNQKFKGKAKLTAVTSASTAYMELSSLTNEDSAVYYCTSLITTAYYFDYWGQGTTLTVSSGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELK	168	168
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAPETGDYGSSYVWYFDVWGTGTTVTVSSGGGSGGGGSGGGGSQLVLTQSSSASFSLGASAKLTCTLSSQHSTYTIEWYQQQPLKPPKYVMELKKDGSHSTGDGIPDRFSGSSSGADRYLSISNIQPEDEAIYICGVGDTIKEQFVFVFGGGTKVTVL	169	169
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSTYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARGKVTRFAYWGQGTLVTVSAGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPPTFGGGTKLEIK	170	170
EVQLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISDGGSYTYYPDNVKGRFTISRDNAKNNLYLQMSHLKSEDTAMYYCARDQDSNWEYFDYWGQGTSLTVSSGGGSGGGGSGGGGSDIQMTQSPSSLSASLGERVSLTCRASQDIGISLNWLQQEPDGTIKRLIYATSSLDSGVPKRFSGSRSGSDYSLTISSLESEDFVDYYCLQYASSPYTFGGGTKLEIK	171	171
QIQLVQSGPELKKPGETVKISCKASGYTFTDYSMHWVRQAPGKGLKWMAWINTETGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARESWDRAMDYWGQGTSVTVSSGGGSGGGGSGGGGSDIQMTQSPASLSASVGETVTITCRASENIYSYLAWYQQKQGKSPQLLVYNAKNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGYYCQHFWGTPYTFGGGTKLEIK	172	172
QVQLQQSGPQLVSPGASVKISCKASGYSFTNYWMHWVKQRPGQGLEWIGMIDPSDSETRLNQQFKDKATLTVDESSSTAYMQLSSPTSEDSAVYYCAIPYYAMDYWGQGTSVTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPLTFGAGTKLELK	173	173
QVQLQQPGSVLVRPGASVKLSCKASGYTFTSSWMHWAKQRPGQGLEWIGEIHPNSGNTNYNEKNKGKATLTVDTSSSTAYVDLSSLTSEDSAVYYCATYYGNYVWYFDVWGAGTSVTVSSGGGSGGGGSGGGGSDIQMTQSPSSLSASLGERVSLTCRASQDIHGYLNLFQQKPGETIKHLIYETSNLDSGVPKRFSGSRSGSDYSLIIGSLESEDFADYYCLQYASSPLTFGAGTKLELK	174	174
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCASYGSSYWYFDVWGTGTTVTVSSGGGSGGGGSGGGGSDIVMTQSHKFMSTSVGDRVSITCKASQD	175	175

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
VGTAVAWYQQKPGQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISNVQSEDL ADYFCQQYSSYPFTFGSGTKLEIK		
QVQLQQPGAELVKPGASVKMSCKASGYTFTSYNMHWVKQTPGQGLEWIGALYS GNGDTSYNQKFKGKATLTADKSSSTAYMQLSSLTSEDSAVYYCARDYYGSSHLWY FDVWGAGTTVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGETVTLTCRSST GAVTTSNYANWVQEKPDHLFTGLIGGTNNRAPGVPARFSGSLIGDKAALTITGAQ TEDEAIYFCALWYSNHLVFGGGTKLTVL	176	176
QVTLKESGPGILQPSQTLSLTCSFSGFSLSTSGMGVSWIRQPSGKGLEWLAHIYWD DDKRYNPSLKSRLTISKDTSRNQVFLKITSVDTADTATYYCARRAHYDYGWYFDVW GAGTTVTVSSGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDY DGDSYMNWYQQKPGQPPKLLIYVASNLESGIPARFRGSGSGTDFTLNIHPVEEED AAIYYCQQSHEDPRTFGGGTKLEIK	177	177
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAGYDYDWYFDV WGTGTTVTVSSGGGSGGGGSGGGGSENVLTQSPAIMSASPGEKVTMTCSASSSV SYMHWYQQKSSTSPKLWIYDTSKLASGVPGRFSGSGSGNSYSLTISSMEAEDVAT YYCFQGSGYPLTFGAGTKLELK	178	178
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARHEDSNYHYFDYW GQGTTLTVFSGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYS NLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSY YCQHFWGTPYTFGGGTKLEIK	179	179
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARQNDSSWAWFAY WGQGTLVTVSAGGGSGGGGSGGGGSETTVTQSPASLSVATGEKVTIRCITSTDID DDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENTLSEDVADY YCLQSDNMPLTFGAGTKLELK	180	180
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCALPYSNYGWYFD VWGTGTTVTVSSGGGSGGGGSGGGGSDIQMTQTTSSLSASLGDRVTISCRASQD ISNYLNWYQQKPDGTVKLLIYYTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATY FCQQGNTLPFTFGSGTKLEIK	181	181
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPS DSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSYWYFDV WGTGTTVTVSSGGGSGGGGSGGGGSDIQMTQTTSSLSASLGDRVTISCSASQGIS NYLNWYQQKPDGTVKLLIYYTSSLHSGVPSRFSGSGSGTDYSLTISNLEPEDIATYYC QQYSKLPYTFGGGTKLEIK	182	182
EVQLQQSGAELVKPGASVKLSCTASGFNIKDYYMHWVKQRTEQGLEWIGRIDPE DGETKYAPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCAAYGNSAWFAYWG QGTLVTVSAGGGSGGGGSGGGGSDIVMTQSPATLSVTPGDRVSLSCRASQSISDY LHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGSDFTLSINSVEPEDVGVYYCQ NGHSFPWTFGGGTKLEIK	183	183
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAG GSTNYNSALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCAREGGYTGYFDVW GAGTTVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGAVTT SNYANWVQEKPDHLFTGLIGGTSYRAPGVPVRFSGSLIGDKAALTITGAQTEDDA MYFCALWYSTHYVFGGGTKVTVL	184	184
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPS DSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAYSNYVPYYAMDY WGQGTSVTVSSGGGSGGGGSGGGGSDIQMTQSPSSLSASLGERVSLTCRASQEIS GYLSWLQQKPDGTIKRLIYAASTLDSGVPKRFSGSRSGSDYSLTISSLESEDFADYYC LQYASYPWTFGGGTKLEIK	185	185
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPG SGSAYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGFDYWGQGTT LTVSSGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSY MNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYC QQSNEDPLPTFGAGTQRELK	186	186
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSGYGYYF DYWGQGTTLTVSSGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASS	187	187

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
SVSYMHWYQQKSGTSPKRWIYDTSKLASGVPVRFSGSGSGTSYSLTISSMEAEDA ATYYCQQWSSNPLTFGAGTKLELK		
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSSYGWY FDVWGTGTTVTVSSGGGSGGGGSGGGGSDIQMTQSPASLSASVGETVTITCRAS GNIHNYLAWYQQKQGKSPQLLVYNAKTLADGVPSRFSGSGSGTQYSLKINSLQPE DFGSYYCQHFWSTPWTFGGGTKLEIK	188	188
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSSYGWY FDVWGTGTTVTVSSGGGSGGGGSGGGGSDIQMTQSPASLSASVGETVTITCRAS ENIYSYLAWYQQKQGKSPQLLVYNAKTLAEGVPSRFSGSGSGTQFSLKINSLQPED FGSYYCQHHYGTPFTFGSGTKLEIK	189	189
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCASDYYGSSYGWY FDVWGTGTTVTVSSGGGSGGGGSGGGGSENVLTQSPAIMSASLGEKVTMSCRA SSSVNYMYWSQQKSDASPKLWIYYTSNLAPGVPPRFSGSGSGNSYSLTISSMEGE DAATYYCQQFTSSLTFGAGTKLELK	190	190
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSSYGWY FDVWGTGTTVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGETVTLTCRSST GAVTTSNYANWVQEKPDHLFTGLIGSTNNRAPGVPARFSGSLIGDKSALTITGAQT EDEAIYFCTLWYSNHWVFGGGTKLTVL	191	191
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCTRDYYGSGYGWY FDVWGTGTTVTVSSGGGSGGGGSGGGGSDIVMTQSHKFMSTSVGDRVSITCKA SQDVGTAVAWYQQKPGQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISNVQS EDLADYFCQQYSSYPFTFGSGTKLEIK	192	192
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPGQGLEWIGMID PSDSETHYNQMFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCATYDGYYRFAY WGQGTLVTVSAGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLQYDEFPPTFGAGTKLELK	193	193
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPGQGLEWIGMID PSDSETHFNQMFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCATYDVYYRFAY WGQGTLVTVSAGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENI YSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFG SYYCQHFWGTPFTFGSGTKLEIK	194	194
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYGNYDYAM DYWGQGTSVTVSSGGGSGGGGSGGGGSDIVMTQSPATLSVTPGDRVSLSCRAS QSISDYLHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGSDFTLSINSVEPEDVG VYYCQNGHSFPYTFGGGTKLEIK	195	195
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYGNYDYAM DYWGQGTSVTVSSGGGSGGGGSGGGGSQAVVTQESALTTSPGETVTLTCRSSTG AVTTSNYANWVQEKPDHLFTGLIGGTNNRAPGVPARFSGSLIGDKAALTITGAQT EDEAIYFCALWYSNHWVFGGGTKLTVL	196	196
EVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCASQLTGTWYYFDYW GQGTTLTVSSGGGSGGGGSGGGGSEIFVTQSPASLSMAIGEKVTIRCITSTDIDDD MNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENMLSEDVADYY CLQSDNLPLTFGAGTKLELK	197	197
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCASQLTGTWYYFDYW GQGTTLTVSSGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTITCSASSSVSY MHWFQQKPGTSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISRMEAEDAATYY CQQRSSYPPTFGSGTKLELK	198	198
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCASQLTGTWYYFDYW	199	199

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
GQGTTLTVSSGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSVSY MHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAATYY CQQWSSNPLTFGSGTKLEIK		
DVKLVESGGGLVKLGGSLKLSCAASGFTFSNYYMSWVRQTPEKRLELVAVINSNG GSTYYPDTVKGRFTISRDNAKNTLYLQMSSLKSEDTALYYCARQEGIGYAMDYWG QGTSVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTLTCSASSSVSSS YLYWYQQKPGSSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAASYFC HQWSSYPPTFGAGTKLELKR	200	200
EVQLQQSGPELVKPGASVKISCKTSGYTFTEYTMHWVKQSHGKSLEWIGGIYPNN GGTSYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVYYCARGGWLLGYWGQG TTLTVSSGGGGSGGGGSGGGGSQLVLTQSSSASFSLGASAKLTCTLSSQHSTYTIE WYQQQPLKPPKYVMELKKDGSHSTGDGIPDRFSGSSSGADRYLSISNIQPEDEAIYI CGVGDTIKEQFVYVFGGGTKVTVLR	201	201
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARDGGIRGAMDYWG QGTSVTVSSGGGGSGGGGSGGGGSDVLMTQTPLSLPVSLGDQASISCRSSQSIVH SNGNTYLEWYLQKPGQSPKLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDL GVYYCFQGSHVPWTFGGGTKLEIKR	202	202
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGS GSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYW GQGTTLTVSSGGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDI NSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIY YCLQYDEFPPTFGAGTKLELKR	203	203
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPE TGGTAYNQKFKGKATLTADKSSSTAYMELRSLTSEDSAVYYCTRNYDYAMDYWG QGTSVTVSSGGGGSGGGGSGGGGSDVVMTQTPLTLSVTIGQPASISCKSSQSLLD SDGKTYLNWLLQRPGQSPKRLIYLVSKLDSGVPDRFTGSGSGTDFTLKISRVEAEDL GVYYCWQGTHFPWTFGGGTKLEIKR	204	204
DVKLVESGGGLVKLGGSLKLSCAASGFTFSSYYMSWVRQTPEKRLELVAVINSNGG STFYPDTVKGRFTISRDNAKNTLYLQMSSLKSEDTALYYCARQEGIGYALDYWGQG TSVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTLPCSASSSVSSSYL YWYQQKPGSSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAASYFCH QWSSYPPTFGAGTKLELKR	205	205
EVKLVESGGDLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVAAISSGG STYYPDSVKGRFTISRDNARNILYLQMSSLRSEDTAMYYCAREREWGVYYGSSLDY WGQGTTLTVSSGGGGSGGGGSGGGGSDVLMTQTPLSLPVSLGDQASISCRSSQS IVYSNGNTYLEWYLQKPGQSPKLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA EDLGVYYCFQGSHVPPTFGGGTKLEIKR	206	206
EVQLQQSGAELVKPGASVKLSCTASGFNIKDTYMHWVKQRPEQGLEWIGRIDPA NGNTKYDPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCARSDGNYDWGQGT LVTVSAGGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYNNLA WYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQ HFWGTPWTFGGGTKLEIKR	207	207
DVKLVESGGGLVKLGGSLKLSCAASGFTFSNYYMSWVRQTPEKRLELVAVINSNG GSTYYPDTVKGRFTISRDNAKNILYLQMSSLKSEDTALYYCARQEGIGYGMDYWG QGTSVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTLTCSASSSVSSS YLYWYQQKPGSSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAASYFC HQWSSYPPTFGAGTKLELKR		208
EVQLVESGGGLVQPKGSLKLSCAASGFTFNTYVMNWVRQAPGKGLEWVARIRSK SDNYATYYADSVKDIFTISRDDSQSMLYLQMNNLKTEDTAMYYCVRHDGVVGFD VWGAGTTVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSAS SSVSYMYWYQQKPGSSPRLLIYDTSNLASGVPVRFSGSGSGTSYSLTISRMEAEDA ATYYCQQWSTYPPITFGAGTKLELKR	209	209
DVQLQESGPGLVKPSQSLSLTCSVTGYSITSGYYWNWIRQFPGNKLEWMGYISYD GSNNYNPSLKNRISITRDTSKNQFFLKLNSVTTEDTATYYCARGGGRGWGQGTLV TVSAGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSVSYMYWY QQKPGSSPRLLIYDTSNLASGVPVRFSGSGSGTSYSLTISRMEAEDAATYYCQQWS SYPFTFGSGTKLEIKR	210	210

TABLE 19-continued

Sequence	Binder Name	SEQ ID NO:
Full Binder Sequences Table 19-Full length sequences		
QIQLVQSGPELKKPGETVKISCKASGYTFTDYSMHWVKQAPGKGLKWMGWINT ETGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARDYYDYYYAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSQIVLSQSPAILSASPGEKVTMTCRASSS VSYMHWYQQKPGSSPKPWIYATSNLASGVPARFSGSGSGTSYSLTISRVEAEDAA TYYCQQWSSNPYTFGGGTKLEIKR	211	211
QIQLVQSGPELKKPGETVKISCKASGYTFTDYSMHWVKQAPGKGLKWMGWINT ETGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARESWDRAMDYW GQGTSVTVSSGGGGSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIY SNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGS YYCQHFWGTPWTFGGGTKLEIKR	212	212
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPEQGLEWIGRIDP YDSETHYNQKFKDKAILTVDKSSSTAYMQLSSLTSEDSAVYYCARIYSDYDGAWFA YWGQGTLVTVSAGGGGSGGGGSGGGGSDIVMTQSHKFMSTSVGDRVSITCKAS QDVSTAVAWYQQKPGQSPKLLIYWASTRHTGVPDRFTGSGSGTDYTLTISSVQAE DLALYYCQQHYSTPWTFGGGTKLEIKR	213	213
EVQLQQSGPELVKPGASVKMSCKASGYTFTDYYMDWVKQSHGESFEWIGRVNP YNGGTSYNQKFKGKATLTVDKSSSTAYMELNSLTSEDSAVYYCARGTVGFAYWGQ GTLVTVSAGGGGSGGGGSGGGGSDVVMTQTPLTLSVTIGQPASISCKSSQSLLDS DGKTYLNWLLQRPGQSPKRLIYLVSKLDSGVPDRFTGSGSGTDFTLKISRVEAEDLG VYYCWQGTHFPWTFGGGTKLEIKR	214	214
EVKLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVASISSGGS TYYPDSVKGRFTISRDNARNILSLQMSSLRSEDTAMYYCAREREWGVFYGSSLDY WGQGTTLTVSSGGGGSGGGGSGGGGSDVLMTQTPLSLPVSLGDQASISCRSSQS IVHSNGNTYLEWYLQKPGQSPKFLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA EDLGVYYCFQGSHVPPTFGGGTKLEIKR	215	215
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWIRQTPEKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARHDDSSYGYFDYW GQGTTLTVSSGGGGSGGGGSGGGGSETTVTQSPASLSVATGEKVTIRCITSTDIDD DMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENTLSEDVADYY CLQSDNMPLTFGGGTKLEIKR	216	216
EVKLVESGGGLVKPGGSLKLSCAASGFTFSNYAMSWVRQTPEKRLEWVASISSGG TTYYPDSVKGRFTISRDNARNILYLQMSSLRSEDTAMYYCARTMPDVWGAGTTVT VSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASLGGKVTITCKASQDINKYIAWYQ HKPGKGPRLLIHYTSTLQPGIPSRFSGSGSGRDYSFSISNLEPEDIATYYCLQYDNLY MYTFGGGTKLEIKR	217	217
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAG GSTNYNSALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCARDTDGYYWAMD YWGQGTSVTVSSGGGGSGGGGSGGGGSDIQMNQSPSSLSASLGDTITITCHASQ NINVWLSWYQQKPGNIPKLLIYKASNLHTGVPSRFSGSGSGTGFTLTISSLQPEDIA TYYCQQGQSYPYTFGGGTKLEIKR	218	218
DVQLQESGPGLVKPSQSLSLTCTVTGYSITSDHAWNWIRQFPGNKLEWMGYISYS GSTTYNPSLKSRISITRDTSKNQFFLQLNSVTTEDTATYYCARKWGDYWGQGTSVT VSSGGGGSGGGGSGGGGSQIVLTQSPALMSASPGEKVTMTCSASSSVSYMYWY QQKPRSSPKPWIYLTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWS SNPPTFGGGTKLEIKR	219	219
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPE TGGTAYNQKFKGKATLTADKSSSTAYMELRSLTSEDSAVYYCTRNYDYALDYWGQ GTSVTVSSGGGGSGGGGSGGGGSDVVMTQTPLTLSVTIGQPASISCKSSQSLLDS DGKTYLNWLLQRPGQSPKRLIYLVSKLDSGVPDRFTGSGSGTDFTLKISRVEAEDLG FYYCWQGTHFPWTFGGGTKLEIKR	220	220
DVQLQESGPGLVKPSQSLSLTCSVTGYSITSGYYWNWIRQFPGNKLEWMGYISYD GSNDYNPSLKNRISITRDTSKNQFFLKLNSVTTEDTATYYCARGGGRGWGQGTLV TVSAGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASLSVSDMYW YQQKPGSSPRLLIYDTSNLASGVPVRFSGSGSGTSYSLTISRMEAEDAATYYCQQW SSYPFTFGSGTKLEIKR	221	221
DVKLVESGGGLVKLGGSLKLSCAASGFTFSNYYMSWVRQTPEKRLELVAVINSNG GSTYYPDTVKGRFTISRDNAKNTLYLQMSSLKSEDTALYYCARQEEIGYAMDYWG QGTSVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTLTCSASSSVSSS YLYWYQQRPGSSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAASYFC HQWSSYPPTFGAGTKLELKR	222	222

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
EVQLQQSGAELVRPGALVKLSCKASGFNIKDYFMHWVKQRPEQGLEWIGWIDPETDNTIYDPKFQGKASITADTSSNTAYLQLSSLTSEDTAVYYCARSGNMGFTYWGQGTLVTVSAGGGGSGGGGSGGGGSENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWYQQKSDASPKLWIYYTSNLAPGVPARFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSPSTFGGGTKLEIKR	223	223
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCASQGGSSWGAMDYWGQGTSVTVSSGGGGSGGGGSGGGGSETTVTQSPASLSMAIGEKVTIRCITNTDIDDDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENMLSEDVADYYCLQSDNLPLTFGAGTKLELKR	224	224
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISNGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARHEITTRFAYWGQGTLVTVSAGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPWTFGGGTKLEIKR	225	225
VQLQESGPGLVKPSQSLSLTCSVTGYSITSGYYWNWIRQFPGNKLEWMGYMSYDGSNNYNPSLKNRISITRDTSKNQFFLKLNSVTTEDTATYYCAREAGYFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIVLTQSPATLSVTPGDSVSLSCRASQSISNNLHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGTDFTLSINSVETEDFGMYFCQQSNSWPFTFGSGTKLEIKR	226	226
EVQLVESGGGLVQPKGSLKLSCAASGFSFNTYAMNWVRQAPGKGLEWVARIRSKSNNYATYYADSVKDRFTISRDDSESMLYLQMNNLKTEDTAMYYCVRQYGYDFDYWGQGTTLTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTLTCSASSSVSSSYLYWYQQKPGSSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAASYFCHQWSSYPPTFGGGTKLEIKR	227	227
EVQLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARHKGVNWDYFDYWGQGTTLTVSSGGGGSGGGGSGGGGSETTVTQSPASLSVATGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENTLSEDVADYYCLQSDNMPLTFGAGTKLELKR	228	228
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPETGGTAYNQKFKGKATLTADKSSSTAYMELRSLTSEDSAVYYCTRGDGNYDSWYFDVWGAGTTVTVSSGGGGSGGGGSGGGGSDVVMTQTPLTLSVTIGQPASISCKSSQSLLDSDGKTYLHWLLQRPGQSPKRLIYLVSKLDSGVPDRFTGSGSGTDFTLKISRVEAEDLAVYYCWQGTHFPWTFGGGTKLEIKR	229	229
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARLPVTTVVFDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASLGGKVTITCKASQDINKYIAWYQHKPGKGPRLLIHYTSTLQPGIPSRFSGSGSGRDYSFSISNLEPEDIATYYCLQYDNLRTFGGGTKLEIKR	230	230
EVQLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARRPVVVPFDYWGQGTTLTVSSGGGGSGGGGSGGGGSQAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIVGTNNRAPGVPARFSGSLIGDKAALTITGAQTEDEAIYFCVLWYSNHLVFGGGTKLTVLG	231	231
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARGWDADYFDYWGQGTTLTVSSGGGGSGGGGSGGGGSNIVMTQSPKSMSMSVGERVTLSCKASENVGTYVSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPPTFGAGTKLELKR	232	232
QVQLQQPGAELVKPGASVKLSCKASGYTFTNYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCTRYDYDDYWGQGTTLTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRASETVDSYGYSFMHWYQQKPGQPPKLLIYRASNLESGIPARFSGSGSRTDFTLTINPVEADDVATYYCQQSNEDPRTFGGGTKLEIKR	233	233
EVQLQQSGPELVKPGASVKISCKASGYTFTDYYMNWVKQSHGKSLEWIGDINPNNGGTSYNQKFKGKATLTVDKSSSTAYMDLRSLTSEDSAVYYCARSELGLYAMDYWGQGTSVTVSSGGGGSGGGGSGGGGSDIVMSQSPSSLAVSVGEKVTMSCKSSQSLLYSTNQKNYLAWYQQKPGQSPKLLLYWASTRESGVPDRFTGSGSGTDFTLTINSVKAEDLAVYYCQQYYSYRTFGGGTKLEIKR	234	234

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLQQSGAELMKPGASVKLSCKATGYTFTGYWIEWVKQRPGHGLEWIGEILPG SGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTTEDSAIYYCARGRIHYFDYWGQG TTLTVSSGGGGSGGGGSGGGGSDVVMTQTPLTLSVTIGQPASISCKSSQSLLDSD GKTYLNWLLQRPGQSPKRLIYLVSKLDSGVPDRFTGSGSGTDFTLKISRVEAEDLGV YYCWQGTHFPFTFGSGTKLEIKR	235	235
QVQLQQSGAELMKPGASVKLSCKATGYTFTGYWIEWVKQRPGHGLEWIGEILPG SGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTTEDSAIYYCARGRIHYFDYWGQG TTLTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCKASQSVDYDG DSYMNWYQQKPGQPPKLLIYAASNLESGIPARFSGSGSGTDFTLNIHPVEEEDAAT YYCQQSNEDPFTFGSGTKLEIKR	236	236
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARKGYGYDWYFDVW GTGTTVTVSSGGGGSGGGGSGGGGSQLVLTQSSSASFSLGASAKLTCTLSSQHSTY TIEWYQQQPLKPPKYVMELKKDGSHSTGDGIPDRFSGSSSGADRYLSISNIQPEDE AIYICGVGDTIKEQFVYVFGGGTKVTVLG	237	237
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGEID PSDSYTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARSSYYYYAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSS VSYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAA TYYCQQWSSNPLTFGAGTKLELKR	238	238
DVQLQESGPGLVKPSQSLSLTCSVTGYSITSGYYWNWIRQFPGNKLEWMGYISYD GSNNYNPSLKNRISITRDTSKNQFFLKLNSVTTEDTATYYCARGGGRDWGQGTTLT VSSGGGGSGGGGSGGGGSEIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWY QQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEF PLTFGAGTKLELKR	239	239
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARGGDYDSYAMDYW GQGTSVTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRASESVD NYGISFMNWFQQKPGQPPKLLIYAASNQGSGVPARFSGSGSGTDFSLNIHPMEE DDTAMYFCQQSKEVPPTFGGGTKLEIKR	240	240
QVQLQQPGAELVKPGASVKMSCKASGYTFTSYWITWVKQRPGQGLEWIGDIYP GSGSTNYNEKFKSKATLTVDTSSSTAYMQLSSLTSEDSAVYYCARESVYDGYSWYF DVWGTGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPATLSVTPGDRVSLSCRA SQSISDYLHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGSDFTLSINSVEPEDV GVYYCQNGHSFPLTFGAGTKLELKR	241	241
EFQLQQSGPELVKPGASVKISCKASGYSFTDYNMNWVKQSNGKSLEWIGVINPNY GTTSYNQKFKGKATLTVDQSSSTAYMQLNSLTSEDSAVYYCASTYDYDDWYFDV WGTGTTVTVSSGGGGSGGGGSGGGGSDVLMTQTPLSLPVSLGDQASISCRSSQS IVHSNGDTYLEWYLQKPGQSPKLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA EDLGVYYCFQGSHVPLTFGAGTKLELKR	242	242
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGEID PSDSYTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARSGNYLYAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSS VSYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAA TYYCQQWSSNPLTFGAGTKLELKR	243	243
EFQLQQSGPELVKPGASVKISCKASGYSFTDYNMNWVKQSNGKSLEWIGVINPNY GTTSYNQKFKGKATLTVDQSSSTAYMQLNSLTSEDSAVYYCAREGTSWYFDVWG TGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPTFLAVTASKKVTISCTASESLYSS KHKVHYLAWYQKKPEQSPKLLIYGASNRYIGVPDRFTGSGSGTDFTLTISSVQVEDL THYYCAQFYSYPYTFGGGTKLEIKR	244	244
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWRG GSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKKGDGYDWYFDV WGTGTTVTVSSGGGGSGGGGSGGGGSQLVLTQSSSASFSLGASAKLTCTLSSQHS TYTIEWYQQQPLKPPKYVMELKKDGSHSTGDGIPDRFSGSSSGADRYLSISNIQPE DEAIYICGVGDTIKEQFVYVFGGGTKVTVLG	245	245
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCAREGNYGSSYDAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRASQS VSTSSYSYMHWYQQKPGQPPKLLIKYASNLESGVPARFSGSGSGTDFTLNIHPVEE EDTATYYCQHSWEIPLTFGAGTKLELKR	246	246

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGEID PSDSYTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARSSNYPYAMDY WGQGTSVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSS VSYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAA TYYCQQWSSNPLTFGAGTKLELKR	247	247
EVQLQQSVAELVRPGASVKLSCTASGFNIKNTYMHWVKQRPEQGLEWIGRIDPA NGNTKYAPKFQGKATITADTSSNTAYLQLSSLTSEDTAIYYCAYYSGLYWGQGTLVT VSAGGGGSGGGGSGGGGSDIQMTQSSSYLSVSLGGRVTITCKASDHINNWLAW YQQKPGNAPRLLISGATSLETGVPSRFSGSGSGKDYTLSITSLQTEDVATYYCQQY WSTPLTFGAGTKLELKR	248	248
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPS DSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARRGQIYYGYSWFA YWGQGTLVTVSAGGGGSGGGGSGGGGSDIQMTQSPASLSASVGETVTITCRASE NIYSYLAWYQQKQGKSPQLLVYNAKTLAEGVPSRFSGSGSGTQFSLKINSLQPEDF GSYYCQHHYGTPYTFGGGTKLEIKR	249	249
EVQLQQSGPELVKPGASVKISCKASGYTFTDYYMNWVKQSHGKSLEWIGDINPN NGGTSYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVYYCARSTVVADWYFDV WGTGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPASLSASVGETVTITCRASEN IYSYLAWYQQKQGKSPQLLVYNAKTLAEGVPSRFSGSGSGTQFSLKINSLQPEDFG SYYCQHHYGTPPTFGGGTKLEIKR	250	250
QVQLQQSGAELARPGASVKLSCKASGYTFTSYGISWVKQRTGQGLEWIGEIYPRS GNTYYNEKFKGKATLTADKSSSTAYMELRSLTSEDSAVYFCARSGSSYGYFDVWGT GTTVTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSVSY MHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAATYY CQQWSSNPPTFGAGTKLELKR	251	251
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARKGGYDAYAMDYW GQGTSVTVSSGGGGSGGGGSGGGGSDIVLTQSPASLAVSLGQRATISCRASESVD NYGISFMNWFQQKPGQPPKLLIYAASNQGSGVPARFSGSGSGTDFSLNIHPMEE DDTAMYFCQQSKEVPPTFGGGTKLEIKR	252	252
EFQLQQSGPELVKPGASVKISCKASGYSFTDYNMNWVKQSNGKSLEWIGVINPNY GTTSYNQKFKGKATLTVDQSSSTAYMQLNSLTSEDSAVYYCAREGFITTVVAVDY WGQGTTLTVSSGGGGSGGGGSGGGGSDIVMTQSPTFLAVTASKKVTISCTASESL YSSKHKVHYLAWYQKKPEQSPKLLIYGASNRYIGVPDRFTGSGSGTDFTLTISSVQV EDLTHYYCAQFYSYPYTFGGGTKLEIKR	253	253
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPE TGGTAYNQKFKGKAILTADKSSSTAYMELRSLTSEDSAVYYCTREGNYDAMDYWG QGTSVTVSSGGGGGGGGSGGGGSQAVVTQESALTTSPGGTVILTCRSSTGAVTT SNYANWVQEKPDHLFTGLIGGTSNRAPGVPVRFSGSLIGDKAALTITGAQTEDDA MYFCALWYSTHYVFGGGTKVTVLG	254	254
QVQLQQPGAELVRPGTSVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGVIDP SDSYTNYNQKFKGKATLTVDTSSSTAYMQLSSLTSEDSAVYYCARWDYYGVDYW GQGTTLTVSSGGGGSGGGGSGGGGSQIVLTQSPAIMSASPGEKVTMTCSASSSIS YMHWYQQKPGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAATY YCHQRSSYPTFGAGTKLELKR	255	255
EVQLVESGGGLVQSGGSLRLSCAASGFTFSGYWMYWVRQAPGKGLEWVSAISP GGGSTYYPDSVKGRFTISRDNAKNTLYLQMNSLEPEDTALYYCASSLTATHTYEYDY WGQGTQVTVSS	256	256
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQCLEWMGWINP NSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGYSGSYSD WGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQS VLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSL QAEDVAVYYCQQYYSTPWTFGCGTKVEIKR	257	9316
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQCLEWMGGIIPL SGAPNYAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGALYNWNDGW FDPWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRA SQDIGDYLAWYQQKPGKAPKLLIYDASSLQSGVPSRFSGSGSGTDFTLTISSLQPED FATYYCQQANSFPLTFGCGTKVEIKR	258	9317

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGASVKVSCKASGYSLITHWMHWVRQAPGQCLEWMGMINPSDGVTYYAQTFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREYYGEGFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYSASNLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGCGTKVEIKR	259	9318
QVQLVQSGAEVKKPGSSVKVSCKASGYTFSDHHVHWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGSSWYLHFQHWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGIASYLAWYQQKPGKAPKLLIYAASTLQPGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQFDSYPITFGQGTKVEIKR	260	9319
QVQLVQSGAEVKKPGASVKVSCKASGGTFSRYGIAWVRQAPGQGLEWMGISYPSDGSTSSAQKLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDRLGDLDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISSWLAWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQGYSTPYIFGQGTKVEIKR	261	9320
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTDYYVHWVRQAPGQGLEWVGWISTFTGNTDYAQNFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDAPLAAAGTDYYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYKASSLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPFTFGQGTRLEIKR	262	9321
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYALSWVRQAPGQGLEWMGIINPSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDPGMDVWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDISNYLNWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGQGTKVEIKR	263	9322
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGYSGSYSDWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPWTFGQGTKVEIKR	264	9323
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYALSWVRQAPGQGLEWMGIINPSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDPGMDVWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDISNYLNWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGQGTKVEIK	265	9324
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGIIDPSGGSTTYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDMGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIK	266	9325
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAFSWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDVGDRGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYKASSLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSFSSPLTFGQGTKVEIK	267	9326
QVQLVQSGAEVKKPGASVKVSCKASGSTFSGYYMHWVRQAPGQGLEWMGWIDPNGGGTQYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDIVHDGTEYFQHWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQNVNTWLAWYQQKPGKAPKLLIYEASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPFTFGQGTKLEIK	268	9327
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYYMHWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDIVHDGTEYFQHWGQGTLVTVSSGGGGGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISDWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCAQHNHYPYTFGQGTRLEIK	269	9328
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPSGGSTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGRDHDAFDIWGQGTMVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYTTPFTFGQGTKVEIK	270	9329

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGASVKVSCKASGFTFTDYGISWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGRSHDAFDIWGQGTMVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRASGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPFTFGQGTKLEIK	271	9330
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGWMNPHSGDTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARWVGTTEYYYYYMDVWGKGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQAIRDDLGWYQQKPGKAPKLLIYDASHLEAGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPITFGQGTRLEIK	272	9331
QVQLVQSGAEVKKPGASVKVSCKASGYTFTDYYLHWVRQAPGQGLEWMGIIDPSGGSTSIAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCATTAYYDFWSGYSMDVWGKGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGVGNDLAWYQQKPGKAPKLLIYAASTLQTGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQASSFPLTFGPGTKVDIK	273	9332
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSHYMHWVRQAPGQGLEWMGIIDPSGGSTSYAQEFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDMDNWNTGYYYYMDVWGKGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQIIGTNLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYTFPVTFGQGTKLEIK	274	9333
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAINWVRQAPGQGLEWMGWVNPNSGDTAYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDQRGGDAWDVWGKGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISTWLAWYQQKPGKAPKLLIYDASSLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPFTFGPGTKVDIK	275	9334
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGIITPSGGSTTYAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDTAGHFDIWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSNNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYGSPLTFGPGTKVDIK	276	9335
QVQLVQSGAEVKKPGASVKVSCKASGGTFRNDVINWVRQAPGQGLEWIGWMNPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDNPDLDGMDVWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSSPPTFGQGTRLEIK	277	9336
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAINWVRQAPGQGLEWLGWISAYNGNTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLVGHFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSSPPTFGGGTKVEIK	278	9337
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGYSGSYSDWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPWTFGQGTKVEIK	279	9338
QVQLVQSGAEVKKPGASVKVSCKASGNTLSSHAISWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDQGSSGTFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRASGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYGSPPTFGQGTRLEIK	280	9339
QVQLVQSGAEVKKPGASVKVSCKASGGTLSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSTDVIDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQDIRNYLNWYQQKPGKAPKLLIYDASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPWTFGQGTKLEIK	281	9340
QVQLVQSGAEVKKPGASVKVSCKASGYIFTSYDINWVRQAPGQGLEWMGWINPNSGDTKYAQNFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGGTVTPTEEYYYYGMDVWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDR	282	9341

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
VTITCQASQDISNYLNWYQQKPGKAPKLLIYNASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLNSYPFTFGGGTKVEIK		
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWISVYNGNTNYAQNLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCASLDDLDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQSISTWLAWYQQKPGKAPKLLIYAASTLRSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCLQHYTYPLTFGQGTRLEIK	283	9342
QVQLVQSGAEVKKPGASVKVSCKASGHTFTSYYIHWVRQAPGQGLEWMGWINPNNGGTHYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDMVRDSAEYFQHWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASEDISTYLAWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSHTIPWTFGQGTRLEIK	284	9343
QVQLVQSGAEVKKPGSSVKVSCKASGYTFITSYIHWVRQAPGQGLEWMGMINPSGGTTTYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDSSGYPIDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASHHISDFLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPYTFGQGTKLEIK	285	9344
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGGIIPLSGAPNYAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGALYNWNDGWFDPWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDIGDYLAWYQQKPGKAPKLLIYDASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPLTFGGGTKVEIK	286	9345
EVQLLESGGGLVQPGGSLRLSCAASGFTVGSWYMSWVRQAPGKGLEWVAGIWYEGSNKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARLGTASLPYFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDIRSYLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYTAPPTFGQGTRLEIK	287	9346
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWVGWINPNRGDTKYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARESGDGFDPWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDISNNLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCLQHNTYPLTFGQGTKLEIK	288	9347
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTNYYIHWVRQAPGQGLEWMGWMNPNSGNTGYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDWPNWFDPWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDISNWLAWYQQKPGKAPKLLIYDASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAISFPLTFGGGTKVEIK	289	9348
QVQLVQSGAEVKKPGASVKVSCKASGYSFTDNYIHWVRQAPGQGLEWMGWIRSDNGETSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREVQLVGFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGIANYLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQADSFPLTFGQGTKVEIK	290	9349
QVQLVQSGAEVKKPGSSVKVSCKASGYTFSDHHVHWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGSSWYLHFQHWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGIASYLAWYQQKPGKAPKLLIYAASTLQPGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQFDSYPITFGQGTKVEIK	291	9350
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAIYWVRQAPGQGLEWMGGIIPIFGTTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKGVDRYNWNDAFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYAASRLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSSIIPFTFGQGTKLEIK	292	9351
QVQLVQSGAEVKKPGASVKVSCKASGYTFTDYYMHWVRQAPGQGLEWMGWIHSNSGGTHSAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARESSGYDSSLDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPYTFGQGTKLEIK	293	9352
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYGISWVRQAPGQGLEWVGWINPNSGDTDYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCTTDPRLDSSDPGYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRAS	294	9353

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
QSIGRWLAWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDF ATYYCQQSYSTPRTFGQGTKVEIK		
QVQLVQSGAEVKKPGASVKVSCKASGGTFGNYGINWVRQAPGQGLEWMGWIS AYNGNTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGGMDVWG QGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSINS WLAWYQQKPGKAPKLLIYDTSSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYC QQTYSTPYTFGQGTKLEIK	295	9354
QVQLVQSGAEVKKPGASVKVSCKASGGTFSRYGIAWVRQAPGQGLEWMGISYP SDGSTSSAQKLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDRLGDLDYWG QGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISS WLAWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYC QQGYSTPYIFGQGTKVEIK	296	9355
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWMN PNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSIVGGYPF DYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRAS QSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFA TYYCQQTDSIPITFGQGTRLEIK	297	9356
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGTITPI FGTTDYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAREGYSSSWHDDAF DIWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRAS QSISSYLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFA TYYCQQSYSIPYTFGQGTKVEIK	298	9357
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGIIDPS GGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDYGLDSW GQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQTIR SYLNWYQQKPGKAPKLLIYKASSLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYC QQTYTIPITFGPGTKVDIK	299	9358
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGW MNPNSGDTGYAQRFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCATGGSDSS GYYYEGYFQHWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGD RVTITCRASQTISNWLAWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTL TISSLQPEDFATYYCQQANSFPPTFGQGTKVEIK	300	9359
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWLGYMSP NSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDKGGYYDSS GYYWYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTIT CRASQYIGSYLNWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQ PEDFATYYCQQVDSYPLTFGGGTKVEIK	301	9360
EVQLLESGGGLVQPGGSLRLSCAASGFSLSSYEMHWVRQAPGKGLEWVSAISSN GGSTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARVGDGDGYNPDF DYWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASISCRSS QSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSNRASGVPDRFSGSGSGTDFTLKISR VEAEDVGVYYCMQGTHWPPTFGQGTKLEIK	302	9361
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYGISWVRQAPGQGLEWMGWIDP TSGATDTAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKDPIVATEVDYW GQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASISCRSSQSLL HSNGYNYLDWYLQKPGQSPQLLIYFGSNRASGVPDRFSGSGSGTDFTLKISRVEAE DVGVYYCMQALQAPVSFGQGTRLEIK	303	9362
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWMS PNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSGAFDIW GQGTMVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSV LSSSYNKNYLAWYQQKPGQPPKLLIYWASSRQSGVPDRFSGSGSGTDFTLTISSLQ AEDVAVYYCQQYYSTPLTFGQGTKVEIK	304	9363
QVQLVQSGAEVKKPGASVKVSCKASGVTISNYAISWVRQAPGQGLEWMGWMN PNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGLLDAFDI WGQGTMVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQ SVSSSSYNKNYLAWYQQKPGQPPKLLIYWASVRESGVPDRFSGSGSGTDFTLTISSL QAEDVAVYYCQQYYSTPITFGQGTRLEIK	305	9364
QVQLVQSGAEVKKPGASVKVSCKASGGTFSRYGITWVRQAPGQGLEWMGWM NPYDGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGGRHHDA	306	9365

TABLE 19-continued

Full Binder Sequences Table 19-Full length sequences		
Sequence	Binder Name	SEQ ID NO:
FDIWGQGTMVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCK STQNVLSSSNNNSYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTL TISSLQAEDVAVYYCQQYYSTPFTFGQGTRLEIK		
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPS GDGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDISNDAFDIW GQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQD IGNYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFAT YYCQQTYNTPLTFGGGTKLEIK	307	9366
QVQLVQSGAEVKKPGASVKVSCKASGYILTGHYMHWVRQAPGQGLEWMGWIS AYNGDTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGSSWDDAF DIWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQA SQDISNYLNWYQQKPGKAPKLLIYEASTLQSGVPSRFSGSGSGTDFTLTISSLQPED FATYYCQQSYSTPFTFGPGTKVDIK	308	9367
EVQLVESGGGLVKPGGSLRLSCAASGFTFSNHYTSWVRQAPGKGLEWVSAIGAG GGTYYADSVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCAREGWNDDVFDIW GQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQD ISTWLAWYQQKPGKAPKLLIYRASTLESGVPSRFSGSGSGTDFTLTISSLQPEDFATY YCQQSYSIPLTFGGGTKVEIK	309	9368
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWVGIINPS AGTTYYAERFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGNFGAFDIWG QGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQNIN NYLNWYQQKPGKAPKLLIYAASRLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYY CQQSYSAPVTFGQGTKLEIK	310	9369
QVQLVQSGAEVKKPGSSVKVSCKASGYSFTTYAITWVRQAPGQGLEWMGEIIPIF GTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDKSGWNYGSGSY NDAFDIWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTI TCRASQNINTWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSL QPEDFATYYCQQAYSFPFTFGPGTKVDIK	311	9370
QVQLVQSGAEVKKPGASVKVSCKASGYAFTGYYMHWVRQAPGQGLEWMGW MNPNSGKTEYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGGLDF DYWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRA SQRIGNYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPED FATYYCQQSYSTPLTFGGGTKVEIK	312	9371
QVQLVQSGAEVKKPGASVKVSCKASGYTFTTYYIHWVRQAPGQGLEWMGWMN PNTGDTGSAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDPAVTPDAF DIWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRA SQSISTYLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDF ATYYCQQSYRTVTFGQGTRLEIK	313	9372
QVQLVQSGAEVKKPGASVKVSCKASGGTLSSYAISWVRQAPGQGLEWMGIIDPS GGGTSYAQKLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAGSLYYYGMDVW GQGTMVTVSSGGGGSGGGGSGGGGSEIVMTQSPATLSVSPGERATLSCRASQSV GSYLAWYQQKPGQAPRLLIYGASTRATGIPARFSGSGSGTEFTLTISSLQSEDFAVY YCQQYDSSSQTFGQGTKVEIK	314	9373
QVQLVQSGAEVKKPGSSVKVSCKASGGTFGSSAISWVRQAPGQGLEWMGGIIPIF GTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKEDDILPPRAFDIW GQGTMVTVSSGGGGSGGGGSGGGGSEIVMTQSPATLSVSPGERATLSCRASRSV STYLAWYQQKPGQAPRLLIYGASTRATGIPARFSGSGSGTEFTLTISSLQSEDFAVYY CQQYDGSPYTFGQGTKLEIK	315	9374
EVQLLESGGGLVQPGGSLRLSCAASGFTFDDYAMHWVRQAPGKGLEWVSGISG GGGVTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARVYSSGWLDAF DIWGQGTMVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASISCRSS QSLLHSNGYNYLDWYLQKPGQSPQLLIYDASNLETGVPDRFSGSGSGTDFTLKISR VEAEDVGVYYCMQALQTPPAFGPGTKVDIK	316	9375
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWISG YNGNTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCASSDVSPDAFDI WGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQS VLSSSYNKNFLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSL QAEDVAVYYCQQYYSAPPTFGQGTKVEIK	317	9376

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGASVKVSCKASGGTQNIYAITWVRQAPGQGLEWVGWVNPNSGNTGYSQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCATPTSSSDDAFDIWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNFLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSDPITFGQGTKLEIK	318	9377
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARASRGDDAFDIWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASARESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSIPIAFGQGTRLEIK	319	9378
QVQLVQSGAEVKKPGASVKVSCKASGIPFTSDDINWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARERYEGGYSSGPGNYYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRDSGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSIPYTFGQGTKLEIK	320	9379
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGWMNPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDDDYGDYPVWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSTSYNKNYLAWYQQKPGQPPKLLIYWASTRASGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYTTPPTFGQGTKVEIK	321	9380
QVQLVQSGAEVKKPGASVKVSCKASGDTFSDHAINWVRQAPGQGLEWMGWMNPKIGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCVYDSSGYDAFDIWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSTSYNRNFLAWYQQKPGQPPKLLIYWASTRQSGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPYTFGQGTKLEIK	322	9381
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGRINPGTGGTDYAHKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARETPSDYYDSSGYYYNDAFDIWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPLTFGGGTKVEIK	323	9382
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAISWVRQAPGQGLEWVGIIIPSGGTNYAQTFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDLGTTFDIWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPTFGQGTKVEIK	324	9383
QVQLVQSGAEVKKPGASVKVSCKASGYTFTAYYLHWVRQAPGQGLEWIGWINPDNDNAYYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDIAVAALAYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYGASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQEADSFPLTFGGGTKVEIK	325	9384
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVAVISYDGSDQYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARQSLYYYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGIRNDLGWYQQKPGKAPKLLIYDASSLHSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPWTFGQGTKLEIK	326	9385
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTDYYVHWVRQAPGQGLEWVGWISTFTGNTDYAQNFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDAPLAAAGTDYYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYKASSLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPFTFGQGTRLEIK	327	9386
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVAFISDDGITKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARDDSSGYGGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSINRWLAWYQQKPGKAPKLLIYSASNLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPLTFGGGTKVEIK	328	9387
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMHWVRQAPGKGLEWVAVISYDGGDKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCASGSLVLGYYYMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSINTWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPFTFGPGTKVDIK	329	9388

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGASVKVSCKASGYTFTNYYIHWVRQAPGQGLEWMGWINPNTGGTDYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCATGGGGSYYDAFDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSIRTWLAWYQQKPGKAPKLLIYDASSLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLNSYPLTFGGGTKVEIK	330	9389
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGRINPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDIGEGYSMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSIRTYLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSAPLTFGGGTKLEIK	331	9390
EVQLLESGGGLVQPGGSLRLSCAASGFTFSNHYTSWVRQAPGKGLEWVAVISYDGSNKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCAREEKYSSSWYVGVDAFDIWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISTYLNWYQQKPGKAPKLLIYAASSLHSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIK	332	9391
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSSAMHWVRQAPGKGLEWISSISGSGDNAYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARDQEDYYYDSSGYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSITTYLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIK	333	9392
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSHAISWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKGDWGIVVVPAAIGAFDIWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYAASSLQSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQARQTPLTFGQGTRLEIK	334	9393
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTAYYMHWVRQAPGQGLEWVGRISPVFGSTTYAQRFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDLGYYDSSGYRYDAFDIWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYGASSLQSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQTLQTPFTFGPGTKVDIK	335	9394
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGGISPMFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKDGWYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSDRASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQTPLTFGPGTKVDIK	336	9395
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYGISWVRQAPGQGLEWMGWINPNSGGTKYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGEAGNLDWYFDLWGRGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQTVFSTSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPLTFGGGTKVEIK	337	9396
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYGISWVRQAPGQGLEWMGWINPNNGDTKYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREDVWYFDLWGRGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKTSQSVFSTSYNRDYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSSPPTFGQGTKVEIK	338	9397
QVQLVQSGAEVKKPGASVKVSCKASGYTFTTYGISWVRQAPGQGLEWMGWISTYDGKTNYAQKLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCALHLGGDWYFDLWGRGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISSWLAWYQQKPGKAPKLLIYDASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPFTFGPGTKVDIK	339	9398
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGWINPNTGATYYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARQHGDYDWYFDLWGRGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYDASNLKTGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSFPTFGGGTKVEIK	340	9399
QVQLVQSGAEVKKPGASVKVSCKASGDTFTTYYVHWVRQAPGQGLEWMGWINPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSGRHWGQGTLVTVSSGGGGSGGGGSGGGGSEIVMTQSPATLSVSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDTSSRATGIPARFSGSGSGTEFTLTISSLQSEDFAVYYCQQYYDTPYTFGQGTKLEIKR*	341	9400

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYGISWVRQAPGQGLEWMGRIIPMLGIANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCVREEVAGANWFDPWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYLASTREPGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPPTFGGGTKLEIKR*	342	9401
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYAMNWVRQAPGQGLEWMGIINPSGGSTSYARKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGDYGSGEFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYVAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPLTFGGGTKVEIKR*	343	9402
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSSYMHWVRQAPGQGLEWMGWMNPRSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARERDDYGDYGWLDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYAAASLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTYSTPWTFGQGTRLEIKR*	344	9403
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLYDSSGYWHYYYYMDVWGKGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDINTYLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSSSFPLTFGQGTKVEIKR*	345	9404
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAFSWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARFSGYDYVDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLYNFPYTFGGGTKVEIKR*	346	9405
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPNGGNTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDVGEDFDLWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISRYLAWYQQKPGKAPKLLIYGASTRESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPLTFGQGTKLEIKR	347	9406
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYYIHWVRQAPGQGLEWLGVINPADGDTTYAQMFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDFDWLFAMDVWGKGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQTLSGWLAWYQQKPGKAPKLLIYGASTLQGGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYYSYPPTFGQGTKVEIKR*	348	9407
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYALNWVRQAPGQGLEWMGRINPNGGTTYYAKNFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKHGDHGFYVWGLWTKGTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCFASQDIINYLNWYQQKPGKAPKLLIYEASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGQGTKVEIKR	349	9408
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGMINPNVGSATYAQRFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREDSGTSWFDPWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYDVFNLGTGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPFTFGQGTRLEIKR*	350	9409
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYYMHWVRQAPGQGLEWMGIINPSDGSTSYAQRFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDDRGSNYYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYMASNLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTNSFPLTFGQGTKLEIKR*	351	9410
QVQLVQSGAEVKKPGASVKVSCKASGYTFTAYYVHWVRQAPGQGLEWMGWMNPNSGTTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSSDYYGDYRADAFDIWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIKR*	352	9411
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGVISPSGDATLYAQSFQGRVTITADESTSTAYMELSSLRSEDTAVYYCVKGLDHWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYN	353	9412

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
YLDWYLQKPGQSPQLLIYLGSNRASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYY CMQALQSPWTFGQGTKLEIKR*		
EVQLLESGGGLVQPGGSLRLSCAASGFSFSDYGMHWVRQAPGKGLEWVSAIGGI GDSTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARMNYGDSNYYYY YGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATI NCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTD FTLTISSLQAEDVAVYYCQQYYSSPLTFGGGTKVEIKR*	354	9413
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYDISWVRQAPGQGLEWMGMISP SDGSTTYAPKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGAVGFDYWG QGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASISCRSSQSLLHS NGYNYLDWYLQKPGQSPQLLIYLGSNRASGVPDRFSGSGSGTDFTLKISRVEAEDV GVYYCMQALQTPPSFGQGTKVEIKR*	355	9414
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYGISWVRQAPGQGLEWMGWINT YSGYTDYAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCTTDDFLSFGYWGQ GTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASESVSTW LAWYQQKPGKAPKLLIYKASRLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQ QSYKTPYTFGQGTKLEIKR*	356	9415
QVQLVQSGAEVKKPGSSVKVSCKASGYMFTDYYIHWVRQAPGQGLEWMGGIIP YFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARSISGSYVLDAFD IWGQGTTVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQ SVLYSSNNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISS LQAEDVAVYYCQQYFTTPLTFGQGTKLEIKR*	357	9416
QVQLVQSGAEVKKPGSSVKVSCKASGYTFNSYGISWVRQAPGQGLEWMGGIIPIF GTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDWGYGDYADDA FDIWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCR ASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPED FATYYCQQSYSTPYTFGQGTKVEIKR*	358	9417
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSNNDINWVRQAPGQGLEWMGWIN PIYGSANYAQNFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAADWRGFDYWG QGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCKSSQSVLS SSYNKNYLAWYQQKPGQPPKLLIYWASTRASGVPDRFSGSGSGTDFTLTISSLQAE DVAVYYCQQYYDTPLTFGQGTKVEIKR*	359	9418
QVQLVQSGAEVKKPGASVKVSCKASGYTFTEYAIHWVRQAPGQGLEWMGRMN PHNGDTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGDYLGYPI DCWGRGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRAS QSISSYLNWYQQKPGKAPKLLIYKASTLESGVPSRFSGSGSGTDFTLTISSLQPEDFA TYYCQQNDSIPITFGQGTRLEIKR*	360	9419
EVQLLESGGGLVQPGGSLRLSCAASGFTFSDYSMSWVRQAPGKGLEWVAAIWQ DGNVKFYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARDGNSGYVFW GQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSIS RWLAWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATY YCLQDYSYPLTFGQGTKVEIKR*	361	9420
QVQLVQSGAEVKKPGASVKVSCKASGYTFTTYYMHWVRQAPGQGLEWMGWIN PNTGDTAYAQKIQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARTAEAVAGLP AFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPDSLAVSLGERATINCK TSQSVFSTSYNRDYLAWYQQKPGQPPKLLIYWASTRAAGVPDRFSGSGSGTDFTL TISSLQAEDVAVYYCQQYYYTSTFGQGTKVEIKR*	362	9421
QVQLVQSGAEVKKPGSSVKVSCKASGGTSNNYAIDWVRQAPGQGLEWMGGIIP LFGTTTYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARVTLYGDYDYWG QGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSINR YLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYC QQANSFPPTFGGGTKVEIKR*	363	9422
QVQLVQSGAEVKKPGASVKVSCKASGYSLITHWMHWVRQAPGQGLEWMGMI NPSDGVTYYAQTFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREYYGEGFD YWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRAS QGISNYLAWYQQKPGKAPKLLIYSASNLQSGVPSRFSGSGSGTDFTLTISSLQPEDF ATYYCQQSYSTPLTFGGGTKVEIKR*	364	9423
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPS GGSTSNAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDTAMDG WGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQS	365	9424

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
IDSYLNWYQQKPGKAPKLLIYKASTLESGVPSRFSGSGSGTDFTLTISSLQPEDFATY YCQQSYSAPLTFGGGTKVEIKR*		
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYYLHWVRQAPGQGLEWMGIITPS GGSTTYAHKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGGLASFDYW GQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQDIS TWLAWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYY CQQVNSDPYTFGQGTRLEIKR*	366	9425
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWMN PNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGGGWAMT DAFDIWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTIT CQASQDISNYLNWYQQKPGKAPKLLIYAASTLESGVPSRFSGSGSGTDFTLTISSLQ PEDFATYYCQQGDSLPLTFGGGTKVEIKR*	367	9426
EVQLVESGGGLVKPGGSLRLSCAASGFTFDDYGMSWVRQAPGKGLEWVSLIYSG GDTYYADSVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCTRKEYYYDSSGYLRLF DYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRAS QGISNYLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDF ATYYCQQSDSFPYTFGQGTKVEIKR*	368	9427
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTDYYMHWVRQAPGQGLEWMGGIN PIFGTSNYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDISGYDYYYG MDVWGQGTTVTVSSGGGGSGGGGSGGGGSEIVMTQSPATLSVSPGERATLSCR ASQSVSTYLAWYQQKPGQAPRLLIYGASTRATGIPARFSGSGSGTEFTLTISSLQSE DFAVYYCQQHDSYPLTFGGGTKVEIKR*	369	9428
QVQLVQSGAEVKKPGASVKVSCKASGGTLNNYAFSWVRQAPGQGLEWMGMID PSDGTIAYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARSDYDFWSGL GGYFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTI TCRASQGIRNDLGWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSL QPEDFATYYCQQANSFPPTFGQGTRLEIKR*	370	9429
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGTIDP NSGGTMFAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSAEWELGG SFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCR ASESISTYLNWYQQKPGKAPKLLIYKASNLESGVPSRFSGSGSGTDFTLTISSLQPED FATYYCQQTDSTFITFGQGTKVEIKR*	371	9430
EVQLLESGGGLVQPGGSLRLSCAASGFTFSNHYTSWVRQAPGKGLEWVSSIGVN GDTYYLDSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCAREGLVFSGRGHWY FDLWGRGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRA SRNIHDYLNWYQQKPGKAPKLLIYAASTLQTGVPSRFSGSGSGTDFTLTISSLQPED FATYYCQQTYSTPPTFGPGTKVDIKR*	372	9431
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGRINP NGGNTSNAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDYEDADFDG WGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQS NDSYLNWYQQKPGKAPKLLIYKASTLESGVPSRFSGSGSGTDFTLTISSLQPEDFAT YYCQQSYSSPLTFGQGTKVEIKR	373	9432
QVQLVQSGAEVKKPGASVKVSCKASGYTFSDHHVHWVRQAPGQGLEWMGW MNPDSGNTGYAQRFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSTSGV DYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRAS QSISDFLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDF ATYYCQQSYSSPYTFGQGTKVEIKR*	374	9433
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVAVISYD GHDQFYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARGEQQLEGFYYY YGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTIT CQASQDISNYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQ PEDFATYYCQQANRFPLTFGQGTKLEIKR*	375	9434
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMHWVRQAPGKGLEWVAVISYD GSKEYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCASDYGDYGTYDYW GQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCQASQDI SNYLNWYQQKPGKAPKLLIYKASNLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATY YCQQSYNFPATFGQGTRLEIKR*	376	9435
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMHWVRQAPGKGLEWVSGISG GGDDTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCAREPLAYCGGDC	377	9436

TABLE 19-continued

Full Binder Sequences
Table 19-Full length sequences

Sequence	Binder Name	SEQ ID NO:
PGGFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIVMTQSPLSLPVTPGEPASIS CRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSNRASGVPDRFSGSGSGTDFTL KISRVEAEDVGVYYCMQGTHWPETFGQGTKVEIKR*		
EVQLLESGGGLVQPGGSLRLSCAASGFTFSDHYMDWVRQAPGKGLEWVSAIGTG GDTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARHEDTAIFLDYWG QGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQSISS YLNWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYC QQSYSTPLTFGQGTKLEIKR*	378	9437
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYYMHWVRQAPGQGLEWMGMIS PSDGSTTYAPKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGYDAWSYG MDVWGQGTMVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITC RASQGISDYLAWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQP EDFATYYCQQSYILPLTFGGGTKVEIKR*	379	9438
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGWM NPNSGNTGYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDGVTGTDY WGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQ DINDFLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFA TYYCQQSYSAPYTFGQGTKLEIKR*	380	9439
EVQLLESGGGLVQPGGSLRLSCAASGFAFSSYVLHWVRQAPGKGLEWVSAISGAG DSTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCAREPTTVTDDWYFD LWGRGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQ SISNWLAWYQQKPGKAPKLLIYAASKLESGVPSRFSGSGSGTDFTLTISSLQPEDFA TYYCQQSYSSPWTFGQGTRLEIKR*	381	9440
EVQLVESGGGLVKPGGSLRLSCAASGFAFSSHWMHWVRQAPGKGLEWVSAISG NGDNSYYADSVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCARDRAPEYFDLW GRGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQGID SWLAWYQQKPGKAPKLLIYAASTLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYY CQQAYSFPLTFGGGTKVEIKR*	382	9441
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWINP NSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDDYGDYGGG MDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCR ASQNIGTWLAWYQQKPGKAPKLLIYRASSLESGVPSRFSGSGSGTDFTLTISSLQPE DFATYYCQQAYSFPWTFGQGTKLEIKR*	383	9442
QVQLVQSGAEVKKPGASVKVSCKASGYTFTDYYMHWVRQAPGQGLEWMGW MNPNSGHTGYAEKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDTSPRY GDGFFDYWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVT ITCRASQNINNWLAWYQQKPGKAPKLLIYKASTLQSGVPSRFSGSGSGTDFTLTISS LQPEDFATYYCQQADSFPPTFGQGTKVEIKR*	384	9443
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMHWVRQAPGKGLEWVAVTSY DGSNKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARESGFSAEYFQH WGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRASQ DISSYLAWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFAT YYCQQLNRYPITFGQGTKVEIKR*	385	9444
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPS GGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARATGLYCSGSCFD YWGQGTLVTVSSGGGGSGGGGSGGGGSDIQMTQSPSSLSASVGDRVTITCRAS QDISNYLAWYQQKPGKAPKLLIYAASILHSGVPSRFSGSGSGTDFTLTISSLQPEDFA TYYCQQYDSSFITFGQGTRLEIKR*	386	9445

TABLE 20

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEIFPGS GHTSFNEKFKGKATFTADTSSNTAYIQLSSLTSEDSAVYYCARRGYGYDEGFDYWG QGTTLTVSS	1	257
QVQLQQSGAELMKPGASVKISCKATGYTLSSYWIEWVKQRPGHGLEWIGEILPGS GSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWG QGTTLTVSS	2	258
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGATSYNQKFKG KATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGEYFDYWGQGTTLTVSS	3	259
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	4	260
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTSYNQKFKG KATFTVDTSSSTAYMHFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	5	261
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	6	262
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGANGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	7	263
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	8	264
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	9	265
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	10	266
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLAVSS	11	267
LVKTGASVKISCKASGYSFTGFYMHWVKQSHGKGLEWIGYISSYNGATGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	12	268
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGATGYNQKFK GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	13	26
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGTTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAYW GQGTLVTVSA	14	270
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPGS GSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWG QGTLVTVSA	15	271
QVQLQQPGAELIKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARPGGYGFVYWG QGTLVTVSA	16	272
EVQLQQSGAELVKPGASVKLSCTPSGFNIKDTSLHWVKQGPEQGLEWIGRIDPAN GNTKYDPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCARGPDDGYFYYYSMD YWGQGTSVTVSS	17	273
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMGWIN TYTGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDMATYFCARKYYDYEFAYW GQGTLVTVSA	18	274
QVQLQESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARHEEANWAWFAY WGQGTLVTVSA	19	275
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQANDTAIYYCASYYGSSRSYWYLDV WGAGTTVTVSS	20	276

TABLE 20-continued

VH Sequences
Table 20-VH sequences

Sequence	Binder Name	SEQ ID NO:
QVKLVESGGDLVKPGGSLKLSCATSGFTFSSYGMSWVRQTPDKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEGTAMYYCARHNYSNWDWFAYWGQGTLVTVSA	21	277
EVQLQQSGAELVKPGASVKLSCTPSGFNIKDTSLHWVKQGPEQGLEWIGRIDPANGNTKYDPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCARGPDDGYFYYYSMDYWGQGTSVTVSS	22	278
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	23	27
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPGQGLEWIGLIDPSDSETHYNQVFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCATYDVYYRFAYWGQGTLVTVSA	24	280
QVQLQQPGAELVRPGASVKLSCKASGYTFTSYWINWVKQRPGQGLEWIGNIYPSDSYTNYNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCTRGNYIDYWGQGTTLTVSS	25	281
QVQLKESGPGLVAPSQSLSIPCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAGGTTNYNSALMSRLSISKDNSKSQVFLKMYSLQTDDTAMYYCARGDGYDDGYAMDYWGQGTSVTVSS	26	282
QVQLQQSGAELVKPGASVKLSCKASGSTFTTYYIYWVKQRPGQGLEWIGEINPSNGGTNFNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCTSYYTHETYYYAMDYWGQGTSVTVSS	27	283
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMQWVKQRPGQGLEWIGEIDPSDSYTNYNQKFKGKATLTVDTSSSTAYMQLSSLTSEDSAVYYCARAEYGYGNYPWFAYWGQGTLVTVSA	28	284
QVQLQQPGAELVKPGASVKVSCKASGYTFTSYWMHWVKQRPGQGLEWIGRIHPSDSDTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAIPYYYGGWYFDVWGTGTTVTVSS	29	285
EVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARLYDAHWDYFDYWGQGTTLTVSS	30	286
QVQLQQPGSVLVRPGASVKLSCKASGYTFTSSWMHWAKQRPGQGLEWIGEIHPNSGNTNYNEKFKGKATLTVDTSSSTAYVDLSSLTSEDSAVYYCAIYYDYDAYYFDYWGQGTTLTVSS	31	287
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCANPYYGYDVGYWGQGTTLTVSS	32	288
EVQLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCTRHDDSSYDWFAYWGQGTLVTVSA	33	289
QVQLQQPGAELVKPGASVKLSCKASGYTFTNYWMHWVKQRPGQGLEWIGMIHPNSGTTNYNEKFKSKATLTVDKSSSTTYMQLISLTSEDSAVYYCARFGDGYHFDYWGQGTTLTVSS	34	290
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMGWINTNTGEPTYAEEFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARWYPYFDYWGQGTTLTVSS	35	291
QVQLQQSGAELAKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGYINPSSGYTKYNQKFKDKATLTADKSSSTAYMQLSSLTYEDSAVYYCARSDGSSGNWYFDVWGTGTTVTVSS	36	292
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYGNYDYAMDYWGQGTSVTVSS	37	293
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKDKATFTADTSSNTAFMQLSSLTSEDSAVYYCARRAYGYDEGFDYWGQGTTLTVSS	38	294

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVRQAPGHGLEWIGEVLPGSGSTSYNEKFKGRATFTADTSSNTAYMQLSSLRSEDSAVYYCARRAYGYDEGFDYWGQGTTVTVSS	39	295
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVRQAPGHGLEWIGEILPGSGRTSYIEKFKGRATFTADTSSNTAYMQLSSLRSEDSAVYYCARRGYGYDEGFDYWGQGTTVTVSS	40	296
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEVLPGSGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDYWGQGTTLTVSS	41	297
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGRTSYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	42	298
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDGGFDYWGQGTTLTVSS	43	299
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSDSTSYNEKFKGKTTFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDYWGQGTTLTVSS	44	300
QVQLQQSGAELMKPGASVKISCKATAYTFSIYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKVKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDGGFDYWGQGTTLTVSS	45	301
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEIFPGSGHTSFNEKFKGKATFTADTSSNTAYIQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	46	302
QVQLQQSGAELMKPGASVKMSCKATGYTFSNYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGSTLTVSS	47	303
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	48	304
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVQQRPGHGLEWIGEILPGSGYTSYIEQFKGKATFTADTSSNTAYMQLGSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	49	305
QVQLQQSGAELMKPGASVKISCKATGYTLSSYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	50	306
QVQLQQSGAELMKPGASVKISCKGTGYTFSSYWIEWVKQRPGHGLEWIGEISPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	51	307
QVQLQQSGAELMKPGASVKISCKATGYTFGTYWIEWVKQRPGHGLEWIGEILPGSGTPNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDAGFDYWGQGTTLTVSS	52	308
QVQLQQSGAELMKPGASVKISFKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTSCNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	53	309
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGRTSYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	54	310
QVQLQQSGAELMKPGASVKMSCKATGYTFSNYWIEWVKQRPGHCLEWIGEILPGSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGSTLTVSS	55	311
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHCLEWIGEILPGSGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	56	312

TABLE 20-continued

VH Sequences
Table 20-VH sequences

Sequence	Binder Name	SEQ ID NO:
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSS	57	313
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGATSYNQKFKGKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGEYFDYWGQGTTLTVSS	58	314
QVQLKQSGPGLVQPSQSLSITCTVSGFSLSSYGVHWVRQSPGKALEWLGVIWRGGSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	59	315
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTSYGVHWVRQSPGKGLEWLGVIWRGGSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDSAIYYCAKNLYGHYVMDYWGQGTSVTVSS	60	316
QVQLKQSGPGPVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWRGGSTDNNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	61	317
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTRYGVHWVRQSPGKGLEWLGVIWRGGSTDHNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	62	318
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTTYGVHWVRQSPGKGLEWLGVIWRGGSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	63	319
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAYWGQGTLVTVSA	64	320
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGTTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAYWGQGTLVTVSA	65	321
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPGSGSTYYNEKFKGKATLTADKSSNTVYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	66	322
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	67	323
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPGSGSSYYNEKFRGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	68	324
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQKTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYIQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	69	325
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSPNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARPGGYGFAYWGQGTLVTVSA	70	326
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSA	71	327
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSA	72	328
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSNYYNEKFKGKAIMTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSA	73	329
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSA	74	330

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVRQAPGQGLEWIGEIYPGSGSSYYNEKFKGRATLTADKSSNTAYMQLSSLRSEDSAVYFCARPGDLGFAYWGQGTLVTVSS	75	331
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSS	76	332
QVQLKESGPGLVAPSQSLSITCTVSGFSLTNYGVHWVRQPPGKGLEWLGVVWAGGITNYNWALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCARGDGYDDGYAMDYWGQGTSVTVSS	77	333
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVLWAGGITNYNSALMSRLSIRKDNSKSQVFLKMYSLHTDDTAMYYCARGDGYDDGYAMDYWGQGTSVTVSS	78	334
QVQLKESGPGLVAPSQSLSIPCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAGGTTNYNSALMSRLSISKDNSKSQVFLKMYSLQTDDTAMYYCARGDGYDDGYAMDYWGQGTSVTVSS	79	335
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNAPLLRYSSAMDYWGQGTSVTVSS	80	336
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYIHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNAPLLRYSSSMDYWGQGTSVTVSS	81	337
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNVALLRYSSAMDYWGQGTSVTVSS	82	338
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSTYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	83	339
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMQWVKQRPGQGLEWIGEIDPSDSYTNYNQKFKGKATLTVDTSSSTAYMQLSSLTSEDSAVYYCARAEYGYGNYPWFAYWGQGTLVTVSA	84	340
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMGWINTYTGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDMATYFCARKYYDYEFAYWGQGTLVTVSA	85	341
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPETGGTAYNQKFKVKAILTADKSSSTAYMELRSLTSEDSAVYYCTRLGDYDVMDYWGQGTSVTVSS	86	342
QVQLQQPGSVLVRPGASVKLSCKASGYTFTSSWMHWAKQRPGQGLEWIGEIHPNSGNTNYNEKFKGKATLTVDTSSSTAYVDLSSLTSEDSAVYYCAIYYDYDAYYFDYWGQGTTLTVSS	87	343
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSNYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCAREEKIYFDYWGQGTTLTVSS	88	344
EVQLQQSGTVLARPGASVKMSCKTSGYTFTSYWMHWIKQRPGQGLEWIGAIYPGNSDTTYNQKFKGKAKLTAVTSASTAYMELSSLTNEDSAVYYCTSLITTAYYFDYWGQGTTLTVSS	89	345
QVQLQESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARHEEANWAWFAYWGQGTLVTVSA	90	346
QVQLQQSGPQLVSPGASVKISCKASGYSFTNYWMHWVKQRPGQGLEWIGMIDPSDSETRLNQQFKDKATLTVDESSSTAYMQLSSPTSEDSAVYYCAIPYYAMDYWGQGTSVTVSS	91	347
QVQLQQPGSVLVRPGASVKLSCKASGYTFTSSWMHWAKQRPGQGLEWIGEIHPNSGNTNYNEKNKGKATLTVDTSSSTAYVDLSSLTSEDSAVYYCATYYGNYVWYFDVWGAGTSVTVSS	92	348

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCASYGSSYWYFDVWGTGTTVTVSS	93	349
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQANDTAIYYCASYYGSSRSYWYLDVWGAGTTVTVSS	94	350
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARQNDSSWAWFAYWGQGTLVTVSA	95	351
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCALPYSNYGWYFDVWGTGTTVTVSS	96	352
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPSDSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSYWYFDVWGTGTTVTVSS	97	353
EVQLQQSGAELVKPGASVKLSCTASGFNIKDYYMHWVKQRTEQGLEWIGRIDPEDGETKYAPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCAAYGNSAWFAYWGQGTLVTVSA	98	354
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVRQAPGKGLEWMGWINTNTGEPTYAEEFKGRFAFSLETSASTAYLQINNLRNEDTATYFCARWYPYFDYWGQGTTVTVSS	99	355
QVQLQQSGPELKKPGETVKISCKASGYTFTNYGMNWVKQAPGKGLKWMGWINTNTGEPTYAEEFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARWYPYFDYWGQGTTLTVSS	100	356
QVQLQQSGAELAKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGYINPSSGYTKYNQKFKDKATLTADKSSSTAYMQLSSLTYEDSAVYYCARSDGSSGNWYFDVWGTGTTVTVSS	101	357
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAGGSTNYNSALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCAREGGYTGYFDVWGAGTTVTVSS	102	358
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPSDSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAYSNYVPYYAMDYWGQGTSVTVSS	103	359
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYGNYDYAMDYWGQGTSVTVSS	104	360
EVQLQQSGAELVRPGALVKLSCKASGFNIKDYFMHWVKQRPEQCLEWIGWIDPETDNTIYDPKFQGKASITADTSSNTAYLQLSSLTSEDTAVYYCARSGNMGFTYWGQGTLVTVSA	105	361
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKCLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCASQGGSSWGAMDYWGQGTSVTVSS	106	362
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKCLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARKGYGYDWYFDVWGTGTTVTVSS	107	363
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQCLEWIGEIDPSDSYTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARSSYYYYAMDYWGQGTSVTVSS	108	364
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDGGFDYWGQGTTLTVSS	109	365
QVQLQQSGAELMKPGASVKISFKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTNYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDYWGQGTTLTVSS	110	366

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSDSTSYNEKFKGKTTFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDYWGQGTTLTVSS	111	367
QVQLQQSGAELMKPGASVKISCKATAYTFSIYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKVKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDGGFDYWGQGTTLTVSS	112	368
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	113	369
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	114	370
QVQLQQSGAELMKPGASVKMSCKATGYTFSNYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGSTLTVSS	115	371
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVQQRPGHGLEWIGEILPGSGYTSYIEQFKGKATFTADTSSNTAYMQLGSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	116	372
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTSYNEKFKDKATFTADTSSNTAFMQLSSLTSEDSAVYYCARRAYGYDEGFDYWGQGTTLTVSS	117	373
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEVLPGSGSTSYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDEGFDYWGQGTTLTVSS	118	374
QVQLQQSGAELMKPGASVKISCKGTGYTFSSYWIEWVKQRPGHGLEWIGEISPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	119	375
QVQLQQSGAELMKPGASVKISCKATGYTFGTYWIEWVKQRPGHGLEWIGEILPGSGTPNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRAYGYDAGFDYWGQGTTLTVSS	120	376
QVQLQQSGAELMKPGASVKISFKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTSCNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	121	377
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGRTSYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	122	378
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGRTSYIEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	123	379
QVQLKQSGPGLVQPSQSLSITCTVSGFSLSSYGVHWVRQSPGKALEWLGVIWRGGSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	124	380
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTSYGVHWVRQSPGKGLEWLGVIWRGGSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDSAIYYCAKNLYGHYVMDYWGQGTSVTVSS	125	381
QVQLKQSGPGPVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWRGGSTDNNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	126	382
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTRYGVHWVRQSPGKGLEWLGVIWRGGSTDHNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	127	383

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTTYGVHWVRQSPGKGLEWLGVIWRGGSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	128	384
QVQLKQSGPGLVQPSQSLSITCTVSGFSVTSYGVHWVRQSPGKGLEWLGVIWRGGSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	129	385
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTRYGVHWVRQSPGKGLEWLGVIWRGGSTDHNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKNLYGHYVMDYWGQGTSVTVSS	130	386
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAYWGQGTLVTVSA	131	387
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAYWGQGTLVTVSA	132	388
QVQLQQPGAELVKPGASVKLSCKASGYTFTTYWMHWVKQRPGQGLEWIGMIHPNSDNTNYNEKFKSKATLTVDKSSSTAYIQLSSLTSEDSAVYYCARWGDGYSFAYWGQGTLVTVSA	133	389
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGNTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARWGDGYSFAYWGQGTLVTVSA	134	390
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPGSGSTYYNEKFKGKATLTADKSSNTVYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	135	391
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	136	392
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQRTGQGLEWIGEIYPGSGSSYYNEKFRGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	137	393
QVQLKQSGPELVKPGASVKMSCKASGYTFTDYVINWVKQKTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYIQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	138	394
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPGQGLEWIGMIDPSDSETHYNQMFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCATYDGYYRFAYWGQGTLVTVSA	139	395
QVQLQQPGAELVRPGASVRLSCKASGYTFTNYWMNWVKQRPGQGLEWIGMIDPSDSETHFNQMFKDKATLTVDKSSSTAYMQVSSLTSEDSAVYYCATYDIYYRFAYWGQGTLVTVSA	140	396
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSDSTNYNEKFKSKATLTVDKSSSTAYMHLSSLTSEDSAVYYCARPGGYGFADWGQGTLVTVSA	141	397
QVQLQQPGAELVKPGASVKLSCKASGYTFTTYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARPGGYGFTYWGQGTLVTVSA	142	398
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSPNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARPGGYGFAYWGQGTLVTVSA	143	399
QVQLQQPGAELVRPGASVKLSCKASGYTFTSYWINWVKQRPGQGLEWIGNIYPSDSYTNYNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCTRGNYIDYWGQGTTLTVSS	144	400
QVQLQQPGAELVRPGASVKLSCKASGYTFTDYWINWVKQRPGQGLEWIGNIYPSDSYTNYNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCTRGNYIDYWGQGTTLTVSS	145	401

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSA	146	402
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSNYYNEKFKGKAIMTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSA	147	403
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARPGDLGFAYWGQGTLVTVSA	148	404
QVQLKESGPGLVAPSQSLSITCTVSGFSLTNYGVHWVRQPPGKGLEWLGVVWAGGITNYNWALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCARGDGYDDGYAMDYWGQGTSVTVSS	149	405
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVLWAGGITNYNSALMSRLSIRKDNSKSQVFLKMYSLHTDDTAMYYCARGDGYDDGYAMDYWGQGTSVTVSS	150	406
QVQLQQSGPQLVSPGASVKISCKASGYSFTSYWMYWVKQRPGQGLEWIGMIDPSDSETRLNQKFKDRATLTVDKSSSTAYMQLSSPTSEDSAVYYCARTRNYWGQGTTLTVSS	151	407
QVQLQQSGPQLVSPGASVKISCKASGYSFTSYWMYWVKQRPGQGLEWIGMIDPSDSETRLNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCARTRNYWGQGTTLTVSS	152	408
QVQLQQSGPQLVSPGASVKISCKASGYSFTSYWMYWVKQRPGQGLEWIGMIDPSDSETRLNQKFKDKATLTVDKSSSTAYMQLSSPTSEDSAVYYCARTRNYWGQGTSLTVSS	153	409
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNAPLLRYSSAMDYWGQGTSVTVSS	154	410
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYIHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNAPLLRYSSSMDYWGQGTSVTVSS	155	411
EVQLQQSGAELVRSGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGWIDPENGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCNVALLRYSSAMDYWGQGTSVTVSS	156	412
QVQLQQSGAELVKPGASVKLSCKASGYTFSNYYVYWVKQRPGQGLEWIGEINPSNGDTNFNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYFCTSYYTHEAYYYAMDCWGQGTSVTVSS	157	413
QVQLQQSGAELVRPGASVKLSCTASGFNIKDYYMHWVKQRPEQGLEWIGRIDPEDGDTEYAPKFQGKATMTADTSSNTAYLQLSSLTSEDTAVYYCTPYSIYDAMDYWGQGTSVTVSS	158	414
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSTYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYWGQGTLVTVSA	159	415
QVQLQQSGAELVRPGVSVKISCKGSGYSFTDYGMHWVKQSHAKSLEWIGVISTYYGDASYNQKFKGKATMTVDKSSSTAYMELARLTSEDSAIYYCARQMDYDYTYYYAMDYWGQGTSVTVSS	160	416
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWNGEIYPGSGSTYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARMDGPWFAYWGQGTLVTVSA	161	417
QVTLKVSGPGILQPSQTLGLACTFSGISLSTSGMGLSWLRQPSGKALEWLASIWNNDNYYNPSLKSRLTISKETSNNQVFLKLTSVDTADSTTYYCAWRPYYRYDSFAYWGQGTLVTVSA	162	418
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPETGGTAYNQKFKVKAILTADKSSSTAYMELRSLTSEDSAVYYCTRLGDYDVMDYWGQGTSVTVSS	163	419

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGEIDPSDSYTNYNQKFKGKSTLTVDKSSSTAYMQLSSLTSEDSAVYYCARAGRYGSSFDYWGQGTTLTVSS	164	420
QVTLKESGPGILQPSQTLSLTCSFSGFSLSTSGMGVSWIRQPSGKGLEWLAHIYWDDDKRYNPSLKSRLTISKDTSRNQGFLKITSVDTADTATYYCAGRPDDYDGAWFPYWGQGTLVTVSA	165	421
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSNYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCAREEKIYFDYWGQGTTLTVSS	166	422
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARYDGYWFDYWGQGTTLTVSS	167	423
EVQLQQSGTVLARPGASVKMSCKTSGYTFTSYWMHWIKQRPGQGLEWIGAIYPGNSDTTYNQKFKGKAKLTAVTSASTAYMELSSLTNEDSAVYYCTSLITTAYYFDYWGQGTTLTVSS	168	424
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAPETGDYGSSYVWYFDVWGTGTTVTVSS	169	425
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGSGSTYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARGKVTRFAYWGQGTLVTVSA	170	426
EVQLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISDGGSYTYYPDNVKGRFTISRDNAKNNLYLQMSHLKSEDTAMYYCARDQDSNWEYFDYWGQGTSLTVSS	171	427
QIQLVQSGPELKKPGETVKISCKASGYTFTDYSMHWVRQAPGKGLKWMAWINTETGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARESWDRAMDYWGQGTSVTVSS	172	428
QVQLQQSGPQLVSPGASVKISCKASGYSFTNYWMHWVKQRPGQGLEWIGMIDPSDSETRLNQQFKDKATLTVDESSSTAYMQLSSPTSEDSAVYYCAIPYYAMDYWGQGTSVTVSS	173	429
QVQLQQPGSVLVRPGASVKLSCKASGYTFTSSWMHWAKQRPGQGLEWIGEIHPNSGNTNYNEKNKGKATLTVDTSSSTAYVDLSSLTSEDSAVYYCATYYGNYVWYFDVWGAGTSVTVSS	174	430
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCASYGSSYWYFDVWGTGTTVTVSS	175	431
QVQLQQPGAELVKPGASVKMSCKASGYTFTSYNMHWVKQTPGQGLEWIGALYSGNGDTSYNQKFKGKATLTADKSSSTAYMQLSSLTSEDSAVYYCARDYYGSSHLWYFDVWGAGTTVTVSS	176	432
QVTLKESGPGILQPSQTLSLTCSFSGFSLSTSGMGVSWIRQPSGKGLEWLAHIYWDDDKRYNPSLKSRLTISKDTSRNQVFLKITSVDTADTATYYCARRAHYDYGWYFDVWGAGTTVTVSS	177	433
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAGYDYDWYFDVWGTGTTVTVSS	178	434
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARHEDSNYHYFDYWGQGTTLTVFS	179	435
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARQNDSSWAWFAYWGQGTLVTVSA	180	436
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHPNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCALPYSNYGWYFDVWGTGTTVTVSS	181	437

TABLE 20-continued

VH Sequences
Table 20-VH sequences

Sequence	Binder Name	SEQ ID NO:
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPS DSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSYWYFDV WGTGTTVTVSS	182	438
EVQLQQSGAELVKPGASVKLSCTASGFNIKDYYMHWVKQRTEQGLEWIGRIDPE DGETKYAPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCAAYGNSAWFAYWG QGTLVTVSA	183	439
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAGG STNYNSALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCAREGGYTGYFDVWG AGTTVTVSS	184	440
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPS DSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCAYSNYVPYYAMDY WGQGTSVTVSS	185	441
QVQLQQSGPELVKPGASVKMSCKASGYTFTDYVISWVKQRTGQGLEWIGEIYPGS GSAYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGFDYWGQGTTL TVSS	186	442
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSGYGYYFD YWGQGTTLTVSS	187	443
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSSYGWYF DVWGTGTTVTVSS	188	444
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSSYGWYF DVWGTGTTVTVSS	189	445
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCASDYYGSSYGWYF DVWGTGTTVTVSS	190	446
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYYGSSYGWYF DVWGTGTTVTVSS	191	447
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCTRDYYGSGYGWYF DVWGTGTTVTVSS	192	448
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPGQGLEWIGMID PSDSETHYNQMFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCATYDGYYRFAY WGQGTLVTVSA	193	449
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPGQGLEWIGMID PSDSETHFNQMFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCATYDVYYRFAY WGQGTLVTVSA	194	450
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYGNYDYAMD YWGQGTSVTVSS	195	451
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGMIHP NSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARDYGNYDYAMD YWGQGTSVTVSS	196	452
EVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCASQLTGTWYYFDYW GQGTTLTVSS	197	453
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCASQLTGTWYYFDYW GQGTTLTVSS	198	454
QVKLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCASQLTGTWYYFDYW GQGTTLTVSS	199	455

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
DVKLVESGGGLVKLGGSLKLSCAASGFTFSNYYMSWVRQTPEKRLELVAVINSNGGSTYYPDTVKGRFTISRDNAKNTLYLQMSSLKSEDTALYYCARQEGIGYAMDYWGQGTSVTVSS	200	456
EVQLQQSGPELVKPGASVKISCKTSGYTFTEYTMHWVKQSHGKSLEWIGGIYPNNGGTSYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVYYCARGGWLLGYWGQGTTLTVSS	201	457
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARDGGIRGAMDYWGQGTSVTVSS	202	458
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYWGQGTTLTVSS	203	45
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPETGGTAYNQKFKGKATLTADKSSSTAYMELRSLTSEDSAVYYCTRNYDYAMDYWGQGTSVTVSS	204	460
DVKLVESGGGLVKLGGSLKLSCAASGFTFSSYYMSWVRQTPEKRLELVAVINSNGGSTFYPDTVKGRFTISRDNAKNTLYLQMSSLKSEDTALYYCARQEGIGYALDYWGQGTSVTVSS	205	461
EVKLVESGGDLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVAAISSGGSTYYPDSVKGRFTISRDNARNILYLQMSSLRSEDTAMYYCAREREWGVYYGSSLDYWGQGTTLTVSS	206	462
EVQLQQSGAELVKPGASVKLSCTASGFNIKDTYMHWVKQRPEQGLEWIGRIDPANGNTKYDPKFQGKATITADTSSNTAYLQLSSLTSEDTAVYYCARSDGNYDWGQGTLVTVSA	207	463
DVKLVESGGGLVKLGGSLKLSCAASGFTFSNYYMSWVRQTPEKRLELVAVINSNGGSTYYPDTVKGRFTISRDNAKNILYLQMSSLKSEDTALYYCARQEGIGYGMDYWGQGTSVTVSS	208	464
EVQLVESGGGLVQPKGSLKLSCAASGFTFNTYVMNWVRQAPGKGLEWVARIRSKSDNYATYYADSVKDIFTISRDDSQSMLYLQMNNLKTEDTAMYYCVRHDGVVGFDVWGAGTTVTVSS	209	465
DVQLQESGPGLVKPSQSLSLTCSVTGYSITSGYYWNWIRQFPGNKLEWMGYISYDGSNNYNPSLKNRISITRDTSKNQFFLKLNSVTTEDTATYYCARGGGRGWGQGTLVTVSA	210	466
QIQLVQSGPELKKPGETVKISCKASGYTFTDYSMHWVKQAPGKGLKWMGWINTETGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARDYYDYYYAMDYWGQGTSVTVSS	211	467
QIQLVQSGPELKKPGETVKISCKASGYTFTDYSMHWVKQAPGKGLKWMGWINTETGEPTYADDFKGRFAFSLETSASTAYLQINNLKNEDTATYFCARESWDRAMDYWGQGTSVTVSS	212	468
QVQLQQPGAELVRPGASVKLSCKASGYTFTNYWMNWVKQRPEQGLEWIGRIDPYDSETHYNQKFKDKAILTVDKSSSTAYMQLSSLTSEDSAVYYCARIYSDYDGAWFAYWGQGTLVTVSA	213	469
EVQLQQSGPELVKPGASVKMSCKASGYTFTDYYMDWVKQSHGESFEWIGRVNPYNGGTSYNQKFKGKATLTVDKSSSTAYMELNSLTSEDSAVYYCARGTVGFAYWGQGTLVTVSA	214	470
EVKLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVASISSGGSTYYPDSVKGRFTISRDNARNILSLQMSSLRSEDTAMYYCAREREWGVFYGSSLDYWGQGTTLTVSS	215	471
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWIRQTPEKRLEWVATISSGGSYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARHDDSSYGYFDYWGQGTTLTVSS	216	472
EVKLVESGGGLVKPGGSLKLSCAASGFTFSNYAMSWVRQTPEKRLEWVASISSGGTTYYPDSVKGRFTISRDNARNILYLQMSSLRSEDTAMYYCARTMPDVWGAGTTVTVSS	217	473

TABLE 20-continued

VH Sequences
Table 20-VH sequences

Sequence	Binder Name	SEQ ID NO:
QVQLKESGPGLVAPSQSLSITCTVSGFSLTSYGVHWVRQPPGKGLEWLGVIWAGG STNYNSALMSRLSISKDNSKSQVFLKMNSLQTDDTAMYYCARDTDGYYWAMDY WGQGTSVTVSS	218	474
DVQLQESGPGLVKPSQSLSLTCTVTGYSITSDHAWNWIRQFPGNKLEWMGYISYS GSTTYNPSLKSRISITRDTSKNQFFLQLNSVTTEDTATYYCARKWGDYWGQGTSVT VSS	219	475
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPE TGGTAYNQKFKGKATLTADKSSSTAYMELRSLTSEDSAVYYCTRNYDYALDYWGQ GTSVTVSS	220	476
DVQLQESGPGLVKPSQSLSLTCSVTGYSITSGYYWNWIRQFPGNKLEWMGYISYD GSNDYNPSLKNRISITRDTSKNQFFLKLNSVTTEDTATYYCARGGGRGWGQGTLVT VSA	221	477
DVKLVESGGGLVKLGGSLKLSCAASGFTFSNYYMSWVRQTPEKRLELVAVINSNG GSTYYPDTVKGRFTISRDNAKNTLYLQMSSLKSEDTALYYCARQEEIGYAMDYWG QGTSVTVSS	222	478
EVQLQQSGAELVRPGALVKLSCKASGFNIKDYFMHWVKQRPEQGLEWIGWIDPE TDNTIYDPKFQGKASITADTSSNTAYLQLSSLTSEDTAVYYCARSGNMGFTYWGQ GTLVTVSA	223	479
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCASQGGSSWGAMDY WGQGTSVTVSS	224	480
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISNGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARHEITTRFAYWGQG TLVTVSA	225	481
VQLQESGPGLVKPSQSLSLTCSVTGYSITSGYYWNWIRQFPGNKLEWMGYMSYD GSNNYNPSLKNRISITRDTSKNQFFLKLNSVTTEDTATYYCAREAGYFDYWGQGTT LTVSS	226	482
EVQLVESGGGLVQPKGSLKLSCAASGFSFNTYAMNWVRQAPGKGLEWVARIRSK SNNYATYYADSVKDRFTISRDDSESMLYLQMNNLKTEDTAMYYCVRQYGYDFDY WGQGTTLTVSS	227	483
EVQLVESGGDLVKPGGSLKLSCAASGFTFSSYGMSWVRQTPDKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLKSEDTAMYYCARHKGVNWDYFDY WGQGTTLTVSS	228	484
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPE TGGTAYNQKFKGKATLTADKSSSTAYMELRSLTSEDSAVYYCTRGDGNYDSWYFD VWGAGTTVTVSS	229	485
EVMLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARLPVTTVVFDYWG QGTTLTVSS	230	486
EVQLVESGGGLVKPGGSLKLSCAASGFTFSSYAMSWVRQTPEKRLEWVATISSGG SYTYYPDSVKGRFTISRDNAKNTLYLQMSSLRSEDTAMYYCARRPVVVPFDYWGQ GTTLTVSS	231	487
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSG GSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARGWDADYFDYWGQ GTTLTVSS	232	488
QVQLQQPGAELVKPGASVKLSCKASGYTFTNYWMHWVKQRPGQGLEWIGMIH PNSGSTNYNEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCTRYDYDDYWGQ GTTLTVSS	233	489
EVQLQQSGPELVKPGASVKISCKASGYTFTDYYMNWVKQSHGKSLEWIGDINPN NGGTSYNQKFKGKATLTVDKSSSTAYMDLRSLTSEDSAVYYCARSELGLYAMDYW GQGTSVTVSS	234	490

TABLE 20-continued

VH Sequences
Table 20-VH sequences

Sequence	Binder Name	SEQ ID NO:
QVQLQQSGAELMKPGASVKLSCKATGYTFTGYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTTEDSAIYYCARGRIHYFDYWGQGTTLTVSS	235	491
QVQLQQSGAELMKPGASVKLSCKATGYTFTGYWIEWVKQRPGHGLEWIGEILPGSGSTNYNEKFKGKATFTADTSSNTAYMQLSSLTTEDSAIYYCARGRIHYFDYWGQGTTLTVSS	236	492
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARKGYGYDWYFDVWGTGTTVTVSS	237	493
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGEIDPSDSYTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARSSYYYYAMDYWGQGTSVTVSS	238	494
DVQLQESGPGLVKPSQSLSLTCSVTGYSITSGYYWNWIRQFPGNKLEWMGYISYDGSNNYNPSLKNRISITRDTSKNQFFLKLNSVTTEDTATYYCARGGGRDWGQGTTLTVSS	239	495
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARGGDYDSYAMDYWGQGTSVTVSS	240	496
QVQLQQPGAELVKPGASVKMSCKASGYTFTSYWITWVKQRPGQGLEWIGDIYPGSGSTNYNEKFKSKATLTVDTSSSTAYMQLSSLTSEDSAVYYCARESVYDGYSWYFDVWGTGTTVTVSS	241	497
EFQLQQSGPELVKPGASVKISCKASGYSFTDYNMNWVKQSNGKSLEWIGVINPNYGTTSYNQKFKGKATLTVDQSSSTAYMQLNSLTSEDSAVYYCASTYDYDDWYFDVWGTGTTVTVSS	242	498
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGEIDPSDSYTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARSGNYLYAMDYWGQGTSVTVSS	243	499
EFQLQQSGPELVKPGASVKISCKASGYSFTDYNMNWVKQSNGKSLEWIGVINPNYGTTSYNQKFKGKATLTVDQSSSTAYMQLNSLTSEDSAVYYCAREGTSWYFDVWGTGTTVTVSS	244	500
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWRGGSTDYNAAFMSRLSITKDNSKSQVFFKMNSLQADDTAIYYCAKKGDGYDWYFDVWGTGTTVTVSS	245	501
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCAREGNYGSSYDAMDYWGQGTSVTVSS	246	502
QVQLQQPGAELVMPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGEIDPSDSYTNYNQKFKGKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARSSNYPYAMDYWGQGTSVTVSS	247	503
EVQLQQSVAELVRPGASVKLSCTASGFNIKNTYMHWVKQRPEQGLEWIGRIDPANGNTKYAPKFQGKATITADTSSNTAYLQLSSLTSEDTAIYYCAYYSGLYWGQGTLVTVSA	248	504
QVQLQQPGAELVRPGSSVKLSCKASGYTFTSYWMHWVKQRPIQGLEWIGNIDPSDSETHYNQKFKDKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARRGQIYYGYSWFAYWGQGTLVTVSA	249	505
EVQLQQSGPELVKPGASVKISCKASGYTFTDYYMNWVKQSHGKSLEWIGDINPNNGGTSYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVYYCARSTVVADWYFDVWGTGTTVTVSS	250	506
QVQLQQSGAELARPGASVKLSCKASGYTFTSYGISWVKQRTGQGLEWIGEIYPRSGNTYYNEKFKGKATLTADKSSSTAYMELRSLTSEDSAVYFCARSGSSYGYFDVWGTGTTVTVSS	251	507
QVQLKQSGPGLVQPSQSLSITCTVSGFSLTSYGVHWVRQSPGKGLEWLGVIWSGGSTDYNAAFISRLSISKDNSKSQVFFKMNSLQADDTAIYYCARKGGYDAYAMDYWGQGTSVTVSS	252	508

TABLE 20-continued

VH Sequences
Table 20-VH sequences

Sequence	Binder Name	SEQ ID NO:
EFQLQQSGPELVKPGASVKISCKASGYSFTDYNMNWVKQSNGKSLEWIGVINPNYGTTSYNQKFKGKATLTVDQSSSTAYMQLNSLTSEDSAVYYCAREGFITTVVAVDYWGQGTTLTVSS	253	509
QVQLQQSGAELVRPGASVTLSCKASGYTFTDYEMHWVKQTPVHGLEWIGAIDPETGGTAYNQKFKGKAILTADKSSSTAYMELRSLTSEDSAVYYCTREGNYDAMDYWGQGTSVTVSS	254	510
QVQLQQPGAELVRPGTSVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGVIDPSDSYTNYNQKFKGKATLTVDTSSSTAYMQLSSLTSEDSAVYYCARWDYYGVDYWGQGTTLTVSS	255	511
EVQLVESGGGLVQSGGSLRLSCAASGFTFSGYWMYWVRQAPGKGLEWVSAISPGGGSTYYPDSVKGRFTISRDNAKNTLYLQMNSLEPEDTALYYCASSLTATHTYEYDYWGQGTQVTVSS	256	256
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQCLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGYSGSYSDWGQGTLVTVSS	257	9447
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQCLEWMGGIIPLSGAPNYAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGALYNWNDGWFDPWGQGTLVTVSS	258	9448
QVQLVQSGAEVKKPGASVKVSCKASGYSLITHWMHWVRQAPGQCLEWMGMINPSDGVTYYAQTFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREYYGEGFDYWGQGTLVTVSS	259	9449
QVQLVQSGAEVKKPGSSVKVSCKASGYTFSDHHVHWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGSSWYLHFQHWGQGTLVTVSS	260	9450
QVQLVQSGAEVKKPGASVKVSCKASGGTFSRYGIAWVRQAPGQGLEWMGISYPSDGSTSSAQKLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDRLGDLDYWGQGTLVTVSS	261	9451
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTDYYVHWVRQAPGQGLEWVGWISTFTGNTDYAQNFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDAPLAAAGTDYYYGMDVWGQGTTVTVSS	262	9452
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYALSWVRQAPGQGLEWMGIINPSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDPGMDVWGQGTLVTVSS	263	9453
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGYSGSYSDWGQGTLVTVSS	264	9454
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYALSWVRQAPGQGLEWMGIINPSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDPGMDVWGQGTLVTVSS	265	9455
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGIIDPSGGSTTYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDMGMDVWGQGTTVTVSS	266	9456
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAFSWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDVGDRGMDVWGQGTTVTVSS	267	9457
QVQLVQSGAEVKKPGASVKVSCKASGSTFSGYYMHWVRQAPGQGLEWMGWIDPNGGGTQYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDIVHDGTEYFQHWGQGTLVTVSS	268	9458
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYYMHWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDIVHDGTEYFQHWGQGTLVTVSS	269	9459
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPSGGSTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGRDHDAFDIWGQGTMVTVSS	270	9460

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGASVKVSCKASGFTFTDYGISWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGRSHDAFDIWGQGTMVTVSS	271	9461
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGWMNPHSGDTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARWVGTTEYYYYYMDVWGKGTTVTVSS	272	9462
QVQLVQSGAEVKKPGASVKVSCKASGYTFTDYYLHWVRQAPGQGLEWMGIIDPSGGSTSIAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCATTAYYDFWSGYSMDVWGKGTTVTVSS	273	9463
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSHYMHWVRQAPGQGLEWMGIIDPSGGSTSYAQEFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDMDNWNTGYYYYMDVWGKGTTVTVSS	274	9464
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAINWVRQAPGQGLEWMGWVNPNSGDTAYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDQRGGDAWDVWGKGTTVTVSS	275	9465
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGIITPSGGSTTYAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDTAGHFDIWGQGTLVTVSS	276	9466
QVQLVQSGAEVKKPGASVKVSCKASGGTFRNDVINWVRQAPGQGLEWIGWMNPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDNPDLDGMDVWGQGTLVTVSS	277	9467
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAINWVRQAPGQGLEWLGWISAYNGNTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLVGHFDYWGQGTLVTVSS	278	9468
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGYSGSYSDWGQGTLVTVSS	279	9469
QVQLVQSGAEVKKPGASVKVSCKASGNTLSSHAISWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDQGSSGTFDYWGQGTLVTVSS	280	9470
QVQLVQSGAEVKKPGASVKVSCKASGGTLSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSTDVIDYWGQGTLVTVSS	281	9471
QVQLVQSGAEVKKPGASVKVSCKASGYIFTSYDINWVRQAPGQGLEWMGWINPNSGDTKYAQNFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGGTVTPTEEYYYYGMDVWGQGTLVTVSS	282	9472
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWISVYNGNTNYAQNLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCASLDDLDYWGQGTLVTVSS	283	9473
QVQLVQSGAEVKKPGASVKVSCKASGHTFTSYYIHWVRQAPGQGLEWMGWINPNNGGTHYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDMVRDSAEYFQHWGQGTLVTVSS	284	9474
QVQLVQSGAEVKKPGSSVKVSCKASGYTFITSYIHWVRQAPGQGLEWMGMINPSGGTTTYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDSSGYPIDYWGQGTLVTVSS	285	9475
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGGIIPLSGAPNYAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGALYNWNDGWFDPWGQGTLVTVSS	286	9476
EVQLLESGGGLVQPGGSLRLSCAASGFTVGSWYMSWVRQAPGKGLEWVAGIWYEGSNKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARLGTASLPYFDYWGQGTLVTVSS	287	9477
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWVGWINPNRGDTKYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARESGDGFDPWGQGTLVTVSS	288	9478

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTNYYIHWVRQAPGQGLEWMGWMNPNSGNTGYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDWPNWFDPWGQGTLVTVSS	289	9479
QVQLVQSGAEVKKPGASVKVSCKASGYSFTDNYIHWVRQAPGQGLEWMGWIRSDNGETSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREVQLVGFDYWGQGTLVTVSS	290	9480
QVQLVQSGAEVKKPGSSVKVSCKASGYTFSDHHVHWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGSSWYLHFQHWGQGTLVTVSS	291	9481
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAIYWVRQAPGQGLEWMGGIIPIFGTTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKGVDRYNWNDAFDYWGQGTLVTVSS	292	9482
QVQLVQSGAEVKKPGASVKVSCKASGYTFTDYYMHWVRQAPGQGLEWMGWIHSNSGGTHSAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARESSGYDSSLDYWGQGTLVTVSS	293	9483
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYGISWVRQAPGQGLEWVGWINPNSGDTDYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCTTDPRLDSSDPGYWGQGTLVTVSS	294	9484
QVQLVQSGAEVKKPGASVKVSCKASGGTFGNYGINWVRQAPGQGLEWMGWISAYNGNTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGGMDVWGQGTLVTVSS	295	9485
QVQLVQSGAEVKKPGASVKVSCKASGGTFSRYGIAWVRQAPGQGLEWMGISYPSDGSTSSAQKLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDRLGDLDYWGQGTLVTVSS	296	9486
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWMNPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSIVGGYPFDYWGQGTLVTVSS	297	9487
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGTITPIFGTTDYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAREGYSSSWHDDAFDIWGQGTLVTVSS	298	9488
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGIIDPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDYGLDSWGQGTLVTVSS	299	9489
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGWMNPNSGDTGYAQRFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCATGGSDSSGYYYEGYFQHWGQGTLVTVSS	300	9490
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWLGYMSPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDKGGYYDSSGYYWYWGQGTLVTVSS	301	9491
EVQLLESGGGLVQPGGSLRLSCAASGFSLSSYEMHWVRQAPGKGLEWVSAISSNGGSTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARVGDGDGYNPDFDYWGQGTLVTVSS	302	9492
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYGISWVRQAPGQGLEWMGWIDPTSGATDTAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKDPIVATEVDYWGQGTLVTVSS	303	9493
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWMSPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSGAFDIWGQGTMVTVSS	304	9494
QVQLVQSGAEVKKPGASVKVSCKASGVTISNYAISWVRQAPGQGLEWMGWMNPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGLLDAFDIWGQGTMVTVSS	305	9495

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGASVKVSCKASGGTFSRYGITWVRQAPGQGLEWMGWMNPYDGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGGRHHDAFDIWGQGTMVTVSS	306	9496
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPSGDGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDISNDAFDIWGQGTMVTVSS	307	9497
QVQLVQSGAEVKKPGASVKVSCKASGYILTGHYMHWVRQAPGQGLEWMGWISAYNGDTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGSSWDDAFDIWGQGTMVTVSS	308	9498
EVQLVESGGGLVKPGGSLRLSCAASGFTFSNHYTSWVRQAPGKGLEWVSAIGAGGGTYYADSVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCAREGWNDDVFDIWGQGTMVTVSS	309	9499
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWVGIINPSAGTTYYAERFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGNFGAFDIWGQGTMVTVSS	310	9500
QVQLVQSGAEVKKPGSSVKVSCKASGYSFTTYAITWVRQAPGQGLEWMGEIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDKSGWNYGSGSYNDAFDIWGQGTMVTVSS	311	9501
QVQLVQSGAEVKKPGASVKVSCKASGYAFTGYYMHWVRQAPGQGLEWMGWMNPNSGKTEYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGGLDFDYWGQGTMVTVSS	312	9502
QVQLVQSGAEVKKPGASVKVSCKASGYTFTTYYIHWVRQAPGQGLEWMGWMNPNTGDTGSAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDPAVTPDAFDIWGQGTMVTVSS	313	9503
QVQLVQSGAEVKKPGASVKVSCKASGGTLSSYAISWVRQAPGQGLEWMGIIDPSGGGTSYAQKLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAGSLYYYGMDVWGQGTMVTVSS	314	9504
QVQLVQSGAEVKKPGSSVKVSCKASGGTFGSSAISWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKEDDILPPRAFDIWGQGTMVTVSS	315	9505
EVQLLESGGGLVQPGGSLRLSCAASGFTFDDYAMHWVRQAPGKGLEWVSGISGGGGVTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARVYSSGWLDAFDIWGQGTMVTVSS	316	9506
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWISGYNGNTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCASSDVSPDAFDIWGQGTTVTVSS	317	9507
QVQLVQSGAEVKKPGASVKVSCKASGGTQNIYAITWVRQAPGQGLEWVGWVNPNSGNTGYSQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCATPTSSSDDAFDIWGQGTTVTVSS	318	9508
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARASRGDDAFDIWGQGTTVTVSS	319	9509
QVQLVQSGAEVKKPGASVKVSCKASGIPFTSDDINWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARERYEGGYSSGPGNYYYGMDVWGQGTTVTVSS	320	9510
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGWMNPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDDDYGDYPVWGQGTTVTVSS	321	9511
QVQLVQSGAEVKKPGASVKVSCKASGDTFSDHAINWVRQAPGQGLEWMGWMNPKIGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCVYDSSGYDAFDIWGQGTTVTVSS	322	9512
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGRINPGTGGTDYAHKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARETPSDYYDSSGYYYNDAFDIWGQGTTVTVSS	323	9513

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAISWVRQAPGQGLEWVGIIIPSGGTNYAQTFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDLGTTFDIWGQGTTVTVSS	324	9514
QVQLVQSGAEVKKPGASVKVSCKASGYTFTAYYLHWVRQAPGQGLEWIGWINPDNDNAYYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDIAVAALAYGMDVWGQGTTVTVSS	325	9515
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVAVISYDGSDQYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARQSLYYYYGMDVWGQGTTVTVSS	326	9516
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTDYYVHWVRQAPGQGLEWVGWISTFTGNTDYAQNFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDAPLAAAGTDYYYGMDVWGQGTTVTVSS	327	9517
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVAFISDDGITKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARDDSSGYGGMDVWGQGTTVTVSS	328	9518
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMHWVRQAPGKGLEWVAVISYDGGDKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCASGSLVLGYYYMDVWGQGTTVTVSS	329	9519
QVQLVQSGAEVKKPGASVKVSCKASGYTFTNYYIHWVRQAPGQGLEWMGWINPNTGGTDYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCATGGGGSYYDAFDVWGQGTTVTVSS	330	9520
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGRINPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDIGEGYSMDVWGQGTTVTVSS	331	9521
EVQLLESGGGLVQPGGSLRLSCAASGFTFSNHYTSWVRQAPGKGLEWVAVISYDGSNKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCAREEKYSSSWYVGVDAFDIWGQGTTVTVSS	332	9522
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSSAMHWVRQAPGKGLEWISSISGSGDNAYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARDQEDYYYDSSGYGMDVWGQGTTVTVSS	333	9523
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSHAISWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKGDWGIVVVPAAIGAFDIWGQGTTVTVSS	334	9524
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTAYYMHWVRQAPGQGLEWVGRISPVFGSTTYAQRFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDLGYYDSSGYRYDAFDIWGQGTTVTVSS	335	9525
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGGISPMFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAKDGWYYGMDVWGQGTTVTVSS	336	9526
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYGISWVRQAPGQGLEWMGWINPNSGGTKYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARGEAGNLDWYFDLWGRGTLVTVSS	337	9527
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYGISWVRQAPGQGLEWMGWINPNNGDTKYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREDVWYFDLWGRGTLVTVSS	338	9528
QVQLVQSGAEVKKPGASVKVSCKASGYTFTTYGISWVRQAPGQGLEWMGWISTYDGKTNYAQKLQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCALHLGGDWYFDLWGRGTLVTVSS	339	9529
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGWINPNTGATYYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARQHGDYDWYFDLWGRGTLVTVSS	340	9530
QVQLVQSGAEVKKPGASVKVSCKASGDTFTTYYVHWVRQAPGQGLEWMGWINPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSGRHWGQGTLVTVSS	341	9531

TABLE 20-continued

VH Sequences
Table 20-VH sequences

Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYGISWVRQAPGQGLEWMGRIIPMLGIANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCVREEVAGANWFDPWGQGTLVTVSS	342	9532
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYAMNWVRQAPGQGLEWMGIINPSGGSTSYARKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGDYGSGEFDYWGQGTLVTVSS	343	9533
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSSYMHWVRQAPGQGLEWMGWMNPRSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARERDDYGDYGWLDYWGQGTLVTVSS	344	9534
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLYDSSGYWHYYYYMDVWGKGTTVTVSS	345	9535
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAFSWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARFSGYDYVDYWGQGTLVTVSS	346	9536
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPNGGNTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDVGEDFDLWGQGTMVTVSS	347	9537
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYYIHWVRQAPGQGLEWLGVINPADGDTTYAQMFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDFDWLFAMDVWGKGTTVTVSS	348	9538
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYALNWVRQAPGQGLEWMGRINPNGGTTYYAKNFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKHGDHGFYVWGLWTKGTVSS	349	9539
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGMINPNVGSATYAQRFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREDSGTSWFDPWGQGTLVTVSS	350	9540
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYYMHWVRQAPGQGLEWMGIINPSDGSTSYAQRFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDDRGSNYYYGMDVWGQGTTVTVSS	351	9541
QVQLVQSGAEVKKPGASVKVSCKASGYTFTAYYVHWVRQAPGQGLEWMGWMNPNSGTTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSSDYYGDYRADAFDIWGQGTMVTVSS	352	9542
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYDINWVRQAPGQGLEWMGVISPSGDATLYAQSFQGRVTITADESTSTAYMELSSLRSEDTAVYYCVKGLDHWGQGTLVTVSS	353	9543
EVQLLESGGGLVQPGGSLRLSCAASGFSFSDYGMHWVRQAPGKGLEWVSAIGGIGDSTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARMNYGDSNYYYYYGMDVWGQGTTVTVSS	354	9544
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYDISWVRQAPGQGLEWMGMISPSDGSTTYAPKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGAVGFDYWGQGTLVTVSS	355	9545
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTSYGISWVRQAPGQGLEWMGWINTYSGYTDYAHKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCTTDDFLSFGYWGQGTLVTVSS	356	9546
QVQLVQSGAEVKKPGSSVKVSCKASGYMFTDYYIHWVRQAPGQGLEWMGGIIPYFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARSISGSYVLDAFDIWGQGTTVTVSS	357	9547
QVQLVQSGAEVKKPGSSVKVSCKASGYTFNSYGISWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDWGYGDYADDAFDIWGQGTMVTVSS	358	9548
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSNNDINWVRQAPGQGLEWMGWINPIYGSANYAQNFQGRVTITADESTSTAYMELSSLRSEDTAVYYCAADWRGFDYWGQGTLVTVSS	359	9549

TABLE 20-continued

VH Sequences
Table 20-VH sequences

Sequence	Binder Name	SEQ ID NO:
QVQLVQSGAEVKKPGASVKVSCKASGYTFTEYAIHWVRQAPGQGLEWMGRMNPHNGDTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREGDYLGYPIDCWGRGTLVTVSS	360	9550
EVQLLESGGGLVQPGGSLRLSCAASGFTFSDYSMSWVRQAPGKGLEWVAAIWQDGNVKFYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARDGNSGYVFWGQGTLVTVSS	361	9551
QVQLVQSGAEVKKPGASVKVSCKASGYTFTTYYMHWVRQAPGQGLEWMGWINPNTGDTAYAQKIQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARTAEAVAGLPAFDYWGQGTLVTVSS	362	9552
QVQLVQSGAEVKKPGSSVKVSCKASGGTSNNYAIDWVRQAPGQGLEWMGGIIPLFGTTTYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARVTLYGDYDYWGQGTLVTVSS	363	9553
QVQLVQSGAEVKKPGASVKVSCKASGYSLITHWMHWVRQAPGQGLEWMGMINPSDGVTYYAQTFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAREYYGEGFDYWGQGTLVTVSS	364	9554
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPSGGSTSNAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDLGDTAMDGWGQGTLVTVSS	365	9555
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYYLHWVRQAPGQGLEWMGIITPSGGSTTYAHKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGGLASFDYWGQGTLVTVSS	366	9556
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWMNPNSGNTGYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARGGGWAMTDAFDIWGQGTMVTVSS	367	9557
EVQLVESGGGLVKPGGSLRLSCAASGFTFDDYGMSWVRQAPGKGLEWVSLIYSGGDTYYADSVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCTRKEYYYDSSGYLRLFDYWGQGTLVTVSS	368	9558
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTDYYMHWVRQAPGQGLEWMGGINPIFGTSNYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDISGYDYYYYGMDVWGQGTTVTVSS	369	9559
QVQLVQSGAEVKKPGASVKVSCKASGGTLNNYAFSWVRQAPGQGLEWMGMIDPSDGTIAYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARSDYDFWSGLGGYFDYWGQGTLVTVSS	370	9560
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGTIDPNSGGTMFAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSAEWELGGSFDYWGQGTLVTVSS	371	9561
EVQLLESGGGLVQPGGSLRLSCAASGFTFSNHYTSWVRQAPGKGLEWVSSIGVNGDTYYLDSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCAREGLVFSGRGHWYFDLWGRGTLVTVSS	372	9562
QVQLVQSGAEVKKPGASVKVSCKASGGTFSNYAISWVRQAPGQGLEWMGRINPNGGNTSNAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDYEDADFDGWGQGTLVTVSS	373	9563
QVQLVQSGAEVKKPGASVKVSCKASGYTFSDHHVHWVRQAPGQGLEWMGWMNPDSGNTGYAQRFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDSTSGVDYWGQGTLVTVSS	374	9564
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVAVISYDGHDQFYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARGEQQLEGFYYYYGMDVWGQGTTVTVSS	375	9565
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMHWVRQAPGKGLEWVAVISYDGSKEYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCASDYGDYGTYDYWGQGTLVTVSS	376	9566

TABLE 20-continued

VH Sequences Table 20-VH sequences		
Sequence	Binder Name	SEQ ID NO:
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMHWVRQAPGKGLEWVSGISGGGDDTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCAREPLAYCGGDCPGGFDYWGQGTLVTVSS	377	9567
EVQLLESGGGLVQPGGSLRLSCAASGFTFSDHYMDWVRQAPGKGLEWVSAIGTGGDTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARHEDTAIFLDYWGQGTLVTVSS	378	9568
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYYMHWVRQAPGQGLEWMGMISPSDGSTTYAPKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDGYDAWSYGMDVWGQGTMVTVSS	379	9569
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTGYYMHWVRQAPGQGLEWMGWMNPNSGNTGYAQKFQGRVTITADESTSTAYMELSSLRSEDTAVYYCARDGVTGTDYWGQGTLVTVSS	380	9570
EVQLLESGGGLVQPGGSLRLSCAASGFAFSSYVLHWVRQAPGKGLEWVSAISGAGDSTYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCAREPTTVTDDWYFDLWGRGTLVTVSS	381	9571
EVQLVESGGGLVKPGGSLRLSCAASGFAFSSHWMHWVRQAPGKGLEWVSAISGNGDNSYYADSVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCARDRAPEYFDLWGRGTLVTVSS	382	9572
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARDDYGDYGGGMDVWGQGTTVTVSS	383	9573
QVQLVQSGAEVKKPGASVKVSCKASGYTFTDYYMHWVRQAPGQGLEWMGWMNPNSGHTGYAEKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCAKDTSPRYGDGFFDYWGQGTLVTVSS	384	9574
EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMHWVRQAPGKGLEWVAVTSYDGSNKYYADSVKGRFTISRDNSKNTLYLQMNSLRAEDTAVYYCARESGFSAEYFQHWGQGTLVTVSS	385	9575
QVQLVQSGAEVKKPGASVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGIINPSGGSTSYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDTAVYYCARATGLYCSGSCFDYWGQGTLVTVSS	386	9576
DLGKKLLEAARAGQDDEVRILMANGADVNAMDHFGFTPLHLAAKVGHLEIVEVLLKYGADVNADDMDGETPLHLAAAIGHLEIVEVLLKNGADVNAHDTWGFTPLHLAASYGHLEIVEVLRKYGADVNAQDKFGETTFDISIDNGNEDLAEILQKLN	387	14114
EVQLVESGGGLVQAGGSLKLSCAASRSILDFNAMGWYRQAPGKQREWVTTIARAGATKYADSVKGRFSISRDNAKNTVYLQMSSLKPEDTATYYCNARVFDLPNDYWGQGTQVTVSS	388	14115
EVQLVESGGGLVQAGGSLRLSCAASGRTSASYSMGWFRQAPGKEREFVAAISWSGDETSYADSVKGRFTIARGNAKNTVYLQMNSLKSEDTAIYYCAGDRWWRPAGLQWDYWGQGTQVTVSS	389	14000
EVQLVESGGGLVQAGGSLKLSCAASRSILDFNAMGWYRQAPGKQREWVTTIARAGATKYADSVKGRFSISRDNAKNTVYLQMSSLKPEDTATYYCNARVFDLPNDYWGQGTQVTVSS	390	14001
EVQLVESGGGSVQPGGSLTLSCGTSGRTFNVMGWFRQAPGKEREFVAAVRWSSTGIYYTQYADSVKSRFTISRDNAKNTVYLEMNSLKPEDTAVYYCAADTYNSNPARWDGYDFRGQGTQVTVSS	391	14002

TABLE 21

VL Sequences Table 21-VL sequences		
Sequence	Binder Name	SEQ ID NO:
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	1	512
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	2	513
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVFAATYLADG VPSRFSGSGSGTQYSLKINSLQSEDFGNYYCQHFWGTPWTFGGGTKLEIK	3	514
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	4	515
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQVLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGSPWTFGGGTKLEIK	5	516
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPRLLVYAATNLAD GVPSRFSGSGSGTQYSLKITSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	6	517
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	7	518
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQVLVYAATNVAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	8	519
DIQMTQSPASLSVSVGETVTITCRASDNIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	9	520
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPRLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	10	521
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	11	522
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVFAATYLADG VPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	12	523
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGSPWTFGGGTKLEIK	13	524
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWFQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTFSSMEAEDAATYYCQQWSSNPLYTFGGGTKLEIK	14	525
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGGGTKLELK	15	526
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTINNLESEDFADYYCLQFYEFPLTFGAGTKLELK	16	527
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	17	528
DIVMTQSPSSLAMSVGQKVTMSCKSSQSLLNSSNQKNYLAWYQQKPGQSPKLLVY FASTRESGVPDRFIGSGSGTDFTLTISSVQAEDLADYFCQQHYSTPLTFGAGTKLELK	18	529
QIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGV PARFSGSGSGTSYSLTISRMEAEDAATYYCQQRSSFPYTFGGGTKLEIK	19	530
SIVMTQTPKFLLVSAGDRVTITCKASQSVSNDVAWYQQKPGQSPKLLIYYASNRYTG VPDRFTGSGYGTDFTFTISTVQAEDLAVYFCQQDYTSLPTFGAGTKLEIK	20	531
DIQMTQSPSSLSASLGDTITITCHASQNINVWLSWYQQKPGNIPKLLIYKASNLHTG VPSRFSGSGSGTGFTLTISSLQPEDIATYYCQQGQSYPLTFGGGTNLEIK	21	532
ETTVTQSPASLSMAIGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNSLRPG VPSRFSSSGYGTDFVFTIENMLSEDVADYYCLQSDNLPLTFGAGTKLELK	22	533
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	23	534
DIQMTQSPSSLSASLGGKVTITCKASQDINKYIAWYQHKPGKGPSLLIHYTSTLQP GIPSRFSGSGSGRDYSFSISNLEPEDIATYYCLQYDNLMYTFGGGTKLEIK	24	535

TABLE 21-continued

VL Sequences Table 21-VL sequences		
Sequence	Binder Name	SEQ ID NO:
ENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWYQQKSDASPKLWIYYTSNLAP GVPGRFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSHTFGGGTKLEIK	25	536
DIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQPPKLLIKYASN LESGVPARFSGSGSGTDFTLNIHPVEEEDTATYYCQHSWEIPLTFGAGTKLELK	26	537
ETTVTQSPASLSVATGEKVSIRCMTSIDIDDDMNWYQQKPGEPPKLLISEGNTLRPG VPSRFSSSGYGTDFVFTIENTLSEDVADYYCLQSDNMPFTFGSGTKLEIK	27	538
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHWVFGGGTKLTVL	28	539
QAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTNNR APGVPARFSGSLIGDKAALTITGAQTEDEAIYFCALWYSNHLFGSGTKVTVL	29	540
QAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTNNR APGVPARFSGSLIGDKAALTITGAQTEDEAIYFCALWYSNHWVFGGGTKLTVL	30	541
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPYTFGGGTKLEIK	31	542
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMYWYQQKPGSSPRLLIYDTSNLASGV PVRFSGSGSGTSYSLTISRMEAEDAATYYCQQWSSYPLTFGAGTKLELK	32	543
ETTVTQSPASLSVATGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPG VPSRFFSGGYGTDFVFTIENTLSEDVADYYCLQSDNMPLMFGAGTKLELK	33	544
QIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQRPGTSPKLWIYSTSNLASGV PARFSGSGSGTSYSLTISRMEAEDAATYYCQQRSTYPTFGGGTKLEIK	34	545
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPYTFGGGTKLEIK	35	546
DIQMTQSPASLSASVGETVTITCRASGNIHNYLAWYQQKQGKSPQLLVYNAKTLAD GVPSRFSGSGSGTQYSLKINSLQPEDFGSYYCQHFWSTPWTFGGGTKLEIK	36	547
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPYTFGGGTKLEIK	37	548
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPLTFGGGTKLEIKR	38	549
DIKMTQSPSSMYASLGERVTITCKASQDINGYLSWYQQKPGKSPQTLLYRANRLVD GVPSRFRGSGSGQDYTLTISSLEYEDMGTYYCLQYDEFPPTFGGGTKLELKR	39	550
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWYQQKPGKSPKTLIYRAKRLVDG VPSRFSGSGSGQDYTLTISSLEYEDMGTYYCLQYDEFPPTFGGGTKLELKR	40	551
DIKMTQSPSSMYASLGERVTITCKASQDINGYLSWFQQKPGKSPQTLLYRANRLVD GVPSRFRGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	41	552
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRAKRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	42	553
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLHYDEFPPTFGAGTKLELKR	43	554
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	44	555
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	45	556
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	46	557
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	47	558

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	48	559
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFHQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	49	560
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	50	561
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	51	562
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGFYYCLQYDEFPPTFGAGTKLELKR	52	563
DIKMTQSPSSIYASLGERVTITCKASQDINSYLNWFQQKPGKSPKTLIYRANRLVDGV PSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGVGTKLELKR	53	564
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	54	565
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGCGTKLELKR	55	566
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGCGTKLELKR	56	567
DIQMTQSPASLSVSVGETVTITCRASDNIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIKR	57	568
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVFAATYLADG VPSRFSGSGSGTQYSLKINSLQSEDFGNYYCQHFWGTPWTFGGGTKLEIKR	58	569
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGAKLELKR	59	570
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPFTFGSGTKLEIKR	60	571
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPFTFGSGTKLEIKR	61	572
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPFTFGSGTKLEIKR	62	573
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPFTFGSGTKLEIKR	63	574
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLYTFGGGTKLEIKR	64	575
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWFQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTFSSMEAEDAATYYCQQWSSNPLYTFGGGTKLEIKR	65	576
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELKR	66	577
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELKR	67	578
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPQLLIYAAS NLQSGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELKR	68	579
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFGGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELKR	69	580
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELKR	70	581
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNKDPLTFGAGTKLELKR	71	582

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNKDPLTFGAGTKLELKR	72	583
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELKR	73	584
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNKDPFTFGAGTKLELKR	74	585
DIVLTQSPSSLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLTIHPVEEEDAATYYCQQSNKDPFTFGGGTKLELKR	75	586
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNKDPFTFGAGTKLELKR	76	587
DIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQAPKLLIKYASN LESGVPARFSGSGSGTDFTLNIHPVEEEDTATYYCQHSWEIPLTFGAGAKLELKR	77	588
DIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQPPKLLIKYASN LESGVPARFSGSGSGTDFTLNVHPVEEEDTATYYCQHSWEIPLTFGAGTKLELKR	78	589
DIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQPPKLLIKYASN LESGVPARFSGSGSGTDFTLNIHPVEEEDTATYYCQHSWEIPLTFGAGTKLELKR	79	590
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVLR	80	591
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVLR	81	592
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVLR	82	593
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELKR	83	594
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHWVFGGGTKLTVLR	84	595
DIVMTQSPSSLAMSVGQKVTMSCKSSQSLLNSSNQKNYLAWYQQKPGQSPKLLVY FASTRESGVPDRFIGSGSGTDFTLTISSVQAEDLADYFCQQHYSTPLTFGAGTKLELK R	85	596
DIQMTQTTSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYYTSRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQDNTLPRTFGGGTKLEIKR	86	597
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPYTFGGGTKLEIKR	87	598
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYRQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPWTFGGGTKLEIKR	88	599
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELKR	89	600
QIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGV PARFSGSGSGTSYSLTISRMEAEDAATYYCQQRSSFPYTFGGGTKLEIKR	90	601
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPLTFGAGTKLELKR	91	602
DIQMTQSPSSLSASLGERVSLTCRASQDIHGYLNLFQQKPGETIKHLIYETSNLDSGV PKRFSGSRSGSDYSLIIGSLESEDFADYYCLQYASSPLTFGAGTKLELKR	92	603
DIVMTQSHKFMSTSVGDRVSITCKASQDVGTAVAWYQQKPGQSPKLLIYWASTRH TGVPDRFTGSGSGTDFTLTISNVQSEDLADYFCQQYSSYPFTFGSGTKLEIKR	93	604
SIVMTQTPKFLLVSAGDRVTITCKASQSVSNDVAWYQQKPGQSPKLLIYYASNRYTG VPDRFTGSGYGTDFTFTISTVQAEDLAVYFCQQDYTSLPTFGAGTKLEIKR	94	605

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
ETTVTQSPASLSVATGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENTLSEDVADYYCLQSDNMPLTFGAGTKLELKR	95	606
DIQMTQTTSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYYTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPFTFGSGTKLEIKR	96	607
DIQMTQTTSSLSASLGDRVTISCSASQGISNYLNWYQQKPDGTVKLLIYYTSSLHSGVPSRFSGSGSGTDYSLTISNLEPEDIATYYCQQYSKLPYTFGGGTKLEIKR	97	608
DIVMTQSPATLSVTPGDRVSLSCRASQSISDYLHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGSDFTLSINSVEPEDVGVYYCQNGHSFPWTFGGGTKLEIKR	98	609
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYTLTISNLESEDFATYYCLQFYEFPYTFGGGTKLEIKR	99	610
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPYTFGGGTKLEIKR	100	611
DIQMTQSPASLSASVGETVTITCRASGNIHNYLAWYQQKQGKSPQLLVYNAKTLADGVPSRFSGSGSGTQYSLKINSLQPEDFGSYYCQHFWSTPWTFGGGTKLEIKR	101	612
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSYRAPGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVLR	102	613
DIQMTQSPSSLSASLGERVSLTCRASQEISGYLSWLQQKPDGTIKRLIYAASTLDSGVPKRFSGSRSGSDYSLTISSLESEDFADYYCLQYASYPWTFGGGTKLEIKR	103	614
DIVMTQSPATLSVTPGDRVSLSCRASQSISDYLHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGSDFTLSINSVEPEDVGVYYCQNGHSFPYTFGGGTKLEIKR	104	615
ENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWYQQKSDASPKLWIYYTSNLAPGVPARFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSPSTFGCGTKLEIKR	105	616
ETTVTQSPASLSMAIGEKVTIRCITNTDIDDDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENMLSEDVADYYCLQSDNLPLTFGCGTKLELKR	106	617
QLVLTQSSSASFSLGASAKLTCTLSSQHSTYTIEWYQQQPLKPPKYVMELKKDGSHSTGDGIPDRFSGSSSGADRYLSISNIQPEDEAIYICGVGDTIKEQFVYVFGCGTKVTVLG	107	618
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLTFGCGTKLELKR	108	619
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLHYDEFPPTFGAGTKLELK	109	620
DIKMTQSPSSMYASLGERVTITCKASQDINSYLNWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	110	621
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	111	622
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	112	623
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCPQYVESPPTFGAGTKLELK	113	624
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	114	625
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	115	626
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFHQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	116	627
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPLTFGGGTKLEIK	117	628
DIKMTQSPSSMYASLGERVTITCKASQDINGYLSWFQQKPGKSPQTLLYRANRLVDGVPSRFRGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	118	629

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	119	630
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGFYYCLQYDEFPPTFGAGTKLELK	120	631
DIKMTQSPSSIYASLGERVTITCKASQDINSYLNWFQQKPGKSPKTLIYRANRLVDGV PSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGVGTKLELK	121	632
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	122	633
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRAKRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	123	634
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGAKLELK	124	635
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPFTFGSGTKLEIK	125	636
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPFTFGSGTKLEIK	126	637
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPFTFGSGTKLEIK	127	638
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPFTFGSGTKLEIK	128	639
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYLIHVRFGSGTKLEIK	129	640
NIVMTQSPKSMSMSVGERVTLSCKASENVVTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYLIHVRFGSGTKLEIK	130	64:
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLYTFGGGTKLEIK	131	642
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPHVHVFGGGTKLEIK	132	643
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLYTFGGGTKLEIK	133	644
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLYTFGGGTKLEIK	134	645
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELK	135	646
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELK	136	647
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPQLLIYAAS NLQSGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELK	137	648
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFGGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELK	138	649
DIQMTQSPSSLSASLGGKVTITCKASQDINKYIAWYQHKPGKGPRLLIHYTSTLQPGI PSRFSGSGSGRDYSFSISNLEPEDIATYYCLQYDILMYTFGGGTKLEIK	139	650
DIQMTQSPSSLSASLGGKVTITCKASQDINKYIAWYQHKPGKGPRLLIHYTSTLQPGI PSRFSGSGSGRDYSFSISNLEPEDIATYYCLQYDILMYTFGGGTKLEIK	140	651
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELK	141	652
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELK	142	653

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELK	143	654
ENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWYQQKSDASPKLWIYYTSNLAP GVPARFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSHTFGGGTKLEIK	144	655
ENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWYQQKSDASPKLWIYYTSNLAP GVPARFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSHTFGGGTKLEIK	145	656
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNKDPLTFGAGTKLELK	146	657
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELK	147	658
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNKDPFTFGAGTKLELK	148	659
DIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQAPKLLIKYASN LESGVPARFSGSGSGTDFTLNIHPVEEEDTATYYCQHSWEIPLTFGAGAKLELK	149	660
DIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQPPKLLIKYASN LESGVPARFSGSGSGTDFTLNVHPVEEEDTATYYCQHSWEIPLTFGAGTKLELK	150	661
DIQMTQTPSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYSTSRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPWTFGGGTKLEIK	151	662
DIQMTQTPSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYSTSRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNALPWTFGGGTKLEIK	152	663
DIQMTQTPSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYSTSRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPWTFGGGTKLEIK	153	664
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVL	154	665
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVL	155	666
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVL	156	667
ETTVTQSPASLSVATGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPG VPSRFSSSGYGTDFVFTIENTLSEDVADYYCLQSDNMPFTFGSGTKLEIK	157	668
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYFGGGTKVTVL	158	669
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLTFGAGTKLELK	159	670
DIVMSQSPSSLAVSVGEKVTMSCKSSQSLLYSTNQKNYLAWYQQKPGQSPKLLIYW ASTRESGVPDRFTGSGSGTDFTLTISSVKAEDLAVYYCQQYYSYPPWTFGGGTKLEIK	160	671
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPPTFGGGTKLEIK	161	672
DIQMTQSPASLAASVGETVTITCRASENIYYSLAWYQQKQGKSPQLLIYNANSLEDG VPSRFSGSGSGTQYSMKINSMQPEDTATYFCKQAYDVPYTFGGGTKLEIK	162	673
DIQMTQTTSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYYTSRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQDNTLPRTFGGGTKLEIK	163	674
DIQMTQTTSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYYTSRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPWTFGGGTKLEIK	164	675
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIK	165	676

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYRQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPWTFGGGTKLEIK	166	677
QAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTNNR APGVPARFSGSLIGDKAALTITGAQTEDEAIYFCALWYSNHWVFGGGTKLTVL	167	678
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELK	168	679
QLVLTQSSSASFSLGASAKLTCTLSSQHSTYTIEWYQQQPLKPPKYVMELKKDGSHS TGDGIPDRFSGSSSGADRYLSISNIQPEDEAIYICGVGDTIKEQFVFVFGGGTKVTVL	169	680
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPPTFGGGTKLEIK	170	681
DIQMTQSPSSLSASLGERVSLTCRASQDIGISLNWLQQEPDGTIKRLIYATSSLDSGV PKRFSGSRSGSDYSLTISSLESEDFVDYYCLQYASSPYTFGGGTKLEIK	171	682
DIQMTQSPASLSASVGETVTITCRASENIYSYLAWYQQKQGKSPQLLVYNAKNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGYYCQHFWGTPYTFGGGTKLEIK	172	683
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDG VPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPLTFGAGTKLELK	173	684
DIQMTQSPSSLSASLGERVSLTCRASQDIHGYLNLFQQKPGETIKHLIYETSNLDSGV PKRFSGSRSGSDYSLIIGSLESEDFADYYCLQYASSPLTFGAGTKLELK	174	685
DIVMTQSHKFMSTSVGDRVSITCKASQDVGTAVAWYQQKPGQSPKLLIYWASTRH TGVPDRFTGSGSGTDFTLTISNVQSEDLADYFCQQYSSYPFTFGSGTKLEIK	175	686
QAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTNNR APGVPARFSGSLIGDKAALTITGAQTEDEAIYFCALWYSNHLVFGGGTKLTVL	176	687
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYVAS NLESGIPARFRGSGSGTDFTLNIHPVEEEDAAIYYCQQSHEDPRTFGGGTKLEIK	177	688
ENVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSSTSPKLWIYDTSKLASG VPGRFSGSGSGNSYSLTISSMEAEDVATYYCFQGSGYPLTFGAGTKLELK	178	689
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLAD GVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPYTFGGGTKLEIK	179	690
ETTVTQSPASLSVATGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPG VPSRFSSSGYGTDFVFTIENTLSEDVADYYCLQSDNMPLTFGAGTKLELK	180	691
DIQMTQTTSSLSASLGDRVTISCRASQDISNYLNWYQQKPDGTVKLLIYYTSRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPFTFGSGTKLEIK	181	692
DIQMTQTTSSLSASLGDRVTISCSASQGISNYLNWYQQKPDGTVKLLIYYTSSLHSGV PSRFSGSGSGTDYSLTISNLEPEDIATYYCQQYSKLPYTFGGGTKLEIK	182	693
DIVMTQSPATLSVTPGDRVSLSCRASQSISDYLHWYQQKSHESPRLLIKYASQSISGIP SRFSGSGSGSDFTLSINSVEPEDVGVYYCQNGHSFPWTFGGGTKLEIK	183	694
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSYRA PGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVL	184	695
DIQMTQSPSSLSASLGERVSLTCRASQEISGYLSWLQQKPDGTIKRLIYAASTLDSGV PKRFSGSRSGSDYSLTISSLESEDFADYYCLQYASYPWTFGGGTKLEIK	185	696
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPLPTFGAGTQRELK	186	697
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPVRFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLTFGAGTKLELK	187	698
DIQMTQSPASLSASVGETVTITCRASGNIHNYLAWYQQKQGKSPQLLVYNAKTLAD GVPSRFSGSGSGTQYSLKINSLQPEDFGSYYCQHFWSTPWTFGGGTKLEIK	188	699
DIQMTQSPASLSASVGETVTITCRASENIYSYLAWYQQKQGKSPQLLVYNAKTLAEG VPSRFSGSGSGTQFSLKINSLQPEDFGSYYCQHHYGTPFTFGSGTKLEIK	189	700

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
ENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWSQQKSDASPKLWIYYTSNLAPGVPPRFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSLTFGAGTKLELK	190	701
QAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGSTNNRAPGVPARFSGSLIGDKSALTITGAQTEDEAIYFCTLWYSNHWVFGGGTKLTVL	191	702
DIVMTQSHKFMSTSVGDRVSITCKASQDVGTAVAWYQQKPGQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISNVQSEDLADYFCQQYSSYPFTFGSGTKLEIK	192	703
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELK	193	704
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPFTFGSGTKLEIK	194	705
DIVMTQSPATLSVTPGDRVSLSCRASQSISDYLHWYQQKSHESPRLLIKYASQSISGIPSRFSGSGSGSDFTLSINSVEPEDVGVYYCQNGHSFPYTFGGGTKLEIK	195	706
QAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTNNRAPGVPARFSGSLIGDKAALTITGAQTEDEAIYFCALWYSNHWVFGGGTKLTVL	196	707
EIFVTQSPASLSMAIGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPGVPSRFSSSGYGTDFVFTIENMLSEDVADYYCLQSDNLPLTFGAGTKLELK	197	708
QIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISRMEAEDAATYYCQQRSSYPPTFGSGTKLELK	198	709
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASGVPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLTFGSGTKLEIK	199	710
QIVLTQSPAIMSASPGEKVTLTCSASSSVSSSYLYWYQQKPGSSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAASYFCHQWSSYPPTFGAGTKLELKR	200	711
QLVLTQSSSASFSLGASAKLTCTLSSQHSTYTIEWYQQQPLKPPKYVMELKKDGSHSTGDGIPDRFSGSSSGADRYLSISNIQPEDEAIYICGVGDTIKEQFVYVFGGGTKVTVLR	201	712
DVLMTQTPLSLPVSLGDQASISCRSSQSIVHSNGNTYLEWYLQKPGQSPKLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDLGVYYCFQGSHVPWTFGGGTKLEIKR	202	713
DIKMTQSPSSMYASLGERVTITCKASQDINSYLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCLQYDEFPPTFGAGTKLELKR	203	714
DVVMTQTPLTLSVTIGQPASISCKSSQSLLDSDGKTYLNWLLQRPGQSPKRLIYLVSKLDSGVPDRFTGSGSGTDFTLKISRVEAEDLGVYYCWQGTHFPWTFGGGTKLEIKR	204	715
QIVLTQSPAIMSASPGEKVTLPCSASSSVSSSYLYWYQQKPGSSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAASYFCHQWSSYPPTFGAGTKLELKR	205	716
DVLMTQTPLSLPVSLGDQASISCRSSQSIVYSNGNTYLEWYLQKPGQSPKLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDLGVYYCFQGSHVPPTFGGGTKLEIKR	206	717
DIQMTQSPASLSVSVGETVTITCRASENIYNNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIKR	207	718
QIVLTQSPAIMSASPGEKVTLTCSASSSVSSSYLYWYQQKPGSSPKLWIYSTSNLASGVPARFSGSGSGTSYSLTISSMEAEDAASYFCHQWSSYPPTFGAGTKLELKR	208	719
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMYWYQQKPGSSPRLLIYDTSNLASGVPVRFSGSGSGTSYSLTISRMEAEDAATYYCQQWSTYPPITFGAGTKLELKR	209	720
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMYWYQQKPGSSPRLLIYDTSNLASGVPVRFSGSGSGTSYSLTISRMEAEDAATYYCQQWSSYPFTFGSGTKLEIKR	210	721
QIVLSQSPAILSASPGEKVTMTCRASSSVSYMHWYQQKPGSSPKPWIYATSNLASGVPARFSGSGSGTSYSLTISRVEAEDAATYYCQQWSSNPYTFGGGTKLEIKR	211	722
DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGKSPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWTFGGGTKLEIKR	212	723
DIVMTQSHKFMSTSVGDRVSITCKASQDVSTAVAWYQQKPGQSPKLLIYWASTRHTGVPDRFTGSGSGTDYTLTISSVQAEDLALYYCQQHYSTPWTFGGGTKLEIKR	213	724

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
DVVMTQTPLTLSVTIGQPASISCKSSQSLLDSDGKTYLNWLLQRPGQSPKRLIYLVSK LDSGVPDRFTGSGSGTDFTLKISRVEAEDLGVYYCWQGTHFPWTFGGGTKLEIKR	214	725
DVLMTQTPLSLPVSLGDQASISCRSSQSIVHSNGNTYLEWYLQKPGQSPKFLIYKVS NRFSGVPDRFSGSGSGTDFTLKISRVEAEDLGVYYCFQGSHVPPTFGGGTKLEIKR	215	726
ETTVTQSPASLSVATGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPG VPSRFSSSGYGTDFVFTIENTLSEDVADYYCLQSDNMPLTFGGGTKLEIKR	216	727
DIQMTQSPSSLSASLGGKVTITCKASQDINKYIAWYQHKPGKGPRLLIHYTSTLQPGI PSRFSGSGSGRDYSFSISNLEPEDIATYYCLQYDNLYMYTFGGGTKLEIKR	217	728
DIQMNQSPSSLSASLGDTITITCHASQNINVWLSWYQQKPGNIPKLLIYKASNLHTG VPSRFSGSGSGTGFTLTISSLQPEDIATYYCQQGQSYPYTFGGGTKLEIKR	218	72
QIVLTQSPALMSASPGEKVTMTCSASSSVSYMYWYQQKPRSSPKPWIYLTSNLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPPTFGGGTKLEIKR	219	730
DVVMTQTPLTLSVTIGQPASISCKSSQSLLDSDGKTYLNWLLQRPGQSPKRLIYLVSK LDSGVPDRFTGSGSGTDFTLKISRVEAEDLGFYYCWQGTHFPWTFGGGTKLEIKR	220	731
QIVLTQSPAIMSASPGEKVTMTCSASLSVSDMYWYQQKPGSSPRLLIYDTSNLASGV PVRFSGSGSGTSYSLTISRMEAEDAATYYCQQWSSYPFTFGSGTKLEIKR	221	732
QIVLTQSPAIMSASPGEKVTLTCSASSSVSSSYLYWYQQRPGSSPKLWIYSTSNLASG VPARFSGSGSGTSYSLTISSMEAEDAASYFCHQWSSYPPTFGAGTKLELKR	222	733
ENVLTQSPAIMSASLGEKVTMSCRASSSVNYMYWYQQKSDASPKLWIYYTSNLAP GVPARFSGSGSGNSYSLTISSMEGEDAATYYCQQFTSSPSTFGGGTKLEIKR	223	734
ETTVTQSPASLSMAIGEKVTIRCITNTDIDDDMNWYQQKPGEPPKLLISEGNTLRPG VPSRFSSSGYGTDFVFTIENMLSEDVADYYCLQSDNLPLTFGAGTKLELKR	224	735
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPWTFGGGTKLEIKR	225	736
DIVLTQSPATLSVTPGDSVSLSCRASQSISNNLHWYQQKSHESPRLLIKYASQSISGIP SRFSGSGSGTDFTLSINSVETEDFGMYFCQQSNSWPFTFGSGTKLEIKR	226	737
QIVLTQSPAIMSASPGEKVTLTCSASSSVSSSYLYWYQQKPGSSPKLWIYSTSNLASG VPARFSGSGSGTSYSLTISSMEAEDAASYFCHQWSSYPPTFGGGTKLEIKR	227	738
ETTVTQSPASLSVATGEKVTIRCITSTDIDDDMNWYQQKPGEPPKLLISEGNTLRPG VPSRFSSSGYGTDFVFTIENTLSEDVADYYCLQSDNMPLTFGAGTKLELKR	228	739
DVVMTQTPLTLSVTIGQPASISCKSSQSLLDSDGKTYLHWLLQRPGQSPKRLIYLVSK LDSGVPDRFTGSGSGTDFTLKISRVEAEDLAVYYCWQGTHFPWTFGGGTKLEIKR	229	740
DIQMTQSPSSLSASLGGKVTITCKASQDINKYIAWYQHKPGKGPRLLIHYTSTLQPGI PSRFSGSGSGRDYSFSISNLEPEDIATYYCLQYDNLRTFGGGTKLEIKR	230	741
QAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQEKPDHLFTGLIVGTNNR APGVPARFSGSLIGDKAALTITGAQTEDEAIYFCVLWYSNHLVFGGGTKLTVLG	231	742
NIVMTQSPKSMSMSVGERVTLSCKASENVGTYVSWYQQKPEQSPKLLIYGASNRYT GVPDRFTGSGSATDFTLTISSVQAEDLADYHCGQSYSYPPTFGAGTKLELKR	232	743
DIVLTQSPASLAVSLGQRATISCRASETVDSYGYSFMHWYQQKPGQPPKLLIYRASN LESGIPARFSGSGSRTDFTLTINPVEADDVATYYCQQSNEDPRTFGGGTKLEIKR	233	744
DIVMSQSPSSLAVSVGEKVTMSCKSSQSLLYSTNQKNYLAWYQQKPGQSPKLLLYW ASTRESGVPDRFTGSGSGTDFTLTINSVKAEDLAVYYCQQYYSYRTFGGGTKLEIKR	234	745
DVVMTQTPLTLSVTIGQPASISCKSSQSLLDSDGKTYLNWLLQRPGQSPKRLIYLVSK LDSGVPDRFTGSGSGTDFTLKISRVEAEDLGVYYCWQGTHFPFTFGSGTKLEIKR	235	746
DIVLTQSPASLAVSLGQRATISCKASQSVDYDGDSYMNWYQQKPGQPPKLLIYAAS NLESGIPARFSGSGSGTDFTLNIHPVEEEDAATYYCQQSNEDPFTFGSGTKLEIKR	236	747

TABLE 21-continued

VL Sequences Table 21-VL sequences		
Sequence	Binder Name	SEQ ID NO:
QLVLTQSSSASFSLGASAKLTCTLSSQHSTYTIEWYQQQPLKPPKYVMELKKDGSHS TGDGIPDRFSGSSSGADRYLSISNIQPEDEAIYICGVGDTIKEQFVYVFGGGTKVTVLG	237	748
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLTFGAGTKLELKR	238	749
EIQMTQSPSSMSASLGDRITITCQATQDIVKNLNWYQQKPGKPPSFLIYYATELAEG VPSRFSGSGSGSDYSLTISNLESEDFADYYCLQFYEFPLTFGAGTKLELKR	239	750
DIVLTQSPASLAVSLGQRATISCRASESVDNYGISFMNWFQQKPGQPPKLLIYAASN QGSGVPARFSGSGSGTDFSLNIHPMEEDDTAMYFCQQSKEVPPTFGGGTKLEIKR	240	751
DIVMTQSPATLSVTPGDRVSLSCRASQSISDYLHWYQQKSHESPRLLIKYASQSISGIP SRFSGSGSGSDFTLSINSVEPEDVGVYYCQNGHSFPLTFGAGTKLELKR	241	752
DVLMTQTPLSLPVSLGDQASISCRSSQSIVHSNGDTYLEWYLQKPGQSPKLLIYKVSN RFSGVPDRFSGSGSGTDFTLKISRVEAEDLGVYYCFQGSHVPLTFGAGTKLELKR	242	753
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLTFGAGTKLELKR	243	754
DIVMTQSPTFLAVTASKKVTISCTASESLYSSKHKVHYLAWYQKKPEQSPKLLIYGAS NRYIGVPDRFTGSGSGTDFTLTISSVQVEDLTHYYCAQFYSYPYTFGGGTKLEIKR	244	755
QLVLTQSSSASFSLGASAKLTCTLSSQHSTYTIEWYQQQPLKPPKYVMELKKDGSHS TGDGIPDRFSGSSSGADRYLSISNIQPEDEAIYICGVGDTIKEQFVYVFGGGTKVTVLG	245	756
DIVLTQSPASLAVSLGQRATISCRASQSVSTSSYSYMHWYQQKPGQPPKLLIKYASN LESGVPARFSGSGSGTDFTLNIHPVEEEDTATYYCQHSWEIPLTFGAGTKLELKR	246	757
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPLTFGAGTKLELKR	247	758
DIQMTQSSSYLSVSLGGRVTITCKASDHINNWLAWYQQKPGNAPRLLISGATSLET GVPSRFSGSGSGKDYTLSITSLQTEDVATYYCQQYWSTPLTFGAGTKLELKR	248	759
DIQMTQSPASLSASVGETVTITCRASENIYSYLAWYQQKQGKSPQLLVYNAKTLAEG VPSRFSGSGSGTQFSLKINSLQPEDFGSYYCQHHYGTPYTFGGGTKLEIKR	249	760
DIQMTQSPASLSASVGETVTITCRASENIYSYLAWYQQKQGKSPQLLVYNAKTLAEG VPSRFSGSGSGTQFSLKINSLQPEDFGSYYCQHHYGTPPTFGGGTKLEIKR	250	761
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMHWYQQKSGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCQQWSSNPPTFGAGTKLELKR	251	762
DIVLTQSPASLAVSLGQRATISCRASESVDNYGISFMNWFQQKPGQPPKLLIYAASN QGSGVPARFSGSGSGTDFSLNIHPMEEDDTAMYFCQQSKEVPPTFGGGTKLEIKR	252	763
DIVMTQSPTFLAVTASKKVTISCTASESLYSSKHKVHYLAWYQKKPEQSPKLLIYGAS NRYIGVPDRFTGSGSGTDFTLTISSVQVEDLTHYYCAQFYSYPYTFGGGTKLEIKR	253	764
QAVVTQESALTTSPGGTVILTCRSSTGAVTTSNYANWVQEKPDHLFTGLIGGTSNR APGVPVRFSGSLIGDKAALTITGAQTEDDAMYFCALWYSTHYVFGGGTKVTVLG	254	765
QIVLTQSPAIMSASPGEKVTMTCSASSSISYMHWYQQKPGTSPKRWIYDTSKLASG VPARFSGSGSGTSYSLTISSMEAEDAATYYCHQRSSYPTFGAGTKLELKR	255	766
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWA STRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPWTFGCGTKVEIKR	257	9577
DIQMTQSPSSLSASVGDRVTITCRASQDIGDYLAWYQQKPGKAPKLLIYDASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPLTFGCGTKVEIKR	258	9578
DIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYSASNLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGCGTKVEIKR	259	9579
DIQMTQSPSSLSASVGDRVTITCRASQGIASYLAWYQQKPGKAPKLLIYAASTLQPG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQFDSYPITFGQGTKVEIKR	260	9580
DIQMTQSPSSLSASVGDRVTITCRASQSISSWLAWYQQKPGKAPKLLIYAASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQGYSTPYIFGQGTKVEIKR	261	9581

TABLE 21-continued

VL Sequences Table 21-VL sequences		
Sequence	Binder Name	SEQ ID NO:
DIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYKASSLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPFTFGQGTRLEIKR	262	9582
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLNWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGQGTKVEIKR	263	9583
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPWTFGQGTKVEIKR	26	9584
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLNWYQQKPGKAPKLLIYDASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGQGTKVEIK	265	9585
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIK	266	9586
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYKASSLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSFSSPLTFGQGTKVEIK	267	9587
DIQMTQSPSSLSASVGDRVTITCRASQNVNTWLAWYQQKPGKAPKLLIYEASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPFTFGQGTKLEIK	268	9588
DIQMTQSPSSLSASVGDRVTITCRASQSISDWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCAQHNHYPYTFGQGTRLEIK	269	9589
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYTTPFTFGQGTKVEIK	270	9590
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRASGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPFTFGQGTKLEIK	271	9591
DIQMTQSPSSLSASVGDRVTITCRASQAIRDDLGWYQQKPGKAPKLLIYDASHLEAGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPITFGQGTRLEIK	272	9592
DIQMTQSPSSLSASVGDRVTITCRASQGVGNDLAWYQQKPGKAPKLLIYAASTLQTGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQASSFPLTFGPGTKVDIK	273	9593
DIQMTQSPSSLSASVGDRVTITCRASQIIGTNLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYTFPVTFGQGTKLEIK	274	9594
DIQMTQSPSSLSASVGDRVTITCRASQSISTWLAWYQQKPGKAPKLLIYDASSLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPFTFGPGTKVDIK	275	9595
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSNNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYGSPLTFGPGTKVDIK	276	9596
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSSPPTFGQGTRLEIK	277	9597
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSSPPTFGGGTKVEIK	278	9598
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPWTFGQGTKVEIK	279	9599
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRASGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYGSPPTFGQGTRLEIK	280	9600
DIQMTQSPSSLSASVGDRVTITCQASQDIRNYLNWYQQKPGKAPKLLIYDASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPWTFGQGTKLEIK	281	9601
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYNASNLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLNSYPFTFGGGTKVEIK	282	9602
DIQMTQSPSSLSASVGDRVTITCQASQSISTWLAWYQQKPGKAPKLLIYAASTLRSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCLQHYTYPLTFGQGTRLEIK	283	9603
DIQMTQSPSSLSASVGDRVTITCRASEDISTYLAWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSHTIPWTFGQGTRLEIK	284	9604
DIQMTQSPSSLSASVGDRVTITCRASHHISDFLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPYTFGQGTKLEIK	285	9605

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
DIQMTQSPSSLSASVGDRVTITCRASQDIGDYLAWYQQKPGKAPKLLIYDASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPLTFGGGTKVEIK	286	9606
DIQMTQSPSSLSASVGDRVTITCRASQDIRSYLAWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYTAPPTFGQGTRLEIK	287	9607
DIQMTQSPSSLSASVGDRVTITCRASQDISNNLNWYQQKPGKAPKLLIYAASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCLQHNTYPLTFGQGTKLEIK	288	9608
DIQMTQSPSSLSASVGDRVTITCRASQDISNWLAWYQQKPGKAPKLLIYDASSLQS GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAISFPLTFGGGTKVEIK	289	9609
DIQMTQSPSSLSASVGDRVTITCRASQGIANYLAWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQADSFPLTFGQGTKVEIK	290	9610
DIQMTQSPSSLSASVGDRVTITCRASQGIASYLAWYQQKPGKAPKLLIYAASTLQPG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQFDSYPITFGQGTKVEIK	291	9611
DIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYAASRLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSSIIPFTFGQGTKLEIK	292	9612
DIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYAASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPYTFGQGTKLEIK	293	9613
DIQMTQSPSSLSASVGDRVTITCRASQSIGRWLAWYQQKPGKAPKLLIYDASNLET GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPRTFGQGTKVEIK	294	9614
DIQMTQSPSSLSASVGDRVTITCRASQSINSWLAWYQQKPGKAPKLLIYDTSSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTYSTPYTFGQGTKLEIK	295	9615
DIQMTQSPSSLSASVGDRVTITCRASQSISSWLAWYQQKPGKAPKLLIYAASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQGYSTPYIFGQGTKVEIK	296	9616
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTDSIPITFGQGTRLEIK	297	9617
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASTLQSGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSIPYTFGQGTKVEIK	29	9618
DIQMTQSPSSLSASVGDRVTITCRASQTIRSYLNWYQQKPGKAPKLLIYKASSLESGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTYTIPITFGPGTKVDIK	299	9619
DIQMTQSPSSLSASVGDRVTITCRASQTISNWLAWYQQKPGKAPKLLIYAASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPPTFGQGTKVEIK	300	9620
DIQMTQSPSSLSASVGDRVTITCRASQYIGSYLNWYQQKPGKAPKLLIYDASNLETG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQVDSYPLTFGGGTKVEIK	301	9621
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSN RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQGTHWPPTFGQGTKLEIK	302	9622
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYFGSN RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQAPVSFGQGTRLEIK	303	9623
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWA SSRQSGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPLTFGQGTKVEIK	304	9624
DIVMTQSPDSLAVSLGERATINCKSSQSVSSSSYNKNYLAWYQQKPGQPPKLLIYW ASVRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPITFGQGTRLEIK	30	9625
DIVMTQSPDSLAVSLGERATINCKSTQNVLSSSNNNSYLAWYQQKPGQPPKLLIYW ASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPFTFGQGTRLEIK	306	9626
DIQMTQSPSSLSASVGDRVTITCQASQDIGNYLNWYQQKPGKAPKLLIYAASSLQS GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTYNTPLTFGGGTKLEIK	607	9627
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYEASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPFTFGPGTKVDIK	608	9628

TABLE 21-continued

VL Sequences Table 21-VL sequences		
Sequence	Binder Name	SEQ ID NO:
DIQMTQSPSSLSASVGDRVTITCQASQDISTWLAWYQQKPGKAPKLLIYRASTLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSIPLTFGGGTKVEIK	609	9629
DIQMTQSPSSLSASVGDRVTITCRASQNINNYLNWYQQKPGKAPKLLIYAASRLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSAPVTFGQGTKLEIK	310	9630
DIQMTQSPSSLSASVGDRVTITCRASQNINTWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPFTFGPGTKVDIK	311	9631
DIQMTQSPSSLSASVGDRVTITCRASQRIGNYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIK	312	9632
DIQMTQSPSSLSASVGDRVTITCRASQSISTYLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYRTVTFGQGTRLEIK	313	9633
EIVMTQSPATLSVSPGERATLSCRASQSVGSYLAWYQQKPGQAPRLLIYGASTRATGIPARFSGSGSGTEFTLTISSLQSEDFAVYYCQQYDSSSQTFGQGTKVEIK	314	9634
EIVMTQSPATLSVSPGERATLSCRASRSVSTYLAWYQQKPGQAPRLLIYGASTRATGIPARFSGSGSGTEFTLTISSLQSEDFAVYYCQQYDGSPYTFGQGTKLEIK	315	9635
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYDASNLETGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQTPPAFGPGTKVDIK	316	9636
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNFLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSAPPTFGQGTKVEIK	317	9637
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNFLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSDPITFGQGTKLEIK	318	9638
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASARESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSIPIAFGQGTRLEIK	319	9639
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWASTRDSGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSIPYTFGQGTKLEIK	320	9640
DIVMTQSPDSLAVSLGERATINCKSSQSVLSTSYNKNYLAWYQQKPGQPPKLLIYWASTRASGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYTTPPTFGQGTKVEIK	321	9641
DIVMTQSPDSLAVSLGERATINCKSSQSVLSTSYNRNFLAWYQQKPGQPPKLLIYWASTRQSGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPYTFGQGTKLEIK	322	9642
DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPLTFGGGTKVEIK	323	9643
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPTFGQGTKVEIK	324	9644
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYGASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQEADSFPLTFGGGTKVEIK	325	9645
DIQMTQSPSSLSASVGDRVTITCRASQGIRNDLGWYQQKPGKAPKLLIYDASSLHSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPWTFGQGTKLEIK	326	9646
DIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYKASSLESGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPFTFGQGTRLEIK	327	9647
DIQMTQSPSSLSASVGDRVTITCRASQSINRWLAWYQQKPGKAPKLLIYSASNLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPLTFGGGTKVEIK	328	9648
DIQMTQSPSSLSASVGDRVTITCRASQSINTWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPFTFGPGTKVDIK	329	9649
DIQMTQSPSSLSASVGDRVTITCRASQSIRTWLAWYQQKPGKAPKLLIYDASSLETGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLNSYPLTFGGGTKVEIK	330	9650
DIQMTQSPSSLSASVGDRVTITCRASQSIRTYLNWYQQKPGKAPKLLIYAASTLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSAPLTFGGGTKLEIK	331	9651
DIQMTQSPSSLSASVGDRVTITCRASQSISTYLNWYQQKPGKAPKLLIYAASSLHSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIK	332	9652

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
DIQMTQSPSSLSASVGDRVTITCRASQSITTYLNWYQQKPGKAPKLLIYAASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIK	333	9653
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYAASS LQSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQARQTPLTFGQGTRLEIK	334	9654
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYGASS LQSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQTLQTPFTFGPGTKVDIK	335	9655
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSD RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQTPLTFGPGTKVDIK	336	9656
DIVMTQSPDSLAVSLGERATINCKSSQTVFSTSYNKNYLAWYQQKPGQPPKLLIYW ASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPLTFGGGTKVEIK	337	9657
DIVMTQSPDSLAVSLGERATINCKTSQSVFSTSYNRDYLAWYQQKPGQPPKLLIYW ASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSSPPTFGQGTKVEIK	338	9658
DIQMTQSPSSLSASVGDRVTITCRASQSISSWLAWYQQKPGKAPKLLIYDASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPFTFGPGTKVDIK	339	9659
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYDASNLKTG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSFPPTFGGGTKVEIK	340	9660
EIVMTQSPATLSVSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDTSSRATGI PARFSGSGSGTEFTLTISSLQSEDFAVYYCQQYYDTPYTFGQGTKLEIKR*	341	9661
DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYLA STREPGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPPTFGGGTKLEIKR*	342	9662
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYVAWYQQKPGQPPKLLIYWA STRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPLTFGGGTKVEIKR*	343	9663
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYAAASLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTYSTPWTFGQGTRLEIKR*	344	9664
DIQMTQSPSSLSASVGDRVTITCRASQDINTYLAWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSSSFPLTFGQGTKVEIKR*	345	9665
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLYNFPYTFGGGTKVEIKR*	346	9666
DIQMTQSPSSLSASVGDRVTITCRASQSISRYLAWYQQKPGKAPKLLIYGASTRESGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPLTFGQGTKLEIKR	347	9667
DIQMTQSPSSLSASVGDRVTITCRASQTLSGWLAWYQQKPGKAPKLLIYGASTLQG GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYYSYPPTFGQGTKVEIKR*	348	9668
DIQMTQSPSSLSASVGDRVTITCFASQDIINYLNWYQQKPGKAPKLLIYEASNLETGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGQGTKVEIKR	349	9669
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYDVFNLGTG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPFTFGQGTRLEIKR*	350	9670
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYMASNLES GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTNSFPLTFGQGTKLEIKR*	351	9671
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYDASNLETGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIKR*	352	9672
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSN RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQSPWTFGQGTKLEIKR*	353	9673
DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYW ASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSSPLTFGGGTKVEIKR *	354	9674
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSN RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQTPPSFGQGTKVEIKR*	355	9675

TABLE 21-continued

VL Sequences Table 21-VL sequences		
Sequence	Binder Name	SEQ ID NO:
DIQMTQSPSSLSASVGDRVTITCRASESVSTWLAWYQQKPGKAPKLLIYKASRLESG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYKTPYTFGQGTKLEIKR*	356	9676
DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYW ASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYFTTPLTFGQGTKLEIKR *	357	9677
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPYTFGQGTKVEIKR*	358	9678
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWA STRASGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYDTPLTFGQGTKVEIKR*	359	9679
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYKASTLESGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQNDSIPITFGQGTRLEIKR*	360	9680
DIQMTQSPSSLSASVGDRVTITCRASQSISRWLAWYQQKPGKAPKLLIYDASNLETG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCLQDYSYPLTFGQGTKVEIKR*	361	9681
DIVMTQSPDSLAVSLGERATINCKTSQSVFSTSYNRDYLAWYQQKPGQPPKLLIYW ASTRAAGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYYTSTFGQGTKVEIKR*	362	9682
DIQMTQSPSSLSASVGDRVTITCRASQSINRYLNWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPPTFGGGTKVEIKR*	363	9683
DIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYSASNLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIKR*	364	9684
DIQMTQSPSSLSASVGDRVTITCRASQSIDSYLNWYQQKPGKAPKLLIYKASTLESGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSAPLTFGGGTKVEIKR*	365	9685
DIQMTQSPSSLSASVGDRVTITCRASQDISTWLAWYQQKPGKAPKLLIYDASNLETG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQVNSDPYTFGQGTRLEIKR*	366	9686
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYAASTLESG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQGDSLPLTFGGGTKVEIKR*	367	9687
DIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSDSFPYTFGQGTKVEIKR*	368	9688
EIVMTQSPATLSVSPGERATLSCRASQSVSTYLAWYQQKPGQAPRLLIYGASTRATG IPARFSGSGSGTEFTLTISSLQSEDFAVYYCQQHDSYPLTFGGGTKVEIKR*	369	9689
DIQMTQSPSSLSASVGDRVTITCRASQGIRNDLGWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPPTFGQGTRLEIKR*	370	9690
DIQMTQSPSSLSASVGDRVTITCRASESISTYLNWYQQKPGKAPKLLIYKASNLESGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTDSTFITFGQGTKVEIKR*	371	9691
DIQMTQSPSSLSASVGDRVTITCRASRNIHDYLNWYQQKPGKAPKLLIYAASTLQTG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTYSTPPTFGPGTKVDIKR*	372	9692
DIQMTQSPSSLSASVGDRVTITCRASQSNDSYLNWYQQKPGKAPKLLIYKASTLESG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPLTFGQGTKVEIKR	373	9693
DIQMTQSPSSLSASVGDRVTITCRASQSISDFLNWYQQKPGKAPKLLIYAASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPYTFGQGTKVEIKR*	374	9694
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANRFPLTFGQGTKLEIKR*	375	9695
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYKASNLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNFPATFGQGTRLEIKR*	376	9696
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSN RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQGTHWPETFGQGTKVEIKR*	377	9697
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYDASNLETGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGQGTKLEIKR*	378	9698
DIQMTQSPSSLSASVGDRVTITCRASQGISDYLAWYQQKPGKAPKLLIYDASNLETG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYILPLTFGGGTKVEIKR*	379	9699

TABLE 21-continued

VL Sequences
Table 21-VL sequences

Sequence	Binder Name	SEQ ID NO:
DIQMTQSPSSLSASVGDRVTITCRASQDINDFLAWYQQKPGKAPKLLIYAASSLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSAPYTFGQGTKLEIKR*	380	9700
DIQMTQSPSSLSASVGDRVTITCRASQSISNWLAWYQQKPGKAPKLLIYAASKLESG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPWTFGQGTRLEIKR*	381	9701
DIQMTQSPSSLSASVGDRVTITCRASQGIDSWLAWYQQKPGKAPKLLIYAASTLESG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPLTFGGGTKVEIKR*	382	9702
DIQMTQSPSSLSASVGDRVTITCRASQNIGTWLAWYQQKPGKAPKLLIYRASSLESG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPWTFGQGTKLEIKR*	383	9703
DIQMTQSPSSLSASVGDRVTITCRASQNINNWLAWYQQKPGKAPKLLIYKASTLQS GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQADSFPPTFGQGTKVEIKR*	384	9704
DIQMTQSPSSLSASVGDRVTITCRASQDISSYLAWYQQKPGKAPKLLIYAASTLQSG VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLNRYPITFGQGTKVEIKR*	385	9705
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYAASILHSGV PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYDSSFITFGQGTRLEIKR*	386	9706

TABLE 22

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

	HCDR1		HCDR2		HCDR3	
Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
109	SYWIE	1280	EILPGSGSTNYNEKFKG	1794	RAYGYDGGFDY	2308
110	SYWIE	1281	EILPGSGSTNYIEKFKG	1795	RAYGYDEGFDY	2309
111	SYWIE	1282	EILPGSDSTSYNEKFKG	1796	RAYGYDEGFDY	2310
112	IYWIE	1283	EILPGSGSTNYNEKVKG	1797	RAYGYDGGFDY	2311
1	SYWIE	1284	EIFPGSGHTSFNEKFKG	1798	RGYGYDEGFDY	2312
113	SYWIE	1285	EILPGSGSTNYNEKFKG	1799	RGYGYDEGFDY	2313
114	SYWIE	1286	EILPGSGSTNYNEKFKG	1800	RGYGYDEGFDY	2314
115	NYWIE	1287	EILPGSGSTSYNEKFKG	1801	RGYGYDEGFDY	2315
23	SYWIE	1288	EILPGSGSTSYNEKFKG	1802	RGYGYDEGFDY	2316
116	SYWIE	1289	EILPGSGYTSYIEQFKG	1803	RGYGYDEGFDY	2317
117	SYWIE	1290	EILPGSGSTSYNEKFKD	1804	RAYGYDEGFDY	2318
2	SYWIE	1291	EILPGSGSTSYNEKFKG	1805	RGYGYDEGFDY	2319
118	SYWIE	1292	EVLPGSGSTSYNEKFKG	1806	RAYGYDEGFDY	2320
119	SYWIE	1293	EISPGSGSTNYNEKFKG	1807	RGYGYDEGFDY	2321
120	TYWIE	1294	EILPGSGTPNYNEKFKG	1808	RAYGYDAGFDY	2322
121	SYWIE	1295	EILPGSGSTSCNEKFKG	1809	RGYGYDEGFDY	2323
122	SYWIE	1296	EILPGSGRTSYIEKFKG	1810	RGYGYDEGFDY	2324
123	SYWIE	1297	EILPGSGRTSYIEKFKG	1811	RGYGYDEGFDY	2325

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
38	SYWIE	1298	EILPGSGSTSYNEKFKD	1812	RAYGYDEGFDY	2326
39	SYWIE	1299	EVLPGSGSTSYNEKFKG	1813	RAYGYDEGFDY	2327
40	SYWIE	1300	EILPGSGRTSYIEKFKG	1814	RGYGYDEGFDY	2328
41	SYWIE	1301	EVLPGSGSTSYNEKFKG	1815	RAYGYDEGFDY	2329
42	SYWIE	1302	EILPGSGRTSYIEKFKG	1816	RGYGYDEGFDY	2330
43	SYWIE	1303	EILPGSGSTNYNEKFKG	1817	RAYGYDGGFDY	2331
44	SYWIE	1304	EILPGSDSTSYNEKFKG	1818	RAYGYDEGFDY	2332
45	IYWIE	1305	EILPGSGSTNYNEKVKG	1819	RAYGYDGGFDY	2333
46	SYWIE	1306	EIFPGSGHTSFNEKFKG	1820	RGYGYDEGFDY	2334
47	NYWIE	1307	EILPGSGSTSYNEKFKG	1821	RGYGYDEGFDY	2335
47	NYWIE	1307	EILPGSGSTSYNEKFKG	1821	RGYGYDEGFDY	2335
48	SYWIE	1308	EILPGSGSTSYNEKFKG	1822	RGYGYDEGFDY	2336
48	SYWIE	1308	EILPGSGSTSYNEKFKG	1822	RGYGYDEGFDY	2336
49	SYWIE	1309	EILPGSGYTSYIEQFKG	1823	RGYGYDEGFDY	2337
50	SYWIE	1310	EILPGSGSTSYNEKFKG	1824	RGYGYDEGFDY	2338
51	SYWIE	1311	EISPGSGSTNYNEKFKG	1825	RGYGYDEGFDY	2339
52	TYWIE	1312	EILPGSGTPNYNEKFKG	1826	RAYGYDAGFDY	2340
53	SYWIE	1313	EILPGSGSTSCNEKFKG	1827	RGYGYDEGFDY	2341
54	SYWIE	1314	EILPGSGRTSYIEKFKG	1828	RGYGYDEGFDY	2342
55	NYWIE	1315	EILPGSGSTSYNEKFKG	1829	RGYGYDEGFDY	2343
56	SYWIE	1316	EILPGSGSTSYNEKFKG	1830	RGYGYDEGFDY	2344
3	GYYMH	1317	YISSYNGATSYNQKFKG	1831	GRYGEYFDY	2345
4	GYYMH	1318	YISSYNGVTGYNQKFKG	1832	GRYGDYFDY	2346
5	GYYMH	1319	YISSYNGVTSYNQKFKG	1833	GRYGDYFDY	2347
6	GYYMH	1320	YISSYNGVTGYNQKFKG	1834	GRYGDYFDY	2348
7	GYYMH	1321	YISSYNGANGYNQKFKG	1835	GRYGDYFDY	2349
8	GYYMH	1322	YISSYNGVTGYNQKFKG	1836	GRYGDYFDY	2350
9	GYYMH	1323	YISSYNGVTGYNQKFKG	1837	GRYGDYFDY	2351
10	GYYMH	1324	YISSYNGVTGYNQKFKG	1838	GRYGDYFDY	2352
11	GYYMH	1325	YISSYNGVTGYNQKFKG	1839	GRYGDYFDY	2353
12	GFYMH	1326	YISSYNGATGYNQKFKG	1840	GRYGDYFDY	2354
13	GYYMH	1327	YISSYNGATGYNQKFKG	1841	GRYGDYFDY	2355
57	GYYMH	1328	YISSYNGVTGYNQKFKG	1842	GRYGDYFDY	2356
58	GYYMH	1329	YISSYNGATSYNQKFKG	1843	GRYGEYFDY	2357
124	SYGVH	1330	VIWRGGSTDYNAAFMS	1844	NLYGHYVMDY	2358
125	SYGVH	1331	VIWRGGSTDYNAAFMS	1845	NLYGHYVMDY	2359
126	SYGVH	1332	VIWRGGSTDNNAAFMS	1846	NLYGHYVMDY	2360

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence							
127	RYGVH	1333	VIWRGGSTDHNAAFMS	1847	NLYGHYVMDY	2361	
128	TYGVH	1334	VIWRGGSTDYNAAFMS	1848	NLYGHYVMDY	2362	
129	SYGVH	1335	VIWRGGSTDYNAAFMS	1849	NLYGHYVMDY	2363	
130	RYGVH	1336	VIWRGGSTDHNAAFMS	1850	NLYGHYVMDY	2364	
59	SYGVH	1337	VIWRGGSTDYNAAFMS	1851	NLYGHYVMDY	2365	
60	SYGVH	1338	VIWRGGSTDYNAAFMS	1852	NLYGHYVMDY	2366	
61	SYGVH	1339	VIWRGGSTDNNAAFMS	1853	NLYGHYVMDY	2367	
62	RYGVH	1340	VIWRGGSTDHNAAFMS	1854	NLYGHYVMDY	2368	
63	TYGVH	1341	VIWRGGSTDYNAAFMS	1855	NLYGHYVMDY	2369	
131	SYWMH	1342	MIHPNSGSTNYNEKFKS	1856	WGDGYSFAY	2370	
132	SYWMH	1343	MIHPNSGSTNYNEKFKS	1857	WGDGYSFAY	2371	
133	TYWMH	1344	MIHPNSDNTNYNEKFKS	1858	WGDGYSFAY	2372	
14	SYWMH	1345	MIHPNSGTTNYNEKFKS	1859	WGDGYSFAY	2373	
134	SYWMH	1346	MIHPNSGNTNYNEKFKS	1860	WGDGYSFAY	2374	
64	SYWMH	1347	MIHPNSGSTNYNEKFKS	1861	WGDGYSFAY	2375	
65	SYWMH	1348	MIHPNSGTTNYNEKFKS	1862	WGDGYSFAY	2376	
135	DYVIN	1349	EIYPGSGSTYYNEKFKG	1863	RGERGPWFAY	2377	
136	DYVIN	1350	EIYPGSGSSYYNEKFKG	1864	RGERGPWFAY	2378	
137	DYVIN	1351	EIYPGSGSSYYNEKFRG	1865	RGERGPWFAY	2379	
138	DYVIN	1352	EIYPGSGSSYYNEKFKG	1866	RGERGPWFAY	2380	
15	DYVIN	1353	EIYPGSGSSYYNEKFKG	1867	RGERGPWFAY	2381	
66	DYVIN	1354	EIYPGSGSTYYNEKFKG	1868	RGERGPWFAY	2382	
67	DYVIN	1355	EIYPGSGSSYYNEKFKG	1869	RGERGPWFAY	2383	
68	DYVIN	1356	EIYPGSGSSYYNEKFRG	1870	RGERGPWFAY	2384	
69	DYVIN	1357	EIYPGSGSSYYNEKFKG	1871	RGERGPWFAY	2385	
24	NYWMN	1358	LIDPSDSETHYNQVFKD	1872	YDVYYRFAY	2386	
139	NYWMN	1359	MIDPSDSETHYNQMFKD	1873	YDGYYRFAY	2387	
140	NYWMN	1360	MIDPSDSETHFNQMFKD	1874	YDIYYRFAY	2388	
16	SYWMH	1361	MIHPNSGSTNYNEKFKS	1875	PGGYGFVY	2389	
141	SYWMH	1362	MIHPNSDSTNYNEKFKS	1876	PGGYGFAD	2390	
142	TYWMH	1363	MIHPNSGSTNYNEKFKS	1877	PGGYGFTY	2391	
143	SYWMH	1364	MIHPNSGSPNYNEKFKS	1878	PGGYGFAY	2392	
70	SYWMH	1365	MIHPNSGSPNYNEKFKS	1879	PGGYGFAY	2393	
25	SYWIN	1366	NIYPSDSYTNYNQKFKD	1880	GNYIDY	2394	
144	SYWIN	1367	NIYPSDSYTNYNQKFKD	1881	GNYIDY	2395	
145	DYWIN	1368	NIYPSDSYTNYNQKFKD	1882	GNYIDY	2396	
146	DYVIS	1369	EIYPGSGSSYYNEKFKG	1883	PGDLGFAY	2397	

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence							
147	DYVIS	1370	EIYPGSGSNYYNEKFKG	1884	PGDLGFAY	2398
148	DYVIS	1371	EIYPGSGSSYYNEKFKG	1885	PGDLGFAY	2399
71	DYVIS	1372	EIYPGSGSSYYNEKFKG	1886	PGDLGFAY	2400
72	DYVIS	1373	EIYPGSGSSYYNEKFKG	1887	PGDLGFAY	2401
73	DYVIS	1374	EIYPGSGSNYYNEKFKG	1888	PGDLGFAY	2402
74	DYVIS	1375	EIYPGSGSSYYNEKFKG	1889	PGDLGFAY	2403
74	DYVIS	1375	EIYPGSGSSYYNEKFKG	1889	PGDLGFAY	2403
75	DYVIS	1376	EIYPGSGSSYYNEKFKG	1890	PGDLGFAY	2404
75	DYVIS	1376	EIYPGSGSSYYNEKFKG	1890	PGDLGFAY	2404
76	DYVIS	1377	EIYPGSGSSYYNEKFKG	1891	PGDLGFAY	2405
149	NYGVH	1378	VVWAGGITNYNWALMS	1892	GDGYDDGYAMDY	2406
150	SYGVH	1379	VLWAGGITNYNSALMS	1893	GDGYDDGYAMDY	2407
26	SYGVH	1380	VIWAGGTTNYNSALMS	1894	GDGYDDGYAMDY	2408
77	NYGVH	1381	VVWAGGITNYNWALMS	1895	GDGYDDGYAMDY	2409
78	SYGVH	1382	VLWAGGITNYNSALMS	1896	GDGYDDGYAMDY	2410
79	SYGVH	1383	VIWAGGTTNYNSALMS	1897	GDGYDDGYAMDY	2411
151	SYWMY	1384	MIDPSDSETRLNQKFKD	1898	TRNY	2412
152	SYWMY	1385	MIDPSDSETRLNQKFKD	1899	TRNY	2413
153	SYWMY	1386	MIDPSDSETRLNQKFKD	1900	TRNY	2414
154	DYYMH	1387	WIDPENGDTEYAPKFQG	1901	PLLRYSSAMDY	2415
155	DYYIH	1388	WIDPENGDTEYAPKFQG	1902	PLLRYSSSMDY	2416
156	DYYMH	1389	WIDPENGDTEYAPKFQG	1903	ALLRYSSAMDY	2417
80	DYYMH	1390	WIDPENGDTEYAPKFQG	1904	PLLRYSSAMDY	2418
81	DYYIH	1391	WIDPENGDTEYAPKFQG	1905	PLLRYSSSMDY	2419
82	DYYMH	1392	WIDPENGDTEYAPKFQG	1906	ALLRYSSAMDY	2420
17	DTSLH	1393	RIDPANGNTKYDPKFQG	1907	GPDDGYFYYYSMD Y	2421
157	NYYVY	1394	EINPSNGDTNFNEKFKS	1908	YYTHEAYYYAMDC	2422
27	TYYIY	1395	EINPSNGGTNFNEKFKS	1909	YYTHETYYYAMDY	2423
158	DYYMH	1396	RIDPEDGDTEYAPKFQG	1910	YSIYDAMDY	2424
159	DYVIS	1397	EIYPGSGSTYYNEKFKG	1911	RGERGPWFAY	2425
83	DYVIS	1398	EIYPGSGSTYYNEKFKG	1912	RGERGPWFAY	2426
160	DYGMH	1399	VISTYYGDASYNQKFKG	1913	QMDYDYTYYYAMD Y	2427
28	SYWMQ	1400	EIDPSDSYTNYNQKFKG	1914	AEYGYGNYPWFAY	2428
84	SYWMQ	1401	EIDPSDSYTNYNQKFKG	1915	AEYGYGNYPWFAY	2429
29	SYWMH	1402	RIHPSDSDTNYNQKFKG	1916	PYYYGGWYFDV	2430
161	DYVIS	1403	EIYPGSGSTYYNEKFKG	1917	MDGPWFAY	2431

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
30	SYGMS	1404	TISSGGSYTYYPDSVKG	1918	LYDAHWDYFDY	2432
162	TSGMG	1405	SIWNNDNYYNPSLKS	1919	RPYYRYDSFAY	2433
18	NYGMN	1406	WINTYTGEPTYADDFKG	1920	KYYDYEFAY	2434
85	NYGMN	1407	WINTYTGEPTYADDFKG	1921	KYYDYEFAY	2435
163	DYEMH	1408	AIDPETGGTAYNQKFKV	1922	LGDYDVMDY	2436
86	DYEMH	1409	AIDPETGGTAYNQKFKV	1923	LGDYDVMDY	2437
164	SYWMH	1410	EIDPSDSYTNYNQKFKG	1924	AGRYGSSFDY	2438
165	TSGMG	1411	HIYWDDDKRYNPSLKS	1925	RPDDYDGAWFPY	2439
31	SSWMH	1412	EIHPNSGNTNYNEKFKG	1926	YYDYDAYYFDY	2440
87	SSWMH	1413	EIHPNSGNTNYNEKFKG	1927	YYDYDAYYFDY	2441
32	SYWMH	1414	MIHPNSGSTNYNEKFKS	1928	PYYGYDVGY	2442
166	DYVIS	1415	EIYPGSGSNYYNEKFKG	1929	EEKIYFDY	2443
88	DYVIS	1416	EIYPGSGSNYYNEKFKG	1930	EEKIYFDY	2444
167	SYWMH	1417	MIHPNSGSTNYNEKFKS	1931	YDGYWFDY	2445
168	SYWMH	1418	AIYPGNSDTTYNQKFKG	1932	LITTAYYFDY	2446
89	SYWMH	1419	AIYPGNSDTTYNQKFKG	1933	LITTAYYFDY	2447
169	SYWMH	1420	MIHPNSGSTNYNEKFKS	1934	ETGDYGSSYVWYF DV	2448
170	DYVIS	1421	EIYPGSGSTYYNEKFKG	1935	GKVTRFAY	2449
171	SYAMS	1422	TISDGGSYTYYPDNVKG	1936	DQDSNWEYFDY	2450
172	DYSMH	1423	WINTETGEPTYADDFKG	1937	ESWDRAMDY	2451
19	SYAMS	1424	TISSGGSYTYYPDSVKG	1938	HEEANWAWFAY	2452
90	SYAMS	1425	TISSGGSYTYYPDSVKG	1939	HEEANWAWFAY	2453
173	NYWMH	1426	MIDPSDSETRLNQQFKD	1940	PYYAMDY	2454
91	NYWMH	1427	MIDPSDSETRLNQQFKD	1941	PYYAMDY	2455
174	SSWMH	1428	EIHPNSGNTNYNEKNKG	1942	YYGNYVWYFDV	2456
92	SSWMH	1429	EIHPNSGNTNYNEKNKG	1943	YYGNYVWYFDV	2457
175	SYWMH	1430	MIHPNSGSTNYNEKFKS	1944	YGSSYWYFDV	2458
93	SYWMH	1431	MIHPNSGSTNYNEKFKS	1945	YGSSYWYFDV	2459
20	SYGVH	1432	VIWSGGSTDYNAAFIS	1946	YYGSSRSYWYLDV	2460
94	SYGVH	1433	VIWSGGSTDYNAAFIS	1947	YYGSSRSYWYLDV	2461
176	SYNMH	1434	ALYSGNGDTSYNQKFK G	1948	DYYGSSHLWYFDV	2462
177	TSGMG	1435	HIYWDDDKRYNPSLKS	1949	RAHYDYGWYFDV	2463
178	SYWMH	1436	MIHPNSGSTNYNEKFKS	1950	YDYDWYFDV	2464
33	SYGMS	1437	TISSGGSYTYYPDSVKG	1951	HDDSSYDWFAY	2465
179	SYGMS	1438	TISSGGSYTYYPDSVKG	1952	HEDSNYHYFDY	2466
34	NYWMH	1439	MIHPNSGTTNYNEKFKS	1953	FGDGYHFDY	2467

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
180	SYGMS	1440	TISSGGSYTYYPDSVKG	1954	QNDSSWAWFAY	2468
95	SYGMS	1441	TISSGGSYTYYPDSVKG	1955	QNDSSWAWFAY	2469
181	SYWMH	1442	MIHPNSGSTNYNEKFKS	1956	PYSNYGWYFDV	2470
96	SYWMH	1443	MIHPNSGSTNYNEKFKS	1957	PYSNYGWYFDV	2471
182	SYWMH	1444	NIDPSDSETHYNQKFKD	1958	DYYGSYWYFDV	2472
97	SYWMH	1445	NIDPSDSETHYNQKFKD	1959	DYYGSYWYFDV	2473
183	DYYMH	1446	RIDPEDGETKYAPKFQG	1960	YGNSAWFAY	2474
98	DYYMH	1447	RIDPEDGETKYAPKFQG	1961	YGNSAWFAY	2475
35	NYGMN	1448	WINTNTGEPTYAEEFKG	1962	WYPYFDY	2476
99	NYGMN	1449	WINTNTGEPTYAEEFKG	1963	WYPYFDY	2477
100	NYGMN	1450	WINTNTGEPTYAEEFKG	1964	WYPYFDY	2478
36	SYWMH	1451	YINPSSGYTKYNQKFKD	1965	SDGSSGNWYFDV	2479
101	SYWMH	1452	YINPSSGYTKYNQKFKD	1966	SDGSSGNWYFDV	2480
184	SYGVH	1453	VIWAGGSTNYNSALMS	1967	EGGYTGYFDV	2481
102	SYGVH	1454	VIWAGGSTNYNSALMS	1968	EGGYTGYFDV	2482
185	SYWMH	1455	NIDPSDSETHYNQKFKD	1969	SNYVPYYAMDY	2483
103	SYWMH	1456	NIDPSDSETHYNQKFKD	1970	SNYVPYYAMDY	2484
186	DYVIS	1457	EIYPGSGSAYYNEKFKG	1971	RGFDY	2485
21	SYGMS	1458	TISSGGSYTYYPDSVKG	1972	HNYSNWDWFAY	2486
187	SYWMH	1459	MIHPNSGSTNYNEKFKS	1973	DYYGSGYGYYFDY	2487
188	SYWMH	1460	MIHPNSGSTNYNEKFKS	1974	DYYGSSYGWYFDV	2488
189	SYWMH	1461	MIHPNSGSTNYNEKFKS	1975	DYYGSSYGWYFDV	2489
190	SYWMH	1462	MIHPNSGSTNYNEKFKS	1976	DYYGSSYGWYFDV	2490
191	SYWMH	1463	MIHPNSGSTNYNEKFKS	1977	DYYGSSYGWYFDV	2491
192	SYWMH	1464	MIHPNSGSTNYNEKFKS	1978	DYYGSGYGWYFD V	2492
193	NYWMN	1465	MIDPSDSETHYNQMFKD	1979	YDGYYRFAY	2493
194	NYWMN	1466	MIDPSDSETHFNQMFKD	1980	YDVYYRFAY	2494
195	SYWMH	1467	MIHPNSGSTNYNEKFKS	1981	DYGNYDYAMDY	2495
104	SYWMH	1468	MIHPNSGSTNYNEKFKS	1982	DYGNYDYAMDY	2496
37	SYWMH	1469	MIHPNSGSTNYNEKFKS	1983	DYGNYDYAMDY	2497
196	SYWMH	1470	MIHPNSGSTNYNEKFKS	1984	DYGNYDYAMDY	2498
197	SYGMS	1471	TISSGGSYTYYPDSVKG	1985	QLTGTWYYFDY	2499
198	SYGMS	1472	TISSGGSYTYYPDSVKG	1986	QLTGTWYYFDY	2500
199	SYGMS	1473	TISSGGSYTYYPDSVKG	1987	QLTGTWYYFDY	2501
22	DTSLH	1474	RIDPANGNTKYDPKFQG	1988	GPDDGYFYYYSMD Y	2502
200	NYYMS	1475	VINSNGGSTYYPDTVKG	1989	QEGIGYAMDY	2503

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
201	EYTMH	1476	GIYPNNGGTSYNQKFKG	1990	GGWLLGY	2504
202	SYGVH	1477	VIWSGGSTDYNAAFIS	1991	DGGIRGAMDY	2505
203	SYWIE	1478	EILPGSGSTNYNEKFKG	1992	RGYGYDEGFDY	2506
204	DYEMH	1479	AIDPETGGTAYNQKFKG	1993	NYDYAMDY	2507
205	SYYMS	1480	VINSNGGSTFYPDTVKG	1994	QEGIGYALDY	2508
206	SYAMS	1481	AISSGGSTYYPDSVKG	1995	EREWGVYYGSSLD Y	2509
207	DTYMH	1482	RIDPANGNTKYDPKFQG	1996	SDGNYD	2510
208	NYYMS	1483	VINSNGGSTYYPDTVKG	1997	QEGIGYGMDY	2511
209	TYVMN	1484	RIRSKSDNYATYYADSV KD	1998	HDGVVGFDV	2512
210	SGYYW	1485	YISYDGSNNYNPSLKN	1999	GGGRG	2513
211	DYSMH	1486	WINTETGEPTYADDFKG	2000	DYYDYYYAMDY	2514
212	DYSMH	1487	WINTETGEPTYADDFKG	2001	ESWDRAMDY	2515
213	NYWMN	1488	RIDPYDSETHYNQKFKD	2002	IYSDYDGAWFAY	2516
214	DYYMD	1489	RVNPYNGGTSYNQKFK G	2003	GTVGFAY	2517
215	SYAMS	1490	SISSGGSTYYPDSVKG	2004	EREWGVFYGSSLD Y	2518
216	SYAMS	1491	TISSGGSYTYYPDSVKG	2005	HDDSSYGYFDY	2519
217	NYAMS	1492	SISSGGTTYYPDSVKG	2006	TMPDV	2520
218	SYGVH	1493	VIWAGGSTNYNSALMS	2007	DTDGYYWAMDY	2521
219	SDHAW	1494	YISYSGSTTYNPSLKS	2008	KWGDY	2522
220	DYEMH	1495	AIDPETGGTAYNQKFKG	2009	NYDYALDY	2523
221	SGYYW	1496	YISYDGSNDYNPSLKN	2010	GGGRG	2524
222	NYYMS	1497	VINSNGGSTYYPDTVKG	2011	QEEIGYAMDY	2525
223	DYFMH	1498	WIDPETDNTIYDPKFQG	2012	SGNMGFTY	2526
105	DYFMH	1499	WIDPETDNTIYDPKFQG	2013	SGNMGFTY	2527
224	SYAMS	1500	TISSGGSYTYYPDSVKG	2014	QGGSSWGAMDY	2528
106	SYAMS	1501	TISSGGSYTYYPDSVKG	2015	QGGSSWGAMDY	2529
225	SYAMS	1502	TISNGGSYTYYPDSVKG	2016	HEITTRFAY	2530
226	SGYYW	1503	YMSYDGSNNYNPSLKN	2017	EAGYFDY	2531
227	TYAMN	1504	RIRSKSNNYATYYADSV KD	2018	QYGYDFDY	2532
228	SYGMS	1505	TISSGGSYTYYPDSVKG	2019	HKGVNWDYFDY	2533
229	DYEMH	1506	AIDPETGGTAYNQKFKG	2020	GDGNYDSWYFDV	2534
230	SYAMS	1507	TISSGGSYTYYPDSVKG	2021	LPVTTVVFDY	2535
231	SYAMS	1508	TISSGGSYTYYPDSVKG	2022	RPVVVPFDY	2536
232	SYGVH	1509	VIWSGGSTDYNAAFIS	2023	GWDADYFDY	2537
233	NYWMH	1510	MIHPNSGSTNYNEKFKS	2024	YDYDDY	2538

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

234	DYYMN	1511	DINPNNGGTSYNQKFKG	2025	SELGLYAMDY	2539
235	GYWIE	1512	EILPGSGSTNYNEKFKG	2026	GRIHYFDY	2540
236	GYWIE	1513	EILPGSGSTNYNEKFKG	2027	GRIHYFDY	2541
237	SYGVH	1514	VIWSGGSTDYNAAFIS	2028	KGYGYDWYFDV	2542
107	SYGVH	1515	VIWSGGSTDYNAAFIS	2029	KGYGYDWYFDV	2543
238	SYWMH	1516	EIDPSDSYTNYNQKFKG	2030	SSYYYYAMDY	2544
108	SYWMH	1517	EIDPSDSYTNYNQKFKG	2031	SSYYYYAMDY	2545
239	SGYYW	1518	YISYDGSNNYNPSLKN	2032	GGGRD	2546
240	SYGVH	1519	VIWSGGSTDYNAAFIS	2033	GGDYDSYAMDY	2547
241	SYWIT	1520	DIYPGSGSTNYNEKFKS	2034	ESVYDGYSWYFDV	2548
242	DYNMN	1521	VINPNYGTTSYNQKFKG	2035	TYDYDDWYFDV	2549
243	SYWMH	1522	EIDPSDSYTNYNQKFKG	2036	SGNYLYAMDY	2550
244	DYNMN	1523	VINPNYGTTSYNQKFKG	2037	EGTSWYFDV	2551
245	SYGVH	1524	VIWRGGSTDYNAAFMS	2038	KGDGYDWYFDV	2552
246	SYGVH	1525	VIWSGGSTDYNAAFIS	2039	EGNYGSSYDAMDY	2553
247	SYWMH	1526	EIDPSDSYTNYNQKFKG	2040	SSNYPYAMDY	2554
248	NTYMH	1527	RIDPANGNTKYAPKFQG	2041	YSGLY	2555
249	SYWMH	1528	NIDPSDSETHYNQKFKD	2042	RGQIYYGYSWFAY	2556
250	DYYMN	1529	DINPNNGGTSYNQKFKG	2043	STVVADWYFDV	2557
251	SYGIS	1530	EIYPRSGNTYYNEKFKG	2044	SGSSYGYFDV	2558
252	SYGVH	1531	VIWSGGSTDYNAAFIS	2045	KGGYDAYAMDY	2559
253	DYNMN	1532	VINPNYGTTSYNQKFKG	2046	EGFITTVVAVDY	2560
254	DYEMH	1533	AIDPETGGTAYNQKFKG	2047	EGNYDAMDY	2561
255	SYWMH	1534	VIDPSDSYTNYNQKFKG	2048	WDYYGVDY	2562
256	GYWMY	1535	AISPGGGSTYYPDSVKG	2049	SLTATHTYEYDY	2563
388	FNAMG	14116	TIARAGATKYADSVKG	14117	RVFDLPNDY	14118
389	SYSMG	9249	AISWSGDETSYADSVKG	9252	DRWWRPAGLQWDY	9255
390	FNAMG	9250	TIARAGATKYADSVKG	9253	RVFDLPNDY	9256
391	VMG		AVRWSSTGIYYTQYADSVKS	9254	DTYNSNPARWDGYDF	9257

	LCDR1		LCDR2		LCDR3	
Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
109	KASQDIN-SYLS	5644	RANRLVD	6154	LHYDEFPPT	6664
110	KASQDIN-SYLN	5645	RANRLVD	6155	LQYDEFPPT	6665
111	KASQDIN-SYLS	5646	RANRLVD	6156	LQYDEFPPT	6666

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

112	KASQDIN-SYLS	5647	RANRLVD	6157	LQYDEFPPT	6667
1	KASQDIN-SYLS	5648	RANRLVD	6158	LQYDEFPPT	6668
113	KASQDIN-SYLS	5649	RANRLVD	6159	PQYVESPPT	6669
114	KASQDIN-SYLS	5650	RANRLVD	6160	LQYDEFPPT	6670
115	KASQDIN-SYLS	5651	RANRLVD	6161	LQYDEFPPT	6671
23	KASQDIN-SYLS	5652	RANRLVD	6162	LQYDEFPPT	6672
116	KASQDIN-SYLS	5653	RANRLVD	6163	LQYDEFPPT	6673
117	KASQDIN-SYLS	5654	RANRLVD	6164	LQYDEFPLT	6674
2	KASQDIN-SYLS	5655	RANRLVD	6165	LQYDEFPPT	6675
118	KASQDIN-GYLS	5656	RANRLVD	6166	LQYDEFPPT	6676
119	KASQDIN-SYLS	5657	RANRLVD	6167	LQYDEFPPT	6677
120	KASQDIN-SYLS	5658	RANRLVD	6168	LQYDEFPPT	6678
121	KASQDIN-SYLN	5659	RANRLVD	6169	LQYDEFPPT	6679
122	KASQDIN-SYLS	5660	RANRLVD	6170	LQYDEFPPT	6680
123	KASQDIN-SYLS	5661	RAKRLVD	6171	LQYDEFPPT	6681
38	ASQDIN-SYLS	5662	RANRLVD	6172	LQYDEFPLT	6682
39	ASQDIN-GYLS	5663	RANRLVD	6173	LQYDEFPPT	6683
40	ASQDIN-SYLS	5664	RAKRLVD	6174	LQYDEFPPT	6684
41	ASQDIN-GYLS	5665	RANRLVD	6175	LQYDEFPPT	6685
42	ASQDIN-SYLS	5666	RAKRLVD	6176	LQYDEFPPT	6686
43	ASQDIN-SYLS	5667	RANRLVD	6177	LHYDEFPPT	6687
44	ASQDIN-SYLS	5668	RANRLVD	6178	LQYDEFPPT	6688
45	ASQDIN-SYLS	5669	RANRLVD	6179	LQYDEFPPT	6689
46	ASQDIN-SYLS	5670	RANRLVD	6180	LQYDEFPPT	6690
47	ASQDIN-SYLS	5671	RANRLVD	6181	LQYDEFPPT	6691

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
48	ASQDIN-SYLS	5672	RANRLVD	6182	LQYDEFPPT	6692
49	ASQDIN-SYLS	5673	RANRLVD	6183	LQYDEFPPT	6693
50	ASQDIN-SYLS	5674	RANRLVD	6184	LQYDEFPPT	6694
51	ASQDIN-SYLS	5675	RANRLVD	6185	LQYDEFPPT	6695
52	ASQDIN-SYLS	5676	RANRLVD	6186	LQYDEFPPT	6696
53	ASQDIN-SYLN	5677	RANRLVD	6187	LQYDEFPPT	6697
54	ASQDIN-SYLS	5678	RANRLVD	6188	LQYDEFPPT	6698
55	ASQDIN-SYLS	5679	RANRLVD	6189	LQYDEFPPT	6699
56	ASQDIN-SYLS	5680	RANRLVD	6190	LQYDEFPPT	6700
3	RASENIYS NLA	5681	AATYLAD	6191	QHFWGTPWT	6701
4	RASENIYS NLA	5682	AATNLAD	6192	QHFWGTPWT	6702
5	RASENIYS NLA	5683	AATNLAD	6193	QHFWGSPWT	6703
6	RASENIYS NLA	5684	AATNLAD	6194	QHFWGTPWT	6704
7	RASENIYS NLA	5685	AATNLAD	6195	QHFWGTPWT	6705
8	RASENIYS NLA	5686	AATNVAD	6196	QHFWGTPWT	6706
9	RAS-DNIYSNLA	5687	AATNLAD	6197	QHFWGTPWT	6707
10	RASENIYS NLA	5688	AATNLAD	6198	QHFWGTPWT	6708
11	RASENIYS NLA	5689	AATNLAD	6199	QHFWGTPWT	6709
12	RASENIYS NLA	5690	AATYLAD	6200	QHFWGTPWT	6710
13	RASENIYS NLA	5691	AATNLAD	6201	QHFWGSPWT	6711
57	AS-DNIYSNLA	5692	AATNLAD	6202	QHFWGTPWT	6712
58	ASENIYSN LA	5693	AATYLAD	6203	QHFWGTPWT	6713
124	KASQDIN-SYLS	5694	RANRLVD	6204	LQYDEFPPT	6714
125	KASEN-VVTYVS	5695	GASNRYT	6205	GQSYSYPFT	6715

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

126	KASEN-VVTYVS	5696	GASNRYT	6206	GQSYSYPFT	6716
127	KASEN-VVTYVS	5697	GASNRYT	6207	GQSYSYPFT	6717
128	KASEN-VVTYVS	5698	GASNRYT	6208	GQSYSYPFT	6718
129	KASEN-VVTYVS	5699	GASNRYT	6209	GQSYSYLIH	6719
130	KASEN-VVTYVS	5700	GASNRYT	6210	GQSYSYLIH	6720
59	ASQDIN-SYLS	5701	RANRLVD	6211	LQYDEFPPT	6721
60	ASEN-VVTYVS	5702	GASNRYT	6212	GQSYSYPFT	6722
61	ASEN-VVTYVS	5703	GASNRYT	6213	GQSYSYPFT	6723
62	ASEN-VVTYVS	5704	GASNRYT	6214	GQSYSYPFT	6724
63	ASEN-VVTYVS	5705	GASNRYT	6215	GQSYSYPFT	6725
131	SASSSVSY MH	5706	DTSKLAS	6216	QQWSSNPLY	6726
132	SASSSVSY MH	5707	DTSKLAS	6217	QQWSSNPHV	6727
133	SASSSVSY MH	5708	DTSKLAS	6218	QQWSSNPLY	6728
14	SASSSVSY MH	5709	DTSKLAS	6219	QQWSSNPLY	6729
134	SASSSVSY MH	5710	DTSKLAS	6220	QQWSSNPLY	6730
64	ASSSVSYM H	5711	DTSKLAS	6221	QQWSSNPLY	6731
65	ASSSVSYM H	5712	DTSKLAS	6222	QQWSSNPLY	6732
135	KASQSVD YD-GDSYMN	5713	AASNLES	6223	QQSNEDPLT	6733
136	KASQSVD YD-GDSYMN	5714	AASNLES	6224	QQSNEDPLT	6734
137	KASQSVD YD-GDSYMN	5715	AASNLQS	6225	QQSNEDPLT	6735
138	KASQSVD YD-GDSYMN	5716	AASNLES	6226	QQSNEDPLT	6736
15	KASQSVD YD-GDSYMN	5717	AASNLES	6227	QQSNEDPLT	6737
66	ASQSVDY DGDSYMN	5718	AASNLES	6228	QQSNEDPLT	6738

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
67	ASQSVDY DGDSYMN	5719	AASNLES	6229	QQSNEDPLT	6739
68	ASQSVDY DGDSYMN	5720	AASNLQS	6230	QQSNEDPLT	6740
69	ASQSVDY DGDSYMN	5721	AASNLES	6231	QQSNEDPLT	6741
24	KASQDINK YIA	5722	YTSTLQP	6232	LQYDNLMYT	6742
139	KASQDINK YIA	5723	YTSTLQP	6233	LQYDILMYT	6743
140	KASQDINK YIA	5724	YTSTLQP	6234	LQYDILMYT	6744
16	QATQDIV-KNLN	5725	YATELAE	6235	LQFYEFPLT	6745
141	QATQDIV-KNLN	5726	YATELAE	6236	LQFYEFPLT	6746
142	QATQDIV-KNLN	5727	YATELAE	6237	LQFYEFPLT	6747
143	QATQDIV-KNLN	5728	YATELAE	6238	LQFYEFPLT	6748
70	ATQDIV-KNLN	5729	YATELAE	6239	LQFYEFPLT	6749
25	RAS-SSVNYMY	5730	YTSNLAP	6240	QQFTSSHTF	6750
144	RAS-SSVNYMY	5731	YTSNLAP	6241	QQFTSSHTF	6751
145	RAS-SSVNYMY	5732	YTSNLAP	6242	QQFTSSHTF	6752
146	KASQSVD YD-GDSYMN	5733	AASNLES	6243	QQSNKDPLT	6753
147	KASQSVD YD-GDSYMN	5734	AASNLES	6244	QQSNEDPLT	6754
148	KASQSVD YD-GDSYMN	5735	AASNLES	6245	QQSNKDPFT	6755
71	ASQSVDY DGDSYMN	5736	AASNLES	6246	QQSNKDPLT	6756
72	ASQSVDY DGDSYMN	5737	AASNLES	6247	QQSNKDPLT	6757
73	ASQSVDY DGDSYMN	5738	AASNLES	6248	QQSNEDPLT	6758
74	ASQSVDY DGDSYMN	5739	AASNLES	6249	QQSNKDPFT	6759
75	ASQSVDY DGDSYMN	5740	AASNLES	6250	QQSNKDPFT	6760
76	ASQSVDY DGDSYMN	5741	AASNLES	6251	QQSNKDPFT	6761
149	RASQSVST SSYSYMH	5742	YASNLES	6252	QHSWEIPLT	6762

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
150	RASQSVST SSYSYMH	5743	YASNLES	6253	QHSWEIPLT	6763
26	RASQSVST SSYSYMH	5744	YASNLES	6254	QHSWEIPLT	6764
77	ASQSVSTS SYSYMH	5745	YASNLES	6255	QHSWEIPLT	6765
78	ASQSVSTS SYSYMH	5746	YASNLES	6256	QHSWEIPLT	6766
79	ASQSVSTS SYSYMH	5747	YASNLES	6257	QHSWEIPLT	6767
151	RASQDISN YLN	5748	STSRLHS	6258	QQGNTLPWT	6768
152	RASQDISN YLN	5749	STSRLHS	6259	QQGNALPWT	6769
153	RASQDISN YLN	5750	STSRLHS	6260	QQGNTLPWT	6770
154	RSST-GAVTTSN YAN	5751	GTSNRAP	6261	ALWYSTHYV	6771
155	RSST-GAVTTSN YAN	5752	GTSNRAP	6262	ALWYSTHYV	6772
156	RSST-GAVTTSN YAN	5753	GTSNRAP	6263	ALWYSTHYV	6773
80	SST-GAVTTSN YAN	5754	GTSNRAP	6264	ALWYSTHYV	6774
81	SST-GAVTTSN YAN	5755	GTSNRAP	6265	ALWYSTHYV	6775
82	SST-GAVTTSN YAN	5756	GTSNRAP	6266	ALWYSTHYV	6776
17	RASENIYS NLA	5757	AATNLAD	6267	QHFWGTPWT	6777
157	IT-STDIDDD MN	5758	EGNTLRP	6268	LQSDNMPFT	6778
27	MTSIDIDD DMN	5759	EGNTLRP	6269	LQSDNMPFT	6779
158	RSST-GAVTTSN YAN	5760	GTSNRAP	6270	ALWYSTHYF	6780
159	KASQSVD YD-GDSYMN	5761	AASNLES	6271	QQSNEDPLT	6781
83	ASQSVDY DGDSYMN	5762	AASNLES	6272	QQSNEDPLT	6782
160	KSS-QSLLYSTN QKNYLA	5763	WASTRES	6273	QQYYSYPPW	6783

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
28	RSST-GAVTTSN YAN	5764	GTSNRAP	6274	ALWYSTHWV	6784
84	SST-GAVTTSN YAN	5765	GTSNRAP	6275	ALWYSTHWV	6785
29	RSST-GAVTTSN YAN	5766	GTNNRAP	6276	ALWYSNHLF	6786
161	KASQSVD YD-GDSYMN	5767	AASNLES	6277	QQSNEDPPT	6787
30	RSST-GAVTTSN YAN	5768	GTNNRAP	6278	ALWYSNHWV	6788
162	RASENIYY SLA	5769	NANSLED	6279	KQAYDVPYT	6789
18	KSS-QSLLNSSN QKNYLA	5770	FASTRES	6280	QQHYSTPLT	6790
85	SSQSLLNS SNQKNYL A	5771	FASTRES	6281	QQHYSTPLT	6791
163	RASQDISN YLN	5772	YTSRLHS	6282	QQDNTLPRT	6792
86	ASQDISNY LN	5773	YTSRLHS	6283	QQDNTLPRT	6793
164	RASQDISN YLN	5774	YTSRLHS	6284	QQGNTLPWT	6794
165	RASENIYS NLA	5775	AATNLAD	6285	QHFWGTPWT	6795
31	QATQDIV-KNLN	5776	YATELAE	6286	LQFYEFPYT	6796
87	ATQDIV-KNLN	5777	YATELAE	6287	LQFYEFPYT	6797
32	SASSSVSY MY	5778	DTSNLAS	6288	QQWSSYPLT	6798
166	KASQSVD YD-GDSYMN	5779	AASNLES	6289	QQSNEDPWT	6799
88	ASQSVDY DGDSYMN	5780	AASNLES	6290	QQSNEDPWT	6800
167	RSST-GAVTTSN YAN	5781	GTNNRAP	6291	ALWYSNHWV	6801
168	QATQDIV-KNLN	5782	YATELAE	6292	LQFYEFPLT	6802
89	ATQDIV-KNLN	5783	YATELAE	6293	LQFYEFPLT	6803
169	TLSSQHST YTIEWYQ Q	5784	GSHSTGD	6294	GVGDTIKEQ	6804

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

170	KASQSVD YD-GDSYMN	5785	AASNLES	6295	QQSNEDPPT	6805
171	RASQDIGI SLN	5786	ATSSLDS	6296	LQYASSPYT	6806
172	RASENIYS YL	5787	YNAKNLA	6297	QHFWGTPYT	6807
19	SASSSVSY MH	5788	STSNLAS	6298	QQRSSFPYT	6808
90	ASSSVSYM H	5789	STSNLAS	6299	QQRSSFPYT	6809
173	KASQDIN-SYLS	5790	RANRLVD	6300	LQYDEFPLT	6810
91	ASQDIN-SYLS	5791	RANRLVD	6301	LQYDEFPLT	6811
174	RASQDIH GYLN	5792	ETSNLDS	6302	LQYASSPLT	6812
92	ASQDIHGY LN	5793	ETSNLDS	6303	LQYASSPLT	6813
175	KASQDVG TAVA	5794	WASTRHT	6304	QQYSSYPFT	6814
93	ASQDVG-TAVA	5795	WASTRHT	6305	QQYSSYPFT	6815
20	KASQSVS NDVA	5796	YASNRYT	6306	QQDYTSLPT	6816
94	ASQSVSN-DVA	5797	YASNRYT	6307	QQDYTSLPT	6817
176	RSST-GAVTTSN YAN	5798	GTNNRAP	6308	ALWYSNHLV	6818
177	KASQSVD YD-GDSYMN	5799	VASNLES	6309	QQSHEDPRT	6819
178	SASSSVSY MH	5800	DTSKLAS	6310	FQGSGYPLT	6820
33	IT-STDIDDD MN	5801	EGNTLRP	6311	LQSDNMPLM	6821
179	RASENIYS NLA	5802	AATNLAD	6312	QHFWGTPYT	6822
34	SASSSVSY MH	5803	STSNLAS	6313	QQRSTYPTF	6823
180	IT-STDIDDD MN	5804	EGNTLRP	6314	LQSDNMPLT	6824
95	TSTDIDDD MN	5805	EGNTLRP	6315	LQSDNMPLT	6825
181	RASQDISN YLN	5806	YTSRLHS	6316	QQGNTLPFT	6826
96	ASQDISNY LN	5807	YTSRLHS	6317	QQGNTLPFT	6827

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
182	SAS-QGISNYLN	5808	YTSSLHS	6318	QQYSKLPYT	6828
97	ASQGISNY LN	5809	YTSSLHS	6319	QQYSKLPYT	6829
183	RASQSISD YLH	5810	YASQSIS	6320	QNGHSFPWT	6830
98	ASQSISDY LH	5811	YASQSIS	6321	QNGHSFPWT	6831
35	QATQDIV-KNLN	5812	YATELAE	6322	LQFYEFPYT	6832
99	ATQDIV-KNLN	5813	YATELAE	6323	LQFYEFPYT	6833
100	ATQDIV-KNLN	5814	YATELAE	6324	LQFYEFPYT	6834
36	RASGNIH NYLA	5815	NAKTLAD	6325	QHFWSTPWT	6835
101	AS-GNIHNYLA	5816	NAKTLAD	6326	QHFWSTPWT	6836
184	RSST-GAVTTSN YAN	5817	GTSYRAP	6327	ALWYSTHYV	6837
102	SST-GAVTTSN YAN	5818	GTSYRAP	6328	ALWYSTHYV	6838
185	RASQEIS-GYLS	5819	AASTLDS	6329	LQYASYPWT	6839
103	ASQEIS-GYLS	5820	AASTLDS	6330	LQYASYPWT	6840
186	KASQSVD YD-GDSYMN	5821	AASNLES	6331	QQSNEDPLP	6841
21	HASQNIN VWLS	5822	KASNLHT	6332	QQGQSYPLT	6842
187	SASSSVSY MH	5823	DTSKLAS	6333	QQWSSNPLT	6843
188	RASGNIH NYLA	5824	NAKTLAD	6334	QHFWSTPWT	6844
189	RASENIYS YLA	5825	NAKTLAE	6335	QHHYGTPFT	6845
190	RAS-SSVNYMY	5826	YTSNLAP	6336	QQFTSSLTF	6846
191	RSST-GAVTTSN YAN	5827	STNNRAP	6337	TLWYSNHWV	6847
192	KASQDVG TAVA	5828	WASTRHT	6338	QQYSSYPFT	6848
193	KASQDIN-SYLS	5829	RANRLVD	6339	LQYDEFPPT	6849
194	RASENIYS NLA	5830	AATNLAD	6340	QHFWGTPFT	6850
195	RASQSISD YLH	5831	YASQSIS	6341	QNGHSFPYT	6851

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

104	ASQSISDY LH	5832	YASQSIS	6342	QNGHSFPYT	6852
37	RASENIYS NLA	5833	AATNLAD	6343	QHFWGTPYT	6853
196	RSST-GAVTTSN YAN	5834	GTNNRAP	6344	ALWYSNHWV	6854
197	IT-STDIDDD MN	5835	EGNTLRP	6345	LQSDNLPLT	6855
198	SASSSVSY MH	5836	STSNLAS	6346	QQRSSYPPT	6856
199	SASSSVSY MH	5837	DTSKLAS	6347	QQWSSNPLT	6857
22	IT-STDIDDD MN	5838	EGNSLRP	6348	LQSDNLPLT	6858
200	SASSSVSS SYLY	5839	STSNLAS	6349	HQWSSYPPT	6859
201	TLSSQHST YTIEWYQ Q	5840	GSHSTGD	6350	GVGDTIKEQ	6860
202	RSSQSIVH SNGNTYLE	5841	KVSNRFS	6351	FQGSHVPWT	6861
203	KASQDIN-SYLS	5842	RANRLVD	6352	LQYDEFPPT	6862
204	KSS-QSLLDSDG KTYLN	5843	LVSKLDS	6353	WQGTHFPWT	6863
205	SASSSVSS SYLY	5844	STSNLAS	6354	HQWSSYPPT	6864
206	RSSQSIVY SNGNTYLE	5845	KVSNRFS	6355	FQGSHVPPT	6865
207	RASENI-YNNLA	5846	AATNLAD	6356	QHFWGTPWT	6866
208	SASSSVSS SYLY	5847	STSNLAS	6357	HQWSSYPPT	6867
209	SASSSVSY MY	5848	DTSNLAS	6358	QQWSTYPPI	6868
210	SASSSVSY MY	5849	DTSNLAS	6359	QQWSSYPFT	6869
211	RAS-SSVSYMH	5850	ATSNLAS	6360	QQWSSNPYT	6870
212	RASENIYS NLA	5851	AATNLAD	6361	QHFWGTPWT	6871
213	KASQDVS TAVA	5852	WASTRHT	6362	QQHYSTPWT	6872
214	KSS-QSLLDSDG KTYLN	5853	LVSKLDS	6363	WQGTHFPWT	6873

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
215	RSSQSIVH SNGNTYLE	5854	KVSNRFS	6364	FQGSHVPPT	6874
216	IT- STDIDDD MN	5855	EGNTLRP	6365	LQSDNMPLT	6875
217	KASQDINK YIA	5856	YTSTLQP	6366	LQYDNLYMY	6876
218	HASQNIN VWLS	5857	KASNLHT	6367	QQGQSYPYT	6877
219	SASSSVSY MY	5858	LTSNLAS	6368	QQWSSNPPT	6878
220	KSS- QSLLDSDG KTYLN	5859	LVSKLDS	6369	WQGTHFPWT	6879
221	SASLSVSD MY	5860	DTSNLAS	6370	QQWSSYPFT	6880
222	SASSSVSS SYLY	5861	STSNLAS	6371	HQWSSYPPT	6881
223	RAS- SSVNYMY	5862	YTSNLAP	6372	QQFTSSPST	6882
105	ASSSVNY MY	5863	YTSNLAP	6373	QQFTSSPST	6883
224	IT- NTDIDDD MN	5864	EGNTLRP	6374	LQSDNLPLT	6884
106	TNTDIDDD MN	5865	EGNTLRP	6375	LQSDNLPLT	6885
225	KASQSVD YD- GDSYMN	5866	AASNLES	6376	QQSNEDPWT	6886
226	RASQSISN NLH	5867	YASQSIS	6377	QQSNSWPFT	6887
227	SASSSVSS SYLY	5868	STSNLAS	6378	HQWSSYPPT	6888
228	IT- STDIDDD MN	5869	EGNTLRP	6379	LQSDNMPLT	6889
229	KSS- QSLLDSDG KTYLH	5870	LVSKLDS	6380	WQGTHFPWT	6890
230	KASQDINK YIA	5871	YTSTLQP	6381	LQYDNLRTF	6891
231	RSST- GAVTTSN YAN	5872	GTNNRAP	6382	VLWYSNHLV	6892
232	KASEN- VGTYVS	5873	GASNRYT	6383	GQSYSYPPT	6893
233	RASET- VDSYGYSF MH	5874	RASNLES	6384	QQSNEDPRT	6894
234	KSS- QSLLYSTN QKNYLA	5875	WASTRES	6385	QQYYSYRTF	6895

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
235	KSS-QSLLDSDG KTYLN	5876	LVSKLDS	6386	WQGTHFPFT	6896
236	KASQSVD YD-GDSYMN	5877	AASNLES	6387	QQSNEDPFT	6897
237	TLSSQHST YTIEWYQ Q	5878	GSHSTGD	6388	GVGDTIKEQ	6898
107	LSSQHSTY TIEWYQQ	5879	GSHSTGD	6389	GVGDTIKEQFVYV	6899
238	SASSSVSY MH	5880	DTSKLAS	6390	QQWSSNPLT	6900
108	ASSSVSYM H	5881	DTSKLAS	6391	QQWSSNPLT	6901
239	QATQDIV-KNLN	5882	YATELAE	6392	LQFYEFPLT	6902
240	RAS-ESVDNY-GISFMN	5883	AASNQGS	6393	QQSKEVPPT	6903
241	RASQSISD YLH	5884	YASQSIS	6394	QNGHSFPLT	6904
242	RSSQSIVH SNGDTYLE	5885	KVSNRFS	6395	FQGSHVPLT	6905
243	SASSSVSY MH	5886	DTSKLAS	6396	QQWSSNPLT	6906
244	TASESLYS-SKHKVHYL A	5887	GASNRYI	6397	AQFYSYPYT	6907
245	TLSSQHST YTIEWYQ Q	5888	GSHSTGD	6398	GVGDTIKEQ	6908
246	RASQSVST SSYSYMH	5889	YASNLES	6399	QHSWEIPLT	6909
247	SASSSVSY MH	5890	DTSKLAS	6400	QQWSSNPLT	6910
248	KASDHIN NWLA	5891	GATSLET	6401	QQYWSTPLT	6911
249	RASENIYS YLA	5892	NAKTLAE	6402	QHHYGTPYT	6912
250	RASENIYS YLA	5893	NAKTLAE	6403	QHHYGTPPT	6913
251	SASSSVSY MH	5894	DTSKLAS	6404	QQWSSNPPT	6914
252	RAS-ESVDNY-GISFMN	5895	AASNQGS	6405	QQSKEVPPT	6915
253	TASESLYS-SKHKVHYL A	5896	GASNRYI	6406	AQFYSYPYT	6916
254	RSST-GAVTTSN YAN	5897	GTSNRAP	6407	ALWYSTHYV	6917

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

255	SASSSI-SYMH	5898	DTSKLAS	6408	HQRSSYPTF	6918

	HCDR1		HCDR2		HCDR3	
Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
265	SYALS	9968	IINPSGGTNYAQKFQG	10230	DLGDPGMDV	10492
266	NYAIS	9969	IIDPSGGSTTYAQKFQG	10231	DLGDMGMDV	10493
267	NYAFS	9970	IINPSGGSTSYAQKFQG	10232	DVGDRGMDV	10494
263	SYALS	9971	IINPSGGTNYAQKFQG	10233	DLGDPGMDV	10495
268	GYYMH	9972	WIDPNGGGTQYAQKFQG	10234	DIVHDGTEYFQH	10496
269	SYYMH	9973	IINPSGGSTSYAQKFQG	10235	DIVHDGTEYFQH	10497
270	SYAIS	9974	IINPSGGSTNYAQKFQG	10236	EGRDHDAFDI	10498
271	DYGIS	9975	IINPSGGSTSYAQKFQG	10237	EGRSHDAFDI	10499
272	GYYMH	9976	WMNPHSGDTGYAQKFQG	10238	WVGTTEYYYYYYMDV	10500
273	DYYLH	9977	IIDPSGGSTSIAQKFQG	10239	TAYYDFWSGYSMDV	10501
274	SHYMH	9978	IIDPSGGSTSYAQEFQG	10240	DMDNWNTGYYYYMD V	10502
275	SYAIN	9979	WVNPNSGDTAYAQKFQG	10241	DQRGGDAWDV	10503
276	NYAIS	9980	IITPSGGSTTYAHKFQG	10242	DTAGHFDI	10504
277	NDVIN	9981	WMNPNSGNTGYAQKFQG	10243	DNPDLDGMDV	10505
278	SYAIN	9982	WISAYNGNTNYAQKFQG	10244	DLVGHFDY	10506
279	SYAIS	9983	WINPNSGGTNYAQKFQG	10245	DGYSGSYSD	10507
264	SYAIS	9984	WINPNSGGTNYAQKFQG	10246	DGYSGSYSD	10508
257	SYAIS	9985	WINPNSGGTNYAQKFQG	10247	DGYSGSYSD	10509
280	SHAIS	9986	IINPSGGSTSYAQKFQG	10248	DQGSSGTFDY	10510
281	SYAIS	9987	WINPNSGGTNYAQKFQG	10249	DSTDVIDY	10511
282	SYDIN	9988	WINPNSGDTKYAQNFQG	10250	DGGTVTPTEEYYYYG-MDV	10512
283	SYAIS	9989	WISVYNGNTNYAQNLQG	10251	LDDLDY	10513
284	SYYIH	9990	WINPNNGGTHYAQKFQG	10252	DMVRDSAEYFQH	10514
285	TSYIH	9991	MINPSGGTTTYAQKFQG	10253	DSSGYPIDY	10515
286	SYDIN	9992	GIIPLSGAPNYAHKFQG	10254	GALYNWNDGWFDP	10516
258	SYDIN	9993	GIIPLSGAPNYAHKFQG	10255	GALYNWNDGWFDP	10517
287	SWYMS	9994	GIWYEGSNKYYADSVKG	10256	LGTASLPYFDY	10518
288	GYYMH	9995	WINPNRGDTKYAQKFQG	10257	ESGDGFDP	10519
289	NYYIH	9996	WMNPNSGNTGYAQKFQG	10258	DWPNWFDP	10520
290	DNYIH	9997	WIRSDNGETSYAQKFQG	10259	EVQLVGFDY	10521
291	DHHVH	9998	GIIPIFGTANYAQKFQG	10260	GSSWYLHFQH	10522
260	DHHVH	9999	GIIPIFGTANYAQKFQG	10261	GSSWYLHFQH	10523

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

292	SYAIY	10000	GIIPIFGTTNYAQKFQG	10262	GVDRYNWNDAFDY	10524
293	DYYMH	10001	WIHSNSGGTHSAQKFQG	10263	ESSGYDSSLDY	10525
294	SYGIS	10002	WINPNSGDTDYAQKFQG	10264	DPRLDSSDPGY	10526
295	NYGIN	10003	WISAYNGNTNYAQKFQG	10265	GGMDV	10527
296	RYGIA	10004	ISYPSDGSTSSAQKLQG	10266	DRLGDLDY	10528
261	RYGIA	10005	ISYPSDGSTSSAQKLQG	10267	DRLGDLDY	10529
297	SYAIS	10006	WMNPNSGNTGYAQKFQG	10268	DSIVGGYPFDY	10530
298	SYDIN	10007	TITPIFGTTDYAQKFQG	10269	EGYSSSWHDDAFDI	10531
299	NYAIS	10008	IIDPSGGSTSYAQKFQG	10270	DLGDYGLDS	10532
300	GYYMH	10009	WMNPNSGDTGYAQRFQG	10271	GGSDSSGYYYEGYFQH	10533
301	SYAIS	10010	YMSPNSGNTGYAQKFQG	10272	DKGGYYDSSGYYWY	10534
302	SYEMH	10011	AISSNGGSTYYADSVKG	10273	VGDGDGYNPDFDY	10535
303	SYGIS	10012	WIDPTSGATDTAHKFQG	10274	DPIVATEVDY	10536
304	SYAIS	10013	WMSPNSGNTGYAQKFQG	10275	DSGAFDI	10537
305	NYAIS	10014	WMNPNSGNTGYAQKFQG	10276	EGLLDAFDI	10538
306	RYGIT	10015	WMNPYDGNTGYAQKFQG	10277	GGRHHDAFDI	10539
307	SYAIS	10016	IINPSGDGTNYAQKFQG	10278	DISNDAFDI	10540
308	GHYMH	10017	WISAYNGDTNYAQKFQG	10279	GSSWDDAFDI	10541
309	NHYTS	10018	AIGAGGGTYYADSVKG	10280	EGWNDDVFDI	10542
310	SYAIS	10019	IINPSAGTTYYAERFQG	10281	DGNFGAFDI	10543
311	TYAIT	10020	EIIPIFGTANYAQKFQG	10282	DKSGWNYGSGSYN-DAFDI	10544
312	GYYMH	10021	WMNPNSGKTEYAQKFQG	10283	DGGLDFDY	10545
313	TYYIH	10022	WMNPNTGDTGSAQKFQG	10284	DPAVTPDAFDI	10546
314	SYAIS	10023	IIDPSGGGTSYAQKLQG	10285	SLYYYGMDV	10547
315	SSAIS	10024	GIIPIFGTANYAQKFQG	10286	EDDILPPRAFDI	10548
316	DYAMH	10025	GISGGGGVTYYADSVKG	10287	VYSSGWLDAFDI	10549
317	SYAIS	10026	WISGYNGNTNYAQKFQG	10288	SDVSPDAFDI	10550
318	IYAIT	10027	WVNPNSGNTGYSQKFQG	10289	PTSSSDDAFDI	10551
319	SYAIS	10028	WINPNSGGTNYAQKFQG	10290	ASRGDDAFDI	10552
320	SDDIN	10029	IINPSGGSTSYAQKFQG	10291	ERYEGGY-SSGPGNYYYGMDV	10553
321	NYAIS	10030	WMNPNSGNTGYAQKFQG	10292	DDDYGDYPV	10554
322	DHAIN	10031	WMNPKIGNTGYAQKFQG	10293	DSSGYDAFDI	10555
323	SYDIN	10032	RINPGTGGTDYAHKFQG	10294	ETPSDYYDSSGYYYN-DAFDI	10556
324	SYAIS	10033	IIIPSGGTNYAQTFQG	10295	DLGTTFDI	10557
325	AYYLH	10034	WINPDNDNAYYAQKFQG	10296	DIAVAALAYGMDV	10558

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
326	SYAMS	10035	VISYDGSDQYYADSVKG	10297	QSLYYYYGMDV	10559
327	DYYVH	10036	WISTFTGNTDYAQNFQG	10298	DAPLAAAGTDYYYG-MDV	10560
262	DYYVH	10037	WISTFTGNTDYAQNFQG	10299	DAPLAAAGTDYYYG-MDV	10561
328	SYAMS	10038	FISDDGITKYYADSVKG	10300	DDSSGYGGMDV	10562
329	SYAMH	10039	VISYDGGDKYYADSVKG	10301	GSLVLGYYYMDV	10563
330	NYYIH	10040	WINPNTGGTDYAQKFQG	10302	GGGGSYYDAFDV	10564
331	SYAIS	10041	RINPNSGNTGYAQKFQG	10303	DIGEGYSMDV	10565
332	NHYTS	10042	VISYDGSNKYYADSVKG	10304	EEKYSSS-WYVGVDAFDI	10566
333	SSAMH	10043	SISGSGDNAYYADSVKG	10305	DQEDYYYDSSGYGMDV	10567
334	SHAIS	10044	GIIPIFGTANYAQKFQG	10306	GDWGIVVVPAAI-GAFDI	10568
335	AYYMH	10045	RISPVFGSTTYAQRFQG	10307	DLGYYDSSGYRYDAFDI	10569
336	SYDIN	10046	GISPMFGTANYAQKFQG	10308	DGWYYGMDV	10570
337	SYGIS	10047	WINPNSGGTKYAQKFQG	10309	GEAGNLDWYFDL	10571
338	NYGIS	10048	WINPNNGDTKYAQKFQG	10310	EDVWYFDL	10572
339	TYGIS	10049	WISTYDGKTNYAQKLQG	10311	HLGGDWYFDL	10573
340	GYYMH	10050	WINPNTGATYYAQKFQG	10312	QHGDYDWYFDL	10574
341	TYYVH	10051	WINPNSGNTGYAQKFQG	10313	DSGRH	10575
342	SYGIS	10052	RIIPMLGIANYAQKFQG	10314	EEVAGANWFDP	10576
343	SYAMN	10053	IINPSGGSTSYARKFQG	10315	EGDYGSGEFDY	10577
344	SSYMH	10054	WMNPRSGNTGYAQKFQG	10316	ERDDYGDYGWLDY	10578
345	GYYMH	10055	IINPSGGSTSYAQKFQG	10317	DLYDSSGY-WHYYYYMDV	10579
346	SYAFS	10056	WINPNSGGTNYAQKFQG	10318	FSGYDYVDY	10580
347	SYAIS	10057	IINPNGGNTSYAQKFQG	10319	DVGEDFDL	10581
348	SYYIH	10058	VINPADGDTTYAQMFQG	10320	DFDWLFAMDV	10582
349	NYALN	10059	RINPNGGTTYYAKNFQG	10321	HGDHGFYV	10583
350	SYAIS	10060	MINPNVGSATYAQRFQG	10322	EDSGTSWFDP	10584
351	SYYMH	10061	IINPSDGSTSYAQRFQG	10323	DDRGSNYYYGMDV	10585
352	AYYVH	10062	WMNPNSGTTGYAQKFQG	10324	DSSDYYGDYRADAFDI	10586
353	SYDIN	10063	VISPSGDATLYAQSFQG	10325	GLDH	10587
354	DYGMH	10064	AIGGIGDSTYYADSVKG	10326	MNYGDSNYYYYYG-MDV	10588
355	SYDIS	10065	MISPSDGSTTYAPKFQG	10327	GAVGFDY	10589
356	SYGIS	10066	WINTYSGYTDYAHKFQG	10328	DDFLSFGY	10590
357	DYYIH	10067	GIIPYFGTANYAQKFQG	10329	SISGSYVLDAFDI	10591
358	SYGIS	10068	GIIPIFGTANYAQKFQG	10330	DWGYGDYADDAFDI	10592

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

359	NNDIN	10069	WINPIYGSANYAQNFQG	10331	DWRGFDY	10593
360	EYAIH	10070	RMNPHNGDTGYAQKFQG	10332	EGDYLGYPIDC	10594
361	DYSMS	10071	AIWQDGNVKFYADSVKG	10333	DGNSGYVF	10595
362	TYYMH	10072	WINPNTGDTAYAQKIQG	10334	TAEAVAGLPAFDY	10596
363	NYAID	10073	GIIPLFGTTTYAQKFQG	10335	VTLYGDYDY	10597
364	THWMH	10074	MINPSDGVTYYAQTFQG	10336	EYYGEGFDY	10598
259	THWMH	10075	MINPSDGVTYYAQTFQG	10337	EYYGEGFDY	10599
365	SYAIS	10076	IINPSGGSTSNAQKFQG	10338	DLGDTAMDG	10600
366	SYYLH	10077	IITPSGGSTTYAHKFQG	10339	DGGLASFDY	10601
367	SYAIS	10078	WMNPNSGNTGYAQKFQG	10340	GGGWAMTDAFDI	10602
368	DYGMS	10079	LIYSGGDTYYADSVKG	10341	KEYYYDSSGYLRLFDY	10603
369	DYYMH	10080	GINPIFGTSNYAQKFQG	10342	DISGYDYYYYGMDV	10604
370	NYAFS	10081	MIDPSDGTIAYAQKFQG	10343	SDYDFWSGLGGYFDY	10605
371	SYAIS	10082	TIDPNSGGTMFAQKFQG	10344	DSAEWELGGSFDY	10606
372	NHYTS	10083	SIGVNGDTYYLDSVKG	10345	EGLVFSGRGHWYFDL	10607
373	NYAIS	10084	RINPNGGNTSNAQKFQG	10346	DYEDADFDG	10608
374	DHHVH	10085	WMNPDSGNTGYAQRFQG	10347	DSTSGVDY	10609
375	SYAMS	10086	VISYDGHDQFYADSVKG	10348	GEQQLEGFYYYYGMDV	10610
376	SYWMH	10087	VISYDGSKEYYADSVKG	10349	DYGDYGTYDY	10611
377	SYWMH	10088	GISGGGDDTYYADSVKG	10350	EPLAYCGGDCPGGFDY	10612
378	DHYMD	10089	AIGTGGDTYYADSVKG	10351	HEDTAIFLDY	10613
379	SYYMH	10090	MISPSDGSTTYAPKFQG	10352	DGYDAWSYGMDV	10614
380	GYYMH	10091	WMNPNSGNTGYAQKFQG	10353	DGVTGTDY	10615
381	SYVLH	10092	AISGAGDSTYYADSVKG	10354	EPTTVTDDWYFDL	10616
382	SHWMH	10093	AISGNGDNSYYADSVKG	10355	DRAPEYFDL	10617
383	SYAIS	10094	WINPNSGGTNYAQKFQG	10356	DDYGDYGGGMDV	10618
384	DYYMH	10095	WMNPNSGHTGYAEKFQG	10357	DTSPRYGDGFFDY	10619
385	SYWMH	10096	VTSYDGSNKYYADSVKG	10358	ESGFSAEYFQH	10620
386	SYAIS	10097	IINPSGGSTSYAQKFQG	10359	ATGLYCSGSCFDY	10621

	LCDR1		LCDR2		LCDR3	
Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
265	RASQDISN YLN	12194	DASNLET	12454	QQSYSTPLT	12714
266	RASQSIS- SYLN	12195	AASSLQS	12455	QQSYSTPLT	12715
267	QASQDIS NYLN	12196	KASSLET	12456	QQSFSSPLT	12716
263	RASQDISN YLN	12197	DASNLET	12457	QQSYSTPLT	12717

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

268	RASQNVN TWLA	12198	EASSLQS	12458	QQANSFPFT	12718
269	RASQSISD WLA	12199	AASSLQS	12459	AQHNHYPYT	12719
270	KSS-QSVLSSSY NKNYLA	12200	WASTRES	12460	QQYYTTPFT	12720
271	KSS-QSVLSSSY NKNYLA	12201	WASTRAS	12461	QQYYSTPFT	12721
272	RAS-QAIRDDLG	12202	DASHLEA	12462	QQANSFPIT	12722
273	RASQGVG NDLA	12203	AASTLQT	12463	QQASSFPLT	12723
274	RASQI-IGTNLA	12204	AASSLQS	12464	QQSYTFPVT	12724
275	RASQSIST-WLA	12205	DASSLES	12465	QQSYSTPFT	12725
276	KSS-QSVLSSS-NNKNYLA	12206	WASTRES	12466	QQYYGSPLT	12726
277	KSS-QSVLSSSY NKNYLA	12207	WASTRES	12467	QQYYSSPPT	12727
278	KSS-QSVLSSSY NKNYLA	12208	WASTRES	12468	QQYYSSPPT	12728
279	KSS-QSVLSSSY NKNYLA	12209	WASTRES	12469	QQYYSTPWT	12729
264	KSS-QSVLSSSY NKNYLA	12210	WASTRES	12470	QQYYSTPWT	12730
257	KSS-QSVLSSSY NKNYLA	12211	WASTRES	12471	QQYYSTPWT	12731
280	KSS-QSVLSSSY NKNYLA	12212	WASTRAS	12472	QQYYGSPPT	12732
281	QASQDIR NYLN	12213	DASTLQS	12473	QQAYSFPWT	12733
282	QASQDIS NYLN	12214	NASNLET	12474	QQLNSYPFT	12734
283	QASQSIST WLA	12215	AASTLRS	12475	LQHYTYPLT	12735
284	RASEDI-STYLA	12216	AASTLQS	12476	QQSHTIPWT	12736
285	RASH-HISDFLN	12217	AASTLQS	12477	QQSYSSPYT	12737
286	RASQDIG DYLA	12218	DASSLQS	12478	QQANSFPLT	12738

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
258	RASQDIG DYLA	12219	DASSLQS	12479	QQANSFPLT	12739
287	RASQDIRS YLA	12220	AASSLQS	12480	QQSYTAPPT	12740
288	RASQDIS-NNLN	12221	AASTLQS	12481	LQHNTYPLT	12741
289	RASQDIS-NWLA	12222	DASSLQS	12482	QQAISFPLT	12742
290	RASQGIA NYLA	12223	AASSLQS	12483	QQADSFPLT	12743
291	RASQGIAS YLA	12224	AASTLQP	12484	QQFDSYPIT	12744
260	RASQGIAS YLA	12225	AASTLQP	12485	QQFDSYPIT	12745
292	RASQGISN YLA	12226	AASRLQS	12486	QQSSIIPFT	12746
293	RASQGISN YLA	12227	AASTLQS	12487	QQAYSFPYT	12747
294	RASQSIGR WLA	12228	DASNLET	12488	QQSYSTPRT	12748
295	RASQSINS WLA	12229	DTSSLQS	12489	QQTYSTPYT	12749
296	RASQSISS WLA	12230	AASTLQS	12490	QQGYSTPYI	12750
261	RASQSISS WLA	12231	AASTLQS	12491	QQGYSTPYI	12751
297	RASQSIS-SYLN	12232	AASSLQS	12492	QQTDSIPIT	12752
298	RASQSIS-SYLN	12233	AASTLQS	12493	QQSYSIPYT	12753
299	RASQTIRS YLN	12234	KASSLES	12494	QQTYTIPIT	12754
300	RASQTISN WLA	12235	AASTLQS	12495	QQANSFPPT	12755
301	RASQYIGS YLN	12236	DASNLET	12496	QQVDSYPLT	12756
302	RSSQSLLH SNGYNYL D	12237	LGSNRAS	12497	MQGTHWPPT	12757
303	RSSQSLLH SNGYNYL D	12238	FGSNRAS	12498	MQALQAPVS	12758
304	KSS-QSVLSSSY NKNYLA	12239	WASSRQS	12499	QQYYSTPLT	12759
305	KSS-QSVSSSSY NKNYLA	12240	WASVRES	12500	QQYYSTPIT	12760
306	KSTQNVLS SSNNN-SYLA	12241	WASTRES	12501	QQYYSTPFT	12761

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

307	QASQDIG NYLN	12242	AASSLQS	12502	QQTYNTPLT	12762
308	QASQDIS NYLN	12243	EASTLQS	12503	QQSYSTPFT	12763
309	QASQDIST WLA	12244	RASTLES	12504	QQSYSIPLT	12764
310	RASQNIN- NYLN	12245	AASRLQS	12505	QQSYSAPVT	12765
311	RASQNINT WLA	12246	AASSLQS	12506	QQAYSFPFT	12766
312	RASQRIGN YLN	12247	AASSLQS	12507	QQSYSTPLT	12767
313	RASQSIST YLN	12248	AASTLQS	12508	QQSYRTVT	12768
314	RASQSVG SYLA	12249	GASTRAT	12509	QQYDSSSQT	12769
315	RASRSVST YLA	12250	GASTRAT	12510	QQYDGSPYT	12770
316	RSSQSLLH SNGYNYL D	12251	DASNLET	12511	MQALQTPPA	12771
317	KSS- QSVLSSSY NKNFLA	12252	WASTRES	12512	QQYYSAPPT	12772
318	KSS- QSVLSSSY NKNFLA	12253	WASTRES	12513	QQYYSDPIT	12773
319	KSS- QSVLSSSY NKNYLA	12254	WASARES	12514	QQYYSIPIA	12774
320	KSS- QSVLSSSY NKNYLA	12255	WASTRDS	12515	QQYYSIPYT	12775
321	KSS- QSVLSTSY NKNYLA	12256	WASTRAS	12516	QQYYTTPPT	12776
322	KSS- QSVLSTSY NRNFLA	12257	WASTRQS	12517	QQYYSTPYT	12777
323	KSS- QSVLYSSN NKNYLA	12258	WASTRES	12518	QQYYSTPLT	12778
324	QASQDIS NYLN	12259	AASSLQS	12519	QQSYSTPT	12779
325	QASQDIS NYLN	12260	GASTLQS	12520	QEADSFPLT	12780
326	RASQGIRN DLG	12261	DASSLHS	12521	QQAYSFPWT	12781
327	RASQGISN YLA	12262	KASSLES	12522	QQSYNTPFT	12782
262	RASQGISN YLA	12263	KASSLES	12523	QQSYNTPFT	12783

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
328	RASQSINR WLA	12264	SASNLQS	12524	QQSYNTPLT	12784
329	RASQSINT WLA	12265	AASSLQS	12525	QQANSFPFT	12785
330	RASQSIRT-WLA	12266	DASSLET	12526	QQLNSYPLT	12786
331	RASQSIRT YLN	12267	AASTLQS	12527	QQSYSAPLT	12787
332	RASQSIST YLN	12268	AASSLHS	12528	QQSYSTPLT	12788
333	RASQSIT-TYLN	12269	AASTLQS	12529	QQSYSTPLT	12789
334	RSSQSLLH SNGYNYL D	12270	AASSLQS	12530	MQARQTPLT	12790
335	RSSQSLLH SNGYNYL D	12271	GASSLQS	12531	MQTLQTPFT	12791
336	RSSQSLLH SNGYNYL D	12272	LGSDRAS	12532	MQALQTPLT	12792
337	KSSQTVF-STSYN-KNYLA	12273	WASTRES	12533	QQYYSTPLT	12793
338	KTSQSVF-STSYN-RDYLA	12274	WASTRES	12534	QQYYSSPPT	12794
339	RASQSISS WLA	12275	DASTLQS	12535	QQSYSTPFT	12795
340	RASQSIS-SYLN	12276	DASNLKT	12536	QQSYSFPT	12796
341	RASQSVSS YLA	12277	DTSSRAT	12537	QQYYDTPYT	12797
342	KSS-QSVLYSSN NKNYLA	12278	LASTREP	12538	QQYYSTPPT	12798
343	KSS-QSVLSSSY NKNYVA	12279	WASTRES	12539	QQYYSTPLT	12799
344	QASQDIS NYLN	12280	AAASLQS	12540	QQTYSTPWT	12800
345	RASQDIN-TYLA	12281	AASSLQS	12541	QQSSSFPLT	12801
346	QASQDIS NYLN	12282	AASSLQS	12542	QQLYNFPYT	12802
347	RASQSISR YLA	12283	GASTRES	12543	QQSYNTPLT	12803
348	RASQTLSG WLA	12284	GASTLQG	12544	QQYYSYPPT	12804
349	FASQDI-INYLN	12285	EASNLET	12545	QQSYSTPLT	12805
350	RASQSIS-SYLN	12286	DVFNLGT	12546	QQSYSSPFT	12806

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme
Table 22-Kabat CDR Sequence

351	QASQDIS NYLN	12287	MASNLES	12547	QQTNSFPLT	12807
352	RASQSIS- SYLN	12288	DASNLET	12548	QQSYSTPLT	12808
353	RSSQSLLH SNGYNYL D	12289	LGSNRAS	12549	MQALQSPWT	12809
354	KSS- QSVLYSSN NKNYLA	12290	WASTRES	12550	QQYYSSPLT	12810
355	RSSQSLLH SNGYNYL D	12291	LGSNRAS	12551	MQALQTPPS	12811
356	RAS- ESVST- WLA	12292	KASRLES	12552	QQSYKTPYT	12812
357	KSS- QSVLYSSN NKNYLA	12293	WASTRES	12553	QQYFTTPLT	12813
358	RASQSIS- SYLN	12294	AASSLQS	12554	QQSYSTPYT	12814
359	KSS- QSVLSSSY NKNYLA	12295	WASTRAS	12555	QQYYDTPLT	12815
360	RASQSIS- SYLN	12296	KASTLES	12556	QQNDSIPIT	12816
361	RASQSISR WLA	12297	DASNLET	12557	LQDYSYPLT	12817
362	KTSQSVF- STSYN- RDYLA	12298	WASTRAA	12558	QQYYYTST	12818
363	RASQSIN- RYLN	12299	AASSLQS	12559	QQANSFPPT	12819
364	RASQGISN YLA	12300	SASNLQS	12560	QQSYSTPLT	12820
259	RASQGISN YLA	12301	SASNLQS	12561	QQSYSTPLT	12821
365	RASQSIDS YLN	12302	KASTLES	12562	QQSYSAPLT	12822
366	RASQDIST WLA	12303	DASNLET	12563	QQVNSDPYT	12823
367	QASQDIS NYLN	12304	AASTLES	12564	QQGDSLPLT	12824
368	RASQGISN YLA	12305	AASSLQS	12565	QQSDSFPYT	12825
369	RASQSVST YLA	12306	GASTRAT	12566	QQHDSYPLT	12826
370	RASQGIRN DLG	12307	AASSLQS	12567	QQANSFPPT	12827
371	RAS- ESISTYLN	12308	KASNLES	12568	QQTDSTFIT	12828

TABLE 22 -continued

CDRs using the Kabat Numbering Scheme Table 22-Kabat CDR Sequence						
372	RASRNIHD YLN	12309	AASTLQT	12569	QQTYSTPPT	12829
373	RASQSND SYLN	12310	KASTLES	12570	QQSYSSPLT	12830
374	RASQSISD FLN	12311	AASTLQS	12571	QQSYSSPYT	12831
375	QASQDIS NYLN	12312	AASSLQS	12572	QQANRFPLT	12832
376	QASQDIS NYLN	12313	KASNLQS	12573	QQSYNFPAT	12833
377	RSSQSLLH SNGYNYL D	12314	LGSNRAS	12574	MQGTHWPET	12834
378	RASQSIS- SYLN	12315	DASNLET	12575	QQSYSTPLT	12835
379	RASQGISD YLA	12316	DASNLET	12576	QQSYILPLT	12836
380	RASQDIN DFLA	12317	AASSLQS	12577	QQSYSAPYT	12837
381	RASQSISN WLA	12318	AASKLES	12578	QQSYSSPWT	12838
382	RASQGIDS WLA	12319	AASTLES	12579	QQAYSFPLT	12839
383	RASQNIGT WLA	12320	RASSLES	12580	QQAYSFPWT	12840
384	RASQNIN NWLA	12321	KASTLQS	12581	QQADSFPPT	12841
385	RASQDIS- SYLA	12322	AASTLQS	12582	QQLNRYPIT	12842
386	RASQDISN YLA	12323	AASILHS	12583	QQYDSSFIT	12843

TABLE 23

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name	HCDR1		HCDR2		HCDR3	
	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
109	GYTFSSY	3077	LPGSGS	3591	ARRAYGYDGGFDY	4105
110	GYTFSSY	3078	LPGSGS	3592	ARRAYGYDEGFDY	4106
111	GYTFSSY	3079	LPGSDS	3593	ARRAYGYDEGFDY	4107
112	AYTFSIY	3080	LPGSGS	3594	ARRAYGYDGGFDY	4108
1	GYTFSSY	3081	FPGSGH	3595	ARRGYGYDEGFDY	4109
113	GYTFSSY	3082	LPGSGS	3596	ARRGYGYDEGFDY	4110

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
114	GYTFSSY	3083	LPGSGS	3597	ARRGYGYDEGFDY	4111
115	GYTFSNY	3084	LPGSGS	3598	ARRGYGYDEGFDY	4112
23	GYTFSSY	3085	LPGSGS	3599	ARRGYGYDEGFDY	4113
116	GYTFSSY	3086	LPGSGY	3600	ARRGYGYDEGFDY	4114
117	GYTFSSY	3087	LPGSGS	3601	ARRAYGYDEGFDY	4115
2	GYTLSSY	3088	LPGSGS	3602	ARRGYGYDEGFDY	4116
118	GYTFSSY	3089	LPGSGS	3603	ARRAYGYDEGFDY	4117
119	GYTFSSY	3090	SPGSGS	3604	ARRGYGYDEGFDY	4118
120	GYTFGTY	3091	LPGSGT	3605	ARRAYGYDAGFDY	4119
121	GYTFSSY	3092	LPGSGS	3606	ARRGYGYDEGFDY	4120
122	GYTFSSY	3093	LPGSGR	3607	ARRGYGYDEGFDY	4121
123	GYTFSSY	3094	LPGSGR	3608	ARRGYGYDEGFDY	4122
38	GYTFSSY	3095	LPGSGS	3609	ARRAYGYDEGFDY	4123
39	GYTFSSY	3096	LPGSGS	3610	ARRAYGYDEGFDY	4124
40	GYTFSSY	3097	LPGSGR	3611	ARRGYGYDEGFDY	4125
41	GYTFSSY	3098	LPGSGS	3612	ARRAYGYDEGFDY	4126
42	GYTFSSY	3099	LPGSGR	3613	ARRGYGYDEGFDY	4127
43	GYTFSSY	3100	LPGSGS	3614	ARRAYGYDGGFDY	4128
44	GYTFSSY	3101	LPGSDS	3615	ARRAYGYDEGFDY	4129
45	AYTFSIY	3102	LPGSGS	3616	ARRAYGYDGGFDY	4130
46	GYTFSSY	3103	FPGSGH	3617	ARRGYGYDEGFDY	4131
47	GYTFSNY	3104	LPGSGS	3618	ARRGYGYDEGFDY	4132
47	GYTFSNY	3104	LPGSGS	3618	ARRGYGYDEGFDY	4132
48	GYTFSSY	3105	LPGSGS	3619	ARRGYGYDEGFDY	4133
48	GYTFSSY	3105	LPGSGS	3619	ARRGYGYDEGFDY	4133
49	GYTFSSY	3106	LPGSGY	3620	ARRGYGYDEGFDY	4134
50	GYTLSSY	3107	LPGSGS	3621	ARRGYGYDEGFDY	4135
51	GYTFSSY	3108	SPGSGS	3622	ARRGYGYDEGFDY	4136
52	GYTFGTY	3109	LPGSGT	3623	ARRAYGYDAGFDY	4137
53	GYTFSSY	3110	LPGSGS	3624	ARRGYGYDEGFDY	4138
54	GYTFSSY	3111	LPGSGR	3625	ARRGYGYDEGFDY	4139
55	GYTFSNY	3112	LPGSGS	3626	ARRGYGYDEGFDY	4140
56	GYTFSSY	3113	LPGSGS	3627	ARRGYGYDEGFDY	4141
3	GYSFTGY	3114	SSYNGA	3628	ARGRYGEYFDY	4142
4	GYSFTGY	3115	SSYNGV	3629	ARGRYGDYFDY	4143
5	GYSFTGY	3116	SSYNGV	3630	ARGRYGDYFDY	4144

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
6	GYSFTGY	3117	SSYNGV	3631	ARGRYGDYFDY	4145
7	GYSFTGY	3118	SSYNGA	3632	ARGRYGDYFDY	4146
8	GYSFTGY	3119	SSYNGV	3633	ARGRYGDYFDY	4147
9	GYSFTGY	3120	SSYNGV	3634	ARGRYGDYFDY	4148
10	GYSFTGY	3121	SSYNGV	3635	ARGRYGDYFDY	4149
11	GYSFTGY	3122	SSYNGV	3636	ARGRYGDYFDY	4150
12	GYSFTGF	3123	SSYNGA	3637	ARGRYGDYFDY	4151
13	GYSFTGY	3124	SSYNGA	3638	ARGRYGDYFDY	4152
57	GYSFTGY	3125	SSYNGV	3639	ARGRYGDYFDY	4153
58	GYSFTGY	3126	SSYNGA	3640	ARGRYGEYFDY	4154
124	GFSLSSY	3127	WRGGS	3641	AKNLYGHYVMDY	4155
125	GFSVTSY	3128	WRGGS	3642	AKNLYGHYVMDY	4156
126	GFSLTSY	3129	WRGGS	3643	AKNLYGHYVMDY	4157
127	GFSLTRY	3130	WRGGS	3644	AKNLYGHYVMDY	4158
128	GFSVTTY	3131	WRGGS	3645	AKNLYGHYVMDY	4159
129	GFSVTSY	3132	WRGGS	3646	AKNLYGHYVMDY	4160
130	GFSLTRY	3133	WRGGS	3647	AKNLYGHYVMDY	4161
59	GFSLSSY	3134	WRGGS	3648	AKNLYGHYVMDY	4162
60	GFSVTSY	3135	WRGGS	3649	AKNLYGHYVMDY	4163
61	GFSLTSY	3136	WRGGS	3650	AKNLYGHYVMDY	4164
62	GFSLTRY	3137	WRGGS	3651	AKNLYGHYVMDY	4165
63	GFSVTTY	3138	WRGGS	3652	AKNLYGHYVMDY	4166
131	GYTFTSY	3139	HPNSGS	3653	ARWGDGYSFAY	4167
132	GYTFTSY	3140	HPNSGS	3654	ARWGDGYSFAY	4168
133	GYTFTTY	3141	HPNSDN	3655	ARWGDGYSFAY	4169
14	GYTFTSY	3142	HPNSGT	3656	ARWGDGYSFAY	4170
134	GYTFTSY	3143	HPNSGN	3657	ARWGDGYSFAY	4171
64	GYTFTSY	3144	HPNSGS	3658	ARWGDGYSFAY	4172
65	GYTFTSY	3145	HPNSGT	3659	ARWGDGYSFAY	4173
135	GYTFTDY	3146	YPGSGS	3660	ARRGERGPWFAY	4174
136	GYTFTDY	3147	YPGSGS	3661	ARRGERGPWFAY	4175
137	GYTFTDY	3148	YPGSGS	3662	ARRGERGPWFAY	4176
138	GYTFTDY	3149	YPGSGS	3663	ARRGERGPWFAY	4177
15	GYTFTDY	3150	YPGSGS	3664	ARRGERGPWFAY	4178
66	GYTFTDY	3151	YPGSGS	3665	ARRGERGPWFAY	4179

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
67	GYTFTDY	3152	YPGSGS	3666	ARRGERGPWFAY	4180
68	GYTFTDY	3153	YPGSGS	3667	ARRGERGPWFAY	4181
69	GYTFTDY	3154	YPGSGS	3668	ARRGERGPWFAY	4182
24	GYTFTNY	3155	DPSDSE	3669	ATYDVYYRFAY	4183
139	GYTFTNY	3156	DPSDSE	3670	ATYDGYYRFAY	4184
140	GYTFTNY	3157	DPSDSE	3671	ATYDIYYRFAY	4185
16	GYTFTSY	3158	HPNSGS	3672	ARPGGYGFVY	4186
141	GYTFTSY	3159	HPNSDS	3673	ARPGGYGFAD	4187
142	GYTFTTY	3160	HPNSGS	3674	ARPGGYGFTY	4188
143	GYTFTSY	3161	HPNSGS	3675	ARPGGYGFAY	4189
70	GYTFTSY	3162	HPNSGS	3676	ARPGGYGFAY	4190
25	GYTFTSY	3163	YPSDSY	3677	TRGNYIDY	4191
144	GYTFTSY	3164	YPSDSY	3678	TRGNYIDY	4192
145	GYTFTDY	3165	YPSDSY	3679	TRGNYIDY	4193
146	GYTFTDY	3166	YPGSGS	3680	ARPGDLGFAY	4194
147	GYTFTDY	3167	YPGSGS	3681	ARPGDLGFAY	4195
148	GYTFTDY	3168	YPGSGS	3682	ARPGDLGFAY	4196
71	GYTFTDY	3169	YPGSGS	3683	ARPGDLGFAY	4197
72	GYTFTDY	3170	YPGSGS	3684	ARPGDLGFAY	4198
73	GYTFTDY	3171	YPGSGS	3685	ARPGDLGFAY	4199
74	GYTFTDY	3172	YPGSGS	3686	ARPGDLGFAY	4200
74	GYTFTDY	3172	YPGSGS	3686	ARPGDLGFAY	4200
75	GYTFTDY	3173	YPGSGS	3687	ARPGDLGFAY	4201
75	GYTFTDY	3173	YPGSGS	3687	ARPGDLGFAY	4201
76	GYTFTDY	3174	YPGSGS	3688	ARPGDLGFAY	4202
149	GFSLTNY	3175	WAGGI	3689	ARGDGYDDGYAMDY	4203
150	GFSLTSY	3176	WAGGI	3690	ARGDGYDDGYAMDY	4204
26	GFSLTSY	3177	WAGGT	3691	ARGDGYDDGYAMDY	4205
77	GFSLTNY	3178	WAGGI	3692	ARGDGYDDGYAMDY	4206
78	GFSLTSY	3179	WAGGI	3693	ARGDGYDDGYAMDY	4207
79	GFSLTSY	3180	WAGGT	3694	ARGDGYDDGYAMDY	4208
151	GYSFTSY	3181	DPSDSE	3695	ARTRNY	4209
152	GYSFTSY	3182	DPSDSE	3696	ARTRNY	4210
153	GYSFTSY	3183	DPSDSE	3697	ARTRNY	4211
154	GFNIKDY	3184	DPENGD	3698	NAPLLRYSSAMDY	4212
155	GFNIKDY	3185	DPENGD	3699	NAPLLRYSSSMDY	4213

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
156	GFNIKDY	3186	DPENGD	3700	NVALLRYSSAMDY	4214
80	GFNIKDY	3187	DPENGD	3701	NAPLLRYSSAMDY	4215
81	GFNIKDY	3188	DPENGD	3702	NAPLLRYSSSMDY	4216
82	GFNIKDY	3189	DPENGD	3703	NVALLRYSSAMDY	4217
17	GFNIKDT	3190	DPANGN	3704	ARGPDDGYFYYYSMDY	4218
157	GYTFSNY	3191	NPSNGD	3705	TSYYTHEAYYYAMDC	4219
27	GSTFTTY	3192	NPSNGG	3706	TSYYTHETYYYAMDY	4220
158	GFNIKDY	3193	DPEDGD	3707	TPYSIYDAMDY	4221
159	GYTFTDY	3194	YPGSGS	3708	ARRGERGPWFAY	4222
83	GYTFTDY	3195	YPGSGS	3709	ARRGERGPWFAY	4223
160	GYSFTDY	3196	STYYGD	3710	ARQMDYDYTYYYAMDY	4224
28	GYTFTSY	3197	DPSDSY	3711	ARAEYGYGNYPWFAY	4225
84	GYTFTSY	3198	DPSDSY	3712	ARAEYGYGNYPWFAY	4226
29	GYTFTSY	3199	HPSDSD	3713	AIPYYYGGWYFDV	4227
161	GYTFTDY	3200	YPGSGS	3714	ARMDGPWFAY	4228
30	GFTFSSY	3201	SSGGSY	3715	ARLYDAHWDYFDY	4229
162	GISLSTSGM	3202	WNND	3716	AWRPYYRYDSFAY	4230
18	GYTFTNY	3203	NTYTGE	3717	ARKYYDYEFAY	4231
85	GYTFTNY	3204	NTYTGE	3718	ARKYYDYEFAY	4232
163	GYTFTDY	3205	DPETGG	3719	TRLGDYDVMDY	4233
86	GYTFTDY	3206	DPETGG	3720	TRLGDYDVMDY	4234
164	GYTFTSY	3207	DPSDSY	3721	ARAGRYGSSFDY	4235
165	GFSLSTSGM	3208	YWDDD	3722	AGRPDDYDGAWFPY	4236
31	GYTFTSS	3209	HPNSGN	3723	AIYYDYDAYYFDY	4237
87	GYTFTSS	3210	HPNSGN	3724	AIYYDYDAYYFDY	4238
32	GYTFTSY	3211	HPNSGS	3725	ANPYYGYDVGY	4239
166	GYTFTDY	3212	YPGSGS	3726	AREEKIYFDY	4240
88	GYTFTDY	3213	YPGSGS	3727	AREEKIYFDY	4241
167	GYTFTSY	3214	HPNSGS	3728	ARYDGYWFDY	4242
168	GYTFTSY	3215	YPGNSD	3729	TSLITTAYYFDY	4243
89	GYTFTSY	3216	YPGNSD	3730	TSLITTAYYFDY	4244
169	GYTFTSY	3217	HPNSGS	3731	APETGDYGSSYVWYFDV	4245
170	GYTFTDY	3218	YPGSGS	3732	ARGKVTRFAY	4246
171	GFTFSSY	3219	SDGGSY	3733	ARDQDSNWEYFDY	4247
172	GYTFTDY	3220	NTETGE	3734	ARESWDRAMDY	4248

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
19	GFTFSSY	3221	SSGGSY	3735	ARHEEANWAWFAY	4249
90	GFTFSSY	3222	SSGGSY	3736	ARHEEANWAWFAY	4250
173	GYSFTNY	3223	DPSDSE	3737	AIPYYAMDY	4251
91	GYSFTNY	3224	DPSDSE	3738	AIPYYAMDY	4252
174	GYTFTSS	3225	HPNSGN	3739	ATYYGNYVWYFDV	4253
92	GYTFTSS	3226	HPNSGN	3740	ATYYGNYVWYFDV	4254
175	GYTFTSY	3227	HPNSGS	3741	ASYGSSYWYFDV	4255
93	GYTFTSY	3228	HPNSGS	3742	ASYGSSYWYFDV	4256
20	GFSLTSY	3229	WSGGS	3743	ASYYGSSRSYWYLDV	4257
94	GFSLTSY	3230	WSGGS	3744	ASYYGSSRSYWYLDV	4258
176	GYTFTSY	3231	YSGNGD	3745	ARDYYGSSHLWYFDV	4259
177	GFSLSTSGM	3232	YWDDD	3746	ARRAHYDYGWYFDV	4260
178	GYTFTSY	3233	HPNSGS	3747	AGYDYDWYFDV	4261
33	GFTFSSY	3234	SSGGSY	3748	TRHDDSSYDWFAY	4262
179	GFTFSSY	3235	SSGGSY	3749	ARHEDSNYHYFDY	4263
34	GYTFTNY	3236	HPNSGT	3750	ARFGDGYHFDY	4264
180	GFTFSSY	3237	SSGGSY	3751	ARQNDSSWAWFAY	4265
95	GFTFSSY	3238	SSGGSY	3752	ARQNDSSWAWFAY	4266
181	GYTFTSY	3239	HPNSGS	3753	ALPYSNYGWYFDV	4267
96	GYTFTSY	3240	HPNSGS	3754	ALPYSNYGWYFDV	4268
182	GYTFTSY	3241	DPSDSE	3755	ARDYYGSYWYFDV	4269
97	GYTFTSY	3242	DPSDSE	3756	ARDYYGSYWYFDV	4270
183	GFNIKDY	3243	DPEDGE	3757	AAYGNSAWFAY	4271
98	GFNIKDY	3244	DPEDGE	3758	AAYGNSAWFAY	4272
35	GYTFTNY	3245	NTNTGE	3759	ARWYPYFDY	4273
99	GYTFTNY	3246	NTNTGE	3760	ARWYPYFDY	4274
100	GYTFTNY	3247	NTNTGE	3761	ARWYPYFDY	4275
36	GYTFTSY	3248	NPSSGY	3762	ARSDGSSGNWYFDV	4276
101	GYTFTSY	3249	NPSSGY	3763	ARSDGSSGNWYFDV	4277
184	GFSLTSY	3250	WAGGS	3764	AREGGYTGYFDV	4278
102	GFSLTSY	3251	WAGGS	3765	AREGGYTGYFDV	4279
185	GYTFTSY	3252	DPSDSE	3766	AYSNYVPYYAMDY	4280
103	GYTFTSY	3253	DPSDSE	3767	AYSNYVPYYAMDY	4281
186	GYTFTDY	3254	YPGSGS	3768	ARRGFDY	4282
21	GFTFSSY	3255	SSGGSY	3769	ARHNYSNWDWFAY	4283
187	GYTFTSY	3256	HPNSGS	3770	ARDYYGSGYGYYFDY	4284

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
188	GYTFTSY	3257	HPNSGS	3771	ARDYYGSSYGWYFDV	4285
189	GYTFTSY	3258	HPNSGS	3772	ARDYYGSSYGWYFDV	4286
190	GYTFTSY	3259	HPNSGS	3773	ASDYYGSSYGWYFDV	4287
191	GYTFTSY	3260	HPNSGS	3774	ARDYYGSSYGWYFDV	4288
192	GYTFTSY	3261	HPNSGS	3775	TRDYYGSGYGWYFDV	4289
193	GYTFTNY	3262	DPSDSE	3776	ATYDGYYRFAY	4290
194	GYTFTNY	3263	DPSDSE	3777	ATYDVYYRFAY	4291
195	GYTFTSY	3264	HPNSGS	3778	ARDYGNYDYAMDY	4292
104	GYTFTSY	3265	HPNSGS	3779	ARDYGNYDYAMDY	4293
37	GYTFTSY	3266	HPNSGS	3780	ARDYGNYDYAMDY	4294
196	GYTFTSY	3267	HPNSGS	3781	ARDYGNYDYAMDY	4295
197	GFTFSSY	3268	SSGGSY	3782	ASQLTGTWYYFDY	4296
198	GFTFSSY	3269	SSGGSY	3783	ASQLTGTWYYFDY	4297
199	GFTFSSY	3270	SSGGSY	3784	ASQLTGTWYYFDY	4298
22	GFNIKDT	3271	DPANGN	3785	ARGPDDGYFYYYSMDY	4299
200	GFTFSNY	3272	NSNGGS	3786	ARQEGIGYAMDY	4300
201	GYTFTEY	3273	YPNNGG	3787	ARGGWLLGY	4301
202	GFSLTSY	3274	WSGGS	3788	ARDGGIRGAMDY	4302
203	GYTFSSY	3275	LPGSGS	3789	ARRGYGYDEGFDY	4303
204	GYTFTDY	3276	DPETGG	3790	TRNYDYAMDY	4304
205	GFTFSSY	3277	NSNGGS	3791	ARQEGIGYALDY	4305
206	GFTFSSY	3278	SSGGS	3792	AREREWGVYYGSSLDY	4306
207	GFNIKDT	3279	DPANGN	3793	ARSDGNYD	4307
208	GFTFSNY	3280	NSNGGS	3794	ARQEGIGYGMDY	4308
209	GFTFNTY	3281	RSKSDNYA	3795	VRHDGVVGFDV	4309
210	GYSITSGY	3282	SYDGS	3796	ARGGGRG	4310
211	GYTFTDY	3283	NTETGE	3797	ARDYYDYYYAMDY	4311
212	GYTFTDY	3284	NTETGE	3798	ARESWDRAMDY	4312
213	GYTFTNY	3285	DPYDSE	3799	ARIYSDYDGAWFAY	4313
214	GYTFTDY	3286	NPYNGG	3800	ARGTVGFAY	4314
215	GFTFSSY	3287	SSGGS	3801	AREREWGVFYGSSLDY	4315
216	GFTFSSY	3288	SSGGSY	3802	ARHDDSSYGYFDY	4316
217	GFTFSNY	3289	SSGGT	3803	ARTMPDV	4317
218	GFSLTSY	3290	WAGGS	3804	ARDTDGYYWAMDY	4318
219	GYSITSDH	3291	SYSGS	3805	ARKWGDY	4319

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
220	GYTFTDY	3292	DPETGG	3806	TRNYDYALDY	4320
221	GYSITSGY	3293	SYDGS	3807	ARGGGRG	4321
222	GFTFSNY	3294	NSNGGS	3808	ARQEEIGYAMDY	4322
223	GFNIKDY	3295	DPETDN	3809	ARSGNMGFTY	4323
105	GFNIKDY	3296	DPETDN	3810	ARSGNMGFTY	4324
224	GFTFSSY	3297	SSGGSY	3811	ASQGGSSWGAMDY	4325
106	GFTFSSY	3298	SSGGSY	3812	ASQGGSSWGAMDY	4326
225	GFTFSSY	3299	SNGGSY	3813	ARHEITTRFAY	4327
226	GYSITSGY	3300	SYDGS	3814	AREAGYFDY	4328
227	GFSFNTY	3301	RSKSNNYA	3815	VRQYGYDFDY	4329
228	GFTFSSY	3302	SSGGSY	3816	ARHKGVNWDYFDY	4330
229	GYTFTDY	3303	DPETGG	3817	TRGDGNYDSWYFDV	4331
230	GFTFSSY	3304	SSGGSY	3818	ARLPVTTVVFDY	4332
231	GFTFSSY	3305	SSGGSY	3819	ARRPVVVPFDY	4333
232	GFSLTSY	3306	WSGGS	3820	ARGWDADYFDY	4334
233	GYTFTNY	3307	HPNSGS	3821	TRYDYDDY	4335
234	GYTFTDY	3308	NPNNGG	3822	ARSELGLYAMDY	4336
235	GYTFTGY	3309	LPGSGS	3823	ARGRIHYFDY	4337
236	GYTFTGY	3310	LPGSGS	3824	ARGRIHYFDY	4338
237	GFSLTSY	3311	WSGGS	3825	ARKGYGYDWYFDV	4339
107	GFSLTSY	3312	WSGGS	3826	ARKGYGYDWYFDV	4340
238	GYTFTSY	3313	DPSDSY	3827	ARSSYYYYAMDY	4341
108	GYTFTSY	3314	DPSDSY	3828	ARSSYYYYAMDY	4342
239	GYSITSGY	3315	SYDGS	3829	ARGGGRD	4343
240	GFSLTSY	3316	WSGGS	3830	ARGGDYDSYAMDY	4344
241	GYTFTSY	3317	YPGSGS	3831	ARESVYDGYSWYFDV	4345
242	GYSFTDY	3318	NPNYGT	3832	ASTYDYDDWYFDV	4346
243	GYTFTSY	3319	DPSDSY	3833	ARSGNYLYAMDY	4347
244	GYSFTDY	3320	NPNYGT	3834	AREGTSWYFDV	4348
245	GFSLTSY	3321	WRGGS	3835	AKKGDGYDWYFDV	4349
246	GFSLTSY	3322	WSGGS	3836	AREGNYGSSYDAMDY	4350
247	GYTFTSY	3323	DPSDSY	3837	ARSSNYPYAMDY	4351
248	GFNIKNT	3324	DPANGN	3838	AYYSGLY	4352
249	GYTFTSY	3325	DPSDSE	3839	ARRGQIYYGYSWFAY	4353
250	GYTFTDY	3326	NPNNGG	3840	ARSTVVADWYFDV	4354
251	GYTFTSY	3327	YPRSGN	3841	ARSGSSYGYFDV	4355

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
252	GFSLTSY	3328	WSGGS	3842	ARKGGYDAYAMDY	4356
253	GYSFTDY	3329	NPNYGT	3843	AREGFITTVVAVDY	4357
254	GYTFTDY	3330	DPETGG	3844	TREGNYDAMDY	4358
255	GYTFTSY	3331	DPSDSY	3845	ARWDYYGVDY	4359
256	GFTFSGY	3332	SPGGGS	3846	ASSLTATHTYEYDY	4360

	LCDR1		LCDR2		LCDR3	
	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
109	KASQDIN-SYLS	5644	KTLIYRANRLVD	7505	LHYDEFPPT	6664
110	KASQDIN-SYLN	5645	KTLIYRANRLVD	7506	LQYDEFPPT	6665
111	KASQDIN-SYLS	5646	KTLIYRANRLVD	7507	LQYDEFPPT	6666
112	KASQDIN-SYLS	5647	KTLIYRANRLVD	7508	LQYDEFPPT	6667
1	KASQDIN-SYLS	5648	KTLIYRANRLVD	7509	LQYDEFPPT	6668
113	KASQDIN-SYLS	5649	KTLIYRANRLVD	7510	PQYVESPPT	6669
114	KASQDIN-SYLS	5650	KTLIYRANRLVD	7511	LQYDEFPPT	6670
115	KASQDIN-SYLS	5651	KTLIYRANRLVD	7512	LQYDEFPPT	6671
23	KASQDIN-SYLS	5652	KTLIYRANRLVD	7513	LQYDEFPPT	6672
116	KASQDIN-SYLS	5653	KTLIYRANRLVD	7514	LQYDEFPPT	6673
117	KASQDIN-SYLS	5654	KTLIYRANRLVD	7515	LQYDEFPLT	6674
2	KASQDIN-SYLS	5655	KTLIYRANRLVD	7516	LQYDEFPPT	6675
118	KASQDIN-GYLS	5656	QTLLYRANRLVD	7517	LQYDEFPPT	6676
119	KASQDIN-SYLS	5657	KTLIYRANRLVD	7518	LQYDEFPPT	6677
120	KASQDIN-SYLS	5658	KTLIYRANRLVD	7519	LQYDEFPPT	6678
121	KASQDIN-SYLN	5659	KTLIYRANRLVD	7520	LQYDEFPPT	6679
122	KASQDIN-SYLS	5660	KTLIYRANRLVD	7521	LQYDEFPPT	6680
123	KASQDIN-SYLS	5661	KTLIYRAKRLVD	7522	LQYDEFPPT	6681

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
38	KASQDIN-SYLS	7174	KTLIYRANRLVD	7523	LQYDEFPLT	6682
39	KASQDIN-GYLS	7175	QTLLYRANRLVD	7524	LQYDEFPPT	6683
40	KASQDIN-SYLS	7176	KTLIYRAKRLVD	7525	LQYDEFPPT	6684
41	KASQDIN-GYLS	7177	QTLLYRANRLVD	7526	LQYDEFPPT	6685
42	KASQDIN-SYLS	7178	KTLIYRAKRLVD	7527	LQYDEFPPT	6686
43	KASQDIN-SYLS	7179	KTLIYRANRLVD	7528	LHYDEFPPT	6687
44	KASQDIN-SYLS	7180	KTLIYRANRLVD	7529	LQYDEFPPT	6688
45	KASQDIN-SYLS	7181	KTLIYRANRLVD	7530	LQYDEFPPT	6689
46	KASQDIN-SYLS	7182	KTLIYRANRLVD	7531	LQYDEFPPT	6690
47	KASQDIN-SYLS	7183	KTLIYRANRLVD	7532	LQYDEFPPT	6691
48	KASQDIN-SYLS	7184	KTLIYRANRLVD	7533	LQYDEFPPT	6692
49	KASQDIN-SYLS	7185	KTLIYRANRLVD	7534	LQYDEFPPT	6693
50	KASQDIN-SYLS	7186	KTLIYRANRLVD	7535	LQYDEFPPT	6694
51	KASQDIN-SYLS	7187	KTLIYRANRLVD	7536	LQYDEFPPT	6695
52	KASQDIN-SYLS	7188	KTLIYRANRLVD	7537	LQYDEFPPT	6696
53	KASQDIN-SYLN	7189	KTLIYRANRLVD	7538	LQYDEFPPT	6697
54	KASQDIN-SYLS	7190	KTLIYRANRLVD	7539	LQYDEFPPT	6698
55	KASQDIN-SYLS	7191	KTLIYRANRLVD	7540	LQYDEFPPT	6699
56	KASQDIN-SYLS	7192	KTLIYRANRLVD	7541	LQYDEFPPT	6700
3	RASENIYS NLA	5681	QLLVFAATYLAD	7542	QHFWGTPWT	6701
4	RASENIYS NLA	5682	QLLVYAATNLAD	7543	QHFWGTPWT	6702
5	RASENIYS NLA	5683	QVLVYAATNLAD	7544	QHFWGSPWT	6703
6	RASENIYS NLA	5684	RLLVYAATNLAD	7545	QHFWGTPWT	6704
7	RASENIYS NLA	5685	QLLVYAATNLAD	7546	QHFWGTPWT	6705

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
8	RASENIYS NLA	5686	QVLVYAATNVAD	7547	QHFWGTPWT	6706
9	RAS-DNIYSNLA	5687	QLLVYAATNLAD	7548	QHFWGTPWT	6707
10	RASENIYS NLA	5688	RLLVYAATNLAD	7549	QHFWGTPWT	6708
11	RASENIYS NLA	5689	QLLVYAATNLAD	7550	QHFWGTPWT	6709
12	RASENIYS NLA	5690	QLLVFAATYLAD	7551	QHFWGTPWT	6710
13	RASENIYS NLA	5691	QLLVYAATNLAD	7552	QHFWGSPWT	6711
57	RAS-DNIYSNLA	7193	QLLVYAATNLAD	7553	QHFWGTPWT	6712
58	RASENIYS NLA	7194	QLLVFAATYLAD	7554	QHFWGTPWT	6713
124	KASQDIN-SYLS	5694	KTLIYRANRLVD	7555	LQYDEFPPT	6714
125	KASEN-VVTYVS	5695	KLLIYGASNRYT	7556	GQSYSYPFT	6715
126	KASEN-VVTYVS	5696	KLLIYGASNRYT	7557	GQSYSYPFT	6716
127	KASEN-VVTYVS	5697	KLLIYGASNRYT	7558	GQSYSYPFT	6717
128	KASEN-VVTYVS	5698	KLLIYGASNRYT	7559	GQSYSYPFT	6718
129	KASEN-VVTYVS	5699	KLLIYGASNRYT	7560	GQSYSYLIHVR	7761
130	KASEN-VVTYVS	5700	KLLIYGASNRYT	7561	GQSYSYLIHVR	7762
59	KASQDIN-SYLS	7195	KTLIYRANRLVD	7562	LQYDEFPPT	6721
60	KASEN-VVTYVS	7196	KLLIYGASNRYT	7563	GQSYSYPFT	6722
61	KASEN-VVTYVS	7197	KLLIYGASNRYT	7564	GQSYSYPFT	6723
62	KASEN-VVTYVS	7198	KLLIYGASNRYT	7565	GQSYSYPFT	6724
63	KASEN-VVTYVS	7199	KLLIYGASNRYT	7566	GQSYSYPFT	6725
131	SASSSVSY MH	5706	KRWIYDTSKLAS	7567	QQWSSNPLYT	7763
132	SASSSVSY MH	5707	KRWIYDTSKLAS	7568	QQWSSNPHVHV	7764
133	SASSSVSY MH	5708	KRWIYDTSKLAS	7569	QQWSSNPLYT	7765
14	SASSSVSY MH	5709	KRWIYDTSKLAS	7570	QQWSSNPLYT	7766

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
134	SASSSVSY MH	5710	KRWIYDTSKLAS	7571	QQWSSNPLYT	7767
64	SASSSVSY MH	7200	KRWIYDTSKLAS	7572	QQWSSNPLY	6731
65	SASSSVSY MH	7201	KRWIYDTSKLAS	7573	QQWSSNPLY	6732
135	KASQSVD YD-GDSYMN	5713	KLLIYAASNLES	7574	QQSNEDPLT	6733
136	KASQSVD YD-GDSYMN	5714	KLLIYAASNLES	7575	QQSNEDPLT	6734
137	KASQSVD YD-GDSYMN	5715	QLLIYAASNLQS	7576	QQSNEDPLT	6735
138	KASQSVD YD-GDSYMN	5716	KLLIYAASNLES	7577	QQSNEDPLT	6736
15	KASQSVD YD-GDSYMN	5717	KLLIYAASNLES	7578	QQSNEDPLT	6737
66	KASQSVD YD-GDSYMN	7202	KLLIYAASNLES	7579	QQSNEDPLT	6738
67	KASQSVD YD-GDSYMN	7203	KLLIYAASNLES	7580	QQSNEDPLT	6739
68	KASQSVD YD-GDSYMN	7204	QLLIYAASNLQS	7581	QQSNEDPLT	6740
69	KASQSVD YD-GDSYMN	7205	KLLIYAASNLES	7582	QQSNEDPLT	6741
24	KASQDINK YIA	5722	SLLIHYTSTLQP	7583	LQYDNLMYT	6742
139	KASQDINK YIA	5723	RLLIHYTSTLQP	7584	LQYDILMYT	6743
140	KASQDINK YIA	5724	RLLIHYTSTLQP	7585	LQYDILMYT	6744
16	QATQDIV-KNLN	5725	SFLIYYATELAE	7586	LQFYEFPLT	6745
141	QATQDIV-KNLN	5726	SFLIYYATELAE	7587	LQFYEFPLT	6746
142	QATQDIV-KNLN	5727	SFLIYYATELAE	7588	LQFYEFPLT	6747
143	QATQDIV-KNLN	5728	SFLIYYATELAE	7589	LQFYEFPLT	6748
70	QATQDIV-KNLN	7206	SFLIYYATELAE	7590	LQFYEFPLT	6749

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
25	RAS-SSVNYMY	5730	KLWIYYTSNLAP	7591	QQFTSSHT	7768
144	RAS-SSVNYMY	5731	KLWIYYTSNLAP	7592	QQFTSSHT	7769
145	RAS-SSVNYMY	5732	KLWIYYTSNLAP	7593	QQFTSSHT	7770
146	KASQSVD YD-GDSYMN	5733	KLLIYAASNLES	7594	QQSNKDPLT	6753
147	KASQSVD YD-GDSYMN	5734	KLLIYAASNLES	7595	QQSNEDPLT	6754
148	KASQSVD YD-GDSYMN	5735	KLLIYAASNLES	7596	QQSNKDPFT	6755
71	KASQSVD YD-GDSYMN	7207	KLLIYAASNLES	7597	QQSNKDPLT	6756
72	KASQSVD YD-GDSYMN	7208	KLLIYAASNLES	7598	QQSNKDPLT	6757
73	KASQSVD YD-GDSYMN	7209	KLLIYAASNLES	7599	QQSNEDPLT	6758
74	KASQSVD YD-GDSYMN	7210	KLLIYAASNLES	7600	QQSNKDPFT	6759
75	KASQSVD YD-GDSYMN	7211	KLLIYAASNLES	7601	QQSNKDPFT	6760
76	KASQSVD YD-GDSYMN	7212	KLLIYAASNLES	7602	QQSNKDPFT	6761
149	RASQSVST SSYSYMH	5742	KLLIKYASNLES	7603	QHSWEIPLT	6762
150	RASQSVST SSYSYMH	5743	KLLIKYASNLES	7604	QHSWEIPLT	6763
26	RASQSVST SSYSYMH	5744	KLLIKYASNLES	7605	QHSWEIPLT	6764
77	RASQSVST SSYSYMH	7213	KLLIKYASNLES	7606	QHSWEIPLT	6765
78	RASQSVST SSYSYMH	7214	KLLIKYASNLES	7607	QHSWEIPLT	6766
79	RASQSVST SSYSYMH	7215	KLLIKYASNLES	7608	QHSWEIPLT	6767
151	RASQDISN YLN	5748	KLLIYSTSRLHS	7609	QQGNTLPWT	6768
152	RASQDISN YLN	5749	KLLIYSTSRLHS	7610	QQGNALPWT	6769
153	RASQDISN YLN	5750	KLLIYSTSRLHS	7611	QQGNTLPWT	6770

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name							
154	RSST-GAVTTSN YAN	5751	TGLIGGTSNRAP	7612	ALWYSTHYV	6771
155	RSST-GAVTTSN YAN	5752	TGLIGGTSNRAP	7613	ALWYSTHYV	6772
156	RSST-GAVTTSN YAN	5753	TGLIGGTSNRAP	7614	ALWYSTHYV	6773
80	RSST-GAVTTSN YAN	7216	TGLIGGTSNRAP	7615	ALWYSTHYV	6774
81	RSST-GAVTTSN YAN	7217	TGLIGGTSNRAP	7616	ALWYSTHYV	6775
82	RSST-GAVTTSN YAN	7218	TGLIGGTSNRAP	7617	ALWYSTHYV	6776
17	RASENIYS NLA	5757	QLLVYAATNLAD	7618	QHFWGTPWT	6777
157	IT-STDIDDD MN	5758	KLLISEGNTLRP	7619	LQSDNMPFT	6778
27	MTSIDIDD DMN	5759	KLLISEGNTLRP	7620	LQSDNMPFT	6779
158	RSST-GAVTTSN YAN	5760	TGLIGGTSNRAP	7621	ALWYSTHY	7771
159	KASQSVD YD-GDSYMN	5761	KLLIYAASNLES	7622	QQSNEDPLT	6781
83	KASQSVD YD-GDSYMN	7219	KLLIYAASNLES	7623	QQSNEDPLT	6782
160	KSS-QSLLYSTN QKNYLA	5763	KLLIYWASTRES	7624	QQYYSYPPWT	7772
28	RSST-GAVTTSN YAN	5764	TGLIGGTSNRAP	7625	ALWYSTHWV	6784
84	RSST-GAVTTSN YAN	7220	TGLIGGTSNRAP	7626	ALWYSTHWV	6785
29	RSST-GAVTTSN YAN	5766	TGLIGGTNNRAP	7627	ALWYSNHL	7773
161	KASQSVD YD-GDSYMN	5767	KLLIYAASNLES	7628	QQSNEDPPT	6787
30	RSST-GAVTTSN YAN	5768	TGLIGGTNNRAP	7629	ALWYSNHWV	6788

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
162	RASENIYY SLA	5769	QLLIYNANSLED	7630	KQAYDVPYT	6789
18	KSS-QSLLNSSN QKNYLA	5770	KLLVYFASTRES	7631	QQHYSTPLT	6790
85	KSS-QSLLNSSN QKNYLA	7221	KLLVYFASTRES	7632	QQHYSTPLT	6791
163	RASQDISN YLN	5772	KLLIYYTSRLHS	7633	QQDNTLPRT	6792
86	RASQDISN YLN	7222	KLLIYYTSRLHS	7634	QQDNTLPRT	6793
164	RASQDISN YLN	5774	KLLIYYTSRLHS	7635	QQGNTLPWT	6794
165	RASENIYS NLA	5775	QLLVYAATNLAD	7636	QHFWGTPWT	6795
31	QATQDIV-KNLN	5776	SFLIYYATELAE	7637	LQFYEFPYT	6796
87	QATQDIV-KNLN	7223	SFLIYYATELAE	7638	LQFYEFPYT	6797
32	SASSSVSY MY	5778	RLLIYDTSNLAS	7639	QQWSSYPLT	6798
166	KASQSVD YD-GDSYMN	5779	KLLIYAASNLES	7640	QQSNEDPWT	6799
88	KASQSVD YD-GDSYMN	7224	KLLIYAASNLES	7641	QQSNEDPWT	6800
167	RSST-GAVTTSN YAN	5781	TGLIGGTNNRAP	7642	ALWYSNHWV	6801
168	QATQDIV-KNLN	5782	SFLIYYATELAE	7643	LQFYEFPLT	6802
89	QATQDIV-KNLN	7225	SFLIYYATELAE	7644	LQFYEFPLT	6803
169	TLSSQHST YTIE	7226	KYVMELKKDGSH STGD	7645	GVGDTIKEQFVFV	7774
170	KASQSVD YD-GDSYMN	5785	KLLIYAASNLES	7646	QQSNEDPPT	6805
171	RASQDIGI SLN	5786	KRLIYATSSLDS	7647	LQYASSPYT	6806
172	RASENIYS YLA	7227	QLLVYNAKNLAD	7648	QHFWGTPYT	6807
19	SASSSVSY MH	5788	KLWIYSTSNLAS	7649	QQRSSFPYT	6808
90	SASSSVSY MH	7228	KLWIYSTSNLAS	7650	QQRSSFPYT	6809
173	KASQDIN-SYLS	5790	KTLIYRANRLVD	7651	LQYDEFPLT	6810

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
91	KASQDIN-SYLS	7229	KTLIYRANRLVD	7652	LQYDEFPLT	6811
174	RASQDIH GYLN	5792	KHLIYETSNLDS	7653	LQYASSPLT	6812
92	RASQDIH GYLN	7230	KHLIYETSNLDS	7654	LQYASSPLT	6813
175	KASQDVG TAVA	5794	KLLIYWASTRHT	7655	QQYSSYPFT	6814
93	KASQDVG TAVA	7231	KLLIYWASTRHT	7656	QQYSSYPFT	6815
20	KASQSVS NDVA	5796	KLLIYYASNRYT	7657	QQDYTSLPT	6816
94	KASQSVS NDVA	7232	KLLIYYASNRYT	7658	QQDYTSLPT	6817
176	RSST-GAVTTSN YAN	5798	TGLIGGTNNRAP	7659	ALWYSNHLV	6818
177	KASQSVD YD-GDSYMN	5799	KLLIYVASNLES	7660	QQSHEDPRT	6819
178	SASSSVSY MH	5800	KLWIYDTSKLAS	7661	FQGSGYPLT	6820
33	IT-STDIDDD MN	5801	KLLISEGNTLRP	7662	LQSDNMPLM	6821
179	RASENIYS NLA	5802	QLLVYAATNLAD	7663	QHFWGTPYT	6822
34	SASSSVSY MH	5803	KLWIYSTSNLAS	7664	QQRSTYPT	7775
180	IT-STDIDDD MN	5804	KLLISEGNTLRP	7665	LQSDNMPLT	6824
95	IT-STDIDDD MN	7233	KLLISEGNTLRP	7666	LQSDNMPLT	6825
181	RASQDISN YLN	5806	KLLIYYTSRLHS	7667	QQGNTLPFT	6826
96	RASQDISN YLN	7234	KLLIYYTSRLHS	7668	QQGNTLPFT	6827
182	SAS-QGISNYLN	5808	KLLIYYTSSLHS	7669	QQYSKLPYT	6828
97	SAS-QGISNYLN	7235	KLLIYYTSSLHS	7670	QQYSKLPYT	6829
183	RASQSISD YLH	5810	RLLIKYASQSIS	7671	QNGHSFPWT	6830
98	RASQSISD YLH	7236	RLLIKYASQSIS	7672	QNGHSFPWT	6831
35	QATQDIV-KNLN	5812	SFLIYYATELAE	7673	LQFYEFPYT	6832

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
99	QATQDIV-KNLN	7237	SFLIYYATELAE	7674	LQFYEFPYT	6833
100	QATQDIV-KNLN	7238	SFLIYYATELAE	7675	LQFYEFPYT	6834
36	RASGNIH NYLA	5815	QLLVYNAKTLAD	7676	QHFWSTPWT	6835
101	RASGNIH NYLA	7239	QLLVYNAKTLAD	7677	QHFWSTPWT	6836
184	RSST-GAVTTSN YAN	5817	TGLIGGTSYRAP	7678	ALWYSTHYV	6837
102	RSST-GAVTTSN YAN	7240	TGLIGGTSYRAP	7679	ALWYSTHYV	6838
185	RASQEIS-GYLS	5819	KRLIYAASTLDS	7680	LQYASYPWT	6839
103	RASQEIS-GYLS	7241	KRLIYAASTLDS	7681	LQYASYPWT	6840
186	KASQSVD YD-GDSYMN	5821	KLLIYAASNLES	7682	QQSNEDPLPT	7776
21	HASQNIN VWLS	5822	KLLIYKASNLHT	7683	QQGQSYPLT	6842
187	SASSSVSY MH	5823	KRWIYDTSKLAS	7684	QQWSSNPLT	6843
188	RASGNIH NYLA	5824	QLLVYNAKTLAD	7685	QHFWSTPWT	6844
189	RASENIYS YLA	5825	QLLVYNAKTLAE	7686	QHHYGTPFT	6845
190	RAS-SSVNYMY	5826	KLWIYYTSNLAP	7687	QQFTSSLT	7777
191	RSST-GAVTTSN YAN	5827	TGLIGSTNNRAP	7688	TLWYSNHWV	6847
192	KASQDVG TAVA	5828	KLLIYWASTRHT	7689	QQYSSYPFT	6848
193	KASQDIN-SYLS	5829	KTLIYRANRLVD	7690	LQYDEFPPT	6849
194	RASENIYS NLA	5830	QLLVYAATNLAD	7691	QHFWGTPFT	6850
195	RASQSISD YLH	5831	RLLIKYASQSIS	7692	QNGHSFPYT	6851
104	RASQSISD YLH	7242	RLLIKYASQSIS	7693	QNGHSFPYT	6852
37	RASENIYS NLA	5833	QLLVYAATNLAD	7694	QHFWGTPYT	6853
196	RSST-GAVTTSN YAN	5834	TGLIGGTNNRAP	7695	ALWYSNHWV	6854

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
197	IT-STDIDDD MN	5835	KLLISEGNTLRP	7696	LQSDNLPLT	6855
198	SASSSVSY MH	5836	KLWIYSTSNLAS	7697	QQRSSYPPT	6856
199	SASSSVSY MH	5837	KRWIYDTSKLAS	7698	QQWSSNPLT	6857
22	IT-STDIDDD MN	5838	KLLISEGNSLRP	7699	LQSDNLPLT	6858
200	SASSSVSS SYLY	5839	KLWIYSTSNLAS	7700	HQWSSYPPT	6859
201	TLSSQHST YTIE	7243	KYVMELKKDGSH STGD	7701	GVGDTIKEQFVYV	7778
202	RSSQSIVH SNGNTYLE	5841	KLLIYKVSNRFS	7702	FQGSHVPWT	6861
203	KASQDIN-SYLS	5842	KTLIYRANRLVD	7703	LQYDEFPPT	6862
204	KSS-QSLLDSDG KTYLN	5843	KRLIYLVSKLDS	7704	WQGTHFPWT	6863
205	SASSSVSS SYLY	5844	KLWIYSTSNLAS	7705	HQWSSYPPT	6864
206	RSSQSIVY SNGNTYLE	5845	KLLIYKVSNRFS	7706	FQGSHVPPT	6865
207	RASENI-YNNLA	5846	QLLVYAATNLAD	7707	QHFWGTPWT	6866
208	SASSSVSS SYLY	5847	KLWIYSTSNLAS	7708	HQWSSYPPT	6867
209	SASSSVSY MY	5848	RLLIYDTSNLAS	7709	QQWSTYPPIT	7779
210	SASSSVSY MY	5849	RLLIYDTSNLAS	7710	QQWSSYPFT	6869
211	RAS-SSVSYMH	5850	KPWIYATSNLAS	7711	QQWSSNPYT	6870
212	RASENIYS NLA	5851	QLLVYAATNLAD	7712	QHFWGTPWT	6871
213	KASQDVS TAVA	5852	KLLIYWASTRHT	7713	QQHYSTPWT	6872
214	KSS-QSLLDSDG KTYLN	5853	KRLIYLVSKLDS	7714	WQGTHFPWT	6873
215	RSSQSIVH SNGNTYLE	5854	KFLIYKVSNRFS	7715	FQGSHVPPT	6874
216	IT-STDIDDD MN	5855	KLLISEGNTLRP	7716	LQSDNMPLT	6875
217	KASQDINK YIA	5856	RLLIHYTSTLQP	7717	LQYDNLYMYT	7780

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
218	HASQNIN VWLS	5857	KLLIYKASNLHT	7718	QQGQSYPYT	6877
219	SASSSVSY MY	5858	KPWIYLTSNLAS	7719	QQWSSNPPT	6878
220	KSS-QSLLDSDG KTYLN	5859	KRLIYLVSKLDS	7720	WQGTHFPWT	6879
221	SASLSVSD MY	5860	RLLIYDTSNLAS	7721	QQWSSYPFT	6880
222	SASSSVSS SYLY	5861	KLWIYSTSNLAS	7722	HQWSSYPPT	6881
223	RAS-SSVNYMY	5862	KLWIYYTSNLAP	7723	QQFTSSPST	6882
105	RAS-SSVNYMY	7244	KLWIYYTSNLAP	7724	QQFTSSPST	6883
224	IT-NTDIDDD MN	5864	KLLISEGNTLRP	7725	LQSDNLPLT	6884
106	IT-NTDIDDD MN	7245	KLLISEGNTLRP	7726	LQSDNLPLT	6885
225	KASQSVD YD-GDSYMN	5866	KLLIYAASNLES	7727	QQSNEDPWT	6886
226	RASQSISN NLH	5867	RLLIKYASQSIS	7728	QQSNSWPFT	6887
227	SASSSVSS SYLY	5868	KLWIYSTSNLAS	7729	HQWSSYPPT	6888
228	IT-STDIDDD MN	5869	KLLISEGNTLRP	7730	LQSDNMPLT	6889
229	KSS-QSLLDSDG KTYLH	5870	KRLIYLVSKLDS	7731	WQGTHFPWT	6890
230	KASQDINK YIA	5871	RLLIHYTSTLQP	7732	LQYDNLRT	7781
231	RSST-GAVTTSN YAN	5872	TGLIVGTNNRAP	7733	VLWYSNHLV	6892
232	KASEN-VGTYVS	5873	KLLIYGASNRYT	7734	GQSYSYPPT	6893
233	RASET-VDSYGYSF MH	5874	KLLIYRASNLES	7735	QQSNEDPRT	6894
234	KSS-QSLLYSTN QKNYLA	5875	KLLLYWASTRES	7736	QQYYSYRT	7782
235	KSS-QSLLDSDG KTYLN	5876	KRLIYLVSKLDS	7737	WQGTHFPFT	6896

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
236	KASQSVD YD-GDSYMN	5877	KLLIYAASNLES	7738	QQSNEDPFT	6897
237	TLSSQHST YTIE	7246	KYVMELKKDGSH STGD	7739	GVGDTIKEQFVYV	7783
107	TLSSQHST YTIE	7247	KYVMELKKDGSH STGD	7740	GVGDTIKEQFVYV	6899
238	SASSSVSY MH	5880	KRWIYDTSKLAS	7741	QQWSSNPLT	6900
108	SASSSVSY MH	7248	KRWIYDTSKLAS	7742	QQWSSNPLT	6901
239	QATQDIV-KNLN	5882	SFLIYYATELAE	7743	LQFYEFPLT	6902
240	RAS-ESVDNY-GISFMN	5883	KLLIYAASNQGS	7744	QQSKEVPPT	6903
241	RASQSISD YLH	5884	RLLIKYASQSIS	7745	QNGHSFPLT	6904
242	RSSQSIVH SNGDTYLE	5885	KLLIYKVSNRFS	7746	FQGSHVPLT	6905
243	SASSSVSY MH	5886	KRWIYDTSKLAS	7747	QQWSSNPLT	6906
244	TASESLYS-SKHKVHYL A	5887	KLLIYGASNRYI	7748	AQFYSYPYT	6907
245	TLSSQHST YTIE	7249	KYVMELKKDGSH STGD	7749	GVGDTIKEQFVYV	7784
246	RASQSVST SSYSYMH	5889	KLLIKYASNLES	7750	QHSWEIPLT	6909
247	SASSSVSY MH	5890	KRWIYDTSKLAS	7751	QQWSSNPLT	6910
248	KASDHIN NWLA	5891	RLLISGATSLET	7752	QQYWSTPLT	6911
249	RASENIYS YLA	5892	QLLVYNAKTLAE	7753	QHHYGTPYT	6912
250	RASENIYS YLA	5893	QLLVYNAKTLAE	7754	QHHYGTPPT	6913
251	SASSSVSY MH	5894	KRWIYDTSKLAS	7755	QQWSSNPPT	6914
252	RAS-ESVDNY-GISFMN	5895	KLLIYAASNQGS	7756	QQSKEVPPT	6915
253	TASESLYS-SKHKVHYL A	5896	KLLIYGASNRYI	7757	AQFYSYPYT	6916

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
254	RSST-GAVTTSN YAN	5897	TGLIGGTSNRAP	7758	ALWYSTHYV	6917
255	SASSSI-SYMH	5898	KRWIYDTSKLAS	7759	HQRSSYPT	7785
	HCDR1		HCDR2		HCDR3	
	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
265	GGTFSSY	10885	NPSGG	11147	ARDLGDPGMDV	11409
266	GGTFSNY	10886	DPSGGS	11148	ARDLGDMGMDV	11410
267	GGTFSNY	10887	NPSGGS	11149	ARDVGDRGMDV	11411
263	GGTFSSY	10888	NPSGG	11150	ARDLGDPGMDV	11412
268	GSTFSGY	10889	DPNGGG	11151	AKDIVHDGTEYFQH	11413
269	GYTFTSY	10890	NPSGGS	11152	AKDIVHDGTEYFQH	11414
270	GGTFSSY	10891	NPSGGS	11153	AREGRDHDAFDI	11415
271	GFTFTDY	10892	NPSGGS	11154	AREGRSHDAFDI	11416
272	GYTFTGY	10893	NPHSGD	11155	ARWVGTTEYYYYYMDV	11417
273	GYTFTDY	10894	DPSGGS	11156	ATTAYYDFWSGYSMDV	11418
274	GYTFTSH	10895	DPSGGS	11157	ARDMDNWNTGYYYYMDV	11419
275	GGTFSSY	10896	NPNSGD	11158	ARDQRGGDAWDV	11420
276	GGTFSNY	10897	TPSGGS	11159	ARDTAGHFDI	11421
277	GGTFRND	10898	NPNSGN	11160	ARDNPDLDGMDV	11422
278	GGTFSSY	10899	SAYNGN	11161	ARDLVGHFDY	11423
279	GGTFSSY	10900	NPNSGG	11162	ARDGYSGSYSD	11424
264	GGTFSSY	10901	NPNSGG	11163	ARDGYSGSYSD	11425
257	GGTFSSY	10902	NPNSGG	11164	ARDGYSGSYSD	11426
280	GNTLSSH	10903	NPSGGS	11165	ARDQGSSGTFDY	11427
281	GGTLSSY	10904	NPNSGG	11166	ARDSTDVIDY	11428
282	GYIFTSY	10905	NPNSGD	11167	ARDGGTVTPTEEYYYYG-MDV	11429
283	GGTFSSY	10906	SVYNGN	11168	ASLDDLDY	11430
284	GHTFTSY	10907	NPNNGG	11169	ARDMVRDSAEYFQH	11431
285	GYTFITS	10908	NPSGGT	11170	ARDSSGYPIDY	11432
286	GYTFTSY	10909	IPLSGA	11171	ARGALYNWNDGWFDP	11433
258	GYTFTSY	10910	IPLSGA	11172	ARGALYNWNDGWFDP	11434
287	GFTVGSW	10911	WYEGSN	11173	ARLGTASLPYFDY	11435
288	GYTFTGY	10912	NPNRGD	11174	ARESGDGFDP	11436
289	GYTFTNY	10913	NPNSGN	11175	ARDWPNWFDP	11437

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
290	GYSFTDN	10914	RSDNGE	11176	AREVQLVGFDY	11438
291	GYTFSDH	10915	IPIFGT	11177	ARGSSWYLHFQH	11439
260	GYTFSDH	10916	IPIFGT	11178	ARGSSWYLHFQH	11440
292	GGTFSSY	10917	IPIFGT	11179	AKGVDRYNWNDAFDY	11441
293	GYTFTDY	10918	HSNSGG	11180	ARESSGYDSSLDY	11442
294	GGTFSSY	10919	NPNSGD	11181	TTDPRLDSSDPGY	11443
295	GGTFGNY	10920	SAYNGN	11182	ARGGMDV	11444
296	GGTFSRY	10921	YPSDGS	11183	ARDRLGDLDY	11445
261	GGTFSRY	10922	YPSDGS	11184	ARDRLGDLDY	11446
297	GGTFSSY	10923	NPNSGN	11185	ARDSIVGGYPFDY	11447
298	GYTFTSY	10924	TPIFGT	11186	AREGYSSSWHDDAFDI	11448
299	GGTFSNY	10925	DPSGGS	11187	ARDLGDYGLDS	11449
300	GYTFTGY	10926	NPNSGD	11188	ATGGSDSSGYYYEGYFQH	11450
301	GGTFSSY	10927	SPNSGN	11189	ARDKGGYYDSSGYYWY	11451
302	GFSLSSY	10928	SSNGGS	11190	ARVGDGDGYNPDFDY	11452
303	GYTFTSY	10929	DPTSGA	11191	AKDPIVATEVDY	11453
304	GGTFSSY	10930	SPNSGN	11192	ARDSGAFDI	11454
305	GVTISNY	10931	NPNSGN	11193	AREGLLDAFDI	11455
306	GGTFSRY	10932	NPYDGN	11194	ARGGRHHDAFDI	11456
307	GGTFSSY	10933	NPSGDG	11195	ARDISNDAFDI	11457
308	GYILTGH	10934	SAYNGD	11196	ARGSSWDDAFDI	11458
309	GFTFSNH	10935	GAGGG	11197	AREGWNDDVFDI	11459
310	GGTFSSY	10936	NPSAGT	11198	ARDGNFGAFDI	11460
311	GYSFTTY	10937	IPIFGT	11199	ARDKSGWNYGSGSYN-DAFDI	11461
312	GYAFTGY	10938	NPNSGK	11200	ARDGGLDFDY	11462
313	GYTFTTY	10939	NPNTGD	11201	AKDPAVTPDAFDI	11463
314	GGTLSSY	10940	DPSGGG	11202	AGSLYYYGMDV	11464
315	GGTFGSS	10941	IPIFGT	11203	AKEDDILPPRAFDI	11465
316	GFTFDDY	10942	SGGGGV	11204	ARVYSSGWLDAFDI	11466
317	GGTFSSY	10943	SGYNGN	11205	ASSDVSPDAFDI	11467
318	GGTQNIY	10944	NPNSGN	11206	ATPTSSSDDAFDI	11468
319	GGTFSSY	10945	NPNSGG	11207	ARASRGDDAFDI	11469
320	GIPFTSD	10946	NPSGGS	11208	ARERYEGGY-SSGPGNYYYGMDV	11470
321	GGTFSNY	10947	NPNSGN	11209	ARDDDYGDYPV	11471
322	GDTFSDH	10948	NPKIGN	11210	VYDSSGYDAFDI	11472

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
323	GYTFTSY	10949	NPGTGG	11211	ARETPSDYYDSSGYYYN-DAFDI	11473
324	GGTFSSY	10950	IPSGG	11212	ARDLGTTFDI	11474
325	GYTFTAY	10951	NPDNDN	11213	AKDIAVAALAYGMDV	11475
326	GFTFSSY	10952	SYDGSD	11214	ARQSLYYYYGMDV	11476
327	GYTFTDY	10953	STFTGN	11215	ARDAPLAAAGTDYYYG-MDV	11477
262	GYTFTDY	10954	STFTGN	11216	ARDAPLAAAGTDYYYG-MDV	11478
328	GFTFSSY	10955	SDDGIT	11217	ARDDSSGYGGMDV	11479
329	GFTFSSY	10956	SYDGGD	11218	ASGSLVLGYYYMDV	11480
330	GYTFTNY	10957	NPNTGG	11219	ATGGGGSYYDAFDV	11481
331	GGTFSSY	10958	NPNSGN	11220	ARDIGEGYSMDV	11482
332	GFTFSNH	10959	SYDGSN	11221	AREEKYSSSWY-VGVDAFDI	11483
333	GFTFSSS	10960	SGSGDN	11222	ARDQEDYYYDSSGYG-MDV	11484
334	GGTFSSH	10961	IPIFGT	11223	AKGDW-GIVVVPAAIGAFDI	11485
335	GYTFTAY	10962	SPVFGS	11224	ARDLGYYDSSGYRYDAF DI	11486
336	GYTFTSY	10963	SPMFGT	11225	AKDGWYYGMDV	11487
337	GGTFSSY	10964	NPNSGG	11226	ARGEAGNLDWYFDL	11488
338	GGTFSNY	10965	NPNNGD	11227	AREDVWYFDL	11489
339	GYTFTTY	10966	STYDGK	11228	ALHLGGDWYFDL	11490
340	GYTFTGY	10967	NPNTGA	11229	ARQHGDYDWYFDL	11491
341	GDTFTTY	10968	NPNSGN	11230	ARDSGRH	11492
342	GGTFSSY	10969	IPMLGI	11231	VREEVAGANWFDP	11493
343	GYTFTSY	10970	NPSGGS	11232	AREGDYGSGEFDY	11494
344	GYTFTSS	10971	NPRSGN	11233	ARERDDYGDYGWLDY	11495
345	GYTFTGY	10972	NPSGGS	11234	ARDLYDSSGY-WHYYYYMDV	11496
346	GGTFSSY	10973	NPNSGG	11235	ARFSGYDYVDY	11497
347	GGTFSSY	10974	NPNGGN	11236	ARDVGEDFDL	11498
348	GYTFTSY	10975	NPADGD	11237	ARDFDWLFAMDV	11499
349	GGTFSNY	10976	NPNGGT	11238	AKHGDHGFYV	11500
350	GGTFSSY	10977	NPNVGS	11239	AREDSGTSWFDP	11501
351	GYTFTSY	10978	NPSDGS	11240	ARDDRGSNYYYGMDV	11502
352	GYTFTAY	10979	NPNSGT	11241	ARDSSDYYGDYRADAFDI	11503

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
353	GYTFTSY	10980	SPSGDA	11242	VKGLDH	11504
354	GFSFSDY	10981	GGIGDS	11243	ARMNYGDSNYYYYYG-MDV	11505
355	GYTFTSY	10982	SPSDGS	11244	ARGAVGFDY	11506
356	GYTFTSY	10983	NTYSGY	11245	TTDDFLSFGY	11507
357	GYMFTDY	10984	IPYFGT	11246	ARSISGSYVLDAFDI	11508
358	GYTFNSY	10985	IPIFGT	11247	ARDWGYGDYADDAFDI	11509
359	GGTFSNN	10986	NPIYGS	11248	AADWRGFDY	11510
360	GYTFTEY	10987	NPHNGD	11249	AREGDYLGYPIDC	11511
361	GFTFSDY	10988	WQDGNV	11250	ARDGNSGYVF	11512
362	GYTFTTY	10989	NPNTGD	11251	ARTAEAVAGLPAFDY	11513
363	GGTSNNY	10990	IPLFGT	11252	ARVTLYGDYDY	11514
364	GYSLITH	10991	NPSDGV	11253	AREYYGEGFDY	11515
259	GYSLITH	10992	NPSDGV	11254	AREYYGEGFDY	11516
365	GGTFSSY	10993	NPSGGS	11255	ARDLGDTAMDG	11517
366	GYTFTSY	10994	TPSGGS	11256	ARDGGLASFDY	11518
367	GGTFSSY	10995	NPNSGN	11257	ARGGGWAMTDAFDI	11519
368	GFTFDDY	10996	YSGGD	11258	TRKEYYYDSSGYLRLFDY	11520
369	GYTFTDY	10997	NPIFGT	11259	ARDISGYDYYYYGMDV	11521
370	GGTLNNY	10998	DPSDGT	11260	ARSDYDFWSGLGGYFDY	11522
371	GGTFSSY	10999	DPNSGG	11261	ARDSAEWELGGSFDY	11523
372	GFTFSNH	11000	GVNGD	11262	AREGLVFSGRGHWYFDL	11524
373	GGTFSNY	11001	NPNGGN	11263	ARDYEDADFDG	11525
374	GYTFSDH	11002	NPDSGN	11264	ARDSTSGVDY	11526
375	GFTFSSY	11003	SYDGHD	11265	ARGEQQLEGFYYYYG-MDV	11527
376	GFTFSSY	11004	SYDGSK	11266	ASDYGDYGTYDY	11528
377	GFTFSSY	11005	SGGGDD	11267	AREPLAYCGG-DCPGGFDY	11529
378	GFTFSDH	11006	GTGGD	11268	ARHEDTAIFLDY	11530
379	GYTFTSY	11007	SPSDGS	11269	ARDGYDAWSYGMDV	11531
380	GYTFTGY	11008	NPNSGN	11270	ARDGVTGTDY	11532
381	GFAFSSY	11009	SGAGDS	11271	AREPTTVTDDWYFDL	11533
382	GFAFSSH	11010	SGNGDN	11272	ARDRAPEYFDL	11534
383	GGTFSSY	11011	NPNSGG	11273	ARDDYGDYGGGMDV	11535
384	GYTFTDY	11012	NPNSGH	11274	AKDTSPRYGDGFFDY	11536
385	GFTFSSY	11013	SYDGSN	11275	ARESGFSAEYFQH	11537

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
386	GGTFSSY	11014	NPSGGS	11276	ARATGLYCSGSCFDY	11538
388	RSILDFN	14119	ARAGA	14120	RVFDLPNDY	14121
	LCDR1		LCDR2		LCDR3	
	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
265	RASQDISN YLN	12194	KLLIYDASNLET	13104	QQSYSTPLT	12714
266	RASQSIS-SYLN	12195	KLLIYAASSLQS	13105	QQSYSTPLT	12715
267	QASQDIS NYLN	12196	KLLIYKASSLET	13106	QQSFSSPLT	12716
263	RASQDISN YLN	12197	KLLIYDASNLET	13107	QQSYSTPLT	12717
268	RASQNVN TWLA	12198	KLLIYEASSLQS	13108	QQANSFPFT	12718
269	RASQSISD WLA	12199	KLLIYAASSLQS	13109	AQHNHYPYT	12719
270	KSS-QSVLSSSY NKNYLA	12200	KLLIYWASTRES	13110	QQYYTTPFT	12720
271	KSS-QSVLSSSY NKNYLA	12201	KLLIYWASTRAS	13111	QQYYSTPFT	12721
272	RAS-QAIRDDLG	12202	KLLIYDASHLEA	13112	QQANSFPIT	12722
273	RASQGVG NDLA	12203	KLLIYAASTLQT	13113	QQASSFPLT	12723
274	RASQI-IGTNLA	12204	KLLIYAASSLQS	13114	QQSYTFPVT	12724
275	RASQSIST-WLA	12205	KLLIYDASSLES	13115	QQSYSTPFT	12725
276	KSS-QSVLSSS-NNKNYLA	12206	KLLIYWASTRES	13116	QQYYGSPLT	12726
277	KSS-QSVLSSSY NKNYLA	12207	KLLIYWASTRES	13117	QQYYSSPPT	12727
278	KSS-QSVLSSSY NKNYLA	12208	KLLIYWASTRES	13118	QQYYSSPPT	12728
279	KSS-QSVLSSSY NKNYLA	12209	KLLIYWASTRES	13119	QQYYSTPWT	12729
264	KSS-QSVLSSSY NKNYLA	12210	KLLIYWASTRES	13120	QQYYSTPWT	12730
257	KSS-QSVLSSSY NKNYLA	12211	KLLIYWASTRES	13121	QQYYSTPWT	12731

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
280	KSS-QSVLSSSYNKNYLA	12212	KLLIYWASTRAS	13122	QQYYGSPPT	12732
281	QASQDIRNYLN	12213	KLLIYDASTLQS	13123	QQAYSFPWT	12733
282	QASQDISNYLN	12214	KLLIYNASNLET	13124	QQLNSYPFT	12734
283	QASQSISTWLA	12215	KLLIYAASTLRS	13125	LQHYTYPLT	12735
284	RASEDI-STYLA	12216	KLLIYAASTLQS	13126	QQSHTIPWT	12736
285	RASH-HISDFLN	12217	KLLIYAASTLQS	13127	QQSYSSPYT	12737
286	RASQDIGDYLA	12218	KLLIYDASSLQS	13128	QQANSFPLT	12738
258	RASQDIGDYLA	12219	KLLIYDASSLQS	13129	QQANSFPLT	12739
287	RASQDIRSYLA	12220	KLLIYAASSLQS	13130	QQSYTAPPT	12740
288	RASQDIS-NNLN	12221	KLLIYAASTLQS	13131	LQHNTYPLT	12741
289	RASQDIS-NWLA	12222	KLLIYDASSLQS	13132	QQAISFPLT	12742
290	RASQGIANYLA	12223	KLLIYAASSLQS	13133	QQADSFPLT	12743
291	RASQGIASYLA	12224	KLLIYAASTLQP	13134	QQFDSYPIT	12744
260	RASQGIASYLA	12225	KLLIYAASTLQP	13135	QQFDSYPIT	12745
292	RASQGISNYLA	12226	KLLIYAASRLQS	13136	QQSSIIPFT	12746
293	RASQGISNYLA	12227	KLLIYAASTLQS	13137	QQAYSFPYT	12747
294	RASQSIGRWLA	12228	KLLIYDASNLET	13138	QQSYSTPRT	12748
295	RASQSINSWLA	12229	KLLIYDTSSLQS	13139	QQTYSTPYT	12749
296	RASQSISSWLA	12230	KLLIYAASTLQS	13140	QQGYSTPYI	12750
261	RASQSISSWLA	12231	KLLIYAASTLQS	13141	QQGYSTPYI	12751
297	RASQSIS-SYLN	12232	KLLIYAASSLQS	13142	QQTDSIPIT	12752
298	RASQSIS-SYLN	12233	KLLIYAASTLQS	13143	QQSYSIPYT	12753
299	RASQTIRSYLN	12234	KLLIYKASSLES	13144	QQTYTIPIT	12754

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
300	RASQTISN WLA	12235	KLLIYAASTLQS	13145	QQANSFPPT	12755
301	RASQYIGS YLN	12236	KLLIYDASNLET	13146	QQVDSYPLT	12756
302	RSSQSLLH SNGYNYL D	12237	QLLIYLGSNRAS	13147	MQGTHWPPT	12757
303	RSSQSLLH SNGYNYL D	12238	QLLIYFGSNRAS	13148	MQALQAPVS	12758
304	KSS- QSVLSSSY NKNYLA	12239	KLLIYWASSRQS	13149	QQYYSTPLT	12759
305	KSS- QSVSSSSY NKNYLA	12240	KLLIYWASVRES	13150	QQYYSTPIT	12760
306	KSTQNVLS SSNNN- SYLA	12241	KLLIYWASTRES	13151	QQYYSTPFT	12761
307	QASQDIG NYLN	12242	KLLIYAASSLQS	13152	QQTYNTPLT	12762
308	QASQDIS NYLN	12243	KLLIYEASTLQS	13153	QQSYSTPFT	12763
309	QASQDIST WLA	12244	KLLIYRASTLES	13154	QQSYSIPLT	12764
310	RASQNIN- NYLN	12245	KLLIYAASRLQS	13155	QQSYSAPVT	12765
311	RASQNINT WLA	12246	KLLIYAASSLQS	13156	QQAYSFPFT	12766
312	RASQRIGN YLN	12247	KLLIYAASSLQS	13157	QQSYSTPLT	12767
313	RASQSIST YLN	12248	KLLIYAASTLQS	13158	QQSYRTVT	12768
314	RASQSVG SYLA	12249	RLLIYGASTRAT	13159	QQYDSSSQT	12769
315	RASRSVST YLA	12250	RLLIYGASTRAT	13160	QQYDGSPYT	12770
316	RSSQSLLH SNGYNYL D	12251	QLLIYDASNLET	13161	MQALQTPPA	12771
317	KSS- QSVLSSSY NKNFLA	12252	KLLIYWASTRES	13162	QQYYSAPPT	12772
318	KSS- QSVLSSSY NKNFLA	12253	KLLIYWASTRES	13163	QQYYSDPIT	12773
319	KSS- QSVLSSSY NKNYLA	12254	KLLIYWASARES	13164	QQYYSIPIA	12774
320	KSS- QSVLSSSY NKNYLA	12255	KLLIYWASTRDS	13165	QQYYSIPYT	12775

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
321	KSS-QSVLSTSY NKNYLA	12256	KLLIYWASTRAS	13166	QQYYTTPPT	12776
322	KSS-QSVLSTSY NRNFLA	12257	KLLIYWASTRQS	13167	QQYYSTPYT	12777
323	KSS-QSVLYSSN NKNYLA	12258	KLLIYWASTRES	13168	QQYYSTPLT	12778
324	QASQDIS NYLN	12259	KLLIYAASSLQS	13169	QQSYSTPT	12779
325	QASQDIS NYLN	12260	KLLIYGASTLQS	13170	QEADSFPLT	12780
326	RASQGIRN DLG	12261	KLLIYDASSLHS	13171	QQAYSFPWT	12781
327	RASQGISN YLA	12262	KLLIYKASSLES	13172	QQSYNTPFT	12782
262	RASQGISN YLA	12263	KLLIYKASSLES	13173	QQSYNTPFT	12783
328	RASQSINR WLA	12264	KLLIYSASNLQS	13174	QQSYNTPLT	12784
329	RASQSINT WLA	12265	KLLIYAASSLQS	13175	QQANSFPFT	12785
330	RASQSIRT-WLA	12266	KLLIYDASSLET	13176	QQLNSYPLT	12786
331	RASQSIRT YLN	12267	KLLIYAASTLQS	13177	QQSYSAPLT	12787
332	RASQSIST YLN	12268	KLLIYAASSLHS	13178	QQSYSTPLT	12788
333	RASQSIT-TYLN	12269	KLLIYAASTLQS	13179	QQSYSTPLT	12789
334	RSSQSLLH SNGYNYL D	12270	QLLIYAASSLQS	13180	MQARQTPLT	12790
335	RSSQSLLH SNGYNYL D	12271	QLLIYGASSLQS	13181	MQTLQTPFT	12791
336	RSSQSLLH SNGYNYL D	12272	QLLIYLGSDRAS	13182	MQALQTPLT	12792
337	KSSQTVF-STSYN-KNYLA	12273	KLLIYWASTRES	13183	QQYYSTPLT	12793
338	KTSQSVF-STSYN-RDYLA	12274	KLLIYWASTRES	13184	QQYYSSPPT	12794
339	RASQSISS WLA	12275	KLLIYDASTLQS	13185	QQSYSTPFT	12795
340	RASQSIS-SYLN	12276	KLLIYDASNLKT	13186	QQSYSFPT	12796

TABLE 23-continued

CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences						
Binder Name						
341	RASQSVSS YLA	12277	RLLIYDTSSRAT	13187	QQYYDTPYT	12797
342	KSS-QSVLYSSN NKNYLA	12278	KLLIYLASTREP	13188	QQYYSTPPT	12798
343	KSS-QSVLSSSY NKNYVA	12279	KLLIYWASTRES	13189	QQYYSTPLT	12799
344	QASQDIS NYLN	12280	KLLIYAAASLQS	13190	QQTYSTPWT	12800
345	RASQDIN-TYLA	12281	KLLIYAASSLQS	13191	QQSSSFPLT	12801
346	QASQDIS NYLN	12282	KLLIYAASSLQS	13192	QQLYNFPYT	12802
347	RASQSISR YLA	12283	KLLIYGASTRES	13193	QQSYNTPLT	12803
348	RASQTLSG WLA	12284	KLLIYGASTLQG	13194	QQYYSYPPT	12804
349	FASQDI-INYLN	12285	KLLIYEASNLET	13195	QQSYSTPLT	12805
350	RASQSIS-SYLN	12286	KLLIYDVFNLGT	13196	QQSYSSPFT	12806
351	QASQDIS NYLN	12287	KLLIYMASNLES	13197	QQTNSFPLT	12807
352	RASQSIS-SYLN	12288	KLLIYDASNLET	13198	QQSYSTPLT	12808
353	RSSQSLLH SNGYNYL D	12289	QLLIYLGSNRAS	13199	MQALQSPWT	12809
354	KSS-QSVLYSSN NKNYLA	12290	KLLIYWASTRES	13200	QQYYSSPLT	12810
355	RSSQSLLH SNGYNYL D	12291	QLLIYLGSNRAS	13201	MQALQTPPS	12811
356	RAS-ESVST-WLA	12292	KLLIYKASRLES	13202	QQSYKTPYT	12812
357	KSS-QSVLYSSN NKNYLA	12293	KLLIYWASTRES	13203	QQYFTTPLT	12813
358	RASQSIS-SYLN	12294	KLLIYAASSLQS	13204	QQSYSTPYT	12814
359	KSS-QSVLSSSY NKNYLA	12295	KLLIYWASTRAS	13205	QQYYDTPLT	12815
360	RASQSIS-SYLN	12296	KLLIYKASTLES	13206	QQNDSIPIT	12816
361	RASQSISR WLA	12297	KLLIYDASNLET	13207	LQDYSYPLT	12817

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
362	KTSQSVF-STSYN-RDYLA	12298	KLLIYWASTRAA	13208	QQYYYTST	12818
363	RASQSIN-RYLN	12299	KLLIYAASSLQS	13209	QQANSFPPT	12819
364	RASQGISN YLA	12300	KLLIYSASNLQS	13210	QQSYSTPLT	12820
259	RASQGISN YLA	12301	KLLIYSASNLQS	13211	QQSYSTPLT	12821
365	RASQSIDS YLN	12302	KLLIYKASTLES	13212	QQSYSAPLT	12822
366	RASQDIST WLA	12303	KLLIYDASNLET	13213	QQVNSDPYT	12823
367	QASQDIS NYLN	12304	KLLIYAASTLES	13214	QQGDSLPLT	12824
368	RASQGISN YLA	12305	KLLIYAASSLQS	13215	QQSDSFPYT	12825
369	RASQSVST YLA	12306	RLLIYGASTRAT	13216	QQHDSYPLT	12826
370	RASQGIRN DLG	12307	KLLIYAASSLQS	13217	QQANSFPPT	12827
371	RAS-ESISTYLN	12308	KLLIYKASNLES	13218	QQTDSTFIT	12828
372	RASRNIHD YLN	12309	KLLIYAASTLQT	13219	QQTYSTPPT	12829
373	RASQSND SYLN	12310	KLLIYKASTLES	13220	QQSYSSPLT	12830
374	RASQSISD FLN	12311	KLLIYAASTLQS	13221	QQSYSSPYT	12831
375	QASQDIS NYLN	12312	KLLIYAASSLQS	13222	QQANRFPLT	12832
376	QASQDIS NYLN	12313	KLLIYKASNLQS	13223	QQSYNFPAT	12833
377	RSSQSLLH SNGYNYL D	12314	QLLIYLGSNRAS	13224	MQGTHWPET	12834
378	RASQSIS-SYLN	12315	KLLIYDASNLET	13225	QQSYSTPLT	12835
379	RASQGISD YLA	12316	KLLIYDASNLET	13226	QQSYILPLT	12836
380	RASQDIN DFLA	12317	KLLIYAASSLQS	13227	QQSYSAPYT	12837
381	RASQSISN WLA	12318	KLLIYAASKLES	13228	QQSYSSPWT	12838
382	RASQGIDS WLA	12319	KLLIYAASTLES	13229	QQAYSFPLT	12839
383	RASQNIGT WLA	12320	KLLIYRASSLES	13230	QQAYSFPWT	12840

TABLE 23-continued

CDRs using the Chothia Numbering Scheme
Table 5-Chothia CDR Sequences

Binder Name						
384	RASQNIN NWLA	12321	KLLIYKASTLQS	13231	QQADSFPPT	12841
385	RASQDIS-SYLA	12322	KLLIYAASTLQS	13232	QQLNRYPIT	12842
386	RASQDISN YLA	12323	KLLIYAASILHS	13233	QQYDSSFIT	12843

TABLE 24

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
	HCDR1		HCDR2		HCDR3	
109	GYTFSSYW	4361	ILPGSGST	4875	ARRAYGYDGGFDY	4105
110	GYTFSSYW	4362	ILPGSGST	4876	ARRAYGYDEGFDY	4106
111	GYTFSSYW	4363	ILPGSDST	4877	ARRAYGYDEGFDY	4107
112	AYTFSIYW	4364	ILPGSGST	4878	ARRAYGYDGGFDY	4108
1	GYTFSSYW	4365	IFPGSGHT	4879	ARRGYGYDEGFDY	4109
113	GYTFSSYW	4366	ILPGSGST	4880	ARRGYGYDEGFDY	4110
114	GYTFSSYW	4367	ILPGSGST	4881	ARRGYGYDEGFDY	4111
115	GYTFSNYW	4368	ILPGSGST	4882	ARRGYGYDEGFDY	4112
23	GYTFSSYW	4369	ILPGSGST	4883	ARRGYGYDEGFDY	4113
116	GYTFSSYW	4370	ILPGSGYT	4884	ARRGYGYDEGFDY	4114
117	GYTFSSYW	4371	ILPGSGST	4885	ARRAYGYDEGFDY	4115
2	GYTLSSYW	4372	ILPGSGST	4886	ARRGYGYDEGFDY	4116
118	GYTFSSYW	4373	VLPGSGST	4887	ARRAYGYDEGFDY	4117
119	GYTFSSYW	4374	ISPGSGST	4888	ARRGYGYDEGFDY	4118
120	GYTFGTYW	4375	ILPGSGTP	4889	ARRAYGYDAGFDY	4119
121	GYTFSSYW	4376	ILPGSGST	4890	ARRGYGYDEGFDY	4120
122	GYTFSSYW	4377	ILPGSGRT	4891	ARRGYGYDEGFDY	4121
123	GYTFSSYW	4378	ILPGSGRT	4892	ARRGYGYDEGFDY	4122
38	GYTFSSYW	4379	ILPGSGST	4893	ARRAYGYDEGFDY	4123
39	GYTFSSYW	4380	VLPGSGST	4894	ARRAYGYDEGFDY	4124
40	GYTFSSYW	4381	ILPGSGRT	4895	ARRGYGYDEGFDY	4125
41	GYTFSSYW	4382	VLPGSGST	4896	ARRAYGYDEGFDY	4126
42	GYTFSSYW	4383	ILPGSGRT	4897	ARRGYGYDEGFDY	4127
43	GYTFSSYW	4384	ILPGSGST	4898	ARRAYGYDGGFDY	4128
44	GYTFSSYW	4385	ILPGSDST	4899	ARRAYGYDEGFDY	4129
45	AYTFSIYW	4386	ILPGSGST	4900	ARRAYGYDGGFDY	4130
46	GYTFSSYW	4387	IFPGSGHT	4901	ARRGYGYDEGFDY	4131
47	GYTFSNYW	4388	ILPGSGST	4902	ARRGYGYDEGFDY	4132
47	GYTFSNYW	4388	ILPGSGST	4902	ARRGYGYDEGFDY	4132
48	GYTFSSYW	4389	ILPGSGST	4903	ARRGYGYDEGFDY	4133
48	GYTFSSYW	4389	ILPGSGST	4903	ARRGYGYDEGFDY	4133
49	GYTFSSYW	4390	ILPGSGYT	4904	ARRGYGYDEGFDY	4134
50	GYTLSSYW	4391	ILPGSGST	4905	ARRGYGYDEGFDY	4135
51	GYTFSSYW	4392	ISPGSGST	4906	ARRGYGYDEGFDY	4136
52	GYTFGTYW	4393	ILPGSGTP	4907	ARRAYGYDAGFDY	4137
53	GYTFSSYW	4394	ILPGSGST	4908	ARRGYGYDEGFDY	4138
54	GYTFSSYW	4395	ILPGSGRT	4909	ARRGYGYDEGFDY	4139
55	GYTFSNYW	4396	ILPGSGST	4910	ARRGYGYDEGFDY	4140
56	GYTFSSYW	4397	ILPGSGST	4911	ARRGYGYDEGFDY	4141
3	GYSFTGYY	4398	ISSYNGAT	4912	ARGRYGEYFDY	4142
4	GYSFTGYY	4399	ISSYNGVT	4913	ARGRYGDYFDY	4143
5	GYSFTGYY	4400	ISSYNGVT	4914	ARGRYGDYFDY	4144
6	GYSFTGYY	4401	ISSYNGVT	4915	ARGRYGDYFDY	4145
7	GYSFTGYY	4402	ISSYNGAN	4916	ARGRYGDYFDY	4146
8	GYSFTGYY	4403	ISSYNGVT	4917	ARGRYGDYFDY	4147
9	GYSFTGYY	4404	ISSYNGVT	4918	ARGRYGDYFDY	4148
10	GYSFTGYY	4405	ISSYNGVT	4919	ARGRYGDYFDY	4149
11	GYSFTGYY	4406	ISSYNGVT	4920	ARGRYGDYFDY	4150

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
12	GYSFTGFY	4407	ISSYNGAT	4921	ARGRYGDYFDY	4151
13	GYSFTGYY	4408	ISSYNGAT	4922	ARGRYGDYFDY	4152
57	GYSFTGYY	4409	ISSYNGVT	4923	ARGRYGDYFDY	4153
58	GYSFTGYY	4410	ISSYNGAT	4924	ARGRYGEYFDY	4154
124	GFSLSSYG	4411	IWRGGST	4925	AKNLYGHYVMDY	4155
125	GFSVTSYG	4412	IWRGGST	4926	AKNLYGHYVMDY	4156
126	GFSLTSYG	4413	IWRGGST	4927	AKNLYGHYVMDY	4157
127	GFSLTRYG	4414	IWRGGST	4928	AKNLYGHYVMDY	4158
128	GFSVTTYG	4415	IWRGGST	4929	AKNLYGHYVMDY	4159
129	GFSVTSYG	4416	IWRGGST	4930	AKNLYGHYVMDY	4160
130	GFSLTRYG	4417	IWRGGST	4931	AKNLYGHYVMDY	4161
59	GFSLSSYG	4418	IWRGGST	4932	AKNLYGHYVMDY	4162
60	GFSVTSYG	4419	IWRGGST	4933	AKNLYGHYVMDY	4163
61	GFSLTSYG	4420	IWRGGST	4934	AKNLYGHYVMDY	4164
62	GFSLTRYG	4421	IWRGGST	4935	AKNLYGHYVMDY	4165
63	GFSVTTYG	4422	IWRGGST	4936	AKNLYGHYVMDY	4166
131	GYTFTSYW	4423	IHPNSGST	4937	ARWGDGYSFAY	4167
132	GYTFTSYW	4424	IHPNSGST	4938	ARWGDGYSFAY	4168
133	GYTFTTYW	4425	IHPNSDNT	4939	ARWGDGYSFAY	4169
14	GYTFTSYW	4426	IHPNSGTT	4940	ARWGDGYSFAY	4170
134	GYTFTSYW	4427	IHPNSGNT	4941	ARWGDGYSFAY	4171
64	GYTFTSYW	4428	IHPNSGST	4942	ARWGDGYSFAY	4172
65	GYTFTSYW	4429	IHPNSGTT	4943	ARWGDGYSFAY	4173
135	GYTFTDYV	4430	IYPGSGST	4944	ARRGERGPWFAY	4174
136	GYTFTDYV	4431	IYPGSGSS	4945	ARRGERGPWFAY	4175
137	GYTFTDYV	4432	IYPGSGSS	4946	ARRGERGPWFAY	4176
138	GYTFTDYV	4433	IYPGSGSS	4947	ARRGERGPWFAY	4177
15	GYTFTDYV	4434	IYPGSGSS	4948	ARRGERGPWFAY	4178
66	GYTFTDYV	4435	IYPGSGST	4949	ARRGERGPWFAY	4179
67	GYTFTDYV	4436	IYPGSGSS	4950	ARRGERGPWFAY	4180
68	GYTFTDYV	4437	IYPGSGSS	4951	ARRGERGPWFAY	4181
69	GYTFTDYV	4438	IYPGSGSS	4952	ARRGERGPWFAY	4182
24	GYTFTNYW	4439	IDPSDSET	4953	ATYDVYYRFAY	4183
139	GYTFTNYW	4440	IDPSDSET	4954	ATYDGYYRFAY	4184
140	GYTFTNYW	4441	IDPSDSET	4955	ATYDIYYRFAY	4185
16	GYTFTSYW	4442	IHPNSGST	4956	ARPGGYGFVY	4186
141	GYTFTSYW	4443	IHPNSDST	4957	ARPGGYGFAD	4187
142	GYTFTTYW	4444	IHPNSGST	4958	ARPGGYGFTY	4188
143	GYTFTSYW	4445	IHPNSGSP	4959	ARPGGYGFAY	4189
70	GYTFTSYW	4446	IHPNSGSP	4960	ARPGGYGFAY	4190
25	GYTFTSYW	4447	IYPSDSYT	4961	TRGNYIDY	4191
144	GYTFTSYW	4448	IYPSDSYT	4962	TRGNYIDY	4192
145	GYTFTDYW	4449	IYPSDSYT	4963	TRGNYIDY	4193
146	GYTFTDYV	4450	IYPGSGSS	4964	ARPGDLGFAY	4194
147	GYTFTDYV	4451	IYPGSGSN	4965	ARPGDLGFAY	4195
148	GYTFTDYV	4452	IYPGSGSS	4966	ARPGDLGFAY	4196
71	GYTFTDYV	4453	IYPGSGSS	4967	ARPGDLGFAY	4197
72	GYTFTDYV	4454	IYPGSGSS	4968	ARPGDLGFAY	4198
73	GYTFTDYV	4455	IYPGSGSN	4969	ARPGDLGFAY	4199
74	GYTFTDYV	4456	IYPGSGSS	4970	ARPGDLGFAY	4200
74	GYTFTDYV	4456	IYPGSGSS	4970	ARPGDLGFAY	4200
75	GYTFTDYV	4457	IYPGSGSS	4971	ARPGDLGFAY	4201
75	GYTFTDYV	4457	IYPGSGSS	4971	ARPGDLGFAY	4201
76	GYTFTDYV	4458	IYPGSGSS	4972	ARPGDLGFAY	4202
149	GFSLTNYG	4459	VWAGGIT	4973	ARGDGYDDGYAMDY	4203
150	GFSLTSYG	4460	LWAGGIT	4974	ARGDGYDDGYAMDY	4204
26	GFSLTSYG	4461	IWAGGTT	4975	ARGDGYDDGYAMDY	4205
77	GFSLTNYG	4462	VWAGGIT	4976	ARGDGYDDGYAMDY	4206
78	GFSLTSYG	4463	LWAGGIT	4977	ARGDGYDDGYAMDY	4207
79	GFSLTSYG	4464	IWAGGTT	4978	ARGDGYDDGYAMDY	4208
151	GYSFTSYW	4465	IDPSDSET	4979	ARTRNY	4209
152	GYSFTSYW	4466	IDPSDSET	4980	ARTRNY	4210
153	GYSFTSYW	4467	IDPSDSET	4981	ARTRNY	4211
154	GFNIKDYY	4468	IDPENGDT	4982	NAPLLRYSSAMDY	4212
155	GFNIKDYY	4469	IDPENGDT	4983	NAPLLRYSSSMDY	4213
156	GFNIKDYY	4470	IDPENGDT	4984	NVALLRYSSAMDY	4214
80	GFNIKDYY	4471	IDPENGDT	4985	NAPLLRYSSAMDY	4215
81	GFNIKDYY	4472	IDPENGDT	4986	NAPLLRYSSSMDY	4216
82	GFNIKDYY	4473	IDPENGDT	4987	NVALLRYSSAMDY	4217
17	GFNIKDTS	4474	IDPANGNT	4988	ARGPDDGYFYYYSMDY	4218
157	GYTFSNYY	4475	INPSNGDT	4989	TSYYTHEAYYYAMDC	4219

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
27	GSTFTTYY	4476	INPSNGGT	4990	TSYYTHETYYYAMDY	4220
158	GFNIKDYY	4477	IDPEDGDT	4991	TPYSIYDAMDY	4221
159	GYTFTDYV	4478	IYPGSGST	4992	ARRGERGPWFAY	4222
83	GYTFTDYV	4479	IYPGSGST	4993	ARRGERGPWFAY	4223
160	GYSFTDYG	4480	ISTYYGDA	4994	ARQMDYDYTYYYAMDY	4224
28	GYTFTSYW	4481	IDPSDSYT	4995	ARAEYGYGNYPWFAY	4225
84	GYTFTSYW	4482	IDPSDSYT	4996	ARAEYGYGNYPWFAY	4226
29	GYTFTSYW	4483	IHPSDSDT	4997	AIPYYYGGWYFDV	4227
161	GYTFTDYV	4484	IYPGSGST	4998	ARMDGPWFAY	4228
30	GFTFSSYG	4485	ISSGGSYT	4999	ARLYDAHWDYFDY	4229
162	GISLSTSGMG	4486	IWNNDN	5000	AWRPYYRYDSFAY	4230
18	GYTFTNYG	4487	INTYTGEP	5001	ARKYYDYEFAY	4231
85	GYTFTNYG	4488	INTYTGEP	5002	ARKYYDYEFAY	4232
163	GYTFTDYE	4489	IDPETGGT	5003	TRLGDYDVMDY	4233
86	GYTFTDYE	4490	IDPETGGT	5004	TRLGDYDVMDY	4234
164	GYTFTSYW	4491	IDPSDSYT	5005	ARAGRYGSSFDY	4235
165	GFSLSTSGMG	4492	IYWDDDK	5006	AGRPDDYDGAWFPY	4236
31	GYTFTSSW	4493	IHPNSGNT	5007	AIYYDYDAYYFDY	4237
87	GYTFTSSW	4494	IHPNSGNT	5008	AIYYDYDAYYFDY	4238
32	GYTFTSYW	4495	IHPNSGST	5009	ANPYYGYDVGY	4239
166	GYTFTDYV	4496	IYPGSGSN	5010	AREEKIYFDY	4240
88	GYTFTDYV	4497	IYPGSGSN	5011	AREEKIYFDY	4241
167	GYTFTSYW	4498	IHPNSGST	5012	ARYDGYWFDY	4242
168	GYTFTSYW	4499	IYPGNSDT	5013	TSLITTAYYFDY	4243
89	GYTFTSYW	4500	IYPGNSDT	5014	TSLITTAYYFDY	4244
169	GYTFTSYW	4501	IHPNSGST	5015	APETGDYGSSYVWYFDV	4245
170	GYTFTDYV	4502	IYPGSGST	5016	ARGKVTRFAY	4246
171	GFTFSSYA	4503	ISDGGSYT	5017	ARDQDSNWEYFDY	4247
172	GYTFTDYS	4504	INTETGEP	5018	ARESWDRAMDY	4248
19	GFTFSSYA	4505	ISSGGSYT	5019	ARHEEANWAWFAY	4249
90	GFTFSSYA	4506	ISSGGSYT	5020	ARHEEANWAWFAY	4250
173	GYSFTNYW	4507	IDPSDSET	5021	AIPYYAMDY	4251
91	GYSFTNYW	4508	IDPSDSET	5022	AIPYYAMDY	4252
174	GYTFTSSW	4509	IHPNSGNT	5023	ATYYGNYVWYFDV	4253
92	GYTFTSSW	4510	IHPNSGNT	5024	ATYYGNYVWYFDV	4254
175	GYTFTSYW	4511	IHPNSGST	5025	ASYGSSYWYFDV	4255
93	GYTFTSYW	4512	IHPNSGST	5026	ASYGSSYWYFDV	4256
20	GFSLTSYG	4513	IWSGGST	5027	ASYYGSSRSYWYLDV	4257
94	GFSLTSYG	4514	IWSGGST	5028	ASYYGSSRSYWYLDV	4258
176	GYTFTSYN	4515	LYSGNGDT	5029	ARDYYGSSHLWYFDV	4259
177	GFSLSTSGMG	4516	IYWDDDK	5030	ARRAHYDYGWYFDV	4260
178	GYTFTSYW	4517	IHPNSGST	5031	AGYDYDWYFDV	4261
33	GFTFSSYG	4518	ISSGGSYT	5032	TRHDDSSYDWFAY	4262
179	GFTFSSYG	4519	ISSGGSYT	5033	ARHEDSNYHYFDY	4263
34	GYTFTNYW	4520	IHPNSGTT	5034	ARFGDGYHFDY	4264
180	GFTFSSYG	4521	ISSGGSYT	5035	ARQNDSSWAWFAY	4265
95	GFTFSSYG	4522	ISSGGSYT	5036	ARQNDSSWAWFAY	4266
181	GYTFTSYW	4523	IHPNSGST	5037	ALPYSNYGWYFDV	4267
96	GYTFTSYW	4524	IHPNSGST	5038	ALPYSNYGWYFDV	4268
182	GYTFTSYW	4525	IDPSDSET	5039	ARDYYGSYWYFDV	4269
97	GYTFTSYW	4526	IDPSDSET	5040	ARDYYGSYWYFDV	4270
183	GFNIKDYY	4527	IDPEDGET	5041	AAYGNSAWFAY	4271
98	GFNIKDYY	4528	IDPEDGET	5042	AAYGNSAWFAY	4272
35	GYTFTNYG	4529	INTNTGEP	5043	ARWYPYFDY	4273
99	GYTFTNYG	4530	INTNTGEP	5044	ARWYPYFDY	4274
100	GYTFTNYG	4531	INTNTGEP	5045	ARWYPYFDY	4275
36	GYTFTSYW	4532	INPSSGYT	5046	ARSDGSSGNWYFDV	4276
101	GYTFTSYW	4533	INPSSGYT	5047	ARSDGSSGNWYFDV	4277
184	GFSLTSYG	4534	IWAGGST	5048	AREGGYTGYFDV	4278
102	GFSLTSYG	4535	IWAGGST	5049	AREGGYTGYFDV	4279
185	GYTFTSYW	4536	IDPSDSET	5050	AYSNYVPYYAMDY	4280
103	GYTFTSYW	4537	IDPSDSET	5051	AYSNYVPYYAMDY	4281
186	GYTFTDYV	4538	IYPGSGSA	5052	ARRGFDY	4282
21	GFTFSSYG	4539	ISSGGSYT	5053	ARHNYSNWDWFAY	4283
187	GYTFTSYW	4540	IHPNSGST	5054	ARDYYGSGYGYYFDY	4284
188	GYTFTSYW	4541	IHPNSGST	5055	ARDYYGSSYGWYFDV	4285
189	GYTFTSYW	4542	IHPNSGST	5056	ARDYYGSSYGWYFDV	4286
190	GYTFTSYW	4543	IHPNSGST	5057	ASDYYGSSYGWYFDV	4287
191	GYTFTSYW	4544	IHPNSGST	5058	ARDYYGSSYGWYFDV	4288
192	GYTFTSYW	4545	IHPNSGST	5059	TRDYYGSGYGWYFDV	4289
193	GYTFTNYW	4546	IDPSDSET	5060	ATYDGYYRFAY	4290

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
194	GYTFTNYW	4547	IDPSDSET	5061	ATYDVYYRFAY	4291
195	GYTFTSYW	4548	IHPNSGST	5062	ARDYGNYDYAMDY	4292
104	GYTFTSYW	4549	IHPNSGST	5063	ARDYGNYDYAMDY	4293
37	GYTFTSYW	4550	IHPNSGST	5064	ARDYGNYDYAMDY	4294
196	GYTFTSYW	4551	IHPNSGST	5065	ARDYGNYDYAMDY	4295
197	GFTFSSYG	4552	ISSGGSYT	5066	ASQLTGTWYYFDY	4296
198	GFTFSSYG	4553	ISSGGSYT	5067	ASQLTGTWYYFDY	4297
199	GFTFSSYG	4554	ISSGGSYT	5068	ASQLTGTWYYFDY	4298
22	GFNIKDTS	4555	IDPANGNT	5069	ARGPDDGYFYYYSMDY	4299
200	GFTFSNYY	4556	INSNGGST	5070	ARQEGIGYAMDY	4300
201	GYTFTEYT	4557	IYPNNGGT	5071	ARGGWLLGY	4301
202	GFSLTSYG	4558	IWSGGST	5072	ARDGGIRGAMDY	4302
203	GYTFSSYW	4559	ILPGSGST	5073	ARRGYGYDEGFDY	4303
204	GYTFTDYE	4560	IDPETGGT	5074	TRNYDYAMDY	4304
205	GFTFSSYY	4561	INSNGGST	5075	ARQEGIGYALDY	4305
206	GFTFSSYA	4562	ISSGGST	5076	AREREWGVYYGSSLDY	4306
207	GFNIKDTY	4563	IDPANGNT	5077	ARSDGNYD	4307
208	GFTFSNYY	4564	INSNGGST	5078	ARQEGIGYGMDY	4308
209	GFTFNTYV	4565	IRSKSDNYAT	5079	VRHDGVVGFDV	4309
210	GYSITSGYY	4566	ISYDGSN	5080	ARGGGRG	4310
211	GYTFTDYS	4567	INTETGEP	5081	ARDYYDYYYAMDY	4311
212	GYTFTDYS	4568	INTETGEP	5082	ARESWDRAMDY	4312
213	GYTFTNYW	4569	IDPYDSET	5083	ARIYSDYDGAWFAY	4313
214	GYTFTDYY	4570	VNPYNGGT	5084	ARGTVGFAY	4314
215	GFTFSSYA	4571	ISSGGST	5085	AREREWGVFYGSSLDY	4315
216	GFTFSSYA	4572	ISSGGSYT	5086	ARHDDSSYGYFDY	4316
217	GFTFSNYA	4573	ISSGGTT	5087	ARTMPDV	4317
218	GFSLTSYG	4574	IWAGGST	5088	ARDTDGYYWAMDY	4318
219	GYSITSDHA	4575	ISYSGST	5089	ARKWGDY	4319
220	GYTFTDYE	4576	IDPETGGT	5090	TRNYDYALDY	4320
221	GYSITSGYY	4577	ISYDGSN	5091	ARGGGRG	4321
222	GFTFSNYY	4578	INSNGGST	5092	ARQEEIGYAMDY	4322
223	GFNIKDYF	4579	IDPETDNT	5093	ARSGNMGFTY	4323
105	GFNIKDYF	4580	IDPETDNT	5094	ARSGNMGFTY	4324
224	GFTFSSYA	4581	ISSGGSYT	5095	ASQGGSSWGAMDY	4325
106	GFTFSSYA	4582	ISSGGSYT	5096	ASQGGSSWGAMDY	4326
225	GFTFSSYA	4583	ISNGGSYT	5097	ARHEITTRFAY	4327
226	GYSITSGYY	4584	MSYDGSN	5098	AREAGYFDY	4328
227	GFSFNTYA	4585	IRSKSNNYAT	5099	VRQYGYDFDY	4329
228	GFTFSSYG	4586	ISSGGSYT	5100	ARHKGVNWDYFDY	4330
229	GYTFTDYE	4587	IDPETGGT	5101	TRGDGNYDSWYFDV	4331
230	GFTFSSYA	4588	ISSGGSYT	5102	ARLPVTTVVFDY	4332
231	GFTFSSYA	4589	ISSGGSYT	5103	ARRPVVVPFDY	4333
232	GFSLTSYG	4590	IWSGGST	5104	ARGWDADYFDY	4334
233	GYTFTNYW	4591	IHPNSGST	5105	TRYDYDDY	4335
234	GYTFTDYY	4592	INPNNGGT	5106	ARSELGLYAMDY	4336
235	GYTFTGYW	4593	ILPGSGST	5107	ARGRIHYFDY	4337
236	GYTFTGYW	4594	ILPGSGST	5108	ARGRIHYFDY	4338
237	GFSLTSYG	4595	IWSGGST	5109	ARKGYGYDWYFDV	4339
107	GFSLTSYG	4596	IWSGGST	5110	ARKGYGYDWYFDV	4340
238	GYTFTSYW	4597	IDPSDSYT	5111	ARSSYYYYAMDY	4341
108	GYTFTSYW	4598	IDPSDSYT	5112	ARSSYYYYAMDY	4342
239	GYSITSGYY	4599	ISYDGSN	5113	ARGGGRD	4343
240	GFSLTSYG	4600	IWSGGST	5114	ARGGDYDSYAMDY	4344
241	GYTFTSYW	4601	IYPGSGST	5115	ARESVYDGYSWYFDV	4345
242	GYSFTDYN	4602	INPNYGTT	5116	ASTYDYDDWYFDV	4346
243	GYTFTSYW	4603	IDPSDSYT	5117	ARSGNYLYAMDY	4347
244	GYSFTDYN	4604	INPNYGTT	5118	AREGTSWYFDV	4348
245	GFSLTSYG	4605	IWRGGST	5119	AKKGDGYDWYFDV	4349
246	GFSLTSYG	4606	IWSGGST	5120	AREGNYGSSYDAMDY	4350
247	GYTFTSYW	4607	IDPSDSYT	5121	ARSSNYPYAMDY	4351
248	GFNIKNTY	4608	IDPANGNT	5122	AYYSGLY	4352
249	GYTFTSYW	4609	IDPSDSET	5123	ARRGQIYYGYSWFAY	4353
250	GYTFTDYY	4610	INPNNGGT	5124	ARSTVVADWYFDV	4354
251	GYTFTSYG	4611	IYPRSGNT	5125	ARSGSSYGYFDV	4355
252	GFSLTSYG	4612	IWSGGST	5126	ARKGGYDAYAMDY	4356
253	GYSFTDYN	4613	INPNYGTT	5127	AREGFITTVVAVDY	4357
254	GYTFTDYE	4614	IDPETGGT	5128	TREGNYDAMDY	4358
255	GYTFTSYW	4615	IDPSDSYT	5129	ARWDYYGVDY	4359
256	GFTFSGYW	4616	ISPGGGST	5130	ASSLTATHTYEYDY	4360

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
	LCDR1		LCDR2		LCDR3	
97	QGISNY	8066	YTS		QQYSKLPYT	6829
182	QGISNY	8067	YTS		QQYSKLPYT	6828
86	QDISNY	8068	YTS		QQDNTLPRT	6793
96	QDISNY	8069	YTS		QQGNTLPFT	6827
151	QDISNY	8070	STS		QQGNTLPWT	6768
152	QDISNY	8071	STS		QQGNALPWT	6769
153	QDISNY	8072	STS		QQGNTLPWT	6770
163	QDISNY	8073	YTS		QQDNTLPRT	6792
164	QDISNY	8074	YTS		QQGNTLPWT	6794
181	QDISNY	8075	YTS		QQGNTLPFT	6826
202	QSIVHSNGNTY	8076	KVS		FQGSHVPWT	6861
206	QSIVYSNGNTY	8077	KVS		FQGSHVPPT	6865
215	QSIVHSNGNTY	8078	KVS		FQGSHVPPT	6874
242	QSIVHSNGDTY	8079	KVS		FQGSHVPLT	6905
204	QSLLDSDGKTY	8080	LVS		WQGTHFPWT	6863
214	QSLLDSDGKTY	8081	LVS		WQGTHFPWT	6873
220	QSLLDSDGKTY	8082	LVS		WQGTHFPWT	6879
229	QSLLDSDGKTY	8083	LVS		WQGTHFPWT	6890
235	QSLLDSDGKTY	8084	LVS		WQGTHFPFT	6896
162	ENIYYS	8085	NAN		KQAYDVPYT	6789
101	GNIHNY	8086	NAK		QHFWSTPWT	6836
36	GNIHNY	8087	NAK		QHFWSTPWT	6835
188	GNIHNY	8088	NAK		QHFWSTPWT	6844
172	ENIYSY	8089	NAK		QHFWGTPYT	6807
189	ENIYSY	8090	NAK		QHHYGTPFT	6845
249	ENIYSY	8091	NAK		QHHYGTPYT	6912
250	ENIYSY	8092	NAK		QHHYGTPPT	6913
57	DNIYSN	8093	AAT		QHFWGTPWT	6712
58	ENIYSN	8094	AAT		QHFWGTPWT	6713
3	ENIYSN	8095	AAT		QHFWGTPWT	6701
4	ENIYSN	8096	AAT		QHFWGTPWT	6702
5	ENIYSN	8097	AAT		QHFWGSPWT	6703
6	ENIYSN	8098	AAT		QHFWGTPWT	6704
7	ENIYSN	8099	AAT		QHFWGTPWT	6705
8	ENIYSN	8100	AAT		QHFWGTPWT	6706
9	DNIYSN	8101	AAT		QHFWGTPWT	6707
10	ENIYSN	8102	AAT		QHFWGTPWT	6708
11	ENIYSN	8103	AAT		QHFWGTPWT	6709
12	ENIYSN	8104	AAT		QHFWGTPWT	6710
13	ENIYSN	8105	AAT		QHFWGSPWT	6711
17	ENIYSN	8106	AAT		QHFWGTPWT	6777
165	ENIYSN	8107	AAT		QHFWGTPWT	6795
179	ENIYSN	8108	AAT		QHFWGTPYT	6822
194	ENIYSN	8109	AAT		QHFWGTPFT	6850
37	ENIYSN	8110	AAT		QHFWGTPYT	6853
207	ENIYNN	8111	AAT		QHFWGTPWT	6866
212	ENIYSN	8112	AAT		QHFWGTPWT	6871
248	DHINNW	8113	GAT		QQYWSTPLT	6911
38	QDINSY	8114	RAN		LQYDEFPLT	6682
39	QDINGY	8115	RAN		LQYDEFPPT	6683
40	QDINSY	8116	RAK		LQYDEFPPT	6684
41	QDINGY	8117	RAN		LQYDEFPPT	6685
42	QDINSY	8118	RAK		LQYDEFPPT	6686
43	QDINSY	8119	RAN		LHYDEFPPT	6687
44	QDINSY	8120	RAN		LQYDEFPPT	6688
45	QDINSY	8121	RAN		LQYDEFPPT	6689
46	QDINSY	8122	RAN		LQYDEFPPT	6690
47	QDINSY	8123	RAN		LQYDEFPPT	6691
48	QDINSY	8124	RAN		LQYDEFPPT	6692
49	QDINSY	8125	RAN		LQYDEFPPT	6693
50	QDINSY	8126	RAN		LQYDEFPPT	6694
51	QDINSY	8127	RAN		LQYDEFPPT	6695
52	QDINSY	8128	RAN		LQYDEFPPT	6696
53	QDINSY	8129	RAN		LQYDEFPPT	6697
54	QDINSY	8130	RAN		LQYDEFPPT	6698
55	QDINSY	8131	RAN		LQYDEFPPT	6699
56	QDINSY	8132	RAN		LQYDEFPPT	6700
91	QDINSY	8133	RAN		LQYDEFPLT	6811
109	QDINSY	8134	RAN		LHYDEFPPT	6664

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
110	QDINSY	8135	RAN		LQYDEFPPT	6665
111	QDINSY	8136	RAN		LQYDEFPPT	6666
112	QDINSY	8137	RAN		LQYDEFPPT	6667
1	QDINSY	8138	RAN		LQYDEFPPT	6668
113	QDINSY	8139	RAN		PQYVESPPT	6669
114	QDINSY	8140	RAN		LQYDEFPPT	6670
115	QDINSY	8141	RAN		LQYDEFPPT	6671
23	QDINSY	8142	RAN		LQYDEFPPT	6672
116	QDINSY	8143	RAN		LQYDEFPPT	6673
117	QDINSY	8144	RAN		LQYDEFPLT	6674
2	QDINSY	8145	RAN		LQYDEFPPT	6675
118	QDINGY	8146	RAN		LQYDEFPPT	6676
119	QDINSY	8147	RAN		LQYDEFPPT	6677
120	QDINSY	8148	RAN		LQYDEFPPT	6678
121	QDINSY	8149	RAN		LQYDEFPPT	6679
122	QDINSY	8150	RAN		LQYDEFPPT	6680
123	QDINSY	8151	RAK		LQYDEFPPT	6681
124	QDINSY	8152	RAN		LQYDEFPPT	6714
173	QDINSY	8153	RAN		LQYDEFPLT	6810
193	QDINSY	8154	RAN		LQYDEFPPT	6849
203	QDINSY	8155	RAN		LQYDEFPPT	6862
70	QDIVKN	8156	YAT		LQFYEFPLT	6749
87	QDIVKN	8157	YAT		LQFYEFPYT	6797
89	QDIVKN	8158	YAT		LQFYEFPLT	6803
99	QDIVKN	8159	YAT		LQFYEFPYT	6833
100	QDIVKN	8160	YAT		LQFYEFPYT	6834
16	QDIVKN	8161	YAT		LQFYEFPLT	6745
141	QDIVKN	8162	YAT		LQFYEFPLT	6746
142	QDIVKN	8163	YAT		LQFYEFPLT	6747
143	QDIVKN	8164	YAT		LQFYEFPLT	6748
31	QDIVKN	8165	YAT		LQFYEFPYT	6796
168	QDIVKN	8166	YAT		LQFYEFPLT	6802
35	QDIVKN	8167	YAT		LQFYEFPYT	6832
239	QDIVKN	8168	YAT		LQFYEFPLT	6902
21	QNINVW	8169	KAS		QQGQSYPLT	6842
218	QNINVW	8170	KAS		QQGQSYPYT	6877
106	TDIDDD	8171	EGN		LQSDNLPLT	6885
197	TDIDDD	8172	EGN		LQSDNLPLT	6855
22	TDIDDD	8173	EGN		LQSDNLPLT	6858
224	TDIDDD	8174	EGN		LQSDNLPLT	6884
95	TDIDDD	8175	EGN		LQSDNMPLT	6825
157	TDIDDD	8176	EGN		LQSDNMPFT	6778
27	IDIDDD	8177	EGN		LQSDNMPFT	6779
33	TDIDDD	8178	EGN		LQSDNMPLM	6821
180	TDIDDD	8179	EGN		LQSDNMPLT	6824
216	TDIDDD	8180	EGN		LQSDNMPLT	6875
228	TDIDDD	8181	EGN		LQSDNMPLT	6889
24	QDINKY	8182	YTS		LQYDNLMYT	6742
139	QDINKY	8183	YTS		LQYDILMYT	6743
140	QDINKY	8184	YTS		LQYDILMYT	6744
217	QDINKY	8185	YTS		LQYDNLYMYT	7780
230	QDINKY	8186	YTS		LQYDNLRT	7781
240	ESVDNYGISF	8187	AAS		QQSKEVPPT	6903
252	ESVDNYGISF	8188	AAS		QQSKEVPPT	6915
66	QSVDYDGDSY	8189	AAS		QQSNEDPLT	6738
67	QSVDYDGDSY	8190	AAS		QQSNEDPLT	6739
68	QSVDYDGDSY	8191	AAS		QQSNEDPLT	6740
69	QSVDYDGDSY	8192	AAS		QQSNEDPLT	6741
71	QSVDYDGDSY	8193	AAS		QQSNKDPLT	6756
72	QSVDYDGDSY	8194	AAS		QQSNKDPLT	6757
73	QSVDYDGDSY	8195	AAS		QQSNEDPLT	6758
74	QSVDYDGDSY	8196	AAS		QQSNKDPFT	6759
75	QSVDYDGDSY	8197	AAS		QQSNKDPFT	6760
76	QSVDYDGDSY	8198	AAS		QQSNKDPFT	6761
83	QSVDYDGDSY	8199	AAS		QQSNEDPLT	6782
88	QSVDYDGDSY	8200	AAS		QQSNEDPWT	6800
135	QSVDYDGDSY	8201	AAS		QQSNEDPLT	6733
136	QSVDYDGDSY	8202	AAS		QQSNEDPLT	6734
137	QSVDYDGDSY	8203	AAS		QQSNEDPLT	6735
138	QSVDYDGDSY	8204	AAS		QQSNEDPLT	6736
15	QSVDYDGDSY	8205	AAS		QQSNEDPLT	6737

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
146	QSVDYDGDSY	8206	AAS		QQSNKDPLT	6753
147	QSVDYDGDSY	8207	AAS		QQSNEDPLT	6754
148	QSVDYDGDSY	8208	AAS		QQSNKDPFT	6755
159	QSVDYDGDSY	8209	AAS		QQSNEDPLT	6781
161	QSVDYDGDSY	8210	AAS		QQSNEDPPT	6787
166	QSVDYDGDSY	8211	AAS		QQSNEDPWT	6799
170	QSVDYDGDSY	8212	AAS		QQSNEDPPT	6805
177	QSVDYDGDSY	8213	VAS		QQSHEDPRT	6819
186	QSVDYDGDSY	8214	AAS		QQSNEDPLPT	7776
225	QSVDYDGDSY	8215	AAS		QQSNEDPWT	6886
236	QSVDYDGDSY	8216	AAS		QQSNEDPFT	6897
233	ETVDSYGYSF	8217	RAS		QQSNEDPRT	6894
77	QSVSTSSYSY	8218	YAS		QHSWEIPLT	6765
78	QSVSTSSYSY	8219	YAS		QHSWEIPLT	6766
79	QSVSTSSYSY	8220	YAS		QHSWEIPLT	6767
149	QSVSTSSYSY	8221	YAS		QHSWEIPLT	6762
150	QSVSTSSYSY	8222	YAS		QHSWEIPLT	6763
26	QSVSTSSYSY	8223	YAS		QHSWEIPLT	6764
246	QSVSTSSYSY	8224	YAS		QHSWEIPLT	6909
105	SSVNY	8225	YTS		QQFTSSPST	6883
25	SSVNY	8226	YTS		QQFTSSHT	7768
144	SSVNY	8227	YTS		QQFTSSHT	7769
145	SSVNY	8228	YTS		QQFTSSHT	7770
190	SSVNY	8229	YTS		QQFTSSLT	7777
223	SSVNY	8230	YTS		QQFTSSPST	6882
32	SSVSY	8231	DTS		QQWSSYPLT	6798
209	SSVSY	8232	DTS		QQWSTYPPIT	7779
210	SSVSY	8233	DTS		QQWSSYPFT	6869
221	LSVSD	8234	DTS		QQWSSYPFT	6880
90	SSVSY	8235	STS		QQRSSFPYT	6809
19	SSVSY	8236	STS		QQRSSFPYT	6808
34	SSVSY	8237	STS		QQRSTYPT	7775
198	SSVSY	8238	STS		QQRSSYPPT	6856
64	SSVSY	8239	DTS		QQWSSNPLY	6731
65	SSVSY	8240	DTS		QQWSSNPLY	6732
	SSVSY	8241	DTS		QQWSSNPLT	6901
131	SSVSY	8242	DTS		QQWSSNPLYT	7763
132	SSVSY	8243	DTS		QQWSSNPHVHV	7764
133	SSVSY	8244	DTS		QQWSSNPLYT	7765
14	SSVSY	8245	DTS		QQWSSNPLYT	7766
134	SSVSY	8246	DTS		QQWSSNPLYT	7767
187	SSVSY	8247	DTS		QQWSSNPLT	6843
199	SSVSY	8248	DTS		QQWSSNPLT	6857
238	SSVSY	8249	DTS		QQWSSNPLT	6900
243	SSVSY	8250	DTS		QQWSSNPLT	6906
247	SSVSY	8251	DTS		QQWSSNPLT	6910
251	SSVSY	8252	DTS		QQWSSNPPT	6914
178	SSVSY	8253	DTS		FQGSGYPLT	6820
219	SSVSY	8254	LTS		QQWSSNPPT	6878
255	SSISY	8255	DTS		HQRSSYPT	7785
211	SSVSY	8256	ATS		QQWSSNPYT	6870
200	SSVSSSY	8257	STS		HQWSSYPPT	6859
205	SSVSSSY	8258	STS		HQWSSYPPT	6864
208	SSVSSSY	8259	STS		HQWSSYPPT	6867
222	SSVSSSY	8260	STS		HQWSSYPPT	6881
227	SSVSSSY	8261	STS		HQWSSYPPT	6888
98	QSISDY	8262	YAS		QNGHSFPWT	6831
104	QSISDY	8263	YAS		QNGHSFPYT	6852
183	QSISDY	8264	YAS		QNGHSFPWT	6830
195	QSISDY	8265	YAS		QNGHSFPYT	6851
241	QSISDY	8266	YAS		QNGHSFPLT	6904
226	QSISNN	8267	YAS		QQSNSWPFT	6887
60	ENVVTY	8268	GAS		GQSYSYPFT	6722
61	ENVVTY	8269	GAS		GQSYSYPFT	6723
62	ENVVTY	8270	GAS		GQSYSYPFT	6724
63	ENVVTY	8271	GAS		GQSYSYPFT	6725
59	QDINSY	8272	RAN		LQYDEFPPT	6721
125	ENVVTY	8273	GAS		GQSYSYPFT	6715
126	ENVVTY	8274	GAS		GQSYSYPFT	6716
127	ENVVTY	8275	GAS		GQSYSYPFT	6717
128	ENVVTY	8276	GAS		GQSYSYPFT	6718

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
129	ENVVTY	8277	GAS		GQSYSYLIHVR	7761
130	ENVVTY	8278	GAS		GQSYSYLIHVR	7762
232	ENVGTY	8279	GAS		GQSYSYPPT	6893
93	QDVGTA	8280	WAS		QQYSSYPFT	6815
175	QDVGTA	8281	WAS		QQYSSYPFT	6814
192	QDVGTA	8282	WAS		QQYSSYPFT	6848
213	QDVSTA	8283	WAS		QQHYSTPWT	6872
94	QSVSND	8284	YAS		QQDYTSLPT	6817
20	QSVSND	8285	YAS		QQDYTSLPT	6816
244	ESLYSSKHKVHY	8286	GAS		AQFYSYPYT	6907
253	ESLYSSKHKVHY	8287	GAS		AQFYSYPYT	6916
85	QSLLNSSNQKNY	8288	FAS		QQHYSTPLT	6791
18	QSLLNSSNQKNY	8289	FAS		QQHYSTPLT	6790
160	QSLLYSTNQKNY	8290	WAS		QQYYSYPPWT	7772
234	QSLLYSTNQKNY	8291	WAS		QQYYSYRT	7782
171	QDIGIS	8292	ATS		LQYASSPYT	6806
92	QDIHGY	8293	ETS		LQYASSPLT	6813
103	QEISGY	8294	AAS		LQYASYPWT	6840
174	QDIHGY	8295	ETS		LQYASSPLT	6812
185	QEISGY	8296	AAS		LQYASYPWT	6839
29	TGAVTTSNY	8297	GTN		ALWYSNHL	7773
30	TGAVTTSNY	8298	GTN		ALWYSNHWV	6788
167	TGAVTTSNY	8299	GTN		ALWYSNHWV	6801
176	TGAVTTSNY	8300	GTN		ALWYSNHLV	6818
191	TGAVTTSNY	8301	STN		TLWYSNHWV	6847
196	TGAVTTSNY	8302	GTN		ALWYSNHWV	6854
231	TGAVTTSNY	8303	GTN		VLWYSNHLV	6892
80	TGAVTTSNY	8304	GTS		ALWYSTHYV	6774
81	TGAVTTSNY	8305	GTS		ALWYSTHYV	6775
82	TGAVTTSNY	8306	GTS		ALWYSTHYV	6776
84	TGAVTTSNY	8307	GTS		ALWYSTHWV	6785
102	TGAVTTSNY	8308	GTS		ALWYSTHYV	6838
154	TGAVTTSNY	8309	GTS		ALWYSTHYV	6771
155	TGAVTTSNY	8310	GTS		ALWYSTHYV	6772
156	TGAVTTSNY	8311	GTS		ALWYSTHYV	6773
158	TGAVTTSNY	8312	GTS		ALWYSTHY	7771
28	TGAVTTSNY	8313	GTS		ALWYSTHWV	6784
184	TGAVTTSNY	8314	GTS		ALWYSTHYV	6837
254	TGAVTTSNY	8315	GTS		ALWYSTHYV	6917
107	SQHSTYT	8316	LKKDGSH	8826	GVGDTIKEQFVYV	6899
169	SQHSTYT	8317	LKKDGSH	8827	GVGDTIKEQFVFV	7774
201	SQHSTYT	8318	LKKDGSH	8828	GVGDTIKEQFVYV	7778
237	SQHSTYT	8319	LKKDGSH	8829	GVGDTIKEQFVYV	7783
245	SQHSTYT	8320	LKKDGSH	8830	GVGDTIKEQFVYV	7784
	HCDR1		HCDR2		HCDR3	
265	GGTFSSYA	11540	INPSGGT	11802	ARDLGDPGMDV	11409
266	GGTFSNYA	11541	IDPSGGST	11803	ARDLGDMGMDV	11410
267	GGTFSNYA	11542	INPSGGST	11804	ARDVGDRGMDV	11411
263	GGTFSSYA	11543	INPSGGT	11805	ARDLGDPGMDV	11412
268	GSTFSGYY	11544	IDPNGGGT	11806	AKDIVHDGTEYFQH	11413
269	GYTFTSYY	11545	INPSGGST	11807	AKDIVHDGTEYFQH	11414
270	GGTFSSYA	11546	INPSGGST	11808	AREGRDHDAFDI	11415
271	GFTFTDYG	11547	INPSGGST	11809	AREGRSHDAFDI	11416
272	GYTFTGYY	11548	MNPHSGDT	11810	ARWVGTTEYYYYYMDV	11417
273	GYTFTDYY	11549	IDPSGGST	11811	ATTAYYDFWSGYSMDV	11418
274	GYTFTSHY	11550	IDPSGGST	11812	ARDMDNWNTGYYYYMDV	11419
275	GGTFSSYA	11551	VNPNSGDT	11813	ARDQRGGDAWDV	11420
276	GGTFSNYA	11552	ITPSGGST	11814	ARDTAGHFDI	11421
277	GGTFRNDV	11553	MNPNSGNT	11815	ARDNPDLDGMDV	11422
278	GGTFSSYA	11554	ISAYNGNT	11816	ARDLVGHFDY	11423
279	GGTFSSYA	11555	INPNSGGT	11817	ARDGYSGSYSD	11424
264	GGTFSSYA	11556	INPNSGGT	11818	ARDGYSGSYSD	11425
257	GGTFSSYA	11557	INPNSGGT	11819	ARDGYSGSYSD	11426
280	GNTLSSHA	11558	INPSGGST	11820	ARDQGSSGTFDY	11427
281	GGTLSSYA	11559	INPNSGGT	11821	ARDSTDVIDY	11428
282	GYIFTSYD	11560	INPNSGDT	11822	ARDGGTVTPTEEYYYYG-MDV	11429
283	GGTFSSYA	11561	ISVYNGNT	11823	ASLDDLDY	11430
284	GHTFTSYY	11562	INPNNGGT	11824	ARDMVRDSAEYFQH	11431

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
285	GYTFITSY	11563	INPSGGTT	11825	ARDSSGYPIDY	11432
286	GYTFTSYD	11564	IIPLSGAP	11826	ARGALYNWNDGWFDP	11433
258	GYTFTSYD	11565	IIPLSGAP	11827	ARGALYNWNDGWFDP	11434
287	GFTVGSWY	11566	IWYEGSNK	11828	ARLGTASLPYFDY	11435
288	GYTFTGYY	11567	INPNRGDT	11829	ARESGDGFDP	11436
289	GYTFTNYY	11568	MNPNSGNT	11830	ARDWPNWFDP	11437
290	GYSFTDNY	11569	IRSDNGET	11831	AREVQLVGFDY	11438
291	GYTFSDHH	11570	IIPIFGTA	11832	ARGSSWYLHFQH	11439
260	GYTFSDHH	11571	IIPIFGTA	11833	ARGSSWYLHFQH	11440
292	GGTFSSYA	11572	IIPIFGTT	11834	AKGVDRYNWNDAFDY	11441
293	GYTFTDYY	11573	IHSNSGGT	11835	ARESSGYDSSLDY	11442
294	GGTFSSYG	11574	INPNSGDT	11836	TTDPRLDSSDPGY	11443
295	GGTFGNYG	11575	ISAYNGNT	11837	ARGGMDV	11444
296	GGTFSRYG	11576	SYPSDGST	11838	ARDRLGDLDY	11445
261	GGTFSRYG	11577	SYPSDGST	11839	ARDRLGDLDY	11446
297	GGTFSSYA	11578	MNPNSGNT	11840	ARDSIVGGYPFDY	11447
298	GYTFTSYD	11579	ITPIFGTT	11841	AREGYSSSWHDDAFDI	11448
299	GGTFSNYA	11580	IDPSGGST	11842	ARDLGDYGLDS	11449
300	GYTFTGYY	11581	MNPNSGDT	11843	ATGGSDSSGYYYEGYFQH	11450
301	GGTFSSYA	11582	MSPNSGNT	11844	ARDKGGYYDSSGYYWY	11451
302	GFSLSSYE	11583	ISSNGGST	11845	ARVGDGDGYNPDFDY	11452
303	GYTFTSYG	11584	IDPTSGAT	11846	AKDPIVATEVDY	11453
304	GGTFSSYA	11585	MSPNSGNT	11847	ARDSGAFDI	11454
305	GVTISNYA	11586	MNPNSGNT	11848	AREGLLDAFDI	11455
306	GGTFSRYG	11587	MNPYDGNT	11849	ARGGRHHDAFDI	11456
307	GGTFSSYA	11588	INPSGDGT	11850	ARDISNDAFDI	11457
308	GYILTGHY	11589	ISAYNGDT	11851	ARGSSWDDAFDI	11458
309	GFTFSNHY	11590	IGAGGGT	11852	AREGWNDDVFDI	11459
310	GGTFSSYA	11591	INPSAGTT	11853	ARDGNFGAFDI	11460
311	GYSFTTYA	11592	IIPIFGTA	11854	ARDKSGWNYGSGSYN-DAFDI	11461
312	GYAFTGYY	11593	MNPNSGKT	11855	ARDGGLDFDY	11462
313	GYTFTTYY	11594	MNPNTGDT	11856	AKDPAVTPDAFDI	11463
314	GGTLSSYA	11595	IDPSGGGT	11857	AGSLYYYGMDV	11464
315	GGTFGSSA	11596	IIPIFGTA	11858	AKEDDILPPRAFDI	11465
316	GFTFDDYA	11597	ISGGGGVT	11859	ARVYSSGWLDAFDI	11466
317	GGTFSSYA	11598	ISGYNGNT	11860	ASSDVSPDAFDI	11467
318	GGTQNIYA	11599	VNPNSGNT	11861	ATPTSSSDDAFDI	11468
319	GGTFSSYA	11600	INPNSGGT	11862	ARASRGDDAFDI	11469
320	GIPFTSDD	11601	INPSGGST	11863	ARERYEGGY-SSGPGNYYYGMDV	11470
321	GGTFSNYA	11602	MNPNSGNT	11864	ARDDDYGDYPV	11471
322	GDTFSDHA	11603	MNPKIGNT	11865	VYDSSGYDAFDI	11472
323	GYTFTSYD	11604	INPGTGGT	11866	ARETPSDYYDSSGYYYN-DAFDI	11473
324	GGTFSSYA	11605	IPSGGT	11867	ARDLGTTFDI	11474
325	GYTFTAYY	11606	INPDNDNA	11868	AKDIAVAALAYGMDV	11475
326	GFTFSSYA	11607	ISYDGSDQ	11869	ARQSLYYYYGMDV	11476
327	GYTFTDYY	11608	ISTFTGNT	11870	ARDAPLAAAGTDYYYG-MDV	11477
262	GYTFTDYY	11609	ISTFTGNT	11871	ARDAPLAAAGTDYYYG-MDV	11478
328	GFTFSSYA	11610	ISDDGITK	11872	ARDDSSGYGGMDV	11479
329	GFTFSSYA	11611	ISYDGGDK	11873	ASGSLVLGYYYMDV	11480
330	GYTFTNYY	11612	INPNTGGT	11874	ATGGGGSYYDAFDV	11481
331	GGTFSSYA	11613	INPNSGNT	11875	ARDIGEGYSMDV	11482
332	GFTFSNHY	11614	ISYDGSNK	11876	AREEKYSSSWYVGVDAFDI	11483
333	GFTFSSSA	11615	ISGSGDNA	11877	ARDQEDYYYDSSGYGMDV	11484
334	GGTFSSHA	11616	IIPIFGTA	11878	AKGDWGIVVVPAAIGAFDI	11485
335	GYTFTAYY	11617	ISPVFGST	11879	ARDLGYYDSSGYRYDAFDI	11486
336	GYTFTSYD	11618	ISPMFGTA	11880	AKDGWYYGMDV	11487
337	GGTFSSYG	11619	INPNSGGT	11881	ARGEAGNLDWYFDL	11488
338	GGTFSNYG	11620	INPNNGDT	11882	AREDVWYFDL	11489
339	GYTFTTYG	11621	ISTYDGKT	11883	ALHLGGDWYFDL	11490
340	GYTFTGYY	11622	INPNTGAT	11884	ARQHGDYDWYFDL	11491
341	GDTFTTYY	11623	INPNSGNT	11885	ARDSGRH	11492
342	GGTFSSYG	11624	IIPMLGIA	11886	VREEVAGANWFDP	11493
343	GYTFTSYA	11625	INPSGGST	11887	AREGDYGSGEFDY	11494
344	GYTFTSSY	11626	MNPRSGNT	11888	ARERDDYGDYGWLDY	11495
345	GYTFTGYY	11627	INPSGGST	11889	ARDLYDSSGY-WHYYYYMDV	11496

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
346	GGTFSSYA	11628	INPNSGGT	11890	ARFSGYDYVDY	11497
347	GGTFSSYA	11629	INPNGGNT	11891	ARDVGEDFDL	11498
348	GYTFTSYY	11630	INPADGDT	11892	ARDFDWLFAMDV	11499
349	GGTFSNYA	11631	INPNGGTT	11893	AKHGDHGFYV	11500
350	GGTFSSYA	11632	INPNVGSA	11894	AREDSGTSWFDP	11501
351	GYTFTSYY	11633	INPSDGST	11895	ARDDRGSNYYYGMDV	11502
352	GYTFTAYY	11634	MNPNSGTT	11896	ARDSSDYYGDYRADAFDI	11503
353	GYTFTSYD	11635	ISPSGDAT	11897	VKGLDH	11504
354	GFSFSDYG	11636	IGGIGDST	11898	ARMNYGDSNYYYYYG-MDV	11505
355	GYTFTSYD	11637	ISPSDGST	11899	ARGAVGFDY	11506
356	GYTFTSYG	11638	INTYSGYT	11900	TTDDFLSFGY	11507
357	GYMFTDYY	11639	IIPYFGTA	11901	ARSISGSYVLDAFDI	11508
358	GYTFNSYG	11640	IIPIFGTA	11902	ARDWGYGDYADDAFDI	11509
359	GGTFSNND	11641	INPIYGSA	11903	AADWRGFDY	11510
360	GYTFTEYA	11642	MNPHNGDT	11904	AREGDYLGYPIDC	11511
361	GFTFSDYS	11643	IWQDGNVK	11905	ARDGNSGYVF	11512
362	GYTFTTYY	11644	INPNTGDT	11906	ARTAEAVAGLPAFDY	11513
363	GGTSNNYA	11645	IIPLFGTT	11907	ARVTLYGDYDY	11514
364	GYSLITHW	11646	INPSDGVT	11908	AREYYGEGFDY	11515
259	GYSLITHW	11647	INPSDGVT	11909	AREYYGEGFDY	11516
365	GGTFSSYA	11648	INPSGGST	11910	ARDLGDTAMDG	11517
366	GYTFTSYY	11649	ITPSGGST	11911	ARDGGLASFDY	11518
367	GGTFSSYA	11650	MNPNSGNT	11912	ARGGGWAMTDAFDI	11519
368	GFTFDDYG	11651	IYSGGDT	11913	TRKEYYYDSSGYLRLFDY	11520
369	GYTFTDYY	11652	INPIFGTS	11914	ARDISGYDYYYYGMDV	11521
370	GGTLNNYA	11653	IDPSDGTI	11915	ARSDYDFWSGLGGYFDY	11522
371	GGTFSSYA	11654	IDPNSGGT	11916	ARDSAEWELGGSFDY	11523
372	GFTFSNHY	11655	IGVNGDT	11917	AREGLVFSGRGHWYFDL	11524
373	GGTFSNYA	11656	INPNGGNT	11918	ARDYEDADFDG	11525
374	GYTFSDHH	11657	MNPDSGNT	11919	ARDSTSGVDY	11526
375	GFTFSSYA	11658	ISYDGHDQ	11920	ARGEQQLEGFYYYYGMDV	11527
376	GFTFSSYW	11659	ISYDGSKE	11921	ASDYGDYGTYDY	11528
377	GFTFSSYW	11660	ISGGGDDT	11922	AREPLAYCGGDCPGGFDY	11529
378	GFTFSDHY	11661	IGTGGDT	11923	ARHEDTAIFLDY	11530
379	GYTFTSYY	11662	ISPSDGST	11924	ARDGYDAWSYGMDV	11531
380	GYTFTGYY	11663	MNPNSGNT	11925	ARDGVTGTDY	11532
381	GFAFSSYV	11664	ISGAGDST	11926	AREPTTVTDDWYFDL	11533
382	GFAFSSHW	11665	ISGNGDNS	11927	ARDRAPEYFDL	11534
383	GGTFSSYA	11666	INPNSGGT	11928	ARDDYGDYGGGMDV	11535
384	GYTFTDYY	11667	MNPNSGHT	11929	AKDTSPRYGDGFFDY	11536
385	GFTFSSYW	11668	TSYDGSNK	11930	ARESGFSAEYFQH	11537
386	GGTFSSYA	11669	INPSGGST	11931	ARATGLYCSGSCFDY	11538
388	RSILDFNA	14122	IARAGAT	14123	NARVFDLPNDY	14124
	LCDR1		LCDR2		LCDR3	
265	QDISNY	13364	DAS		QQSYSTPLT	12714
266	QSISSY	13365	AAS		QQSYSTPLT	12715
267	QDISNY	13366	KAS		QQSFSSPLT	12716
263	QDISNY	13367	DAS		QQSYSTPLT	12717
268	QNVNTW	13368	EAS		QQANSFPFT	12718
269	QSISDW	13369	AAS		AQHNHYPYT	12719
272	QAIRDD	13370	DAS		QQANSFPIT	12722
273	QGVGND	13371	AAS		QQASSFPLT	12723
274	QIIGTN	13372	AAS		QQSYTFPVT	12724
275	QSISTW	13373	DAS		QQSYSTPFT	12725
281	QDIRNY	13374	DAS		QQAYSFPWT	12733
282	QDISNY	13375	NAS		QQLNSYPFT	12734
283	QSISTW	13376	AAS		LOHYTYPLT	12735
284	EDISTY	13377	AAS		QQSHTIPWT	12736
285	HHISDF	13378	AAS		QQSYSSPYT	12737
286	QDIGDY	13379	DAS		QQANSFPLT	12738
258	QDIGDY	13380	DAS		QQANSFPLT	12739
287	QDIRSY	13381	AAS		QQSYTAPPT	12740
288	QDISNN	13382	AAS		LOHNTYPLT	12741
289	QDISNW	13383	DAS		QQAISFPLT	12742
290	QGIANY	13384	AAS		QQADSFPLT	12743
291	QGIASY	13385	AAS		QQFDSYPIT	12744
260	QGIASY	13386	AAS		QQFDSYPIT	12745
292	QGISNY	13387	AAS		QQSSIIPFT	12746

TABLE 24-continued

CDRs using the IMGT Numbering Scheme Table 6-IMGT CDR Sequences						
Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
293	QGISNY	13388	AAS		QQAYSFPYT	12747
294	QSIGRW	13389	DAS		QQSYSTPRT	12748
295	QSINSW	13390	DTS		QQTYSTPYT	12749
296	QSISSW	13391	AAS		QQGYSTPYI	12750
261	QSISSW	13392	AAS		QQGYSTPYI	12751
297	QSISSY	13393	AAS		QQTDSIPIT	12752
298	QSISSY	13394	AAS		QQSYSIPYT	12753
299	QTIRSY	13395	KAS		QQTYTIPIT	12754
300	QTISNW	13396	AAS		QQANSFPPT	12755
301	QYIGSY	13397	DAS		QQVDSYPLT	12756
307	QDIGNY	13398	AAS		QQTYNTPLT	12762
308	QDISNY	13399	EAS		QQSYSTPFT	12763
309	QDISTW	13400	RAS		QQSYSIPLT	12764
310	QNINNY	13401	AAS		QQSYSAPVT	12765
311	QNINTW	13402	AAS		QQAYSFPFT	12766
312	QRIGNY	13403	AAS		QQSYSTPLT	12767
313	QSISTY	13404	AAS		QQSYRTVT	12768
324	QDISNY	13405	AAS		QQSYSTPT	12779
325	QDISNY	13406	GAS		QEADSFPLT	12780
326	QGIRND	13407	DAS		QQAYSFPWT	12781
327	QGISNY	13408	KAS		QQSYNTPFT	12782
262	QGISNY	13409	KAS		QQSYNTPFT	12783
328	QSINRW	13410	SAS		QQSYNTPLT	12784
329	QSINTW	13411	AAS		QQANSFPFT	12785
330	QSIRTW	13412	DAS		QQLNSYPLT	12786
331	QSIRTY	13413	AAS		QQSYSAPLT	12787
332	QSISTY	13414	AAS		QQSYSTPLT	12788
333	QSITTY	13415	AAS		QQSYSTPLT	12789
339	QSISSW	13416	DAS		QQSYSTPFT	12795
340	QSISSY	13417	DAS		QQSYSFPT	12796
344	QDISNY	13418	AAA		QQTYSTPWT	12800
345	QDINTY	13419	AAS		QQSSSFPLT	12801
346	QDISNY	13420	AAS		QQLYNFPYT	12802
347	QSISRY	13421	GAS		QQSYNTPLT	12803
348	QTLSGW	13422	GAS		QQYYSYPPT	12804
349	QDIINY	13423	EAS		QQSYSTPLT	12805
350	QSISSY	13424	DVF		QQSYSSPFT	12806
351	QDISNY	13425	MAS		QQTNSFPLT	12807
352	QSISSY	13426	DAS		QQSYSTPLT	12808
356	ESVSTW	13427	KAS		QQSYKTPYT	12812
358	QSISSY	13428	AAS		QQSYSTPYT	12814
360	QSISSY	13429	KAS		QQNDSIPIT	12816
361	QSISRW	13430	DAS		LODYSYPLT	12817
363	QSINRY	13431	AAS		QQANSFPPT	12819
364	QGISNY	13432	SAS		QQSYSTPLT	12820
259	QGISNY	13433	SAS		QQSYSTPLT	12821
365	QSIDSY	13434	KAS		QQSYSAPLT	12822
366	QDISTW	13435	DAS		QQVNSDPYT	12823
367	QDISNY	13436	AAS		QQGDSLPLT	12824
368	QGISNY	13437	AAS		QQSDSFPYT	12825
370	QGIRND	13438	AAS		QQANSFPPT	12827
371	ESISTY	13439	KAS		QQTDSTFIT	12828
372	RNIHDY	13440	AAS		QQTYSTPPT	12829
373	QSNDSY	13441	KAS		QQSYSSPLT	12830
374	QSISDF	13442	AAS		QQSYSSPYT	12831
375	QDISNY	13443	AAS		QQANRFPLT	12832
376	QDISNY	13444	KAS		QQSYNFPAT	12833
378	QSISSY	13445	DAS		QQSYSTPLT	12835
379	QGISDY	13446	DAS		QQSYILPLT	12836
380	QDINDF	13447	AAS		QQSYSAPYT	12837
381	QSISNW	13448	AAS		QQSYSSPWT	12838
382	QGIDSW	13449	AAS		QQAYSFPLT	12839
383	QNIGTW	13450	RAS		QQAYSFPWT	12840
384	QNINNW	13451	KAS		QQADSFPPT	12841
385	QDISSY	13452	AAS		QQLNRYPIT	12842
386	QDISNY	13453	AAS		QQYDSSFIT	12843
302	QSLLHSNGYNY	13454	LGS		MQGTHWPPT	12757
303	QSLLHSNGYNY	13455	FGS		MQALQAPVS	12758
316	QSLLHSNGYNY	13456	DAS		MQALQTPPA	12771
334	QSLLHSNGYNY	13457	AAS		MQARQTPLT	12790
335	QSLLHSNGYNY	13458	GAS		MQTLQTPFT	12791

TABLE 24-continued

CDRs using the IMGT Numbering Scheme
Table 6-IMGT CDR Sequences

Binder Name	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:	Sequence	SEQ ID NO:
336	QSLLHSNGYNY	13459	LGS		MQALQTPLT	12792
353	QSLLHSNGYNY	13460	LGS		MQALQSPWT	12809
355	QSLLHSNGYNY	13461	LGS		MQALQTPPS	12811
377	QSLLHSNGYNY	13462	LGS		MQGTHWPET	12834
314	QSVGSY	13463	GAS		QQYDSSSQT	12769
315	RSVSTY	13464	GAS		QQYDGSPYT	12770
341	QSVSSY	13465	DTS		QQYYDTPYT	12797
369	QSVSTY	13466	GAS		QQHDSYPLT	12826
270	QSVLSSSYNKNY	13467	WAS		QQYYTTPFT	12720
271	QSVLSSSYNKNY	13468	WAS		QQYYSTPFT	12721
276	QSVLSSSNNKNY	13469	WAS		QQYYGSPLT	12726
277	QSVLSSSYNKNY	13470	WAS		QQYYSSPPT	12727
278	QSVLSSSYNKNY	13471	WAS		QQYYSSPPT	12728
279	QSVLSSSYNKNY	13472	WAS		QQYYSTPWT	12729
264	QSVLSSSYNKNY	13473	WAS		QQYYSTPWT	12730
257	QSVLSSSYNKNY	13474	WAS		QQYYSTPWT	12731
280	QSVLSSSYNKNY	13475	WAS		QQYYGSPPT	12732
304	QSVLSSSYNKNY	13476	WAS		QQYYSTPLT	12759
305	QSVSSSSYNKNY	13477	WAS		QQYYSTPIT	12760
306	QNVLSSSNNNSY	13478	WAS		QQYYSTPFT	12761
317	QSVLSSSYNKNF	13479	WAS		QQYYSAPPT	12772
318	QSVLSSSYNKNF	13480	WAS		QQYYSDPIT	12773
319	QSVLSSSYNKNY	13481	WAS		QQYYSIPIA	12774
320	QSVLSSSYNKNY	13482	WAS		QQYYSIPYT	12775
321	QSVLSTSYNKNY	13483	WAS		QQYYTTPPT	12776
322	QSVLSTSYNRNF	13484	WAS		QQYYSTPYT	12777
323	QSVLYSSNNKNY	13485	WAS		QQYYSTPLT	12778
337	QTVFSTSYNKNY	13486	WAS		QQYYSTPLT	12793
338	QSVFSTSYNRDY	13487	WAS		QQYYSSPPT	12794
342	QSVLYSSNNKNY	13488	LAS		QQYYSTPPT	12798
343	QSVLSSSYNKNY	13489	WAS		QQYYSTPLT	12799
354	QSVLYSSNNKNY	13490	WAS		QQYYSSPLT	12810
357	QSVLYSSNNKNY	13491	WAS		QQYFTTPLT	12813
359	QSVLSSSYNKNY	13492	WAS		QQYYDTPLT	12815
362	QSVFSTSYNRDY	13493	WAS		QQYYYTST	12818

TABLE 25

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
9258	MVVILDKRCY CNLLILILMI SECSVGILHY EKLSKIGLVK GVTRKYKIKS NPLIKDIVIK MIPNVSNMSQ CTGSVMENYK TRLNGILTPI KGALEIYKNN THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMIN NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNTYSRLED RRVRPTSSGD LYYIGT	Nipah virus NiV-F with signal sequence (aa 1-546) Uniprot Q9IH63
9259	ILHY EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK MIPNVSNMSQ CTGSVMENYK TRLNGILTPI KGALEIYKNN THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL	Nipah virus NiV-F F0 (aa 27-546)

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
	LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDEDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMIN NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNTYSRLED RRVRPTSSGD LYYIGT	
9260	ILHYEKLSKIGLVKGVTRKYKIKSNPLTKDIVIKMIPNVS NMSQCTGSVMENYKTRLNGILTPIKGALEIYKNNTHDLVG DVR	Nipah virus NiV-F F2 (aa 27-109)
9261	LAGVIMAGVAIGIATAAQITAGVALYEAMKNADNINKLKS SIESTNEAVVKLQETAEKTVYVLTALQDYINTNLVPTIDK ISCKQTELSLDLALSKYLSDLLFVFGPNLQDPVSNSMTIQ AISQAFGGNYETLLRTLGYATEDFDDLLESDSITGQIIYV DLSSYYIIVRVYFPILTEIQQAYIQELLPVSFNNDNSEWI SIVPNFILVRNTLISNIEIGFCLITKRSVICNQDYATPMT NNMRECLIGSTEKCPRELVVSSHVPRFALSNGVLFANCIS VTCQCQTTGRAISQSGEQTLLMIDNTTCPTAVLGNVIISL GKYLGSVNYNSEGIAIGPPVFTDKVDISSQISSMNQSLQQ SKDYIKEAQRLLDTVNPSLISMLSMIILYVLSIASLCIGL ITFISFIIVEKKRNTYSRLEDRRVRPTSSGDLYYIGT	Nipah virus NiV F F1 (aa 110-546)
9262	ILHY EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK MIPNVSNMSQ CTGSVMENYK TRLNGILTPI KGALEIYKNN THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGE CLITKRSVIC NQDYATPMIN NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNTGT	Nipah virus NiV-F F0 (aa T234 truncation 525-544)
9263	LAGVIMAGVAIGIATAAQITAGVALYEAMKNADNINKLKS SIESTNEAVVKLQETAEKTVYVLTALQDYINTNLVPTIDK ISCKQTELSLDLALSKYLSDLLFVFGPNLQDPVSNSMTIQ AISQAFGGNYETLLRTLGYATEDFDDLLESDSITGQIIYV DLSSYYIIVRVYFPILTEIQQAYIQELLPVSFNNDNSEWI SIVPNFILVRNTLISNIEIGFCLITKRSVICNQDYATPMT NNMRECLTGSTEKCPRELVVSSHVPRFALSNGVLFANCIS VTCQCQTTGRAISQSGEQTLLMIDNTTCPTAVLGNVIISL GKYLGSVNYNSEGIAIGPPVFTDKVDISSQISSMNQSLQQ SKDYIKEAQRLLDTVNPSLISMLSMIILYVLSIASLCIGL ITFISFIIVEKKRNTGT	Nipah virus NiV F F1 (aa 110-546) truncation (aa 525-544)
9264	ILHY EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK MIPNVSNMSQ CTGSVMENYK TRINGILIPI KGALEIYKNQ THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGE CLITKRSVIC NQDYATPMIN NMRECLIGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF	Nipah virus NiV-F F0 T234 truncation (aa 525-544) AND mutation on N-linked glycosylation site

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
	TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNTGT	
9265	MVVILDKRCY CNLLILILMI SECSVGILHY EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK MIPNVSNMSQ CTGSVMENYK TRLNGILTPI KGALEIYKNN THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMTN NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNT	Truncated NiV fusion glycoprotein (FcDelta22) at cytoplasmic tail (with signal sequence)
9266	MGPAENKKVR FENTTSDKGK IPSKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein (602 aa)
9267	MGKVR FENTTSDKGK IPSKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDINAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein Truncated Δ5
9268	MGNTTSDKGK IPSKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD	NiVG protein attachment glycoprotein Truncated Δ10

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
	PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	
9269	MGKGK IPSKVIKSYY GIMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGICITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRIEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein Truncated Δ15
9270	MGSKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRIEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein Truncated Δ20
9271	MGSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN	NiVG protein attachment glycoprotein Truncated Δ25

TABLE 25-continued

G and F Proteins		
SEQ ID NO:	SEQUENCE	ANNOTATION
	DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	
9272	MGTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein Truncated Δ30
9273	MKKINEGLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PAICAEGVYN DAFLIDRINW ISAGVFLDSN ATAANPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QCT	NiVG protein attachment glycoprotein Truncated and mutated (E501A, W504A, Q530A, E533A) NIV G protein (Gc Δ 34)
9274	MATQEVRLKC LLCGIIVLVL SLEGLGILHY EKLSKIGLVK GITRKYKIKS NPLTKDIVIK MIPNVSNVSK CTGTVMENYK SRLIGILSPI KGAIELYNNN THDLVGDVKL AGVVMAGIAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDQI SCKQTELALD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SIAGQIVYVD LSSYYIIVRV YFPILTEIQQ AYVQELLPVS FNNDNSEWIS IVPNFVLIRN TLISNIEVKY CLITKKSVIC NQDYATPMTA SVRECLIGST DKCPRELVVS SHVPRFALSG GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCTTV VLGNIIISLG KYLGSINYNS ESIAVGPPVY TDKVDISSQI SSMNQSLQQS KDYIKEAQKI LDTVNPSLIS MLSMIILYVL SIAALCIGLI TFISFVIVEK KRGNYSRLDD RQVRPVSNGD LYYIGT	Hendra virus F protein Uniprot 089342 (with signal sequence)

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
9275	MMADSKLVSL NNNLSGKIKD QGKVIKNYYG TMDIKKINDG LLDSKILGAF NTVIALLGSI IIIVMNIMII QNYTRTTDNQ ALIKESLQSV QQQIKALTDK IGTEIGPKVS LIDTSSTITI PANIGLLGSK ISQSTSSINE NVNDKCKFTL PPLKIHECNI SCPNPLPFRE YRPISQGVSD LVGLPNQICL QKTTSTILKP RLISYTLPIN TREGVCITDP LLAVDNGFFA YSHLEKIGSC TRGIAKQRII GVGEVLDRGD KVPSMFMTNV WTPPNPSTIH HCSSTYHEDF YYTLCAVSHV GDPILNSTSW TESLSLIRLA VRPKSDSGDY NQKYIAITKV ERGKYDKVMP YGPSGIKQGD TLYFPAVGFL PRTEFQYNDS NCPIIHCKYS KAENCRLSMG VNSKSHYILR SGLLKYNLSL GGDIILQFIE IADNRLTIGS PSKIYNSLGQ PVFYQASYSW DTMIKLGDVD TVDPLRVQWR NNSVISRPGQ SQCPRFNVCP EVCWEGTYND AFLIDRLNWV SAGVYLNSNQ TAENPVFAVF KDNEILYQVP LAEDDTNAQK TITDCFLLEN VIWCISLVEI YDTGDSVIRP KLFAVKIPAQ CSES	Hendra virus G protein Uniprot 089343
9276	MVVILDKRCY CNLLILILMI SECSVGILHY EKLSKIGLVK GVTRKYKIKS NPLIKDIVIK MIPNVSNMSQ CTGSVMENYK TRINGILTPI KGALEIYKNN THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGE CLITKRSVIC NQDYATPMIN NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNTGT	Nipah virus NiV-F F0 T234 truncation (aa 525-544) (with signal sequence)
9277	MVVILDKRCY CNLLILILMI SECSVGILHY EKLSKIGLVK GVTRKYKIKS NPLIKDIVIK MIPNVSNMSQ CTGSVMENYK TRLNGILTPI KGALEIYKNQ THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMIN NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNTGT	Nipah virus NiV-F F0 T234 truncation (aa 525-544) AND mutation on N-linked glycosylation site (with signal sequence)
9278	MVVILDKRCY CNLLILILMI SECSVGILHY EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK MIPNVSNMSQ CTGSVMENYK TRLNGILTPI KGALEIYKNN THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS	Truncated NiV fusion glycoprotein (FcDelta22) at cytoplasmic tail (with signal sequence)

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
	FNNDNSEWIS IVPNFILVRN TLISNIEIGE CLITKRSVIC NQDYATPMIN NMRECLIGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVE TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNT	
9279	MKKINEGLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QCT	NiVG protein attachment glycoprotein Truncated (Gc Δ 34)
9280	ILHY EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK MIPNVSNMSQ CTGSVMENYK TRLNGILTPI KGALEIYKNN THDLVGDVRL AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMIN NMRECLIGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNT	Truncated mature NiV fusion glycoprotein (FcDelta22) at cytoplasmic tail
9281	MSNKRITVLIIISYTLFYLNNAAIVGFDFDKLNKIGVVQG RVLNYKIKGDPMTKDLVLKFIPNIVNITECVREPLSRYNE TVRRLLLPIHNMLGLYLNNTNAKMTGLMIAGVIMGGIAIG IATAAQITAGFALYEAKKNTENIQKLIDSIMKTQDSIDKL TDSVGTSILILNKLQTYINNQLVPNLELLSCRQNKIEFDL MLTKYLVDLMTVIGPNINNPVNKDMTIQSLSLLFDGNYDI MMSELGYTPQDFLDLIESKSITGQIIYVDMENLYVVIRTY LPTLIEVPDAQIYEFNKITMSSNGGEYLSTIPNFILIRGN YMSNIDVATCYMTKASVICNQDYSLPMSQNLRSCYQGETE YCPVEAVIASHSPRFALINGVIFANCINTICRCQDNGKTI TQNINQFVSMIDNSTCNDVMVDKFTIKVGKYMGRKDINNI NIQIGPQIIIDKVDLSNEINKMNQSLKDSIFYLREAKRIL DSVNISLISPSVQLFLIIISVLSFIILLIIIVYLYCKSKH SYKYNKFIDDPDYYNDYKRERINGKASKSNNIYYVGD	gb: JQ001776:6129-8166|Organism: Cedar virus|Strain Name: CG1a|Protein Name: fusion glycoprotein|Gene Symbol: F (with signal sequence)
9282	MALNKNMFSSLFLGYLLVYATTVQSSIHYDSLSKVGVIKG LTYNYKIKGSPSTKLMVVKLIPNIDSVKNCTQKQYDEYKN LVRKALEPVKMAIDTMLNNVKSGNNKYRFAGAIMAGVALG VATAATVTAGIALHRSNENAQAIANMKSAIQNTNEAVKQL QLANKQTLAVIDTIRGEINNNIIPVINQLSCDTIGLSVGI RLTQYYSEIITAFGPALQNPVNTRITIQAISSVENGNFDE	gb: NC_025352:5950-8712|Organism: Mojiang virus|Strain Name: Tongguan1|Protein Name: fusion protein|Gene

TABLE 25-continued

<table>
<tr><th colspan="3">G and F Proteins</th></tr>
<tr><th>SEQ ID NO:</th><th>SEQUENCE</th><th>ANNOTATION</th></tr>
<tr><td></td><td>LLKIMGYTSGDLYEILHSELIRGNIIDVDVDAGYIALEIE
FPNLTLVPNAVVQELMPISYNIDGDEWVTLVPRFVLTRIT
LLSNIDTSRCTITDSSVICDNDYALPMSHELIGCLQGDTS
KCAREKVVSSYVPKFALSDGLVYANCLNTICRCMDTDTPI
SQSLGATVSLLDNKRCSVYQVGDVLISVGSYLGDGEYNAD
NVELGPPIVIDKIDIGNQLAGINQTLQEAEDYIEKSEEFL
KGVNPSIITLGSMVVLYIFMILIAIVSVIALVLSIKLTVK
GNVVRQQFTYTQHVPSMENINYVSH</td><td>Symbol: F (with signal
sequence)</td></tr>
<tr><td>9283</td><td>MKKKTDNPTISKRGHNHSRGIKSRALLRETDNYSNGLIVE
NLVRNCHHPSKNNLNYTKTQKRDSTIPYRVEERKGHYPKI
KHLIDKSYKHIKRGKRRNGHNGNIITIILLLILILKTQMS
EGAIHYETLSKIGLIKGITREYKVKGTPSSKDIVIKLIPN
VTGLNKCTNISMENYKEQLDKILIPINNIIELYANSTKSA
PGNARFAGVIIAGVALGVAAAAQITAGIALHEARQNAERI
NLLKDSISATNNAVAELQEATGGIVNVITGMQDYININLV
PQIDKLQCSQIKTALDISLSQYYSEILTVFGPNLQNPVTT
SMSIQAISQSFGGNIDLLLNLLGYTANDLLDLLESKSITG
QITYINLEHYFMVIRVYYPIMTTISNAYVQELIKISFNVD
GSEWVSLVPSYILIRNSYLSNIDISECLITKNSVICRHDF
AMPMSYTLKECLTGDTEKCPREAVVTSYVPRFAISGGVIY
ANCLSTTCQCYQTGKVIAQDGSQTLMMIDNQTCSIVRIEE
ILISTGKYLGSQEYNTMHVSVGNPVFTDKLDITSQISNIN
QSIEQSKFYLDKSKAILDKINLNLIGSVPISILFIIAILS
LILSIITFVIVMIIVRRYNKYTPLINSDPSSRRSTIQDVY
IIPNPGEHSIRSAARSIDRDRD</td><td>gb: NC_025256:6865-
8853|Organism: Bat
Paramyxovirus
Eid_hel/GH-
M74a/GHA/2009|Strain
Name: BatPV/Eid_hel/
GH-M74a/GHA/2009|
Protein
Name: fusion
protein|Gene
Symbol: F (with signal
sequence)</td></tr>
<tr><td>9284</td><td>(GGGGGS)n wherein n is 1 to 6</td><td>Peptide Linker</td></tr>
<tr><td>9285</td><td>MPAENKKVRFENTTSDKGKIPSKVIKSYYGTMDIKKINEG
LLDSKILSAFNTVIALLGSIVIIVMNIMIIQNYTRSTDNQ
AVIKDALQGIQQQIKGLADKIGTEIGPKVSLIDTSSTITI
PANIGLLGSKISQSTASINENVNEKCKFTLPPLKIHECNI
SCPNPLPFREYRPQTEGVSNLVGLPNNICLQKTSNQILKP
KLISYTLPVVGQSGTCITDPLLAMDEGYFAYSHLERIGSC
SRGVSKQRIIGVGEVLDRGDEVPSLFMTNVWTPPNPNTVY
HCSAVYNNEFYYVLCAVSTVGDPILNSTYWSGSLMMTRLA
VKPKSNGGGYNQHQLALRSIEKGRYDKVMPYGPSGIKQGD
TLYFPAVGFLVRTEFKYNDSNCPITKCQYSKPENCRLSMG
IRPNSHYILRSGLLKYNLSDGENPKVVFIEISDQRLSIGS
PSKIYDSLGQPVFYQASFSWDTMIKFGDVLTVNPLVVNWR
NNTVISRPGQSQCPRFNTCPEICWEGVYNDAFLIDRINWI
SAGVFLDSNQTAENPVFTVFKDNEILYRAQLASEDTNAQK
TITNCFLLKNKIWCISLVEIYDTGDNVIRPKLFAVKIPEQ
CT</td><td>gb: AF212302|Organism:
Nipah virus|Strain
Name: UNKNOWN-
AF212302|Protein
Name: attachment
glycoprotein|Gene
Symbol: G
(Uniprot Q9IH62)</td></tr>
<tr><td>9286</td><td>MLSQLQKNYLDNSNQQGDKMNNPDKKLSVNFNPLELDKGQ
KDLNKSYYVKNKNYNVSNLLNESLHDIKFCIYCIFSLLII
ITIINIITISIVITRLKVHEENNGMESPNLQSIQDSLSSL
TNMINTEITPRIGILVTATSVTLSSSINYVGTKTNQLVNE
LKDYITKSCGFKVPELKLHECNISCADPKISKSAMYSTNA
YAELAGPPKIFCKSVSKDPDFRLKQIDYVIPVQQDRSICM
NNPLLDISDGFFTYIHYEGINSCKKSDSFKVLLSHGEIVD
RGDYRPSLYLLSSHYHPYSMQVINCVPVTCNQSSFVFCHI
SNNTKTLDNSDYSSDEYYITYFNGIDRPKTKKIPINNMTA
DNRYIHFTFSGGGGVCLGEEFIIPVTTVINTDVFTHDYCE
SFNCSVQTGKSLKEICSESLRSPINSSRYNLNGIMIISQN
NMTDFKIQLNGITYNKLSFGSPGRLSKTLGQVLYYQSSMS
WDTYLKAGFVEKWKPFTPNWMNNTVISRPNQGNCPRYHKC
PEICYGGTYNDIAPLDLGKDMYVSVILDSDQLAENPEITV
FNSTTILYKERVSKDELNTRSTTTSCFLFLDEPWCISVLE
TNRFNGKSIRPEIYSYKIPKYC</td><td>gb: JQ001776:8170-
10275|Organism: Cedar
virus|Strain
Name: CG1a|Protein
Name: attachment
glycoprotein|Gene
Symbol: G</td></tr>
<tr><td>9287</td><td>MPQKTVEFINMNSPLERGVSTLSDKKTLNQSKITKQGYFG
LGSHSERNWKKQKNQNDHYMTVSIMILEILVVLGIMENLI
VLTMVYYQNDNINQRMAELTSNITVLNLNLNQLINKIQRE
IIPRITLIDTATTITIPSAITYILATLITRISELLPSINQ
KCEFKTPTLVLNDCRINCTPPLNPSDGVKMSSLATNLVAH
GPSPCRNFSSVPTIYYYRIPGLYNRTALDERCILNPRLTI
SSTKFAYVHSEYDKNCTRGFKYYELMTFGEILEGPEKEPR
MFSRSFYSPTNAVNYHSCTPIVTVNEGYFLCLECTSSDPL</td><td>gb: NC_025256:9117-
11015|Organism: Bat
Paramyxovirus
Eid_hel/GH-
M74a/GHA/2009|Strain
Name: BatPV/Eid_hel/
GH-
M74a/GHA/2009|Protein</td></tr>
</table>

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
	YKANLSNSTFHLVILRHNKDEKIVSMPSFNLSTDQEYVQI IPAEGGGTAESGNLYFPCIGRLLHKRVTHPLCKKSNCSRT DDESCLKSYYNQGSPQHQVVNCLIRIRNAQRDNPTWDVIT VDLTNTYPGSRSRIFGSFSKPMLYQSSVSWHTLLQVAEIT DLDKYQLDWLDTPYISRPGGSECPFGNYCPTVCWEGTYND VYSLTPNNDLFVTVYLKSEQVAENPYFAIFSRDQILKEFP LDAWISSARTTTISCFMFNNEIWCIAALEITRLNDDIIRP IYYSFWLPTDCRTPYPHTGKMTRVPLRSTYNY	Name: glycoprotein| Gene Symbol: G
9288	MATNRDNTITSAEVSQEDKVKKYYGVETAEKVADSISGNK VFILMNTLLILTGAIITITLNITNLTAAKSQQNMLKIIQD DVNAKLEMFVNLDQLVKGEIKPKVSLINTAVSVSIPGQIS NLQTKFLQKYVYLEESITKQCTCNPLSGIFPTSGPTYPPT DKPDDDTTDDDKVDTTIKPIEYPKPDGCNRTGDHFTMEPG ANFYTVPNLGPASSNSDECYTNPSFSIGSSIYMFSQEIRK TDCTAGEILSIQIVLGRIVDKGQQGPQASPLLVWAVPNPK IINSCAVAAGDEMGWVLCSVTLTAASGEPIPHMEDGEWLY KLEPDTEVVSYRITGYAYLLDKQYDSVFIGKGGGIQKGND LYFQMYGLSRNRQSFKALCEHGSCLGTGGGGYQVLCDRAV MSFGSEESLITNAYLKVNDLASGKPVIIGQTFPPSDSYKG SNGRMYTIGDKYGLYLAPSSWNRYLRFGITPDISVRSTTW LKSQDPIMKILSTCTNTDRDMCPEICNTRGYQDIFPLSED SEYYTYIGITPNNGGTKNFVAVRDSDGHIASIDILQNYYS ITSATISCFMYKDEIWCIAITEGKKQKDNPQRIYAHSYKI RQMCYNMKSATVTVGNAKNITIRRY	gb: NC_025352:8716-11257|Organism: Mojiang virus|Strain Name: Tongguan1| Protein Name: attachment glycoprotein|Gene Symbol: G
9289	FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDIGDNVIR PKLFAVKIPE QC	NivG protein attachment glycoprotein Without cytoplasmic tail Uniprot Q9IH62
9290	FNTVIALLGSI IIIVMNIMII QNYTRTTDNQ ALIKESLQSV QQQIKALTDK IGTEIGPKVS LIDTSSTITI PANIGLLGSK ISQSTSSINE NVNDKCKFTL PPLKIHECNI SCPNPLPFRE LVGLPNQICL QKITSTILKP RLISYTLPIN TREGVCITDP LLAVDNGFFA YSHLEKIGSC TRGIAKQRII GVGEVLDRGD KVPSMFMTNV WTPPNPSTIH HCSSTYHEDF YYTLCAVSHV GDPILNSTSW TESLSLIRLA VRPKSDSGDY NQKYIAITKV ERGKYDKVMP YGPSGIKQGD TLYFPAVGFL PRTEFQYNDS NCPIIHCKYS KAENCRLSMG VNSKSHYILR SGLLKYNLSL GGDIILQFIE IADNRLTIGS PSKIYNSLGQ PVFYQASYSW DTMIKLGDVD TVDPLRVQWR NNSVISRPGQ SQCPRFNVCP EVCWEGTYND AFLIDRLNWV SAGVYLNSNQ TAENPVFAVF KDNEILYQVP LAEDDTNAQK TITDCFLLEN VIWCISLVEI YDIGDSVIRP KLFAVKIPAQ CSES	Hendra virus G protein Uniprot O89343 Without cytoplasmic tail YRPISQGVSD
9291	MVVILDKRCY CNLLILILMI SECSVG	Signal sequence
9292	GGGGGS	Peptide linker

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
9293	(GGGGS)n wherein n is 1 to 10	Peptide linker
9294	GGGGS	Peptide linker
9295	PAENKKVR FENTTSDKGK IPSKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein (602 aa) Without N-terminal methionine
9296	KVR FENTTSDKGK IPSKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein Truncated Δ5 Without N-terminal methionine
9297	NTTSDKGK IPSKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein Truncated Δ10 Without N-terminal methionine
9298	KGK IPSKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN	NiVG protein attachment glycoprotein Truncated Δ15

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
	ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDINAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	Without N-terminal methionine
9299	SKVIKSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRFNTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDINAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein Truncated Δ20 Without N-terminal methionine
9300	SYY GTMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDINAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	NiVG protein attachment glycoprotein Truncated Δ25 Without N-terminal methionine
9301	TMDIKKINE GLLDSKILSA FNTVIALLGS IVIIVMNIMI IQNYTRSTDN QAVIKDALQG IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLEMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG	NiVG protein attachment glycoprotein Truncated Δ30 Without N-terminal methionine

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
	QPVFYQASFS WDTMIKFGDV LTVNPLVVNW	
	RNNTVISRPG QSQCPRENTC PEICWEGVYN	
	DAFLIDRINW ISAGVELDSN QTAENPVFTV	
	FKDNEILYRA QLASEDTNAQ KTITNCFLLK	
	NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC	
9302	KKINEGLLDSKILSA FNTVIALLGS IVIIVMNIMI	NiVG protein
	IQNYTRSTDN QAVIKDALQG IQQQIKGLAD	attachment
	KIGTEIGPKV SLIDTSSTIT IPANIGLLGS	glycoprotein
	KISQSTASIN ENVNEKCKFT LPPLKIHECN	Truncated and
	ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC	mutated
	LQKTSNQILK PKLISYTLPV VGQSGTCITD	(E501 A, W504A,
	PLLAMDEGYF AYSHLERIGS CSRGVSKQRI	Q530A, E533A) NIV G
	IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV	protein (Gc Δ 34)
	YHCSAVYNNE FYYVLCAVST VGDPILNSTY	Without N-terminal
	WSGSLMMTRL AVKPKSNGGG YNQHQLALRS	methionine
	IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF	
	LVRTEFKYND SNCPITKCQY SKPENCRLSM	
	GIRPNSHYIL RSGLLKYNLS DGENPKVVFI	
	EISDQRLSIG SPSKIYDSLG QPVFYQASFS	
	WDTMIKFGDV LTVNPLVVNW RNNTVISRPG	
	QSQCPRENTC PAICAEGVYN DAFLIDRINW	
	ISAGVFLDSN ATAANPVFTV FKDNEILYRA	
	QLASEDTNAQ KTITNCFLLK NKIWCISLVE	
	IYDTGDNVIR PKLFAVKIPE QCT	
9303	MADSKLVSL NNNLSGKIKD QGKVIKNYYG	Hendra virus G protein
	TMDIKKINDG LLDSKILGAF	Uniprot O89343
	NTVIALLGSI IIIVMNIMII QNYTRTTDNQ	Without N-terminal
	ALIKESLQSV QQQIKALTDK IGTEIGPKVS	methionine
	LIDTSSTITI PANIGLLGSK ISQSTSSINE	
	NVNDKCKFTL	
	PPLKIHECNI SCPNPLPFRE YRPISQGVSD	
	LVGLPNQICL QKTTSTILKP RLISYTLPIN	
	TREGVCITDP LLAVDNGFFA YSHLEKIGSC	
	TRGIAKQRII GVGEVLDRGD KVP SMFMTNV	
	WTPPNPSTIH HCSSTYHEDF YYTLCAVSHV	
	GDPILNSTSW TESLSLIRLA VRPKSDSGDY	
	NQKYIAITKV ERGKYDKVMP	
	YGPSGIKQGD TLYFPAVGFL PRTEFQYNDS	
	NCPIIHCKYS KAENCRLSMG	
	VNSKSHYILR SGLLKYNLSL GGDIILQFIE	
	IADNRLTIGS PSKIYNSLGQ PVFYQASYSW	
	DTMIKLGDVD TVDPLRVQWR NNSVISRPGQ	
	SQCPRFNVCP	
	EVCWEGTYND AFLIDRLNWV SAGVYLNSNQ	
	TAENPVFAVF KDNEILYQVP LAEDDINAQK	
	TITDCFLLEN VIWCISLVEI YDTGDSVIRP	
	KLFAVKIPAQ CSES	
9304	KKINEGLLDSKILSA FNTVIALLGS IVIIVMNIMI	NiVG protein
	IQNYTRSTDN QAVIKDALQG IQQQIKGLAD	attachment
	KIGTEIGPKV SLIDTSSTIT IPANIGLLGS	glycoprotein
	KISQSTASIN ENVNEKCKFT LPPLKIHECN	Truncated (Gc Δ 34)
	ISCPNPLPER EYRPQTEGVS NLVGLPNNIC	Without N-terminal
	LQKTSNQILK PKLISYTLPV VGQSGICITD	methionine
	PLLAMDEGYF AYSHLERIGS CSRGVSKQRI	
	IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV	
	YHCSAVYNNE FYYVLCAVST VGDPILNSTY	
	WSGSLMMTRL AVKPKSNGGG YNQHQLALRS	
	IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF	
	LVRTEFKYND SNCPITKCQY SKPENCRLSM	
	GIRPNSHYIL RSGLLKYNLS DGENPKVVFI	
	EISDQRLSIG SPSKIYDSLG QPVFYQASFS	
	WDTMIKFGDV LTVNPLVVNW RNNTVISRPG	
	QSQCPRENTC PEICWEGVYN DAFLIDRINW	
	ISAGVFLDSN QTAENPVFTV FKDNEILYRA	
	QLASEDTNAQ KTITNCFLLK NKIWCISLVE	
	IYDTGDNVIR PKLFAVKIPE QCT	

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
9305	LSQLQKNYLDNSNQQGDKMNNPDKKLSVNFNPLELDKGQK DLNKSYYVKNKNYNVSNLLNESLHDIKFCIYCIFSLLIII TIINIITISIVITRLKVHEENNGMESPNLQSIQDSLSSLT NMINTEITPRIGILVTATSVTLSSSINYVGTKTNQLVNEL KDYITKSCGFKVPELKLHECNISCADPKISKSAMYSTNAY AELAGPPKIFCKSVSKDPDFRLKQIDYVIPVQQDRSICMN NPLLDISDGFFTYIHYEGINSCKKSDSFKVLLSHGEIVDR GDYRPSLYLLSSHYHPYSMQVINCVPVTCNQSSFVFCHIS NNTKTLDNSDYSSDEYYITYFNGIDRPKTKKIPINNMTAD NRYIHFTFSGGGGVCLGEEFIIPVTTVINTDVFTHDYCES FNCSVQTGKSLKEICSESLRSPINSSRYNLNGIMIISQNN MTDFKIQLNGITYNKLSFGSPGRLSKTLGQVLYYQSSMSW DTYLKAGFVEKWKPFTPNWMNNTVISRPNQGNCPRYHKCP EICYGGTYNDIAPLDLGKDMYVSVILDSDQLAENPEITVF NSTTILYKERVSKDELNTRSTTTSCFLFLDEPWCISVLET NRFNGKSIRPEIYSYKIPKYC	gb: JQ001776:8170-10275|Organism: Cedar virus|Strain Name: CG1a|Protein Name: attachment glycoprotein|Gene Symbol: G Without N-terminal methionine
9306	PQKTVEFINMNSPLERGVSTLSDKKTLNQSKITKQGYFGL GSHSERNWKKQKNQNDHYMTVSTMILEILVVLGIMENLIV LTMVYYQNDNINQRMAELISNITVLNLNLNQLINKIQREI IPRITLIDTATTITIPSAITYILATLTTRISELLPSINQK CEFKTPTLVLNDCRINCTPPLNPSDGVKMSSLATNLVAHG PSPCRNFSSVPTIYYYRIPGLYNRTALDERCILNPRLTIS STKFAYVHSEYDKNCTRGFKYYELMTFGEILEGPEKEPRM FSRSFYSPTNAVNYHSCTPIVTVNEGYFLCLECTSSDPLY KANLSNSTFHLVILRHNKDEKIVSMPSFNLSTDQEYVQII PAEGGGTAESGNLYFPCIGRLLHKRVTHPLCKKSNCSRID DESCLKSYYNQGSPQHQVVNCLIRIRNAQRDNPTWDVITV DLTNTYPGSRSRIFGSFSKPMLYQSSVSWHILLQVAEITD LDKYQLDWLDTPYISRPGGSECPFGNYCPTVCWEGTYNDV YSLTPNNDLFVTVYLKSEQVAENPYFAIFSRDQILKEFPL DAWISSARTTTISCFMFNNEIWCIAALEITRLNDDIIRPI YYSFWLPTDCRTPYPHTGKMTRVPLRSTYNY	gb: NC_025256:9117-11015|Organism: Bat Paramyxovirus Eid_hel/GH-M74a/GHA/2009|Strain Name: BatPV/Eid_hel/GH-M74a/GHA/2009|Protein Name: glycoprotein|Gene Symbol: G Without N-terminal methionine
9307	ATNRDNTITSAEVSQEDKVKKYYGVETAEKVADSISGNKV FILMNTLLILTGAIITITLNITNLTAAKSQQNMLKIIQDD VNAKLEMFVNLDQLVKGEIKPKVSLINTAVSVSIPGQISN LQTKFLQKYVYLEESITKQCTCNPLSGIFPTSGPTYPPTD KPDDDTTDDDKVDTTIKPIEYPKPDGCNRTGDHFTMEPGA NFYTVPNLGPASSNSDECYINPSFSIGSSIYMFSQEIRKT DCTAGEILSIQIVLGRIVDKGQQGPQASPLLVWAVPNPKI INSCAVAAGDEMGWVLCSVTLTAASGEPIPHMEDGFWLYK LEPDTEVVSYRITGYAYLLDKQYDSVFIGKGGGIQKGNDL YFQMYGLSRNRQSFKALCEHGSCLGTGGGGYQVLCDRAVM SFGSEESLITNAYLKVNDLASGKPVIIGQTFPPSDSYKGS NGRMYTIGDKYGLYLAPSSWNRYLRFGITPDISVRSTTWL KSQDPIMKILSTCTNTDRDMCPEICNTRGYQDIFPLSEDS EYYTYIGITPNNGGTKNFVAVRDSDGHIASIDILQNYYSI TSATISCFMYKDEIWCIAITEGKKQKDNPQRIYAHSYKIR QMCYNMKSATVTVGNAKNITIRRY	gb: NC_025352:8716-11257|Organism: Mojiang virus|Strain Name: Tongguan1|Protein Name: attachment glycoprotein|Gene Symbol: G Without N-terminal methionine
9308	DFDKLNKIGVVQGRVLNYKIKGDPMTKDLVLKFIPNIVNI TECVREPLSRYNETVRRLLLPIHNMLGLYLNNTNAKMTGL MIAGVIMGGIAIGIATAAQITAGFALYEAKKNTENIQKLT DSIMKTQDSIDKLTDSVGTSILILNKLQTYINNQLVPNLE LLSCRQNKIEFDLMLTKYLVDLMTVIGPNINNPVNKDMTI QSLSLLFDGNYDIMMSELGYTPQDFLDLIESKSITGQIIY VDMENLYVVIRTYLPTLIEVPDAQIYEFNKITMSSNGGEY LSTIPNFILIRGNYMSNIDVATCYMTKASVICNQDYSLPM SQNLRSCYQGETEYCPVEAVIASHSPRFALINGVIFANCI NTICRCQDNGKTITQNINQFVSMIDNSTCNDVMVDKFTIK VGKYMGRKDINNINIQIGPQIIIDKVDLSNEINKMNQSLK DSIFYLREAKRILDSVNISLISPSVQLFLIIISVLSFIIL LIIIVYLYCKSKHSYKYNKFIDDPDYYNDYKRERINGKAS KSNNIYYVGD	gb: JQ001776:6129-8166|Organism: Cedar virus|Strain Name: CG1a|Protein Name: fusion glycoprotein|Gene Symbol: F (without signal sequence)
9309	SRALLRETDNYSNGLIVENLVRNCHHPSKNNLNYTKTQKR DSTIPYRVEERKGHYPKIKHLIDKSYKHIKRGKRRNGHNG NIITIILLLILILKTQMSEGAIHYETLSKIGLIKGITREY KVKGTPSSKDIVIKLIPNVTGLNKCTNISMENYKEQLDKI LIPINNIIELYANSTKSAPGNARFAGVIIAGVALGVAAAA QITAGIALHEARQNAERINLLKDSISATNNAVAELQEATG	gb: NC_025256:6865-8853|Organism: Bat Paramyxovirus Eid_hel/GH-M74a/GHA/2009|Strain Name: BatPV/Eid_hel/

TABLE 25-continued

G and F Proteins

SEQ ID NO:	SEQUENCE	ANNOTATION
	GIVNVITGMQDYINTNLVPQIDKLQCSQIKTALDISLSQY YSEILTVFGPNLQNPVTTSMSIQAISQSFGGNIDLLLNLL GYTANDLLDLLESKSITGQITYINLEHYFMVIRVYYPIMT TISNAYVQELIKISFNVDGSEWVSLVPSYILIRNSYLSNI DISECLITKNSVICRHDFAMPMSYTLKECLTGDTEKCPRE AVVTSYVPRFAISGGVIYANCLSTTCQCYQTGKVIAQDGS QTLMMIDNQTCSIVRIEEILISTGKYLGSQEYNTMHVSVG NPVFTDKLDITSQISNINQSIEQSKFYLDKSKAILDKINL NLIGSVPISILFIIAILSLILSIITFVIVMIIVRRYNKYT PLINSDPSSRRSTIQDVYIIPNPGEHSIRSAARSIDRDRD	GH-M74a/GHA/2009|Protein Name: fusion protein|Gene Symbol: F (without signal sequence)
9310	ILHY EKLSKIGLVK GITRKYKIKS NPLTKDIVIK MIPNVSNVSK CTGTVMENYK SRLIGILSPI KGAIELYNNN THDLVGDVKL AGVVMAGIAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDQI SCKQTELALD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SIAGQIVYVD LSSYYIIVRV YFPILTEIQQ AYVQELLPVS FNNDNSEWIS IVPNFVLIRN TLISNIEVKY CLITKKSVIC NQDYATPMTA SVRECLIGST DKCPRELVVS SHVPRFALSG GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCTTV VLGNIIISLG KYLGSINYNS ESIAVGPPVY TDKVDISSQI SSMNQSLQQS KDYIKEAQKI LDTVNPSLIS MLSMIILYVL SIAALCIGLI TFISFVIVEK KRGNYSRLDD RQVRPVSNGD LYYIGT	Hendra virus F protein Uniprot O89342 (without signal sequence)
9311	IHYDSLSKVGVIKGLTYNYKIKGSPSTKLMVVKLIPNIDS VKNCTQKQYDEYKNLVRKALEPVKMAIDTMLNNVKSGNNK YRFAGAIMAGVALGVATAATVTAGIALHRSNENAQAIANM KSAIQNTNEAVKQLQLANKQTLAVIDTIRGEINNNIIPVI NQLSCDTIGLSVGIRLTQYYSEIITAFGPALQNPVNTRIT IQAISSVENGNFDELLKIMGYTSGDLYEILHSELIRGNII DVDVDAGYIALEIEFPNLILVPNAVVQELMPISYNIDGDE WVTLVPRFVLTRTILLSNIDTSRCTITDSSVICDNDYALP MSHELIGCLQGDTSKCAREKVVSSYVPKFALSDGLVYANC LNTICRCMDTDTPISQSLGATVSLLDNKRCSVYQVGDVLI SVGSYLGDGEYNADNVELGPPIVIDKIDIGNQLAGINQTL QEAEDYIEKSEEFLKGVNPSIITLGSMVVLYIFMILIAIV SVIALVLSIKLTVKGNVVRQQFTYTQHVP SMENINYVSH	gb: NC_025352:5950-8712|Organism: Mojiang virus|Strain Name: Tongguan1| Protein Name: fusion protein|Gene Symbol: F (without signal sequence)
9312	(GGGS)n wherein n is 1 to 10	Peptide linker
9313	GGGGSGGGGSGGGGS	Peptide linker
9314	TTAASGSSGGSSSGA	Peptide linker
9315	GSTSGSGKPGSGEGSTKG	Peptide linker

SEQUENCE LISTING

The patent application contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US20250281536A1). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

1. An antibody or antigen binding fragment thereof that specifically binds CD4, comprising a heavy chain variable region (VH) and/or a light chain variable region (VL), wherein the heavy chain variable region comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3), and the light chain variable region comprises three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3, respectively, comprise:
a) SEQ ID NOs: 1280, 1794, 2308, 5644, 6154, 6664, respectively;
b) SEQ ID NOs: 1281, 1795, 2309, 5645, 6155, 6665, respectively;
c) SEQ ID NOs: 1282, 1796, 2310, 5646, 6156, 6666, respectively;
d) SEQ ID NOs: 1283, 1797, 2311, 5647, 6157, 6667, respectively;
e) SEQ ID NOs: 1284, 1798, 2312, 5648, 6158, 6668, respectively;
f) SEQ ID NOs: 1285, 1799, 2313, 5649, 6159, 6669, respectively;
g) SEQ ID NOs: 1286, 1800, 2314, 5650, 6160, 6670, respectively;
h) SEQ ID NOs: 1287, 1801, 2315, 5651, 6161, 6671, respectively;
i) SEQ ID NOs: 1288, 1802, 2316, 5652, 6162, 6672, respectively;
j) SEQ ID NOs: 1289, 1803, 2317, 5653, 6163, 6673, respectively;
k) SEQ ID NOs: 1290, 1804, 2318, 5654, 6164, 6674, respectively;
l) SEQ ID NOs: 1291, 1805, 2319, 5655, 6165, 6675, respectively;
m) SEQ ID NOs: 1292, 1806, 2320, 5656, 6166, 6676, respectively;
n) SEQ ID NOs: 1293, 1807, 2321, 5657, 6167, 6677, respectively;
o) SEQ ID NOs: 1294, 1808, 2322, 5658, 6168, 6678, respectively;
p) SEQ ID NOs: 1295, 1809, 2323, 5659, 6169, 6679, respectively;
q) SEQ ID NOs: 1296, 1810, 2324, 5660, 6170, 6680, respectively;
r) SEQ ID NOs: 1297, 1811, 2325, 5661, 6171, 6681, respectively;
s) SEQ ID NOs: 1298, 1812, 2326, 5662, 6172, 6682, respectively;
t) SEQ ID NOs: 1299, 1813, 2327, 5663, 6173, 6683, respectively;
u) SEQ ID NOs: 1300, 1814, 2328, 5664, 6174, 6684, respectively;
v) SEQ ID NOs: 1301, 1815, 2329, 5665, 6175, 6685, respectively;
w) SEQ ID NOs: 1302, 1816, 2330, 5666, 6176, 6686, respectively;
x) SEQ ID NOs: 1303, 1817, 2331, 5667, 6177, 6687, respectively;
y) SEQ ID NOs: 1304, 1818, 2332, 5668, 6178, 6688, respectively;
z) SEQ ID NOs: 1305, 1819, 2333, 5669, 6179, 6689, respectively;
aa) SEQ ID NOs: 1306, 1820, 2334, 5670, 6180, 6690, respectively;
bb) SEQ ID NOs: 1307, 1821, 2335, 5671, 6181, 6691, respectively;
cc) SEQ ID NOs: 1308, 1822, 2336, 5672, 6182, 6692, respectively;
dd) SEQ ID NOs: 1309, 1823, 2337, 5673, 6183, 6693, respectively;
ee) SEQ ID NOs: 1310, 1824, 2338, 5674, 6184, 6694, respectively;
ff) SEQ ID NOs: 1311, 1825, 2339, 5675, 6185, 6695, respectively;
gg) SEQ ID NOs: 1312, 1826, 2340, 5676, 6186, 6696, respectively;
hh) SEQ ID NOs: 1313, 1827, 2341, 5677, 6187, 6697, respectively;
ii) SEQ ID NOs: 1314, 1828, 2342, 5678, 6188, 6698, respectively;
jj) SEQ ID NOs: 1315, 1829, 2343, 5679, 6189, 6699, respectively;
kk) SEQ ID NOs: 1316, 1830, 2344, 5680, 6190, 6700, respectively;
ll) SEQ ID NOs: 1317, 1831, 2345, 5681, 6191, 6701, respectively;
mm) SEQ ID NOs: 1318, 1832, 2346, 5682, 6192, 6702, respectively;
nn) SEQ ID NOs: 1319, 1833, 2347, 5683, 6193, 6703, respectively;
oo) SEQ ID NOs: 1320, 1834, 2348, 5684, 6194, 6704, respectively;
pp) SEQ ID NOs: 1321, 1835, 2349, 5685, 6195, 6705, respectively;
qq) SEQ ID NOs: 1322, 1836, 2350, 5686, 6196, 6706, respectively;
rr) SEQ ID NOs: 1323, 1837, 2351, 5687, 6197, 6707, respectively;
ss) SEQ ID NOs: 1324, 1838, 2352, 5688, 6198, 6708, respectively;

tt) SEQ ID NOs: 1325, 1839, 2353, 5689, 6199, 6709, respectively;
uu) SEQ ID NOs: 1326, 1840, 2354, 5690, 6200, 6710, respectively;
vv) SEQ ID NOs: 1327, 1841, 2355, 5691, 6201, 6711, respectively;
ww) SEQ ID NOs: 1328, 1842, 2356, 5692, 6202, 6712, respectively;
xx) SEQ ID NOs: 1329, 1843, 2357, 5693, 6203, 6713, respectively;
yy) SEQ ID NOs: 1330, 1844, 2358, 5694, 6204, 6714, respectively;
zz) SEQ ID NOs: 1331, 1845, 2359, 5695, 6205, 6715, respectively;
aaa) SEQ ID NOs: 1332, 1846, 2360, 5696, 6206, 6716, respectively;
bbb) SEQ ID NOs: 1333, 1847, 2361, 5697, 6207, 6717, respectively;
ccc) SEQ ID NOs: 1334, 1848, 2362, 5698, 6208, 6718, respectively;
ddd) SEQ ID NOs: 1335, 1849, 2363, 5699, 6209, 6719, respectively;
eee) SEQ ID NOs: 1336, 1850, 2364, 5700, 6210, 6720, respectively;
fff) SEQ ID NOs: 1337, 1851, 2365, 5701, 6211, 6721, respectively;
ggg) SEQ ID NOs: 1338, 1852, 2366, 5702, 6212, 6722, respectively;
hhh) SEQ ID NOs: 1339, 1853, 2367, 5703, 6213, 6723, respectively;
iii) SEQ ID NOs: 1340, 1854, 2368, 5704, 6214, 6724, respectively;
jjj) SEQ ID NOs: 1341, 1855, 2369, 5705, 6215, 6725, respectively;
kkk) SEQ ID NOs: 1342, 1856, 2370, 5706, 6216, 6726, respectively;
lll) SEQ ID NOs: 1343, 1857, 2371, 5707, 6217, 6727, respectively;
mmm) SEQ ID NOs: 1344, 1858, 2372, 5708, 6218, 6728, respectively;
nnn) SEQ ID NOs: 1345, 1859, 2373, 5709, 6219, 6729, respectively;
ooo) SEQ ID NOs: 1346, 1860, 2374, 5710, 6220, 6730, respectively;
ppp) SEQ ID NOs: 1347, 1861, 2375, 5711, 6221, 6731, respectively;
qqq) SEQ ID NOs: 1348, 1862, 2376, 5712, 6222, 6732, respectively;
rrr) SEQ ID NOs: 1349, 1863, 2377, 5713, 6223, 6733, respectively;
sss) SEQ ID NOs: 1350, 1864, 2378, 5714, 6224, 6734, respectively;
ttt) SEQ ID NOs: 1351, 1865, 2379, 5715, 6225, 6735, respectively;
uuu) SEQ ID NOs: 1352, 1866, 2380, 5716, 6226, 6736, respectively;
vvv) SEQ ID NOs: 1353, 1867, 2381, 5717, 6227, 6737, respectively;
www) SEQ ID NOs: 1354, 1868, 2382, 5718, 6228, 6738, respectively;
xxx) SEQ ID NOs: 1355, 1869, 2383, 5719, 6229, 6739, respectively;
yyy) SEQ ID NOs: 1356, 1870, 2384, 5720, 6230, 6740, respectively;
zzz) SEQ ID NOs: 1357, 1871, 2385, 5721, 6231, 6741, respectively;
aaaa) SEQ ID NOs: 1358, 1872, 2386, 5722, 6232, 6742, respectively;
bbbb) SEQ ID NOs: 1359, 1873, 2387, 5723, 6233, 6743, respectively;
cccc) SEQ ID NOs: 1360, 1874, 2388, 5724, 6234, 6744, respectively;
dddd) SEQ ID NOs: 1361, 1875, 2389, 5725, 6235, 6745, respectively;
eeee) SEQ ID NOs: 1362, 1876, 2390, 5726, 6236, 6746, respectively;
ffff) SEQ ID NOs: 1363, 1877, 2391, 5727, 6237, 6747, respectively;
gggg) SEQ ID NOs: 1364, 1878, 2392, 5728, 6238, 6748, respectively;
hhhh) SEQ ID NOs: 1365, 1879, 2393, 5729, 6239, 6749, respectively;
iiii) SEQ ID NOs: 1366, 1880, 2394, 5730, 6240, 6750, respectively;
jjjj) SEQ ID NOs: 1367, 1881, 2395, 5731, 6241, 6751, respectively;
kkkk) SEQ ID NOs: 1368, 1882, 2396, 5732, 6242, 6752, respectively;
llll) SEQ ID NOs: 1369, 1883, 2397, 5733, 6243, 6753, respectively;
mmmm) SEQ ID NOs: 1370, 1884, 2398, 5734, 6244, 6754, respectively;
nnnn) SEQ ID NOs: 1371, 1885, 2399, 5735, 6245, 6755, respectively;
oooo) SEQ ID NOs: 1372, 1886, 2400, 5736, 6246, 6756, respectively;
pppp) SEQ ID NOs: 1373, 1887, 2401, 5737, 6247, 6757, respectively;
qqqq) SEQ ID NOs: 1374, 1888, 2402, 5738, 6248, 6758, respectively;
rrrr) SEQ ID NOs: 1375, 1889, 2403, 5739, 6249, 6759, respectively;
ssSS) SEQ ID NOs: 1376, 1890, 2404, 5740, 6250, 6760, respectively;
tttt) SEQ ID NOs: 1377, 1891, 2405, 5741, 6251, 6761, respectively;
uuuu) SEQ ID NOs: 1378, 1892, 2406, 5742, 6252, 6762, respectively;
vvvv) SEQ ID NOs: 1379, 1893, 2407, 5743, 6253, 6763, respectively;
wwww) SEQ ID NOs: 1380, 1894, 2408, 5744, 6254, 6764, respectively;
xxxx) SEQ ID NOs: 1381, 1895, 2409, 5745, 6255, 6765, respectively;
yyyy) SEQ ID NOs: 1382, 1896, 2410, 5746, 6256, 6766, respectively;
zzzz) SEQ ID NOs: 1383, 1897, 2411, 5747, 6257, 6767, respectively;
aaaaa) SEQ ID NOs: 1384, 1898, 2412, 5748, 6258, 6768, respectively;
bbbbb) SEQ ID NOs: 1385, 1899, 2413, 5749, 6259, 6769, respectively;
ccccc) SEQ ID NOs: 1386, 1900, 2414, 5750, 6260, 6770, respectively;
ddddd) SEQ ID NOs: 1387, 1901, 2415, 5751, 6261, 6771, respectively;
eeeee) SEQ ID NOs: 1388, 1902, 2416, 5752, 6262, 6772, respectively;

fffff) SEQ ID NOs: 1389, 1903, 2417, 5753, 6263, 6773, respectively;
ggggg) SEQ ID NOs: 1390, 1904, 2418, 5754, 6264, 6774, respectively;
hhhhh) SEQ ID NOs: 1391, 1905, 2419, 5755, 6265, 6775, respectively;
iiiii) SEQ ID NOs: 1392, 1906, 2420, 5756, 6266, 6776, respectively;
jjjjj) SEQ ID NOs: 1393, 1907, 2421, 5757, 6267, 6777, respectively;
kkkkk) SEQ ID NOs: 1394, 1908, 2422, 5758, 6268, 6778, respectively;
lllll) SEQ ID NOs: 1395, 1909, 2423, 5759, 6269, 6779, respectively;
mmmmm) SEQ ID NOs: 1396, 1910, 2424, 5760, 6270, 6780, respectively;
nnnnn) SEQ ID NOs: 1397, 1911, 2425, 5761, 6271, 6781, respectively;
ooooo) SEQ ID NOs: 1398, 1912, 2426, 5762, 6272, 6782, respectively;
ppppp) SEQ ID NOs: 1399, 1913, 2427, 5763, 6273, 6783, respectively;
qqqqq) SEQ ID NOs: 1400, 1914, 2428, 5764, 6274, 6784, respectively;
rrrrr) SEQ ID NOs: 1401, 1915, 2429, 5765, 6275, 6785, respectively;
sssss) SEQ ID NOs: 1402, 1916, 2430, 5766, 6276, 6786, respectively;
ttttt) SEQ ID NOs: 1403, 1917, 2431, 5767, 6277, 6787, respectively;
uuuuu) SEQ ID NOs: 1404, 1918, 2432, 5768, 6278, 6788, respectively;
vvvvv) SEQ ID NOs: 1405, 1919, 2433, 5769, 6279, 6789, respectively;
wwwww) SEQ ID NOs: 1406, 1920, 2434, 5770, 6280, 6790, respectively;
xxxxx) SEQ ID NOs: 1407, 1921, 2435, 5771, 6281, 6791, respectively;
yyyyy) SEQ ID NOs: 1408, 1922, 2436, 5772, 6282, 6792, respectively;
zzzzz) SEQ ID NOs: 1409, 1923, 2437, 5773, 6283, 6793, respectively;
aaaaaa) SEQ ID NOs: 1410, 1924, 2438, 5774, 6284, 6794, respectively;
bbbbbb) SEQ ID NOs: 1411, 1925, 2439, 5775, 6285, 6795, respectively;
cccccc) SEQ ID NOs: 1412, 1926, 2440, 5776, 6286, 6796, respectively;
dddddd) SEQ ID NOs: 1413, 1927, 2441, 5777, 6287, 6797, respectively;
eeeeee) SEQ ID NOs: 1414, 1928, 2442, 5778, 6288, 6798, respectively;
ffffff) SEQ ID NOs: 1415, 1929, 2443, 5779, 6289, 6799, respectively;
gggggg) SEQ ID NOs: 1416, 1930, 2444, 5780, 6290, 6800, respectively;
hhhhhh) SEQ ID NOs: 1417, 1931, 2445, 5781, 6291, 6801, respectively;
iiiiii) SEQ ID NOs: 1418, 1932, 2446, 5782, 6292, 6802, respectively;
jjjjjj) SEQ ID NOs: 1419, 1933, 2447, 5783, 6293, 6803, respectively;
kkkkkk) SEQ ID NOs: 1420, 1934, 2448, 5784, 6294, 6804, respectively;
llllll) SEQ ID NOs: 1421, 1935, 2449, 5785, 6295, 6805, respectively;
mmmmmm) SEQ ID NOs: 1422, 1936, 2450, 5786, 6296, 6806, respectively;
nnnnnn) SEQ ID NOs: 1423, 1937, 2451, 5787, 6297, 6807, respectively;
oooooo) SEQ ID NOs: 1424, 1938, 2452, 5788, 6298, 6808, respectively;
pppppp) SEQ ID NOs: 1425, 1939, 2453, 5789, 6299, 6809, respectively;
qqqqqq) SEQ ID NOs: 1426, 1940, 2454, 5790, 6300, 6810, respectively;
rrrrrr) SEQ ID NOs: 1427, 1941, 2455, 5791, 6301, 6811, respectively;
ssssss) SEQ ID NOs: 1428, 1942, 2456, 5792, 6302, 6812, respectively;
tttttt) SEQ ID NOs: 1429, 1943, 2457, 5793, 6303, 6813, respectively;
uuuuuu) SEQ ID NOs: 1430, 1944, 2458, 5794, 6304, 6814, respectively;
vvvvvv) SEQ ID NOs: 1431, 1945, 2459, 5795, 6305, 6815, respectively;
wwwwww) SEQ ID NOs: 1432, 1946, 2460, 5796, 6306, 6816, respectively;
xxxxxx) SEQ ID NOs: 1433, 1947, 2461, 5797, 6307, 6817, respectively;
yyyyyy) SEQ ID NOs: 1434, 1948, 2462, 5798, 6308, 6818, respectively;
zzzzzz) SEQ ID NOs: 1435, 1949, 2463, 5799, 6309, 6819, respectively;
aaaaaaa) SEQ ID NOs: 1436, 1950, 2464, 5800, 6310, 6820, respectively;
bbbbbbb) SEQ ID NOs: 1437, 1951, 2465, 5801, 6311, 6821, respectively;
ccccccc) SEQ ID NOs: 1438, 1952, 2466, 5802, 6312, 6822, respectively;
ddddddd) SEQ ID NOs: 1439, 1953, 2467, 5803, 6313, 6823, respectively;
eeeeeee) SEQ ID NOs: 1440, 1954, 2468, 5804, 6314, 6824, respectively;
fffffff) SEQ ID NOs: 1441, 1955, 2469, 5805, 6315, 6825, respectively;
ggggggg) SEQ ID NOs: 1442, 1956, 2470, 5806, 6316, 6826, respectively;
hhhhhhh) SEQ ID NOs: 1443, 1957, 2471, 5807, 6317, 6827, respectively;
iiiiiii) SEQ ID NOs: 1444, 1958, 2472, 5808, 6318, 6828, respectively;
jjjjjjj) SEQ ID NOs: 1445, 1959, 2473, 5809, 6319, 6829, respectively;
kkkkkkk) SEQ ID NOs: 1446, 1960, 2474, 5810, 6320, 6830, respectively;
lllllll) SEQ ID NOs: 1447, 1961, 2475, 5811, 6321, 6831, respectively;
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ppppppp) SEQ ID NOs: 1451, 1965, 2479, 5815, 6325, 6835, respectively;
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wwwwwww) SEQ ID NOs: 1458, 1972, 2486, 5822, 6332, 6842, respectively;
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iiiiiiii) SEQ ID NOs: 1470, 1984, 2498, 5834, 6344, 6854, respectively;
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hhhhhhhhh) SEQ ID NOs: 1495, 2009, 2523, 5859, 6369, 6879, respectively;
iiiiiiiii) SEQ ID NOs: 1496, 2010, 2524, 5860, 6370, 6880, respectively;
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kkkkkkkkk) SEQ ID NOs: 1498, 2012, 2526, 5862, 6372, 6882, respectively;
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uuuuuuuuu) SEQ ID NOs: 1508, 2022, 2536, 5872, 6382, 6892, respectively;
vvvvvvvvv) SEQ ID NOs: 1509, 2023, 2537, 5873, 6383, 6893, respectively;
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zzzzzzzzz) SEQ ID NOs: 1513, 2027, 2541, 5877, 6387, 6897, respectively;
aaaaaaaaaa) SEQ ID NOs: 1514, 2028, 2542, 5878, 6388, 6898, respectively;
bbbbbbbbbb) SEQ ID NOs: 1515, 2029, 2543, 5879, 6389, 6899, respectively;
cccccccccc) SEQ ID NOs: 1516, 2030, 2544, 5880, 6390, 6900, respectively;

dddddddddd) SEQ ID NOs: 1517, 2031, 2545, 5881, 6391, 6901, respectively;
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oooooooooo) SEQ ID NOs: 1528, 2042, 2556, 5892, 6402, 6912, respectively;
pppppppppp) SEQ ID NOs: 1529, 2043, 2557, 5893, 6403, 6913, respectively;
qqqqqqqqqq) SEQ ID NOs: 1530, 2044, 2558, 5894, 6404, 6914, respectively;
rrrrrrrrrr) SEQ ID NOs: 1531, 2045, 2559, 5895, 6405, 6915, respectively;
ssssssssss) SEQ ID NOs: 1532, 2046, 2560, 5896, 6406, 6916, respectively;
tttttttttt) SEQ ID NOs: 1533, 2047, 2561, 5897, 6407, 6917, respectively;
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xxxxxxxxxx) SEQ ID NOs: 9970, 10232, 10494, 12196, 12456, 12716, respectively;
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ccccccccccc) SEQ ID NOs: 9975, 10237, 10499, 12201, 12461, 12721, respectively;
ddddddddddd) SEQ ID NOs: 9976, 10238, 10500, 12202, 12462, 12722, respectively;
eeeeeeeeeee) SEQ ID NOs: 9977, 10239, 10501, 12203, 12463, 12723, respectively;
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hhhhhhhhhhh) SEQ ID NOs: 9980, 10242, 10504, 12206, 12466, 12726, respectively;
iiiiiiiiiii) SEQ ID NOs: 9981, 10243, 10505, 12207, 12467, 12727, respectively;
jjjjjjjjjjj) SEQ ID NOs: 9982, 10244, 10506, 12208, 12468, 12728, respectively;
kkkkkkkkkkk) SEQ ID NOs: 9983, 10245, 10507, 12209, 12469, 12729, respectively;
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wwwwwwwwwww) SEQ ID NOs: 9995, 10257, 10519, 12221, 12481, 12741, respectively;
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yyyyyyyyyyy) SEQ ID NOs: 9997, 10259, 10521, 12223, 12483, 12743, respectively;
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cccccccccccc) SEQ ID NOs: 10001, 10263, 10525, 12227, 12487, 12747, respectively;
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hhhhhhhhhhhh) SEQ ID NOs: 10006, 10268, 10530, 12232, 12492, 12752, respectively;
iiiiiiiiiiii) SEQ ID NOs: 10007, 10269, 10531, 12233, 12493, 12753, respectively;
jjjjjjjjjjjj) SEQ ID NOs: 10008, 10270, 10532, 12234, 12494, 12754, respectively;
kkkkkkkkkkkk) SEQ ID NOs: 10009, 10271, 10533, 12235, 12495, 12755, respectively;
llllllllllll) SEQ ID NOs: 10010, 10272, 10534, 12236, 12496, 12756, respectively;
mmmmmmmmmmmm) SEQ ID NOs: 10011, 10273, 10535, 12237, 12497, 12757, respectively;
nnnnnnnnnnnn) SEQ ID NOs: 10012, 10274, 10536, 12238, 12498, 12758, respectively;
oooooooooooo) SEQ ID NOs: 10013, 10275, 10537, 12239, 12499, 12759, respectively;

pppppppppppp) SEQ ID NOs: 10014, 10276, 10538, 12240, 12500, 12760, respectively;
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ssssssssssss) SEQ ID NOs: 10017, 10279, 10541, 12243, 12503, 12763, respectively;
tttttttttttt) SEQ ID NOs: 10018, 10280, 10542, 12244, 12504, 12764, respectively;
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vvvvvvvvvvvv) SEQ ID NOs: 10020, 10282, 10544, 12246, 12506, 12766, respectively;
wwwwwwwwwwww) SEQ ID NOs: 10021, 10283, 10545, 12247, 12507, 12767, respectively;
xxxxxxxxxxxx) SEQ ID NOs: 10022, 10284, 10546, 12248, 12508, 12768, respectively;
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aaaaaaaaaaaaa) SEQ ID NOs: 10025, 10287, 10549, 12251, 12511, 12771, respectively;
bbbbbbbbbbbbb) SEQ ID NOs: 10026, 10288, 10550, 12252, 12512, 12772, respectively;
ccccccccccccc) SEQ ID NOs: 10027, 10289, 10551, 12253, 12513, 12773, respectively;
ddddddddddddd) SEQ ID NOs: 10028, 10290, 10552, 12254, 12514, 12774, respectively;
eeeeeeeeeeeee) SEQ ID NOs: 10029, 10291, 10553, 12255, 12515, 12775, respectively;
fffffffffffff) SEQ ID NOs: 10030, 10292, 10554, 12256, 12516, 12776, respectively;
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hhhhhhhhhhhhh) SEQ ID NOs: 10032, 10294, 10556, 12258, 12518, 12778, respectively;
iiiiiiiiiiiii) SEQ ID NOs: 10033, 10295, 10557, 12259, 12519, 12779, respectively;
jjjjjjjjjjjjj) SEQ ID NOs: 10034, 10296, 10558, 12260, 12520, 12780, respectively;
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cccccccccccccc) SEQ ID NOs: 10053, 10315, 10577, 12279, 12539, 12799, respectively;
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aaaaaaaaaaaaaaa) SEQ ID NOs: 10077, 10339, 10601, 12303, 12563, 12823, respectively;

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ccccccccccccccc) SEQ ID NOs: 10079, 10341, 10603, 12305, 12565, 12825, respectively;
ddddddddddddddd) SEQ ID NOs: 10080, 10342, 10604, 12306, 12566, 12826, respectively;
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rrrrrrrrrrrrrrr) SEQ ID NOs: 10094, 10356, 10618, 12320, 12580, 12840, respectively;
sssssssssssssss) SEQ ID NOs: 10095, 10357, 10619, 12321, 12581, 12841, respectively;
ttttttttttttttt) SEQ ID NOs: 10096, 10358, 10620, 12322, 12582, 12842, respectively;
uuuuuuuuuuuuuuu) SEQ ID NO: 10097, 10359, 10621, 12323, 12583, 12843, respectively; or
wherein the HCDR1, HCDR2, and HCDR3, respectively, comprise:
vvvvvvvvvvvvvvv) SEQ ID NOs: 1535, 2049, 2563, respectively;
wwwwwwwwwwwwwww) SEQ ID NOs: 9249, 9252, 9255, respectively;
xxxxxxxxxxxxxxx) SEQ ID NOs: 9250, 9253, 9256, respectively; or
yyyyyyyyyyyyyyy) wherein the HCDR2 and HCDR3 comprise SEQ ID NOs: 9254 and 9257, respectively.

2-4. (canceled)

5. The antibody or antigen binding fragment thereof of claim **1**, comprising a VH having an amino acid sequence with at least 90% identity to a sequence selected from SEQ ID NOs: 303, 304, 331, 9469, 9476, or 9554, and a VL having an amino acid sequence with at least 90% identity to a sequence selected from SEQ ID NOs: 558, 559, 586, 9599, 9606, or 9684.

6-7. (canceled)

8. The antibody or antigen binding fragment thereof of claim **1**, wherein the antigen binding fragment is a Fab, Fab', $F(ab')_2$, Fd, scFv, $(scFv)_2$, scFv-Fc, sdAb, VHH, or Fv fragment.

9. The antibody or antigen binding fragment thereof of claim **8**, wherein the antigen binding fragment is a scFv comprising a linker connecting the VH and VL, wherein the linker comprises an amino acid sequence selected from the group SEQ ID NOs: 9312-9315.

10-15. (canceled)

16. An isolated polynucleotide encoding the antibody or antigen binding fragment thereof of claim **1**.

17. An isolated vector comprising the polynucleotide of claim **16**.

18. An isolated host cell comprising the polynucleotide of claim **16**, and/or the vector of claim **17**.

19. A fusion protein comprising a glycoprotein G (G protein), hemagglutinin (H protein), or hemagglutinin-neuraminidase (HN protein) of the Paramyxoviridae family, or a biologically active portion thereof and at least one antibody or antigen binding fragment thereof of claim **1**, wherein the antibody or antigen binding fragment is fused to the C-terminus of the G protein or the biologically active portion thereof.

20. The fusion protein of claim **19**, wherein the antibody or antigen binding fragment thereof is fused to the G protein via a peptide linker comprising a sequence selected from the group GS, GGS, GGGS (SEQ ID NO: 14125), GGGGS (SEQ ID NO: 9294), GGGGGS (SEQ ID NO: 9292), and combinations thereof.

21-33. (canceled)

34. The fusion protein of claim **19**, wherein the G protein or a biologically active portion thereof is a Henipavirus G protein or a functionally active variant or a biologically active portion thereof comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295, or wherein the functionally active variant or a biologically active portion thereof lacks up to 40 contiguous amino acid residues at or near the N-terminus of a wild-type NiV-G protein SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295.

35-51. (canceled)

52. The fusion protein of claim **19**, in which the protein is pseudotyped onto a lentiviral particle.

53. (canceled)

54. A fusosome comprising at least one antibody or antigen binding fragment thereof that specifically binds CD4 of claim **1** and at least one fusogen.

55-58. (canceled)

59. A viral vector comprising:
i) a F protein molecule or biologically active portion thereof of the Paramyxoviridae family;
ii) an envelope glycoprotein G (G protein), hemagglutinin (H protein), or hemagglutinin-neuraminidase (HN Protein) of the Paramyxoviridae family, or a biologically active portion thereof; and
iii) at least one antibody or antigen binding fragment thereof of claim **1**,
wherein the antibody or antigen binding fragment thereof is attached to the C-terminus of the G protein or the biologically active portion thereof.

60-63. (canceled)

64. The viral vector of claim **59**, wherein the F protein or the biologically active portion thereof is a Henipavirus F protein or a functionally active variant or biologically active portion thereof comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 9259, 9265, or 9278, or fragments thereof lacking about 20 contiguous amino acid residues from the C-terminus.

65-72. (canceled)

73. The viral vector of claim **59**, wherein the F-protein or the biologically active portion thereof comprises an F1 subunit or a fusogenic portion thereof, wherein the F1 subunit comprises an amino acid sequence having at least 80% sequence identity to SEQ ID NO:9261.

74. The viral vector of claim **73**, wherein the F1 subunit is a proteolytically cleaved portion of the F0 precursor.

75-101. (canceled)

102. A method for selectively modulating the activity of CD4+ T cells, comprising contacting a viral vector according to claim **59** with cells comprising CD4+ T cells.

103-104. (canceled)

105. A method of delivering an exogenous agent to a subject, comprising administering to the subject a viral vector according to claim **59**, wherein the viral vector further comprises an exogenous agent.

106. The method of claim **105**, wherein the exogenous agent encodes a therapeutic agent or a diagnostic agent.

107-109. (canceled)

110. A method of treating cancer in a subject, comprising administering to the subject a viral vector according to claim **59**, wherein the viral vector further comprises an exogenous agent.

111. The method of claim **110**, wherein the exogenous agent encodes a therapeutic agent or a diagnostic agent.

112-114. (canceled)

115. A method of transducing a cell that expresses CD4, comprising contacting the cell with the viral vector of claim **59**.

116-126. (canceled)

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